

UNIT & DIMENSION, BASIC MATHS AND VECTOR

KEY CONCEPT

PHYSICAL QUANTITIES AND UNITS

Physical quantities :

All quantities that can be measured are called physical quantities. eg. time, length, mass, force, work done, etc. In physics we study about physical quantities and their inter relationship.

Measurement :

Measurement is the comparison of a quantity with a standard of the same physical quantity.

Classification :

Physical quantities can be classified on the following bases :

I. Based on their directional properties

- Scalars:** The physical quantities which have only magnitude but no direction are called *scalar quantities*.
Ex. mass, density, volume, time, etc.
- Vectors :** The physical quantities which have both magnitude and direction and obey laws of vector algebra are called *vector quantities*. **Ex.** displacement, force, velocity, etc.

II. Based on their dependency

- 1. Fundamental or base quantities :** A set of physical quantities which are completely independent of each other and all other physical quantities can be expressed in terms of these physical quantities is called Set of Fundamental Quantities.
- 2. Derived quantities :** The quantities which can be expressed in terms of the fundamental quantities are known as *derived quantities*. **Ex.** Speed (= distance/time), volume, acceleration, force, pressure, etc.

Physical quantities can also be classified as dimensional and dimensionless quantities or constants and variables.

- Ex.** Classify the quantities displacement, mass, force, time, speed, velocity, acceleration, moment of inertia, pressure and work under the following categories :

- (a) base and scalar (b) base and vector (c) derived and scalar (d) derived and vector

- Ans.** (a) mass, time (b) displacement (c) speed, pressure, work
(d) force, velocity, acceleration

Units of Physical Quantities

The chosen reference standard of measurement in multiples of which, a physical quantity is expressed is called the *unit* of that quantity. Four basic properties of units are :

1. They must be well defined.
2. They should be easily available and reproducible.
3. They should be invariable e.g. step as a unit of length is not invariable.
4. They should be accepted to all.

System of Units :

- 1. FPS or British Engineering system :**

In this system length, mass and time are taken as fundamental quantities and their base units are foot (ft), pound (lb) and second (s) respectively.

2. CGS or Gaussian system :

In this system the fundamental quantities are length, mass and time and their respective units are centimetre (cm), gram (g) and second (s).

3. MKS system :

In this system also the fundamental quantities are length, mass and time but their fundamental units are metre (m), kilogram (kg) and second (s) respectively.

4. International system (SI) of units :

This system is modification over the MKS system and so it is also known as *Rationalised MKS system*. Besides the three base units of MKS system four fundamental and two supplementary units are also included in this system.

Classification of Units : The units of physical quantities can be classified as follows :

1. Fundamental or base units :

The units of fundamental quantities are called *base units*. In SI there are seven base units.

SI BASE QUANTITIES AND THEIR UNITS

S.No.	Physical quantity	SI unit	Symbol
1.	Length	metre	m
2.	Mass	kilogram	kg
3.	Time	second	s
4.	Temperature	Kelvin	K
5.	Electric current	ampere	A
6.	Luminous intensity	candela	cd
7.	Amount of substance	mole	mol

2. Derived units :

The units of derived quantities or the units that can be expressed in terms of the base units are called *derived units*

Ex. Unit of speed = $\frac{\text{unit of distance}}{\text{unit of time}} = \frac{\text{metre}}{\text{second}} = \text{ms}^{-1}$

Some derived units are named in honour of great scientists.

- Unit of force – newton (N)
- Unit of frequency – hertz (Hz) etc.

UNITS OF SOME PHYSICAL QUANTITIES IN DIFFERENT SYSTEMS

Type of Physical Quantity	Physical Quantity	CGS (Originated in France)	MKS (Originated in France)	FPS (Originated in Britain)
Fundamental	Length	cm	m	ft
	Mass	g	kg	lb
	Time	s	s	s
Derived	Force	dyne	newton(N)	poundal
	Work or Energy	erg	joule(J)	ft-poundal
	Power	erg/s	watt(W)	ft-poundal/s

Dimensions :

Dimensions of a physical quantity are the powers (or exponents) to which the base quantities are raised to represent that quantity. To make it clear, consider the physical quantity force. As we shall learn later, force is equal to mass times acceleration. Acceleration is change in velocity divided by time interval. Velocity is length divided by time interval. Thus,

force = mass $\times$ acceleration

$$= \text{mass} \times \frac{\text{velocity}}{\text{time}} = \text{mass} \times \frac{\text{length} / \text{time}}{\text{time}} = \text{mass} \times \text{length} \times (\text{time})^{-2}$$

Thus, the dimensions of force are 1 in mass, 1 in length and -2 in time. The dimensions in all other base quantities are zero.

1. Dimensional formula :

The physical quantity that is expressed in terms of the base quantities is enclosed in square brackets to remind that the equation is among the dimensions and not among the magnitudes. Thus above equation may be written as $[\text{force}] = \text{MLT}^{-2}$.

Such an expression for a physical quantity in terms of the base quantities is called the dimensional formula. Thus, the dimensional formula of force is MLT^{-2} . The two versions given below are equivalent and are used interchangeably.

- (a) The dimensional formula of force is MLT^{-2} .
(b) The dimensions of force are 1 in mass, 1 in length and -2 in time.

The *dimensional formula* of any physical quantity is that expression which represents how and which of the base quantities are included in that quantity.

Ex. Dimensional formula of mass is $[M^1 L^0 T^0]$ and that of speed (= distance/time) is $[M^0 L^1 T^{-1}]$

2. Applications of dimensional analysis :

- (i)** To convert a physical quantity from one system of units to the other :

This is based on a fact that *magnitude of a physical quantity remains same whatever system is used for measurement*

i.e. magnitude = numeric value (n) $\times$

unit (u) = constant or $n_1 u_1 = n_2 u_2$

So if a quantity is represented by $[M^a L^b T^c]$

$$\text{Then } \boxed{n_2 = n_1 \left[\frac{u_1}{u_2} \right] = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c}$$

n_2 = numerical value in II system
 n_1 = numerical value in I system
 M_1 = unit of mass in I system
 M_2 = unit of mass in II system
 L_1 = unit of length in I system
 L_2 = unit of length in II system
 T_1 = unit of time in I system
 T_2 = unit of time in II system

Ex. $1\text{ m} = 100\text{ cm} = 3.28\text{ ft} = 39.4\text{ inch}$

(SI) (CGS) (FPS)

Ex. The acceleration due to gravity is 9.8 m s^{-2} . Give its value in ft s^{-2}

Sol. As $1\text{ m} = 3.28\text{ ft}$ $\therefore 9.8\text{ m/s}^2 = 9.8 \times 3.28\text{ ft/s}^2 = 32.14\text{ ft/s}^2 \approx 32\text{ ft/s}^2$

Ex. Convert 1 newton (SI unit of force) into dyne (CGS unit of force)

Sol. The dimensional equation of force is $[F] = [M^1 L^1 T^{-2}]$

Therefore if n_1, u_1 , and n_2, u_2 corresponds to SI & CGS units respectively, then

$$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^{-1} \left[\frac{L_1}{L_2} \right]^{-1} \left[\frac{T_1}{T_2} \right]^{-2} = 1 \left[\frac{\text{kg}}{\text{g}} \right] \left[\frac{\text{m}}{\text{cm}} \right] \left[\frac{\text{s}}{\text{s}} \right]^{-2} = 1 \times 1000 \times 100 \times 1 = 10^5$$

$$\therefore 1 \text{ newton} = 10^5 \text{ dyne.}$$

Q. The value of Gravitational constant G in MKS system is $6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$.
What will be its value in CGS system ?

Ans. $6.67 \times 10^{-8} \text{ cm}^3/\text{g s}^2$

(ii) To check the dimensional correctness of a given physical relation :

If in a given relation, the terms on both the sides have the same dimensions, then the relation is dimensionally correct. This is known as the *principle of homogeneity of dimensions*.

Ex. Check the accuracy of the relation $T = 2\pi\sqrt{\frac{L}{g}}$ for a simple pendulum using dimensional analysis.

Sol. The dimensions of LHS = the dimension of $T = [M^0 L^0 T^1]$

The dimensions of RHS = $\left(\frac{\text{dimensions of length}}{\text{dimensions of acceleration}} \right)^{1/2}$ ($\because 2\pi$ is a dimensionless constant)

$$= \left(\frac{L}{LT^{-2}} \right)^{1/2} = (T^2)^{1/2} = [T] = [M^0 L^0 T^1]$$

Since the dimensions are same on both the sides, the relation is correct.

(iii) To derive relationship between different physical quantities :

Using the same principle of homogeneity of dimensions new relations among physical quantities can be derived if the dependent quantities are known.

Ex. It is known that the time of revolution T of a satellite around the earth depends on the universal gravitational constant G , the mass of the earth M , and the radius of the circular orbit R . Obtain an expression for T using dimensional analysis.

Sol. We have $[T] = [G]^a [M]^b [R]^c$
 $[M]^0 [L]^0 [T]^1 = [M]^{-a} [L]^{3a} [T]^{-2a} \times [M]^b \times [L]^c = [M]^{b-a} [L]^{c+3a} [T]^{-2a}$

Comparing the exponents

$$\text{For } [T] : 1 = -2a \Rightarrow a = -\frac{1}{2}$$

$$\text{For } [M] : 0 = b - a \Rightarrow b = a = -\frac{1}{2}$$

$$\text{For } [L] : 0 = c + 3a \Rightarrow c = -3a = \frac{3}{2}$$

$$\text{Putting the values we get } T = G^{-1/2} M^{-1/2} R^{3/2} = \sqrt{\frac{R^3}{GM}}$$

$$\text{So the actual expression is } T = 2\pi\sqrt{\frac{R^3}{GM}}$$

Limitations of this method :

- In Mechanics the formula for a physical quantity depending on more than three physical quantities cannot be derived. It can only be checked.
- This method can be used only if the dependency is of multiplication type. The formulae containing exponential, trigonometrical and logarithmic functions can't be derived using this method. Formulae containing more than one term which are added or subtracted like $s = ut + at^2/2$ also can't be derived.

- (iii) The relation derived from this method gives no information about the dimensionless constants.
- (iv) If dimensions are given, physical quantity may not be unique as many physical quantities have the same dimensions.
- (v) It gives no information whether a physical quantity is a scalar or a vector.

Units and Dimensions of Physical Quantities			
Quantity	Common Symbol	SI unit	Dimension
Displacement	s	METRE (m)	L
Mass	m, M	KILOGRAM (kg)	M
Time	t	SECOND (s)	T
Area	A	m ²	L ²
Volume	V	m ³	L ³
Density	ρ	kg/m ³	M/L ³
Velocity	v, u	m/s	L/T
Acceleration	a	m/s ²	L/T ²
Force	F	newton (N)	ML/T ²
Work	W	joule (J) (=N – m)	ML ² /T ²
Energy	E, U, K	joule (J)	ML ² /T ²
Power	P	watt (W) (=J/s)	ML ² /T ³
Momentum	p	kg-m/s	ML/T
Gravitational constant	G	N-m ² /kg ²	L ³ /MT ²
Angle	θ , ϕ	radian	
Angular velocity	ω	radian/s	T ⁻¹
Angular acceleration	α	radian/s ²	T ⁻²
Angular momentum	L	kg-m ² /s	ML ² /T
Moment of inertia	I	kg-m ²	ML ²
Torque	τ	N-m	ML ² /T ²
Angular frequency	ω	radian/s	T ⁻¹
Frequency	ν	hertz (Hz)	T ⁻¹
Period	T	s	T
Young's modulus	Y	N/m ²	M/LT ²
Bulk modulus	B	N/m ²	M/LT ²
Shear modulus	η	N/m ²	M/LT ²
Surface tension	S	N/m	M/T ²
Coefficient of viscosity	η	N-s/m ²	M/LT
Pressure	P, p	N/m ² , Pa	M/LT ²
Wavelength	λ	m	L
Intensity of wave	I	W/m ²	M/T ³
Temperature	T	KELVIN (K)	K
Specific heat capacity	c	J/kg–K	L ² /T ² K
Stefan's constant	σ	W/m ² –K ⁴	M/T ³ K ⁴
Heat	Q	J	ML ² /T ²
Thermal conductivity	K	W/m–K	ML/T ³ K

Basic Mathematics used in physics

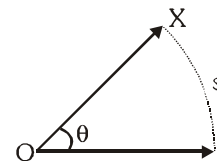
Plane-angle

It is measure of change in direction.

If a line rotates in a plane about one of its ends, the other end sweeps an arc.

Angle (θ) between two orientation of the line is defined by ratio of the arc

$$\text{length}(s) \text{ to length of the line}(r) \quad \theta = \frac{s}{r} \text{ radian}$$

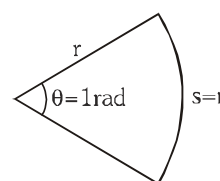


Angles measured in anticlockwise and clockwise directions are usually taken positive and negative respectively.

Angle is measured in radians (rad) or degrees. One radian is the angle subtended at the centre of a circle by an arc of the circle, whose length is equal to the radius of the circle.

$$\pi \text{ rad} = 180^\circ \quad \pi = 3.1415 = \frac{22}{7}$$

$$1^\circ = 60' \text{ (minute)}, \quad 1' \text{ (minute)} = 60'' \text{ (sec)}$$

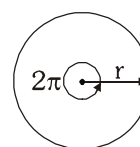


Example

Write expression for circumference of a circle of radius 'r'.

Solution

$$s = (\text{Total angle about a point}) r = 2\pi r$$

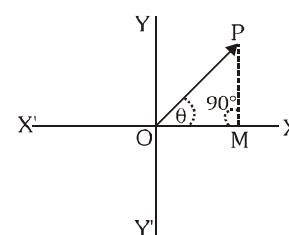


Trigonometrical ratios (or T-ratios)

Let two fixed lines XOX' and YOY' intersecting at right angles to each other at point O.

- Point O is called origin.
- Line XOX' is known as x-axis and YOY' as y-axis.
- Regions XOY, YOX', X'OY' and Y'OX are called I, II, III and IV quadrant respectively.

Consider a line OP making angle θ with OX as shown. Line PM is perpendicular drawn from P on OX. In the right angled triangle OPM, side OP is called hypotenuse, the side OM adjacent to angle θ is called base and the side PM opposite to angle θ is called the perpendicular.



Following ratios of the sides of a right angled triangle are known as trigonometrical ratios or T-ratio

$$\sin \theta = \frac{\text{perpendicular}}{\text{hypotenuse}} = \frac{MP}{OP} \quad \cos \theta = \frac{\text{base}}{\text{hypotenuse}} = \frac{OM}{OP} \quad \tan \theta = \frac{\text{perpendicular}}{\text{base}} = \frac{MP}{OM}$$

$$\cot \theta = \frac{\text{base}}{\text{perpendicular}} = \frac{OM}{MP} \quad \sec \theta = \frac{\text{hypotenuse}}{\text{base}} = \frac{OP}{OM} \quad \text{cosec } \theta = \frac{\text{hypotenuse}}{\text{perpendicular}} = \frac{OP}{MP}$$

$$\text{cosec } \theta = \frac{1}{\sin \theta} \quad \sec \theta = \frac{1}{\cos \theta} \quad \cot \theta = \frac{1}{\tan \theta}$$

Some trigonometric identities

$$\sin^2 \theta + \cos^2 \theta = 1 \quad 1 + \tan^2 \theta = \sec^2 \theta \quad 1 + \cot^2 \theta = \text{cosec}^2 \theta$$

Example

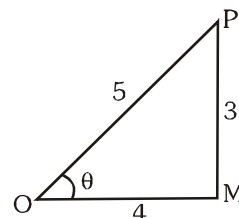
Given $\sin \theta = \frac{3}{5}$. Find all the other T-ratios, if θ lies in the first quadrant.

Solution

In ΔOMP , $\sin \theta = \frac{3}{5}$ So $MP = 3$ and $OP = 5 \therefore OM = \sqrt{(5)^2 - (3)^2} = \sqrt{25 - 9} = \sqrt{16} = 4$

$$\text{Now } \cos \theta = \frac{OM}{OP} = \frac{4}{5} \quad \tan \theta = \frac{MP}{OM} = \frac{3}{4} \quad \cot \theta = \frac{OM}{MP} = \frac{4}{3}$$

$$\sec \theta = \frac{OP}{OM} = \frac{5}{4} \quad \operatorname{cosec} \theta = \frac{OP}{MP} = \frac{5}{3}$$



T-ratios of some commonly used angles

Angle (θ)	0 rad	$\frac{\pi}{6}$ rad	$\frac{\pi}{4}$ rad	$\frac{\pi}{3}$ rad	$\frac{\pi}{2}$ rad
	0°	30°	45°	60°	90°
$\sin \theta$	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
$\cos \theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
$\tan \theta$	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	∞

$\sin(90^\circ - \theta) = \cos \theta$ $\cos(90^\circ - \theta) = \sin \theta$ $\tan(90^\circ - \theta) = \cot \theta$	$\sin(180^\circ - \theta) = \sin \theta$ $\cos(180^\circ - \theta) = -\cos \theta$ $\tan(180^\circ - \theta) = -\tan \theta$	$\sin(360^\circ - \theta) = -\sin \theta$ $\cos(360^\circ - \theta) = \cos \theta$ $\tan(360^\circ - \theta) = -\tan \theta$
$\sin(90^\circ + \theta) = \cos \theta$ $\cos(90^\circ + \theta) = -\sin \theta$ $\tan(90^\circ + \theta) = -\cot \theta$	$\sin(180^\circ + \theta) = -\sin \theta$ $\cos(180^\circ + \theta) = -\cos \theta$ $\tan(180^\circ + \theta) = \tan \theta$	$\sin(-\theta) = -\sin \theta$ $\cos(-\theta) = \cos \theta$ $\tan(-\theta) = -\tan \theta$

- When θ is very small we can use following approximations : $\cos \theta \approx 1$

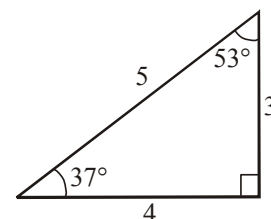
$$\left. \begin{array}{l} \sin \theta \approx \theta \\ \tan \theta \approx \theta \end{array} \right\} \text{ If } \theta \text{ is in radians}$$

$$\tan \theta \approx \sin \theta.$$

- In the given right angled triangle we have very commonly used T-ratios

$$\sin 37^\circ = \frac{3}{5} \quad \cos 37^\circ = \frac{4}{5} \quad \tan 37^\circ = \frac{3}{4}$$

$$\sin 53^\circ = \frac{4}{5} \quad \cos 53^\circ = \frac{3}{5} \quad \tan 53^\circ = \frac{4}{3}$$



Example

Find the value of
(iv) $\cos 120^\circ$

(i) $\cos (-60^\circ)$

(ii) $\tan 210^\circ$

(iii) $\sin 300^\circ$

Solution

(i) $\cos (-60^\circ) = \cos 60^\circ = \frac{1}{2}$

(ii) $\tan 210^\circ = \tan (180^\circ + 30^\circ) = \tan 30^\circ = \frac{1}{\sqrt{3}}$

(iii) $\sin 300^\circ = \sin (270^\circ + 30^\circ) = -\cos 30^\circ = -\frac{\sqrt{3}}{2}$

(iv) $\cos 120^\circ = \cos (180^\circ - 60^\circ) = -\cos 60^\circ = -\frac{1}{2}$

VECTORS

Precise description of laws of physics and physical phenomena requires expressing them in form of mathematical equations. In doing so we encounter several physical quantities, some of them have only magnitude and other have direction in addition to magnitude. Quantities of the former kind are referred as scalars and the latter as vectors and mathematical operations with vectors are collectively known as vector analysis.

Vectors

A vector has both magnitude and sense of direction, and follows triangle law of vector addition. For example, displacement, velocity, and force are vectors.

Vector quantities are usually denoted by putting an arrow over the corresponding letter, as $\vec{A}$ or $\vec{a}$. Sometimes in print work (books) vector quantities are usually denoted by boldface letters as **A** or **a**.

Magnitude of a vector $\vec{A}$ is a positive scalar and written as $|\vec{A}|$ or A .

Geometrical Representation of Vectors.

A vector is represented by a directed straight line, having the magnitude and direction of the quantity represented by it.

e.g. if we want to represent a force of 5 N acting 45° N of E

(i) We choose direction coordinates.

(ii) We choose a convenient scale like $1 \text{ cm} \equiv 1 \text{ N}$

(iii) We draw a line of length equal in magnitude and in the direction of vector to the chosen quantity.

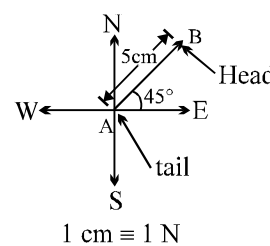
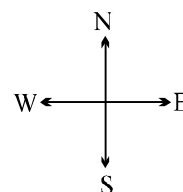
(iv) We put arrow in the direction of vector.

$\vec{AB}$

Magnitude of vector:

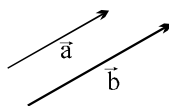
$$|\vec{AB}| = 5 \text{ N}$$

By definition magnitude of a vector quantity is scalar and is always positive.



3. TERMINOLOGY OF VECTORS

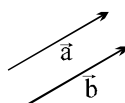
Parallel vector: If two vectors have same direction, they are parallel to each other. They may be located anywhere in the space.



Antiparallel vectors: When two vectors are in opposite direction they are said to be antiparallel vectors.

Equality of vectors: When two vectors have equal magnitude and are in same direction and represent the same physical quantity, they are equal.

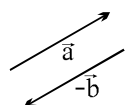
i.e. $\vec{a} = \vec{b}$



Thus when two parallel vectors have same magnitude they are equal. (Their initial point & terminal point may not be same)

Negative of a vector: When a vector have equal magnitude and is in opposite direction, it is said to be negative vector of the former.

i.e. $\vec{a} = -\vec{b}$ or $\vec{b} = -\vec{a}$

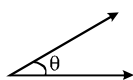


Thus when two antiparallel vectors have same magnitude they are negative of each other.

Remark : Vector shifting is allowed without change in their direction.

2. Angle Between two Vectors

It is the smaller angle formed when the initial points or the terminal points of the two vectors are brought together. It should be noted that $0^\circ \leq \theta \leq 180^\circ$.

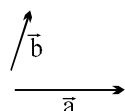


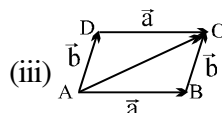
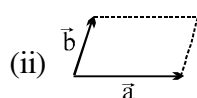
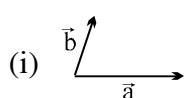
3. Addition Of Vectors:

Parallelogram law of addition:

Steps:

- (i) Keep two vectors such that their tails coincide.
- (ii) Draw parallel vectors to both of them considering both of them as sides of a parallelogram.
- (iii) Then the diagonal drawn from the point where tails coincide represents the sum of two vectors, with its tail at point of coincidence of the two vectors.

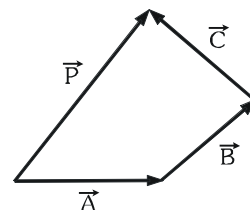




$$\vec{AC} = \vec{a} + \vec{b}$$

Addition of more than two Vectors

The triangle law can be extended to define addition of more than two vectors. Accordingly, if vectors to be added are drawn in head to tail fashion, resultant is defined by a vector drawn from the tail of the first vector to the head of the last vector. This is also known as the ***polygon rule for vector addition***.



Operation of addition of three vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$ and their resultant $\vec{P}$ are shown in figure.

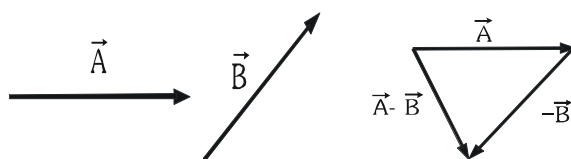
$$\vec{A} + \vec{B} + \vec{C} = \vec{P}$$

Here it is not necessary that three or more vectors and their resultant are coplanar. In fact, the vectors to be added and their resultant may be in different planes. However if all the vectors to be added are coplanar, their resultant must also be in the same plane containing the vectors.

Subtraction of Vectors

A vector opposite in direction but equal in magnitude to another vector $\vec{A}$ is known as negative vector of $\vec{A}$. It is written as $-\vec{A}$. Addition of a vector and its negative vector results a vector of zero magnitude, which is known as a null vector. A null vector is denoted by arrowed zero ($\vec{0}$).

The idea of negative vector explains operation of subtraction as addition of negative vector. Accordingly to subtract a vector from another consider vectors $\vec{A}$ and $\vec{B}$ shown in the figure. To subtract $\vec{B}$ from $\vec{A}$, the negative vector $-\vec{B}$ is added to $\vec{A}$ according to the triangle law as shown in figure-II.



If two vectors $\vec{a}$ & $\vec{b}$ are represented by $\vec{OA}$ & $\vec{OB}$ then their sum $\vec{a} + \vec{b}$ is a vector represented by $\vec{OC}$, where OC is the diagonal of the parallelogram OACB.

- $\vec{a} + \vec{b} = \vec{b} + \vec{a}$ (commutative)
- $(\vec{a} + \vec{b}) + \vec{c} = \vec{a} + (\vec{b} + \vec{c})$ (associativity)
- $\vec{a} + \vec{0} = \vec{a} = \vec{0} + \vec{a}$
- $\vec{a} + (-\vec{a}) = \vec{0} = (-\vec{a}) + \vec{a}$
- $|\vec{a} + \vec{b}| \leq |\vec{a}| + |\vec{b}|$
- $|\vec{a} - \vec{b}| \geq ||\vec{a}| - |\vec{b}||$
- $|\vec{a} \pm \vec{b}| = \sqrt{|\vec{a}|^2 + |\vec{b}|^2 \pm 2|\vec{a}||\vec{b}|\cos\theta}$ where θ is the angle between the vectors

Some Important Results :

(1) If $\theta = 0^\circ \Rightarrow \vec{a} \parallel \vec{b}$

then, $|\vec{R}| = |\vec{a}| + |\vec{b}|$ & $|\vec{R}|$ is maximum

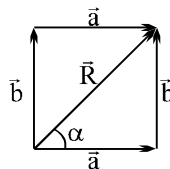
(2) If $\theta = \pi \Rightarrow \vec{a}$ anti $\parallel \vec{b}$

then, $|\vec{R}| = ||\vec{a}| - |\vec{b}||$ & $|\vec{R}|$ is minimum

(3) If $\theta = \pi/2 \Rightarrow \vec{a} \perp \vec{b}$

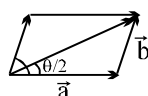
$$R = \sqrt{a^2 + b^2}$$

& $\tan \alpha = b/a$ (α is angle made by $\vec{R}$ with $\vec{a}$)



(4) $|\vec{a}| = |\vec{b}| = a$

$$|\vec{R}| = 2a \cos \theta/2 \text{ \& } \alpha = \theta/2$$



(5) If $|\vec{a}| = |\vec{b}| = a$ & $\theta = 120^\circ$

then $|\vec{R}| = a$

4. Multiplication Of A Vector By A Scalar:

If $\vec{a}$ is a vector & m is a scalar, then $m\vec{a}$ is a vector parallel to $\vec{a}$ whose modulus is $|m|$ times that of $\vec{a}$.

This multiplication is called **SCALAR MULTIPLICATION**. If $\vec{a}$ and $\vec{b}$ are vectors & m, n are scalars, then:

$$m(\vec{a}) = (\vec{a}) \quad m = m\vec{a}$$

$$m(n\vec{a}) = n(m\vec{a}) = (mn)\vec{a}$$

$$(m+n)\vec{a} = m\vec{a} + n\vec{a}$$

$$m(a + \vec{b}) = m\vec{a} + m\vec{b}$$

Resolution of a Vector into Components

Following laws of vector addition, a vector can be represented as a sum of two (in two-dimensional space) or three (in three-dimensional space) vectors each along predetermined directions. These directions are called axes and parts of the original vector along these axes are called components of the vector.

UNIT VECTOR:

A unit vector is a vector of magnitude of 1, with no units. Its only purpose is to point, i.e. to describe a direction in space.

A unit vector in direction of vector $\vec{A}$ is represented as $\hat{A}$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|}$$

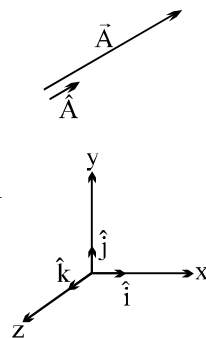
or $\vec{A}$ can be expressed in terms of a unit vector in its direction i.e. $\vec{A} = |\vec{A}| \hat{A}$

Unit Vectors along three coordinates axes:—

unit vector along x-axis is $\hat{i}$

unit vector along y-axis is $\hat{j}$

unit vector along z-axis is $\hat{k}$



Cartesian components in two dimensions

If a vector is resolved into its components along mutually perpendicular directions, the components are called Cartesian or rectangular components.

In figure is shown, a vector $\vec{A}$ resolved into its Cartesian components $\vec{A}_x$ and $\vec{A}_y$ along the x and y-axis. Magnitudes A_x and A_y of these components are given by the following equation.

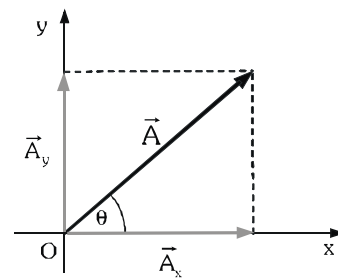
$$A_x = A \cos \theta \quad \text{and} \quad A_y = A \sin \theta$$

$$\vec{A} = A_x \hat{i} + A_y \hat{j}$$

$$A = \sqrt{A_x^2 + A_y^2}$$

Here $\hat{i}$ and $\hat{j}$ are the unit vectors for x and y coordinates respectively.

Mathematical operations e.g. addition, subtraction, differentiation and integration can be performed independently on these components. This is why in most of the problems use of Cartesian components becomes desirable.

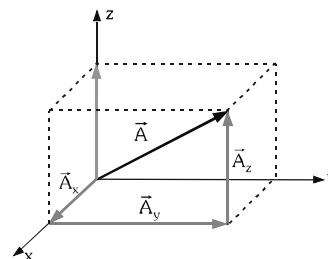


Cartesian components in three dimensions

A vector $\vec{A}$ resolved into its three Cartesian components one along each of the directions x, y, and z-axis is shown in the figure.

$$\vec{A} = \vec{A}_x + \vec{A}_y + \vec{A}_z = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2}$$



Product of Vectors

In all physical situation, whose description involve product of two vectors, only two categories are observed. One category where product is also a vector involves multiplication of magnitudes of two vectors and sine of the angle between them, while the other category where product is a scalar involves multiplication of magnitudes of two vectors and cosine of the angle between them. Accordingly, we define two kinds of product operation. The former category is known as vector or cross product and the latter category as scalar or dot product.

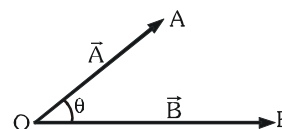
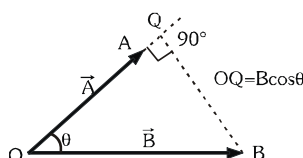
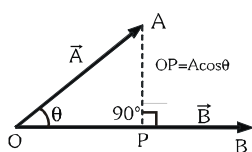
Scalar or dot product of two vectors

The scalar product of two vectors $\vec{A}$ and $\vec{B}$ equals to the product of their magnitudes and the cosine of the angle θ between them.

$$\vec{A} \cdot \vec{B} = AB \cos \theta = OA \cdot OB \cdot \cos \theta$$

The above equation can also be written in the following ways.

$$\vec{A} \cdot \vec{B} = (A \cos \theta) B = OP \cdot OB \quad \vec{A} \cdot \vec{B} = A (B \cos \theta) = OA \cdot OQ$$



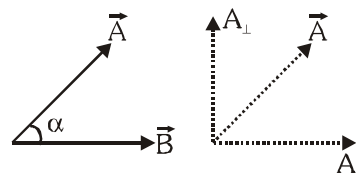
Above two equations and figures, suggest a scalar product as product of magnitude of the one vector and magnitude of the component of another vector in the direction of the former vector.

KEY POINTS

- Dot product of two vectors is commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$
- If two vectors are perpendicular, their dot product is zero. $\vec{A} \cdot \vec{B} = 0$, if $\vec{A} \perp \vec{B}$
- Dot product of a vector by itself is known as self-product. $\vec{A} \cdot \vec{A} = A^2 \Rightarrow A = \sqrt{\vec{A} \cdot \vec{A}}$
- The angle between the vectors $\theta = \cos^{-1} \left(\frac{\vec{A} \cdot \vec{B}}{AB} \right)$

- (a) Component of $\vec{A}$ in direction of $\vec{B}$

$$\vec{A}_{\parallel} = (|\vec{A}| \cos \theta) \hat{B} = |\vec{A}| \left(\frac{\vec{A} \cdot \vec{B}}{|\vec{A}| |\vec{B}|} \right) \hat{B} = \left(\frac{\vec{A} \cdot \vec{B}}{|\vec{B}|} \right) \hat{B} = (\vec{A} \cdot \hat{B}) \hat{B}$$



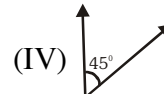
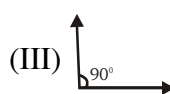
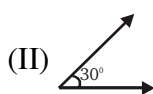
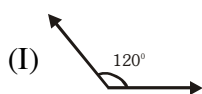
- (b) Component of $\vec{A}$ perpendicular to $\vec{B}$: $\vec{A}_{\perp} = \vec{A} - \vec{A}_{\parallel}$

- Dot product of Cartesian unit vectors: $\hat{i} \cdot \hat{i} = \hat{j} \cdot \hat{j} = \hat{k} \cdot \hat{k} = 1$
 $\hat{i} \cdot \hat{j} = \hat{j} \cdot \hat{k} = \hat{k} \cdot \hat{i} = 0$
- If $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$, their dot product is given by

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

Solved Examples

- Two displacement vectors of same magnitude are arranged in the following manner



Magnitude of resultant is maximum for

(A) case I

(B) case II

(C) case III

(D) case IV

Ans. (B)

Sol. Magnitude of Resultant of $\vec{A}$ and $\vec{B} = \sqrt{A^2 + B^2 + 2AB \cos \theta}$ which is maximum for $\theta = 30^\circ$

- Two vectors $\vec{P}$ and $\vec{Q}$ are added, the magnitude of resultant is 15 units. If $\vec{Q}$ is reversed and added to $\vec{P}$ resultant has a magnitude $\sqrt{113}$ units. The resultant of $\vec{P}$ and a vector perpendicular to $\vec{P}$ and equal in magnitude to $\vec{Q}$ has a magnitude

(A) 13 units

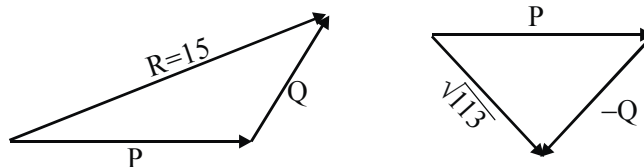
(B) 17 units

(C) 19 units

(D) 20 units

Ans. (A)

- Sol.** $P^2 + Q^2 + 2PQ\cos\theta = 225 \dots(i)$
 $P^2 + Q^2 - 2PQ\cos\theta = 113 \dots(ii)$
 By adding (i) & (ii) $2(P^2 + Q^2) = 338$
 $P^2 + Q^2 = 169 \Rightarrow \sqrt{P^2 + Q^2} = 13$

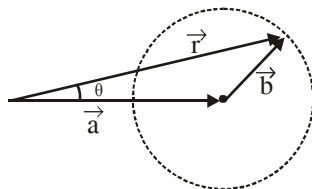


3. Three forces are acting on a body to make it in equilibrium, which set **can not** do it?
 (A) 3 N, 3 N, 7 N (B) 2 N, 3 N, 6 N (C) 2 N, 1 N, 1 N (D) 8 N, 6 N, 1 N

Ans. (A, B, D)

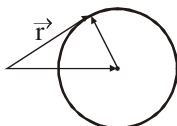
Sol. They must form a triangle. ($a + b \geq c$)

4. Keeping one vector constant, if direction of other to be added in the first vector is changed continuously, tip of the resultant vector describes a circle. In the following figure vector $\vec{a}$ is kept constant. When vector $\vec{b}$ added to $\vec{a}$ changes its direction, the tip of the resultant vector $\vec{r} = \vec{a} + \vec{b}$ describes circle of radius b with its center at the tip of vector $\vec{a}$. Maximum angle between vector $\vec{a}$ and the resultant $\vec{r} = \vec{a} + \vec{b}$ is



- (A) $\tan^{-1}\left(\frac{b}{r}\right)$ (B) $\tan^{-1}\left(\frac{b}{\sqrt{a^2 - b^2}}\right)$ (C) $\cos^{-1}(r/a)$ (D) $\cos^{-1}(a/r)$

Ans. (A, B, C)

Sol.  Angle between $\vec{r}$ and $\vec{b}$ is maximum when $\vec{r}$ is tangent to circle.

5. If $\vec{A} = 2\hat{i} + \hat{j} + \hat{k}$ and $\vec{B} = 10\hat{i} + 5\hat{j} + 5\hat{k}$, if the magnitude of component of $(\vec{B} - \vec{A})$ along $\vec{A}$ is $4\sqrt{x}$. Then x will be.

Ans. 6

Sol. $\vec{r} = \vec{B} - \vec{A} = 4(2\hat{i} + \hat{j} + \hat{k})$

$$r \cos \theta = \frac{\vec{r} \cdot \vec{A}}{|\vec{A}|} = \frac{4(4+1+1)}{\sqrt{6}} = 4\sqrt{6}$$

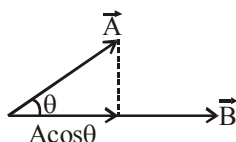
$$x = 6$$

6. The component of $\vec{A} = \hat{i} + \hat{j} + 5\hat{k}$ perpendicular to $\vec{B} = 3\hat{i} + 4\hat{j}$ is

- (A) $-\frac{4}{25}\hat{i} + \frac{3}{25}\hat{j} + 5\hat{k}$ (B) $-\frac{8}{25}\hat{i} - \frac{6}{25}\hat{j} + 5\hat{k}$ (C) $\frac{4}{25}\hat{i} - \frac{3}{25}\hat{j} + 5\hat{k}$ (D) $+\frac{8}{25}\hat{i} - \frac{6}{25}\hat{j} + 5\hat{k}$

Ans. (C)

Sol.



$$\vec{A}_{\parallel} = A \cos \theta = A \left(\frac{\vec{A} \cdot \vec{B}}{AB} \right)$$

$$= \frac{\vec{A} \cdot \vec{B}}{B} = \frac{3+4}{5} = \frac{7}{5}$$

$$\vec{A}_{\parallel} = \frac{7}{5} \left(\frac{3\hat{i} + 4\hat{j}}{5} \right) = \frac{7}{25} (3\hat{i} + 4\hat{j})$$

$$\vec{A}_{\parallel} = \frac{21}{25} \hat{i} + \frac{28}{25} \hat{j}$$

$$\vec{A}_{\perp} = (\hat{i} + \hat{j} + 5\hat{k}) - \left(\frac{21}{25} \hat{i} + \frac{28}{25} \hat{j} \right)$$

$$= \frac{4}{25} \hat{i} - \frac{3}{25} \hat{j} + 5\hat{k}$$

ALGEBRA : SOME USEFUL FORMULAE

Quadratic equation and its solution

An algebraic equation of second order (highest power of the variable is equal to 2) is called a quadratic equation. General quadratic equation is $ax^2 + bx + c = 0$. The general solution of the above quadratic

equation or value of variable x is $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\Rightarrow x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ and } x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

Example

Solve $2x^2 + 5x - 12 = 0$

Solution

By comparison with the standard quadratic equation $a = 2$, $b = 5$ and $c = -12$

$$x = \frac{-5 \pm \sqrt{(5)^2 - 4 \times 2 \times (-12)}}{2 \times 2} = \frac{-5 \pm \sqrt{121}}{4} = \frac{-5 \pm 11}{4} = \frac{+6}{4}, \frac{-16}{4} \Rightarrow x = \frac{3}{2}, -4$$

Binomial approximation

In case, x is very small, then terms containing higher powers of x can be neglected. In such a case,

$$(1 + x)^n = 1 + nx$$

$$\text{Also } (1 + x)^{-n} = 1 - nx \text{ and } (1 - x)^n = 1 - nx \text{ and } (1 - x)^{-n} = 1 + nx$$

Exponential Expansion

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \text{ and } e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \dots$$

Componendo and dividendo theorem :

If $\frac{p}{q} = \frac{a}{b}$ then by componendo and dividendo theorem $\frac{p+q}{p-q} = \frac{a+b}{a-b}$

Determinant

$$D = \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc, \text{ For example } \begin{vmatrix} -3 & 3 \\ -5 & 1 \end{vmatrix} = 12, \quad \begin{vmatrix} 2 & -4 \\ -3 & 3 \end{vmatrix} = -6$$

$$D = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} = a_1 \begin{vmatrix} b_2 & b_3 \\ c_2 & c_3 \end{vmatrix} - a_2 \begin{vmatrix} b_1 & b_3 \\ c_1 & c_3 \end{vmatrix} + a_3 \begin{vmatrix} b_1 & b_2 \\ c_1 & c_2 \end{vmatrix}$$

Example

$$\begin{vmatrix} 5 & 4 & 3 \\ 2 & 1 & 6 \\ 7 & 8 & 9 \end{vmatrix} = 5 \begin{vmatrix} 1 & 6 \\ 8 & 9 \end{vmatrix} - 4 \begin{vmatrix} 2 & 6 \\ 7 & 9 \end{vmatrix} + 3 \begin{vmatrix} 2 & 1 \\ 7 & 8 \end{vmatrix} = 5(9 - 48) - 4(18 - 42) + 3(16 - 7) = -72$$

Logarithm

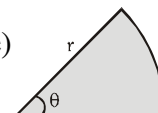
$$\log_e x = \ln x \text{ (base } e) \quad \log x = \log_{10} x \text{ (base } 10)$$

(a) Product formula $\log mn = \log m + \log n$ (b) Quotient formula $\log \frac{m}{n} = \log m - \log n$

(c) Power formula $\log m^n = n \log m$

GEOMETRY : SOME USEFUL FORMULAE**Formulae for determination of area :**

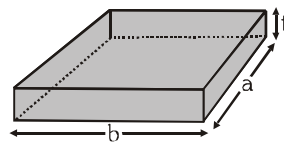
- Area of a square = (side)²
- Area of rectangle = length $\times$ breadth
- Area of a triangle = $(1/2) \times \text{base} \times \text{height}$
- Area of a trapezoid = $(1/2) \times (\text{distance between parallel sides}) \times (\text{sum of parallel sides})$
- Area enclosed by a circle = πr^2 (where r = radius)
- Area of a sector a circle = $\frac{1}{2} \theta r^2$ (where r = is radius and θ is angle subtended at a centre)
- Area of ellipse = πab (where a and b are semi major and semi minor axis respectively)
- Surface area of a sphere = $4\pi r^2$ (where r = radius)
- Area of a parallelogram = base $\times$ height
- Area of curved surface of cylinder = $2\pi r\ell$ (where r = radius and ℓ = length)



- Area of whole surface of cylinder = $2\pi r(r + \ell)$ (where ℓ = length)
- Surface area of a cube = $6(\text{side})^2$
- Total surface area of a cone = $\pi r^2 + \pi r \ell$ [where $\pi r \ell = \pi r \sqrt{r^2 + h^2}$ = lateral area (h=height)]

Formulae for determination of volume :

- Volume of a rectangular slab = length $\times$ breadth $\times$ height = abt
- Volume of a cube = $(\text{side})^3$
- Volume of a sphere = $\frac{4}{3} \pi r^3$ (where r = radius)
- Volume of a cylinder = $\pi r^2 \ell$ (where r = radius and ℓ = length)
- Volume of a cone = $\frac{1}{3} \pi r^2 h$ (where r = radius and h = height)



EXERCISE (S-1)

Units & Dimensions

1. A particle is in a unidirectional potential field where the potential energy (U) of a particle depends on the x -coordinate given by $U_x = k(1 - \cos ax)$ & k and ' a ' are constants. Find the physical dimensions of ' a ' & k .

UD0001

2. The equation for the speed of sound in a gas states that $v = \sqrt{\gamma k_B T / m}$. Speed v is measured in m/s, γ is a dimensionless constant, T is temperature in kelvin (K), and m is mass in kg. Find the SI unit for the Boltzmann constant, k_B ?

UD0002

3. The time period (T) of a spring mass system depends upon mass (m) & spring constant (k) & length of the spring (ℓ) $\left[k = \frac{\text{Force}}{\text{length}} \right]$. Find the relation among T , m , ℓ & k using dimensional method.

UD0003

4. The distance moved by a particle in time t from centre of a ring under the influence of its gravity is given by $x = a \sin \omega t$, where a & ω are constants. If ω is found to depend on the radius of the ring (r), its mass (m) and universal gravitational constant (G). Using dimensional analysis find an expression for ω in terms of r , m and G .

UD0004

5. A satellite is orbiting around a planet. Its orbital velocity (v_0) is found to depend upon

(A) Radius of orbit (R)(B) Mass of planet (M)(C) Universal gravitation constant (G)Using dimensional analysis find an expression relating orbital velocity (v_0) to the above physical quantities.

UD0005

6. Assume that the largest stone of mass ' m ' that can be moved by a flowing river depends upon the velocity of flow v , the density d & the acceleration due to gravity g . If ' m ' varies as the K^{th} power of the velocity of flow, then find the value of K .

UD0006

7. Given $\vec{F} = \frac{\vec{a}}{t}$ where symbols have their usual meaning. The dimensions of a is.

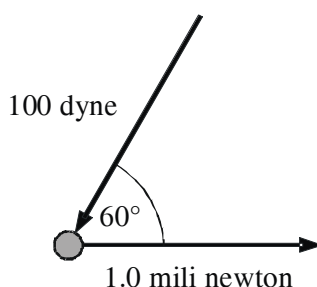
UD0007

Addition of vectors

8. A block is applied two forces of magnitude 5N each. One force is acting towards East and the other acting along 60° North of East. The resultant of the two forces (in N) is of magnitude :-

UD0008

9. Two forces act on a particle simultaneously as shown in the figure. Find net force in milli newton on the particle. [Dyne is the CGS unit of force]

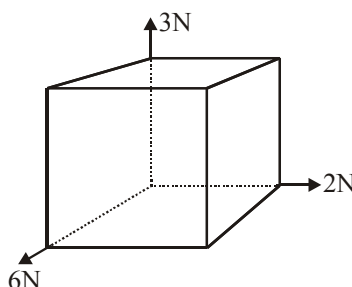


UD0009

- 10.** The maximum and minimum magnitudes of the resultant of two forces are 35 N and 5 N respectively. Find the magnitude of resultant force when act orthogonally to each other.

UD0010

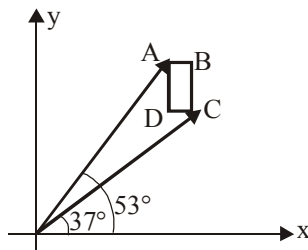
- 11.** Three forces of magnitudes 2 N, 3 N and 6 N act at corners of a cube along three sides as shown in figure. Find the resultant of these forces in N.



UD0011

Resolution of vectors and unit vector

- 12.** The farm house shown in figure has rectangular shape and has sides parallel to the chosen x and y axes. The position vector of corner A is 125 m at 53° and corner C is 100 m at 37° from x axis. Find the length of the fencing required in meter.

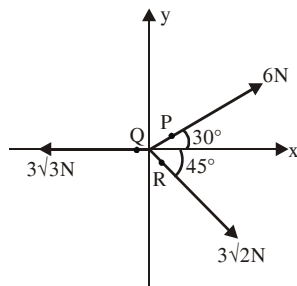


UD0012

- 13.** Vector B has x, y and z components of 4.00, 6.00 and 3.00 units, respectively. Calculate the magnitude of B and the angles that B makes with the coordinates axes.

UD0013

14. Three ants P, Q and R are pulling a grain with forces of magnitude 6N , $3\sqrt{3}\text{N}$ and $3\sqrt{2}\text{N}$ as shown in the figure. Find the magnitude of resultant force (in N) acting on the grain.



UD0014

15. Three boys are pushing horizontally a box placed on horizontal table. One is pushing towards north with a $5\sqrt{3}\text{N}$ force. The second is pushing towards east and third pushes with a force 10N such that the box is in equilibrium. Find the magnitude of the force, second boy is applying in newton.

UD0015

Scalar product of vectors

16. Consider the two vectors : $\vec{L} = \hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{l} = 4\hat{i} + 5\hat{j} + 6\hat{k}$. Find the value of the scalar α such that the vector $\vec{L} - \alpha\vec{l}$ is perpendicular to $\vec{L}$.

UD0016

17. Find components of vector $\vec{a} = \hat{i} + \hat{j} + 3\hat{k}$ in directions parallel to and perpendicular to vector $\vec{b} = \hat{i} + \hat{j}$.

UD0017

18. (a) Calculate $\vec{r} = \vec{a} - \vec{b} + \vec{c}$ where $\vec{a} = 5\hat{i} + 4\hat{j} - 6\hat{k}$, $\vec{b} = -2\hat{i} + 2\hat{j} + 3\hat{k}$ and $\vec{c} = 4\hat{i} + 3\hat{j} + 2\hat{k}$.
(b) Calculate the angle between $\vec{r}$ and the z-axis.

(c) Find the angle between $\vec{a}$ and $\vec{b}$

UD0018

19. If the velocity of a particle is $(2\hat{i} + 3\hat{j} - 4\hat{k})$ and its acceleration is $(-\hat{i} + 2\hat{j} + \hat{k})$ and angle between them is $\frac{n\pi}{4}$. The value of n is.

UD0019

Method of approximation

20. Quito, a city in Ecuador and Kampala, a city situated in Uganda both lie on the Equator. The longitude of Quito is $82^\circ 30'$ W and that of Kampala is $37^\circ 30'$ E. What is the distance from Quito to Kampala.

(a) along the shortest surface path

(b) along a direct (through-the-Earth) path? (The radius of the Earth is $6.4 \times 10^6\text{m}$)

UD0020

21. Use the approximation $(1+x)^n \approx 1 + nx$, $|x| \ll 1$, to find approximate value for

(a) $\sqrt{99}$

(b) $\frac{1}{1.01}$

UD0021

22. Use the small angle approximations to find approximate values for (A) $\sin 8^\circ$ and (B) $\tan 5^\circ$

UD0022

EXERCISE (S-2)

1. The equation of state for a real gas at high temperature is given by $P = \frac{nRT}{V-b} - \frac{a}{T^{1/2}V(V+b)}$ where n , P , V & T are number of moles, pressure, volume & temperature respectively & R is the universal gas constant. Find the dimensions of constant a in the above equation.

UD0023

2. If Energy (E), velocity (v) and time (T) are fundamental units. What will be the dimension of surface tension?

UD0024

3. In system called the star system we have 1 star kilogram = 10^{20} kg. 1 starmeter = 10^8 m, 1 starsecond = 10^3 s then calculate the value of 1 joule in this system.

UD0025

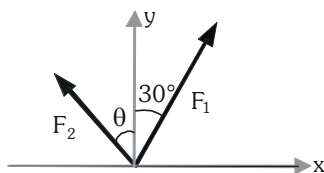
4. A vector $\vec{A}$ of length 10 units makes an angle of 60° with the vector $\vec{B}$ of length 6 units. Find the magnitude of the vector difference $\vec{A} - \vec{B}$ & the angle it makes with vector $\vec{A}$.

UD0026

5. A bird is at a point $P(4, -1, -5)$ and sees two points $P_1(-1, -1, 0)$ and $P_2(3, -1, -3)$. At time $t = 0$, it starts flying with a constant speed of 10 m/s to be in line with points P_1 and P_2 in minimum possible time t . Find t , if all coordinates are in kilometers.

UD0027

6. In the figure, F_1 and F_2 , the two unknown forces give a resultant force of $80\sqrt{3}$ N along the y -axis. It is required that F_2 must have minimum magnitude. Find the magnitudes of F_1 and F_2 .



UD0028

7. A particle is displaced from $A \equiv (2, 2, 4)$ to $B \equiv (5, -3, -1)$. A constant force of 34 N acts in the direction of $\vec{AP}$, where $P \equiv (10, 2, -11)$. (Coordinates are in m).

(i) Find the $\vec{F}$. (ii) Find the work done by the force to cause the displacement.

UD0029

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Units & Dimensions

- The dimensional formula for which of the following pair is not the same
 (A) impulse and momentum (B) torque and work
 (C) stress and pressure (D) momentum and angular momentum
UD0030
- Which of the following can be a set of fundamental quantities
 (A) length, velocity, time (B) momentum, mass, velocity
 (C) force, mass, velocity (D) momentum, time, frequency
UD0031
- Which of the following functions of A and B may be performed if A and B possess different dimensions?
 (A) $\frac{A}{B}$ (B) $A + B$ (C) $A - B$ (D) None
UD0032
- The velocity v of a particle at time t is given by $v = at + \frac{b}{t+c}$, where a , b and c are constants. The dimensions of a , b and c are respectively :-
 (A) LT^{-2} , L and T (B) L^2 , T and LT^2 (C) LT^2 , LT and L (D) L , LT and T^2
UD0033
- If area (A), velocity (v), and density (ρ) are base units, then the dimensional formula of force can be represented as :-
 (A) $Av\rho$ (B) $Av^2\rho$ (C) $Av\rho^2$ (D) $A^2v\rho$
UD0034
- Density of wood is 0.5 g/cc in the CGS system of units. The corresponding value in MKS units is :-
 (A) 500 (B) 5 (C) 0.5 (D) 5000
UD0035
- In a book, the answer for a particular question is expressed as $b = \frac{ma}{k} \left[\sqrt{1 + \frac{2k\ell}{ma}} \right]$ here m represents mass, a represents acceleration, ℓ represents length. The unit of b should be :-
 (A) m/s (B) m/s^2 (C) meter (D) $/\text{sec}$
UD0036
- The frequency f of vibrations of a mass m suspended from a spring of spring constant k is given by $f = Cm^x k^y$, where C is a dimensionless constant. The values of x and y are, respectively
 (A) $\frac{1}{2}, \frac{1}{2}$ (B) $-\frac{1}{2}, -\frac{1}{2}$ (C) $\frac{1}{2}, -\frac{1}{2}$ (D) $-\frac{1}{2}, \frac{1}{2}$
UD0037

9. If force, acceleration and time are taken as fundamental quantities, then the dimensions of length will be:
 (A) FT^2 (B) $F^{-1}A^2T^{-1}$ (C) FA^2T (D) AT^2 UD0038
10. In a particular system the units of length, mass and time are chosen to be 10 cm, 10 g and 0.1 s respectively. The unit of force in this system will be equal to :-
 (A) 0.1 N (B) 1 N (C) 10 N (D) 100 N UD0039
11. The units of three physical quantities x, y and z are gcm^2s^{-5} , gs^{-1} and cms^{-2} respectively. The relation among the units of x, y and z is :-
 (A) $z = x^2y$ (B) $y^2 = xz$ (C) $x = yz^2$ (D) $x = y^2z$ UD0040
12. The angle subtended by the moon's diameter at a point on the earth is about 0.50° . Use this and the fact that the moon is about 384000 km away to find the approximate diameter of the moon.



- (A) 192000 km (B) 3350 km (C) 1600 km (D) 1920 km

UD0041

13. **Statement 1 :** Method of dimensions cannot tell whether an equation is correct.
and
Statement 2 : A dimensionally incorrect equation may be correct.
 (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

UD0042

14. The equation of the stationary wave is $y = 2A \sin\left(\frac{2\pi ct}{\lambda}\right) \cos\left(\frac{2\pi x}{\lambda}\right)$ which of the following statement is wrong?
 (A) The unit of ct is same as that of λ . (B) The unit of x is same as that of λ .
 (C) The unit of $\frac{2\pi c}{\lambda}$ is same as that of $\frac{2\pi x}{\lambda t}$ (D) The unit of $\frac{c}{\lambda}$ is same as that of $\frac{x}{\lambda}$

UD0043

15. Due to some unknown interaction, force F experienced by a particle is given by the following equation.

$$\vec{F} = -\frac{A}{r^3} \vec{r}$$

Where A is positive constant and r distance of the particle from origin of a coordinate system. Dimensions of constant A are :-

- (A) ML^2T^{-2} (B) ML^3T^{-2} (C) ML^4T^{-2} (D) ML^0T^0

UD0044

Addition of vectors

16. Three concurrent forces of the same magnitude are in equilibrium. What is the angle between the force? Also name the triangle formed by the force as sides :-

(A) 60° equilateral triangle (B) 120° equilateral triangle
(C) $120^\circ, 30^\circ, 30^\circ$ an isosceles triangle (D) 120° an obtuse angled triangle

UD0045

17. The resultant of two forces, one double the other in magnitude is perpendicular to the smaller of the two forces. The angle between two forces is :-

(A) 150° (B) 90° (C) 60° (D) 120°

UD0046

18. The resultant of two forces acting at an angle of 120° is 10 kg wt and is perpendicular to one of the forces. That force is :

(A) $10\sqrt{3}$ kg wt (B) $20\sqrt{3}$ kg wt (C) 10 kg wt (D) $\frac{10}{\sqrt{3}}$ kg wt

UD0047

19. If the resultant of two forces of magnitudes P and Q acting at a point at an angle of 60° is $\sqrt{7} Q$, then P/Q is :-

(A) 1 (B) $\frac{3}{2}$ (C) 2 (D) 4

UD0048

20. There are two force vectors, one of 5N and other of 12N at what angle the two vectors be added to get resultant vector of 17N, 7N and 13N respectively.

(A) $0^\circ, 180^\circ$ and 90° (B) $0^\circ, 90^\circ$ and 180°
(C) $0^\circ, 90^\circ$ and 90° (D) $180^\circ, 0^\circ$ and 90°

UD0049

21. A body placed in free space, is simultaneously acted upon by three forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$. The body is in equilibrium and the forces $\vec{F}_1$ and $\vec{F}_2$ are known to be 36 N due north and 27 N due east respectively.

Which of the following best describes the force $\vec{F}_3$?

(A) 36 N due south. (B) 53 N due 60° south of east
(C) 45 N due 53° south of west (D) 45 N due 37° north of west

UD0050

22. The ratio of maximum and minimum magnitudes of the resultant of two vectors $\vec{A}$ and $\vec{B}$ is 3 : 2. The relation between A and B is

(A) $A = 5B$ (B) $5A = B$ (C) $A = 3B$ (D) $A = 4B$

UD0051

23. Find the resultant of the following two vectors $\vec{A}$ and $\vec{B}$.

$\vec{A}$: 40 units due east and ; $\vec{B}$: 25 units 37° north of west

(A) 25 units 37° north of west (B) 25 units 37° north of east
(C) 40 units 53° north of west (D) 40 units 53° north of east

UD0052

- 24.** Two vectors $\vec{a}$ and $\vec{b}$ add to give a resultant $\vec{c} = \vec{a} + \vec{b}$. In which of these cases angle between $\vec{a}$ and $\vec{b}$ is maximum: (a, b, c represent the magnitudes of respective vectors)
- (A) $c = a + b$ (B) $c^2 = a^2 + b^2$ (C) $c = a - b$ (D) can not be determined

UD0053

- 25.** If the angle between the unit vectors $\hat{a}$ and $\hat{b}$ is 60° , then $|\hat{a} - \hat{b}|$ is :-
 (A) 0 (B) 1 (C) 2

UD0054

- 26.** A man moves towards 3 m north then 4 m towards east and finally 5 m towards 37° south of west. His displacement from origin is :-
 (A) $5\sqrt{2}\text{m}$ (B) 0m (C) 1 m (D) 12 m

UD0055

Resolution of vectors and unit vector

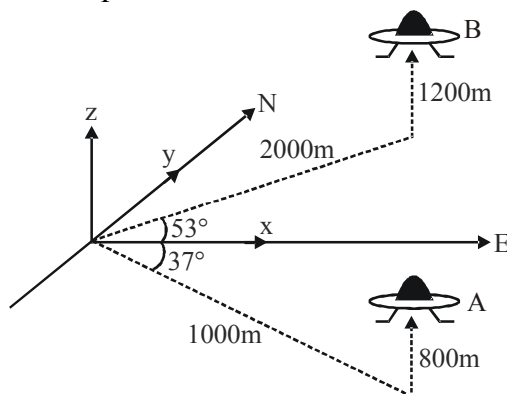
- 27.** The projection of a vector, $\vec{r} = 3\hat{i} + \hat{j} + 2\hat{k}$, on the x-y plane has magnitude :—
 (A) 3 (B) 4 (C) $\sqrt{14}$ (D)

UD0056

28. A bird moves from point $(1, -2)$ to $(4, 2)$. If the speed of the bird is 10 m/s, then the velocity vector of the bird is
- (A) $5(\hat{i} - 2\hat{j})$ (B) $5(4\hat{i} + 2\hat{j})$ (C) $0.6\hat{i} + 0.8\hat{j}$ (D) $6\hat{i} + 8\hat{j}$

UD0057

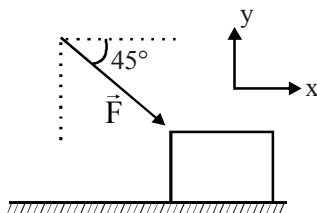
- 29.** Personnel at an air post control tower track a UFO. At 11:02 am it was located at position A and at 11:12 am it was located at position B. Displacement vector of UFO is :



- (A) $400\hat{i} + 2200\hat{j} + 400\hat{k}$ (B) $1200\hat{i} + 1000\hat{j} + 800\hat{k}$
(C) $2000\hat{i} + 2200\hat{j} + 2000\hat{k}$ (D) $400\hat{i} + 1000\hat{j} + 400\hat{k}$

UD0058

- 30.** A person pushes a box kept on a horizontal surface with force of 100 N. In unit vector notation force $\vec{F}$ can be expressed as :



- (A) $100(\hat{i} + \hat{j})$ (B) $100(\hat{i} - \hat{j})$ (C) $50\sqrt{2}(\hat{i} - \hat{j})$ (D) $50\sqrt{2}(\hat{i} + \hat{j})$

UD0059

31. For the given vector $\vec{A} = 3\hat{i} - 4\hat{j} + 10\hat{k}$, the ratio of magnitude of its component on the x-y plane and the component on z-axis is :-

(A) 2 (B) $\frac{1}{2}$ (C) 1 (D) None

UD0060

32. After firing, a bullet is found to move at an angle of 37° to horizontal. Its acceleration is 10 m/s^2 downwards. Find the component of acceleration in the direction of the velocity.

(A) -6 m/s^2 (B) -4 m/s^2 (C) -8 m/s^2 (D) -5 m/s^2

UD0061

Scalar product of vectors

33. In a methane (CH_4) molecule each hydrogen atom is at a corner of a regular tetrahedron with the carbon atom at the centre. In coordinates where one of the C-H bonds is in the direction of $\hat{i} + \hat{j} + \hat{k}$, an adjacent C-H bond in the $\hat{i} - \hat{j} - \hat{k}$ direction. Then angle between these two bonds :-

(A) $\cos^{-1}\left(-\frac{2}{3}\right)$ (B) $\cos^{-1}\left(\frac{2}{3}\right)$ (C) $\cos^{-1}\left(-\frac{1}{3}\right)$ (D) $\cos^{-1}\left(\frac{1}{3}\right)$

UD0062

34. If $\vec{a}$ and $\vec{b}$ are two unit vectors such that $\vec{a} + 2\vec{b}$ and $5\vec{a} - 4\vec{b}$ are perpendicular to each other then the angle between $\vec{a}$ and $\vec{b}$ is :-

(A) 45° (B) 60° (C) $\cos^{-1}\left(\frac{1}{3}\right)$ (D) $\cos^{-1}\left(\frac{2}{7}\right)$

UD0063

35. The velocity of a particle is $\vec{v} = 6\hat{i} + 2\hat{j} - 2\hat{k}$. The component of the velocity of a particle parallel to vector $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ is :-

(A) $6\hat{i} + 2\hat{j} + 2\hat{k}$ (B) $2\hat{i} + 2\hat{j} + 2\hat{k}$ (C) $\hat{i} + \hat{j} + \hat{k}$ (D) $6\hat{i} + 2\hat{j} - 2\hat{k}$

UD0064

36. A particle moves from a position $3\hat{i} + 2\hat{j} - 6\hat{k}$ to a position $14\hat{i} + 13\hat{j} + 9\hat{k}$ in m and a uniform force of $4\hat{i} + \hat{j} + 3\hat{k}$ N acts on it. The work done by the force is :-

(A) 200 J (B) 100 J (C) 300 J (D) 500 J

UD0065

37. Which of the following is perpendicular to $\hat{i} - \hat{j} - \hat{k}$?

(A) $\hat{i} + \hat{j} + \hat{k}$ (B) $-\hat{i} + \hat{j} + \hat{k}$ (C) $\hat{i} + \hat{j} - \hat{k}$ (D) none of these

UD0066

MULTIPLE CORRECT TYPE QUESTIONS

38. Which of the following statements about the sum of the two vectors $\vec{A}$ and $\vec{B}$, is/are correct?

(A) $|\vec{A} + \vec{B}| \leq A + B$ (B) $|\vec{A} + \vec{B}| \geq A + B$ (C) $|\vec{A} + \vec{B}| \geq |\vec{A} - \vec{B}|$ (D) $|\vec{A} + \vec{B}| \geq |A - B|$

UD0067

39. Priya says that the sum of two vectors by the parallelogram method is $\vec{R} = 5\hat{i}$. Subhangi says it is $\vec{R} = \hat{i} + 4\hat{j}$. Both used the parallelogram method, but one used the wrong diagonal. Which one of the vector pairs below contains the original two vectors?

- (A) $\vec{A} = +3\hat{i} - 2\hat{j}$; $\vec{B} = -2\hat{i} + 2\hat{j}$ (B) $\vec{A} = -3\hat{i} - 2\hat{j}$; $\vec{B} = +2\hat{i} + 2\hat{j}$
(C) $\vec{A} = +3\hat{i} + 2\hat{j}$; $\vec{B} = +2\hat{i} - 2\hat{j}$ (D) $\vec{A} = +3\hat{i} + 2\hat{j}$; $\vec{B} = -2\hat{i} + 2\hat{j}$

UD0068

40. For the equation $x = AC \sin(Bt) + D e^{(Ct)}$, where x and t represent position and time respectively, which of the following is/are **CORRECT** :-

- (A) Dimension of AC is LT^{-1} (B) Dimension of B is T^{-1}
(C) Dimension of AC and D are same (D) Dimension of C is T^{-1}

UD0069

COMPREHENSION TYPE QUESTIONS

Paragraph for Question no. 41 to 43

In a certain system of absolute units the acceleration produced by gravity in a body falling freely is denoted by 5, the kinetic energy of a 500 kg shot moving with velocity 400 metres per second is denoted by 2000 & its momentum by 100.

41. The unit of length is :-

- (A) 15 m (B) 50 m (C) 25 m (D) 100 m

UD0070

42. The unit of time is :-

- (A) 10 s (B) 20 s (C) 5 s (D) 15 s

UD0070

43. The unit of mass is :-

- (A) 200 kg (B) 400 kg (C) 800 kg (D) 1200 kg

UD0070

Paragraph for Question Nos. 44 and 45

For any particle moving with some velocity ($\vec{v}$) & acceleration ($\vec{a}$), tangential acceleration & normal acceleration are defined as follows

Tangential acceleration – The component of acceleration in the direction of velocity.

Normal acceleration – The component of acceleration in the direction perpendicular to velocity.

If at a given instant, velocity & acceleration of a particle are given by.

$$\vec{v} = 4\hat{i} + 3\hat{j}$$

$$\vec{a} = 10\hat{i} + 15\hat{j} + 20\hat{k}$$

44. Find the tangential acceleration of the particle at the given instant :-

- (A) $17(4\hat{i} + 3\hat{j})$ (B) $\frac{17}{5}(4\hat{i} + 3\hat{j})$ (C) $17(4\hat{i} - 3\hat{j})$ (D) $\frac{17}{5}(4\hat{i} - 3\hat{j})$

UD0071

45. Find the normal acceleration of the particle at the given instant :-

- (A) $\frac{-9\hat{i} + 12\hat{j} + 50\hat{k}}{5}$ (B) $\frac{9\hat{i} - 12\hat{j} - 50\hat{k}}{5}$ (C) $\frac{-18\hat{i} + 24\hat{j} + 100\hat{k}}{5}$ (D) $\frac{18\hat{i} - 24\hat{j} - 100\hat{k}}{5}$

UD0071

MATRIX MATCH TYPE QUESTION

46. Two particles A and B start from origin of a coordinate system towards point P (10, 20) and Q (20, 10) respectively with speed $5\sqrt{5}$ each. Both continue their motion for 10 s and then stop. There after particle B moves towards particle A with speed $2\sqrt{2}$ and after particle B meets particle A, they both return to origin following a straight line path with speed $5\sqrt{5}$. Match the items of column-I with suitable items of Column-II.

Column-I

- (A) Initial velocity vector of A
 (B) Initial velocity of B
 (C) Velocity vector of B while it moves towards A
 (D) Velocity vector of A and B while they return to origin

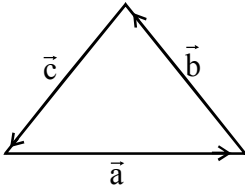
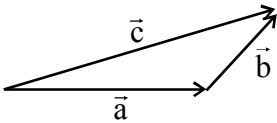
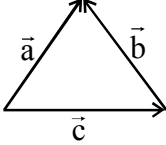
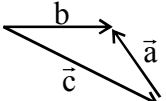
Column-II

- (P) $(-5\hat{i} - 10\hat{j})$
 (Q) $(5\hat{i} + 10\hat{j})$
 (R) $(10\hat{i} + 5\hat{j})$
 (S) $(2\hat{i} - 2\hat{j})$
 (T) $(-2\hat{i} + 2\hat{j})$

UD0072

47. Column-I show vector diagram relating three vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$. Match the vector equation in column-II, with vector diagram in column-I :

Column-I

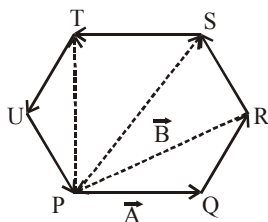
- (A) 
- (B) 
- (C) 
- (D) 

Column-II

- (P) $\vec{a} - (\vec{b} + \vec{c}) = 0$
 (Q) $\vec{b} - \vec{c} = \vec{a}$
 (R) $\vec{a} + \vec{b} = -\vec{c}$
 (S) $\vec{a} + \vec{b} = \vec{c}$

UD0073

48. In a regular hexagon two vectors $\overrightarrow{PQ} = \vec{A}$, $\overrightarrow{RP} = \vec{B}$. Express other vector's in term of them :-



Column-I

- (A) $\overrightarrow{PS}$
(B) $\overrightarrow{PT}$
(C) $\overrightarrow{RS}$
(D) $\overrightarrow{TS}$

Column-II

- (P) $-2\vec{B} - 3\vec{A}$
(Q) $-\vec{B} - \vec{A}$
(R) $-\vec{B} - 2\vec{A}$
(S) $-2(\vec{B} + \vec{A})$
(T) $\vec{A}$

UD0074

49. Show a vector $\vec{a}$ at angle θ as shown in the figure column-II. Show its unit vector representation.

Column-I

- (A)
- (B)
- (C)
- (D)

Column-II

- (P) $\vec{a} = a \sin \theta \hat{i} + a \cos \theta \hat{j}$
(Q) $\vec{a} = -a \cos \theta \hat{i} + a \sin \theta \hat{j}$
(R) $\vec{a} = -a \sin \theta \hat{i} - a \cos \theta \hat{j}$
(S) $\vec{a} = a \cos \theta \hat{i} - a \sin \theta \hat{j}$

UD0075

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. In a certain system of units, 1 unit of time is 5 sec, 1 unit of mass is 20 kg and 1 unit of length is 10 m. In this system, one unit of power will correspond to :-

(A) 16 watts (B) $\frac{1}{16}$ watts (C) 25 watts (D) none of these

UD0076

2. If the unit of length be doubled then the numerical value of the universal gravitation constant G will become (with respect to present value)

(A) Double (B) Half (C) 8 times (D) $1/8$ times

UD0077

3. If in a system, the force of attraction between two point masses of 1 kg each situated 1 km apart is taken as a unit of force and is called notwen (newton written in reverse order) & if $G = 6.67 \times 10^{-11} \text{ N-m}^2 \text{ kg}^{-2}$ in SI units then which of the following is true?

(A) 1 notwen = 6.67×10^{-11} newton (B) 1 newton = 6.67×10^{-17} notwen
(C) 1 notwen = 6.67×10^{-17} newton (D) 1 newton = 6.67×10^{-12} notwen

UD0078

4. In two different systems of units an acceleration is represented by the same number, while a velocity is represented by numbers in the ratio 1 : 3. The ratios of unit of length and time are respectively

(A) 3, 9 (B) 9, 3 (C) 1, 1 (D) None of these

UD0079

5. **Statement-1 :** Whenever the unit of measurement of a quantity is changed, its numerical value changes. **and**

Statement-2 : Smaller the unit of measurement smaller is its numerical value.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

UD0080

6. Forces proportional to AB, BC and 2CA act along the sides of triangle ABC in order. Their resultant represented in magnitude and direction as

(A) CA (B) AC (C) BC (D) CB

UD0081

7. A man rows a boat with a speed of 18 km/hr in north–west direction. The shoreline makes an angle of 15° south of west. Obtain the component of the velocity of the boat along the shoreline.

(A) 9 km/hr
(B) $18\frac{\sqrt{3}}{2}$ km/hr
(C) $18 \cos (15^\circ)$ km/hr
(D) $18 \cos (75^\circ)$ km/hr

UD0082

- 8. Statement 1 :** Unit vector has a unit though its magnitude is one and

Statement 2 : Unit vector is obtained by dividing a vector by its own magnitude.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
- (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
- (C) Statement-1 is true, statement-2 is false.
- (D) Statement-1 is false, statement-2 is true.

UD0083

9. A vector of magnitude 10 m in the direction 37° south of west has its initial point at (5 m, 2 m). If positive x-axis represents the east and positive y-axis the north, the coordinates of its terminal point are

(A) $(-3 \text{ m}, -4 \text{ m})$ (B) $(3 \text{ m}, 4 \text{ m})$ (C) $(-4 \text{ m}, 6 \text{ m})$ (D) $(-4 \text{ m}, -6 \text{ m})$

UD0084

- 10.** A plumber steps down 1 m out of his truck and walks 50 m east and then 25 m south, and then takes an elevator to the basement of the building 9 m below street level. If the east, the north and the upward direction are represented by the positive x, y and z-axes, which one of the following represents displacement (meters) of the plumber?

(A) $50\hat{i} - 25\hat{j} - 9\hat{k}$ (B) $50\hat{i} + 25\hat{j} - 9\hat{k}$
(C) $50\hat{i} - 25\hat{j} - 10\hat{k}$ (D) $50\hat{i} + 25\hat{j} - 10\hat{k}$

UD0085

- 11.** A body moves in anticlockwise direction on a circular path in the x-y plane. The radius of the circular path is 5 m and its centre is at the origin. In a certain interval of time, displacement of the body is observed to be 6 m in the positive y-direction. Which of the following is true?

(A) Its initial position vector is $5\hat{i}$ m. (B) Its initial position vector is $(-3\hat{i} + 4\hat{j})$ m.

(C) Its final position vector is $(4\hat{i} + 3\hat{j})$ m. (D) Its final position vector is $6\hat{j}$ m.

UD0086

- 12.** A boy A is standing $20\sqrt{3}$ m away in a direction 30° north of east from his friend B. Another boy C standing somewhere east of B can reach A, if he walks in a direction 60° north of east. In a Cartesian coordinate system with its x-axis towards the east, the position of C with respect to A is

(A) $(-20\hat{i} + -10\hat{j})$ m

(B) $(-10\hat{i} - 10\sqrt{3}\hat{j})$ m

(C) $(10\hat{i} + 10\sqrt{3}\hat{j})$ m

(D) It depends on where we chose the origin.

UD0087

13. Find the component of $\vec{r}$ in the direction of $\vec{a}$:-

(A) $\frac{(\vec{r} \cdot \vec{a})\vec{a}}{a^2}$ (B) $\frac{(\vec{r} \cdot \vec{a})\vec{a}}{a}$ (C) $\frac{(\vec{r} \cdot \vec{a})\hat{r}}{r}$ (D) $\frac{(\vec{r} \cdot \vec{a})\hat{r}}{r^2}$

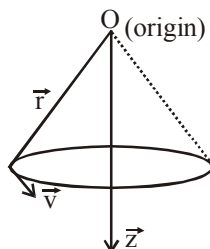
UD0088

14. Consider three vectors $\vec{A} = \hat{i} + \hat{j} - 2\hat{k}$, $\vec{B} = \hat{i} - \hat{j} + \hat{k}$ and $\vec{C} = 2\hat{i} - 3\hat{j} + 4\hat{k}$. A vector $\vec{X}$ of the form $\alpha\vec{A} + \beta\vec{B}$ (α and β are numbers) is perpendicular to $\vec{C}$. The ratio of α and β is

(A) 1:1 (B) 2:1 (C) -1:1 (D) 3:1

UD0089

15. A string connected with bob is suspended by the point (O) such that it sweeps out conical surface in horizontal plane. Here $\vec{r}$ is the position vector of bob, $\vec{v}$ is its velocity and $\vec{z}$ is the axis of swept cone as shown. Select **INCORRECT** statement :-



(A) $\vec{r} \cdot \vec{z}$ is always zero

(B) $\vec{r} \cdot \vec{v}$ is always zero

(C) $\vec{z} \cdot \vec{v}$ is always constant

(D) $\vec{r} \cdot \vec{z}$ is always non zero constant

UD0090

16. x-component of a vector $\vec{A}$ is twice of its y-component and $\sqrt{2}$ times of its z-component. Find out the angle made by the vector from y-axis.

(A) $\cos^{-1}\left(\frac{2}{\sqrt{7}}\right)$ (B) $\cos^{-1}\left(\frac{1}{\sqrt{7}}\right)$ (C) $\cos^{-1}\left(\frac{1}{\sqrt{6}}\right)$ (D) $\cos^{-1}\left(\frac{2}{\sqrt{6}}\right)$

UD0091

17. Given the vectors $\vec{A} = 2\hat{i} + 3\hat{j} - \hat{k}$; $\vec{B} = 3\hat{i} - 2\hat{j} - 2\hat{k}$ & $\vec{C} = p\hat{i} + p\hat{j} + 2p\hat{k}$. Find the angle between $(\vec{A} - \vec{B})$ & $\vec{C}$

(A) $\theta = \cos^{-1}\left(\frac{2}{\sqrt{3}}\right)$ (B) $\theta = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ (C) $\theta = \cos^{-1}\left(\frac{\sqrt{2}}{3}\right)$ (D) none of these

UD0092

MULTIPLE CORRECT TYPE QUESTIONS

18. Four forces acting on a particle keep it in equilibrium, then :-

(A) the force must be coplanar.
(B) the forces cannot be coplanar.
(C) the forces may or may not be coplanar.
(D) if three of these forces are coplanar, so must be the fourth.

UD0093

19. A man is standing at point $(x = 5 \text{ m}, y = 0)$. Then he walks along straight line to $(x = 0, y = 5 \text{ m})$. A second man walks from the same initial position along the x -axis to the origin and then along the y -axis to $(x = 0, y = 5 \text{ m})$. Mark the **CORRECT** statement(s) :
- (A) Displacement vector of first man and second man are equal
 (B) Distance travelled by second man is greater
 (C) Magnitude of displacement of second man is same that of first man but direction is different
 (D) Magnitude of displacement is $\sqrt{50} \text{ m}$ for 2nd man.

UD0094

20. The vector $i + xj + 3k$ is rotated through an angle θ and doubled in magnitude, then it becomes $4i + (4x - 2)j + 2k$. The values of x are

- (A) $-\frac{2}{3}$ (B) $\frac{1}{3}$ (C) $\frac{2}{3}$ (D) 2

UD0095

21. The value of $|\vec{A} + \vec{B} - \vec{C} + \vec{D}|$ can be zero if :-

- (A) $|\vec{A}| = 5, |\vec{B}| = 3, |\vec{C}| = 4, |\vec{D}| = 13$ (B) $|\vec{A}| = 2\sqrt{2}, |\vec{B}| = 2, |\vec{C}| = 2, |\vec{D}| = 5$
 (C) $|\vec{A}| = 2\sqrt{2}, |\vec{B}| = 2, |\vec{C}| = 2, |\vec{D}| = 10$ (D) $|\vec{A}| = 5, |\vec{B}| = 4, |\vec{C}| = 3, |\vec{D}| = 8$

UD0096

22. The four pairs of force vectors are given, which pairs of force vectors cannot be added to give a resultant vector of magnitude 10 N?

- (A) 2N, 13 N (B) 5N, 16 N
 (C) 7N, 8N (D) 100N, 105 N

UD0097

23. Select **CORRECT** statement(s) for three vectors $\vec{a} = -3\hat{i} + 2\hat{j} - \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} + 5\hat{k}$ and $\vec{c} = 2\hat{i} + \hat{j} - 4\hat{k}$

- (A) The above vectors can form triangle.
 (B) Component of $\vec{a}$ along $\vec{c}$ is 3.
 (C) $\vec{a}$ makes angle $\cos^{-1} \sqrt{\frac{2}{7}}$ with y -axis.
 (D) A vector having magnitude twice the vector $\vec{a}$ and anti parallel to vector $\vec{b}$ is $\sqrt{\frac{2}{5}}(-2\hat{i} + 6\hat{j} - 10\hat{k})$

UD0098

24. If a vector $\vec{P}$ makes an angle α, β, γ with x, y, z axis respectively then it can be represented as

$\vec{P} = P[\cos \alpha \hat{i} + \cos \beta \hat{j} + \cos \gamma \hat{k}]$. Choose the **CORRECT** option(s) :-

- (A) $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$ (B) $\vec{P} \cdot \vec{P} = P^2$
 (C) $\vec{P} \cdot (\hat{i} - \hat{k}) = P(\cos \alpha - \cos \gamma)$ (D) $\vec{P} \cdot \hat{i} = \cos \alpha$

UD0099

COMPREHENSION TYPE QUESTIONS

Paragraph for Question no. 25 to 27

A boy lost in a jungle finds a note. In the note was written the following things.

Displacements

1. 300 m 53° South of East.
2. 400 m 37° North of East
3. 500 m North
4. $500\sqrt{2}$ m North-West
5. 500 m South

He starts walking at constant speed 2 m/s following these displacements in the given order.

25. How far and in which direction is he from the starting point after 5 min. and 50 s?

- (A) 500 m due East (B) 500 m due West
(C) 700 m due South-West (D) 700 m due North-East

UD0100

26. How far and in which direction is he from the starting point after 10 minutes?

- (A) $500\sqrt{2}$ m due North (B) 1200 m due North-East
(C) $500\sqrt{2}$ m due North-East (D) 900 m due 37° North of East

UD0100

27. How far and in which direction has he finally displaced after all the displacements in the note?

- (A) $500\sqrt{2}$ m due North-East (B) 500 m due North
(C) 866 m due North-West (D) $500\sqrt{3}$ m due 60° North of West

UD0100

Paragraph for Question Nos. 28 to 30

A physical quantity is a physical property of a phenomenon, body, or substance, that can be quantified by measurement.

The magnitude of the components of a vector are to be considered dimensionally distinct. For example, rather than an undifferentiated length unit L , we may represent length in the x direction as L_x , and so forth. This requirement stems ultimately from the requirement that each component of a physically meaningful equation (scalar or vector) must be dimensionally consistent. As an example, suppose we wish to calculate the drift S of a swimmer crossing a river flowing with velocity V_x and of width D and he is swimming in direction perpendicular to the river flow with velocity V_y relative to river,

assuming no use of directed lengths, the quantities of interest are then V_x , V_y both dimensioned as $\frac{L}{T}$,

S the drift and D width of river both having dimension L . With these four quantities, we may conclude that the equation for the drift S may be written: $S \propto V_x^a V_y^b D^c$

Or dimensionally $L = \left(\frac{L}{T}\right)^{a+b} \times (L)^c$ from which we may deduce that $a + b + c = 1$ and $a + b = 0$, which leaves one of these exponents undetermined. If, however, we use directed length dimensions, then V_x will

be dimensioned as $\frac{L_x}{T}$, V_y as $\frac{L_y}{T}$, S as L_x and D as L_y . The dimensional equation becomes :

$L_x = \left(\frac{L_x}{T}\right)^a \left(\frac{L_y}{T}\right)^b (L_y)^c$ and we may solve completely as $a = 1$, $b = -1$ and $c = 1$. The increase in deductive power gained by the use of directed length dimensions is apparent.

28. Which of the following is not a physical quantity

- (A) Height of a boy (B) Weight of a boy
(C) Fever of a boy (D) Speed of a running boy

UD0101

29. From the concept of directed dimension what is the formula for a range (R) of a cannon ball when it is fired with vertical velocity component V_y and a horizontal velocity component V_x assuming it is fired on a flat surface. [Range also depends upon acceleration due to gravity, g and k is numerical constant]

(A) $R = \frac{k(V_x V_y)}{g}$ (B) $R = \frac{k(V_x)^2}{g}$ (C) $R = \frac{k(V_x)^3}{V_y g}$ (D) $R = \frac{k(V_y)^3}{V_x g}$

UD0101

30. A conveyer belt of width D is moving along x-axis with velocity V. A man moving with velocity U on the belt in the direction perpendicular to the belt's velocity with respect to belt wants to cross the belt. The correct expression for the drift (S) suffered by man is given by (k is numerical constant)

(A) $S = k \frac{UD}{V}$ (B) $S = k \frac{VD}{U}$ (C) $S = k \frac{U^2 D}{V^2}$ (D) $S = k \frac{V^2 D}{U^2}$

UD0101

**MATCHING LIST TYPE ($4 \times 4 \times 4$) MULTIPLE OPTION CORRECT
(THREE COLUMNS AND FOUR ROWS)**

Answer Q.31, Q.32 and Q.33 by appropriately matching the information given in the three columns of the following table.

L, M and T are units of length, Mass and Time respectively in a system of units.

Column-1	Column-2	Column-3
(I) L = 10 m	(i) M = 100 gm	(P) T = 0.1 sec
(II) L = 10 cm	(ii) M = 10 kg	(Q) T = 10 ms
(III) L = 0.1 mm	(iii) M = 10 gm	(R) T = 10 sec
(IV) L = 1 km	(iv) M = 1 tonne	(S) T = 0.01 sec

31. In which of the following combinations unit of force is 10^6 dyne.

- (A) (IV) (i) (P) (B) (II) (iii) (S) (C) (III) (iv) (P) (D) (I) (ii) (Q)

UD0102

32. In which of the following system, unit of energy is 10^9 erg

- (A) (III) (i) (S) (B) (IV) (iii) (R) (C) (II) (iv) (Q) (D) (I) (iii) (P)

UD0103

33. In which of the following system, unit for coefficient of viscosity is 100 poiseuille

- (A) (III) (ii) (S) (B) (II) (i) (Q) (C) (III) (iii) (R) (D) (IV) (iv) (P)

UD0104

MATRIX MATCH TYPE QUESTION

34. In a new system of units known as RMP, length is measured in 'retem', mass is measured in 'marg' and time is measured in 'pal'.

$$100 \text{ retem} = 1.0 \text{ meter}$$

$$1.0 \text{ marg} = 10^{-3} \text{ kilogram}$$

$$10 \text{ pal} = 1.0 \text{ second}$$

In the given table some unit conversion factors are given. Suggest suitable match.

Column-I

- (A) One SI unit of force
(B) One SI unit of potential energy
(C) One SI unit of power
(D) One SI unit of momentum

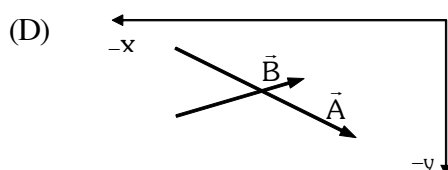
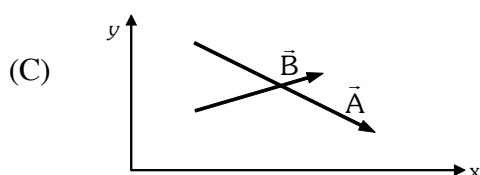
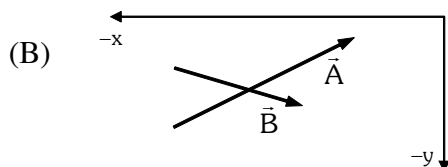
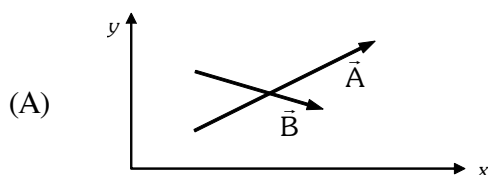
Column-II

- (P) 10^2 units of RMP
(Q) 10^3 units of RMP
(R) 10^4 units of RMP
(S) 10^5 units of RMP
(T) 10^6 units of RMP

UD0105

35. Refer the following table, where in the first column four pairs of two vectors are shown and in the second column some possible outcomes of basic mathematical operation on these vectors are given. Suggest suitable match(s).

Column-I

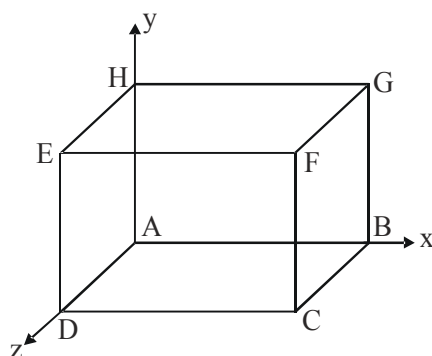


Column-II

- (P) X-component of $\vec{A} + \vec{B}$ is positive
(Q) Y-component of $\vec{A} + \vec{B}$ is negative
(R) X-component of $\vec{A} - \vec{B}$ is positive
(S) Y-component of $\vec{A} - \vec{B}$ is negative
(T) $\vec{B} \cdot \vec{A}$ is positive

UD0106

36. Figure shows a cube of edge length a .



Column-I

- (A) The angle between AF and x-axis
(B) Angle between AF and DG
(C) Angle between AE and AG

Column-II

- (P) 60°
(Q) $\cos^{-1} \frac{1}{3}$
(R) $\cos^{-1} \frac{1}{\sqrt{3}}$
(S) $\cos^{-1} \sqrt{\frac{2}{3}}$

UD0107

EXERCISE (J-M)

1. An ideal gas enclosed in a vertical cylindrical container supports a freely moving piston of mass M . The piston and the cylinder have equal cross sectional area A . When the piston is in equilibrium, the volume of the gas is V_0 and its pressure is P_0 . The piston is slightly displaced from the equilibrium position and released. Assuming that the system is completely isolated from its surrounding, the piston executes a simple harmonic motion with frequency. [JEE Main-2013]

(1) $\frac{1}{2\pi} \frac{A\gamma P_0}{V_0 M}$ (2) $\frac{1}{2\pi} \frac{V_0 M P_0}{A^2 \gamma}$ (3) $\frac{1}{2\pi} \sqrt{\frac{A^2 \gamma P_0}{M V_0}}$ (4) $\frac{1}{2\pi} \sqrt{\frac{M V_0}{A \gamma P_0}}$

UD0108

EXERCISE (J-A)

1. Match List I with List II and select the correct answer using the codes given below the lists :

List I	
P.	Boltzmann constant
Q.	Coefficient of viscosity
R.	Planck constant
S.	Thermal conductivity

List II	
1.	$[ML^2T^{-1}]$
2.	$[ML^{-1}T^{-1}]$
3.	$[MLT^{-3}K^{-1}]$
4.	$[ML^2T^{-2}K^{-1}]$

[JEE Advanced-2013]

Codes :

	P	Q	R	S
(A)	3	1	2	4
(B)	3	2	1	4
(C)	4	2	1	3
(D)	4	1	2	3

UD0109

2. To find the distance d over which a signal can be seen clearly in foggy conditions, a railways engineer uses dimensional analysis and assumes that the distance depends on the mass density ρ of the fog, intensity (power/area) S of the light from the signal and its frequency f . The engineer finds that d is proportional to $S^{1/n}$. The value of n is.

[JEE Advanced-2014]

UD0110

3. In terms of potential difference V , electric current I , permittivity ϵ_0 , permeability μ_0 and speed of light c , the dimensionally correct equation(s) is(are)

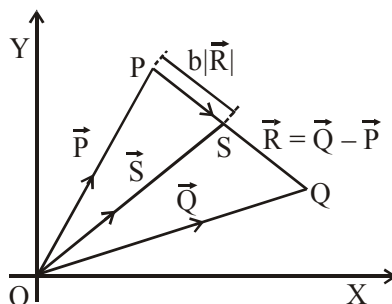
[JEE Advanced-2015]

- (A) $\mu_0 I^2 = \epsilon_0 V^2$ (B) $\epsilon_0 I = \mu_0 V$ (C) $I = \epsilon_0 c V$ (D) $\mu_0 c I = \epsilon_0 V$

UD0111

4. Three vectors $\vec{P}$, $\vec{Q}$ and $\vec{R}$ are shown in the figure. Let S be any point on the vector $\vec{R}$. The distance between the points P and S is $b|\vec{R}|$. The general relation among vectors $\vec{P}$, $\vec{Q}$ and $\vec{S}$ is :

[JEE Advanced-2017]



- (A) $\vec{S} = (1-b)\vec{P} + b^2\vec{Q}$ (B) $\vec{S} = (b-1)\vec{P} + b\vec{Q}$
 (C) $\vec{S} = (1-b)\vec{P} + b\vec{Q}$ (D) $\vec{S} = (1-b^2)\vec{P} + b\vec{Q}$

UD0112

5. Two vectors $\vec{A}$ and $\vec{B}$ are defined as $\vec{A} = a\hat{i}$ and $\vec{B} = a(\cos \omega t \hat{i} + \sin \omega t \hat{j})$, where a is a constant and $\omega = \pi/6 \text{ rad s}^{-1}$. If $|\vec{A} + \vec{B}| = \sqrt{3}|\vec{A} - \vec{B}|$ at time $t = \tau$ for the first time, the value of τ , in seconds, is _____.

[JEE Advanced-2018]

UD0113

PARAGRAPH "X"

In electromagnetic theory, the electric and magnetic phenomena are related to each other. Therefore, the dimensions of electric and magnetic quantities must also be related to each other. In the questions below, $[E]$ and $[B]$ stand for dimensions of electric and magnetic fields respectively, while $[\epsilon_0]$ and $[\mu_0]$ stand for dimensions of the permittivity and permeability of free space respectively. $[L]$ and $[T]$ are dimensions of length and time respectively. All the quantities are given in SI units.

(There are two questions based on Paragraph "X", the question given below is one of them)

6. The relation between $[E]$ and $[B]$ is :- [JEE Advanced-2018]
 (A) $[E] = [B][L][T]$ (B) $[E] = [B][L]^{-1}[T]$ (C) $[E] = [B][L][T]^{-1}$ (D) $[E] = [B][L]^{-1}[T]^{-1}$

UD0114

PARAGRAPH "X"

In electromagnetic theory, the electric and magnetic phenomena are related to each other. Therefore, the dimensions of electric and magnetic quantities must also be related to each other. In the questions below, $[E]$ and $[B]$ stand for dimensions of electric and magnetic fields respectively, while $[\epsilon_0]$ and $[\mu_0]$ stand for dimensions of the permittivity and permeability of free space respectively. $[L]$ and $[T]$ are dimensions of length and time respectively. All the quantities are given in SI units.

(There are two questions based on Paragraph "X", the question given below is one of them)

7. The relation between $[\epsilon_0]$ and $[\mu_0]$ is :- [JEE Advanced-2018]
 (A) $[\mu_0] = [\epsilon_0][L]^2[T]^{-2}$ (B) $[\mu_0] = [\epsilon_0][L]^{-2}[T]^2$
 (C) $[\mu_0] = [\epsilon_0]^{-1}[L]^2[T]^{-2}$ (D) $[\mu_0] = [\epsilon_0]^{-1}[L]^{-2}[T]^2$

UD0114

8. Let us consider a system of units in which mass and angular momentum are dimensionless. If length has dimension of L , which of the following statement (s) is/are correct ? [JEE Advanced-2019]

- (1) The dimension of force is L^{-3}
 (2) The dimension of energy is L^{-2}
 (3) The dimension of power is L^{-5}
 (4) The dimension of linear momentum is L^{-1}

UD0115

ANSWER KEY

EXERCISE (S-1)

1. Ans. L^{-1}, ML^2T^{-2}
2. Ans. $kg \cdot m^2 \cdot s^{-2} \cdot K^{-1}$
3. Ans. $T = a\sqrt{\frac{m}{k}}$
4. Ans. $\omega = K\sqrt{\frac{GM}{r^3}}$
5. Ans. $v_0 = k\sqrt{\frac{GM}{R}}$
6. Ans. $K = 6$
7. Ans. $[MLT^{-1}]$
8. Ans. $\sqrt{75}$
9. Ans. 1
10. Ans. 25
11. Ans. 7
12. Ans. 90 m
13. Ans. $\cos\alpha = \left(\frac{4}{\sqrt{61}}\right), \cos\beta = \left(\frac{6}{\sqrt{61}}\right), \cos\gamma = \left(\frac{3}{\sqrt{61}}\right), \text{magnitude} = \sqrt{61}$
14. Ans. 3
15. Ans. 5
16. Ans. $\frac{7}{16}$
17. Ans. $\hat{i} + \hat{j}, 3\hat{k}$
18. Ans. (a) $11\hat{i} + 5\hat{j} - 7\hat{k}$, (b) $\cos^{-1}\left(\frac{-7}{\sqrt{195}}\right)$, (c) $\cos^{-1}\left(\frac{-20}{\sqrt{1309}}\right)$
19. Ans. 2
20. Ans. (a) $\frac{2\pi}{3} \times 6.4 \times 10^6 m$, (b) $\sqrt{3} \times 6.4 \times 10^6 m$
21. Ans. (a) 9.95, (b) 0.99
22. Ans. 0.14, 0.09

EXERCISE (S-2)

1. Ans. $ML^5T^{-2}K^{1/2}$
2. Ans. $[S] = Ev^{-2}T^{-2}$
3. Ans. 10^{-30} star joule
4. Ans. $2\sqrt{19}, \cos^{-1}\frac{7}{2\sqrt{19}}$ or $\tan^{-1}\frac{3\sqrt{3}}{7}$
5. Ans. 100 s
6. Ans. 120 N, $40\sqrt{3}$ N
7. Ans. $16\hat{i} - 30\hat{k}, 198J$

EXERCISE (O-1)

1. Ans. (D)
2. Ans. (C)
3. Ans. (A)
4. Ans. (A)
5. Ans. (B)
6. Ans. (A)
7. Ans. (C)
8. Ans. (D)
9. Ans. (D)
10. Ans. (A)
11. Ans. (C)
12. Ans. (B)
13. Ans. (C)
14. Ans. (D)
15. Ans. (B)
16. Ans. (B)
17. Ans. (D)
18. Ans. (D)
19. Ans. (C)
20. Ans. (A)
21. Ans. (C)
22. Ans. (A)
23. Ans. (B)
24. Ans. (C)
25. Ans. (B)
26. Ans. (B)
27. Ans. (D)
28. Ans. (D)
29. Ans. (A)
30. Ans. (C)
31. Ans. (B)
32. Ans. (A)
33. Ans. (C)
34. Ans. (B)
35. Ans. (B)
36. Ans. (B)
37. Ans. (D)
38. Ans. (A,D)
39. Ans. (C,D)
40. Ans. (B, C)
41. Ans. (B)
42. Ans. (C)
43. Ans. (A)
44. Ans. (B)
45. Ans. (C)
46. Ans. (A) $\rightarrow$ (Q); (B) $\rightarrow$ (R); (C) $\rightarrow$ (T); (D) $\rightarrow$ (P)
47. Ans. (A)-R; (B)-S; (C)-P; (D)-Q
48. Ans. (A) $\rightarrow$ (S); (B) $\rightarrow$ (P); (C) $\rightarrow$ (R); (D) $\rightarrow$ (T)
49. Ans. (A)-S; (B)-P; (C)-Q; (D)-R

EXERCISE (O-2)

1. Ans. (A) 2. Ans. (D) 3. Ans. (C) 4. Ans. (B) 5. Ans. (C) 6. Ans. (A)
 7. Ans. (A) 8. Ans. (D) 9. Ans. (A) 10. Ans. (C) 11. Ans. (C) 12. Ans. (B)
 13. Ans. (A) 14. Ans. (A) 15. Ans. (A) 16. Ans. (B) 17. Ans. (C) 18. Ans. (C, D)
 19. Ans. (A,B,D) 20. Ans. (A, D) 21. Ans. (B,D) 22. Ans. (A,B) 23. Ans. (A,C,D)
 24. Ans. (A, B, C) 25. Ans. (A) 26. Ans. (C) 27. Ans. (B) 28. Ans. (C)
 29. Ans. (A) 30. Ans. (B) 31. Ans. (B,C) 32. Ans. (B,D) 33. Ans. (B)
 34. Ans. (A) $\rightarrow$ (Q); (B) $\rightarrow$ (S); (C) $\rightarrow$ (R); (D) $\rightarrow$ (R)
 35. Ans. (A) $\rightarrow$ (P,R,T); (B) $\rightarrow$ (P,R,T); (C) $\rightarrow$ (P,Q,R,S,T); (D) $\rightarrow$ (P,Q,R,S,T)
 36. Ans. (A) R (B) Q (C) P

EXERCISE (J-M)

1. Ans. (3)

EXERCISE (J-A)

1. Ans. (C) 2. Ans. 3 3. Ans. (A,C) 4. Ans. (C) 5. Ans. 2 [1.99, 2.01]
 6. Ans. (C) 7. Ans. (D) 8. Ans. (1,2,4)

EXERCISE (S)

1. How many significant figures are given in the following quantities ?

(A) 343 g (B) 2.20 (C) 1.103 N (D) 0.4142 s
(E) 0.0145 m (F) 1.0080 V (G) 9.1×10^4 km (H) 1.124×10^{-3} V

ER0001

2. Perform the following operations:

(A) $703 + 7 + 0.66$ (B) 2.21×0.3 (C) 12.4×84 (D) $14.28/0.714$

ER0002

3. Solve with due regard to significant digits

(i) $\sqrt{6.5 - 6.32}$ (ii) $\frac{2.91 \times 0.3842}{0.080}$

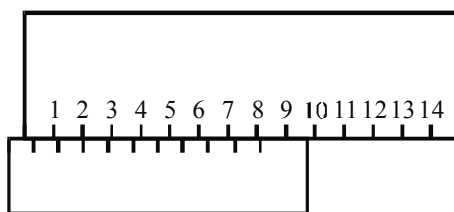
ER0003

4. The main scale of a vernier calipers reads in millimeter and its vernier is divided into 10 divisions which coincide with 9 divisions of the main scale. When the two jaws of the instrument touch each other the seventh division of the vernier scale coincide with a scale division and the zero of the vernier lies to the right of the zero of main scale. Furthermore, when a cylinder is tightly placed along its length between the two jaws, the zero of the vernier scale lies slightly to the left of 3.2 cm; and the fourth vernier division coincides with a scale division. Calculate the measured length of the cylinder.

ER0004

5. The VC shown in the diagram has zero error in it (as you can see). It is given that $9 \text{ msd} = 10 \text{ vsd}$.

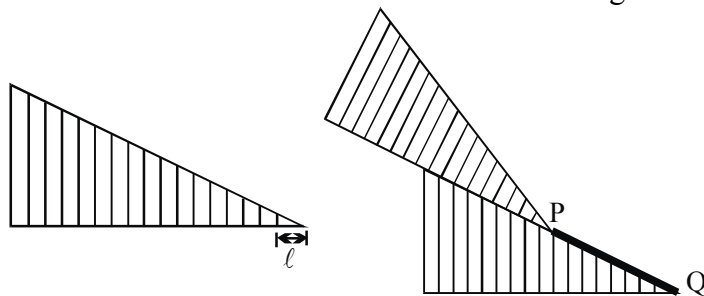
(i) What is the magnitude of the zero error? ($1 \text{ msd} = 1 \text{ mm}$)



(ii) The observed reading of the length of a rod measured by this VC comes out to be 5.4 mm. If the vernier had been error free then reading of main scale would be ____ and the coinciding division of vernier scale would be ____.

ER0005

6. Consider a home made vernier scale as shown in the figure.



In this diagram, we are interested in measuring the length of the line PQ. If both the inclines are identical and their angles are equal to θ then what is the least count of the instrument.

ER0065

7. The pitch of a screw gauge is 0.5 mm and there are 50 divisions on the circular scale. In measuring the thickness of a metal plate, there are five divisions on the pitch scale (or main scale) and thirty fourth division coincides with the reference line. Calculate the thickness of the metal plate.

ER0006

8. The pitch of a screw gauge is 1 mm and there are 50 divisions on its cap. When nothing is put in between the studs, 44th division of the circular scale coincides with the reference line zero of the main scale is not visible. When a glass plate is placed between the studs, the main scale reads three divisions and the circular scale reads 26 divisions. Calculate the thickness of the plate.

ER0007

9. In a given optical bench, a needle of length 10 cm is used to estimate bench error. The object needle, image needle & lens holder have their reading as shown.

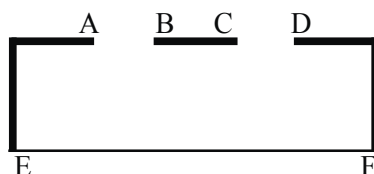
$$x_0 = 1.1 \text{ cm} \quad x_I = 21.0 \text{ cm} \quad x_L = 10.9 \text{ cm}$$

Estimate the bench errors which are present in image needle holder and object needle holder. Also find the focal length of the convex lens when.

$$x_0 = 0.6 \text{ cm} \quad x_I = 22.5 \text{ cm} \quad x_L = 11.4 \text{ cm}$$

ER0066

10. Make the appropriate connections in the meter bridge set up shown. Resistance box is connected between _____. Unknown resistance is connected between _____. Battery is connected between _____.



ER0008

11. A body travels uniformly a distance of (13.8 ± 0.2) m in time (4.0 ± 0.3) sec. Calculate its velocity.

ER0009

12. Consider $S = x \cos(\theta)$ for $x = (2.0 \pm 0.2)$ cm, $\theta = 53 \pm 2^\circ$. Find S.

ER0010

13. Two resistance R_1 and R_2 are connected in (i) series and (ii) parallel. What is the equivalent resistance with limit of possible percentage error in each case of $R_1 = 5.0 \pm 0.2 \Omega$ and $R_2 = 10.0 \pm 0.1 \Omega$.

ER0011

14. 5.74 gm of a substance occupies a volume of 1.2 cm^3 . Calculate its density with due regard for significant figures.

ER0012

15. The time period of oscillation of a simple pendulum is given by $T = 2\pi\sqrt{l/g}$. The length of the pendulum is measured as $l = 10.0 \pm 0.1$ cm and the time period as $T = 0.50 \pm 0.02$ s. Determine percentage error in the value of g.

ER0013

16. In a Searle's experiment, the diameter of the wire as measured by a screw gauge of least count 0.001 cm is 0.050 cm. The length, measured by a scale of least count 0.1 cm, is 110.0 cm. When a weight of 50 N is suspended from the wire, the extension is measured to be 0.125 cm by a micrometer of least count 0.001 cm. Find the maximum error in the measurement of Young's modulus of the material of the wire from these data.

[JEE 2004]

ER0014

17. The pitch of a screw gauge is 1 mm and there are 100 divisions on the circular scale. While measuring the diameter of a wire, the linear scale reads 1 mm and 47th division on the circular scale coincides with the reference line. The length of the wire is 5.6 cm. Find the curved surface area (in cm²) of the wire in appropriate number of significant figures. **[JEE 2004]**
ER0015
18. A physical quantity P is related to four observables A, B, C and D as $P = 4\pi^2 A^3 B^2 / (\sqrt{C} D)$. The percentage error of the measurement in A, B, C and D are 1%, 3% and 2%, 4% respectively. Determine the percentage error & absolute error in the quantity P. Value of P is calculated 3.763. **ER0067**
19. A glass prism of angle $A = 60^\circ$ gives minimum angle of deviation $\theta = 30^\circ$ with the max. error of 1° when a beam of parallel light passed through the prism during an experiment. Find the permissible error in the measurement of refractive index μ of the material of the prism. **ER0068**
20. In a vernier calipers the main scale and the vernier scale are made up of different materials. When the room temperature increases by $\Delta T^\circ\text{C}$, it is found the reading of the instrument remains the same. Earlier it was observed that the front edge of the wooden rod placed for measurement crossed the Nth main scale division and (N + 2) msd coincided with the 2nd vsd. Initially, 10 vsd coincided with 9 msd. If coefficient of linear expansion of the main scale is α_1 and that of the vernier scale is α_2 then what is the value of α_1 / α_2 ? (Ignore the expansion of the rod on heating) **ER0069**
21. In a vernier callipers, n divisions of its main scale match with (n + 1) divisions on its vernier scale. Each division of the main scale is a units. Using the vernier principle, calculate its least count. **[JEE 2003]**
ER0070
22. The period of oscillation of a simple pendulum in an experiment is recorded as 2.63 sec, 2.56 sec, 2.42 sec, 2.71 sec and 2.80 sec respectively. Find the time period, the absolute error in each observation, average absolute error and the percentage error. **ER0016**
23. The side of a cube is measured by vernier callipers (10 divisions of a vernier scale coincide with 9 divisions of main scale, where 1 division of main scale is 1 mm). The main scale reads 10 mm and first division of vernier scale coincides with the main scale. Mass of the cube is 2.736 g. Find the density of the cube in appropriate significant figures. **[JEE 2005]**
ER0017

EXERCISE (O)

1. A wire has a mass 0.3 ± 0.003 g, radius 0.5 ± 0.005 mm and length 6 ± 0.06 cm. The maximum percentage error in the measurement of its density is :- [JEE 2004]

(A) 1 (B) 2 (C) 3 (D) 4

ER0018

2. The edge of a cube is $a = 1.2 \times 10^{-2}$ m. Then its volume will be recorded as : [JEE 2003]

(A) $1.7 \times 10^{-6} \text{ m}^3$ (B) $1.70 \times 10^{-6} \text{ m}^3$
(C) $1.70 \times 10^{-7} \text{ m}^3$ (D) $1.78 \times 10^{-6} \text{ m}^3$

ER0019

3. A vernier callipers having 1 main scale division = 0.1 cm is designed to have a least count of 0.02 cm. If n be the number of divisions on vernier scale and m be the length of vernier scale, then :-

(A) $n = 10$, $m = 0.5$ cm (B) $n = 9$, $m = 0.4$ cm
(C) $n = 10$, $m = 0.8$ cm (D) $n = 10$, $m = 0.2$ cm

ER0020

4. In the Searle's experiment, after every step of loading, why should we wait for two minutes before taking the readings? (More than one correct.)

(A) So that the wire can have its desired change in length.
(B) So that the wire can attain room temperature.
(C) So that vertical oscillations can get subsided.
(D) So that the wire has no change in its radius.

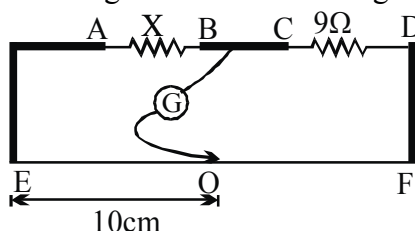
ER0021

5. In a meter bridge set up, which of the following should be the properties of the one meter long wire?

(A) High resistivity and low temperature coefficient
(B) Low resistivity and low temperature coefficient
(C) Low resistivity and high temperature coefficient
(D) High resistivity and high temperature coefficient

ER0022

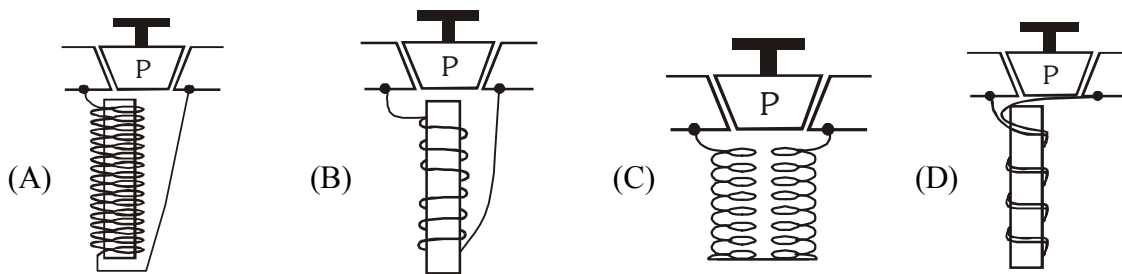
6. Consider the MB shown in the diagram, let the resistance X have temperature coefficient α_1 and the resistance from the RB have the temperature coefficient α_2 . Let the reading of the meter scale be 10 cm from the LHS. If the temperature of the two resistance increase by small temperature ΔT then what is the shift in the position of the null point? Neglect all the other changes in the bridge due to temperature rise.



(A) $9(\alpha_1 - \alpha_2)\Delta T$ (B) $9(\alpha_1 + \alpha_2)\Delta T$ (C) $\frac{1}{9}(\alpha_1 + \alpha_2)\Delta T$ (D) $\frac{1}{9}(\alpha_1 - \alpha_2)\Delta T$

ER0023

7. Identify which of the following diagrams represent the internal construction of the coils wound in a resistance box or PO box?



ER0024

8. In a meter bridge experiment, we try to obtain the null point at the middle. This
 (A) reduces systematic error as well as random error.
 (B) reduces systematic error but not the random error.
 (C) reduces random error but not the systematic error
 (D) reduces neither systematic error nor the random error

ER0025

9. An approximate value of number of seconds in an year is $\pi \times 10^7$. Determine the % error in this value
 (A) 0.5% (B) 8% (C) 4% (D) 15%

ER0071

10. In a Searle's experiment for determination of Young's Modulus, when a load of 50 kg is added to a 3 meter long wire micrometer screw having pitch 1 mm needs to be given a quarter turn in order to restore the horizontal position of spirit level. Young's modulus of the wire if its cross sectional area is 10^{-5} m^2 is
 (A) $6 \times 10^{11} \text{ N/m}^2$ (B) $1.5 \times 10^{11} \text{ N/m}^2$ (C) $3 \times 10^{11} \text{ N/m}^2$ (D) None

ER0026

11. On the basis of detail given about two measuring instruments, select the correct statement.

- (i) Vernier callipers having main scale division = 0.05 cm and Vernier scale division = $\frac{2.45}{50}$ cm.
 (ii) Screw gauge having pitch 0.5 mm and its circular scale division measures 0.01 mm.
 (A) Both the instruments have same least count.
 (B) Least count of screw gauge is lesser than that of vernier callipers.
 (C) Both the instruments have same least count but screw gauge is more precise.
 (D) Both the instruments have different least count and screw gauge is more precise.

ER0027

12. A student obtained following observations in an experiment of meter bridge to find unknown resistance of given wire :

S.No.	R	ℓ	$100 - \ell$	$S = \left(\frac{100 - \ell}{\ell} \right) R$
1	2Ω	43	57	2.65
2	3Ω	52	48	2.77
3	4Ω	58	42	2.89
4	5Ω	69	31	2.25

The most accurate value of unknown resistance will be

- (A) 2.65Ω (B) 2.77Ω (C) 2.89Ω (D) 2.25Ω

ER0028

13. In which of the following instruments used in the lab there exists an error of random category known as back lash error

(i) Screw gauge (ii) Spherometer (iii) Searle's apparatus (iv) Vernier callipers
 (A) (i) & (ii) only (B) (i), (ii) & (iii) only (C) (i) only (D) all four

ER0029

14. In Searle's apparatus, when experimental wire is loaded and unloaded, the air bubble in spirit level gets shifted.

(A) towards reference wire while loading & towards experimental wire while unloading
 (B) towards experimental wire while loading & towards reference wire while unloading
 (C) towards experimental wire, both the times, during loading & unloading
 (D) towards reference wire, both the times during loading & unloading

ER0030

15. Accuracy and precision are _____ (i) _____ and these are respectively linked with _____ (ii) & _____ (iii) _____. Fill the blanks above in correct order.

(A) (i) same, (ii) systematic error, (iii) random error
 (B) (i) different, (ii) systematic error (iii) random error
 (C) (i) same, (ii) random error, (iii) systematic error
 (D) (i) different, (ii) random error, (iii) systematic error

ER0031

16. The vernier of a circular scale is divided into 30 divisions, which coincides with 29 main scale divisions. If each main scale division is $(1/2)^\circ$, the least count of the instrument is

(A) $0.1'$ (B) $1'$ (C) $10'$ (D) $30'$

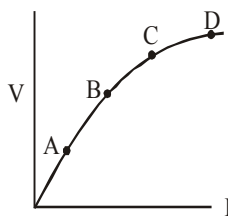
ER0032

17. When the gap is closed without placing any object in the screw gauge whose least count is 0.005 mm, the 5th division on its circular scale coincides with the reference line on main scale, and when a small sphere is placed reading on main scale advances by 4 divisions, whereas circular scale reading advances by five times to the corresponding reading when no object was placed. There are 200 divisions on the circular scale. The radius of the sphere is

(A) 4.10 mm (B) 4.05 mm (C) 2.10 mm (D) 2.05 mm

ER0033

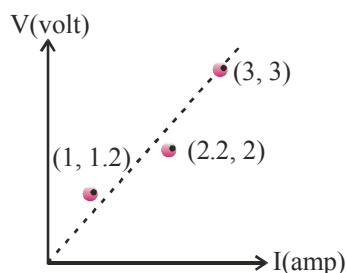
18. Variation of current passing through a conductor as the voltage supplied across its ends as varied is shown in the adjoining diagram. If the resistance (R) is determined at the points A, B, C and D we will find that -



(A) $R_C = R_D$ (B) $R_B > R_A$ (C) $R_C > R_B$ (D) $R_A > R_C$

ER0072

19. In the measurement of resistance of a wire using Ohm's law, the plot between V and I is drawn as shown. The resistance of the wire is -

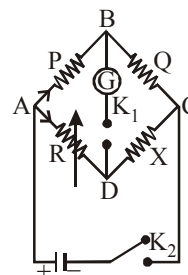


- (A) 0.833Ω (B) 0.9Ω (C) 1Ω (D) None of these

ER0034

20. In Wheatstone bridge experiment as shown in figure -

- (A) Key K_1 should be pressed first and then K_2
 (B) Key K_2 should be pressed first and then K_1
 (C) any key can be pressed in any order
 (D) both keys should be pressed simultaneously.



ER0035

21. In a metre bridge experiment null point is obtained at 20 cm from one end of the wire when resistance X is balanced against another resistance Y . If $X < Y$, then where will be the new position of the null point from the same end, if one decide to balance a resistance of $4X$ against Y -

- (A) 50 cm (B) 80 cm (C) 40 cm (D) 70 cm

ER0036

22. In a resonance column method, resonance occurs at two successive level of $l_1 = 30.7$ cm and $l_2 = 63.2$ cm using a tuning fork of $f = 512$ Hz. What is the maximum error in measuring speed of sound using relations $v = f\lambda$ & $\lambda = 2(l_2 - l_1)$

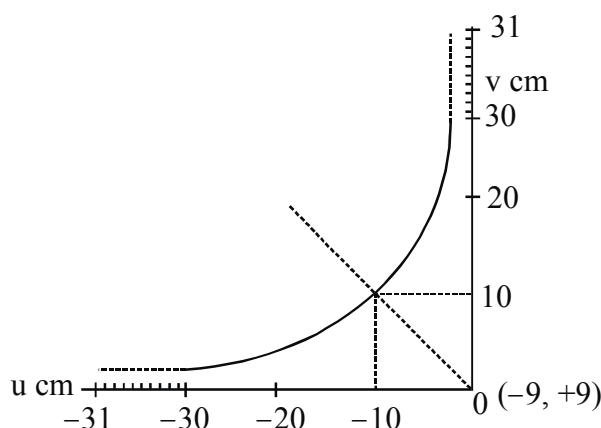
[JEE 2005]

- (A) 256 cm/sec (B) 92 cm/sec (C) 128 cm/sec (D) 102.4 cm/sec

ER0037

23. Graph of position of image vs position of point object from a convex lens is shown. Then, focal length of the lens is

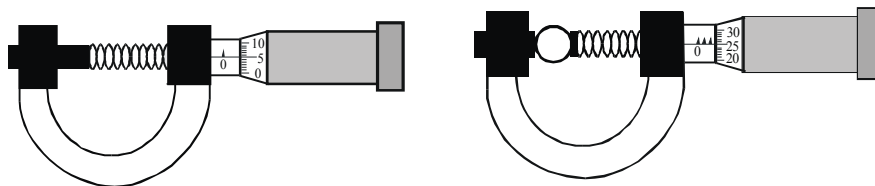
[JEE 2006]



- (A) 0.50 ± 0.05 cm (B) 0.50 ± 0.10 cm (C) 5.00 ± 0.05 cm (D) 5.00 ± 0.10 cm

ER0038

24. The circular divisions of shown screw gauge are 50. It moves 0.5 mm on main scale in one rotation. The diameter of the ball is [JEE 2006]



- (A) 2.25 mm (B) 2.20 mm (C) 1.20 mm (D) 1.25 mm

ER0073

25. A student performs an experiment for determination of $g \left(= \frac{4\pi^2 l}{T^2} \right)$ $l \approx 1$ m and he commits an error of Δl . For the experiment takes the time of n oscillations with the stop watch of least count ΔT and he commits a human error of 0.1 sec. For which of the following data, the measurement of g will be most accurate?

	Δl	ΔT	n	Amplitude of oscillation
(A)	5 mm	0.2 sec	10	5 mm
(B)	5 mm	0.2 sec	20	5 mm
(C)	5 mm	0.1 sec	20	1 mm
(D)	1 mm	0.1 sec	50	1 mm

[JEE 2006]

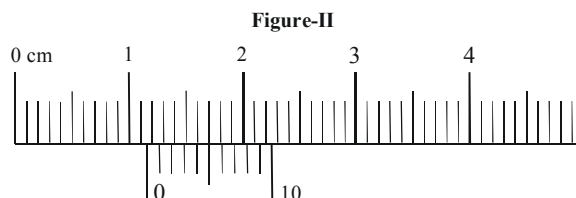
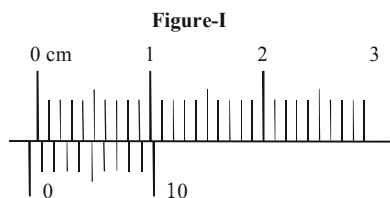
ER0039

26. In an experiment to determine the focal length (f) of a concave mirror by the u - v method, a student places the object pin A on the principal axis at a distance x from the pole P. The student looks at the pin and its inverted image from a distance keeping his/her eye in line with PA. When the student shifts his/her eye towards left, the image appears to the right of the object pin. Then, [JEE 2007]

- (A) $x < f$ (B) $f < x < 2f$ (C) $x = 2f$ (D) $x > 2f$

ER0040

27. In ordinary Vernier calipers, 10th division of the Vernier scale coincides with 9th division of the main scale. In a specially designed Vernier calipers the Vernier scale is so constructed that 10th division on it coincides with 11th division on the main scale. Each division on the main scale equals to 1 mm. The calipers have a zero error as shown in the figure-I. When the Vernier caliper is used to measure a length, the concerned portion of its scale is shown in figure-II.



- (A) Zero error in the calipers has magnitude 0.7 mm. (B) The length being measured is 1.08 cm.
(C) The length being measured is 1.22 cm. (D) Zero error in the calipers has magnitude 0.3 mm

ER0041

28. A student performs an experiment to determine the Young's modulus of a wire, exactly 2 m long, by Searle's method. In a particular reading, the student measures the extension in the length of the wire to be 0.8 mm with an uncertainty of ± 0.05 mm at a load of exactly 1.0 kg. The student also measures the diameter of the wire to be 0.4 mm with an uncertainty of ± 0.01 mm. Take $g = 9.8 \text{ m/s}^2$ (exact). The Young's modulus obtained from the reading is [JEE 2007]

- (A) $(2.0 \pm 0.3) \times 10^{11} \text{ N/m}^2$ (B) $(2.0 \pm 0.2) \times 10^{11} \text{ N/m}^2$
(C) $(2.0 \pm 0.1) \times 10^{11} \text{ N/m}^2$ (D) $(2.0 \pm 0.05) \times 10^{11} \text{ N/m}^2$

ER0042

EXERCISE-JM

1. An experiment is performed to find the refractive index of glass using a travelling microscope. In this experiment distances are measured by [AIEEE - 2008]

- (1) a vernier scale provided on the microscope
- (2) a standard laboratory scale
- (3) a meter scale provided on the microscope
- (4) a screw gauge provided on the microscope

ER0043

2. Two full turns of the circular scale of gauge cover a distance of 1 mm on scale. The total number of divisions on circular scale is 50. Further, it is found that screw gauge has a zero error of -0.03 mm. While measuring the diameter of a thin wire a student notes the main scale reading of 3 mm and the number of circular scale division in line, with the main scale as 35. The diameter of the wire is

[AIEEE - 2008]

- (1) 3.32 mm
- (2) 3.73 mm
- (3) 3.67 mm
- (4) 3.38 mm

ER0044

3. In an experiment the angles are required to be measured using an instrument 29 divisions of the main scale exactly coincide with the 30 divisions of the vernier scale. If the smallest division of the main scale is half-a-degree ($=0.5^\circ$), then the least count of the instrument is :- [AIEEE - 2009]

- (1) One degree
- (2) Half degree
- (3) One minute
- (4) Half minute

ER0045

4. In an optics experiment, with the position of the object fixed, a student varies the position of a convex lens and for each position, the screen is adjusted to get a clear image of the object. A graph between the object distance u and the image distance v , from the lens, is plotted using the same scale for the two axes. A straight line passing through the origin and making an angle of 45° with the x -axis meets the experimental curve at P. The coordinates of P will be :- [AIEEE - 2009]

- (1) (f, f)
- (2) $(4f, 4f)$
- (3) $(2f, 2f)$
- (4) $\left(\frac{f}{2}, \frac{f}{2}\right)$

ER0046

5. The respective number of significant figures for the numbers 23.023, 0.0003 and 2.1×10^{-3} are :- [AIEEE - 2010]

- (1) 4, 4, 2
- (2) 5, 1, 2
- (3) 5, 1, 5
- (4) 5, 5, 2

ER0047

6. A screw gauge gives the following reading when used to measure the diameter of a wire.

Main scale reading : 0 mm.

Circular scale reading : 52 divisions

Given that 1 mm on main scale corresponds to 100 divisions of the circular scale.

The diameter of wire from the above data is :-

[AIEEE - 2011]

- (1) 0.026 cm
- (2) 0.005 cm
- (3) 0.52 cm
- (4) 0.052 cm

ER0048

7. A spectrometer gives the following reading when used to measure the angle of a prism. Main scale reading : 58.5 degree
Vernier scale reading : 09 divisions
Given that 1 division on main scale corresponds to 0.5 degree. Total divisions on the vernier scale is 30 and match with 29 divisions of the main scale. The angle of the prism from the above data :
[AIEEE - 2012]
(1) 59 degree (2) 58.59 degree (3) 58.77 degree (4) 58.65 degree
ER0049
8. Resistance of a given wire is obtained by measuring the current flowing in it and the voltage difference applied across it. If the percentage errors in the measurement of the current and the voltage difference are 3% each, then error in the value of resistance of the wire is :-
[AIEEE - 2012]
(1) 3% (2) 6% (3) zero (4) 1%
ER0050
9. The current voltage relation of diode is given by $I = (e^{1000V/T} - 1)$ mA, where the applied voltage V is in volts and the temperature T is in degree Kelvin. If a student makes an error measuring ± 0.01 V while measuring the current of 5 mA at 300 K, what will be error in the value of current in mA ?
[JEE-Main 2014]
(1) 0.5 mA (2) 0.05 mA (3) 0.2 mA (4) 0.02 mA
ER0051
10. A student measured the length of a rod and wrote it as 3.50 cm. Which instrument did he use to measure it ?
[JEE-Main 2014]
(1) A screw gauge having 100 divisions in the circular scale and pitch as 1 mm.
(2) A screw gauge having 50 divisions in the circular scale and pitch as 1 mm.
(3) A meter scale
(4) A vernier calliper where the 10 divisions in vernier scale matches with 9 division in main scale and main scale has 10 divisions in 1 cm.
ER0052
11. The period of oscillation of a simple pendulum is $T = 2\pi\sqrt{\frac{L}{g}}$. Measured value of L is 20.0 cm known to 1 mm accuracy and time for 100 oscillations of the pendulum is found to be 90s using a wrist watch of 1s resolution. The accuracy in the determination of g is :
[JEE-Main 2015]
(1) 1% (2) 5% (3) 2% (4) 3%
ER0053
12. A screw gauge with a pitch of 0.5 mm and a circular scale with 50 divisions is used to measure the thickness of a thin sheet of Aluminium. Before starting the measurement, it is found that when the two jaws of the screw gauge are brought in contact, the 45th division coincides with the main scale line and that the zero of the main scale is barely visible. What is the thickness of the sheet if the main scale reading is 0.5 mm and the 25th division coincides with the main scale line ?
[JEE-Main 2016]
(1) 0.50 mm (2) 0.75 mm (3) 0.80 mm (4) 0.70 mm
ER0054

13. A student measures the time period of 100 oscillations of a simple pendulum four times. The data set is 90 s, 91 s, 95 s and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be :- [JEE-Main 2016]

(1) 92 ± 3 s (2) 92 ± 2 s (3) 92 ± 5.0 s (4) 92 ± 1.8 s

ER0055

14. To know the resistance G of a galvanometer by half deflection method, a battery of emf V_E and resistance R is used to deflect the galvanometer by angle θ . If a shunt of resistance S is needed to get half deflection then G , R and S are related by the equation : [JEE-Mains (Online) 2016]

(1) $2S(R+G) = RG$ (2) $S(R+G) = RG$ (3) $2G = S$ (4) $2S = G$

ER0056

15. The following observations were taken for determining surface tension T of water by capillary method:

Diameter of capillary, $D = 1.25 \times 10^{-2}$ m

rise of water, $h = 1.45 \times 10^{-2}$ m

Using $g = 9.80$ m/s² and the simplified relation $T = \frac{r h g}{2} \times 10^3$ N/m, the possible error in surface tension is closest to : [JEE-Main 2017]

(1) 2.4% (2) 10% (3) 0.15% (4) 1.5%

ER0057

16. The density of a material in the shape of a cube is determined by measuring three sides of the cube and its mass. If the relative errors in measuring the mass and length are respectively 1.5% and 1%, the maximum error in determining the density is :- [JEE-Main 2018]

(1) 3.5 % (2) 4.5 % (3) 6 % (4) 2.5 %

ER0058

EXERCISE - JA

1. Students I, II and III perform an experiment for measuring the acceleration due to gravity (g) using pendulum. They use different lengths of the pendulum and/or record time for different number of oscillations. The observations are shown in the table. **[JEE 2008]**

[JEE 2008]

Least count for length = 0.1 cm

Least count for time = 0.1 s

Student	Length of the Pendulum (cm)	Number of oscilltions (n)	Total time for (n) oscillations (s)	Time period (s)
I	64.0	8	128.0	16.0
II	64.0	4	64.0	16.0
III	20.0	4	36.0	9.0

If E_I , E_{II} and E_{III} are the percentage error in g , i.e., $\left(\frac{\Delta g}{g} \times 100\right)$ for students I, II and III, respectively,

- (A) $E_I = 0$ (B) E_I is minimum
(C) $E_I = E_{II}$ (D) E_{II} is maximum

ER0074

2. A student uses a simple pendulum of exactly 1m length to determine g , the acceleration due to gravity. He uses a stopwatch with the least count of 1 sec for this and records 40 seconds for 20 oscillations. For this observation, which of the following statement(s) is (are) true? **[JEE 2010]**

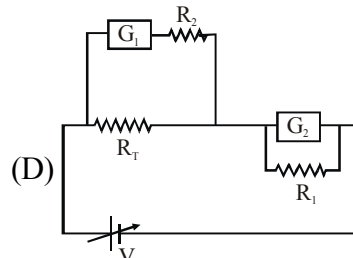
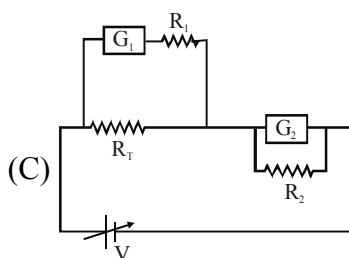
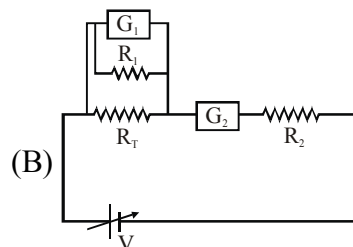
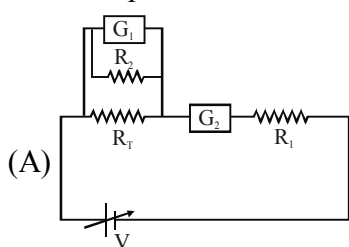
[JEE 2010]

- (A) Error ΔT in measuring T , the time period is 0.05 seconds
(B) Error ΔT in measuring T , the time period is 1 second
(C) Percentage error in the determination of g is 5%
(D) Percentage error in the determination of g is 2.5%

ER0075

3. To verify Ohm's law, a student is provided with a test resistor R_T , a high resistance R_1 , a small resistance R_2 , two identical galvanometers G_1 and G_2 , and a variable voltage source V . The correct circuit to carry out the experiment is :— **[JEE 2010]**

[JEE 2010]

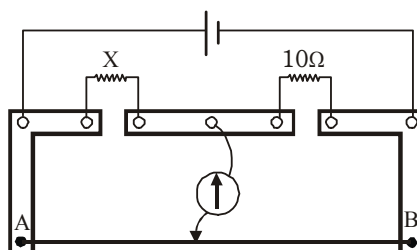


ER0076

4. A Vernier callipers has 1 mm marks on the main scale. It has 20 equal divisions on the Vernier scale which match with 16 main scale divisions. For this Vernier calipers, the least count is [JEE 2010]
 (A) 0.02 mm (B) 0.05 mm (C) 0.1 mm (D) 0.2 mm

ER0077

5. A meter bridge is set-up as shown, to determine an unknown resistance 'X' using a standard 10 ohm resistor. The galvanometer shows null point when tapping-key is at 52 cm mark. The end-corrections are 1 cm and 2 cm respectively for the ends A and B. The determined value of 'X' is [JEE 2011]



- (A) 10.2 ohm (B) 10.6 ohm (C) 10.8 ohm (D) 11.1 ohm

ER0078

6. The density of a solid ball is to be determined in an experiment. The diameter of the ball is measured with a screw gauge, whose pitch is 0.5 mm and there are 50 divisions on the circular scale. The reading on the main scale is 2.5 mm and that on the circular scale is 20 divisions. If the measured mass of the ball has a relative error of 2%, the relative percentage error in the density is [JEE 2011]
 (A) 0.9% (B) 2.4% (C) 3.1% (D) 4.2%

ER0079

7. In the determination of Young's modulus $\left(Y = \frac{4MLg}{\pi \ell d^2}\right)$ by using Searle's method, a wire of length $L = 2\text{m}$

and diameter $d = 0.5\text{ mm}$ is used. For a load $M = 2.5\text{ kg}$, an extension $\ell = 0.25\text{ mm}$ in the length of the wire is observed. Quantities d and ℓ are measured using a screw gauge and a micrometer, respectively. They have the same pitch of 0.5 mm. The number of divisions on their circular scale is 100. The contributions to the maximum probable error of the Y measurement

- (A) due to the errors in the measurements of d and ℓ are the same.
 (B) due to the error in the measurement of d is twice that due to the error in the measurement of ℓ .
 (C) due to the error in the measurement of ℓ is twice that due to the error in the measurement of d .
 (D) due to the error in the measurement of d is four times that due to the error in the measurement of ℓ .

[JEE 2012]

ER0080

8. The diameter of a cylinder is measured using a Vernier callipers with no zero error. It is found that the zero of the Vernier scale lies between 5.10 cm and 5.15 cm of the main scale. The Vernier scale has 50 divisions equivalent to 2.45 cm. The 24th division of the Vernier scale exactly coincides with one of the main scale divisions. The diameter of the cylinder is :- [JEE-Advance 2013]

- (A) 5.112 cm (B) 5.124 cm (C) 5.136 cm (D) 5.148 cm

ER0081

9. Using the expression $2d \sin \theta = \lambda$, one calculates the values of d by measuring the corresponding angles θ in the range 0 to 90° . The wavelength λ is exactly known and the error in θ is constant for all values of θ . As θ increases from 0° :-

[JEE-Advance 2013]

- (A) the absolute error in d remains constant
- (B) the absolute error in d increases
- (C) the fractional error in d remains constant
- (D) the fractional error in d decreases

ER0082

10. During Searle's experiment, zero of the Vernier scale lies between 3.20×10^{-2} m and 3.25×10^{-2} m of the main scale. The 20^{th} division of the Vernier scale exactly coincides with one of the main scale divisions. When an additional load of 2 kg is applied to the wire, the zero of the Vernier scale still lies between 3.20×10^{-2} m and 3.25×10^{-2} m of the main scale but now the 45^{th} division of Vernier scale coincides with one of the main scale divisions. The length of the thin metallic wire is 2m and its cross-sectional area is 8×10^{-7} m². The least count of the Vernier scale is 1.0×10^{-5} m. The maximum percentage error in the Young's modulus of the wire is.

[JEE-Advance 2014]

ER0083

11. Consider a Vernier callipers in which each 1 cm on the main scale is divided into 8 equal divisions and a screw gauge with 100 divisions on its circular scale. In the Vernier callipers, 5 divisions of the Vernier scale coincide with 4 divisions on the main scale and in the screw gauge, one complete rotation of the circular scale moves it by two divisions on the linear scale. Then :

[JEE-Advance 2015]

- (A) If the pitch of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.01 mm.
- (B) If the pitch of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.005 mm.
- (C) If the least count of the linear scale of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.01 mm.
- (D) If the least count of the linear scale of the screw gauge is twice the least count of the Vernier callipers, the least count of the screw gauge is 0.005 mm.

ER0084

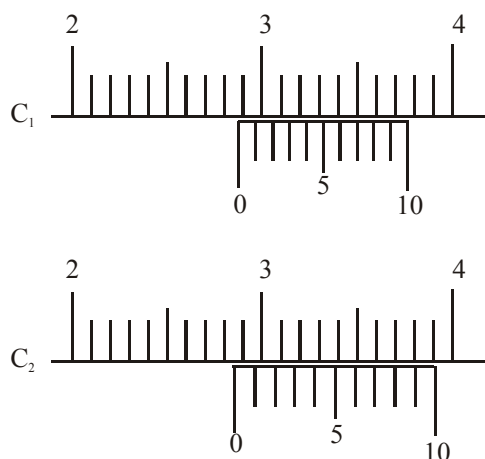
12. The energy of a system as a function of time t is given as $E(t) = A^2 \exp(-\alpha t)$, where $\alpha = 0.2 \text{ s}^{-1}$. The measurement of A has an error of 1.25%. If the error in the measurement of time is 1.50%, the percentage error in the value of $E(t)$ at $t = 5$ s is.

[JEE-Advance 2015]

ER0085

13. There are two vernier calipers both of which have 1 cm divided into 10 equal divisions on the main scale. The Vernier scale of one of the calipers (C_1) has 10 equal divisions that correspond to 9 main scale divisions. The Vernier scale of the other caliper (C_2) has 10 equal divisions that correspond to 11 main scale divisions. The readings of the two calipers are shown in the figure. The measured values (in cm) by calipers C_1 and C_2 respectively, are

[JEE-Advance 2016]



- (A) 2.87 and 2.86 (B) 2.87 and 2.87 (C) 2.87 and 2.83 (D) 2.85 and 2.82

ER0086

14. In an experiment to determine the acceleration due to gravity g , the formula used for the time period of a periodic motion is $T = 2\pi \sqrt{\frac{7(R-r)}{5g}}$. The values of R and r are measured to be (60 ± 1) mm and (10 ± 1) mm, respectively. In five successive measurements, the time period is found to be 0.52 s, 0.56 s, 0.57 s, 0.54 s and 0.59 s. The least count of the watch used for the measurement of time period is 0.01 s. Which of the following statement(s) is(are) true? [JEE-Advance 2016]

- (A) The error in the measurement of r is 10%
 (B) The error in the measurement of T is 3.57 %
 (C) The error in the measurement of T is 2%
 (D) The error in the determined value of g is 11%

ER0087

15. A person measures the depth of a well by measuring the time interval between dropping a stone and receiving the sound of impact with the bottom of the well. The error in his measurement of time is $\delta T = 0.01$ second and he measures the depth of the well to be $L = 20$ meters. Take the acceleration due to gravity $g = 10 \text{ ms}^{-2}$ and the velocity of sound is 300 ms^{-1} . Then the fractional error in the measurement, $\delta L/L$, is closest to [JEE-Advance 2017]
- (A) 0.2 % (B) 5 % (C) 3 % (D) 1 %

ER0088

PARAGRAPH "A"

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the expansion at the first power of the error. For example, consider the relation $z = x/y$. If the errors in x , y and z are Δx , Δy and Δz , respectively, then [JEE-Advance 2018]

$$z \pm \Delta z = \frac{x \pm \Delta x}{y \pm \Delta y} = \frac{x}{y} \left(1 \pm \frac{\Delta x}{x} \right) \left(1 \pm \frac{\Delta y}{y} \right)^{-1}$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y} \right)^{-1}$, to first power in $\Delta y/y$, is $1 \mp (\Delta y/y)$. The relative errors in independent variables are always added. So the error in z will be

$$\Delta z = z \left(\frac{\Delta x}{x} + \frac{\Delta y}{y} \right).$$

The above derivation makes the assumption that $\frac{\Delta x}{x} \ll 1, \frac{\Delta y}{y} \ll 1$. Therefore, the higher powers of these quantities are neglected.

(There are two questions based on Paragraph "A", the question given below is one of them)

16. Consider the ratio $r = \frac{(1-a)}{(1+a)}$ to be determined by measuring a dimensionless quantity a . If the error in the measurement of a is Δa ($\Delta a/a \ll 1$), then what is the error Δr in determining r ?

- (A) $\frac{\Delta a}{(1+a)^2}$ (B) $\frac{2\Delta a}{(1+a)^2}$
(C) $\frac{2\Delta a}{(1-a^2)}$ (D) $\frac{2a\Delta a}{(1-a^2)}$

ER0089

PARAGRAPH "A"

If the measurement errors in all the independent quantities are known, then it is possible to determine the error in any dependent quantity. This is done by the use of series expansion and truncating the expansion at the first power of the error. For example, consider the relation $z = x/y$. If the errors in x , y and z are Δx , Δy and Δz , respectively, then

$$z \pm \Delta z = \frac{x \pm \Delta x}{y \pm \Delta y} = \frac{x}{y} \left(1 \pm \frac{\Delta x}{x} \right) \left(1 \pm \frac{\Delta y}{y} \right)^{-1}.$$

The series expansion for $\left(1 \pm \frac{\Delta y}{y} \right)^{-1}$, to first power in $\Delta y/y$, is $1 \mp (\Delta y/y)$. The relative errors in independent variables are always added. So the error in z will be

$$\Delta z = z \left(\frac{\Delta x}{x} + \frac{\Delta y}{y} \right).$$

The above derivation makes the assumption that $\frac{\Delta x}{x} \ll 1, \frac{\Delta y}{y} \ll 1$. Therefore, the higher powers of these quantities are neglected.

(There are two questions based on Paragraph "A", the question given below is one of them)

17. In an experiment the initial number of radioactive nuclei is 3000. It is found that 1000 ± 40 nuclei decayed in the first 1.0 s. For $|x| \ll 1$, $\ln(1+x) = x$ up to first power in x . The error $\Delta \lambda$, in the determination of the decay constant λ , in s^{-1} , is :-
(A) 0.04 (B) 0.03 (C) 0.02 (D) 0.01

ER0090

18. A steel wire of diameter 0.5 mm and Young's modulus $2 \times 10^{11} \text{ N m}^{-2}$ carries a load of mass M . The length of the wire with the load is 1.0 m. A vernier scale with 10 divisions is attached to the end of this wire. Next to the steel wire is a reference wire to which a main scale, of least count 1.0 mm, is attached. The 10 divisions of the vernier scale correspond to 9 divisions of the main scale. Initially, the zero of vernier scale coincides with the zero of main scale. If the load on the steel wire is increased by 1.2 kg, the vernier scale division which coincides with a main scale division is..... Take $g = 10 \text{ ms}^{-2}$ and $\pi = 3.2$.

[JEE-Advance 2018]

ER0091

19. An optical bench has 1.5 m long scale having four equal divisions in each cm. While measuring the focal length of a convex lens, the lens is kept at 75 cm mark of the scale and the object pin is kept at 45 cm mark. The image of the object pin on the other side of the lens overlaps with image pin that is kept at 135 cm mark. In this experiment, the percentage error in the measurement of the focal length of the lens is_____.

[JEE-Advance 2019]

ER0092

ANSWER KEY

EXERCISE (S)

1. Ans. (A) 3, (B) 3, (C) 4, (D) 4, (E) 3, (F) 5, (G) 2, (H) 4
 2. Ans. (A) 711, (B) 0.7, (C) 1.0×10^3 , (D) 20.0
 3. Ans. (i) 0.4, (ii) 14 4. Ans. 3.07 cm 5. Ans. (i) $x = -0.7$ msd, (ii) 6, 1
 6. Ans. L.C. = $l \left[\frac{1 - \cos \theta}{\cos \theta} \right]$ 7. Ans. 2.84 mm 8. Ans. $R_t = 3.64$ mm
 9. Ans. 5.5 ± 0.05 cm 10. Ans. CD, AB, EF 11. Ans. $v = (3.4 \pm 0.4)$ m/s
 12. Ans. $S = (1.2 \pm 0.18)$ cm 13. Ans. $R_g = 15.0 \Omega \pm 2\%$, $R_p = 3.3 \Omega \pm 3\%$
 14. Ans. 4.8 g/cm^3 15. Ans. 9% 16. Ans. $\Delta Y = 0.0489Y = 1.1 \times 10^{10} \text{ N/m}^2$
 17. Ans. 2.6 cm^2 (in two significant figures) 18. Ans. 14%, 0.53
 19. Ans. $5\pi/18\%$ 20. Ans. $1.8/(N+2)$ 21. Ans. $\frac{a}{n+1}$
 22. Ans. $T = 2.62 \text{ s}$, 0.01 s , -0.06 s , -0.20 s , 0.09 s , 0.18 s , Average absolute error = 0.11 s , 4.2%
 23. Ans. 2.66 g/cm^3

EXERCISE (O)

1. Ans. (D) 2. Ans. (A) 3. Ans. (C) 4. Ans. (A,B,C) 5. Ans. (A) 6. Ans. (A)
 7. Ans. (D) 8. Ans. (A) 9. Ans. (A) 10. Ans. (A) 11. Ans. (A) 12. Ans. (B)
 13. Ans. (B) 14. Ans. (A) 15. Ans. (B) 16. Ans. (B) 17. Ans. (D) 18. Ans. (D)
 19. Ans. (C) 20. Ans. (B) 21. Ans. (A) 22. Ans. (D) 23. Ans. (C) 24. Ans. (C)
 25. Ans. (D) 26. Ans. (B) 27. Ans. (A,C) 28. Ans. (A, B)

EXERCISE-JM

1. Ans. (1) 2. Ans. (4) 3. Ans. (3) 4. Ans. (3) 5. Ans. (2) 6. Ans. (4)
 7. Ans. (4) 8. Ans. (2) 9. Ans. (3) 10. Ans. (4) 11. Ans. (4) 12. Ans. (3)
 13. Ans. (2) 14. Ans. (2) 15. Ans. (4) 16. Ans. (2)

EXERCISE - JA

1. Ans. (B) 2. Ans. (A,C) 3. Ans. (C) 4. Ans. (D) 5. Ans. (B) 6. Ans. (C)
 7. Ans. (A) 8. Ans. (B) 9. Ans. (D) 10. Ans. 4 11. Ans. (B,C) 12. Ans. 4
 13. Ans. (C) 14. Ans. (A, B, D) 15. Ans. (D) 16. Ans. (B) 17. Ans. (C)
 18. Ans. 3 [2.99, 3.01] 19. Ans. (1.35 to 1.45)

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2	KINEMATICS-1D & CALCULUS	71-120
3	KINEMATICS-2D	121-170

PHYSICS



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Important Notes

[illegible]

NEWTON'S LAWS OF MOTION & FRICTION

KEY CONCEPTS

FORCE

A push or pull that one object exerts on another.

FORCES IN NATURE

There are four fundamental forces in nature :

1. Gravitational force
2. Electromagnetic force
3. Strong nuclear force
4. Weak force

TYPES OF FORCES ON MACROSCOPIC OBJECTS

(a) Field Forces or Range Forces :

These are the forces in which contact between two objects is not necessary.

Ex. (i) Gravitational force between two bodies. (ii) Electrostatic force between two charges.

(b) Contact Forces :

Contact forces exist only as long as the objects are touching each another.

Ex. (i) Normal forces. (ii) Frictional force

(c) Attachment to Another Body :

Tension (T) in a string and spring force ($F = kx$) comes in this group.

NEWTON'S FIRST LAW OF MOTION (OR GALILEO'S LAW OF INERTIA)

Every body continues in its state of rest or uniform motion in a straight line unless compelled by an external force to change that state.

Definition of force from Newton's first law of motion "Force is that push or pull which changes or tends to change the state of rest or of uniform motion in a straight line".

Inertia : Inertia is the property of the body due to which body oppose the change of its state. Inertia of a body is measured by mass of the body. $\boxed{\text{inertia} \propto \text{mass}}$

TYPES OF INERTIA

Inertia of rest : It is the inability of a body to change, its state of rest by itself.

Examples :

- When we shake a branch of a mango tree, the mangoes fall down.
- When a bus or train starts suddenly the passengers sitting inside tends to fall backwards.
- The dust particles in a blanket fall off when it is beaten with a stick.
- When a stone hits a window pane, the glass is broken into a number of pieces whereas if the high speed bullet strikes the pane, it leaves a clean hole.

Inertia of motion : It is the inability of a body to change its state of uniform motion by itself.

Examples :

- When a bus or train stop suddenly, a passenger sitting inside tends to fall forward.
- A person jumping out of a speeding train may fall forward.
- A ball thrown upwards in a running train continues to move along with the train.

Inertia of direction : It is the inability of a body to change its direction of motion by itself.

Examples :

- When a car rounds a curve suddenly, the person sitting inside is thrown outwards.
- Rotating wheels of vehicle throw out mud, mudguard over the wheels stop this mud.
- A body released from a balloon rising up, continues to move in the direction of balloon.

NEWTON'S SECOND LAW OF MOTION

Rate of change of momentum of a body is directly proportional to the external force applied on it and the change in momentum takes place in the direction of force

$$\vec{F} \propto \frac{d\vec{p}}{dt} \quad \text{or} \quad \vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}(m\vec{v}) = m \frac{d\vec{v}}{dt} + \vec{v} \frac{dm}{dt}$$

$$\text{if } m = \text{constant then } \vec{F} = m \frac{d\vec{v}}{dt} = m\vec{a}$$

- Newton's Second Law Provides the Definition of the Concept of Force.
- **Definition of the 1 Newton (N) :-** If an object of mass one kilogram has an acceleration of 1ms^{-2} relative to an inertial reference frame, then the net force exerted on the object is one newton.

CONSEQUENCES OF NEWTON'S II LAW OF MOTION

- **Concept of inertial mass :** From Newton's II law of motion $a = \frac{F}{M}$
i.e., the magnitude of acceleration produced by a given body is inversely proportional to mass i.e. greater the mass, smaller is the acceleration produced in the body. Thus, mass is the measure of inertia of the body. The mass given by above equation is therefore called the inertial mass.
- **An accelerated motion is the result of application of the force :**
There may be two types of accelerated motion :
(i) When only the magnitude of velocity of the body changes : In this types of motion the force is applied along the direction of motion or opposite to the direction of motion.
(ii) When only the direction of motion of the body changes : In this case the force is applied at right angles to the direction of motion of the body, e.g. uniform circular motion.
- **Acceleration produced in the body depends only on its mass and not on the final or initial velocity.**

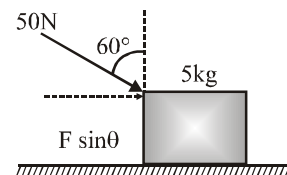
Ex. A force $\vec{F} = (6\hat{i} - 8\hat{j} + 10\hat{k})\text{N}$ produces acceleration of 1ms^{-2} in a body. Calculate the mass of the body.

Sol. $\therefore$ Acceleration $a = \frac{|\vec{F}|}{m}$ $\therefore$ mass $m = \frac{|\vec{F}|}{a} = \frac{\sqrt{6^2 + 8^2 + 10^2}}{1} = 10\sqrt{2}\text{ kg}$

Ex. A force of 50 N acts in the direction as shown in figure. The block of mass 5kg, resting on a smooth horizontal surface. Find out the acceleration of the block.

Sol. Horizontal component of the force = $F \sin \theta = 50 \sin 60^\circ = \frac{50\sqrt{3}}{2}$ N

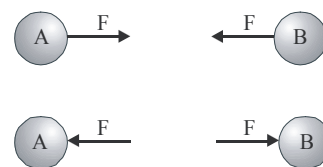
$$\text{acceleration of the block } a = \frac{F \sin \theta}{m} = \frac{50\sqrt{3}}{2} \times \frac{1}{5} = 5\sqrt{3} \text{ m/s}^2$$



NEWTON'S THIRD LAW OF MOTION

The first and second laws are statements about a single object, whereas the third law is a statement about two objects.

- According to this law, every action has equal and opposite reaction. Action and reaction act on different bodies and they are simultaneous. There can be no reaction without action.
- If an object A exerts a force F on an object B, then B exerts an equal and opposite force $(-F)$ on A.
- Action and reaction never cancel each other, since they act on different bodies.
- First law :** If no net force acts on a particle, then it is possible to select a set of reference frames, called inertial reference frames, observed from which the particle moves without any change in velocity.
- Second law :** Observed from an inertial reference frame, the net force on a particle is proportional



The forces between two objects A and B are equal and opposite, whether they are attractive or repulsive.

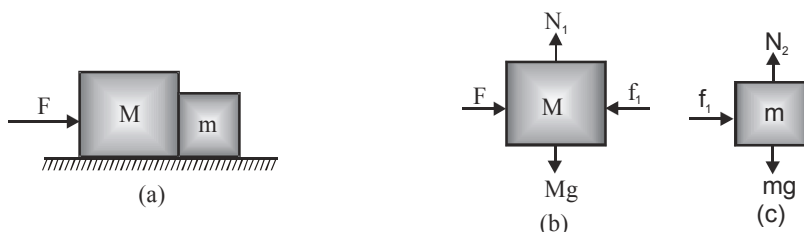
to the time rate of change of its linear momentum: $\frac{d(m\vec{v})}{dt}$

- Third law :** Whenever a particle A exerts a force on another particle B, B simultaneously exerts a force on A with the same magnitude in the opposite direction.

FREE BODY DIAGRAM

A diagram showing all external forces acting on an object is called "Free Body Diagram" (F.B.D.) In a specific problem, first we are required to choose a body and then we find the number of forces acting on it, and all the forces are drawn on the body, considering it as a point mass. The resulting diagram is known as free body diagram (FBD).

For example, if two bodies of masses m and M are in contact and a force F on M is applied from the left as shown in figure (a), the free body diagrams of M and m will be as shown in figure (b) and (c).



Important Point :

Two forces in Newton's third law never occur in the same free-body diagram. This is because a free-body diagram shows forces acting on a single object, and the action-reaction pair in Newton's third law always act on different objects.

MOTION OF BODIES IN CONTACT**Case I :**

When two bodies of masses m_1 and m_2 are kept on the frictionless surface and a force F is applied on one body, then the force with which one body presses the other at the point of contact is called Force of Contact. These two bodies will move with same acceleration a .

- (i) When the force F acts on the body with mass m_1 as shown in fig. (1) $F = (m_1 + m_2) a$.
If the force exerted by m_2 on m_1 is f_1 (force of contact) then for body m_1 : $(F - f_1) = m_1 a$

for body m_2 : $f_1 = m_2 a$

$$\Rightarrow \text{action of } m_1 \text{ on } m_2 : f_1 = \frac{m_2 F}{m_1 + m_2}$$

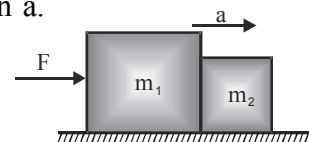


Fig.(1) : When the force F acts on mass m_1 .

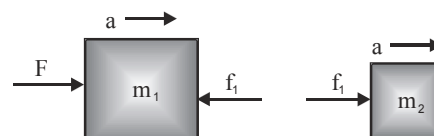


Fig. 1(a) : F.B.D. representation of action and reaction forces.

- (ii) When the force F acts on the body with mass m_2 as shown in figure 2

$$F = (m_1 + m_2) a$$

for body with mass m_2

$$F - f_2 = m_2 a$$

for body m_1 , $f_2 = m_1 a$

$$\Rightarrow \text{action on } m_1, f_2 = \frac{m_1 F}{m_1 + m_2}$$

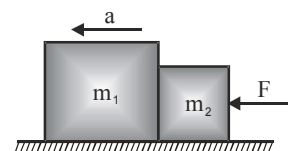


Fig. (2) : When the force F acts on mass m_2 .

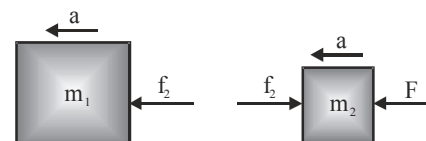


Fig. 2 (a) : F.B.D. representation of action and reaction forces.

Case II :

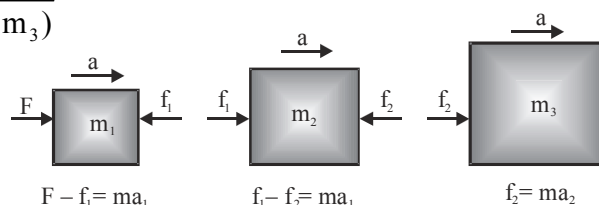
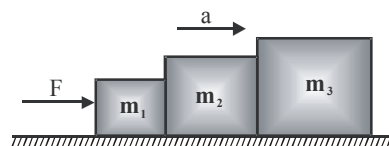
Three bodies of masses m_1 , m_2 and m_3 placed one after another and in contact with each other.

Suppose a force F is applied horizontally on mass m_1

$$\text{then } F = (m_1 + m_2 + m_3) a \Rightarrow a = \frac{F}{(m_1 + m_2 + m_3)}$$

$$f_1 = \frac{(m_2 + m_3)F}{(m_1 + m_2 + m_3)} \text{ (action on both } m_2 \text{ and } m_3)$$

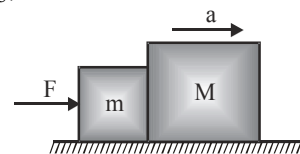
$$\text{and } f_2 = \frac{m_3 F}{(m_1 + m_2 + m_3)} \text{ (action on } m_3 \text{ alone)}$$



when the force F is applied on m_3 , then

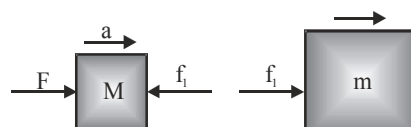
$$f_1 = \frac{m_1 F}{(m_1 + m_2 + m_3)} \text{ (action on } m_1 \text{ alone) and } f_2 = \frac{(m_1 + m_2) F}{(m_1 + m_2 + m_3)} \text{ (action on } m_1 \text{ and } m_2)$$

Ex. Two blocks of mass $m = 2 \text{ kg}$ and $M = 5 \text{ kg}$ are in contact on a frictionless table. A horizontal force $F (= 35 \text{ N})$ is applied to m . Find the force of contact between the block, will the force of contact remain same if F is applied to M ?



Sol. As the blocks are rigid under the action of a force F , both will move with same acceleration

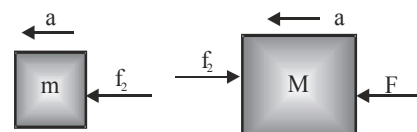
$$a = \frac{F}{m + M} = \frac{35}{2 + 5} = 5 \text{ m/s}^2$$



force of contact $f_1 = Ma = 5 \times 5 = 25 \text{ N}$

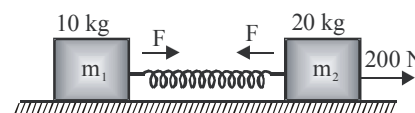
If the force is applied to M then its action on m will be

$$f_2 = ma = 2 \times 5 = 10 \text{ N.}$$



From this problem it is clear that acceleration does not depend on the fact that whether the force is applied to m or M , but force of contact does.

Ex. Two masses 10 kg and 20 kg respectively are connected by a massless spring as shown in figure. Force of 200 N acts on the 20 kg mass. At the instant shown in figure the 10 kg mass has acceleration of 12 m/s^2 , what is the acceleration of 20 kg mass?



Sol. Equation of motion for m_1 is $F = m_1 a_1 = 10 \times 12 = 120 \text{ N}$.

Force on 10 kg -mass is 120 N to the right. As action and reaction are equal and opposite, the reaction force F on 20 kg mass $F = 120 \text{ N}$ to the left.

$\therefore$ Equation of motion for m_2 is $200 - F = 20 a_2$

$$\Rightarrow 200 - 120 = 20 a_2 \quad \Rightarrow 20 a_2 = 80 \quad \Rightarrow a_2 = \frac{80}{20} = 4 \text{ ms}^{-2}$$

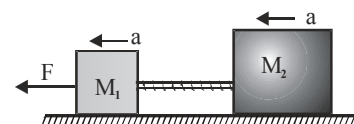
SYSTEM OF MASSES TIED BY STRINGS

Tension in a String :

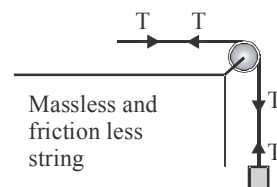
It is an intermolecular force between the atoms of a string, which acts or reacts when the string is stretched.

Important points about the tension in a string :

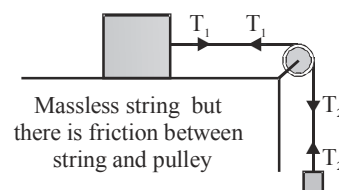
- Force of tension act on a body in the direction away from the point of contact or tied ends of the string.
- String is assumed to be inextensible so that the magnitude of accelerations of any number of masses connected through strings is always same.



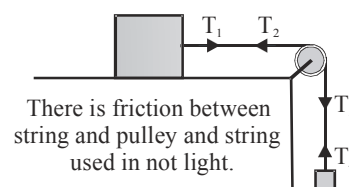
- If the string is extensible the acceleration of different masses connected through it will be different until the string can stretch.
- String is massless and frictionless so that tension throughout the string remains same.



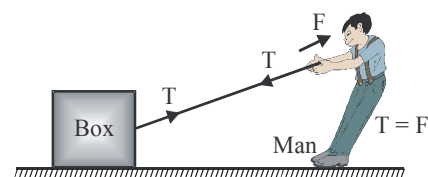
- If the string is massless but not frictionless, at every contact tension changes.



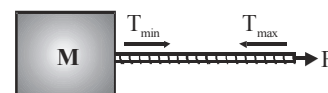
- If the string is not light, tension at each point will be different depending on the acceleration of the string.



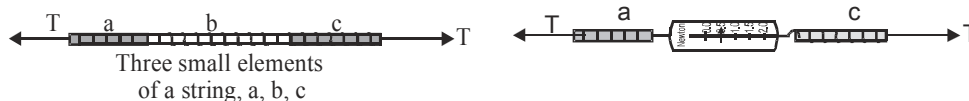
- If a force is directly applied on a string as say man is pulling a tied string from the other end with some force the tension will be equal to the applied force irrespective of the motion of the pulling agent, irrespective of whether the box will move or not, man will move or not.



- String is assumed to be massless unless stated, hence tension in it everywhere remains the same and equal to applied force. However, if a string has a mass, tension at different points will be different being maximum (= applied force) at the end through which force is applied and minimum at the other end connected to a body.
- In order to produce tension in a string two equal and opposite stretching forces must be applied. The tension thus produced is equal in magnitude to either applied force (i.e., $T = F$) and is directed inwards opposite to F . Here it must be noted that a string can never be compressed like a spring.



- If string is cut so that element b is replaced by a spring scale (the rest of the string being undisturbed), the scale reads the tension T .



- Every string can bear a maximum tension, i.e. if the tension in a string is continuously increased it will break if the tension is increased beyond a certain limit. The maximum tension which a string can bear without breaking is called "breaking strength". It is finite for a string and depends on its material and dimensions.

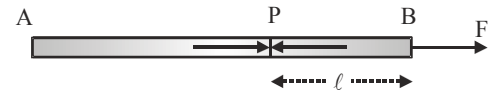
Ex. A uniform rope of length L is pulled by a constant force F . What is the tension in the rope at a distance ℓ from the end where it is applied?

Sol. Let mass of rope is M and T be tension in the rope at

point P , then. Acceleration of rope, $a = \frac{F}{M}$

Equation of motion of part PB is $F - T = (m\ell) a$

$$\Rightarrow T = F - (m\ell) a = F - \left(\frac{M}{L}\right) (\ell) \left(\frac{F}{M}\right) = \left[1 - \frac{\ell}{L}\right] F$$



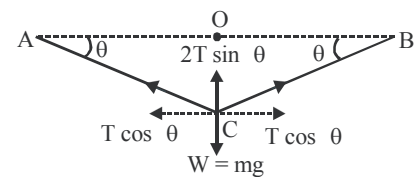
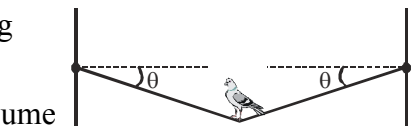
Ex. A bird with mass m perches at the middle of a stretched string

Show that the tension in the string is give by $T = \frac{mg}{2\sin\theta}$. Assume

that each half of the string is straight.

Sol. Initial position of wire = AOB . Final position of wire = ACB

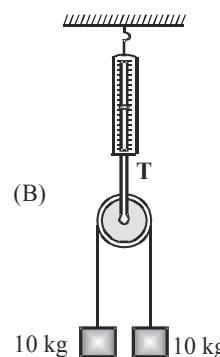
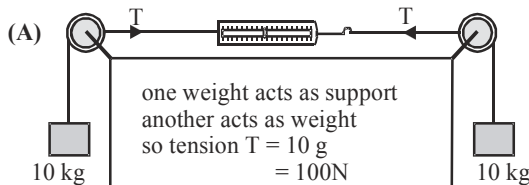
Let θ be the angle made by wire with horizontal, which is very small. Resolving tension T of string in horizontal and vertical directions, we note that the horizontal components cancel while vertical components add and balance the weight.



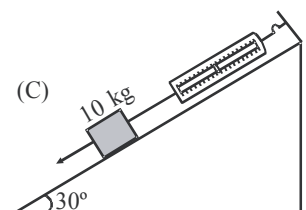
For equilibrium $2T \sin\theta = W = mg \Rightarrow T = \frac{W}{2\sin\theta} = \frac{mg}{2\sin\theta}$

Ex. The system shown in figure are in equilibrium. If the spring balance is calibrated in newtons, what does it record in each case? ($g = 10 \text{ m/s}^2$)

Sol.



$$T = 2 \times 10 \times g = 2 \times 10 \times 10 = 200\text{N}$$



$$T = 10 \times 10 \sin 30^\circ = 10 \times 10 \times \frac{1}{2} = 50\text{N}$$

MOTION OF BODIES CONNECTED BY STRINGS

Case : I

Two bodies :

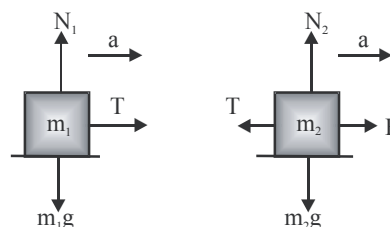
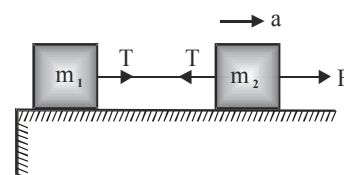
Let us consider the case of two bodies of masses m_1 and m_2 connected by a thread and placed on a smooth horizontal surface as shown in figure. A force F is applied on the body of mass m_2 in forward direction as shown. Our aim is to consider the acceleration of the system and the tension T in the thread. The forces acting separately on two bodies are also shown in the figure:

From figure $T = m_1 a$

and $F - T = m_2 a$

$$\Rightarrow F = (m_1 + m_2)a$$

$$\Rightarrow a = \frac{F}{m_1 + m_2} \quad \& \quad T = \frac{m_1 F}{m_1 + m_2}$$



Case II:

Three bodies

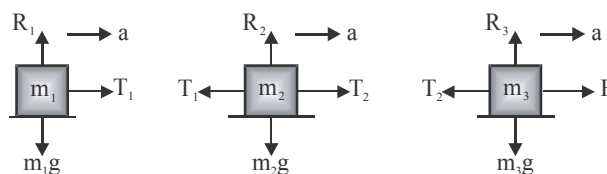
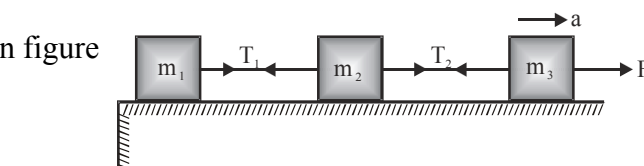
In case of three bodies, the situation is shown in figure

$$\text{Acceleration } a = \frac{F}{m_1 + m_2 + m_3}$$

$$T_1 = m_1 a = \frac{m_1 F}{m_1 + m_2 + m_3}$$

$$\therefore \text{ for block of mass } m_3 \quad F - T_2 = m_3 a$$

$$\therefore T_2 = F - \frac{m_3 F}{m_1 + m_2 + m_3} = \frac{(m_1 + m_2)F}{m_1 + m_2 + m_3}$$



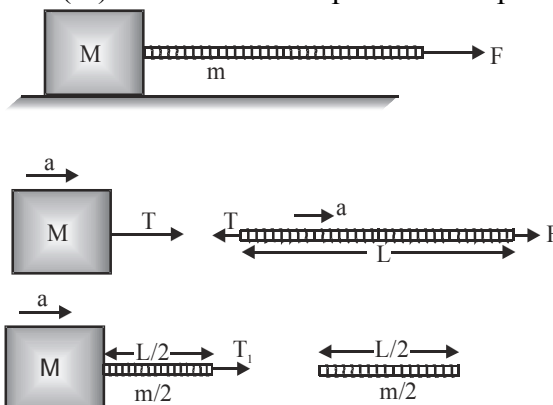
Ex. A block of mass M is pulled along a horizontal frictionless surface by a rope of mass m as shown in fig. A horizontal force F is applied to one end of the rope. Find (i) The acceleration of the rope and block (ii) The force that the rope exerts on the block. (iii) Tension in the rope at its midpoint.

Sol. (i) Acceleration $a = \frac{F}{(m + M)}$

(ii) Force exerted by rope $T = Ma = \frac{M.F}{(m + M)}$

(iii) $T_1 = \left(\frac{m}{2} + M \right) a = \left(\frac{m + 2M}{2} \right) \left(\frac{F}{m + M} \right)$

Tension in rope at midpoint $T_1 = \frac{(m + 2M)F}{2(m + M)}$

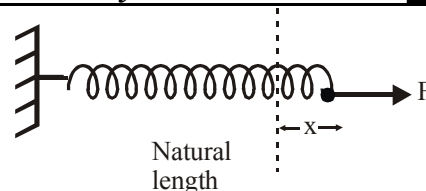


Spring Force (According to Hooke's law) :

In equilibrium $F = kx$

k is spring constant

Note : Spring force is non impulsive in nature.



Ex. If the lower spring is cut, find acceleration of the blocks, immediately after cutting the spring.

Sol. Initial stretches

$$x_{\text{upper}} = \frac{3mg}{k} \quad \& \quad x_{\text{lower}} = \frac{mg}{k}$$

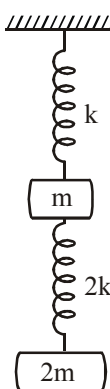
On cutting the lower spring, by virtue of non-impulsive nature of spring the stretch in upper spring remains same. Thus,

Lower block :

$$2mg = 2ma \Rightarrow a = g$$

Upper block :

$$k \left(\frac{3mg}{k} \right) - mg = ma \Rightarrow a = 2g$$



FRAME OF REFERENCE

It is a conveniently chosen co-ordinate system which describes the position and motion of a body in space.

INERTIAL AND NON-INERTIAL FRAMES OF REFERENCE

Inertial frames of reference :

A reference frame which is either at rest or in uniform motion along the straight line. A non-accelerating frame of reference is called an inertial frame of reference.

- All the fundamental laws of physics have been formulated in respect of inertial frame of reference.
- All the fundamental laws of physics can be expressed as to have the same mathematical form in all the inertial frames of reference.
- The mechanical and optical experiments performed in an inertial frame in any direction will always yield the same results. It is called isotropic property of the inertial frame of reference.

Examples of inertial frames of reference :

- A frame of reference remaining fixed w.r.t. distant stars is an inertial frame of reference.
- A space-ship moving in outer space without spinning and with its engine cut-off is also inertial frame of reference.
- For practical purposes, a frame of reference fixed to the earth can be considered as an inertial frame. Strictly speaking, such a frame of reference is not an inertial frame of reference, because the motion of earth around the sun is accelerated motion due to its orbital and rotational motion. However, due to negligibly small effects of rotation and orbital motion, the motion of earth may be assumed to be uniform and hence a frame of reference fixed to it may be regarded as inertial frame of reference.

Non-inertial frame of reference :

An accelerating frame of reference is called a non-inertial frame of reference.

Newton's laws of motion are not directly applicable in such frames, before application we must add pseudo force.

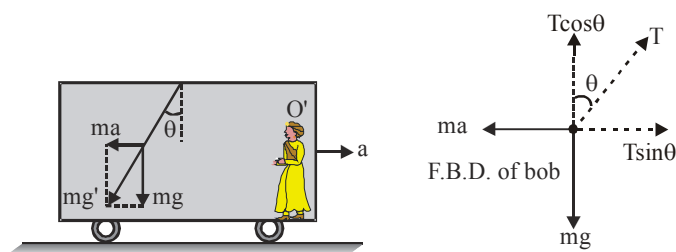
Note : A rotating frame of references is a non-inertial frame of reference, because it is also an accelerating one due to its centripetal acceleration.

PSEUDO FORCE

The force on a body due to acceleration of non-inertial frame is called fictitious or apparent or pseudo force and is given by $\vec{F} = -m\vec{a}_0$, where $\vec{a}_0$ is acceleration of non-inertial frame with respect to an inertial frame and m is mass of the particle or body. The direction of pseudo force must be opposite to the direction of acceleration of the non-inertial frame.

- When we draw the free body diagram of a mass, with respect to an **inertial frame of reference** we apply only the real forces (forces which are actually acting on the mass).
- But when the free body diagram is drawn from a non-inertial frame of reference a pseudo force (in addition to all real forces) has to be applied to make the equation $\vec{F} = m\vec{a}$ to be valid in this frame also.

Ex. A pendulum of mass m is suspended from the ceiling of a train moving with an acceleration ' a ' as shown in figure. Find the angle θ in equilibrium position.



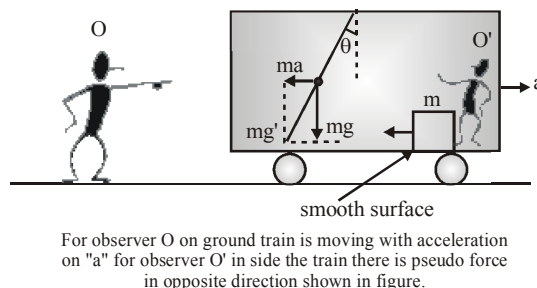
Sol. Non-inertial frame of reference (Train)

F.B.D. of bob w.r.t. train. (real forces + pseudo force) : with respect to train, bob is in equilibrium

$$\therefore \Sigma F_y = 0 \Rightarrow T \cos \theta = mg \quad \text{and}$$

$$\Sigma F_x = 0 \Rightarrow T \sin \theta = ma$$

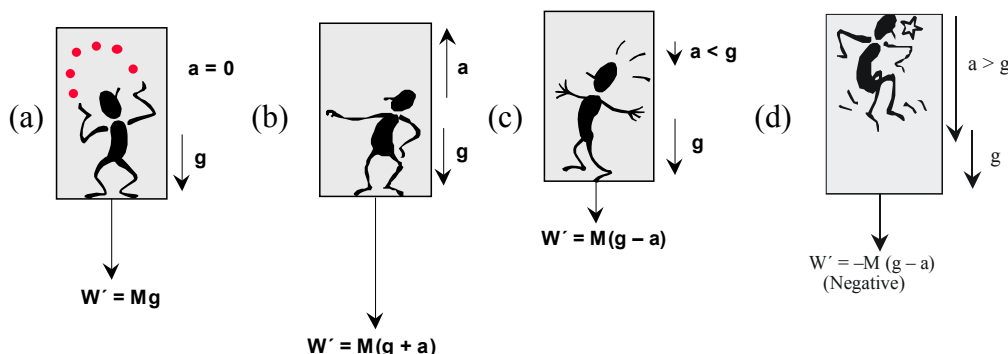
$$\Rightarrow \tan \theta = \frac{a}{g} \Rightarrow \theta = \tan^{-1} \left(\frac{a}{g} \right)$$



MOTION IN A LIFT

The weight of a body is simply the force exerted by earth on the body. If body is on an accelerated platform, the body experiences fictitious force, so the weight of the body appears changed and this new weight is called apparent weight. Let a man of weight $W = Mg$ be standing in a lift.

We consider the following cases :



Case (a) :

If the lift moving with constant velocity v upwards or downwards.

In this case there is no accelerated motion hence no pseudo force experienced by observer inside the lift.

So apparent weight $W' = Mg$ (Actual weight).

Case (b) :

If the lift is accelerated upward with constant acceleration a .

Then net forces acting on the man are (i) weight $W = Mg$ downward (ii) fictitious force $F_0 = Ma$ downward.

So apparent weight $W' = W + F_0 = Mg + Ma = M(g + a)$

Case (c) :

If the lift is accelerated downward with acceleration $a < g$

Then fictitious force $F_0 = Ma$ acts upward while weight of man $W = Mg$ always acts downward.

So apparent weight $W' = W + F_0 = Mg - Ma = M(g - a)$

Special Case : If $a = g$ then $W' = 0$ (condition of weightlessness).

Thus, in a freely falling lift the man will experience weightlessness.

Case (d) :

If lift accelerates downward with acceleration $a > g$:

Then as in Case c . Apparent weight $W' = M(g - a)$ is negative, i.e., the man will be accelerated upward and will stay at the ceiling of the lift.

Ex. A spring weighing machine inside a stationary lift reads 50 kg when a man stands on it. What would happen to the scale reading if the lift is moving upward with (i) constant velocity, and (ii) constant acceleration?

Sol. (i) In the case of constant velocity of lift, there is no fictitious force; therefore the apparent weight = actual weight. Hence the reading of machine is 50 kgwt.

(ii) In this case the acceleration is upward, the fictitious force ma acts downward, therefore apparent weight is more than actual weight i.e. $W' = m(g + a)$.

Hence scale shows a reading = $m(g + a) = \frac{mg\left(1 + \frac{a}{g}\right)}{g} = \left(50 + \frac{50a}{g}\right)$ kg wt.

Ex. Two objects of equal mass rest on the opposite pans of an arm balance. Does the scale remain balanced when it is accelerated up or down in a lift?

Sol. Yes, since both masses experience equal fictitious forces in magnitude as well as direction.

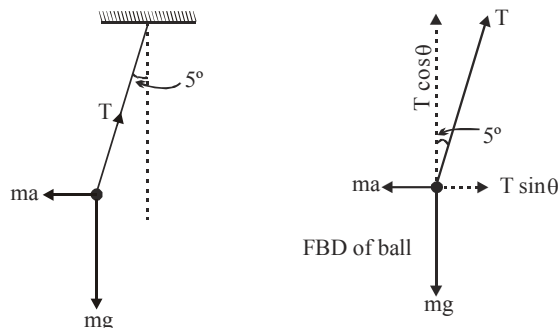
Ex. A passenger on a large ship sailing in a quiet sea hangs a ball from the ceiling of her cabin by means of a long thread. Whenever the ship accelerates, she notes that the pendulum ball lags behind the point of suspension and so the pendulum no longer hangs vertically. How large is the ship's acceleration when the pendulum stands at an angle of 5° to the vertical?

Sol. The ball is accelerated by the force $T \sin 5^\circ$.

Therefore $T \sin 5^\circ = ma$.

Vertical component $\Sigma F = 0$, so $T \cos 5^\circ = mg$.

By solving $a = g \tan 5^\circ = 0.0875 \text{ g}$
 $= 0.86 \text{ ms}^{-2}$.



Ex. A 12 kg monkey climbs a light rope as shown in figure. The rope passes over a pulley and is attached to a 16 kg bunch of bananas. Mass and friction in the pulley are negligible so that the pulley's only effect is to reverse the direction of the rope. What is the maximum acceleration the monkey can have without lifting the bananas?

Sol. Effective weight of monkey

As per given condition

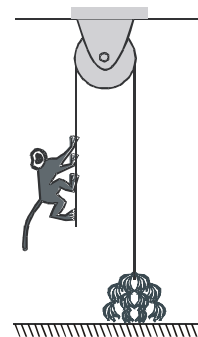
$$W_m = M_m(g + a)$$

$$\mathbf{W}_m^{\text{III}} = \mathbf{W}_b^{\text{III}}$$

$$\Rightarrow M_m (g + a) = M_b g$$

$$\Rightarrow a = \frac{(M_b - M_m) g}{M_m} = \left(\frac{16 - 12}{12} \right) \times 9.8$$

$$= \frac{9.8}{3} = 3.26 \text{ m/s}^2$$



PULLEY SYSTEM

A single fixed pulley changes the direction of force only and in general, assumed to be massless and frictionless.

It is clear from example given below.

Ex. A block of mass 25 kg is raised by a 50 kg man in two different ways as shown in figure. What is the action on the floor by the man in the two cases ? If the floor yields to a normal force of 700 N, which mode should be the man adopt to lift the block without the floor yielding ?

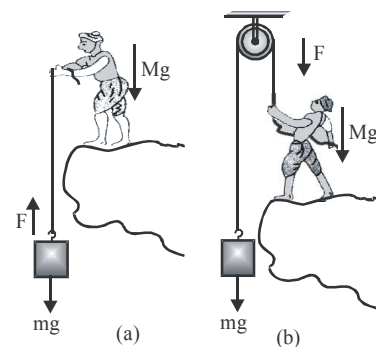
Sol. Mass of the block, $m = 25 \text{ kg}$; mass of the man, $M = 50 \text{ kg}$

Force applied to lift the block $F = mg = 25 \times 9.8 = 245 \text{ N}$

Weight of the man, $M g = 50 \times 9.8 = 490 \text{ N}$

(a) When the block is raised by the man by applying force F in upward direction, reaction equal and opposite to F will act on the floor in addition to the weight of the man.

$$\therefore \text{action on the floor} \quad \text{Mg} + \text{F} = 490 + 245 = 735 \text{ N}$$



- (b) When the block is raised by the mass applying force F over the rope (passed over the pulley) in downward direction, reaction equal and opposite to F will act on the floor,
 $\therefore$ action on the floor $Mg - F = 490 - 425 = 245 \text{ N}$
 floor yields to a normal force of 700 N , the mode (b) should be adopted by the man to lift block.

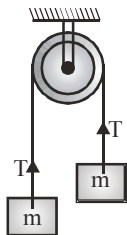
Some cases of pulley

I Case

$$m_1 = m_2 = m$$

$$\text{Tension in the string } T = mg$$

$$\text{Acceleration 'a' = zero}$$



Reaction at the suspension of the pulley

$$R = 2T = 2mg.$$

$$\text{Acceleration} = \frac{\text{net pulling force}}{\text{total mass to be pulled}}$$

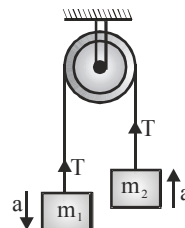
II Case

$$m_1 > m_2$$

$$\text{now for mass } m_1, m_1g - T = m_1a$$

$$\text{for mass } m_2, T - m_2g = m_2a$$

$$a = \frac{(m_1 - m_2)}{(m_1 + m_2)}g \text{ and } T = \frac{2m_1m_2}{(m_1 + m_2)}g$$



$$\text{Tension} = \frac{2 \times \text{Product of masses}}{\text{Sum of two masses}}g$$

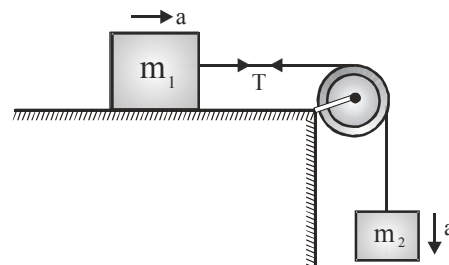
$$\text{Reaction at the suspension of pulley } R = 2T = \frac{4m_1m_2g}{(m_1 + m_2)}$$

III Case :

$$\text{For mass } m_1 : T = m_1a$$

$$\text{For mass } m_2 : m_2g - T = m_2a$$

$$\text{acceleration } a = \frac{m_2g}{(m_1 + m_2)} \text{ and } T = \frac{m_1m_2}{(m_1 + m_2)}g$$



IV Case :

$$(m_1 > m_2)$$

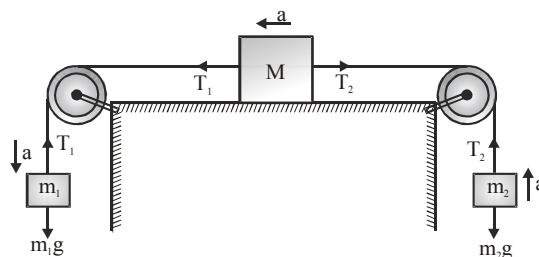
$$\text{For } m_1, m_1g - T_1 = m_1a$$

$$\text{For } m_2, T_2 - m_2g = m_2a$$

$$\text{For } M, T_1 - T_2 = Ma$$

$$\Rightarrow a = \frac{(m_1 - m_2)}{(m_1 + m_2 + M)}g,$$

$$T_1 = \frac{(2m_2 + M)m_1g}{m_1 + m_2 + M}, T_2 = \frac{(2m_1 + M)m_2g}{m_1 + m_2 + M}$$



V Case :

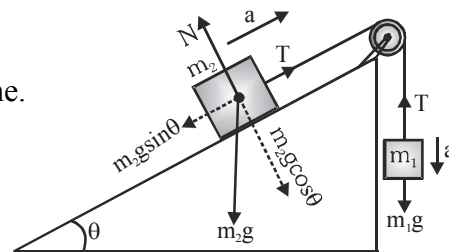
Mass suspended over a pulley from another on an inclined plane.

$$\text{For mass } m_1 : m_1g - T = m_1a$$

$$\text{For mass } m_2 : T - m_2g \sin\theta = m_2a$$

$$\text{acceleration } a = \frac{(m_1 - m_2 \sin\theta)}{(m_1 + m_2)}g$$

$$T = \frac{m_1m_2(1 + \sin\theta)}{(m_1 + m_2)}g$$

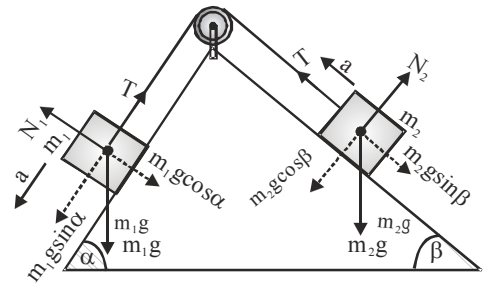


VI Case :

Masses m_1 and m_2 are connected by a string passing over a pulley ($m_1 > m_2$)

$$\text{Acceleration } a = \frac{(m_1 \sin \alpha - m_2 \sin \beta)}{(m_1 + m_2)} g$$

$$\text{Tension } T = \frac{m_1 m_2 (\sin \alpha + \sin \beta)}{(m_1 + m_2)} g$$

**VII Case :**

For mass m_1 : $T_1 - m_1 g = m_1 a$

For mass m_2 : $m_2 g + T_2 - T_1 = m_2 a$

For mass m_3 : $m_3 g - T_2 = m_3 a$

$$\text{Acceleration } a = \frac{(m_2 + m_3 - m_1)}{(m_1 + m_2 + m_3)} g$$

we can calculate tensions T_1 and T_2 from above equations

VIII Case :

From case (iii)

$$\text{tension } T = \frac{m_1 m_2}{(m_1 + m_2)} g$$

If x is the extension in the spring,

then $T = kx$

$$x = \frac{T}{k} = \frac{m_1 m_2 g}{k(m_1 + m_2)}$$

Ex. In the system shown in figure all surface are smooth, string is massless and inextensible. Find:

- acceleration of the system
- tension in the string and
- extension in the spring if force constant of spring is $k = 50 \text{ N/m}$ (Take $g = 10 \text{ m/s}^2$)

Sol. (a) In this case net pulling force $= m_c g + m_b g = 50 \text{ N}$ and total mass to be pulled is $(1 + 2 + 3) \text{ kg} = 6 \text{ kg}$.

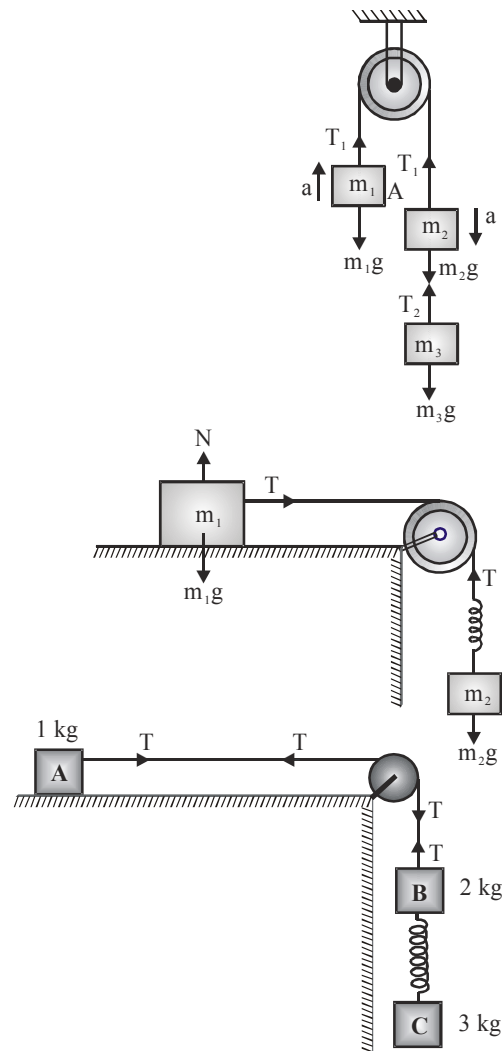
$$\therefore \text{Acceleration of the system is } a = \frac{50}{6} \text{ ms}^{-2}$$

(b) Free body diagram of 1 kg block gives $T = ma = (1) \left(\frac{50}{6} \right) \text{ N} = \frac{50}{6} \text{ N}$

(c) Free body diagram of 3 kg block gives

$$30 - kx = ma \quad \text{but} \quad ma = 3 \times \frac{50}{6} = 25 \text{ N}$$

$$x = \frac{30 - 25}{k} = \frac{5}{50} = 0.1 \text{ m} = 10 \text{ cm}$$



Ex. In the adjacent figure, masses of A, B and C are 1 kg, 3 kg and 2 kg respectively.

Find : (a) the acceleration of the system and

(b) tensions in the string

Neglect friction. ($g = 10 \text{ ms}^{-2}$)

Sol. (a) In this case net pulling force

$$\begin{aligned} &= m_A g \sin 60^\circ + m_B g \sin 60^\circ - m_C g \sin 30^\circ \\ &= (m_A + m_B) g \sin 60^\circ - m_C g \sin 30^\circ \\ &= (1 + 3) \times 10 \times \frac{\sqrt{3}}{2} - 2 \times 10 \times \frac{1}{2} \\ &= 20\sqrt{3} - 10 = 20 \times 1.732 - 10 = 24.64 \text{ N} \end{aligned}$$

Total mass being pulled = $1 + 3 + 2 = 6 \text{ kg}$

$$\therefore \text{Acceleration of the system } a = \frac{24.64}{6} = 4.1 \text{ m/s}^2$$

(b) For the tension T_1 in the string between A and B, $m_A g \sin 60^\circ - T_1 = (m_A) a$

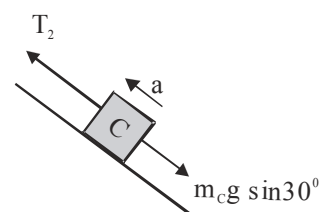
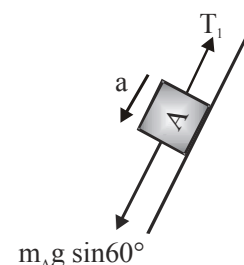
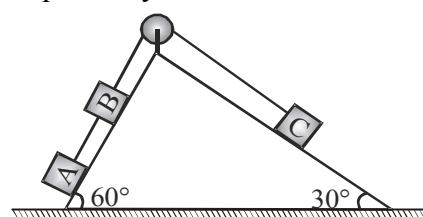
$$\therefore T_1 = m_A g \sin 60^\circ - m_A a = m_A (g \sin 60^\circ - a)$$

$$\Rightarrow T_1 = (1) \left(10 \times \frac{\sqrt{3}}{2} - 4.1 \right) = 4.56 \text{ N}$$

For the tension T_2 in the string between B and C.

$$T_2 - m_C g \sin 30^\circ = m_C a$$

$$\Rightarrow T_2 = m_C (a + g \sin 30^\circ) = 2 \left[4.1 + 10 \left(\frac{1}{2} \right) \right] = 18.2 \text{ N}$$



Ex.. Consider the situation shown in figure. Both the pulleys and the strings are light and all the surfaces are frictionless. Calculate

(a) the acceleration of mass M,

(b) tension in the string PQ and

(c) force exerted by the clamp on the pulley P.

Sol. As pulley Q is not fixed so if it moves a distance d the length of string between P and Q will change by $2d$ (d from above and d from below), i.e., M will move $2d$. This in turn implies that if a is the

acceleration of M the acceleration of Q and of $2M$ will be $\frac{a}{2}$.

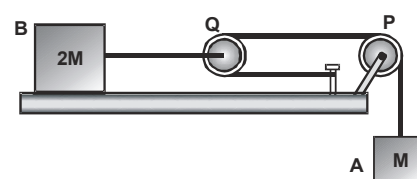
Now if we consider the motion of mass M , it is accelerated downward, so

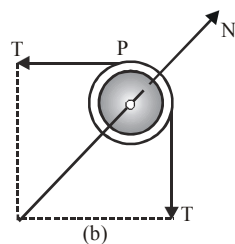
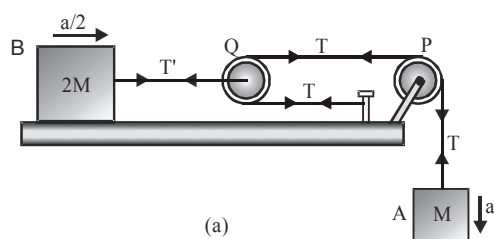
$$T = M(g - a) \dots\dots(i)$$

And for the motion of Q

$$2T - T' = 0 \times \frac{a}{2} = 0 \Rightarrow T' = 2T \dots\dots(ii)$$

$$\text{And for the motion of mass } 2M \quad T' = 2M \left(\frac{a}{2} \right), \Rightarrow T' = Ma \dots\dots(iii)$$





- (a) From equation (ii) and (iii) as $T = \frac{1}{2} Ma$, so equation (i) reduces to

$$T = \frac{1}{2} Ma = M(g - a) \Rightarrow a = \frac{2}{3} g$$

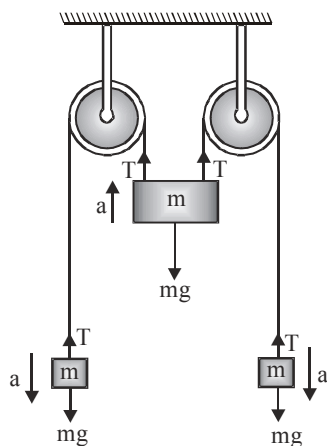
- (b) So the acceleration of mass M is $\frac{2}{3} g$ while tension in the string PQ from equation (1) will be

$$T = M(g - \frac{2}{3} g) = \frac{1}{3} Mg$$

- (c) Now from figure (b), it is clear that force on pulley by the clamp will be equal and opposite to the resultant of T and T at 90° to each other, i.e.,

$$(N_2) = \sqrt{T^2 + T^2} = \sqrt{2} T = \frac{\sqrt{2}}{3} Mg$$

Ex. Consider the double Atwood's machine as shown in the figure



- (a) What is acceleration of the masses?
 (b) What is the tension in each string?

Sol. (a) Here the system behaves as a rigid system, therefore every part of the system will move with same acceleration. Thus Applying newton's law

$$mg - T = ma \quad \dots(1)$$

$$2T - mg = ma \quad \dots(2)$$

Doubling the first equation and adding

$$mg = 3 ma \Rightarrow \text{acceleration } a = \frac{1}{3} g$$

- (b) Tension in the string $T = m(g - a) = m\left(g - \frac{g}{3}\right) = \frac{2}{3} mg$

Ex. Consider the system of masses and pulleys shown in fig. with massless string and frictionless pulleys.

(a) Give the necessary relation between masses m_1 and m_2 such that system is in equilibrium and does not move.

(b) If $m_1 = 6$ kg and $m_2 = 8$ kg, calculate the magnitude and direction of the acceleration of m_1 .

Sol. (a) Applying newton's law $m_2g - 2T = 0$ (because there is no acceleration) and $T - m_1g = 0$

$$\Rightarrow (m_2 - 2m_1)g = 0 \Rightarrow m_2 = 2m_1$$

(b) If the upwards acceleration of m_1 is a ,

then acceleration of m_2 is $\frac{a}{2}$ downwards

$$\text{for mass } m_2: m_2g - 2T = m_2\left(\frac{a}{2}\right) \Rightarrow 2m_2g - 4T = m_2a$$

$$\text{for mass } m_1: T - m_1g = m_1a$$

$$\Rightarrow a = \left(\frac{2m_2 - 4m_1}{m_2 + 4m_1}\right)g = \frac{2(8 - 12)}{8 + 24}g = -\frac{g}{4}$$

Negative sign shows that acceleration is opposite to considered direction i.e.

it is downwards for m_1 and upwards for m_2 .

Ex. In the given figure If $T_1 = 2T_2 = 50$ N then find the value of T .

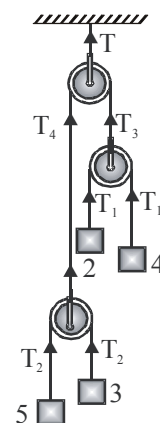
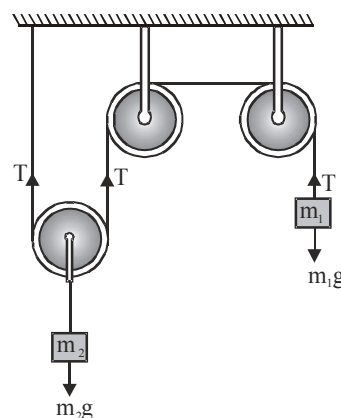
Sol. As given in figure,

$$T_3 = 2T_1 = 2(2T_2) = 4T_2$$

$$\text{and } T_4 = 2T_2$$

$$\therefore T = T_3 + T_4 = 4T_2 + 2T_2 = 6T_2$$

$$= 6 \times \frac{50}{2} = 150 \text{ N}$$



CONSTRAINT RELATIONS

These equations establish the relation between accelerations (or velocities) of different masses attached by string(s). Normally number of constraint equations are equal to number of strings in the system under consideration.

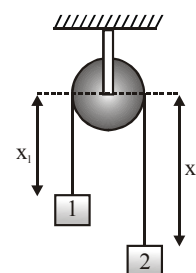
Ex. Find the relation between acceleration of 1 and 2.

Sol. At any instant of time let x_1 and x_2 be the displacements of 1 and 2 from a fixed line. Then $x_1 + x_2 = \text{constant}$

Differentiating w.r.t. time, $v_1 + v_2 = 0$

Again differentiating w.r.t. time, $a_1 + a_2 = 0 \Rightarrow a_1 = -a_2$

So acceleration of 1 and 2 are equal but in opposite directions.



Ex. At certain moment of time, velocities of 1 and 2 both are 1 ms^{-1} upwards. Find the velocity of 3 at that moment.

Sol. $x_1 + x_4 = \ell_1$ (length of first string)

$x_2 - x_4 + x_3 - x_4 = \ell_2$ (length of second string)

$$\Rightarrow v_1 + v_4 = 0 \text{ \& } v_2 + v_3 - 2v_4 = 0$$

$$\Rightarrow v_2 + v_3 + 2v_1 = 0$$

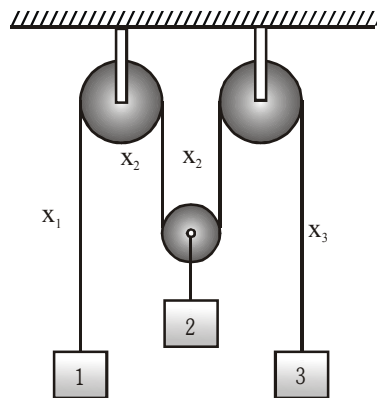
Taking upward direction as positive

$$v_1 = v_2 = 1$$

$$\text{so } 1 + v_3 + 2 \times 1 = 0 \Rightarrow v_3 = -3 \text{ ms}^{-1}$$

i.e. velocity of block 3 is 3 ms^{-1} downwards.

Ex. Find the relation between acceleration of blocks a_1 , a_2 and a_3 .

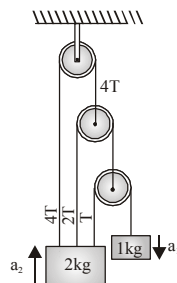


Sol. $x_1 + 2x_2 + x_3 = \ell$

$$v_1 + 2v_2 + v_3 = 0$$

$$a_1 + 2a_2 + a_3 = 0$$

Ex. Using constraint equation. Find the relation between a_1 and a_2 .

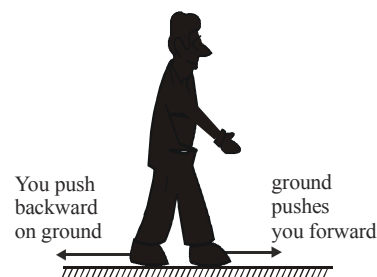


Sol. For this system $a_1 T = a_2(4T + 2T + T) \Rightarrow a_1 = 7a_2$

FRICTION

INTRODUCTION

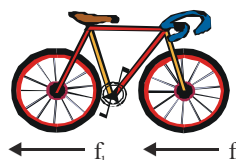
Friction is the force of two surfaces in contact, or the force of a medium acting on a moving object. (i.e. air on aircraft.). Frictional forces may also exist between surfaces when there is no relative motion. Frictional forces arise due to molecular interactions. In some cases friction acts as a supporting force and in some cases it acts as opposing force.



- **Supporting :** Walking process can only take place because there is friction between the shoes and ground.
- **Opposing :** When a block slides over a surface the force of friction acts as an opposing force in the opposite direction of the motion
- **Both Supporting and Opposing :**
- **Pedaling :** When cyclist pedals the friction force on rear wheel acts as a supporting force and on front wheel as a opposing force.



Force of friction
when pedaling



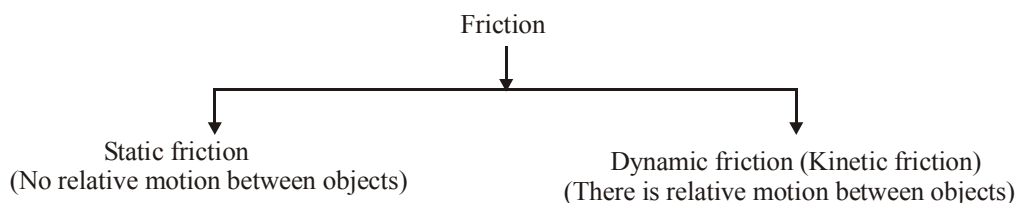
Force of friction
when not pedaling

- **Non-Pedaling :** When cyclist not pedals the friction force on rear wheel & front wheel act as a opposing force.

CAUSE OF FRICTION

- **Old View :** When two bodies are in contact with each other, the irregularities in the surface of one body set interlocked in the irregularities of another surface. This locking opposes the tendency of motion.
- **Modern View :** Friction arises on account of strong atomic or molecular forces of attraction between the two surfaces at the point of actual contact.

TYPES OF FRICTION



STATIC FRICTION

- It is the frictional force which is effective before motion starts between two planes in contact with each other.
- Its nature is self adjusting.
- Numerical value of static friction is equal to external force which creates the tendency of motion of body.
- Maximum value of static friction is called limiting friction. $0 \leq f_s \leq \mu_s N$, $\vec{f}_s = -\vec{F}_{\text{applied}}$

LAWS OF LIMITING FRICTION

- The magnitude of the force of limiting friction (F) between any two bodies in contact is directly proportional to the normal reaction (N) between them $F \propto N$
- The direction of the force of limiting friction is always opposite to the direction in which one body is on the verge of moving over the other.
- The force of limiting friction is independent of the apparent contact area, as long as normal reaction between the two bodies in contact remains the same.
- Limiting friction between any two bodies in contact depends on the nature of material of the surfaces in contact and their roughness and smoothness.
- Its value is more than to other types of friction force.

DYNAMIC FRICTION

The friction opposing the relative motion between two bodies is called dynamic or kinetic friction

$$\vec{f}_k = -(\mu_k N)$$

- This is always slightly less than the limiting friction

COEFFICIENT OF FRICTION

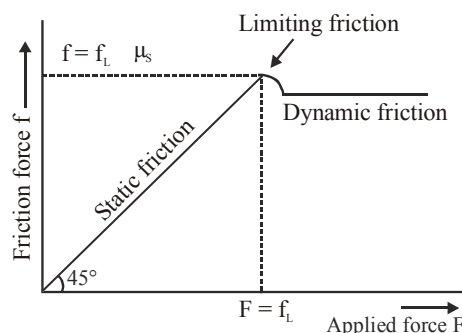
The frictional coefficient is a dimensionless scalar value which describes the ratio of the force of friction between two bodies and the force pressing them together.

- Static friction coefficient $\mu_s = \frac{F}{N}$
- Sliding friction coefficient $\mu_k = \frac{F_k}{N}$

The values of μ_s and μ_k depend on the nature of both the surfaces in contact.

GRAPH BETWEEN APPLIED FORCE AND FORCE OF FRICTION

If we slowly increase the force with which we are pulling the box, graph shows that the friction force increases with our force upto a certain critical value, f_L , the box suddenly begins to move, and as soon as it starts moving, a smaller force is required to maintain its motion as in motion friction is reduced. The friction value from 0 to f_L is known as static friction, which balances the external force on the body and prevent it from sliding. The value f_L is the maximum limit up to which the static friction acts is known as limiting friction, after which body starts sliding and friction reduces to kinetic friction.



- When two highly polished surfaces are pressed hard, then a situation similar to welding occurs. It is called **cold welding**.
- When two copper plates are highly polished and placed in contact with each other, then instead of decreasing, the force of friction increases. This arises due to the fact that for two highly polished surfaces in contact, the number of molecules coming in contact increases and as a result the cohesive/adhesive forces increases. This in turn, increases the force of friction.

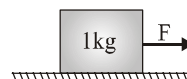
Net contact force is the resultant of normal reaction and frictional force.

APPROXIMATE COEFFICIENTS OF FRICTION

Materials	Coefficient of static friction, μ_s	Coefficient of kinetic friction, μ_k
Steel on steel	0.74	0.57
Aluminum on steel	0.61	0.47
Copper on steel	0.53	0.36
Copper on cast iron	1.05	0.29
Brass on steel	0.51	0.44
Teflon on teflon	0.04	0.04
Rubber on concrete (dry)	1.0	0.8
Rubber on concrete (wet)	0.30	0.25

Ex. A block of mass 1 kg is at rest on a rough horizontal surface having coefficient of static friction 0.2 and kinetic friction 0.15, find the frictional forces if a horizontal force,

- (a) $F = 1\text{ N}$
 (b) $F = 1.96\text{ N}$
 (c) $F = 2.5\text{ N}$, is applied on a block



Sol. Maximum force of friction $f_{\max} = 0.2 \times 1 \times 9.8 \text{ N} = 1.96 \text{ N}$

- (a) for $F_{\text{ext}} = 1 \text{ N}$, $F_{\text{ext}} < f_{\text{max}}$
So, body is in rest means static friction is present and hence $f_s = F_{\text{ext}} = 1 \text{ N}$
- (b) for $F_{\text{ext}} = 1.96 \text{ N}$, $F_{\text{ext}} = f_{\text{max}} = 1.96 \text{ N}$ so $f = 1.96 \text{ N}$
- (c) for $F_{\text{ext}} = 2.5 \text{ N}$, so $F_{\text{ext}} > f_{\text{max}}$

now body is in moving condition

$$\therefore f_{\max} = f_k = \mu_k N = \mu_k mg = 0.15 \times 1 \times 9.8 = 1.47 \text{ N}$$

Ex. Length of a chain is L and coefficient of static friction is μ . Calculate the maximum length of the chain which can be hang from the table without sliding.

Sol. Let y be the maximum length of the chain can be hold out side the table without sliding.

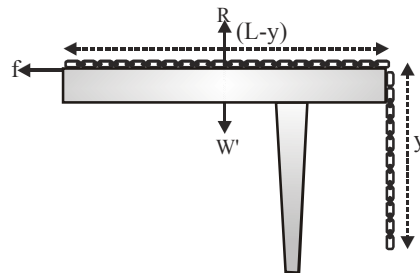
Length of chain on the table = $(L - y)$

$$\text{Weight of part of the chain on table } W' = \frac{M}{L}(L-y)g$$

Weight of hanging part of the chain $W = \frac{M}{L} y g$

For equilibrium : limiting force of friction = weight of hanging part of the chain

$$\mu R = W \Rightarrow \mu W' = W \Rightarrow \mu \frac{M}{L}(L-y)g = \frac{M}{L}yg \Rightarrow \mu L - \mu y = y \Rightarrow y = \frac{\mu L}{1+\mu}$$

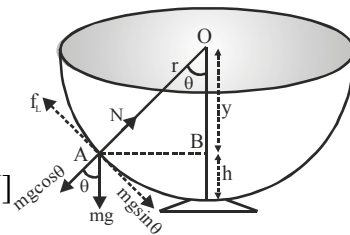


Ex. If the coefficient of friction between an insect and bowl is μ and the radius of the bowl is r , find the maximum height to which the insect can crawl up in the bowl.

Sol. The insect will crawl up the bowl till the component of its weight along the bowl is balanced by limiting frictional force. So, resolving weight perpendicular to the bowl and along the bowl,

$$N = mg \cos \theta, f_L = mg \sin \theta \Rightarrow \tan \theta = \frac{f_L}{N} \Rightarrow \tan \theta = \mu \quad [\because f_L = \mu N]$$

$$\Rightarrow \sqrt{\frac{r^2 - y^2}{y}} = \mu \Rightarrow y = \frac{r}{\sqrt{1 + \mu^2}} \quad \text{So } h = r - y = r \left[1 - \frac{1}{\sqrt{1 + \mu^2}} \right]$$



Ex. A body of mass M is kept on a rough horizontal surface (friction coefficient $= \mu$). A person is trying to pull the body by applying a horizontal force F , but the body is not moving. What is the force by the surface on A.

Sol. Let f is the force of friction and N is the normal reaction,

$$\text{then the net force by the surface on the body is } F = \sqrt{N^2 + f^2}$$

Let the applied force is F' (varying), applied horizontally then $f \leq \mu_s N$ (adjustable with $f = F'$).

Now if F' is zero, $f = 0$ and $F_{\min} = N = Mg$

and when F' is increased to maximum value permissible for no motion. $f = \mu_s N$,

$$\text{giving } F_{\max} = \sqrt{N^2 + \mu_s^2 N^2} = Mg \sqrt{1 + \mu_s^2}$$

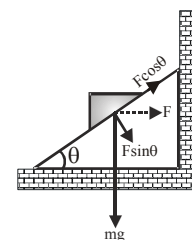
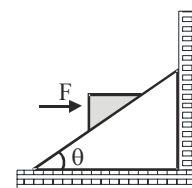
$$\text{therefore we can write } Mg \leq F \leq Mg \sqrt{1 + \mu_s^2}$$

Ex. A block rest on a rough inclined plane as shown in fig. A horizontal force F is applied to it (a) Find out the force of reaction, (b) Can the force of friction be zero if yes when? and (c) Assuming that friction is not zero find its magnitude and direction of its limiting value.

Sol. (a) $N = mg \cos \theta + F \sin \theta$

(b) Yes, if $mg \sin \theta = F \cos \theta$

(c) $f = \mu R = \mu (mg \cos \theta + F \sin \theta)$; up the plane if the body has tendency to slide down and down the plane if the body has tendency to move up.

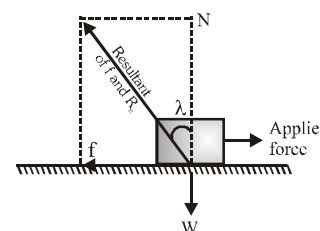


ANGLE OF FRICTION

The angle of friction is the angle which the resultant of limiting friction f_s and normal reaction N makes with the normal reaction. It is

$$\text{represented by } \lambda \quad \tan \lambda = \frac{f_s}{N} = \frac{\mu N}{N} = \mu$$

- For smooth surface $\lambda = 0$



ANGLE OF REPOSE (θ)

If a body is placed on an inclined plane and if its angle of inclination is gradually increased, then at some angle of inclination θ the body will just on the point to slide down. The angle is called angle of repose (θ).

$$F_s = mg \sin \theta \text{ and } N = mg \cos \theta$$

so
$$\frac{F_s}{N} = \tan \theta \Rightarrow \mu = \tan \theta$$

Relation between angle of friction (λ) and angle of repose (θ)

$$\tan \lambda = \mu \text{ and } \mu = \tan \theta, \text{ hence } \tan \lambda = \tan \theta \Rightarrow \theta = \lambda$$

Thus, angle of repose = angle of friction

Ex. A block of mass 2 kg slides down an inclined plane which makes an angle of 30° with the horizontal.

The coefficient of friction between the block and the surface is $\frac{\sqrt{3}}{2}$.

- What force must be applied to the block so that the block moves down the plane without acceleration?
- What force should be applied to the block so that it can move up without any acceleration?

Sol. Make a 'free-body' diagram of the block. Take the force of friction opposite to the direction of motion.

- Project forces along and perpendicular to the plane

perpendicular to plane $N = mg \cos \theta$

along the plane $F + mg \sin \theta - f = 0$

($\because$ there is no acceleration along the plane)

$$F + mg \sin \theta - \mu N = 0 \Rightarrow F + mg \sin \theta = \mu mg \cos \theta$$

$$F = mg (\mu \cos \theta - \sin \theta) = 2 \times 9.8 \left(\frac{\sqrt{3}}{2} \cos 30^\circ - \sin 30^\circ \right)$$

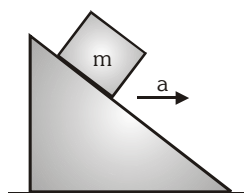
$$= 19.6 \left(\frac{\sqrt{3}}{2} \times \frac{\sqrt{3}}{2} - \frac{1}{2} \right) = 19.6 \left(\frac{3}{4} - \frac{1}{2} \right) = 4.9 \text{ N}$$

- This time the direction of F is reversed and that of the frictional force is also reversed.

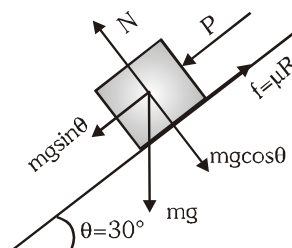
$$\therefore N = mg \cos \theta ; F = mg \sin \theta + f$$

$$\Rightarrow F = mg (\mu \cos \theta + \sin \theta) = 19.6 \left(\frac{3}{4} + \frac{1}{2} \right) = 24.5 \text{ N}$$

Ex. A block of mass 1 kg sits on an incline as shown in figure.



- What must be the frictional force between block and incline if the block is not to slide along the incline when the incline is accelerating to the right at 3 m/s^2 ?
- What is the least value μ_s can have for this to happen?

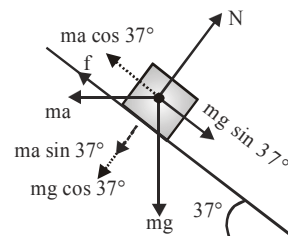


Sol. $N = m(g \cos 37^\circ + a \sin 37^\circ) = 1(9.8 \times 0.8 + 3 \times 0.6) = 9.64 \text{ N}$

$$mg \sin 37^\circ = ma \cos 37^\circ + f$$

(a) $f = 1(9.8 \times 0.6 - 3 \times 0.8) = 3.48$

(b) $\therefore f = \mu N \quad \therefore \mu = \frac{f}{N} = \frac{3.48}{9.64} = 0.36$

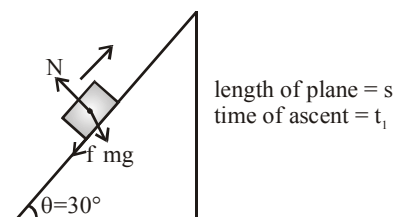


Ex. A body of mass $5 \times 10^{-3} \text{ kg}$ is launched up on a rough inclined plane making an angle of 30° with the horizontal. Obtain the coefficient of friction between the body and the plane if the time of ascent is half of the time of descent.

Sol. For upward motion: upward retardation $a_1 = \frac{\mu N + mg \sin \theta}{m}$

$$a_1 = \mu g \cos 30^\circ + g \sin 30^\circ = (\sqrt{3}\mu + 1) \frac{g}{2}$$

$$\therefore s = \frac{1}{2} a_1 t_1^2 \quad \therefore t_1 = \sqrt{\frac{2s}{a_1}} = \sqrt{\frac{4s}{(\sqrt{3}\mu + 1)g}}$$



For downward motion: downward acceleration $a_2 = \frac{mg \sin \theta - \mu N}{m}$

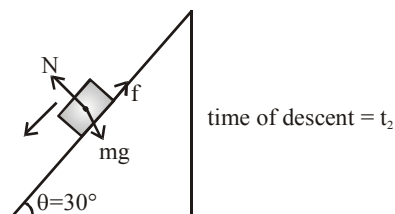
$$a_2 = g \sin 30^\circ - g \cos 30^\circ = (1 - \sqrt{3}\mu) \frac{g}{2}$$

$$\Rightarrow t_2 = \sqrt{\frac{2s}{a_2}} = \sqrt{\frac{4s}{(1 - \sqrt{3}\mu)g}}$$

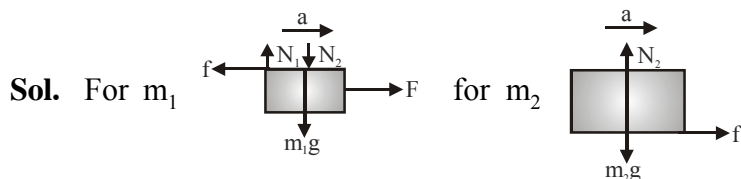
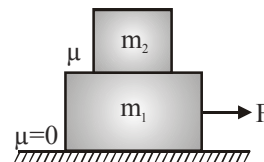
Now according to question $2t_1 = t_2$

$$\Rightarrow 2\sqrt{\frac{4s}{(\sqrt{3}\mu + 1)g}} = \sqrt{\frac{4s}{(1 - \sqrt{3}\mu)g}}$$

$$\Rightarrow \frac{1 - \sqrt{3}\mu}{1 + \sqrt{3}\mu} = \frac{1}{4} \Rightarrow \mu = \frac{\sqrt{3}}{5}$$



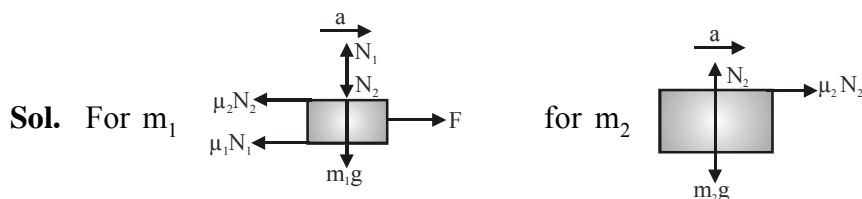
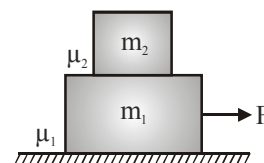
Ex. When force F applied on m_1 and there is no friction between m_1 and surface and the coefficient of friction between m_1 and m_2 is μ . What should be the minimum value of F so that there is no relative motion between m_1 and m_2



For system acceleration $a = \frac{F}{m_1 + m_2}$

For m_2 $f = m_2 a \Rightarrow \mu m_2 g = m_2 \left(\frac{F}{m_1 + m_2} \right) \Rightarrow F_{\min} = \mu(m_1 + m_2) g$

Ex. When force F applied on m_1 and the coefficient of friction between m_1 and surface is μ_1 and the coefficient of friction between m_1 and m_2 is μ_2 . What should be the minimum value of F so that there is no relative motion between m_1 and m_2

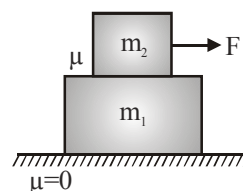


For system $a = \frac{F - \mu_1(m_1 + m_2)g}{m_1 + m_2}$

For m_2 , $\mu_2(m_2g) = m_2 a = m_2 \left(\frac{F - \mu_1(m_1 + m_2)g}{m_1 + m_2} \right)$

$\Rightarrow F_{\min} = (m_1 + m_2)(\mu_1 + \mu_2)g$

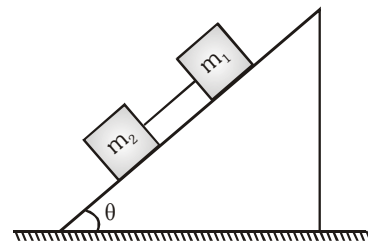
Ex. When force F applied on m_2 and there is no friction between m_1 and surface and the coefficient of friction between m_1 and m_2 is μ . What should be the minimum value of F so that there is no relative motion between m_1 and m_2



for system : acceleration $= \frac{F}{m_1 + m_2}$

for m_1 : $\mu m_2 g = m_1 a = m_1 \left(\frac{F}{m_1 + m_2} \right)$, $F_{\min} = (m_1 + m_2) \left(\frac{\mu m_2 g}{m_1} \right)$

Ex. Two blocks with masses $m_1 = 1$ kg and $m_2 = 2$ kg are connected by a string and slide down a plane inclined at an angle $\theta = 45^\circ$ with the horizontal. The coefficient of sliding friction between m_1 and plane is $\mu_1 = 0.4$, and that between m_2 and plane is $\mu_2 = 0.2$. Calculate the common acceleration of the two blocks and the tension in the string.



Sol. As $\mu_2 < \mu_1$, block m_2 has greater acceleration than m_1 if we separately consider the motion of blocks. But they are connected so they move together as a system with common acceleration. So acceleration of the blocks :

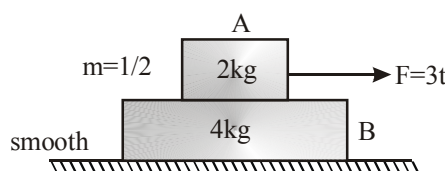
$$a = \frac{(m_1 + m_2)g \sin \theta - \mu_1 m_1 g \cos \theta - \mu_2 m_2 g \cos \theta}{m_1 + m_2}$$

$$= \frac{(1+2)(10)\left(\frac{1}{\sqrt{2}}\right) - 0.4 \times 1 \times 10 \times \frac{1}{\sqrt{2}} - 0.2 \times 2 \times 10 \times \frac{1}{\sqrt{2}}}{1+2} = \frac{22}{3\sqrt{2}} \text{ ms}^{-2}$$

For block m_2 : $m_2 g \sin \theta - \mu_2 m_2 g \cos \theta - T = m_2 a \Rightarrow T = m_2 g \sin \theta - \mu_2 m_2 g \cos \theta - m_2 a$

$$= 2 \times 10 \times \frac{1}{\sqrt{2}} - 0.2 \times 2 \times 10 \times \frac{1}{\sqrt{2}} - 2 \times \frac{22}{3\sqrt{2}} = \frac{2}{3\sqrt{2}} \text{ N}$$

Ex. For shown situation draw a graph showing accelerations of A and B on y-axis and time on x-axis. ($g=10\text{ms}^{-2}$)



Sol. Limiting friction between A & B, $f_L = \mu m_A g = \left(\frac{1}{2}\right)(2)(10) = 10 \text{ N}$

Block B moves due to friction only. So maximum acceleration of B,

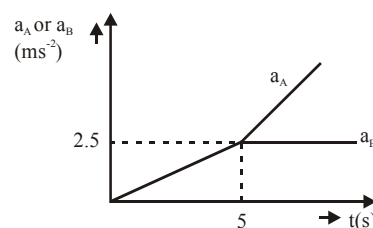
$$a_{\max} = \frac{f_L}{m_B} = \frac{10}{4} = 2.5 \text{ ms}^{-2}.$$

So both the blocks move together till the common acceleration becomes 2.5 ms^{-2} , after that accelerations of B will become constant while that of A will go on increasing. Slipping will start between A & B at 2.5 ms^{-2}

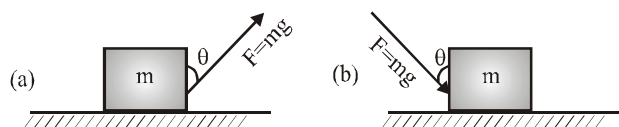
$$\Rightarrow 2.5 = \frac{F}{m_A + m_B} = \frac{3t}{6} \Rightarrow t = 5 \text{ s}$$

$$\text{Hence for } t \leq 5 \text{ s, } a_A = a_B = \frac{F}{m_A + m_B} = \frac{3t}{6} = \frac{t}{2}$$

$$\text{and for } t > 5 \text{ s } a_B = 2.5 \text{ ms}^{-2}, a_A = \frac{F - f_L}{m_A} = \frac{3t - 10}{2} = \frac{3}{2}t - 5$$



Ex. A block of mass m rests on a rough horizontal surface as shown in figure (a) and (b). Coefficient of friction between block and surface is μ . A force $F = mg$ acting at an angle θ with the vertical side of the block. Find the condition for which block will move along the surface.



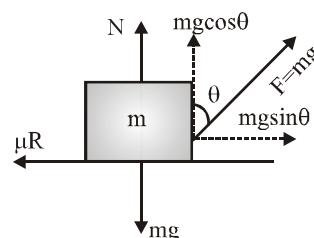
Sol. For (a) : normal reaction $N = mg - mg \cos \theta$, frictional force $= \mu N = \mu(mg - mg \cos \theta)$

Now block can be pulled when : Horizontal component of force $\geq$ frictional force

$$\text{i.e. } mg \sin \theta \geq \mu(mg - mg \cos \theta)$$

$$\text{or } 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \geq \mu(1 - \cos \theta)$$

$$\text{or } 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \geq 2\mu \sin^2 \frac{\theta}{2} \quad \text{or} \quad \cot \frac{\theta}{2} \geq \mu$$

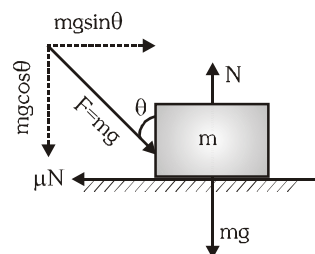


For (b) : Normal reaction $N = mg + mg \cos \theta = mg(1 + \cos \theta)$

Hence, block can be pushed along the horizontal surface when horizontal component of force $\geq$ frictional force

$$\text{i.e. } mg \sin \theta \geq \mu mg(1 + \cos \theta)$$

$$\text{or } 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} \geq \mu \times 2 \cos^2 \frac{\theta}{2} \Rightarrow \tan \frac{\theta}{2} \geq \mu$$



Ex. A body of mass m rests on a horizontal floor with which it has a coefficient of static friction μ . It is desired to make the body move by applying the minimum possible force F . Find the magnitude of F and the direction in which it has to be applied.

Sol. Let the force F be applied at an angle θ with the horizontal as shown in figure.

For vertical equilibrium,

$$R + F \sin \theta = mg \Rightarrow N = mg - F \sin \theta \quad (i)$$

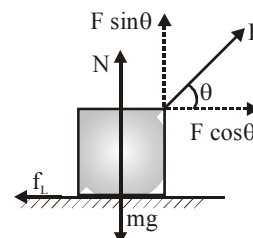
$$\text{for horizontal motion } F \cos \theta \geq f_L \Rightarrow F \cos \theta \geq \mu N \quad [\text{as } f_L = \mu N] \quad (ii)$$

substituting value of R from equation (i) in (ii),

$$F \cos \theta \geq \mu(mg - F \sin \theta) \Rightarrow F \geq \frac{\mu mg}{(\cos \theta + \mu \sin \theta)} \quad (iii)$$

For the force F to be minimum $(\cos \theta + \mu \sin \theta)$ must be maximum,

maximum value of $\cos \theta + \mu \sin \theta$ is $\sqrt{1 + \mu^2}$ so that $F_{\min} = \frac{\mu mg}{\sqrt{1 + \mu^2}}$ with $\theta = \tan^{-1}(\mu)$



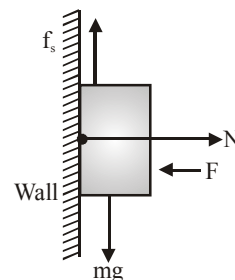
Ex. A book of 1 kg is held against a wall by applying a perpendicular force F . If $\mu_s = 0.2$ then what is the minimum value of F ?

Sol. The situation is shown in fig. The forces acting on the book are—

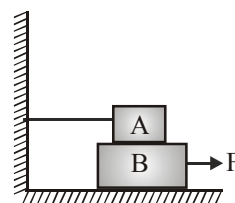
For book to be at rest it is essential that $Mg = f_s$

But $f_{s \max} = \mu_s N$ and $N = F$

$$\therefore Mg = \mu_s F \Rightarrow F = \frac{Mg}{\mu_s} = \frac{1 \times 9.8}{0.2} = 49 \text{ N}$$



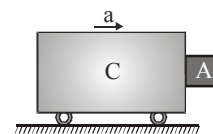
Ex. A is a 100 kg block and B is a 200 kg block. As shown in fig., the block A is attached to a string tied to a wall. The coefficient of friction between A and B is 0.2 and the coefficient of friction between B and floor is 0.3. Then calculate the minimum force required to move the block B. (take $g = 10 \text{ m/s}^2$).



Sol. When B is tied to move, by applying a force F , then the frictional forces acting on the block B are f_1 and f_2 with limiting values, $f_1 = (\mu_s)_A m_A g$ and $f_2 = (\mu_s)_B (m_A + m_B)g$ then minimum value of F should be (for just tending to move),

$$F = f_1 + f_2 = 0.2 \times 100 g + 0.3 \times 300 g = 110 g = 1100 \text{ N}$$

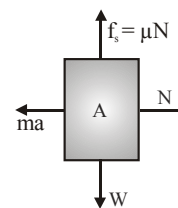
Ex. Consider the figure shown here of a moving cart C. If the coefficient of friction between the block A and the cart is μ , then calculate the minimum acceleration a of the cart C so that the block A does not fall.



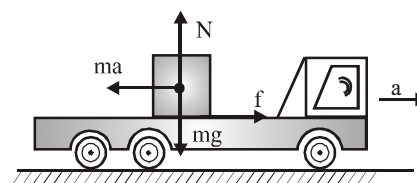
Sol. The forces acting on the block A (in block A's frame (i.e. non inertial frame) are :

For A to be at rest in block A's frame i.e. no fall,

$$\text{we require } W = f_s \Rightarrow mg = \mu(ma) \text{ Thus } a = \frac{g}{\mu}$$



Ex. A block of mass 1kg lies on a horizontal surface in a truck, the coefficient of static friction between the block and the surface is 0.6, What is the force of friction on the block. If the acceleration of the truck is 5 m/s^2 .



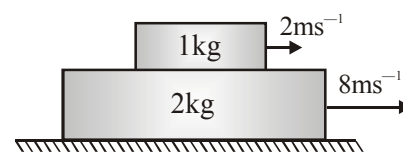
Sol. Fictitious force on the block $F = ma = 1 \times 5 = 5 \text{ N}$

While the limiting friction force

$$F = \mu_s N = \mu_s mg = 0.6 \times 1 \times 9.8 = 5.88 \text{ N}$$

As required force F lesser than limiting friction force. The block will remain at rest in the truck and the force of friction will be equal to 5N and in the direction of acceleration of the truck.

- Ex.** Coefficient of friction between two blocks shown in figure is $\mu = 0.4$. The blocks are given velocities of 2ms^{-1} and 8ms^{-1} in the directions shown in figure. Find
- The time when relative motion between them will stop.
 - The common velocities of blocks upto that instant.
 - Displacement of blocks upto that instant ($g = 10\text{ms}^{-2}$)



- Sol.** (i) Frictional force between two blocks will oppose the relative motion. For 1 kg block friction supports the motion & for 2 kg friction opposes the motion. Let common velocity be v then

$$\text{for 1 kg } v = 2 + a_1 t \text{ where } a_1 = \frac{\mu(1g)}{1} = \frac{0.4 \times 10}{1} = 4\text{ms}^{-2}$$

$$\text{for 2 kg } v = 8 - a_2 t \text{ where } a_2 = \frac{\mu(1g)}{2} = \frac{0.4 \times 10}{2} = 2\text{ms}^{-2} \Rightarrow 2 + 4t = 8 - 2t \Rightarrow 6t = 6 \Rightarrow t = 1\text{s}$$

$$(ii) v = 2 + 4t = 2 + 4 \times 1 = 6\text{ms}^{-1}$$

(iii) Displacement of 1 kg block from rest

$$s = ut + \frac{1}{2}at^2 \Rightarrow s_1 = 2 \times 1 + \frac{1}{2} \times 4 \times 1^2 = 2 + 2 = 4\text{m}$$

Displacement of 2 kg block from rest

$$s = ut + \frac{1}{2}at^2 \Rightarrow s_2 = 8 \times 1 - \frac{1}{2} \times 2 \times 1^2 = 8 - 1 = 7\text{m}$$

Friction is a Necessary Evil :

Friction is a necessary evil. It means it has advantage as well as disadvantages. In other words, friction is not desirable but without friction, we cannot think of survival.

Disadvantages :

- A significant amount of energy of a moving object is wasted in the form of heat energy to overcome the force of friction.
- The force of friction restricts the speed of moving vehicles like buses, trains, aeroplanes, rockets etc.
- The efficiency of machines decreases due to the presence of force of friction.
- The force of friction causes lot of wear and tear in the moving parts of a machine.
- Sometimes, a machine gets burnt due to the friction force between different moving parts.

Advantages :

- The force of friction helps us to move on the surface of earth. In the absence of friction, we cannot think of walking on the surface. That is why, we fall down while moving on a smooth surface.
- The force of friction between the tip of a pen and the surface of paper helps us to write on the paper. It is not possible to write on the glazed paper as there is not force of friction.
- The force of friction between the tyres of a vehicle and the road helps the vehicle to stop when brake is applied. In the absence of friction, the vehicle skid off the road when brake is applied.
- moving belts remain on the rim of a wheel because of friction.
- The force of friction between a chalk and the black board helps us to write on the board.

Thus, we observe that inspect of various disadvantages of the friction, it is very difficult to part with it. So, friction is a necessary evil.

METHODS OF REDUCING FRICTION

As friction causes the wastage of energy so it becomes necessary to reduce the friction. Friction can be reduced by the following methods.

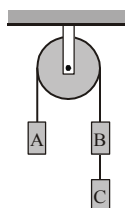
- (i) Polishing the surface. We know, friction between rough surface is much more than between the polished surfaces. So we polish the surface to reduce the friction. The irregularities on the surface are filled with polish and hence the friction decreases.
- (ii) Lubrication. To reduce friction, lubricants like oil or greese are used. When the oil or greese is put in between the two surfaces, the irregularities remain apart and do not interlock tightly. Thus, the surface can move over each other with less friction between them.
- (iii) By providing the streamlined shape. When a body (e.g. bus, train, aeroplane etc.) moves with high speed, air resistance (friction) opposes its motion. The effect of air resistance on the motion of the objects (stated above) is decreased by providing them a streamlined shape.

EXERCISE (S-1)

1. A force F applied to an object of mass m_1 produces an acceleration of 3.00 m/s^2 . The same force applied to a second object of mass m_2 produces an acceleration of 1.00 m/s^2 .
(i) What is the value of the ratio m_1/m_2 ?
(ii) If m_1 and m_2 are combined, find their acceleration under the action of the force F .

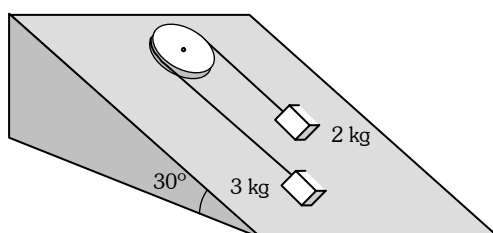
NL0001

2. In the system shown, the blocks A, B and C are of weight $4W$, W and W respectively. The system set free. The tension in the string connecting the blocks B and C is



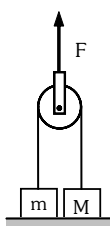
NL0002

3. Two blocks of masses 2.0 kg and 3.0 kg are connected by light inextensible string. The string passes over an ideal pulley pivoted to a fixed axel on a smooth incline plane as shown in the figure. When the blocks are released, find magnitude of their accelerations.



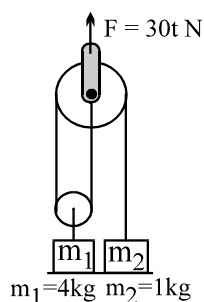
NL0003

4. In the system shown, pulley and strings are ideal. The vertically upward pull F is being increased gradually, find magnitude of F and acceleration of the 5 kg block at the moment the 10 kg block leaves the floor.



NL0004

5. Force F is applied on upper pulley. If $F = 30t$ N where t is time in second. Find the time when m_1 loses contact with floor.

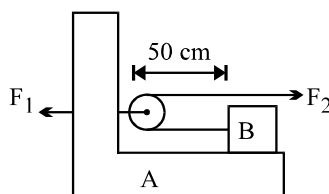


NL0005

6. A 40 kg boy climbs a rope that passes over an ideal pulley. The other end of the rope is attached to a 60 kg weight placed on the ground. What is the maximum upward acceleration the boy can have without lifting the weight? If he climbs the rope with upward acceleration $2g$, with what acceleration the weight will rise up?

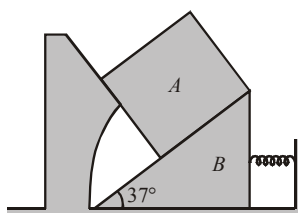
NL0006

7. A 1 kg block B rests as shown on a bracket A of same mass. Constant forces $F_1 = 20$ N and $F_2 = 8$ N start to act at time $t = 0$ when the distance of block B from pulley is 50 cm. Time when block B reaches the pulley is _____.



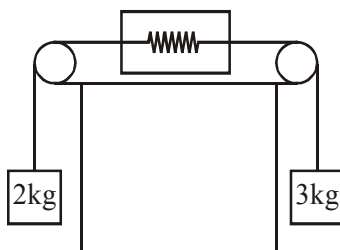
NL0007

8. In the figure shown, all surfaces are smooth and block A and wedge B have mass 10 kg and 20 kg respectively. Find normal reaction between block A & B , spring force and normal reaction of ground on block B . ($g = 10$ m/s²).



NL0008

9. Find the reading of the massless spring balance in the given condition



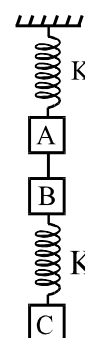
10. The system shown adjacent is in equilibrium. Find the acceleration of the blocks A , B & C all of equal masses m at the instant when

(Assume springs to be ideal)

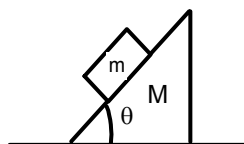
- The spring between ceiling & A is cut.
- The string (inextensible) between A & B is cut.
- The spring between B & C is cut.

Also find the tension in the string when the system is at rest and in the above 3 cases.

NL0009



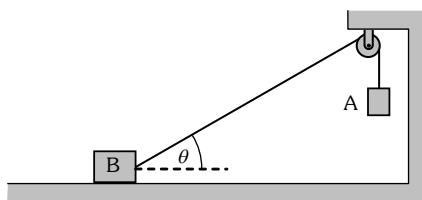
11. A block of mass m lies on wedge of mass M as shown in figure.



With what minimum acceleration must the wedge be moved towards right horizontally so that block m falls freely.

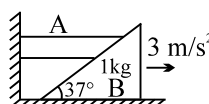
NL0011

12. The block A is moving downward with constant velocity v_0 . Find the velocity of the block B , when the string makes an angle θ with the horizontal



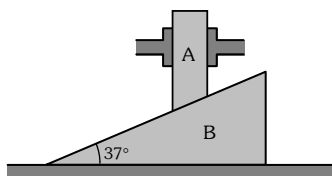
NL0012

13. Find force in newton which mass A exerts on mass B if B is moving towards right with 3 m/s^2 . Also find mass of A . (All surfaces are smooth)



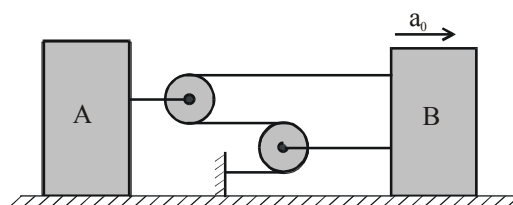
NL0013

14. Rod A can slide in vertical direction pushing the triangular wedge B towards right. The wedge is moving toward right with uniform acceleration a_B . Find acceleration of the rod A.



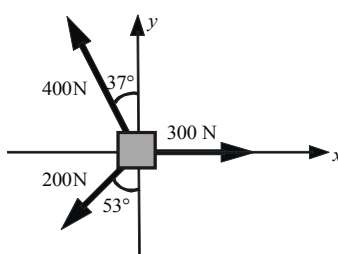
NL0014

15. Calculate the relative acceleration of A w.r.t. B if B is moving with acceleration a_0 towards right.



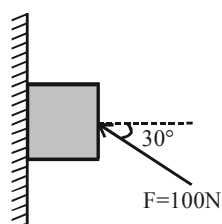
NL0015

16. A block is placed on a rough horizontal plane. Three horizontal forces are applied on the block as shown in the figure. If the block is in equilibrium, find the friction force acting on the block.



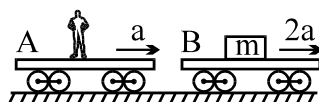
NL0016

17. A force of 100 N is applied on a block of mass 3 kg as shown in figure. The coefficient of friction between the wall and the surface of the block is $1/4$. Calculate frictional force acting on the block.



NL0017

18. Two trolley A and B are moving with accelerations a and $2a$ respectively in the same direction. To an observer in trolley A, the magnitude of pseudo force acting on a block of mass m on the trolley B is



NL0018

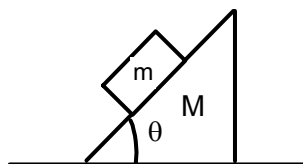
19. A thin rod of length 1 m is fixed in a vertical position inside a train, which is moving horizontally with constant acceleration 4 m/s^2 . A bead can slide on the rod, and friction coefficient between them is $1/2$. If the bead is released from rest at the top of the rod, find the time when it will reach at the bottom. [$g = 10\text{ m/s}^2$]

NL0019

20. A block of mass 1 kg is horizontally thrown with a velocity of 10 m/s on a stationary long plank of mass 2 kg whose surface has $\mu = 0.5$. Plank rests on frictionless surface. Find the time when block comes to rest w.r.t. plank.

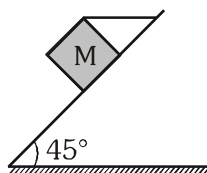
NL0020

21. A block of mass m lies on wedge of mass M as shown in figure. Find the minimum friction coefficient required between wedge M and ground so that it does not move while block m slips down on it.



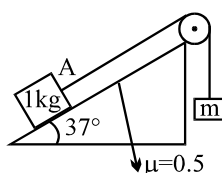
NL0021

22. A block of mass 15 kg is resting on a rough inclined plane as shown in figure. The block is tied up by a horizontal string which has a tension of 50 N . Calculate the minimum coefficient of friction between the block and inclined plane.



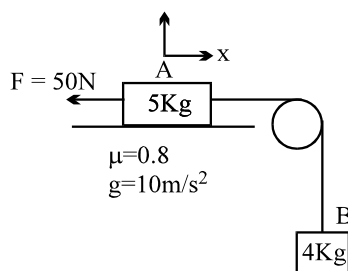
NL0022

23. In the figure, what should be mass m so that block A slides up with a constant velocity?



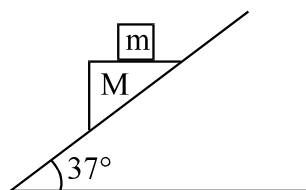
NL0023

24. Find the acceleration of the blocks and magnitude & direction of frictional force between block A and table, if block A is pulled towards left with a force of 50 N.



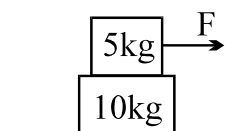
NL0024

25. Block M slides down on frictionless incline as shown. Find the minimum friction coefficient so that m does not slide with respect to M .



NL0025

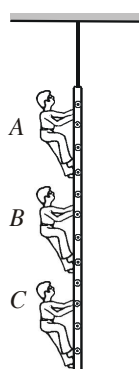
26. Coefficient of friction between 5 kg and 10 kg block is 0.5. If friction between them is 20 N. What is the value of force being applied on 5 kg. The floor is frictionless.



NL0026

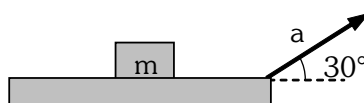
EXERCISE (S-2)

1. A ladder is hanging from ceiling as shown in figure. Three men A , B and C of masses 40 kg , 60 kg , and 50 kg are climbing the ladder. Man A is going up with retardation 2 m/s^2 , C is going up with an acceleration of 1 m/s^2 and man B is going up with a constant speed of 0.5 m/s . Find the tension in the string supporting the ladder. [$g = 9.8\text{ m/s}^2$]



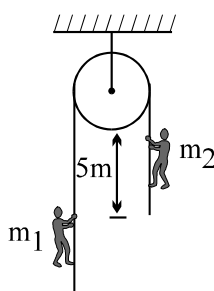
NL0027

2. A box of mass m is placed on a smooth horizontal platform as shown in the figure. The platform is made to move in direction 30° above the horizontal with acceleration a so that the contact force between the box and the platform becomes $3mg/2$. Find the magnitude of the acceleration.



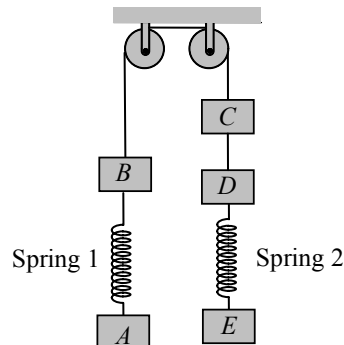
NL0028

3. Two men of masses m_1 and m_2 hold on the opposite ends of a rope passing over a frictionless pulley. The man m_1 climbs up the rope with an acceleration of 1.2 m/s^2 relative to the rope. The man m_2 climbs up the rope with an acceleration of 2 m/s^2 relative to the rope. Find the tension in the rope if $m_1 = 40\text{ kg}$ and $m_2 = 60\text{ kg}$. Also find the time after which they will be at same horizontal level if they start from rest and are initially separated by 5 m .



NL0029

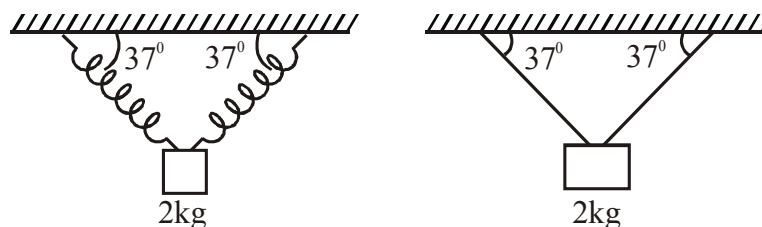
4. The system shown in the figure is initially in equilibrium. A is of mass $2m$ and B, C, D and E are of mass m . Certain actions are performed on the system. Every action has been taken individually when the system is intact. Find the direction and magnitude of acceleration of the blocks after each action of the following actions has been taken



- (i) Spring 1 is cut
 (ii) Spring 2 is cut
 (iii) String between C and D is cut.
 (iv) String between B and C is cut.

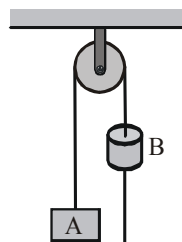
NL0030

5. The blocks are of mass 2 kg shown is in equilibrium. At $t = 0$ right spring in figure (i) and right string in figure (ii) breaks. Find the ratio of instantaneous acceleration of blocks?



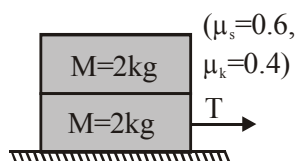
NL0031

6. A 2 kg block A is attached to one end of a light string that passes over an ideal pulley and a 1 kg sleeve B slides down the other part of the string with an acceleration of 5 m/s^2 with respect to the string. Find the acceleration of the block, acceleration of sleeve and tension in the string. [$g = 10\text{ m/s}^2$]



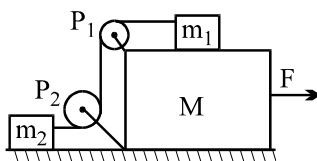
NL0032

7. The coefficient of static and kinetic friction between the two blocks and also between the lower block and the ground are $\mu_s = 0.6$ and $\mu_k = 0.4$. Find the value of tension T applied on the lower block at which the upper block begins to slip relative to lower block.



NL0033

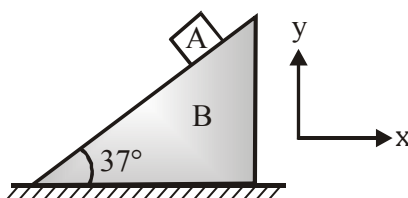
8. In the figure masses m_1 , m_2 and M are 20 kg, 5 kg and 50 kg respectively. The co-efficient of friction between M and ground is zero. The co-efficient of friction between m_1 and M and that between m_2 and ground is 0.3. The pulleys and the string are massless. The string is perfectly horizontal between P_1 and m_1 and also between P_2 and m_2 . The string is perfectly vertical between P_1 and P_2 . An external horizontal force F is applied to the mass M . Take $g = 10 \text{ m/s}^2$.



- (i) Draw a free-body diagram for mass M , clearly showing all the forces.
 (ii) Let the magnitude of the force of friction between m_1 and M be f_1 and that between m_2 and ground be f_2 . For a particular F it is found that $f_1 = 2 f_2$. Find f_1 and f_2 . Write down equations of motion of all the masses. Find F , tension in the string and accelerations of the masses.

NL0034

9. In the figure shown the acceleration of A is, $\vec{a}_A = (15\hat{i} + 15\hat{j}) \text{ m/s}^2$. If A is sliding on B then the acceleration of B is.

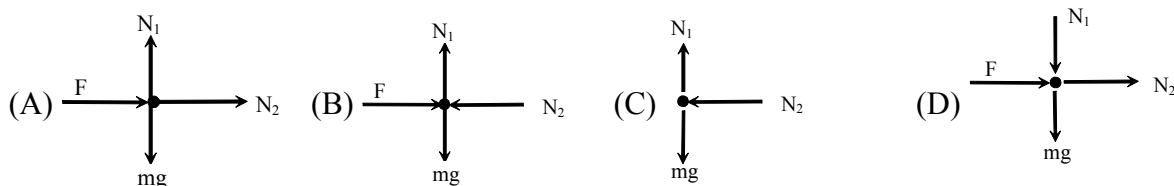
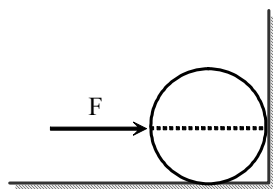


NL0035

EXERCISE (O-1)

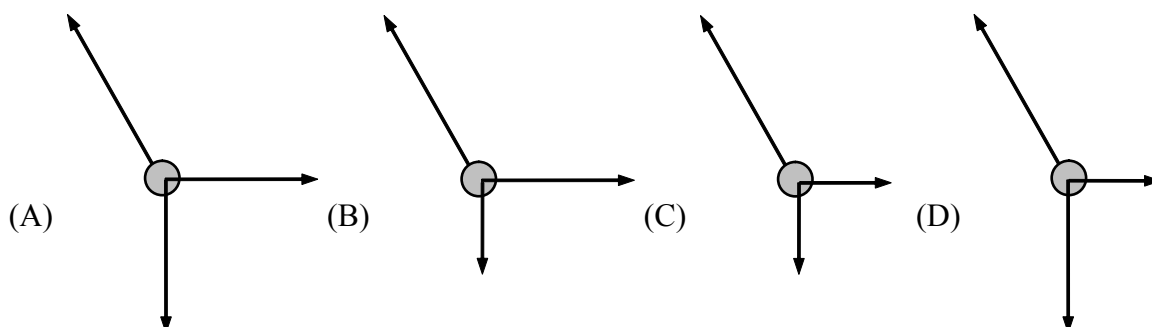
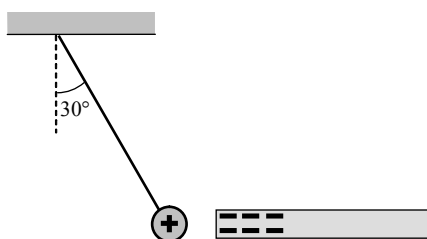
SINGLE CORRECT TYPE QUESTIONS

1. A ball of mass m kept at the corner as shown in the figure, is acted by a horizontal force F . The correct free body diagram of ball is



NL0036

2. A small electrically charged sphere is suspended vertically from a thread. An oppositely charged rod is brought close to the sphere such that the sphere is in equilibrium displaced from the vertical by an angle of 30° . Which one of the following best represents the free body diagram for the sphere?



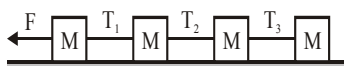
NL0037

3. Under what condition(s) will an object be in equilibrium ?

(A) Only if it is at rest
(B) Only if it is moving with constant velocity
(C) Only if it is moving with constant acceleration
(D) If it is either at rest or moving with constant velocity

NL0038

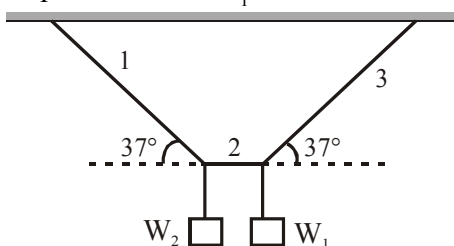
4. Four blocks of same mass connected by cords are pulled by force F on a smooth horizontal surface, as in figure. The tension T_1 , T_2 and T_3 will be



(A) $T_1 = F/4$, $T_2 = 3F/2$, $T_3 = F/4$
(B) $T_1 = F/4$, $T_2 = F/2$, $T_3 = F/2$
(C) $T_1 = 3F/4$, $T_2 = F/2$, $T_3 = F/4$
(D) $T_1 = 3F/4$, $T_2 = F/2$, $T_3 = F/2$

NL0039

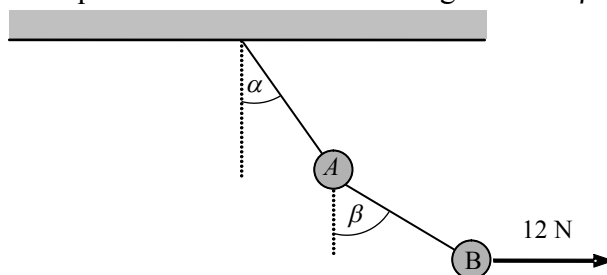
5. In a given figure system is in equilibrium. If $W_1 = 300$ N. Then W_2 is approximately equal to



(A) 500 N (B) 400 N (C) 670 N (D) 300 N

NL0040

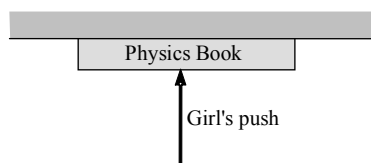
6. Two balls A and B weighing 7 N and 9 N are connected by a light cord. The system is suspended from a fixed support by connecting the ball A with another light cord. The ball B is pulled aside by a horizontal force 12 N and equilibrium is established. Angles α and β respectively are



(A) 30° and 60° (B) 60° and 30° (C) 37° and 53° (D) 53° and 37°

NL0041

7. A girl pushes her physics book up against the horizontal ceiling of her room as shown in the figure. The book weighs 20 N and she pushes upwards with a force of 25 N. The choices below list the magnitudes of the contact force F_{CB} between the ceiling and the book, and F_{BH} between the book and her hand. Select the correct pair.



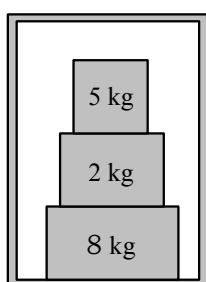
(A) $F_{CB} = 20$ N and $F_{BH} = 25$ N
(B) $F_{CB} = 25$ N and $F_{BH} = 45$ N
(C) $F_{CB} = 5$ N and $F_{BH} = 25$ N
(D) $F_{CB} = 5$ N and $F_{BH} = 45$ N

NL0042

8. Two astronauts A and B connected with a rope stay stationary in free space relative to their spaceship. Mass of A is more than that of B and the rope is straight. Astronaut A starts pulling the rope but astronaut B does not. If you were the third astronaut in the spaceship, what do you observe?
- (A) Astronaut B accelerates towards A and A remains stationary.
 (B) Both accelerate towards each other with equal accelerations of equal modulus.
 (C) Both accelerate towards each other but acceleration of B is greater than that of A.
 (D) Both accelerate towards each other but acceleration of B is smaller than that of A.

NL0043

9. Three boxes are placed in a lift. When acceleration of the lift is 4 m/s^2 , the net force on the 8 kg box is closest to



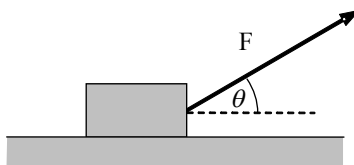
- (A) 80 N (B) 48 N (C) 40 N (D) 32 N

NL0044

10. A man is standing on a weighing machine with a block in his hand. The machine records w . When he takes the block upwards with some acceleration the machine records w_1 . When he takes the block down with some acceleration, the machine records w_2 . Then choose correct option
- (A) $w_1 = w = w_2$ (B) $w_1 < w < w_2$ (C) $w_2 < w < w_1$ (D) $w_2 = w_1 > w$

NL0045

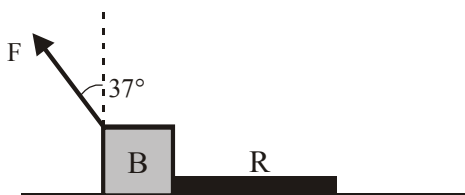
11. A block is being pulled by a force F on a long frictionless level floor. Magnitude of the force is gradually increases from zero until the block lifts off the floor. Immediately before the block leaves the floor, its acceleration is



- (A) $g \cos \theta$ (B) $g \cot \theta$ (C) $g \sin \theta$ (D) $g \tan \theta$

NL0046

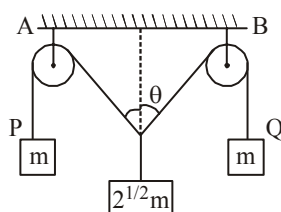
12. A block B is tied to one end of a uniform rope R as shown. The mass of block is 2 kg and that of rope is 1 kg. A force $F = 15$ N is applied at angle 37° with vertical. The tension at the mid-point of rope is



- (A) 1.5 N (B) 2 N (C) 3 N (D) 4.5 N

NL0047

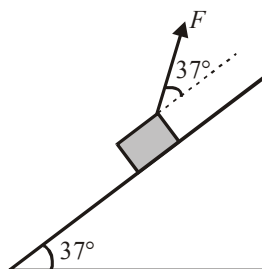
13. The pulleys and strings shown in the figure are smooth and of negligible mass. For the system to remain in equilibrium, the angle θ should be [JEE (Scr) 2001]



- (A) 0° (B) 30° (C) 45° (D) 60°

NL0048

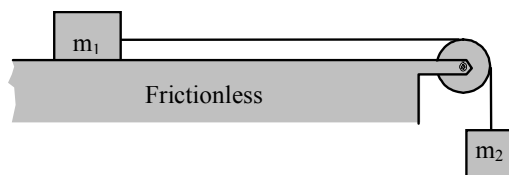
14. A block resting on a smooth inclined plane is acted upon by a force F as shown. If mass of block is 2 kg and $F = 20$ N and $\sin 37^\circ = 3/5$, the acceleration of block is



- (A) 2 m/s^2 (B) 6 m/s^2 (C) 8 m/s^2 (D) zero

NL0049

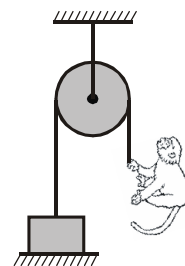
15. In the arrangement shown, the blocks of unequal masses are held at rest. When released, acceleration of the blocks is



- (A) $g/2$. (B) g .
(C) a value between zero and g . (D) a value that could be greater than g .

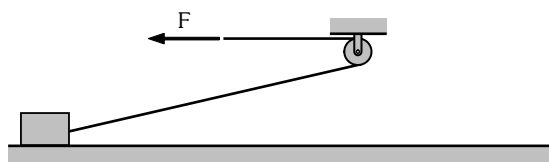
NL0050

16. A monkey weighing 10 kg is climbing up a light rope and frictionless pulley attached to 15 kg mass at other end as in figure. In order to raise the 15 kg mass off the ground the monkey must climb-up
- (A) with constant acceleration $g/3$.
 (B) with an acceleration greater than $g/2$.
 (C) with an acceleration greater than $g/4$.
 (D) It is not possible because weight of monkey is lesser than the block.



NL0051

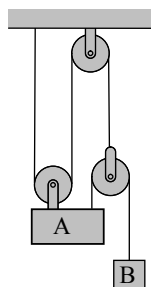
17. A heavy cart is pulled by a constant force F along a horizontal track with the help of a rope that passes over a fixed pulley, as shown in the figure. Assume the tension in the rope and the frictional forces on the cart remain constant and consider motion of the cart until it reaches vertically below the pulley. As the cart moves to the right, its acceleration



- (A) decreases. (B) increases. (C) remains constant. (D) is zero

NL0052

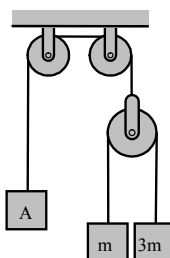
18. In arrangement shown the block A of mass 15 kg is supported in equilibrium by the block B. Mass of the block B is closest to



- (A) 2 kg (B) 3 kg (C) 4 kg (D) 5 kg

NL0053

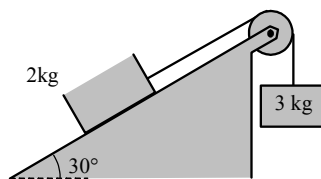
19. In the given figure, find mass of the block A, if it remains at rest, when the system is released from rest. Pulleys and strings are massless. [$g = 10 \text{ m/s}^2$]



- (A) m (B) $2m$ (C) $2.5m$ (D) $3m$

NL0054

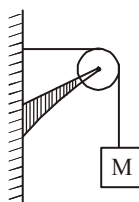
- 20.** In the arrangement shown, the 2 kg block is held to keep the system at rest. The string and pulley are ideal. When the 2 kg block is set free, by what amount the tension in the string changes? [$g = 10 \text{ m/s}^2$]



- (A) Increase of 12 N (B) Decrease of 12 N (C) Increase of 18 N (D) Decrease of 18 N

NL0055

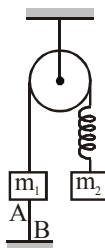
- 21.** A string of negligible mass going over a clamped pulley of mass m supports a block of mass M as shown in the figure. The force on the pulley by the clamp is given **[JEE (Scr) 2001]**



- (A) $\sqrt{2} Mg$ (B) $\sqrt{2} mg$ (C) $\sqrt{(M+m)^2 + m^2} \text{ g}$ (D) $\sqrt{(M+m)^2 + M^2} \text{ g}$

NL0056

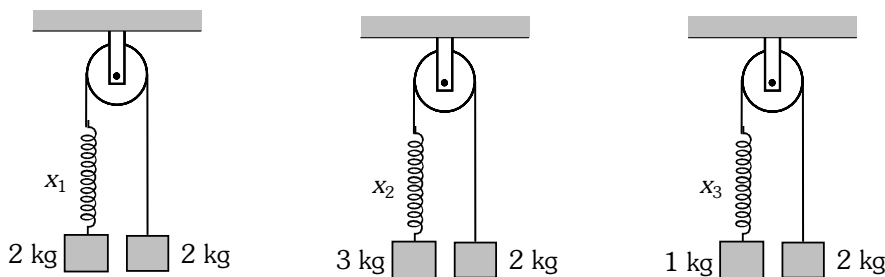
- 22.** In a given figure two masses m_1 & m_2 ($m_2 > m_1$) are at rest in equilibrium position. Find the tension in string AB



- (A) $m_1 g$ (B) $m_2 g$ (C) $(m_1 + m_2) g$ (D) $(m_2 - m_1) g$

NL0057

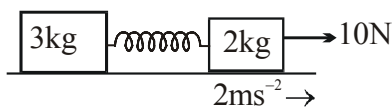
- 23.** Same spring is attached with 2 kg, 3 kg and 1 kg blocks in three different cases as shown. If x_1 , x_2 and x_3 be the extensions in the spring in these three cases, when acceleration of both the blocks have same magnitude, then



- (A) $x_2 > x_3 > x_1$ (B) $x_2 > x_1 > x_3$ (C) $x_3 > x_1 > x_2$ (D) $x_1 > x_2 > x_3$

NL0058

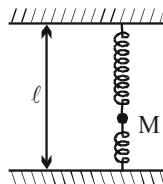
24. Find the acceleration of 3 kg mass when acceleration of 2 kg mass is 2 ms^{-2} as shown in figure.



- (A) 3 ms^{-2} (B) 2 ms^{-2} (C) 0.5 ms^{-2} (D) zero

NL0059

25. A small ball of mass M is held in equilibrium with two identical springs as shown in the figure. Force constant of each spring is k and relaxed length of each spring is $\ell/2$. What is distance between the ball and roof?



- (A) $\frac{\ell}{2} + \frac{Mg}{k}$ (B) $\frac{\ell}{2} - \frac{Mg}{k}$ (C) $\frac{\ell}{2} + \frac{Mg}{2k}$ (D) $\frac{\ell}{2} - \frac{Mg}{2k}$

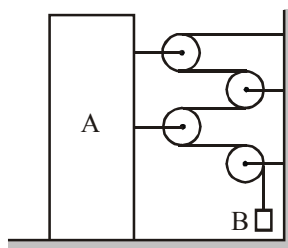
NL0060

26. An elastic spring of relaxed length ℓ_0 and force constant k is cut into two parts of lengths ℓ_1 and ℓ_2 . The force constants of these parts are respectively

- (A) $\frac{k\ell_0}{\ell_1}$ and $\frac{k\ell_0}{\ell_2}$ (B) $\frac{k\ell_1}{\ell_0}$ and $\frac{k\ell_2}{\ell_0}$
(C) $\frac{k\ell_0}{\ell_2}$ and $\frac{k\ell_0}{\ell_1}$ (D) $\frac{k\ell_2}{\ell_0}$ and $\frac{k\ell_1}{\ell_0}$

NL0061

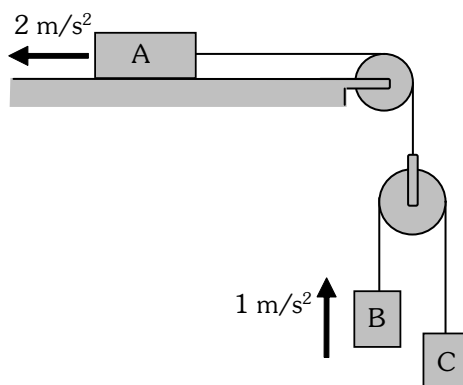
27. Block A is moving away from the wall at a speed v and acceleration a .



- (A) Velocity of B is v with respect to A.
(B) Acceleration of B is a with respect to A.
(C) Acceleration of B is $4a$ with respect to A.
(D) Acceleration of B is $\sqrt{17}a$ with respect to A.

NL0062

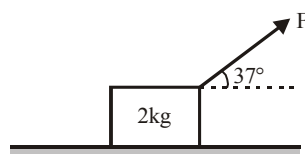
28. In the setup shown, find acceleration of the block C.



- (A) $3 \text{ m/s}^2 \uparrow$ (B) $3 \text{ m/s}^2 \downarrow$ (C) $5 \text{ m/s}^2 \uparrow$ (D) $5 \text{ m/s}^2 \downarrow$

NL0063

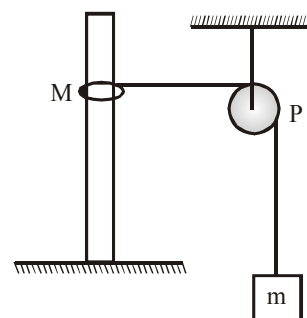
29. A block of mass 2 kg is kept on a rough horizontal floor and pulled with a force F . If the coefficient of friction is 0.5 , then the minimum force required to move the block is :-



- (A) 10 N (B) $\frac{100}{11} \text{ N}$ (C) $\frac{100}{8} \text{ N}$ (D) 20 N

NL0064

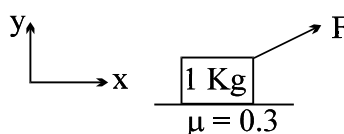
30. In the figure shown a ring of mass M and a block of mass m are in equilibrium. The string is light and pulley P does not offer any friction and coefficient of friction between pole and M is μ . The frictional force offered by the pole on M is



- (A) Mg directed up
(B) μmg directed up
(C) $(M - m)g$ directed down
(D) μmg directed down

NL0065

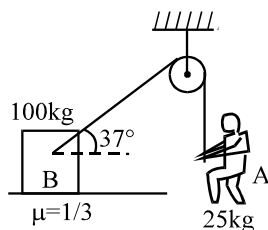
31. A force $\vec{F} = \hat{i} + 4\hat{j}$ acts on block shown. The force of friction acting on the block is :



- (A) $-\hat{i}$ (B) $-1.8\hat{i}$ (C) $-2.4\hat{i}$ (D) $-3\hat{i}$

NL0066

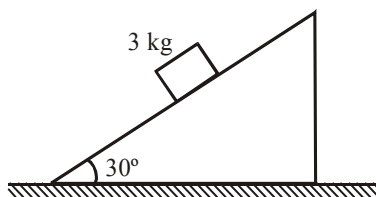
32. Block B of mass 100 kg rests on a rough surface of friction coefficient $\mu = 1/3$. A rope is tied to block B as shown in figure. The maximum acceleration with which boy A of 25 kg can climbs on rope without making block move is :



- (A) $\frac{4g}{3}$ (B) $\frac{g}{3}$ (C) $\frac{g}{2}$ (D) $\frac{3g}{4}$

NL0067

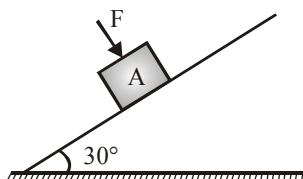
33. A block of mass 3 kg is at rest on a rough inclined plane as shown in the figure. The magnitude of net force exerted by the surface on the block will be ($g = 10\text{ m/s}^2$)



- (A) 26 N (B) 19.5 N (C) 10 N (D) 30 N

NL0068

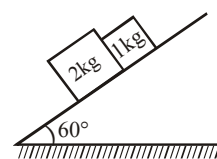
34. A block of mass $m = 2\text{ kg}$ is resting on a rough inclined plane of inclination 30° as shown in figure. The coefficient of friction between the block and the plane is $\mu = 0.5$. What minimum force F should be applied perpendicular to the plane on the block, so that block does not slip on the plane ($g = 10\text{ m/s}^2$)



- (A) zero (B) 6.24 N (C) 2.68 N (D) 4.34 N

NL0069

35. In the figure shown if friction coefficient of block 1 kg and 2 kg with inclined plane is $\mu_1 = 0.5$ and $\mu_2 = 0.4$ respectively, then
 (A) both block will move together.
 (B) both block will move separately.
 (C) there is a non zero contact force between two blocks.
 (D) none of these



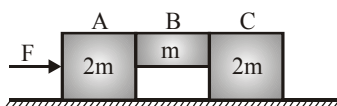
NL0070

36. A block is pushed with some velocity up a rough inclined plane. It stops after ascending few meters and then reverses its direction and returns back to point from where it started. If angle of inclination is 37° and the time to climb up is half of the time to return back then coefficient of friction is

(A) $\frac{9}{20}$ (B) $\frac{7}{5}$ (C) $\frac{7}{12}$ (D) $\frac{5}{7}$

NL0071

37. The system is pushed by a force F as shown in figure. All surfaces are smooth except between B and C . Friction coefficient between B and C is μ . Minimum value of F to prevent block B from downward slipping is :-

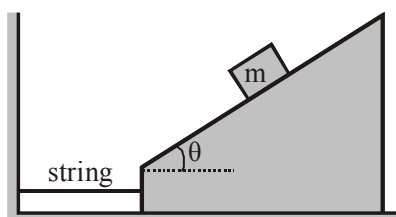


(A) $\left(\frac{3}{2\mu}\right)mg$ (B) $\left(\frac{5}{2\mu}\right)mg$ (C) $\left(\frac{5}{2}\right)\mu mg$ (D) $\left(\frac{3}{2}\right)\mu mg$

NL0072

MULTIPLE CORRECT TYPE QUESTIONS

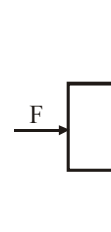
38. Refer the system shown in the figure. Block is sliding down the wedge. All surfaces are frictionless. Find correct statement(s)



(A) Acceleration of block is $g \sin \theta$ (B) Acceleration block is $g \cos \theta$
(C) Tension in the string is $mg \cos^2 \theta$ (D) Tension in the string is $mg \sin \theta \cdot \cos \theta$

NL0073

39. A block of mass 1 kg is held at rest against a rough vertical surface by pushing by a force F horizontally. The coefficient of friction is 0.5. When
- (A) $F = 40$ N, friction on the block is 20 N.
(B) $F = 30$ N, friction on the block is 10 N.
(C) $F = 20$ N, friction on the block is 10 N.
(D) Minimum value of force F to keep block at rest is 20 N.



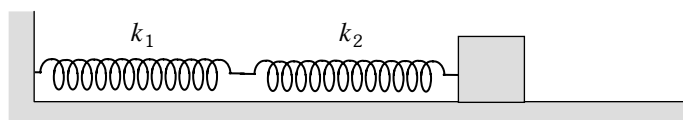
NL0074

- 40.** A block is kept on a rough horizontal surface as shown. Its mass is 2 kg and coefficient of friction between block and surface (μ) = 0.5. A horizontal force F is acting on the block. When
(A) $F = 4$ N, acceleration is zero.



NL0075

- 41.** The mass in the figure can slide on a frictionless surface. When the mass is pulled out, spring 1 is stretched a distance x_1 and spring 2 is stretched a distance x_2 . The spring constants are k_1 and k_2 respectively. Magnitude of spring force pulling back on the mass is



- (A) $k_1 x_1$ (B) $k_2 x_2$ (C) $(k_1 x_1 + k_2 x_2)$ (D) $0.5(k_1 + k_2)(x_1 + x_2)$

NL0076

- 42.** A carpenter of mass 50 kg is standing on a weighing machine placed in a lift of mass 20 kg. A light string is attached to the lift. The string passes over a smooth pulley and the other end is held by the carpenter as shown. When carpenter keeps the lift moving upward with constant velocity :- ($g = 10 \text{ m/s}^2$)



- (A) the reading of weighing machine is 15 kg
(B) the man applies a force of 350 N on the string
(C) net force on the man is 150 N
(D) Net force on the weighing machine is 150 N

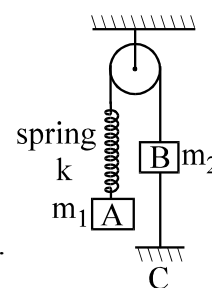
NL0077

43. In the system shown in the figure $m_1 > m_2$. System is held at rest by thread BC. Just after the thread BC is burnt :

- (A) initial acceleration of m_2 will be upwards

- (B) magnitude of initial acceleration of both blocks will be equal to $\left(\frac{m_1 - m_2}{m_1 + m_2} \right) g$.

- (C) initial acceleration of m_1 will be equal to zero
(D) magnitude of initial acceleration of two blocks will be non-zero and unequal.

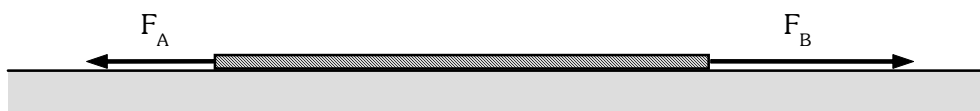


NL0078

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 44 to 47

A uniform rope of mass (m) and length (L) placed on frictionless horizontal ground is being pulled by two forces F_A and F_B at its ends as shown in the figure. As a result, the rope accelerates toward the right.



44. Acceleration (A) of the rope is

(A) zero (B) $a = \frac{F_A + F_B}{m}$ (C) $a = \frac{F_A - F_B}{m}$ (D) $a = \frac{F_B - F_A}{m}$

NL0079

45. Tension (T) at the mid point of the rope is

(A) $T = F_B - F_A$ (B) $T = F_A + F_B$ (C) $T = \frac{1}{2}(F_B - F_A)$ (D) $T = \frac{1}{2}(F_A + F_B)$

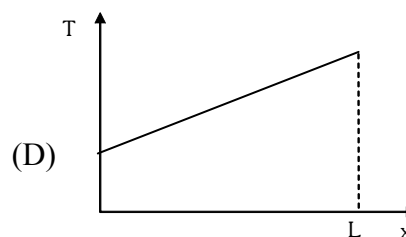
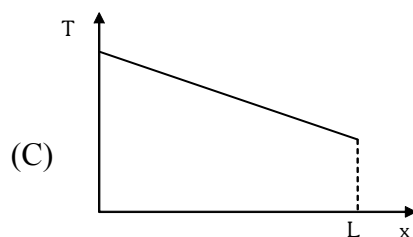
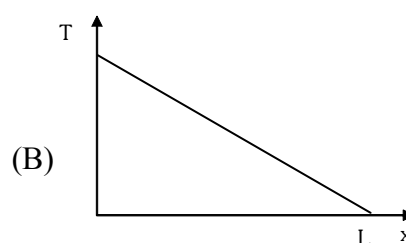
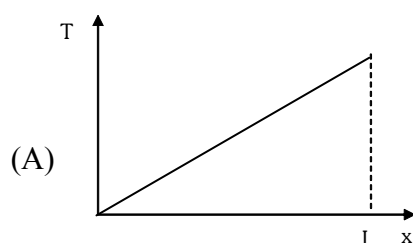
NL0079

46. Expression (T_x) of tension at a point at distance x from the end A is

(A) $T_x = \left(\frac{F_B - F_A}{L}\right)x + F_A$ (B) $T_x = \left(\frac{F_B - F_A}{L}\right)x - F_A$
(C) $T_x = \left(\frac{F_B - F_A}{L}\right)x + F_B$ (D) $T_x = \left(\frac{F_B - F_A}{L}\right)x - F_B$

NL0079

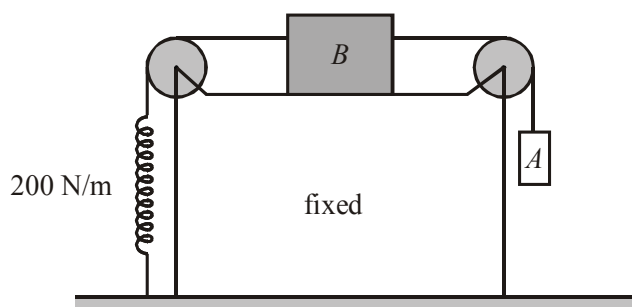
47. Which of the following graph best represents variation in tension at a point on the rope with distance x of the point from the end A ?



NL0079

Paragraph for Question No. 48 to 50

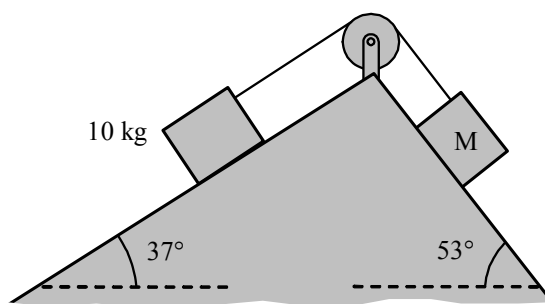
The figure shown blocks A and B are of mass 2 kg and 8 kg and they are connected through strings to a spring connected to ground. The blocks are in equilibrium. ($g = 10\text{ m/s}^2$)



48. The elongation of the spring is
 (A) 1 cm (B) 10 cm (C) 0.1 cm (D) 1 m NL0080
49. Now the block A is pulled downwards by a force gradually increasing to 20 N . The new elongation of spring is :-
 (A) 2 cm (B) 4 cm (C) 20 cm (D) 40 cm NL0080
50. Now the force on A is suddenly removed. The acceleration of block B becomes :-
 (A) 1.0 m/s (B) 2.0 m/s^2 (C) 3.0 m/s^2 (D) 4.0 m/s^2 NL0080

Paragraph for Question No. 51 to 53

The blocks are on frictionless inclined ramp and connected by a massless cord. The cord passes over an ideal pulley. [$g = 10\text{ m/s}^2$]



51. When set free, the 10 kg block slides down the ramp with acceleration of 2 m/s^2 . Mass M is closest to
 (A) 13 kg (B) 8 kg (C) 5 kg (D) 4 kg NL0081
52. When the 10 kg block slides down the ramp with acceleration of 2 m/s^2 , tension in the cord is closest to :-
 (A) 80 N (B) 60 N (C) 40 N (D) 30 N NL0081

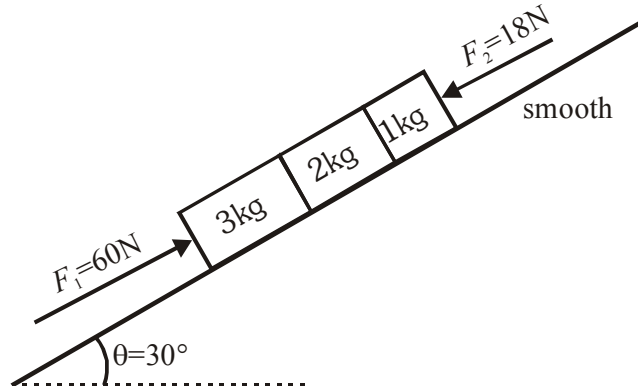
53. What value of M would keep the system at rest.

- (A) 10 kg (B) 8 kg (C) 7.5 kg (D) 6 kg

NL0081

MATRIX MATCH TYPE QUESTION

54. In the diagram shown in figure ($g = 10 \text{ m/s}^2$)



Column I

- (A) Acceleration of 2 kg block in m/s^2
 (B) Net force on 3 kg block in newton
 (C) Normal reaction between 2 kg and 1 kg in newton
 (D) Normal reaction between 3 kg and 2 kg in newton

Column II

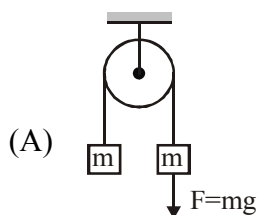
- (P) 8
 (Q) 25
 (R) 2
 (S) 45
 (T) None

NL0082

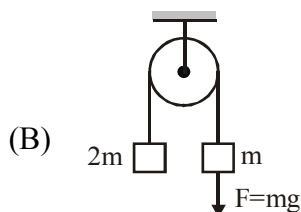
55. Match the situations in column I to the accelerations of blocks in the column II (acceleration due to gravity is g and F is an additional force applied to one of the blocks ?)

Column I

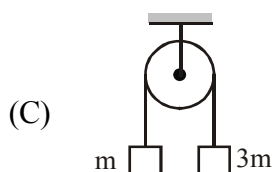
Column II



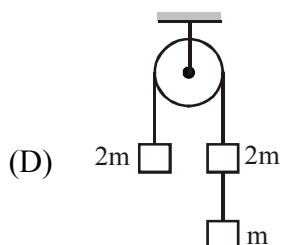
(P) $\frac{g}{5}$



(Q) $\frac{g}{3}$



(R) $\frac{g}{2}$

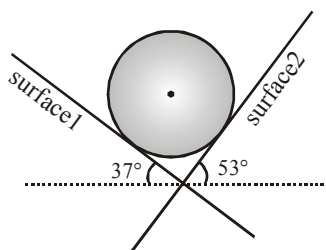


(S) $\frac{2g}{3}$

(T) zero

NL0083

56. A sphere of mass 10 kg is placed in equilibrium in a V shaped groove plane made of two smooth surfaces 1 and 2 as shown in figure. ($g = 10 \text{ ms}^{-2}$)



Column I

Column II

- (A) Normal reaction by Surface 1
(B) Normal reaction by surface 2
(C) Force on sphere by Earth
(D) Net force on sphere

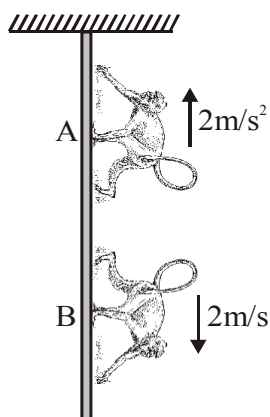
- (P) Zero
(Q) 60 N
(R) 80 N
(S) 100 N
(T) 120 N

NL0084

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. Two monkeys of masses 10 kg and 8 kg are moving along a vertical light rope, the former climbing up with an acceleration of 2 m/s^2 , while the latter coming down with a uniform velocity of 2 m/s . Find tension in the rope at the fixed support.

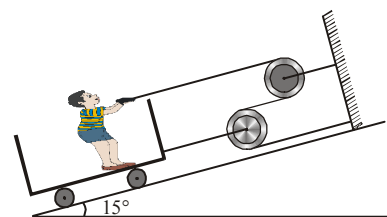


- (A) 180 N (B) 200 N (C) 80 N (D) 216 N

NL0085

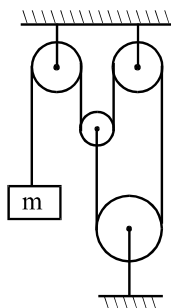
2. A trolley is being pulled up an incline plane by a man sitting on it (as shown in figure). He applies a force of 250 N. If the combined mass of the man and trolley is 100 kg, the acceleration of the trolley will be [$\sin 15^\circ = 0.26$]

- (A) 2.4 m/s^2 (B) 9.4 m/s^2
(C) 6.9 m/s^2 (D) 4.9 m/s^2



NL0086

3. If the string & all the pulleys are ideal, acceleration of mass m is :-



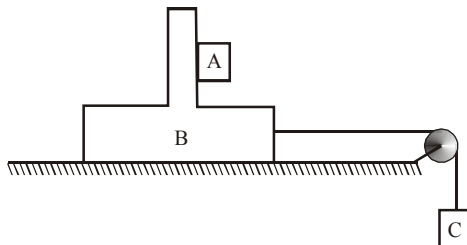
- (A) $\frac{g}{2}$ (B) 0 (C) g (D) dependent on m

NL0087

4. The rear side of a truck is open and a box of mass 20 kg is placed on the truck 4 m away from the open end, $\mu = 0.15$ and $g = 10 \text{ m/s}^2$. The truck starts from rest with an acceleration of 2 m/s^2 on a straight road. The distance moved by the truck when box starts fall down is :-
 (A) 4 m (B) 8 m (C) 16 m (D) 32 m

NL0088

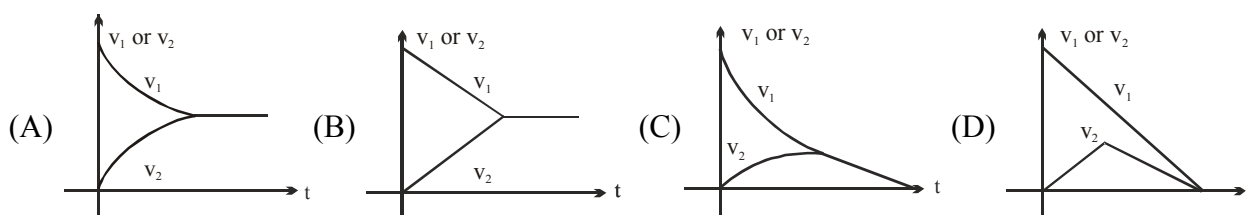
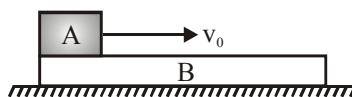
5. In the arrangement shown in the figure, mass of the block B and A is $2m$ and m respectively. Surface between B and floor is smooth. The block B is connected to the block C by means of a string-pulley system. If the whole system is released, then find the minimum value of mass of block C so that A remains stationary w.r.t. B . Coefficient of friction between A and B is μ .



- (A) $\frac{m}{\mu}$ (B) $\frac{2m+1}{\mu+1}$ (C) $\frac{3m}{\mu-1}$ (D) $\frac{6m}{\mu+1}$

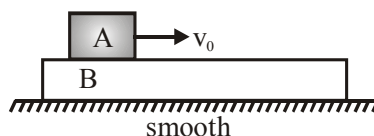
NL0089

6. A block A is placed over a long rough plank B of same mass as shown in figure. The plank is placed over a smooth horizontal surface. At time $t = 0$, block A is given a velocity v_0 in horizontal direction. Let v_1 and v_2 be the velocities of A and B at time t . Then choose the correct graph between v_1 or v_2 and t .



NL0090

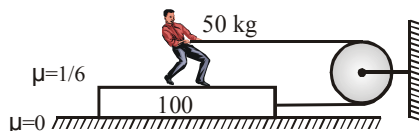
7. A block A of mass m is placed over a plank B of mass $2m$. Plank B is placed over a smooth horizontal surface. The coefficient of friction between A and B is 0.5 . Block A is given a velocity v_0 towards right. Acceleration of B relative to A is :-



- (A) $\frac{g}{2}$ (B) g (C) $\frac{3g}{4}$ (D) zero

NL0091

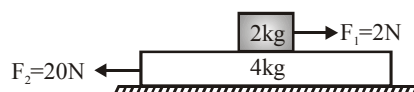
8. A man of mass 50 kg is pulling on a plank of mass 100 kg kept on a smooth floor as shown with force of 100 N. If both man & plank move together, find force of friction acting on man.



- (A) $\frac{100}{3}$ N towards left
(B) $\frac{100}{3}$ N towards right
(C) $\frac{250}{3}$ N towards left
(D) $\frac{250}{3}$ N towards right

NL0092

9. In the arrangement shown in figure, coefficient of friction between the two blocks is $\mu = 1/2$. The force of friction acting between the two blocks is :-



- (A) 8 N
(B) 10 N
(C) 6 N
(D) 4 N

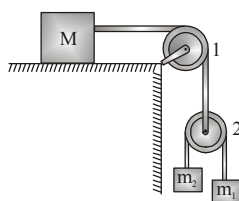
NL0093

10. A flexible chain of weight W hangs between two fixed points A & B which are at the same horizontal level. The inclination of the chain with the horizontal at both the points of support is θ . What is the tension of the chain at the mid point?

- (A) $\frac{W}{2} \cdot \operatorname{cosec} \theta$
(B) $\frac{W}{2} \cdot \tan \theta$
(C) $\frac{W}{2} \cot \theta$
(D) none

NL0094

11. In the arrangement shown in figure $m_1 = 1\text{kg}$, $m_2 = 2\text{kg}$. Pulleys are massless and strings are light. For what value of M the mass m_1 moves with constant velocity (Neglect friction)

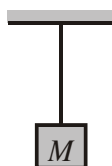


- (A) 6 kg
(B) 4 kg
(C) 8 kg
(D) 10 kg

NL0095

MULTIPLE CORRECT TYPE QUESTIONS

12. Consider a block suspended from a light string as shown in the figure. Which of the following pairs of forces constitute Newton's third law pair?



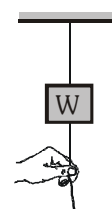
- (A) Force with which string pulls on the ceiling and the force with which string pulls on block
 (B) Force with which string pulls on the block and weight of the block
 (C) Force acting on block due to the earth and force the block exerts on the earth
 (D) Force with which block pulls on string and force with which the string pulls on the block

NL0096

13. If a horizontal support exerts an upward force of 10 N on a block of weight 9.8 N placed on it, which of the following statements is/are correct. Assume acceleration due to gravity to be 9.8 m/s^2 .
 (A) The block exerts a force of 10 N on the support.
 (B) The block exerts a force of 9.8 N on the support.
 (C) The block has an upward acceleration.
 (D) The block has a downward acceleration.

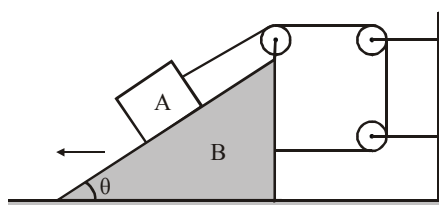
NL0097

14. A block of mass m is suspended from a fixed support with the help of a cord. Another identical cord is attached to the bottom of the block. Which of the following statement is /are true?
 (A) If the lower cord is pulled suddenly, only the upper cord will break.
 (B) If the lower cord is pulled suddenly, only the lower cord will break.
 (C) If pull on the lower cord is increased gradually, only the lower cord will break.
 (D) If pull on the lower cord is increased gradually, only the upper cord will break.



NL0098

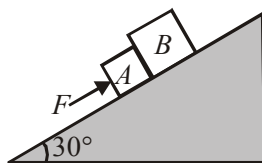
15. A block A and wedge B connected through a string as shown. The wedge B is moving away from the wall with acceleration 2 m/s^2 horizontally and acceleration of block A is vertical upwards. Then



- (A) Acceleration of A with respect to B is 4 m/s^2 .
 (B) Acceleration of A with respect to B is $2\sqrt{3} \text{ m/s}^2$.
 (C) Angle θ is 60° .
 (D) Acceleration of A is $2\sqrt{3} \text{ m/s}^2$.

NL0099

16. Two blocks A and B of mass 2 kg and 4 kg respectively are placed on a smooth inclined plane and 2 kg block is pushed by a force F acting parallel to the plane as shown. If N be the magnitude of contact force applied on B by A, which of the following is/are correct?



- (A) if $F = 0$ N, $N = 10$ N
(B) if $F = 15$ N, $N = 10$ N
(C) If $F = 30$ N, $N = 20$ N
(D) if $F = 45$ N, $N = 30$ N

NL0100

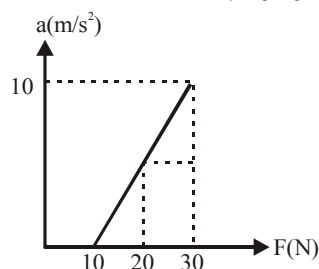
17. A block is kept on a rough surface and applied with a horizontal force as shown which is gradually increasing from zero. The coefficient of static and kinetic friction are $1/\sqrt{3}$ then



- (A) When F is less than the limiting friction, angle made by net force on the block by the surface is less than 30° with vertical.
(B) When the block is just about to move, the angle made by net force by the surface on the block becomes equal to 30° with vertical.
(C) When the block starts to accelerate, the angle made by net force by the surface on the block becomes constant and equal to 30° vertical.
(D) The angle made by net force with vertical on the block by the surface, depends on the mass of the block.

NL0101

18. A block placed on a rough horizontal surface is pushed with a force F acting horizontally on the block. The magnitude of F is increased and acceleration produced is plotted in the graph shown.



- (A) Mass of the block is 2 kg.
(B) Coefficient of friction between block and surface is 0.5.
(C) Limiting friction between block and surface is 10 N.
(D) When $F = 8$ N, friction between block and surface is 10 N.

NL0102

19. A block is placed over a plank. The coefficient of friction between the block and the plank is $\mu = 0.2$. Initially both are at rest, suddenly the plank starts moving with acceleration $a_0 = 4 \text{ m/s}^2$. The displacement of the block in 1s is ($g = 10 \text{ m/s}^2$)

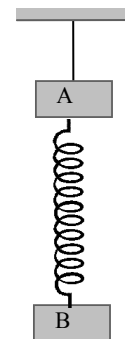
- (A) 1 m relative to ground
(B) 1 m relative to plank
(C) zero relative to plank
(D) 2 m relative to ground

NL0103

20. A block is released from rest from a point on a rough inclined plane of inclination 37° . The coefficient of friction is 0.5.
- (A) The time taken to slide down 9 m on the plane is 3 s.
 (B) The velocity of block after moving 4 m is 4 m/s.
 (C) The block travels equal distances in equal intervals of time.
 (D) The velocity of block increases linearly.

NL0104

21. In the given figure both the blocks have equal mass. When the thread is cut, which of the following statements give correct description immediately after the thread is cut?
- (A) Relative to the block A, acceleration of block B is $2g$ upwards.
 (B) Relative to the block B, acceleration of block A is $2g$ downwards.
 (C) Relative to the ground, accelerations of the blocks A and B are both g downwards.
 (D) Relative to the ground, accelerations of the blocks A and B are $2g$ downwards and zero respectively.

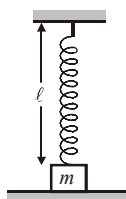


NL0105

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 22 to 24

A block of mass m is placed on a smooth horizontal floor is attached to one end of spring. The other end of the spring is attached to fixed support. When spring is vertical it is relaxed. Now the block is pulled towards right by a force F , which is being increased gradually. When the spring makes angle 53° with the vertical, block leaves the floor.



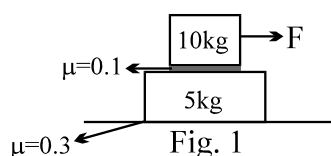
22. When block leaves the table, the normal force on it from table is
- (A) mg (B) zero (C) $\frac{4mg}{3}$ (D) $\frac{3mg}{4}$
23. Force constant of the spring is :-
- (A) $\frac{5mg}{2\ell}$ (B) $\frac{15mg}{8\ell}$ (C) $\frac{5mg}{3\ell}$ (D) $\frac{5mg}{4\ell}$
24. When the block leaves the table, the force F is :-
- (A) $\frac{3mg}{4}$ (B) $\frac{4mg}{3}$ (C) $\frac{3mg}{5}$ (D) $\frac{4mg}{5}$

NL0106

NL0106

NL0106

Paragraph for Question No. 25 to 29



25. When $F = 2\text{N}$, the frictional force between 5 kg block and ground is
(A) 2N (B) 0 (C) 8 N (D) 10 N
26. When $F = 2\text{N}$, the frictional force between 10 kg block and 5 kg block is
(A) 2N (B) 15 N (C) 10 N (D) None
27. The maximum F which will not cause motion of any of the blocks is
(A) 10 N (B) 15 N (C) data insufficient (D) None
28. The maximum acceleration of 5 kg block is :-
(A) 1 m/s^2 (B) 3 m/s^2 (C) 0 (D) None
29. The acceleration of 10 kg block when $F = 30\text{ N}$ is
(A) 2 m/s^2 (B) 3 m/s^2 (C) 1 m/s^2 (D) None

NL0107

NL0107

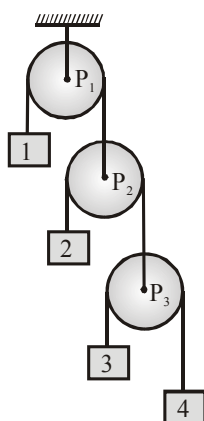
NL0107

NL0107

NL0107

MATRIX MATCH TYPE QUESTION

30. In the figure shown, acceleration of 1 is x (upwards). Acceleration of pulley P_3 , w.r.t. pulley P_2 is y (downwards) and acceleration of 4 w.r.t. to pulley P_3 is z (upwards). Then



Column I

- (A) Absolute acceleration of 2
(B) Absolute acceleration of 3
(C) Absolute acceleration of 4

Column II

- (P) $(y-x)$ downwards
(Q) $(z-x-y)$ upwards
(R) $(x+y+z)$ downwards
(S) None

NL0108

31. Velocity of three particles A, B and C varies with time t as, $\vec{v}_A = (2t\hat{i} + 6\hat{j})$ m/s; $\vec{v}_B = (3\hat{i} + 4\hat{j})$ m/s

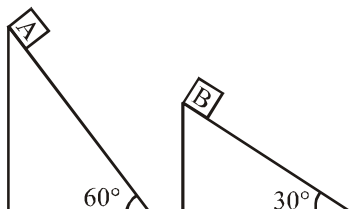
and $\vec{v}_C = (6\hat{i} - 4\hat{j})$ m/s. Regarding the pseudo force match the following table

Column I	Column II
(A) On A as observed by B	(P) Along positive x-direction
(B) On B as observed by C	(Q) Along negative x-direction
(C) On A as observed by C	(R) Along positive y-direction
(D) On C as observed by A	(S) Along negative y-direction
	(T) Zero

NL0109

EXERCISE (JM)

1. Two fixed frictionless inclined planes making an angle 30° and 60° with the vertical are shown in the figure. Two blocks A and B are placed on the two planes. What is the relative vertical acceleration of A with respect to B ? [AIEEE - 2010]



- (1) 4.9 ms^{-2} in vertical direction. (2) 4.9 ms^{-2} in horizontal direction
(3) 9.8 ms^{-2} in vertical direction (4) Zero

NL0110

2. The minimum force required to start pushing a body up a rough (frictional coefficient μ) inclined plane is F_1 while the minimum force needed to prevent it from sliding down is F_2 . If the inclined plane makes an angle θ from the horizontal such that $\tan\theta = 2\mu$ then the ratio $\frac{F_1}{F_2}$ is :- [AIEEE - 2011]

- (1) 4 (2) 1 (3) 2 (4) 3

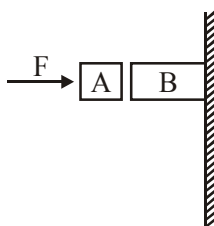
NL0111

3. A block of mass m is placed on a surface with a vertical cross section given by $y = \frac{x^3}{6}$. If the coefficient of friction is 0.5, the maximum height above the ground at which the block can be placed without slipping is :- [JEE-Main-2014]

- (1) $\frac{1}{3}m$ (2) $\frac{1}{2}m$ (3) $\frac{1}{6}m$ (4) $\frac{2}{3}m$

NL0112

4. Given in the figure are two blocks A and B of weight 20 N and 100 N, respectively. These are being pressed against a wall by a force F as shown. If the coefficient of friction between the blocks is 0.1 and between block B and the wall is 0.15, the frictional force applied by the wall on block B is :- [JEE-Main-2015]



- (1) 120 N (2) 150 N (3) 100 N (4) 80 N

NL0113

5. A rocket is fired vertically from the earth with an acceleration of $2g$, where g is the gravitational acceleration. On an inclined plane inside the rocket, making an angle θ with the horizontal, a point object of mass m is kept. The minimum coefficient of friction $\mu_{\min}$ between the mass and the inclined surface such that the mass does not move is:

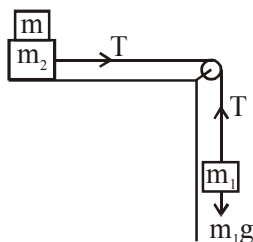
[JEE-Main Online-2016]

- (1) $2 \tan \theta$ (2) $3 \tan \theta$ (3) $\tan \theta$ (4) $\tan 2\theta$

NL0114

6. Two masses $m_1 = 5\text{kg}$ and $m_2 = 10\text{kg}$, connected by an inextensible string over a frictionless pulley, are moving as shown in the figure. The coefficient of friction of horizontal surface is 0.15 . The minimum weight m that should be put on top of m_2 to stop the motion is :-

[JEE-Main-2018]



- (1) 27.3 kg (2) 43.3 kg (3) 10.3 kg (4) 18.3 kg

NL0115

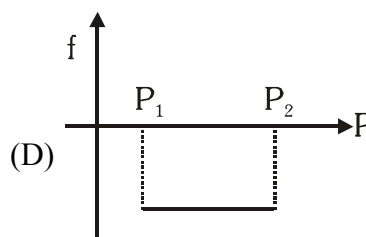
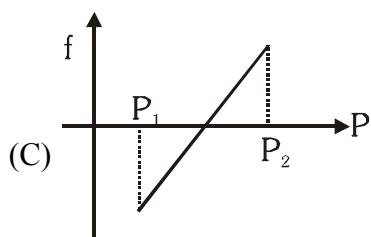
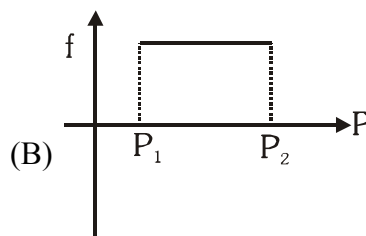
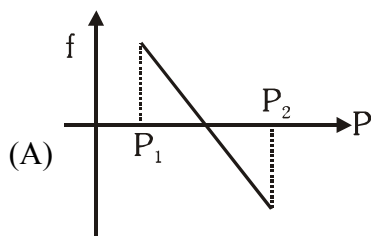
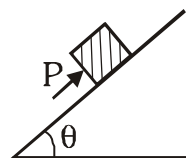
EXERCISE (JA)

1. A piece of wire is bent in the shape of a parabola $y = kx^2$ (y-axis vertical) with a bead of mass m on it. The bead can slide on the wire without friction. It stays at the lowest point of the parabola when the wire is at rest. The wire is now accelerated parallel to the x-axis with a constant acceleration a . The distance of the new equilibrium position of the bead, where the bead can stay at rest with respect to the wire, from the y-axis is [IIT-JEE-2009]

- (A) $\frac{a}{gk}$ (B) $\frac{a}{2gk}$ (C) $\frac{2a}{gk}$ (D) $\frac{a}{4gk}$

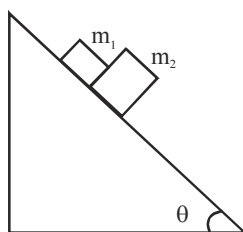
NL0116

2. A block of mass m is on an inclined plane of angle θ . The coefficient of friction between the block and the plane is μ and $\tan\theta > \mu$. The block is held stationary by applying a force P parallel to the plane. The direction of force pointing up the plane is taken to be positive. As P is varied from $P_1 = mg(\sin\theta - \mu\cos\theta)$ to $P_2 = mg(\sin\theta + \mu\cos\theta)$, the frictional force f versus P graph will look like [IIT-JEE-2010]



NL0117

3. A block is moving on an inclined plane making an angle 45° with the horizontal and the coefficient of friction is μ . The force required to just push it up the inclined plane is 3 times the force required to just prevent it from sliding down. If we define $N = 10\mu$, then N is **[IIT-JEE-2011]**
- NL0118**
4. A block of mass $m_1 = 1$ kg another mass $m_2 = 2$ kg, are placed together (see figure) on an inclined plane with angle of inclination θ . Various values of θ are given in List I. The coefficient of friction between the block m_1 and the plane is always zero. The coefficient of static and dynamic friction between the block m_2 and the plane are equal to $\mu = 0.3$. In List II expressions for the friction on block m_2 are given. Match the correct expression of the friction in List II with the angles given in List I, and choose the correct option. The acceleration due to gravity is denoted by g .
- [useful information : $\tan(5.5^\circ) \approx 0.1$; $\tan(11.5^\circ) \approx 0.2$; $\tan(16.5^\circ) \approx 0.3$]** **[IIT-JEE-2014]**



List-I

- (P) $\theta = 5^\circ$
 (Q) $\theta = 10^\circ$
 (R) $\theta = 15^\circ$
 (S) $\theta = 20^\circ$

Code :

- (A) P-1, Q-1, R-1, S-3
(C) P-2, Q-2, R-2, S-4

List-II

- (1) $m_2 g \sin \theta$
- (2) $(m_1 + m_2) g \sin \theta$
- (3) $\mu m_2 g \cos \theta$
- (4) $\mu (m_1 + m_2) g \cos \theta$

- (B) P-2, Q-2, R-2, S-3
(D) P-2, Q-2, R-3, S-3

NL0119

ANSWER KEY

EXERCISE (S-1)

1. Ans. (i) $\frac{m_1}{m_2} = \frac{1}{3}$ (ii) $a = 3/4 \text{ m/s}^2$ 2. Ans. $\frac{4}{3}W$ 3. Ans. $\frac{g}{10} \text{ m/s}^2$ 4. Ans. 200 N, 10 m/s²

5. Ans. 2 sec 6. Ans. 0.5g, g 7. Ans. 0.5 s 8. Ans. 80 N, 48 N, 264 N 9. Ans. 24 N

10. Ans. (i) $a_A = \frac{3g}{2} \downarrow = a_B$; $a_C = 0$; $T = mg/2$; (ii) $a_A = 2g \uparrow$, $a_B = 2g \downarrow$, $a_C = 0$, $T = 0$;

(iii) $a_A = a_B = g/2 \uparrow$, $a_C = g \downarrow$, $T = \frac{3mg}{2}$;

11. Ans. $a = g \cot \theta$ 12. Ans. $v_0/\cos \theta$ 13. Ans. 5N, 16/31 kg 14. Ans. $3a_B/4$

15. Ans. $\frac{a_0}{2}$ 16. Ans. $(100\hat{i} - 200\hat{j}) \text{ N}$ 17. Ans. 20 N vertically downward

18. Ans. (ma) 19. Ans. 1/2 s 20. Ans. 4/3 s 21. Ans. $\mu_{\min} = \frac{m \sin \theta \cos \theta}{m \cos^2 \theta + M}$

22. Ans. 0.5 23. Ans. 1 kg 24. Ans. $10\hat{i}$ 25. Ans. 3/4 26. Ans. 30 N

EXERCISE (S-2)

1. Ans. 1440 N 2. Ans. g m/s² 3. Ans. 556.8 N, 1.47 s

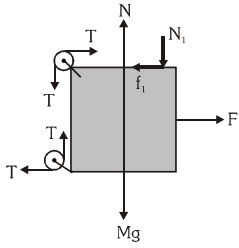
4. Ans. (i) $a_A = g \downarrow$, $a_B = \frac{2g}{3} \uparrow$, $a_C = \frac{2g}{3} \downarrow$, $a_D = \frac{2g}{3} \downarrow$, $a_E = 0$

(ii) $a_A = 0$, $a_B = \frac{g}{3} \downarrow$, $a_C = \frac{g}{3} \uparrow$, $a_D = \frac{g}{3} \uparrow$, $a_E = g \downarrow$

(iii) $a_A = 0$, $a_B = g \downarrow$, $a_C = g \uparrow$, $a_D = 2g \downarrow$, $a_E = 0$

(iv) $a_A = 0$, $a_B = 3g \downarrow$, $a_C = \frac{3g}{2} \downarrow$, $a_D = \frac{3g}{2} \downarrow$, $a_E = 0$

5. Ans. $\frac{25}{24}$ 6. Ans. 5 m/s² downwards, 0 m/s², 10 N 7. Ans. 40 N

8. Ans. (i)  (ii) $a = 3/5 \text{ m/s}^2$, $T = 18 \text{ N}$, $F = 60 \text{ N}$ 9. Ans. $-5 \hat{i} \text{ m/s}^2$

EXERCISE (O-1)**SINGLE CORRECT TYPE QUESTIONS**

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 1. Ans. (B) | 2. Ans. (D) | 3. Ans. (D) | 4. Ans. (C) | 5. Ans. (D) | 6. Ans. (C) |
| 7. Ans. (C) | 8. Ans. (C) | 9. Ans. (D) | 10. Ans. (C) | 11. Ans. (B) | 12. Ans. (A) |
| 13. Ans. (C) | 14. Ans. (A) | 15. Ans. (C) | 16. Ans. (B) | 17. Ans. (A) | 18. Ans. (B) |
| 19. Ans. (D) | 20. Ans. (B) | 21. Ans. (D) | 22. Ans. (D) | 23. Ans. (B) | 24. Ans. (B) |
| 25. Ans. (C) | 26. Ans. (A) | 27. Ans. (D) | 28. Ans. (A) | 29. Ans. (B) | 30. Ans. (A) |
| 31. Ans. (A) | 32. Ans. (B) | 33. Ans. (D) | 34. Ans. (C) | 35. Ans. (B) | 36. Ans. (A) |
| 37. Ans. (B) | | | | | |

MULTIPLE CORRECT TYPE QUESTIONS

- | | | | | | |
|----------------|------------------|----------------|----------------|----------------|----------------|
| 38. Ans. (A,D) | 39. Ans. (B,C,D) | 40. Ans. (A,C) | 41. Ans. (A,B) | 42. Ans. (A,B) | 43. Ans. (A,C) |
|----------------|------------------|----------------|----------------|----------------|----------------|

COMPREHENSION TYPE QUESTIONS

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 44. Ans. (D) | 45. Ans. (D) | 46. Ans. (A) | 47. Ans. (D) | 48. Ans. (B) | 49. Ans. (C) |
| 50. Ans. (B) | 51. Ans. (D) | 52. Ans. (C) | 53. Ans. (C) | | |

MATRIX MATCH TYPE QUESTION

- | | |
|---|---|
| 54. Ans. (A) - (R); (B) - (T); (C) - (Q); (D) - (T) | 55. Ans. (A) - (R); (B) - (T); (C) - (R); (D) - (P) |
| 56. Ans. (A)-R; (B)-Q; (C)-S; (D)-P | |

EXERCISE (O-2)**SINGLE CORRECT TYPE QUESTIONS**

- | | | | | | |
|-------------|-------------|-------------|--------------|--------------|-------------|
| 1. Ans. (B) | 2. Ans. (D) | 3. Ans. (C) | 4. Ans. (C) | 5. Ans. (C) | 6. Ans. (B) |
| 7. Ans. (C) | 8. Ans. (A) | 9. Ans. (A) | 10. Ans. (C) | 11. Ans. (C) | |

MULTIPLE CORRECT TYPE QUESTIONS

- | | | | | |
|--------------------|----------------|------------------|------------------|------------------|
| 12. Ans. (C,D) | 13. Ans. (A,C) | 14. Ans. (B,D) | 15. Ans. (A,C,D) | 16. Ans. (B,C,D) |
| 17. Ans. (A,B,C,D) | | 18. Ans. (A,B,C) | 19. Ans. (A,B) | 20. Ans. (A,B,D) |
| 21. Ans. (A,B,D) | | | | |

COMPREHENSION TYPE QUESTIONS

- | | | | | | |
|--------------|--------------|--------------|--------------|--------------|--------------|
| 22. Ans. (B) | 23. Ans. (A) | 24. Ans. (B) | 25. Ans. (A) | 26. Ans. (A) | 27. Ans. (A) |
| 28. Ans. (C) | 29. Ans. (A) | | | | |

MATRIX MATCH TYPE QUESTION

- | | |
|--|---|
| 30. Ans. (A) - (S) ; (B) - (R) ; (C) - (Q) | 31. Ans. (A) - (T); (B) - (R); (C) - (R); (D) - (Q) |
|--|---|

EXERCISE (JM)

- | | | | | | |
|-------------|-------------|-------------|-------------|-------------|-------------|
| 1. Ans. (1) | 2. Ans. (4) | 3. Ans. (3) | 4. Ans. (1) | 5. Ans. (3) | 6. Ans. (1) |
|-------------|-------------|-------------|-------------|-------------|-------------|

EXERCISE (JA)

- | | | | |
|-------------|-------------|-----------|-------------|
| 1. Ans. (B) | 2. Ans. (A) | 3. Ans. 5 | 4. Ans. (D) |
|-------------|-------------|-----------|-------------|



Chapter Contents

02

Kinematics-1D & Calculus

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Important Notes

[illegible]

KINEMATICS-1D & CALCULUS

KEY CONCEPT

Kinematics

Study of motion of objects without taking into account the factor which cause the motion (i.e. nature of force).

Motion

If a body changes its position with time, it is said to be moving else it is at rest. Motion is always relative to the observer.

Motion is a combined property of the object under study and the observer. There is no meaning of rest or motion without the viewer. In other words absolute motion or rest is meaningless.

- To locate the position of a particle we need a reference frame. A commonly used reference frame is Cartesian coordinate system or simply coordinate system.

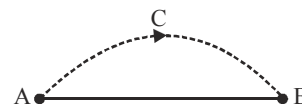
The coordinates (x, y, z) of a particle specify the position of the particle with respect to origin of that frame. If all the three coordinates of the particle remain unchanged as time passes it means the particle is at rest w.r.t. this frame.

- The reference frame is chosen according to the problems.
- If the frame is not mentioned, then ground is taken as the reference frame.

DISTANCE AND DISPLACEMENT

Total length of path covered by the particle, in definite time interval is called distance.

Let a body moves from A to B via C. The length of path ACB is called the distance travelled by the body.



But overall, body is displaced from A to B. A vector from A to B, i.e. $\overline{AB}$ is its displacement vector or displacement that is the minimum distance and directed from initial position to final position.

- Distance is a scalar while displacement is a vector.
- Distance depends on path while displacement is independent of path but depends only on final and initial position.
- For a moving body, distance can't have zero or negative values but displacement may be +ive, -ive or zero.
- For a moving/stationary object distance can't be decreasing.
- If motion is in straight line without change in direction then
distance = |displacement| i.e. magnitude of displacement.
- Magnitude of displacement may be equal or less than distance but never greater than distance.
distance $\geq$ |displacement|

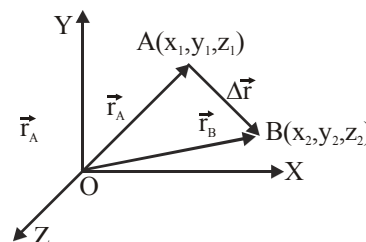
Displacement in terms of position vector

Let a body is displaced from A (x_1, y_1, z_1) to B (x_2, y_2, z_2) then its displacement is given by vector $\overline{AB}$.

From ΔOAB $\vec{r}_A + \Delta\vec{r} = \vec{r}_B \Rightarrow \Delta\vec{r} = \vec{r}_B - \vec{r}_A$

$$\vec{r}_B = x_2\hat{i} + y_2\hat{j} + z_2\hat{k} \text{ and } \vec{r}_A = x_1\hat{i} + y_1\hat{j} + z_1\hat{k}$$

$$\Delta\vec{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k} \Rightarrow \Delta\vec{r} = \Delta x\hat{i} + \Delta y\hat{j} + \Delta z\hat{k}$$

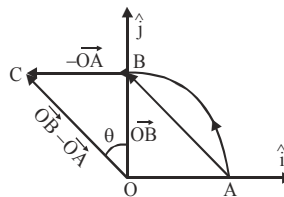


Ex. A particle goes along a quadrant from A to B of a circle radius 10m as shown in fig. Find the direction and magnitude of displacement and distance along path AB.

Sol. Displacement $\vec{AB} = \vec{OB} - \vec{OA} = 10\hat{j} - 10\hat{i}$

$$|\vec{AB}| = \sqrt{10^2 + 10^2} = 10\sqrt{2} \text{ m}$$

$$\text{From } \triangle OBC \tan \theta = \frac{OA}{OB} = \frac{10}{10} = 1 \Rightarrow \theta = 45^\circ$$



Angle between displacement vector $\vec{OC}$ and x-axis $= 90^\circ + 45^\circ = 135^\circ$

$$\text{Distance of path AB} = \frac{1}{4} (\text{circumference}) = \frac{1}{4} (2\pi R) \text{ m} = (5\pi) \text{ m}$$

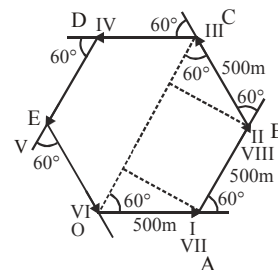
Ex. On an open ground a motorist follows a track that turns to his left by an angle of 60° after every 500 m. Starting from a given turn, specify the displacement of the motorist at the third, sixth and eighth turn. Compare the magnitude of displacement with the total path length covered by the motorist in each case.

Sol. At III turn

$$\text{Displacement} = \vec{OA} + \vec{AB} + \vec{BC} = \vec{OC}$$

$$= 500 \cos 60^\circ + 500 + 500 \cos 60^\circ$$

$$= 500 \times \frac{1}{2} + 500 + 500 \times \frac{1}{2} = 1000 \text{ m from O to C}$$



$$\text{Distance} = 500 + 500 + 500 = 1500 \text{ m. So } \frac{\text{Displacement}}{\text{Distance}} = \frac{1000}{1500} = \frac{2}{3}$$

At VI turn

$\therefore$ initial and final positions are same so displacement = 0 and distance = $500 \times 6 = 3000 \text{ m}$

$$\therefore \frac{\text{Displacement}}{\text{Distance}} = \frac{0}{3000} = 0$$

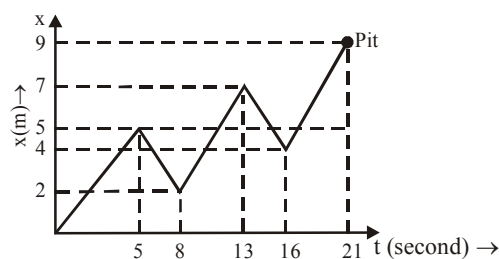
At VIII turn

$$\text{Displacement} = 2(500) \cos \left(\frac{60^\circ}{2} \right) = 1000 \times \cos 30^\circ = 1000 \times \frac{\sqrt{3}}{2} = 500\sqrt{3} \text{ m}$$

$$\text{Distance} = 500 \times 8 = 4000 \text{ m}$$

$$\therefore \frac{\text{Displacement}}{\text{Distance}} = \frac{500\sqrt{3}}{4000} = \frac{\sqrt{3}}{8}$$

Ex. A drunkard walking in a narrow lane takes 5 steps forward and 3 steps backward, followed again by 5 steps forward and 3 steps backward, and so on. Each step is 1m long and requires 1s. Plot the x-t graph of his motion. Determine graphically or otherwise how long the drunkard takes to fall in a pit 9m away from the start.



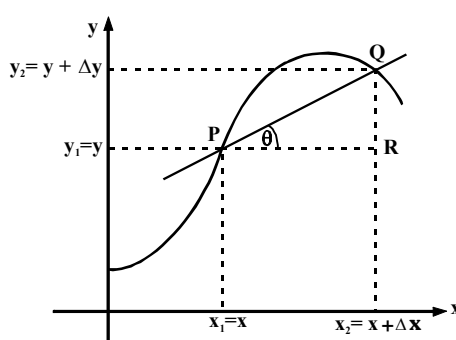
Sol. from $x-t$ graph time taken = 21 s
or $(5m - 3m) + (5m - 3m) + 5m = 9m$
 $\Rightarrow$ total steps = 21 $\Rightarrow$ time = 21 s

DERIVATIVE OF A FUNCTION

Average Rate of Change

Let a function $y = f(x)$ be plotted by an arbitrary graph as shown in the figure. Average rate of change in y with respect to x in an interval $[x_1, x_2]$ is defined as

$$\text{Average rate of change} = \frac{\text{Change in } y}{\text{Change in } x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$



= slope of chord

If both the axes have equal scale

Average rate of change = slope of chord PQ = $\tan \theta$

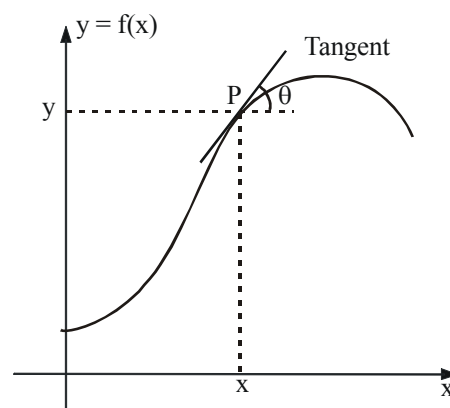
Instantaneous Rate of Change : First derivative

It is defined as the rate of change in y with x at a particular value of x . It is measured graphically by the slope of the tangent drawn to the y - x graph at the point (x, y) and algebraically by the first derivative of the function $y = f(x)$.

Instantaneous rate of change = $\frac{dy}{dx}$ = Slope of the tangent

If both the axes have equal scale then $\frac{dy}{dx} = \tan \theta$

$$\begin{aligned} \text{Instantaneous rate of change} &= \frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x} \\ &= \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} \end{aligned}$$



Derivatives of Commonly Used Functions.

$$\bullet y = \text{constant} \Rightarrow \frac{dy}{dx} = 0$$

$$\bullet y = \cos x \Rightarrow \frac{dy}{dx} = -\sin x$$

$$\bullet y = x^n \Rightarrow \frac{dy}{dx} = nx^{n-1}$$

$$\bullet y = \tan x \Rightarrow \frac{dy}{dx} = \sec^2 x$$

$$\bullet y = e^x \Rightarrow \frac{dy}{dx} = e^x$$

$$\bullet y = \cot x \Rightarrow \frac{dy}{dx} = -\operatorname{cosec}^2 x$$

$$\bullet y = \ln x \Rightarrow \frac{dy}{dx} = \frac{1}{x}$$

$$\bullet y = \operatorname{cosec} x \Rightarrow \frac{dy}{dx} = -\operatorname{cosec} x \cot x$$

$$\bullet y = \sin x \Rightarrow \frac{dy}{dx} = \cos x$$

$$\bullet y = \sec x \Rightarrow \frac{dy}{dx} = \sec x \tan x$$

Method of Differentiation.

If $y = f(x)$, let us denote $\frac{dy}{dx} = f'(x)$

$$\bullet \text{ Sum or Subtraction of two functions } y = f(x) \pm g(x) \Rightarrow \frac{dy}{dx} = f'(x) \pm g'(x)$$

$\bullet$ Product of two functions

$$y = f(x) \cdot g(x) \Rightarrow \frac{dy}{dx} = g(x)f'(x) + f(x)g'(x)$$

$\bullet$ Division of two functions.

$$y = \frac{f(x)}{g(x)} \Rightarrow \frac{dy}{dx} = \frac{g(x)f'(x) - f(x)g'(x)}{\{g(x)\}^2}$$

$\bullet$ Chain Rule

$$y = f\{g(x)\} \Rightarrow \frac{dy}{dx} = g'(x)f'\{g(x)\}$$

Ex. Find $\frac{dy}{dx}$, when (i) $y = \sqrt{x}$ (ii) $y = x^5 + x^4 + 7$ (iii) $y = x^2 + 4x^{-1/2} - 3x^{-2}$

Sol. (i) $\frac{dy}{dx} = \frac{d}{dx}(\sqrt{x}) = \frac{d}{dx}(x^{1/2}) = \frac{1}{2}x^{1/2-1} = \frac{1}{2}x^{-1/2} = \frac{1}{2\sqrt{x}}$

(ii) $\frac{dy}{dx} = \frac{d}{dx}(x^5 + x^4 + 7) = \frac{d}{dx}(x^5) + \frac{d}{dx}(x^4) + \frac{d}{dx}(7) = 5x^4 + 4x^3 + 0 = 5x^4 + 4x^3$

(iii) $\frac{dy}{dx} = \frac{d}{dx}(x^2 + 4x^{-1/2} - 3x^{-2}) = \frac{d}{dx}(x^2) + \frac{d}{dx}(4x^{-1/2}) - \frac{d}{dx}(3x^{-2})$
 $= \frac{d}{dx}(x^2) + 4\frac{d}{dx}(x^{-1/2}) - 3\frac{d}{dx}(x^{-2}) = 2x + 4\left(-\frac{1}{2}\right)x^{-3/2} - 3(-2)x^{-3} = 2x - 2x^{-3/2} + 6x^{-3}$

Second Derivative and it's meaning

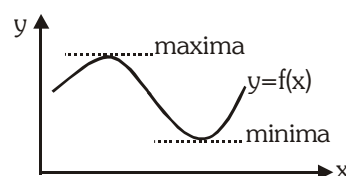
Second derivative of a function $y = f(x)$ is defined as $\frac{d}{dx} \left[\frac{dy}{dx} \right]$. It is obtained by differentiating the function with respect to x two times successively. Geometrically it expresses rate of change in slope of graph of the function.

Maxima & Minima

Maxima & minima of a function $y = f(x)$

for maximum value $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} = \text{negative}$

for minimum value $\frac{dy}{dx} = 0$ and $\frac{d^2y}{dx^2} = \text{positive}$



Ex. Find minimum value of $y = 25x^2 - 10x + 5$.

Sol. For maximum/minimum value $\frac{dy}{dx} = 0 \Rightarrow 50x - 10 = 0 \Rightarrow x = \frac{1}{5}$

Now at $x = \frac{1}{5}$, $\frac{d^2y}{dx^2} = 50$ which is positive. So $y_{\min} = 25\left(\frac{1}{5}\right)^2 - 10\left(\frac{1}{5}\right) + 5 = 1 - 2 + 5 = 4$

INTEGRATION

- This operation enables us to find sum of infinite number of infinitely small quantities.

$$\lim_{\Delta x \rightarrow 0} \sum_{i=1}^{\infty} \Delta x_i = \int dx = x$$

- It is reverse operation of differentiation. If derivative, which is rate of change at a point, is given as

a function $f(x) = \frac{dF(x)}{dx}$, operation of integration enables us to find original function $F(x)$.

$$\int f(x) dx = F(x) + C$$

Here function $f(x)$ is known as integrand, function $F(x)$ as integral and C as constant of integration. Value of C is obtained by substituting initial, final or any other condition (known as boundary conditions) in the above equation.

This interpretation enables us to find integral of those functions whose derivative is known.

Integrand	Integral
$f(x) = \frac{dF(x)}{dx}$	$\int f(x)dx = F(x) + C$
$k = \text{Constant}$	$kx + C$
x^n	$\frac{x^{n+1}}{n+1} + C$ If $n \neq -1$
x^{-1}	$\ln x + C$
e^x	$e^x + C$
$\sin x$	$-\cos x + C$
$\cos x$	$\sin x + C$
$f(ax+b)$	$\frac{F(ax+b)}{a} + C$

Ex. Evaluate the following:

(i) $\int x^{-7} dx$ (ii) $\int x^{p/q} dx$

Sol. (i) $\int x^{-7} dx = \frac{x^{-7+1}}{-7+1} + c = -\frac{1}{6} x^{-6} + c$

(ii) $\int x^{\frac{p}{q}} dx = \frac{x^{\frac{p}{q}+1}}{\frac{p}{q}+1} + c = \frac{q}{p+q} x^{(p+q)/q} + c$

Ex. Evaluate $\int \left(x^2 - \cos x + \frac{1}{x} \right) dx$

Sol. $I = \int x^2 dx - \int \cos x dx + \int \frac{1}{x} dx = \frac{x^{2+1}}{2+1} - \sin x + \log_e x + c = \frac{x^3}{3} - \sin x + \log_e x + c$

SPEED :

Speed is the rate of change of distance with respect to time.

Uniform speed :

An object is said to be moving with a uniform speed, if it covers equal distances in equal intervals of time, irrespective of duration of interval. The uniform speed is shown by straight line in distance–time graph.

For example, suppose a train travels 1000 m in 60 s. The train is said to be moving with uniform speed, if it travels 500 m. in 30s, 250 m in 15s, 125 m in 7.5s and so on.

Non Uniform speed :

An object is said to be moving with a variable speed if it covers equal distances in unequal intervals of time or unequal distances in equal intervals of time, irrespective of duration of interval.

For example, suppose a train travels first 1000 m in 60s next 1000 m in 120s and next 1000m in 50s, then the train is moving with variable speed.

Average speed :

Speed is distance travelled per unit time. Average speed of a trip $v_{av} = \frac{\text{Total travelled distance}}{\text{Total time taken}}$

If a particle travels a distance s in time t_1 to t_2 , the average speed is $v_{av} = \frac{\Delta s}{\Delta t} = \frac{s}{t_2 - t_1}$

Instantaneous speed

The speed at a particular instant is defined as instantaneous speed (or speed) while average speed is defined for a time interval.

If Δt approaches zero, average speed becomes instantaneous speed. $v = \lim_{\Delta t \rightarrow 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$

i.e. instantaneous speed is the time derivative of distance.

If a particle travels distances s_1, s_2, s_3 etc. with speeds v_1, v_2, v_3 etc. respectively, then total travelled distance

$$s = s_1 + s_2 + s_3 + \dots + s_n$$

$$\text{Total time taken } t = \frac{s_1}{v_1} + \frac{s_2}{v_2} + \frac{s_3}{v_3} + \dots + \frac{s_n}{v_n}$$

No sign is needed for distance or speed.
They are always positive quantities.

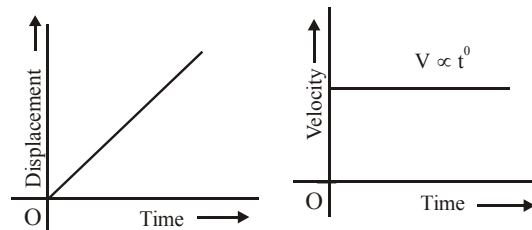
$$\text{Average speed of the trip} = \frac{s_1 + s_2 + s_3 + \dots + s_n}{\left(\frac{s_1}{v_1} + \frac{s_2}{v_2} + \dots + \frac{s_n}{v_n}\right)}$$

VELOCITY :

Velocity is the rate of change of displacement with respect to time.

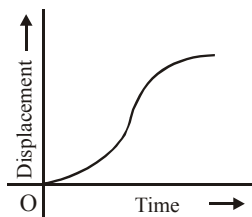
Uniform Velocity :

A body is said to move with uniform velocity, if it covers equal displacements in equal intervals of time, irrespective of duration of interval. When a body is moving with uniform velocity, then the magnitude and direction of the velocity of the body remains same at all points of its path.



Non-uniform Velocity :

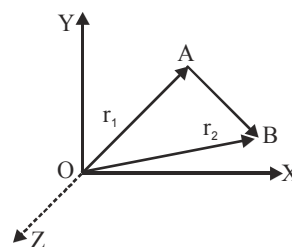
The particle is said to have non-uniform motion if it covers unequal displacements in equal intervals of time, irrespective of duration of interval. In this type of motion velocity does not remain constant.

**Average Velocity**

The average velocity of a particle in a time interval t_1 to t_2 is defined as its displacement divided by the time interval. Let a particle is at a point A at time t_1 and B at time t_2 . Position vectors of A and B are $\vec{r}_1$ and $\vec{r}_2$. The displacement in this time interval is the vector $\vec{AB} = (\vec{r}_2 - \vec{r}_1)$. The average velocity

in this time interval is, $\vec{v}_{av} = \frac{\text{displacement vector}}{\text{time interval}}$

$$\vec{v}_{av} = \frac{\vec{AB}}{t_2 - t_1} = \frac{\vec{r}_2 - \vec{r}_1}{t_2 - t_1} = \frac{\Delta \vec{r}}{\Delta t}$$



here $\vec{AB} = \vec{OB} - \vec{OA} = \vec{r}_2 - \vec{r}_1 = \text{change in position vector.}$

Instantaneous velocity

The velocity of the object at a given instant of time or at a given position

during motion is called instantaneous velocity $\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t} = \frac{d\vec{r}}{dt}$

From fig., the average velocity between points A and B is

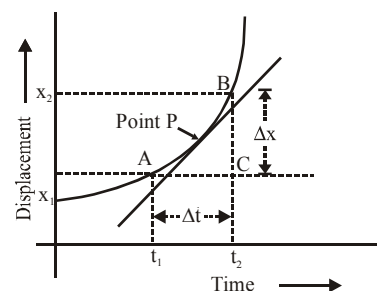
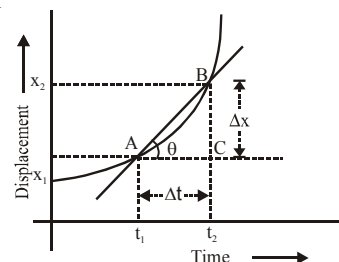
$$v_{av} = \frac{x_2 - x_1}{t_2 - t_1} = \frac{\Delta x}{\Delta t} = \text{slope of secant AB} = \tan \theta$$

Average velocity is equal to slope of straight line joining two points on displacement time graph. If $\Delta t \rightarrow 0$, then average velocity becomes instantaneous velocity.

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} = \text{slope of tangent at P} = \tan \alpha$$

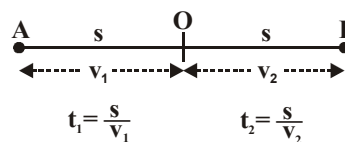
- Magnitude of instantaneous velocity is the instantaneous speed.

Note : When a particle moves with constant velocity, its magnitude of average velocity, its magnitude of instantaneous velocity and its speed all are equal.



Ex. If a particle travels the first half distance with speed v_1 and second half distance with speed v_2 . Find its average speed during journey.

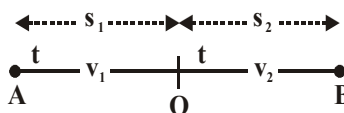
Sol.
$$v_{\text{avg.}} = \frac{s+s}{t_1+t_2} = \frac{2s}{\frac{s}{v_1} + \frac{s}{v_2}} = \frac{2v_1v_2}{v_1+v_2}$$



Note :- Here $v_{\text{avg.}}$ is the harmonic mean of two speeds.

Ex. If a particle travels with speed v_1 during first half time interval and with v_2 speed during second half time interval. Find its average speed during its journey.

Sol. Total distance $= s_1 + s_2 = v_1t + v_2t = (v_1 + v_2)t$



Total time $= t + t = 2t$
$$v_{\text{avg.}} = \frac{s_1 + s_2}{t + t} = \frac{(v_1 + v_2)t}{2t} = \frac{v_1 + v_2}{2}$$

Note :- here $v_{\text{avg.}}$ is arithmetic mean of two speeds.

Ex. A car travels a distance A to B at a speed of 40 km/h and returns to A at a speed of 30 km/h.

- What is the average speed for the whole journey?
- What is the average velocity?

Sol. (i) Let $AB = s$, time taken to go from A to B, $t_1 = \frac{s}{40}$ h

and time taken to go from B to A, $t_2 = \frac{s}{30}$ h

$\therefore$ total time taken $= t_1 + t_2 = \frac{s}{40} + \frac{s}{30} = \frac{(3+4)s}{120} = \frac{7s}{120}$ h

Total distance travelled $= s + s = 2s$

$\therefore$ Average speed $= \frac{\text{total distance travelled}}{\text{total time taken}} = \frac{2s}{\frac{7s}{120}} = \frac{120 \times 2}{7} = 34.3$ km/h.

(ii) Total displacement = zero, since the car returns to the original position.

Therefore, average velocity $= \frac{\text{total displacement}}{\text{time taken}} = \frac{0}{2t} = 0$

Ex. A man walks on a straight road from his home to a market 2.5 km away with a speed of 5 km/h. On reaching the market he instantly turns and walks back with a speed of 7.5 km/h. What is the (a) magnitude of average velocity and (b) average speed of the man, over the interval of time (i) 0 to 30 min. (ii) 0 to 50 min (iii) 0 to 40 min.

Sol. Time taken by man to go from his home to market, $t_1 = \frac{\text{distance}}{\text{speed}} = \frac{2.5}{5} = \frac{1}{2}$ h

Time taken by man to go from market to his home, $t_2 = \frac{2.5}{7.5} = \frac{1}{3}$ h

$$\therefore \text{Total time taken} = t_1 + t_2 = \frac{1}{2} + \frac{1}{3} = \frac{5}{6} \text{ h} = 50 \text{ min.}$$

(i) 0 to 30 min

$$\text{Average velocity} = \frac{\text{displacement}}{\text{time interval}} = \frac{2.5}{\frac{30}{60}} = 5 \text{ km/h towards market}$$

$$\text{Average speed} = \frac{\text{distance}}{\text{time interval}} = \frac{2.5}{\frac{30}{60}} = 5 \text{ km/h}$$

(ii) 0 to 50 min

Total displacement = zero so average velocity = 0

$$\text{So, average speed} = \frac{5}{50/60} = 6 \text{ km/h}$$

Total distance travelled = $2.5 + 2.5 = 5 \text{ km.}$

(iii) 0 to 40 min

Distance moved in 30 min (from home to market) = 2.5 km.

Distance moved in 10 min (from market to home) with speed 7.5 km/h = $7.5 \times \frac{10}{60} = 1.25 \text{ km}$

So, displacement = $2.5 - 1.25 = 1.25 \text{ km (towards market)}$

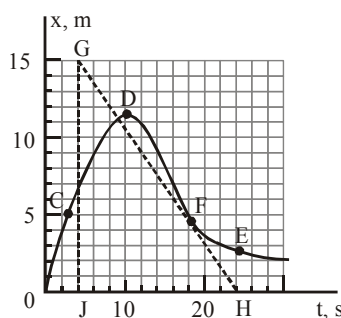
Distance travelled = $2.5 + 1.25 = 3.75 \text{ km}$

$$\text{Average velocity} = \frac{1.25}{\frac{40}{60}} = 1.875 \text{ km/h. (towards market)}$$

$$\text{Average speed} = \frac{3.75}{\frac{40}{60}} = 5.625 \text{ km/h.}$$

Note : Moving body with uniform speed may have variable velocity. e.g. in uniform circular motion speed is constant but velocity is non-uniform.

Ex. Refer to figure for the motion of an object along the x-axis.



What is the instantaneous velocity of the object at (a) F (b) D

- Sol.** (a) The tangent at F is the dashed line GH. Taking triangle GHJ,
 $\Delta t = 24 - 4 = 20 \text{ s}$ $\Delta x = 0 - 15 = -15 \text{ m}$

Hence slope at F is $v_F = \frac{\Delta x}{\Delta t} = \frac{-15 \text{ m}}{20 \text{ s}} = -0.75 \text{ m/s}$

The negative sign tells us that the object is moving in the $-x$ direction.

- (b) At point D slope of curve is zero so $v = \frac{dx}{dt} = 0$.

ACCELERATION :

Acceleration

The acceleration is rate of change of velocity or change in velocity per unit time interval.

Velocity is a vector quantity hence a change in its magnitude or in direction or in both, will change the velocity.

Uniform acceleration :

An object is said to be moving with a uniform acceleration if its velocity changes by equal amounts in equal intervals of time, irrespective of duration of intervals.

Variable acceleration :

An object is said to be moving with a variable acceleration if its velocity changes by unequal amounts in equal intervals of time, irrespective of duration of intervals..

Average Acceleration :

When an object is moving with a variable acceleration, then the average acceleration of the object for the given motion is defined as the ratio of the total change in velocity of the object during motion to the total time taken

$$\text{Average Acceleration} = \frac{\text{Total change in velocity}}{\text{total time taken}}$$

Suppose the velocity of a particle is $\vec{v}_1$ at time t_1 and $\vec{v}_2$ at time t_2 . Then $\vec{a}_{av} = \frac{\vec{v}_2 - \vec{v}_1}{t_2 - t_1} = \frac{\Delta \vec{v}}{\Delta t}$

Instantaneous Acceleration :

The acceleration of the object at a given instant of time or at a given point of motion, is called its instantaneous acceleration. Suppose the velocity of a particle at time $t_1 = t$ is $\vec{v}_1 = \vec{v}$ and becomes

$$\vec{v}_2 = \vec{v} + \Delta \vec{v} \text{ at time } t_2 = t + \Delta t, \text{ Then, } \vec{a}_{av} = \frac{\Delta \vec{v}}{\Delta t}$$

If Δt approaches to zero then the rate of change of velocity will be instantaneous acceleration.

$$\text{Instantaneous acceleration } \vec{a} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{v}}{\Delta t} = \frac{d\vec{v}}{dt} = \frac{d}{dt} \left(\frac{d\vec{r}}{dt} \right) = \frac{d^2 \vec{r}}{dt^2}$$

- Ex.** An athlete takes 2 second to reach the maximum speed of 18 km/h from rest. What is the magnitude of his average acceleration.

- Sol.** Here, Initial velocity $u = 0$, $v = (v_{\max}) = 18 \text{ km/h} = 18 \times \frac{5}{18} = 5 \text{ m/s}$, $t_1 = 0 \text{ s}$, $t_2 = 2 \text{ s}$.

$$a_{av} = \frac{v - u}{t_2 - t_1} = \frac{5.0}{2} = 2.5 \text{ m/s}^2$$

Ex. A car moving with a velocity of 20 ms^{-1} is brought to rest in 5 seconds by applying brakes. Calculate the retardation of the car.

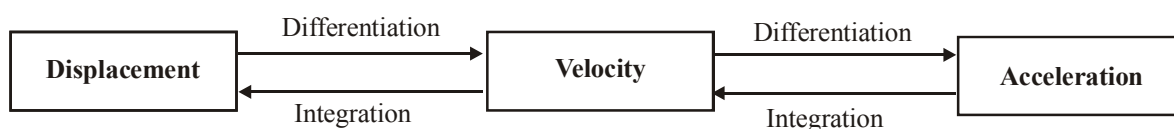
Sol. Here, $u = 20 \text{ ms}^{-1}$, $v = 0$, $t = 5 \text{ s}$. acceleration $a = \frac{v - u}{t} = \frac{(0 - 20)}{5} = -4 \text{ m/s}^2$

–ve acceleration is known as retardation. Thus, retardation of the car = 4 ms^{-2} .

Use of Mathematical Tools in Solving Problems of One-Dimensional Motion

If displacement–time equation is given, we can get velocity–time equation with the help of differentiation. Again, we can get acceleration–time equation with the help of differentiation.

If acceleration–time equation is given, we can get velocity–time equation by integration. From velocity equation, we can get displacement–time equation by integration.



Ex. The velocity of any particle is related with its displacement as; $x = \sqrt{v+1}$, Calculate acceleration at $x = 5 \text{ m}$.

Sol. $\therefore x = \sqrt{v+1} \therefore x^2 = v+1 \Rightarrow v = (x^2-1)$

$$\text{Therefore } a = \frac{dv}{dt} = \frac{d}{dt}(x^2 - 1) = 2x \frac{dx}{dt} = 2x \quad v = 2x(x^2 - 1)$$

$$\text{At } x = 5 \text{ m, } a = 2 \times 5 (25 - 1) = 240 \text{ m/s}^2$$

Ex. The velocity of a particle moving in the positive direction of x -axis varies as $v = \alpha\sqrt{x}$ where α is positive constant. Assuming that at the moment $t = 0$, the particle was located at $x = 0$ find, (i) the time dependence of the velocity and the acceleration of the particle and (ii) the average velocity of the particle averaged over the time that the particle takes to cover first s metres of the path.

Sol. (i) Given that $v = \alpha\sqrt{x}$

$$\Rightarrow \frac{dx}{dt} = \alpha\sqrt{x} \therefore \frac{dx}{\sqrt{x}} = \alpha dt \Rightarrow \int_0^x \frac{dx}{\sqrt{x}} = \int_0^t \alpha dt \quad 2\sqrt{x} = \alpha t \Rightarrow x = (\alpha^2 t^2 / 4)$$

$$\text{Velocity } \frac{dx}{dt} = \frac{1}{2} \alpha^2 t \text{ and Acceleration } \frac{d^2x}{dt^2} = \frac{1}{2} \alpha^2$$

$$(ii) \text{ Time taken to cover first } s \text{ metres } s = \frac{\alpha^2 t^2}{4} \Rightarrow t^2 = \frac{4s}{\alpha^2} \Rightarrow t = \frac{2\sqrt{s}}{\alpha};$$

$$\text{average velocity} = \frac{\text{total displacement}}{\text{total time}} = \frac{s\alpha}{2\sqrt{s}} = \frac{1}{2} \sqrt{s}\alpha$$

Ex. A particle moves in the plane xy with constant acceleration a directed along the negative y -axis. The equation of motion of the particle has the form $y = px - qx^2$ where p and q are positive constants. Find the velocity of the particle at the origin of coordinates.

Sol. Given that $y = px - qx^2$

$$\therefore \frac{dy}{dt} = p \frac{dx}{dt} - q \cdot 2x \frac{dx}{dt} \text{ and } \frac{d^2y}{dt^2} = p \frac{d^2x}{dt^2} - 2q \frac{d^2x}{dt^2} - 2q \left(\frac{dx}{dt} \right)^2 = (p - 2qx) \frac{d^2x}{dt^2} - 2q \left(\frac{dx}{dt} \right)^2$$

$$\therefore \frac{d^2x}{dt^2} = 0 \text{ (no acceleration along } x\text{-axis) and } \frac{d^2y}{dt^2} = -a$$

$$\therefore v_x^2 = \frac{a}{2q} \Rightarrow v_x = \sqrt{\frac{a}{2q}} \text{ Further, } \left(\frac{dy}{dt} \right)_{x=0} = p \frac{dx}{dt} \Rightarrow v_y = p \sqrt{\frac{a}{2q}}$$

$$\text{Now } v = \sqrt{(v_x^2 + v_y^2)} = \sqrt{\left(\frac{a}{2q} + \frac{ap^2}{2q} \right)} \Rightarrow v = \sqrt{\frac{a(p^2 + 1)}{2q}}$$

Ex. A particle moves along a straight line path such that its magnitude of velocity is given by $v = (3t^2 - 6t) \text{ ms}^{-1}$, where t is the time in seconds. If it is initially located at the origin O then determine the magnitude of particle's average velocity and average speed in time interval from $t = 0$ to $t = 4$ s.

Sol. Average velocity = $\frac{\int v dt}{\int dt} = \frac{\int_0^4 (3t^2 - 6t) dt}{\int_0^4 dt} = \frac{(t^3 - 3t^2)_0^4}{(t)_0^4} = 4 \text{ ms}^{-1}$

$$\begin{aligned} \text{Average speed} &= \frac{\int |v| dt}{\int dt} = \frac{\int_0^4 |3t^2 - 6t| dt}{\int_0^4 dt} = \frac{\int_0^2 (6t - 3t^2) dt + \int_2^4 (3t^2 - 6t) dt}{\int_0^4 dt} \\ &= \frac{(3t^2 - t^3)_0^2 + (t^3 - 3t^2)_2^4}{(t)_0^4} = \frac{24}{4} = 6 \text{ ms}^{-1} \end{aligned}$$

Ex. The coordinates a particle moving in a plane are given by $x = 3 \cos 2t$ and $y = 4 \sin 2t$.

(i) Find the equation of the path of the particle.

(ii) Find the angle between $\vec{r}$ and $\vec{v}$ at $t = \frac{\pi}{4}$.

(iii) Prove that acceleration of the particle is always directed towards a fixed point.

Sol. (i) Eliminating t from $x = 3 \cos 2t$ & $y = 4 \sin 2t$. We get $\left(\frac{x}{3} \right)^2 + \left(\frac{y}{4} \right)^2 = 1 \Rightarrow \frac{x^2}{9} + \frac{y^2}{16} = 1$

$$(ii) \quad \vec{r} = x\hat{i} + y\hat{j} = 3 \cos 2t \hat{i} + 4 \sin 2t \hat{j} \Rightarrow \vec{v} = \frac{dx}{dt} \hat{i} + \frac{dy}{dt} \hat{j} = -6 \sin 2t \hat{i} + 8 \cos 2t \hat{j}$$

$$\text{At } t = \frac{\pi}{4}, \vec{r} = 4\hat{j}, \vec{v} = -6\hat{i}$$

$$\text{Angle between } \vec{r} \text{ and } \vec{v} \cos \theta = \frac{\vec{r} \cdot \vec{v}}{rv} = \frac{(4\hat{j}) \cdot (-6\hat{i})}{(4)(6)} = 0 \Rightarrow \theta = \frac{\pi}{2}$$

$$(iii) \quad \vec{a} = \frac{d\vec{v}}{dt} = -12 \cos 2t \hat{i} - 16 \sin 2t \hat{j} = -4(3 \cos 2t \hat{i} + 4 \sin 2t \hat{j}) = -4\vec{r}$$

So acceleration is always directed toward origin (a fixed point)

Ex. A particle moves in a straight line according to the relation $x = \frac{t^3}{3} - \frac{5t^2}{2} + 6t$.

Find the displacement and distance travelled by the particle upto $t = 4$ sec.

Sol. $v = \frac{dx}{dt} = t^2 - 5t + 6$

The particle turns when $v = 0 = t^2 - 5t + 6 \Rightarrow (t-2)(t-3) = 0$ i.e. $t = 2$ sec, 3 secs

$$\text{Displacement} = x(4) - x(0) = \frac{64}{3} - \frac{80}{2} + 24 = \frac{16}{3} \text{ m}$$

$$\begin{aligned} \text{Distance} &= |x(2) - x(0)| + |x(3) - x(2)| + |x(4) - x(3)| \\ &= \left| \frac{8}{3} - 10 + 12 \right| + \left| \left(9 - \frac{45}{2} + 18 - \frac{14}{3} \right) \right| + \left| \frac{16}{3} - \frac{9}{2} \right| = \frac{2}{3} + \frac{1}{6} + \frac{5}{6} = \frac{5}{3} \text{ m} \end{aligned}$$

$$\text{Alter : distance} = \int_0^4 |v| dt$$

Equations of motion (motion with constant acceleration)

If a particle moves with acceleration $\vec{a}$, then by definition $\vec{a} = \frac{d\vec{v}}{dt} \Rightarrow d\vec{v} = \vec{a} dt$. Let at starting ($t=0$)

initial velocity of the particle $\vec{u}$ and at time t its final velocity $= \vec{v}$ then $\int_{\vec{u}}^{\vec{v}} d\vec{v} = \int_0^t \vec{a} dt$

If acceleration is constant

$$\int_{\vec{u}}^{\vec{v}} d\vec{v} = \vec{a} \int_0^t dt \Rightarrow [\vec{v}]_{\vec{u}}^{\vec{v}} = \vec{a} [t]_0^t \Rightarrow \vec{v} - \vec{u} = \vec{a} t \Rightarrow \vec{v} = \vec{u} + \vec{a} t \quad \dots\dots(1)$$

Now by definition of velocity, equation (1) reduces to

$$\vec{v} = \frac{d\vec{s}}{dt} = \vec{u} + \vec{a} t \Rightarrow \int_0^{\vec{s}} d\vec{s} = \int_0^t (\vec{u} + \vec{a} t) dt \Rightarrow \vec{s} = \left[\vec{u} t + \frac{1}{2} \vec{a} t^2 \right]_0^t \Rightarrow \vec{s} = \vec{u} t + \frac{1}{2} \vec{a} t^2 \quad \dots\dots(2)$$

Vertical motion under gravity

If air resistance is neglected and a body is freely moving along vertical line near the earth surface then an acceleration downward which is 9.8m/s^2 or 980 cm/s^2 or 32 ft/s^2 is experienced by the body.

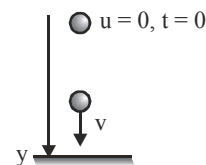
Freely falling bodies from a height h above the ground

Taking initial position as origin and direction of motion (i.e. downward direction) positive y axis, as body is just released/dropped $u = 0$

acceleration along $+Y$ axis $a = g$

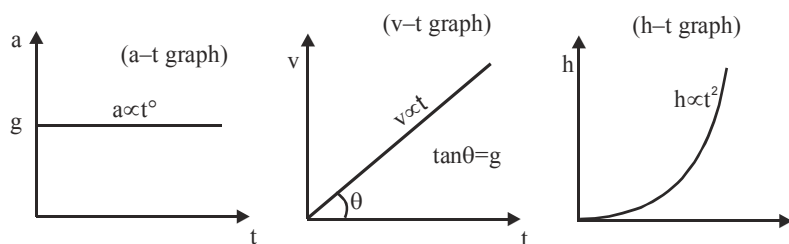
Use equations of motion to describe the motion, i.e.

$$v = u + at, \quad y = ut + \frac{1}{2}at^2, \quad v^2 = u^2 + 2ay$$



Let the body acquires velocity v (downward) after falling a distance h in time t , then

$$v = gt \Rightarrow t = v/g \therefore h = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{\frac{2h}{g}}, \quad v^2 = 2gh \Rightarrow v = \sqrt{2gh}$$



Body is projected vertically upward : With velocity u take initial position as origin and direction of motion

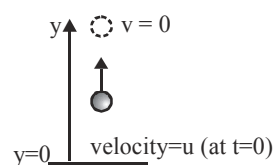
(i.e. vertically upward) as positive y -axis.

$v = 0$ at maximum height, at $t = T$,

$a = -g$ (because directed downward)

Put the values in equation of motion

$$v = u + at \Rightarrow 0 = u - gT \Rightarrow u = gT$$



$$s = ut + \frac{1}{2}at^2 \Rightarrow h = ut - \frac{1}{2}gt^2$$

$$\Rightarrow h_{\max} = uT - \frac{1}{2}gT^2 \Rightarrow h_{\max} = (gT)T - \frac{1}{2}gT^2 = \frac{1}{2}gT^2$$

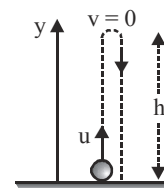
$$v^2 = u^2 + 2as \Rightarrow v^2 = u^2 - 2gh \Rightarrow 0 = u^2 - 2gh_{\max} \Rightarrow u^2 = 2gh_{\max} \Rightarrow u = \sqrt{2gh_{\max}}$$

After attaining maximum height body turns and come back at ground. During complete flight acceleration is constant,

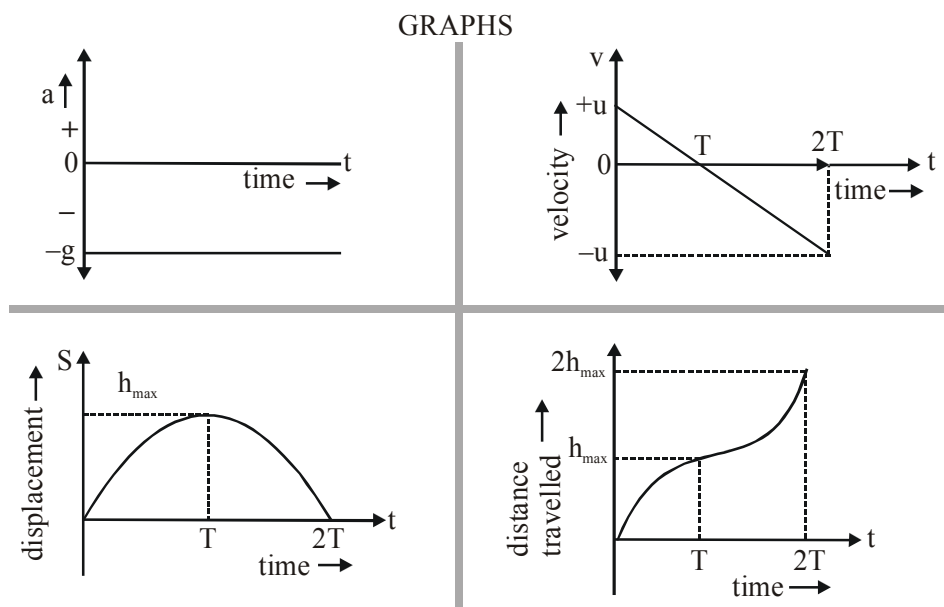
Time taken during up flight and down flight are equal

$$\text{Time for one side } T = \frac{u}{g} \text{ and total flight time} = 2T = \frac{2u}{g}$$

At each equal height from ground speed of body will be same either going up or coming down.



SOME RELATED GRAPHS FOR ABOVE MOTION'S



Ex. A body is freely dropped from a height h above the ground. Find the ratio of distances fallen in first one second, first two seconds, first three seconds, also find the ratio of distances fallen in 1st second, in 2nd second, in 3rd second etc.

Sol. From second equation of motion, i.e. $h = \frac{1}{2}gt^2$ ($h = ut + \frac{1}{2}gt^2$ and $u = 0$)

$$h_1 : h_2 : h_3 \dots = \frac{1}{2}g(1)^2 : \frac{1}{2}g(2)^2 : \frac{1}{2}g(3)^2 = 1^2 : 2^2 : 3^2 \dots = 1 : 4 : 9 \dots$$

Now from the of distance travelled in n^{th} second

$$s_n = u + \frac{1}{2}a(2n-1) \text{ here } u = 0, a = g \Rightarrow s_n = \frac{1}{2}g(2n-1)$$

$$\Rightarrow s_1 : s_2 : s_3 \dots = \frac{1}{2}g(2 \times 1 - 1) : \frac{1}{2}g(2 \times 2 - 1) : \frac{1}{2}g(2 \times 3 - 1) = 1 : 3 : 5 \dots$$

Ex. A rocket is fired vertically up from the ground with a resultant vertical acceleration of 10m/s^2 . The fuel is finished in 1 minute and it continues to move up.

(a) What is the maximum height reached?

(b) After finishing fuel, calculate the time for which it continues its upwards motion. (Take $g = 10\text{m/s}^2$)

Sol. (a) The distance travelled by the rocket during burning interval (1 minute = 60s) in which resultant acceleration 10m/s^2 is vertically upwards will be $h_1 = 0 \times 60 + \frac{1}{2} \times 10 \times 60^2 = 18000\text{m} = 18\text{km}$ and velocity acquired by it will be $v = 0 + 10 \times 60 = 600\text{m/s}$

Now after 1 minute the rocket moves vertically up with initial velocity of 600m/s and acceleration due to

gravity opposes its motion. So, it will go to a height h_2 from this point, till its velocity becomes zero such that

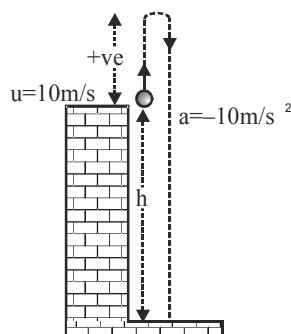
$$0 = (600)^2 - 2gh_2 \Rightarrow h_2 = 18000\text{m} = 18\text{km} [g = 10\text{ms}^{-2}]$$

So the maximum height reached by the rocket from the ground, $H = h_1 + h_2 = 18 + 18 = 36\text{km}$

(b) As after burning of fuel the initial velocity 600 m/s and gravity opposes the motion of rocket, so from 1st equation of motion time taken by it till it velocity $v=0$

$$0 = 600 - gt \Rightarrow t = 60 \text{ s}$$

Ex. A ball is thrown upwards from the top of a tower 40 m high with a velocity of 10 m/s , find the time when it strikes the ground ($g = 10 \text{ m/s}^2$)



Sol. In the problem $u = +10 \text{ m/s}$, $a = -10 \text{ m/s}^2$ and $s = -40 \text{ m}$ (at the point where ball strikes the ground)

Substituting in $s = ut + \frac{1}{2}at^2$

$$-40 = 10t - 5t^2 \Rightarrow 5t^2 - 10t - 40 = 0 \Rightarrow t^2 - 2t - 8 = 0$$

Solving this we have $t = 4 \text{ s}$ and -2 s . Taking the positive value $t = 4 \text{ s}$.

Ex. The acceleration of a particle moving in a straight line varies with its displacement as, $a = 2s$ velocity of the particle is zero at zero displacement. Find the corresponding velocity displacement equation.

Sol. $a = 2s \Rightarrow \frac{dv}{dt} = 2s \Rightarrow \frac{dv}{ds} \cdot \frac{ds}{dt} = 2s \Rightarrow \frac{dv}{ds} \cdot v = 2s$

$$\Rightarrow \int v dv = 2 \int s ds \Rightarrow \left(\frac{v^2}{2} \right)_0^v = 2 \left(\frac{s^2}{2} \right)_0^s$$

$$\Rightarrow \frac{v^2}{2} = s^2 \Rightarrow v = s\sqrt{2}$$

Ex. If a body travels half its total path in the last second of its fall from rest, find :

(a) The time and

(b) height of its fall. Explain the physically unacceptable solution of the quadratic time equation. ($g = 9.8 \text{ m/s}^2$)

Sol. If the body falls a height h in time t , then

$$h = \frac{1}{2}gt^2 \quad [u = 0 \text{ as the body starts from rest}] \quad \dots (1)$$

$$\text{Now, as the distance covered in } (t-1) \text{ second is } h' = \frac{1}{2}g(t-1)^2 \quad \dots (2)$$

So from Equations (1) and (2) distance travelled in the last second.

$$h - h' = \frac{1}{2}gt^2 - \frac{1}{2}g(t-1)^2 \text{ i.e., } h - h' = \frac{1}{2}g(2t-1)$$

But according to given problem as $(h-h') = \frac{h}{2}$

$$\text{i.e., } \left(\frac{1}{2}\right)h = \left(\frac{1}{2}\right)g(2t-1) \text{ or } \left(\frac{1}{2}\right)gt^2 = g(2t-1) \quad [\text{as from equation (1) } h = \left(\frac{1}{2}\right)gt^2]$$

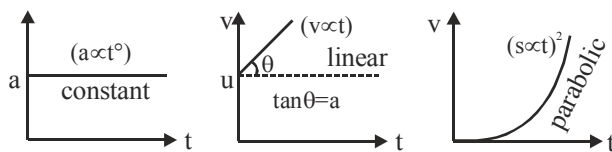
$$\text{or } t^2 - 4t + 2 = 0 \text{ or } t = [4 \pm \sqrt{(4^2 - 4 \times 2)}] / 2 \text{ or } t = 2 \pm \sqrt{2} \Rightarrow t = 0.59 \text{ s or } 3.41 \text{ s}$$

0.59 s is physically unacceptable as it gives the total time t taken by the body to reach ground lesser than one sec while according to the given problem time of motion must be greater than 1 s.

$$\text{so } t = 3.41 \text{ s and } h = \frac{1}{2} \times (9.8) \times (3.41)^2 = 57 \text{ m}$$

Graphs based on 1-D

For constant acceleration, a/t , v/t and s/t curve from equations of motion are –



In case of constant acceleration motion in a straight line, scalar form of equations of motion can be applied and problem becomes fairly simple.

$$\text{As } d\vec{v} = \vec{a}dt \text{ or } [\vec{v}]_{\vec{u}}^{\vec{v}} = \vec{v} - \vec{u} = \int_{t_1}^{t_2} \vec{a}dt = \text{Area between curve and time axis from } t_1 \text{ to } t_2.$$

Area under the curve of $a-t$ graph always gives the change in velocity.

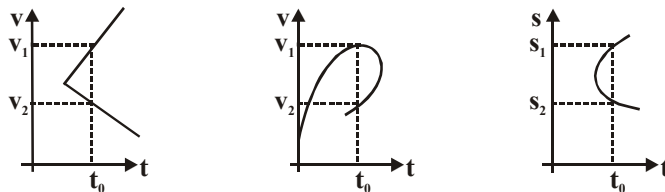
$$\text{Similarly } d\vec{s} = \vec{v}dt \text{ or } \vec{s} = \int_{t_1}^{t_2} \vec{v}dt = \text{Area between curve and time axis from } t_1 \text{ to } t_2.$$

Here $\vec{s}$ is the displacement of particle in time interval t_1 to t_2 , i.e. area under the curve of v/t graph always gives the displacement. If only magnitude of area is taken into account then sum of all area is the total distance travelled by the particle.

- Slopes of $v-t$ or $s-t$ graphs can never be infinite at any point, because infinite slope of $v-t$ graph means infinite acceleration. Similarly, infinite slope of $s-t$ graph means infinite velocity. Hence, the following graphs are not possible.



- At one time, two values of velocity or displacement are not possible. Hence, the following graphs are not acceptable.



- The slope of velocity–time graph of uniform motion is zero.
- When a body is having uniform motion along a straight line in a given direction, the magnitude of the displacement of body is equal to the actual distance travelled by the body in the given time.
- The average and instantaneous velocity in a uniform motion are equal in magnitude.
- In a uniform motion along a straight line, the slope of position–time graph gives the velocity of the body.
- The position–time graph of a body moving along a straight line can never be a straight line parallel to position axis because it will indicate infinite velocity.
- The speed of a body can never be negative
- Medium effects the motion of a body falling freely under gravity due to thrust and viscous drag.

Ex. A car accelerates from rest at a constant rate α for some time, after which it decelerates at a constant rate β , to come to rest. If the total time elapsed is t evaluate (a) the maximum velocity attained and (b) the total distance travelled.

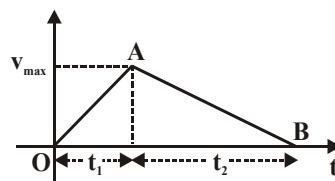
Sol. (a) Let the car accelerates for time t_1 and decelerates for time t_2 then

$$t = t_1 + t_2 \quad \text{.....(i)}$$

and corresponding velocity–time graph will be as shown in. fig.

$$\text{From the graph } \alpha = \text{slope of line AB} = \frac{v_{\max}}{t_1} \Rightarrow t_1 = \frac{v_{\max}}{\alpha}$$

$$\text{and } \beta = -\text{slope of line OB} = \frac{v_{\max}}{t_2} \Rightarrow t_2 = \frac{v_{\max}}{\beta}$$

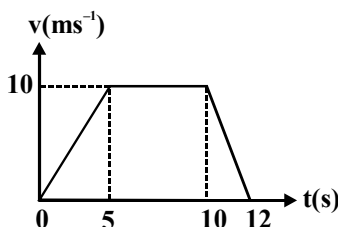


$$\Rightarrow \frac{v_{\max}}{\alpha} + \frac{v_{\max}}{\beta} = t \Rightarrow v_{\max} \left(\frac{\alpha + \beta}{\alpha\beta} \right) = t \Rightarrow v_{\max} = \frac{\alpha\beta t}{\alpha + \beta}$$

$$(b) \text{ Total distance} = \text{area under } v\text{-}t \text{ graph} = \frac{1}{2} \times t \times v_{\max} = \frac{1}{2} \times t \times \frac{\alpha\beta t}{\alpha + \beta} = \frac{1}{2} \left(\frac{\alpha\beta t^2}{\alpha + \beta} \right)$$

Note: This problem can also be solved by using equations of motion ($v = u + at$, etc.).

Ex. Draw displacement time and acceleration – time graph for the given velocity–time graph



Sol. For $0 \leq t \leq 5$ $v \propto t \Rightarrow s \propto t^2$ and $a_1 = \text{constant}$ $\frac{10}{5} = 2 \text{ ms}^{-2}$

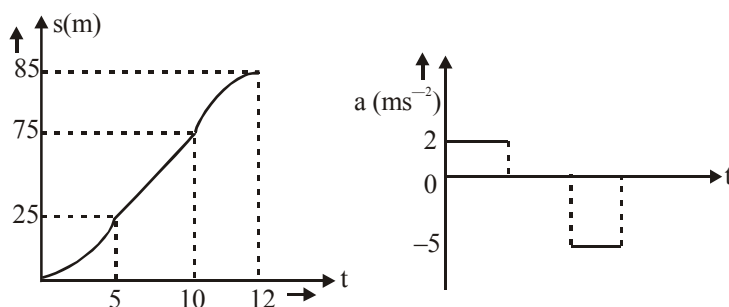
for whole interval $s_1 = \text{Area under the curve} = \frac{1}{2} \times 5 \times 10 = 25 \text{ m}$

For $5 \leq t \leq 10$ $v = 10\text{ms}^{-1}$ $\Rightarrow a = 0$

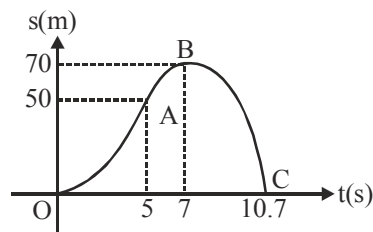
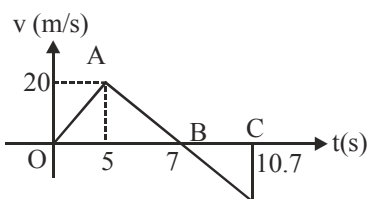
for whole interval $s_2 = \text{Area under the curve} = \frac{1}{2} \times 5 \times 10 = 50 \text{ m}$

For $10 \leq t \leq 12$ v linearly decreases with time $\Rightarrow a_3 = -\frac{10}{2} = -5 \text{ ms}^{-1}$

for whole interval $s_3 = \text{Area under the curve} = \frac{1}{2} \times 2 \times 10 = 10 \text{ m}$



Ex. A rocket is fired upwards vertically with a net acceleration of 4 m/s^2 and initial velocity zero. After 5 seconds its fuel is finished and it decelerates with g . At the highest point its velocity becomes zero. Then it accelerates downwards with acceleration g and return back to ground. Plot velocity–time and displacement–time graphs for the complete journey. Take $g = 10 \text{ m/s}^2$.



In the graphs, $v_A = at_{OA} = (4)(5) = 20 \text{ m/s}$

$$\therefore t_{AB} = \frac{v_A}{g} = \frac{20}{10} = 2s$$

Now, s_{OAB} = area under v-t graph between 0 to 7 s = $\frac{1}{2} (7) (20) = 70 \text{ m}$

$$\text{Now, } s_{\text{OAB}} = s_{\text{BC}} = \frac{1}{2} g t_{\text{BC}}^2$$

$$\therefore t_{BC} = \sqrt{14} = 3.7s$$

Also $s_{OA} = \text{area under } v\text{-}t \text{ graph between OA} = \frac{1}{2} (5) (20) = 50 \text{ m}$

$$v_B = 0 = v_A - gt_{AB}$$

$$\therefore t_{OAB} = (5 + 2)s = 7s$$

$$\therefore 70 = \frac{1}{2} (10) t_{BC}^2$$

$$\therefore t_{OAB} = 7 + 3.7 = 10.7s$$

Ex. At the height of 500m, a particle A is thrown up with $v = 75 \text{ ms}^{-1}$ and particle B is released from rest. Draw, acceleration–time, velocity–time, speed–time and displacement–time graph of each particle.

For particle A :

Time of flight

$$-500 = +75t - \frac{1}{2} \times 10t^2$$

$$\Rightarrow t^2 - 15t - 100 = 0$$

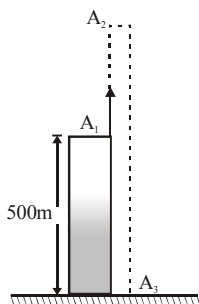
$$\Rightarrow t = 20 \text{ s}$$

Time taken for $A_1 A_2$

$$= 75 - 10t \Rightarrow t = 7.5 \text{ s}$$

Velocity at A_3 , $v = 75 - 10 \times 20 = -125 \text{ ms}^{-1}$

$$\text{Height } A_2 A_1 = \frac{1}{2} (10) (7.5)^2 = 281.25 \text{ m}$$



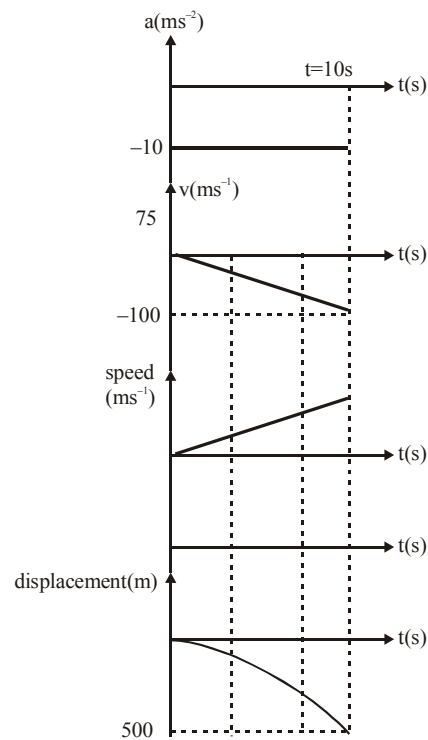
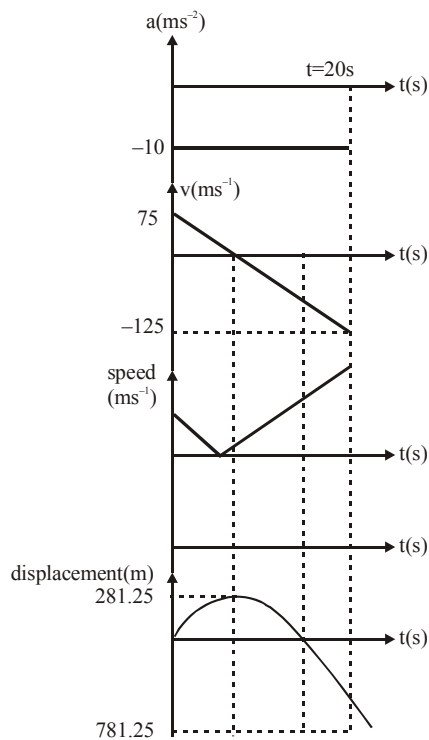
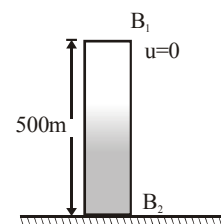
For Particle B

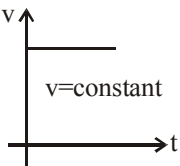
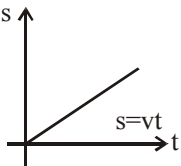
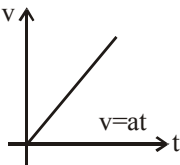
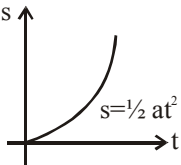
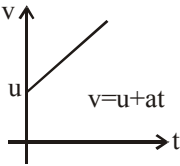
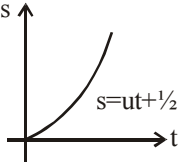
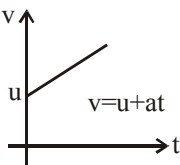
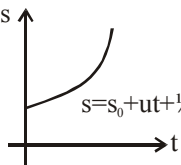
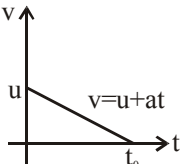
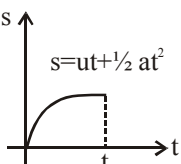
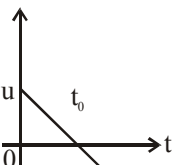
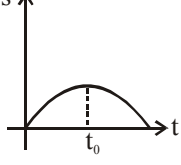
Time of flight

$$500 = \frac{1}{2} (10)t^2 \Rightarrow t = 10 \text{ s}$$

Velocity at B_2

$$v = 0 - (10) (10) = -100 \text{ ms}^{-1}$$

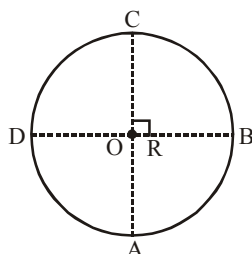


S.N.	Different Cases	v-t graph	s-t graph	Important Points
1.	Uniform motion			(i) Slope of s-t graph = $v = \text{constant}$ (ii) In s-t graph $s = 0$ at $t = 0$
2.	Uniformly accelerated motion with $u = 0$ at $t = 0$			(i) $u = 0$, i.e. $v = 0$ at $t = 0$ (ii) $u = 0$, i.e., slope of s-t graph at $t = 0$, should be zero (iii) a or slope of v-t graph is constant
3.	Uniformly accelerated with $u \neq 0$ at $t = 0$			(i) $u \neq 0$, i.e., v or slope motion of s-t graph at $t = 0$ is not zero (ii) v or slope of s-t graph gradually goes on increasing.
4.	Uniformly accelerated motion with $u \neq 0$ and $s = s_0$ at $t = 0$			(i) $s = s_0$ at $t = 0$
5.	Uniformly retarded motion till velocity becomes zero			(i) Slope of s-t graph at $t = 0$ gives u (ii) Slope of s-t graph at $t = t_0$ becomes zero (iii) In this case u can't be zero.
6.	Uniformly retarded then accelerated in opposite direction			(i) At time $t = t_0$, $v = 0$ or slope of s-t graph is zero (ii) In s-t graph slope or velocity first decreases then increases with opposite sign.

EXERCISE (S-1)

Definitions of kinematics variables

1. A particles starts from point A with constant speed v on a circle of radius R . Find magnitude of average velocity during its journey from :-



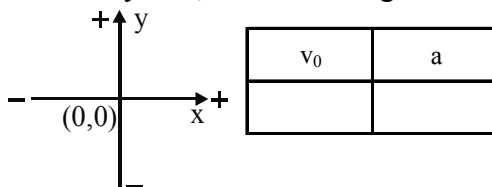
(a) A to B

(b) A to C

(c) A to D

KM0001

2. A particle is moving along x-axis. Initially it is located 5 m left of origin and it is moving away from the origin and slowing down. In this coordinate system, what are the signs of the initial velocity and acceleration.



KM0002

Motion with constant acceleration

3. A car accelerates with uniform rate from rest on a straight road. The distance travelled in the last second of a three second interval from the start is 15 m then find the distance travelled in first second in m.

KM0003

4. A particle moving in one-dimension with constant acceleration of 10 m/s^2 is observed to cover a distance of 100 m during a 4s interval. How far will the particle move in the next 4s?

KM0004

5. A particle starts from rest at $t = 0$ and $x = 0$ to move with a constant acceleration $= +2 \text{ m/s}^2$, for 20 seconds. After that, it moves with -4 m/s^2 for the next 20 seconds. Finally, it moves with positive acceleration for 10 seconds until its velocity becomes zero.

- (a) What is the value of the acceleration in the last phase of motion?
 (b) What is the final x-coordinate of the particle?
 (c) Find the total distance covered by the particle during the whole motion.

KM0005

6. A body moving with uniform acceleration has a velocity of -11 cm/s when its x coordinate is 3.00 cm. If its x coordinate 2 s later is -5 cm , what is the magnitude in cm/s^2 of its acceleration?

KM0006

7. A driver travelling at speed 36 kmh^{-1} sees the light turn red at the intersection. If his reaction time is 0.6s, and then the car can deaccelerate at 4 ms^{-2} . Find the stopping distance of the car.

KM0007

8. The window of the fourth floor of SANKALP building is 5 m high. A man looking out of the window sees an object moving up and down the height of window for 2 sec. Find the height that the object reaches from the top end of the window.

KM0008

9. A body is dropped from a height of 300 m. Exactly at the same instant another body is projected from the ground level vertically up with a velocity of 150 ms^{-1} . Find when they will meet.

KM0009

10. A stone is dropped from the top of a tall cliff, and 1 s later a second stone is thrown vertically downward with a velocity of 20 ms^{-1} . How far below the top of the cliff will the second stone overtake the first?

KM0010

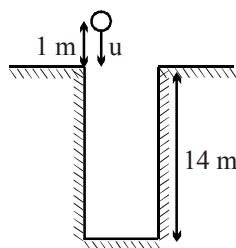
11. Speed of train is increasing linearly with time. The train passes a hut with speed 2 m/s and acquires a speed of 12 m/s after 10 s. What is the speed of the train in m/s , 5 s after passing the hut?

KM0011

12. Two particle A and B are moving in same direction on same straight line. A is ahead of B by 20m. A has constant speed 5 m/sec and B has initial speed 30 m/sec and retardation of 10 m/sec^2 . Then if x (in m) is total distance travelled by B as it meets A for second time. Then value of x will be.

KM0012

13. A boy throws a ball with speed u in a well of depth 14 m as shown. On bounce with bottom of the well the speed of the ball gets halved. What should be the minimum value of u (in m/s) such that the ball may be able to reach his hand again? It is given that his hands are at 1 m height from top of the well while throwing and catching.



KM0013

14. From the top of a tower, a ball is thrown vertically upwards. When the ball reaches h below the tower, its speed is double of what it was at height h above the tower. Find the greatest height attained by the ball from the tower.

KM0014

15. A rocket is fired vertically upwards with initial velocity 40 m/s at the ground level. Its engines then fired and it is accelerated at 2 m/s^2 until it reaches an altitude of 1000 m. At that point the engines shut off and the rocket goes into free-fall. If the velocity (in m/s) just before it collides with the ground is 40α . Then fill the value of α . Disregard air resistance ($g = 10 \text{ m/s}^2$).

KM0015

16. A balloon rises from rest on the ground with constant acceleration $\frac{g}{3}$. A stone is dropped when the balloon has rises to a height 60 metre. The time taken by the stone to reach the ground is.

KM0016

Motion with variable acceleration & calculus

17. The position x of a particle w.r.t. time t along x -axis is given by $x = 9t^2 - t^3$ where x is in metre and t in second. Find
- Maximum speed along $+x$ direction
 - Position of turning point
 - Displacement in first ten seconds
 - Distance travelled in first ten seconds

KM0017

18. The momentum of a particle moving in straight line is given by $p = \ln t + \frac{1}{t}$ (in kg m/s) find the time $t > 0$ at which the net force acting on particle is 0 and it's momentum at that time. [Hint : $F = \frac{dp}{dt}$]
KM0018
19. The velocity of the particle is given as $v = 3t^3 + t - \frac{1}{t^2}$. Calculate the net force acting on the body at time $t = 2$ sec, if the mass of the body is 5 kg.
KM0019
20. A wheel rotates so that the angle of rotation is proportional to the square of time. The first revolution was performed by the wheel for 8 sec. Find the angular velocity ω , 32 sec after the wheel started.
 [Hint: Consider $\theta = kt^2$, find k]
KM0020
21. The charge flowing through a conductor beginning with time $t = 0$ is given by the formula $q = 2t^2 + 3t + 1$ (coulombs). Find the current $i = \frac{dq}{dt}$ at the end of the 5th second.
KM0021
22. The angle θ through which a pulley turns with time t is specified by the function $\theta = t^2 + 3t - 5$. Find the angular velocity $\omega = \frac{d\theta}{dt}$ at $t = 5$ sec.
KM0022
23. The motion of a particle in a straight line is defined by the relation $x = t^4 - 12t^2 - 40$ where x is in meters and t is in sec. Determine the position x , velocity v and acceleration a of the particle at $t = 2$ sec.
KM0023
24. A point moves in a straight line so that its distance from the start in time t is equal to
- $$s = \frac{1}{4} t^4 - 4t^3 + 16t^2.$$
- (a) At what times was the point at its starting position?
 (b) At what times is its velocity equal to zero?
KM0024
25. A body whose mass is 3 kg performs rectilinear motion according to the formula $s = 1 + t + t^2$, where s is measured in centimetres & t in seconds. Determine the kinetic energy $\frac{1}{2}mv^2$ of the body in 5 sec after its start.
KM0025
26. A force of 40N is responsible for the motion of a body governed by the equation $s = 2t + 2t^2$ where s is in meters and t in sec. What is the momentum of the body at $t = 2$ sec?
 [Hint: Find acc. then $m = F/a$ & $p = mv$]
KM0026
27. The angle rotated by a disc is given by $\theta = \frac{2}{3} t^3 - \frac{25}{2} t^2 + 77t + 5$, where θ is in rad and t in seconds.
 (a) Find the times at which the angular velocity of the disc is zero.
 (b) Its angular acceleration at these times.
KM0027
28. The acceleration of a particle starting from rest vary with respect to time is given by $a = (2t - 6)$, where t is in seconds. Find the time (in seconds) at which velocity of particle in negative direction is maximum.
KM0028

- 29.** Acceleration of a particle is defined as $a = (75V^2 - 30V + 3) \text{ (m/s}^2\text{)}$. If the constant speed achieved by the particle is given by V_c , then find the value of $10V_c$.

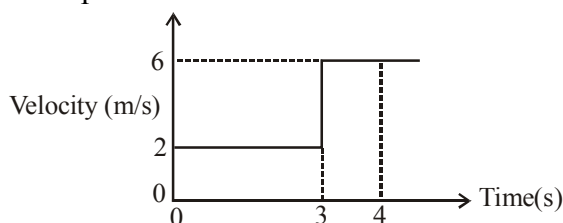
KM0029

- 30.** Position vector of a particle is given by $\vec{r} = 3t^3\hat{i} + 4t\hat{j} + t^2\hat{k}$. Find avg acceleration of particle from $t = 1$ to $t = 2$ sec.

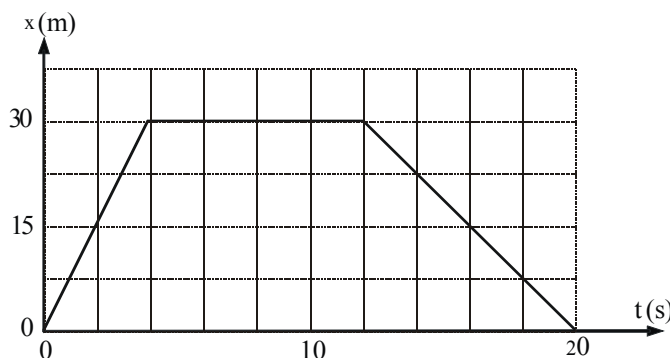
KM0030

Question based on graph

- 31.** In the following graph variation with time (t), in velocity (v) of a particle moving rectilinearly is shown. What is average velocity in m/s of the particle in time interval from 0 s to 4 s?

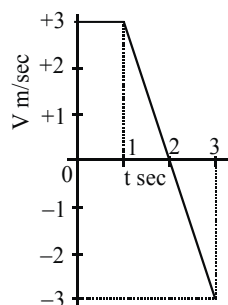
**KM0031**

- 32.** The graph illustrates motion of a bucket being lowered into a well from the top at the instant $t = 0$, down to the water level, filled with water and drawn up again. Here 'x' is the depth. Find the average speed of the bucket in m/s during whole operation.

**KM0032**

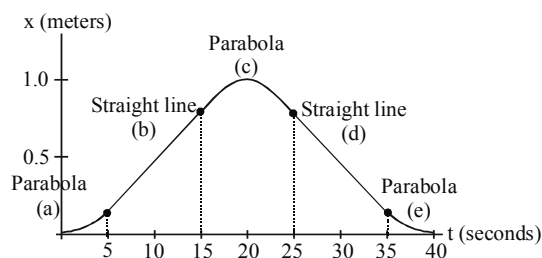
- 33.** A particle moves along a straight line, x . At time $t = 0$, its position is at $x = 0$. The velocity, V , of the object changes as a function of time t , as indicated in the figure; t is in seconds, V in m/sec and x in meters.

- What is x at $t = 3$ sec?
- What is the instantaneous acceleration (in m/sec^2) at $t = 2$ sec?
- What is the average velocity (in m/sec) between $t = 0$ and $t = 3$ sec?
- What is the average speed (in m/sec) between $t = 1$ and $t = 3$ sec?

**KM0033**

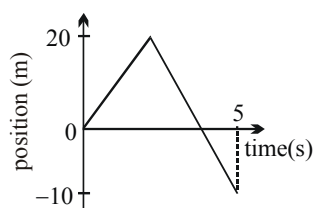
34. The figure below is a displacement vs time plot for the motion of an object, answer questions (i) & (ii) with the letter of appropriate section of the graph.

- (i) Which section represents motion in the forward direction with positive acceleration?
 (ii) Which section represents uniform motion backwards ($-x$ direction)?

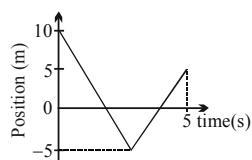


KM0034

35. (a) The diagram shows the position-time graph for a particle moving in a straight line. Find the average velocity for the interval from $t = 0$ to $t = 5$.



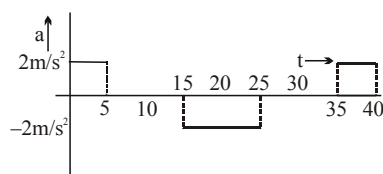
- (b) The diagram shows the position-time graph for a particle moving in a straight line. Find the average speed for the interval from $t = 0$ to $t = 5$.



KM0035

36. Figure shows a graph of acceleration of a particle moving on the x -axis. Plot the following graphs if the particle is at origin and at rest at $t = 0$.

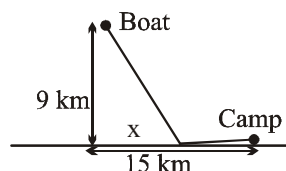
- (i) velocity-time graph (ii) displacement-time graph (iii) distance-time graph.



KM0036

EXERCISE (S-2)

- At a distance $L = 400\text{m}$ from the traffic light, brakes are applied to a locomotive moving at a velocity $v = 54\text{ km/hr}$. Determine the position of the locomotive relative to the traffic light 1 minute after the application of the brakes if its acceleration is -0.3m/sec^2 . **KM0037**
- A particle goes from A to B with a speed of 40km/h and B to C with a speed of 60km/h . If $AB = 6BC$, the average speed in km/h between A and C is. **KM0038**
- A flower pot falls off a window sill and falls past the window below. It takes 0.30 s to pass a window 3.45 m high. How far is the top of the window below upper window sill? **KM0039**
- A juggler performs in a room whose ceiling is 3 m above the level of his hands. He throws a ball vertically upward so that it just reaches the ceiling.
 - With what initial velocity does he throw the ball ?
 - What time is required for the ball to reach the ceiling ?
 He throws a second ball upward with the same initial velocity, at the instant that the first ball is at the ceiling.
 - How long after the second ball is thrown do the two ball pass each other ?
 - When the balls pass each other, how far are they above the juggler's hands ?**KM0040**
- A train, travelling at 20 km/hr is approaching a platform. A bird is sitting on a pole on the platform. When the train is at a distance of 2 km from pole, brakes are applied which produce a uniform deceleration in it to stop it at pole. At that instant the bird flies towards the train at 60 km/hr and after touching the nearest point on the train flies back to the pole and then flies towards the train and continues repeating itself. Calculate how much distance will the bird have flown before the train stops? **KM0041**
- A helicopter takes off along the vertical with an acceleration of 3 m/sec^2 & zero initial velocity. In a certain time, the pilot switches off the engine. At the point of take off, the sound dies away in 30 sec . Determine the velocity of the helicopter at the moment when its engine is switched off, assuming the velocity of sound is 320 m/sec . **KM0042**
- A fishing boat is anchored 9 km away from the nearest point on shore. A messenger must be sent from the fishing boat to a camp, 15 km from the point on shore closest to the boat. If the messenger can walk at a speed of 5 km per hour and can row at 4 km per hour .
 - Form an expression relating time taken to reach the camp t with distance x on shore where he lands.
 - At what point on shore must he land in order to reach the camp in the shortest possible time?



KM0043

8. Two body move from the same point along a straight line. The first body moves with velocity $v = (3t^2 - 6t)$ m/s, the second with velocity $v = (10t + 20)$ m/s. At what instant and at what distance from the initial point will they meet.

KM0044

9. Velocity of a car depends on its distance ℓ from a fixed pole on a straight road as $v = 2\sqrt{\ell}$, where ℓ is in meters and v in m/s. Find acceleration (in m/s^2) when $\ell = 8\text{m}$.

KM0045

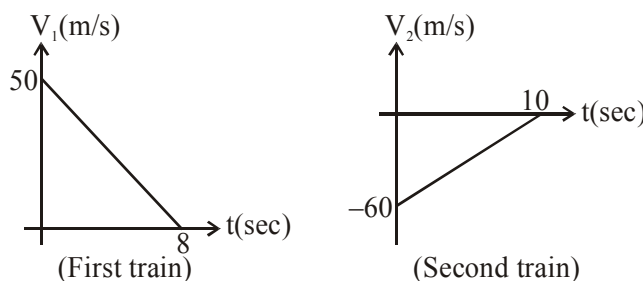
10. A particle is moving with uniform acceleration along x-axis with initial velocity along positive x. At

$t = \frac{3\sqrt{2}}{\sqrt{2}-1}$ s the magnitude of displacement becomes $\frac{1}{3}$ the total distance travelled. By this time the x coordinate of particle is still positive. The instant (in sec) at which displacement becomes zero is

KM0046

11. Two trains are moving in opposite direction on same track. When their separation was 600 m their drivers notice the mistake and start slowing down to avoid collision. Graphs of their velocities as function of time

is as shown. If separation between the drivers when first train stops is x then find the value of $\frac{x}{16}$.



KM0047

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Definitions of kinematics variables

1. In 1.0 sec. a particle goes from point A to point B moving in a semicircle of radius 1.0 m. The magnitude of average velocity is : [JEE '99]



- (A) 3.14 m/sec (B) 2.0 m/sec (C) 1.0 m/sec (D) zero

KM0048

2. An object is tossed vertically into the air with an initial velocity of 8 m/s. Using the sign convention upwards as positive, how does the vertical component of the acceleration a_y of the object (after leaving the hand) vary during the flight of the object?
- (A) On the way up $a_y > 0$, on the way down $a_y > 0$
 (B) On the way up $a_y < 0$, on the way down $a_y > 0$
 (C) On the way up $a_y > 0$, on the way down $a_y < 0$
 (D) On the way up $a_y < 0$, on the way down $a_y < 0$

KM0049

Motion with constant acceleration

3. A body starts from rest and is uniformly accelerated for 30 s. The distance travelled in the first 10 s is x_1 , next 10 s is x_2 and the last 10 s is x_3 . Then $x_1 : x_2 : x_3$ is the same as :-
- (A) 1 : 2 : 4 (B) 1 : 2 : 5 (C) 1 : 3 : 5 (D) 1 : 3 : 9

KM0050

4. If a body starts from rest and travels 120 cm in the 6th second, with constant acceleration then what is the acceleration :
- (A) 0.20 m/s² (B) 0.027 m/s² (C) 0.218 m/s² (D) 0.03 m/s²

KM0051

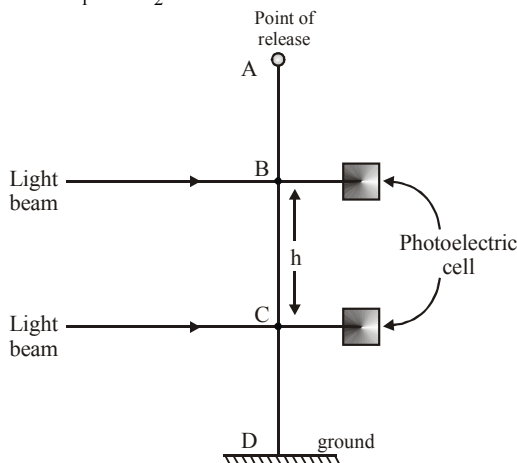
5. A particle travels 10m in first 5 sec and 10m in next 3 sec. Assuming constant acceleration what is the distance travelled in next 2 sec.
- (A) 8.3 m (B) 9.3 m (C) 10.3 m (D) None of above

KM0052

6. The engine of a motorcycle can produce a maximum acceleration 5 m/s². Its brakes can produce a maximum retardation 10 m/s². If motorcyclist start from point A and reach at point B. What is the minimum time in which it can cover if distance between A and B is 1.5 km. (Given : that motorcycle comes to rest at B)
- (A) 30 sec (B) 15 sec (C) 10 sec (D) 5 sec

KM0053

7. The acceleration of free fall at a planet is determined by timing the fall of a steel ball photo-electrically. The ball passes B and C at times t_1 and t_2 after release from A. The acceleration of free fall is given by



- (A) $\frac{2h}{t_2 - t_1}$ (B) $\frac{h}{t_2^2 - t_1^2}$ (C) $\frac{2h}{t_2^2 - t_1^2}$ (D) $\frac{2h}{t_2^2 + t_1^2}$

KM0054

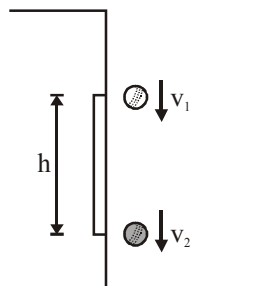
8. A particle has an initial velocity of 9 m/s due east and a constant acceleration of 2 m/s² due west. The distance covered by the particle in the fifth second of its motion is :-
 (A) 0 (B) 0.5 m (C) 2 m (D) none of these

KM0055

9. A physics teacher finds a scrap of paper on which one of his students has written the following equation : $0^2 - 5^2 = 2 \times (-9.8) \times x$; of which of the following problem would this equation be part of the correct solution?
 (A) Find the speed of an object 5 seconds after it was dropped from rest.
 (B) Find the distance of an object has fallen 5 seconds after it was released from rest on Earth.
 (C) Find the height from which a ball when released will strike the ground with a speed of 5 m/s.
 (D) Find the maximum height to which a ball will rise if it is thrown upward with an initial speed of 5 m/s.

KM0056

10. A ball dropped from the top of a building passes past a window of height h in time t . If its speeds at the top and the bottom edges of the window are denoted by v_1 and v_2 respectively, which of the following set of equations are correct?



- (A) $v_2 - v_1 = gt$ and $(v_2 - v_1)t = h$ (B) $v_2 - v_1 = gt$ and $(v_2 + v_1)t = 2h$
 (C) $v_2 + v_1 = gt$ and $(v_2 - v_1)t = h$ (D) None of the above.

KM0057

11. A body falls freely from rest. It covers as much distance in the last second of its motion as covered in the first three seconds. The body has fallen for a time of :
 (A) 3 s (B) 5 s (C) 7 s (D) 9 s **KM0058**
12. A ball is thrown vertically upward with initial velocity 30 m/sec. What will be its position vector at time $t = 5$ sec taking origin at the point of projection, vertical up as positive y-axis and horizontal as x-axis :-
 (A) (0, 25) (B) (0, 20) (C) (0, 45) (D) (0, 5) **KM0059**
13. A particle is thrown upwards from ground. It experiences a constant resistance force which can produce retardation 2 m/s^2 . The ratio of time of ascent to the time of descent is [$g = 10 \text{ m/s}^2$]
 (A) 1 : 1 (B) $\sqrt{\frac{2}{3}}$ (C) $\frac{2}{3}$ (D) $\sqrt{\frac{3}{2}}$ **KM0060**
14. A body of mass 'm' is travelling with a velocity 'u'. When a constant retarding force 'F' is applied, it comes to rest after travelling a distance ' s_1 '. If the initial velocity is '2u', with the same force 'F', the distance travelled before it comes to rest is ' s_2 '. Then
 (A) $s_2 = 2s_1$ (B) $s_2 = \frac{s_1}{2}$ (C) $s_2 = s_1$ (D) $s_2 = 4s_1$ **KM0061**
15. A ball is thrown vertically upward with initial velocity 30 m/sec. What will be its position vector at time $t = 5$ sec, taking origin at 45 m above the point of projection, vertical up as positive y-axis and horizontal as x-axis :-
 (A) (0, -25) (B) (0, -20) (C) (0, -45) (D) (0, -5) **KM0062**

Motion with variable acceleration & calculus

16. If $s = 2t^3 + 3t^2 + 2t + 8$ then the time at which acceleration is zero, is :-
 (A) $t = \frac{1}{2}$ (B) $t = 2$ (C) $t = \frac{1}{2\sqrt{2}}$ (D) Never **KM0063**
17. Velocity of a particle varies with time as $v = 4t$. The displacement of particle between $t = 2$ to $t = 4$ sec, is :-
 (A) 12 m (B) 36 m (C) 24 m (D) 6 m **KM0064**
18. A point mass moves with velocity $v = (5t - t^2) \text{ ms}^{-1}$ in a straight line. Find the distance travelled (i.e. $\int v dt$) in fourth second.
 (A) $\frac{31}{6} \text{ m}$ (B) $\frac{29}{6} \text{ m}$ (C) $\frac{37}{6} \text{ m}$ (D) None of these **KM0065**
19. A particle is projected with velocity v_0 along x-axis. The deceleration on the particle is proportional to the square of the distance from the origin i.e., $a = -\alpha x^2$. The distance at which the particle stops is:-
 (A) $\sqrt{\frac{3v_0}{2\alpha}}$ (B) $\left(\frac{3v_0}{2\alpha}\right)^{\frac{1}{3}}$ (C) $\sqrt{\frac{3v_0^2}{2\alpha}}$ (D) $\left(\frac{3v_0^2}{2\alpha}\right)^{\frac{1}{3}}$ **KM0066**

20. The acceleration vector along x-axis of a particle having initial speed v_0 changes with distance as $a = \sqrt{x}$. The distance covered by the particle, when its speed becomes twice that of initial speed is:-

(A) $\left(\frac{9}{4}v_0\right)^{\frac{4}{3}}$ (B) $\left(\frac{3}{2}v_0\right)^{\frac{4}{3}}$ (C) $\left(\frac{2}{3}v_0\right)^{\frac{4}{3}}$ (D) $2v_0$ **KM0067**

21. For a particle moving in a straight line the position of the particle at time (t) is given by

$x = \frac{t^3}{6} - t^2 - 9t + 18$ m. What is the velocity of the particle when its acceleration is zero :-

(A) 18 m/s (B) -9 m/s (C) -11 m/s (D) 6 m/s **KM0068**

22. A particle moves along a straight line such that at time t its displacement from a fixed point O on the line is $3t^2 - 2$. The velocity of the particle when $t = 2$ is:

(A) 8 ms^{-1} (B) 4 ms^{-1} (C) 12 ms^{-1} (D) 0 **KM0069**

23. Temperature of a body varies with time as $T = (T_0 + \alpha t^2 + \beta \sin t) \text{ K}$, where T_0 is the temperature in Kelvin at $t = 0$ sec. & $\alpha = 2/\pi \text{ K/s}^2$ & $\beta = -4 \text{ K}$, then rate of change of temperature at $t = \pi$ sec. is

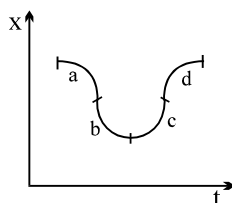
(A) 8 K (B) 8°K (C) 8 K/sec (D) 8°K/sec **KM0070**

24. The velocity of a particle moving on the x-axis is given by $v = x^2 + x$ where v is in m/s and x is in m. Find its acceleration in m/s^2 when passing through the point $x = 2 \text{ m}$

(A) 0 (B) 5 (C) 11 (D) 30 **KM0071**

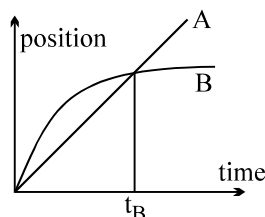
Question based on graph

25. The graph shown is a plot of position versus time. For which labeled region is the velocity positive and the acceleration negative?



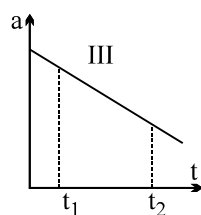
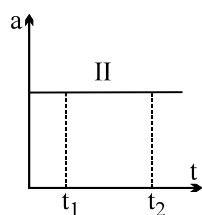
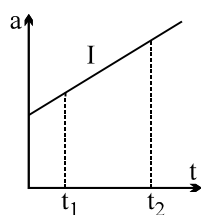
(A) a (B) b (C) c (D) d **KM0072**

26. The graph shows position as a function of time for two trains running on parallel tracks. Which statement is true?



- (A) At time t_B , both trains have the same velocity.
 (B) Both trains have the same velocity at some time after t_B .
 (C) Both trains have the same velocity at some time before t_B .
 (D) Somewhere on the graph, both trains have the same acceleration. **KM0073**

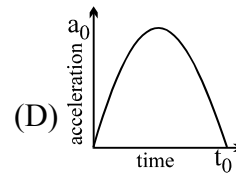
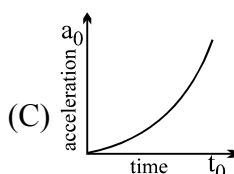
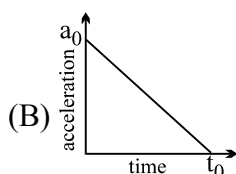
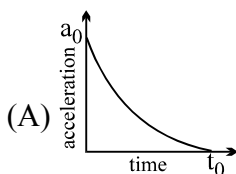
27. Each of the three graphs represents acceleration versus time for an object that already has a positive velocity at time t_1 . Which graphs show an object whose speed is increasing for the entire time interval between t_1 and t_2 ?



- (A) graph I, only
(B) graphs I and II, only
(C) graphs I and III, only
(D) graphs I, II, and III

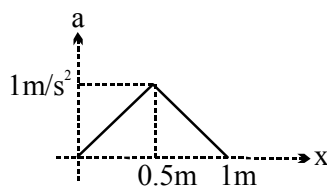
KM0074

28. Acceleration versus time graphs for four objects are shown below. All axes have the same scale. Which object had the greatest change in velocity during the interval?



KM0075

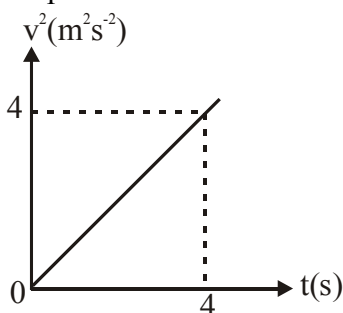
29. A body initially at rest, starts moving along x-axis in such a way so that its acceleration vs displacement plot is as shown in figure. The maximum velocity of particle is :-



- (A) 1 m/s
(B) 6 m/s
(C) 2 m/s
(D) none

KM0076

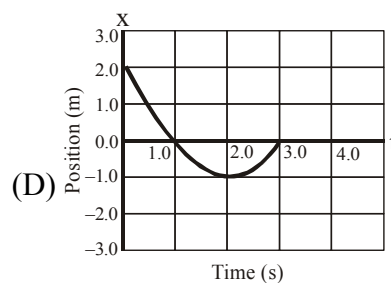
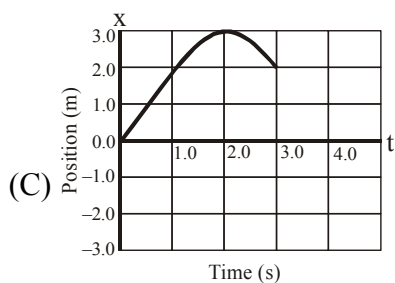
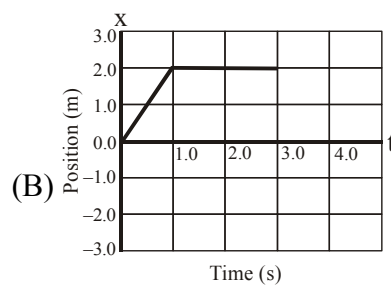
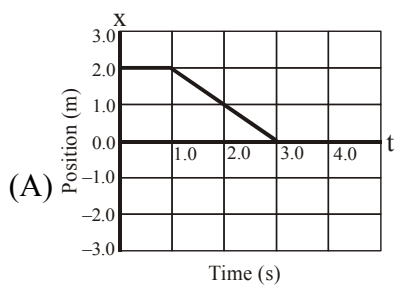
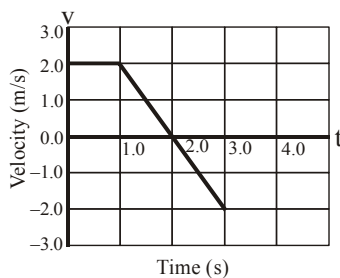
30. A particle is moving along a straight line such that square of its velocity varies with time as shown in the figure. What is the acceleration of the particle at $t = 4$ s?



- (A) 4 m/s²
(B) 1/4 m/s²
(C) 1/2 m/s²
(D) 0

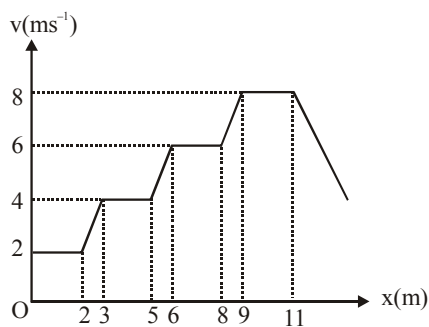
KM0077

31. The graph below shows the velocity of a particle moving in a straight line. At $t = 0$, the particle is located at $x = 0$. Which of the following graphs shows the position of the particle with respect to time, $x(t)$?



KM0078

32. The velocity of a particle that moves in the positive x -direction varies with its position as shown in figure. The acceleration of the particle when $x = 5.5$ m is-



(A) 0

(B) 5 ms^{-2} (C) 10 ms^{-2} (D) 20 ms^{-2}

KM0079

COMPREHENSION TYPE QUESTIONS

Paragraph for Question no. 33 and 34

A particle is moving in a straight line along positive y-axis. Its displacement from origin at any time t is given by $y = 5t^2 - 10t + 5$ where y is in meters and t is in seconds.

33. The velocity at $t = 2s$ will be :

(A) 20 ms^{-1} (B) 10 ms^{-1} (C) 5 ms^{-1} (D) 15 ms^{-1}

KM0080

34. Displacement of particle when its velocity is zero, is

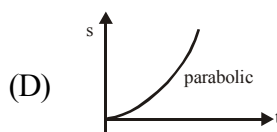
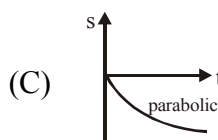
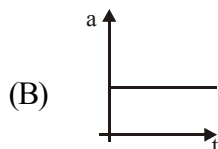
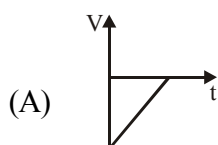
(A) 2.5 m (B) 1.25 m (C) 5m (D) 0m

KM0080

MATRIX MATCH TYPE QUESTION

35. v , a , s and t denote velocity, acceleration, displacement and time respectively. Match the columns :-

Column-I



Column-II

(P) Velocity of the particle is in positive direction, acceleration in negative direction

(Q) Both velocity and acceleration of the particle are in negative directions.

(R) Velocity of the particle is in negative direction and acceleration in positive direction

(S) Velocity and acceleration both in positive direction

(T) Acceleration is constant

KM0081

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. A parachutist jumps out of an airplane and accelerates with gravity for 6 seconds. He then pulls the parachute cord and after a 4 s deceleration period, descends at 10 m/s for 60 seconds, reaching the ground. From what height did the parachutist jump? Assume acceleration due to gravity to be 10 m/s^2 throughout the motion.

(A) 840 m (B) 920 m (C) 980 m (D) 1020 m

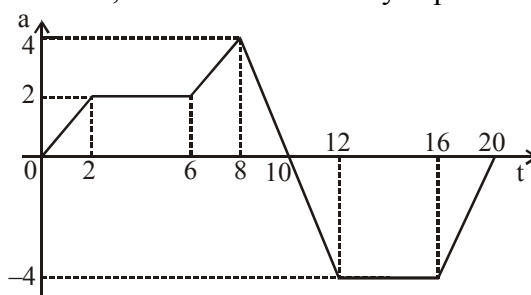
KM0082

2. A train moving with a speed of 60 km/hr is slowed down uniformly to 30 km/hr for repair purposes during running. After this it was accelerated uniformly to reach to its original speed. If the distance covered during constant retardation be 2 km and that covered during constant acceleration be 1 km, find the time lost in the above journey

(A) 1 min (B) 2 min (C) 4 min (D) 5 min

KM0083

3. If initial velocity of particle is 2 m/s, the maximum velocity of particle from $t = 0$ to $t = 20$ sec is :



(A) 20 m/s (B) 18 m/s (C) 22 m/s (D) 24 m/s

KM0084

ASSERTION & REASON

These questions contains, Statement-1 (assertion) and Statement-2 (reason).

- (A) Statement-1 is true, Statement-2 is true ; Statement-2 is correct explanation for Statement-1.
 (B) Statement-1 is true, Statement-2 is true ; Statement-2 is NOT a correct explanation for statement-1.
 (C) Statement-1 is true, Statement-2 is false.
 (D) Statement-1 is false, Statement-2 is true.
 (E) Both Statement-1 and Statement-2 are false.

4. **Statement I :** When velocity of a particle is zero then acceleration of particle is zero.
and

Statement II : Acceleration is equal to rate of change of velocity.

KM0085

- 5. Statement-I :** A particle moves in a straight line with constant acceleration. The average velocity of this particle cannot be zero in any time interval.

and

Statement-II : For a particle moving in straight line with constant acceleration, the average velocity in a time interval is $\frac{u+v}{2}$, where u and v are initial and final velocity of the particle of the given time interval.

KM0086

KM0086

6. A particle moves in a straight line, according to the law $x = 4a \left[t + a \sin \left(\frac{t}{a} \right) \right]$, where x is its position in meters, t in sec. & a is some constants, then the velocity is zero at :-
- (A) $x = 4a^2\pi$ meters (B) $t = \pi$ sec. (C) $t = 0$ sec (D) none

KM0087

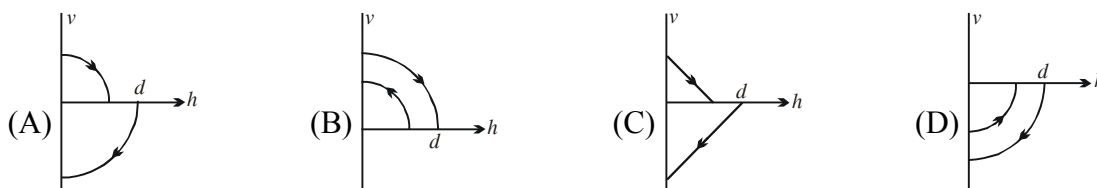
7. A particle moving on the x-axis with constant acceleration has displacements of 6 m from $t = 4$ s to $t = 7$ s and 3 m from $t = 5$ s to $t = 8$ s. The distance covered from $t = 6$ s to $t = 9$ s is
- (A) 1.75 m (B) 2.25 m (C) 3.0 m (D) 0

KM0088

8. A point moves in a straight line so that its displacement is x m at time t sec, given by $x^2 = t^2 + 1$. Its acceleration in m/s^2 at time t sec is :
- (A) $\frac{1}{x}$ (B) $\frac{1}{x} - \frac{1}{x^2}$ (C) $-\frac{t}{x^2}$ (D) $\frac{1}{x^3}$

KM0089

9. A ball is dropped vertically from a height d above the ground, hits the ground and bounces up vertically to a height $d/2$. Neglecting subsequent motion and air resistances, its velocity v varies with the height h above the ground as :- **[IIT-JEE'2000 (Scr)]**

**KM0090**

MULTIPLE CORRECT TYPE QUESTIONS

- 10.** A particle moving along a straight line with uniform acceleration has velocities 7m/s at A and 17 m/s at C. B is the mid point of AC. Then
- (A) The velocity at B is 12 m/s.
- (B) The average velocity between A and B is 10 m/s.
- (C) The ratio of the time to go from A to B to that from B to C is 3 : 2.
- (D) The average velocity between B and C is 15 m/s. **KM0091**
- 11.** A particle moves along the X-axis as $x = u(t-2s) + a(t-2s)^2$
- (A) The initial velocity of the particle is u
- (B) The acceleration of the particle is a
- (C) The acceleration of the particle is 2a
- (D) At $t = 2s$ particle is at the origin. **KM0092**

KM0092

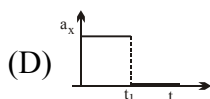
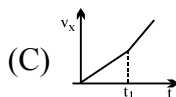
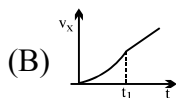
12. The position of a particle with time is given by

$$(x, y) = (8t^2, 3) \text{ for } t \leq t_1$$

$$= (8t_1, 3) \text{ for } t > t_1$$

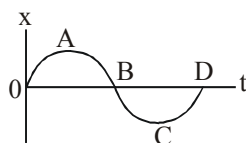
Choose the **CORRECT** alternative.

(A) Particle moves along a straight line parallel to x axis.



KM0093

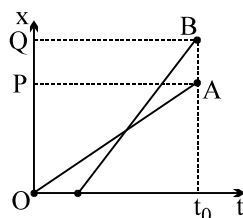
13. A particle has a rectilinear motion and the figure gives its displacement as a function of time. Which of the following statements are true with respect to the motion



- (A) in the motion between O and A the velocity is positive and acceleration is negative
 (B) between A and B the velocity and acceleration are positive
 (C) between B and C the velocity is negative and acceleration is positive
 (D) between C and D the acceleration is positive

KM0094

14. The position-time ($x-t$) graphs for two children A and B returning from their school O to their homes P and Q respectively along straight line path (taken as x axis) are shown in figure below. Choose the **CORRECT** statement (s):



- (A) A lives closer to the school than B
 (B) A starts from the school earlier than B
 (C) A and B have equal average velocities from 0 to t_0 .
 (D) B overtakes A on the way

KM0095

15. A ball is dropped from a building. Somewhere down it crosses a window of length 4 m in 0.5 sec. Speed of ball at top of window is v_1 and at bottom v_2 , then choose the **CORRECT** option(s) ($g = 10 \text{ m/s}^2$) :-

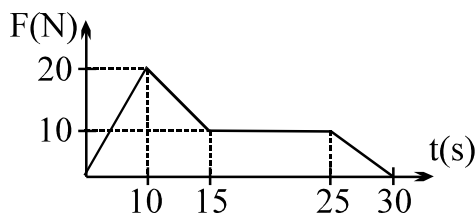
- (A) $v_2 - v_1 = 5 \text{ m/s}$ (B) $v_2 + v_1 = 16 \text{ m/s}$ (C) $\frac{v_2}{v_1} = 9$ (D) $\frac{v_2}{v_1} = \frac{21}{11}$

KM0096

COMPREHENSION TYPE QUESTIONS

Paragraph for Question Nos. 16 to 19

In figure shown, the graph shows the variation of a unidirectional force F acting on a body of mass 10 kg (in gravity free space), with time t . The velocity of the body at $t = 0$ is zero. (Area under F - t curve gives change in momentum)



16. The velocity of the body at $t = 30\text{ s}$ is
(A) 30 m/s (B) 20 m/s (C) 40 m/s (D) none
17. The power of the force at $t = 12\text{ s}$ is (Power = force $\times$ velocity)
(A) 225.0 W (B) 217.6 W (C) 226.7 W (D) none
18. The average acceleration of the body from $t = 0$ to $t = 15\text{ s}$ is :-
(A) 1.25 m/s^2 (B) $4/7\text{ m/s}^2$ (C) $5/6\text{ m/s}^2$ (D) $7/6\text{ m/s}^2$
19. The change in momentum of the body between the time $t = 10\text{ s}$ to 15 s is :-
(A) 100 kg.m/s (B) 75 kg.m/s (C) 125 kg.m/s (D) none

KM0097

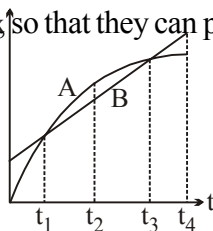
KM0097

KM0097

KM0097

Paragraph for Question nos. 20 to 23

The graph given shows the **POSITION** of two cars, A and B, as a function of time. The cars move along the x -axis on parallel but separate tracks, so that they can pass each other's position without colliding.



20. At which instant in time is car-A overtaking the car-B?
(A) t_1 (B) t_2 (C) t_3 (D) t_4
21. At time t_3 , which car is moving faster?
(A) car A (B) car B (C) same speed (D) None of these
22. At which instant do the two cars have the same velocity?
(A) t_1 (B) t_2 (C) t_3 (D) t_4
23. Which one of the following best describes the motion of car A as shown on the graphs?
(A) speeding up (B) constant velocity
(C) slowing down (D) first speeding up, then slowing down

KM0098

KM0098

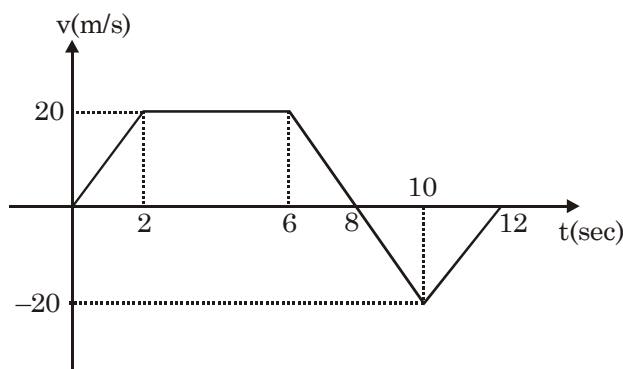
KM0098

KM0098

**MATCHING LIST TYPE ($4 \times 4 \times 4$) SINGLE OPTION CORRECT
(THREE COLUMNS AND FOUR ROWS)**

Answer Q.24, Q.25 and Q.26 by appropriately matching the information given in the three columns of the following table.

The velocity-time graph of an object moving along a straight line is given below.

**Column-I****Time interval**

(I) 0 to 2 sec

(II) 2 to 6 sec

(III) 0 to 10 sec

(IV) 6 to 12 sec

Column-II**Average velocity**

(i) 10 m/s

(ii) $-\frac{10}{3}$ m/s

(iii) 15 m/s

(iv) 20 m/s

Column-III**Average acceleration**

(P) zero

(Q) -2 m/s²(R) $-\frac{10}{3}$ m/s²(S) -4 m/s²

24. Which of the following combination is correctly matched :-

(A) (III) (iv) (S) (B) (III) (i) (R) (C) (III) (i) (S) (D) (III) (i) (Q)

KM0099

25. Which of the following combination is correctly matched :-

(A) (II) (iv) (P) (B) (I) (i) (Q) (C) (II) (iv) (Q) (D) (I) (i) (S)

KM0099

26. Which of the following combination is correctly matched :-

(A) (IV) (ii) (S) (B) (IV) (iii) (R) (C) (IV) (i) (S) (D) (IV) (ii) (R)

KM0099

MATRIX MATCH TYPE QUESTION

27. A balloon starts rising up from ground with constant net acceleration of 10 m/s^2 . After 2 s a particle drops from the balloon. After further 2 s match the following : ($g = 10\text{ m/s}^2$)

Column-I

(A) Height of particle from ground

(B) Speed of particle

(C) Displacement of Particle

(D) Acceleration of particle

Column-II

(P) Zero

(Q) 10 SI units

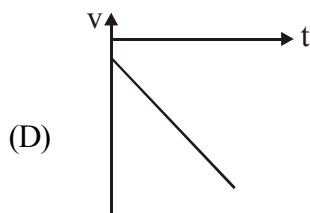
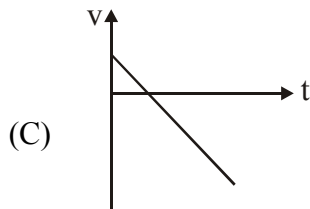
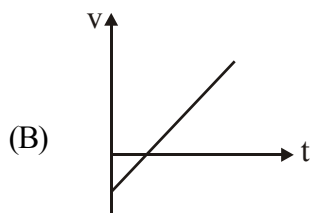
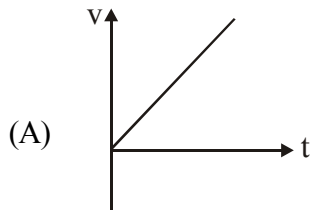
(R) 40 SI units

(S) 20 SI units

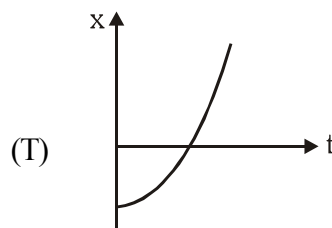
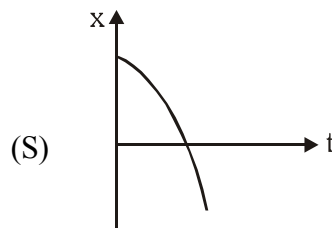
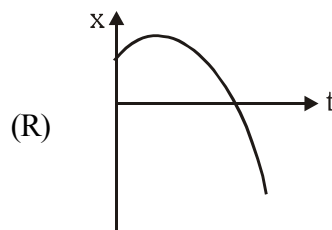
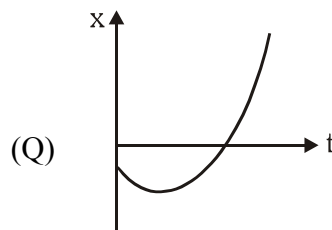
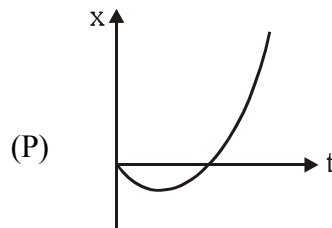
KM0100

28. In the first column of the given table, some velocity-time (v - t) graphs and in the second column some position-time (x - t) graphs are shown. Suggest suitable match or matches.

Column-I



Column-II



KM0101

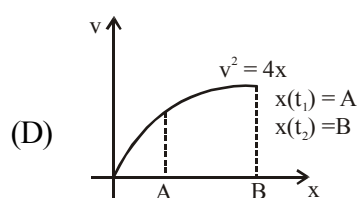
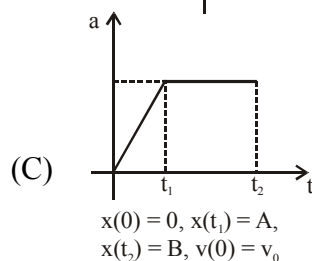
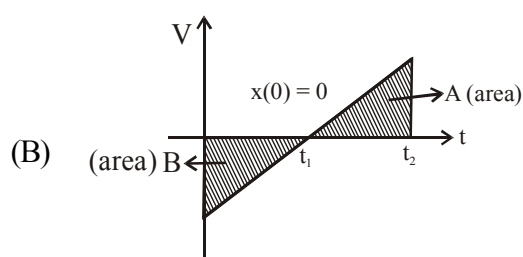
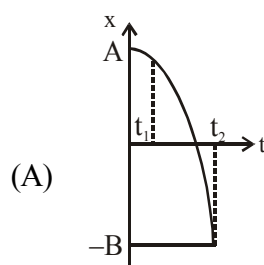
29. Match the column :-

Column-I : Shows graph of One Dimension motion of a particle. Symbols have their usual meaning such as $x(0)$ = initial position, $x(t_1)$ = position at $t = t_1$, $v(0)$ = initial velocity.

Column-II : Shows physical quantities. Displacement and distance are asked for $0 < t < t_2$, and average values are asked for $0 < t < t_2$

V_0, A & B are positive constant

Column-I



Column-II

(P) |Displacement| = Distance

(Q) |Instantaneous velocity| = |Instantaneous speed|

(R) |Average velocity| ≤ Average speed

(S) Instantaneous acceleration = Average acceleration

(T) Displacement = $A - B$ and Distance = $A + B$

KM0102

EXERCISE (J-M)

1. An object, moving with a speed of 6.25 m/s, is decelerated at a rate given by

$$\frac{dv}{dt} = -2.5\sqrt{v}$$

where v is the instantaneous speed. The time taken by the object, to come to rest, would be :-

[AIEEE-2011]

- (1) 4 s (2) 8 s (3) 1 s (4) 2 s

KM0103

2. From a tower of height H , a particle is thrown vertically upwards with a speed u . The time taken by the particle, to hit the ground, is n times that taken by it to reach the highest point of its path. The relation between H , u and n is :

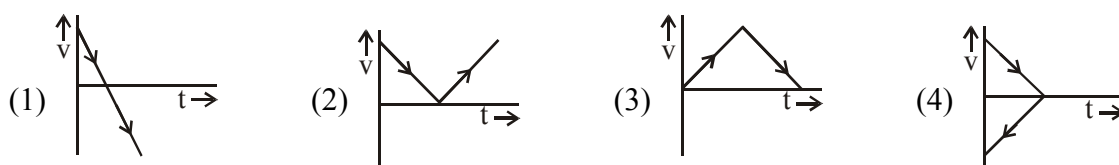
[JEE-Main-2014]

- (1) $2gH = nu^2(n-2)$ (2) $gH = (n-2)u^2$ (3) $2gH = n^2u^2$ (4) $gH = (n-2)^2u^2$

KM0104

3. A body is thrown vertically upwards. Which one of the following graphs correctly represent the velocity vs time?

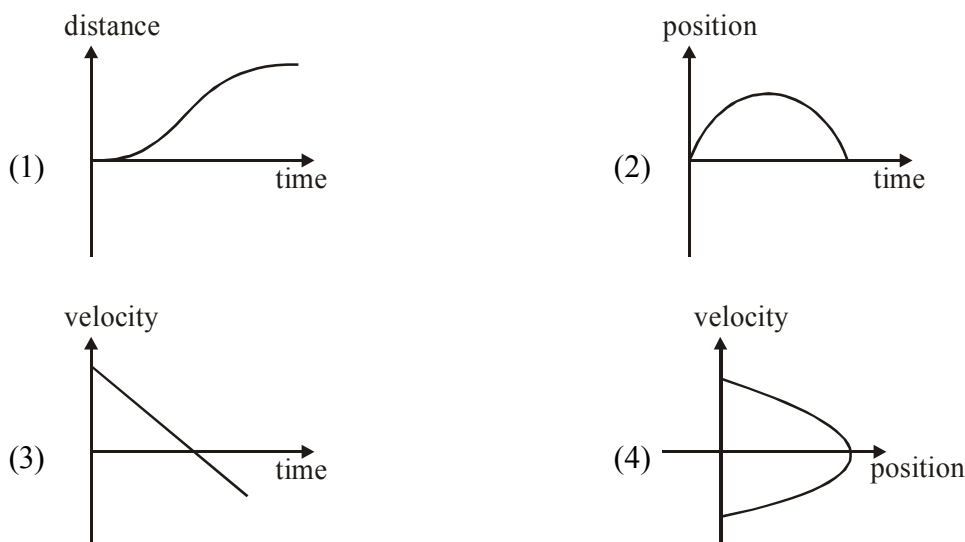
[JEE-Main-2017]



KM0105

4. All the graphs below are intended to represent the same motion. One of them does it incorrectly. Pick it up.

[JEE-Main-2018]



KM0106

EXERCISE (J-A)

1. Consider an expanding sphere of instantaneous radius R whose total mass remains constant. The expansion is such that the *instantaneous* density ρ remains uniform throughout the volume. The rate of fractional change in density $\left(\frac{1}{\rho} \frac{d\rho}{dt}\right)$ is constant. The velocity v of any point on the surface of the expanding sphere is proportional to :
- [JEE Advanced-2017]**

- (A) R^3 (B) $\frac{1}{R}$ (C) R (D) $R^{2/3}$

KM0107

ANSWER KEY

EXERCISE (S-1)

1. Ans. (a) $2\sqrt{2} \frac{v}{\pi}$, (b) $\frac{2v}{\pi}$, (c) $\frac{2\sqrt{2}v}{3\pi}$

2. Ans.

v_0	a
—	+

 $\leftarrow \text{vel}$

Because particle is slowing down so velocity & acceleration are in opposite direction.

3. Ans. 3

4. Ans. 260 m

5. Ans. (a) 4 m/s², (b) 200, (c) 1000 m

6. Ans. 7

7. Ans. 18.5 m

8. Ans. Zero

9. Ans. 2 sec. after body is dropped

10. Ans. $\frac{45}{4}$ m

11. Ans. 7

12. Ans. 50

13. Ans. 30

14. Ans. 5h/3

15. Ans. 4

16. Ans. 6

17. Ans. (a) 27 m/s, (b) 108 m, (c) -100 m, (d) 316 m

18. Ans. 1 kg m/sec.

19. Ans. 186.25 N

20. Ans. 2π rad/sec.

21. Ans. 23 amp

22. Ans. 13 rad/s

23. Ans. -72, -16, 24

24. Ans. (a) 0, 8 sec (b) 0, 4, 8 sec

25. Ans. 1.815×10^5 ergs.

26. Ans. 100 kgm/s

27. Ans. (a) 7, $\frac{11}{2}$ (b) 3, -3

28. Ans. 3

29. Ans. 2

30. Ans. $(27\hat{i} + 2\hat{k}) \text{ m/s}^2$

31. Ans. 3

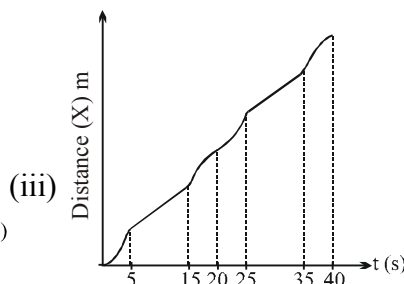
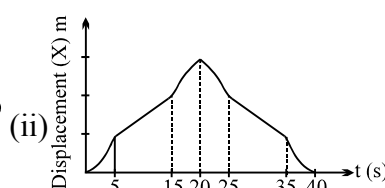
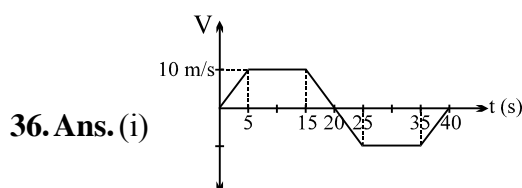
32. Ans. 3

33. Ans. (a) 3 m; (b) -3 m/s²; (c) 1m/s; (d) 3/2 m/s

34. Ans. (i) section (a) as slope = $v = \frac{dx}{dt}$ is positive and increasing.

(ii) section (d) as slope = $v = \frac{dx}{dt}$ is negative and constant.

35. Ans. (a) -2 ms⁻¹ (b) 5 m/s



EXERCISE (S-2)

1. Ans. 25 m 2. Ans. 42 km/hr 3. Ans. 5 m
4. Ans. (a) $\sqrt{60}$ m/sec (b) $\sqrt{\frac{3}{5}}$ sec (c) $\frac{3}{\sqrt{60}}$ sec (d) $\frac{9}{4}$ m 5. Ans. 12 km
6. Ans. 80 m/sec 7. Ans. (i) $t = \frac{\sqrt{x^2 + (9)^2}}{4} + \frac{15-x}{5}$ (ii) 3 km from the camp.
8. Ans. 10 sec ; 700m 9. Ans. 2 10. Ans. 012 11. Ans. 7

EXERCISE (O-1)

1. Ans. (B) 2. Ans. (D) 3. Ans. (C) 4. Ans. (C) 5. Ans. (A) 6. Ans. (A)
7. Ans. (C) 8. Ans. (B) 9. Ans. (D) 10. Ans. (B) 11. Ans. (B) 12. Ans. (A)
13. Ans. (B) 14. Ans. (D) 15. Ans. (B) 16. Ans. (D) 17. Ans. (C) 18. Ans. (A)
19. Ans. (D) 20. Ans. (B) 21. Ans. (C) 22. Ans. (C) 23. Ans. (C) 24. Ans. (D)
25. Ans. (D) 26. Ans. (C) 27. Ans. (D) 28. Ans. (D) 29. Ans. (A) 30. Ans. (B)
31. Ans. (C) 32. Ans. (C) 33. Ans. (B) 34. Ans. (D)
35. Ans. (A) $\rightarrow$ (R,T) ; (B) $\rightarrow$ (T) ; (C) $\rightarrow$ (R, T) ; (D) $\rightarrow$ (S, T)

EXERCISE (O-2)

1. Ans. (B) 2. Ans. (A) 3. Ans. (C) 4. Ans. (D) 5. Ans. (D) 6. Ans. (A)
7. Ans. (B) 8. Ans. (D) 9. Ans. (A) 10. Ans. (B,C,D) 11. Ans. (C,D)
12. Ans. (A,D) 13. Ans. (A,C,D) 14. Ans. (A,B,D) 15. Ans. (A,B,D)
16. Ans. (A) 17. Ans. (B) 18. Ans. (D) 19. Ans. (B) 20. Ans. (A) 21. Ans. (B)
22. Ans. (B) 23. Ans. (C) 24. Ans. (D) 25. Ans. (A) 26. Ans. (D)
27. Ans. (A) - (R); (B) - (P); (C) - (S); (D) - (Q)
28. Ans. (A) $\rightarrow$ (T); (B) $\rightarrow$ (P,Q); (C) $\rightarrow$ (R); (D) $\rightarrow$ (S)
29. Ans. (A) $\rightarrow$ (P,Q,R) ; (B) $\rightarrow$ (Q,R,S,T) ; (C) $\rightarrow$ (P,Q,R) ; (D) $\rightarrow$ (P,Q,R,S)

EXERCISE (JM)

1. Ans. (4) 2. Ans. (1) 3. Ans. (1) 4. Ans. (1)

EXERCISE (JA)

1. Ans. (C)



Chapter Contents

04

KINEMATICS-2D

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Important Notes

[illegible]

KINEMATICS-2D

KEY CONCEPT

MOTION IN TWO AND THREE DIMENSIONS :

When a particle is moving in space then its motion can be broken up in three co-ordinate axis (x, y & z). The motion in these three directions is governed only by velocity & acceleration in that particular direction and is totally independent of the velocities and acceleration in other directions. Lets say a particle is moving in space

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

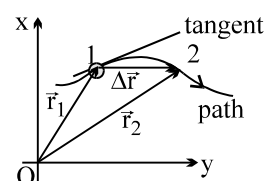
Gives position of particle in space.

VELOCITY

Using the language of calculus, we may write $\vec{v}$ as the derivative

$$\vec{v} = \frac{d\vec{r}}{dt}$$

$$\vec{v} = \frac{d\vec{r}}{dt} = \frac{d}{dt}(x\hat{i} + y\hat{j} + z\hat{k}) = \left(\frac{dx}{dt}\right)\hat{i} + \left(\frac{dy}{dt}\right)\hat{j} + \left(\frac{dz}{dt}\right)\hat{k}$$



where the scalar components of $\vec{v}$ are

$$v_x = \frac{dx}{dt}, v_y = \frac{dy}{dt}, v_z = \frac{dz}{dt}$$

Differentiating $\vec{r}$ w.r.t. time gives us velocity vector of particle at that time.

ACCELERATION

Similarly, if we differentiate $\vec{v}$ w.r.t. time we get acceleration of particle $\vec{a} = \frac{d\vec{v}}{dt}$

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k}$$

where the scalar components of $\vec{a}$ are

$$a_x = \frac{dv_x}{dt}, a_y = \frac{dv_y}{dt}, a_z = \frac{dv_z}{dt}$$

Now, collecting equations of motion relating to x & y axes separately

x-axis

$$v_x = \frac{dx}{dt}$$

$$a_x = \frac{dv_x}{dt}$$

y-axis

$$v_y = \frac{dy}{dt}$$

$$a_y = \frac{dv_y}{dt}$$

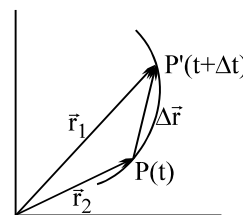
Thus we can see that motion in plane is composed of two straight line motions. **These motions are completely independent of each other.** Only thing connecting them is fact that they are occurring simultaneously.

Velocity is along tangent of path

The direction of the instantaneous velocity $\vec{v}$ of a particle is always tangent to

the particle's path at the particle position. $\vec{v} = \lim_{\Delta t \rightarrow 0} \frac{\Delta \vec{r}}{\Delta t}$

The result is the same in three dimensions:



Ex. A particle with velocity $\vec{v}_0 = -2\hat{i} + 4\hat{j}$ (in meters per second) at $t = 0$ undergoes a constant acceleration $\vec{a}$ of magnitude $a = 3 \text{ m/s}^2$ at an angle $\theta = 127^\circ$ from the positive direction of the x axis. What is the particle's velocity $\vec{v}$ at $t = 5 \text{ sec}$, in unit vector notation?

Sol. We know that $\vec{v} = \vec{v}_0 + \vec{a}t$

now $v_x = v_{0x} + a_x t$ and $v_y = v_{0y} + a_y t$

$$a_x = a \cos \theta = (3 \text{ m/s}^2)(\cos 127^\circ) = -1.80 \text{ m/s}^2$$

$$a_y = a \sin \theta = (3 \text{ m/s}^2)(\sin 127^\circ) = +2.40 \text{ m/s}^2$$

at time $t = 5 \text{ sec}$

$$v_x = -2 \text{ m/s} + (-1.80 \text{ m/s}^2)(5 \text{ sec}) = -11 \text{ m/s}$$

$$v_y = 4 \text{ m/s} + (2.40 \text{ m/s}^2)(5 \text{ sec}) = 16 \text{ m/s}$$

Thus, at $t = 5 \text{ sec}$,

$$\vec{v} = (-11 \text{ m/s})\hat{i} + (16 \text{ m/s})\hat{j} \quad \text{Ans.}$$

Ex. A particle moves in the x - y plane according to the law $x = at$; $y = at(1 - \alpha t)$ where a and α are positive constants and t is time. Find the velocity and acceleration vector. Also find the moment t_0 at which the velocity vector forms angle of 90° with acceleration vector.

Sol. $V_x = a$; $V_y = a - 2a\alpha t \Rightarrow \vec{V} = a\hat{i} + (a - 2a\alpha t)\hat{j}$

$$a_x = 0; a_y = -2a\alpha \Rightarrow \vec{a} = -2a\alpha\hat{j}$$

$$\text{for } 90^\circ, \quad \vec{V} \cdot \vec{a} = 0$$

$$-2a\alpha(a - 2a\alpha t_0) = 0$$

$$1 - 2\alpha t_0 = 0 \Rightarrow t_0 = 1/(2\alpha) \text{ sec.}$$

PROJECTILE MOTION

We next consider a special case of two-dimensional motion: A particle moves in a vertical plane with some initial velocity $\vec{v}_0$ but its acceleration is always the freefall acceleration $\vec{g}$, which is downward. Such a particle is called a projectile (meaning that it is projected or launched) and its motion is called **projectile motion**.

Assumptions:-

Particle remains close to earth's surface, so acceleration due to gravity remains constant.

Air resistance is neglected.

Distance that projectile travels is small so that earth can be treated as plane surface.

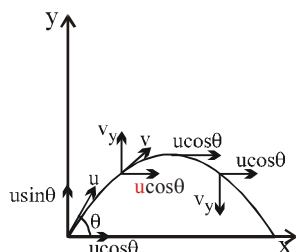
Two straight line motions:-

Our goal here is to analyse projectile motion using the tools for two dimensional motion. This feature allows us to break up a problem involving two dimensional motion into two separate and easier one-dimensional problems,

(a) **The horizontal motion is motion with uniform velocity (no effect of gravity)**

(b) **The vertical motion is motion of uniform acceleration, or freely falling bodies.**

Note: In projectile motion, the horizontal motion and the vertical motion are independent of each other, that is either motion does not affects the other.



Treating as two straight line motions:-

The horizontal Motion(x axis):

Because there is no acceleration in the horizontal direction, the horizontal component v_x of the projectile velocity remains unchanged from its initial value v_{0x} throughout the motion,

The vertical motion(y axis):

The vertical motion is the motion we discussed for a particle in free fall.

As is illustrated in figure, the vertical component behaves just as for a ball thrown vertically upward. It is directed upward initially and its magnitude steadily decreasing to zero, at the maximum height of the path. The vertical velocity component then reverses direction, and its magnitude becomes larger with time.

x-axis

Initial velocity(u_x) = $u \cos \theta$

acceleration(a_x) = 0

Thus, velocity after time t

$v_x = u \cos \theta$

Displacement after time t

$x = u \cos \theta t$

Resultant velocity

$$(\vec{V}_R) = (u \cos \theta)\hat{i} + (u \sin \theta - gt)\hat{j}$$

$$|\vec{V}_R| = \sqrt{u^2 \cos^2 \theta + (u \sin \theta - gt)^2}$$

$$\& \tan \alpha = \frac{u \sin \theta - gt}{u \cos \theta}$$

where α is angle that velocity vector makes with horizontal. Also known as direction or angle of motion

y-axis

Initial velocity(u_y) = $u \sin \theta$

acceleration(a_y) = $-g$

Thus, velocity after time t

$v_y = u \sin \theta - gt$

Displacement after time t

$y = u \sin \theta t - gt^2/2$

Time of flight(T)

$$T = \frac{2u \sin \theta}{g}$$

Considering vertical motion

$$s_y = 0; u_y = u \sin \theta; a_y = -g$$

$$0 = u \sin \theta T - gT^2/2 \Rightarrow T = \frac{2u \sin \theta}{g}$$

Maximum Height(H)

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

Vertical velocity at maximum height $v_y = 0$

$$0 = u^2 \sin^2 \theta - 2gH \Rightarrow H = \frac{u^2 \sin^2 \theta}{2g}$$

Horizontal Range(R)

$$R = \frac{u^2 \sin 2\theta}{g} = \frac{2u_x u_y}{g}$$

$$\text{Total time } T = \frac{2u \sin \theta}{g}$$

Velocity in horizontal direction $u_x = u \cos \theta$ Total displacement in horizontal direction $R = u \cos \theta T$

$$R = \frac{u^2 \sin 2\theta}{g}$$

Note:- For complementary angles i.e. $\theta + \alpha = 90^\circ$, the range is same for same projection speed but maximum height and time of flight are different.

Ex. A body is thrown with initial velocity 10m/sec. at an angle 37° from horizontal. Find
(Take : $g = 10\text{m/s}^2$)

(i) Time of flight

(ii) Maximum height.

(iii) Range

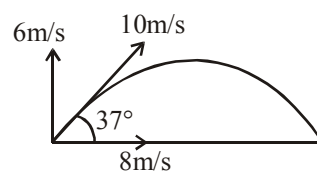
(iv) Position vector at $t = 1$ sec.

Ans. (i) 1.2 sec, (ii) 1.8 m, (iii) 9.6 m, (iv) $(8\hat{i} + \hat{j})$

Sol. (i) Time of flight $T = \frac{2u \sin \theta}{g} = \frac{2 \times 10 \times \frac{3}{5}}{10} = \frac{6}{5} = 1.2 \text{ sec}$

(ii) Maximum height $H = \frac{u^2 \sin^2 \theta}{2g} = \frac{100 \left(\frac{9}{25} \right)}{2 \times 10} = \frac{9}{5} = 1.8 \text{ m}$

(iii) $R = \frac{u^2 \sin 2\theta}{g} = \frac{100 \times 2 \times \frac{3}{5} \times \frac{4}{5}}{10} = \frac{240}{25} = 9.6 \text{ m}$



(iv) $x = 8 \times 1 = 8 \text{ m}$

$$y = 6 \times 1 - \frac{1}{2} \times 10 \times (1)^2 = 1 \text{ m}$$

$$\vec{r} = 8\hat{i} + \hat{j}$$

Maximum Range

$$R = \frac{u^2 \sin 2\theta}{g}$$

for $\theta = 45^\circ$, R is maximum

$$R_{\max} = \frac{u^2}{g}$$

Ex. A person can throw a ball vertically upto maximum height of 20 m. How far can he throw the ball.

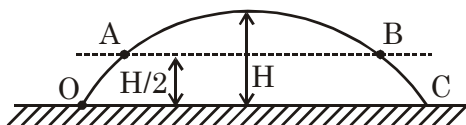
Sol. $H_{\max} = \frac{u^2}{2g}$

$$\therefore u = 20 \text{ m/s}$$

$$R_{\max} = \frac{u^2}{g} = 40 \text{ m}$$

Ex. A particle is projected with a speed u at an angle θ with horizontal. Find the average velocity of projectile for the period during which it crosses half of maximum height.

Sol.



avg. velocity is a vector

Let t_1 is time taken by particle to travel from A to B

First we will find vertical component.

For motion along AB

$$(\vec{V}_y)_{\text{avg}} = \frac{\text{Total Displacement along y direction}}{\text{Total time}} = \frac{0}{t_1} = 0$$

For Horizontal component

$$(\vec{V}_x)_{\text{avg}} = \frac{\text{Total Displacement along x direction}}{\text{Total time}}$$

$$(\vec{V}_x)_{\text{avg}} = \frac{(u \cos \theta)t_1}{t_1} = u \cos \theta$$

So average velocity for motion along AB is $u \cos \theta$

EQUATION OF TRAJECTORY

Lets say point of projection is our origin and horizontal direction is x-axis and vertically upwards is positive y-axis.

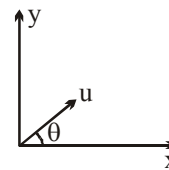
We know $x = u \cos \theta t$

$$\therefore t = \frac{x}{u \cos \theta} \quad \dots (1)$$

$$\text{also } y = u \sin \theta t - \frac{1}{2} g t^2 \quad \dots (2)$$

Putting value of 't' from eq. (1) in eq. (2), we get

$$y = x \tan \theta - \frac{g x^2}{2 u^2 \cos^2 \theta}$$



Ex. A particle is projected with a velocity 10 m/s at an angle 37° to the horizontal. Find the location at which the particle is at a height 1m from point of projection. ($g = 10 \text{ m/s}^2$)

Ans. 1.6 m, 8 m.

Sol. $y = x \tan \theta - \frac{g x^2}{2 u^2 \cos^2 \theta}$
for $y = 1$; $\theta = 37^\circ$; $u = 10 \text{ m/s}$

$$1 = \frac{3}{4} x - \frac{10 x^2}{2 \times 100 \times \left(\frac{16}{25}\right)}$$

$$1 = \frac{3}{4} x - \frac{5}{64} x^2$$

$$5x^2 - 48x + 64 = 0$$

$$5x^2 - 40x - 8x + 64 = 0$$

$$x = 8\text{m}, 1.6\text{m}$$

Ex. We have a hose pipe which disposes water at the speed of 10 ms^{-1} . The safe distance from a building on fire, on ground is 5 m. How high can this water go? (take : $g = 10 \text{ ms}^{-2}$)

Sol. Here we must understand that taking range of projectile as 10m and making projectile hit the building when it is at maximum height is wrong. By doing this we are not achieving maximum y for given $x = 5\text{m}$. This just makes highest point of path to lie on $x = 5$, But there may be other path for which y will be maximum for given x. This problem will be solved by using equation of trajectory by putting $x = 5\text{m}$ and maximising y by varying θ .

$$y = x \tan \theta - \frac{g x^2}{2 u^2 \cos^2 \theta}$$

Putting $x = 5\text{m}$ we get

$$y = 5 \tan \theta - \frac{10 \times 25 \sec^2 \theta}{2 \times 100}$$

$$5 \tan^2 \theta - 20 \tan \theta + (4y + 5) = 0$$

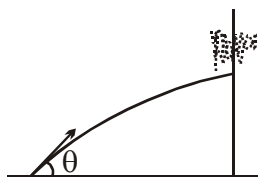
for real roots discriminant must be positive.

$$400 - 4 \times 5 (4y + 5) > 0$$

Solving $3.75 \geq y$

hence maximum $y = 3.75 \text{ m}$

If we have taken range as 10 m then angle of projection will be $\theta = 45^\circ$ corresponding maximum height $H = 2.5\text{m}$ which is smaller than our answer.



- Ex.** A projectile is fired horizontally with a velocity of 98 m/s from the top of a hill 490 m high. Find
 (i) Time to reach ground
 (ii) The horizontal distance from foot of hill to ground.
 (iii) The speed with which it hits the ground.

Ans. (i) 10 sec, (ii) 980 m, (iii) $98\sqrt{2}$ m/s

Sol. $u_y = 0$; $u_x = 98$ m/s
 on y-axis

$$s = ut + \frac{1}{2} at^2$$

$$-490 = 0 - \frac{1}{2} \times 9.8 \times t^2$$

$$t = \sqrt{\frac{490 \times 2}{9.8}} = 10 \text{ sec}$$

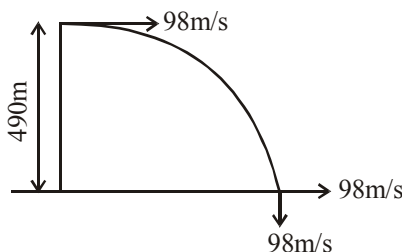
on x-axis for horizontal displacement,

$$R = u_x t = 98 \times 10 = 980 \text{ m}$$

$$v_x = 98 \text{ m/s}$$

$$v_y = 0 - 9.8 \times 10 = -98 \text{ m/s}$$

$$\text{So speed} = 98\sqrt{2} \text{ m/s}$$



- Ex.** A ball is thrown from the top of a tower with an initial velocity of 10m/s at an angle 37° above the horizontal, hits the ground at a distance 16m from the base of tower. Calculate height of tower.
 [$g = 10 \text{ m/s}^2$]

Ans. 8 m

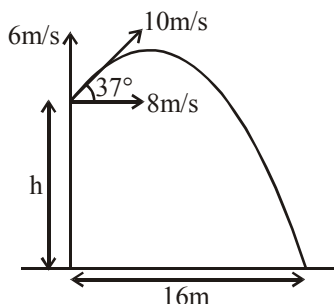
Sol. Time to reach ground is $= \frac{16}{8} = 2 \text{ sec}$

on y-axis (for height of tower)

$$-h = 6 \times 2 - \frac{1}{2} \times 10 \times (2)^2$$

$$-h = 12 - 22$$

$$h = 8 \text{ m}$$



- Ex.** Prithvi missile is fired to destroy an enemy military base situated on same horizontal level, situated 99 km away. The missile rises vertically for 1 km & then for remainder of flight, it follows parabolic path like a free body under earth's gravity, at an angle of 45° . Calculate its velocity at beginning of parabolic path. ($g = 10 \text{ ms}^{-2}$)

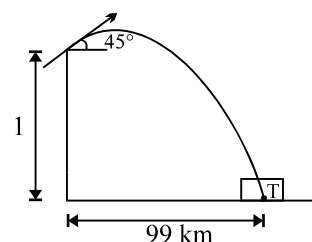
Sol. for horizontal motion time t

$$t = \frac{99 \times 10^3}{u \cos 45^\circ}$$

for vertical

$$-1 \times 10^3 = u \sin 45^\circ t - \frac{1}{2} \times 10 \times t^2$$

$$1 \times 10^3 + \frac{u \sin 45^\circ}{u \sin 45^\circ} \times 99 \times 10^3 = \frac{10}{2} \times \frac{(99 \times 10^3)^2 \times 2}{u^2}$$

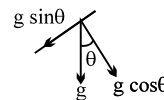
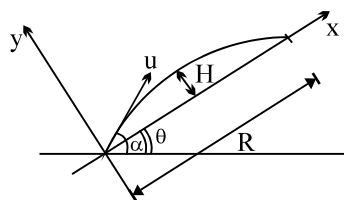


$$u^2 = \frac{(99 \times 10^3)^2 \times 10}{100 \times 10^3}$$

$$u = 99 \times 10^3 \sqrt{\frac{1}{10^4}} = 990 \text{ ms}^{-1}$$

PROJECTION ON INCLINED PLANE

There is an inclined plane making an angle θ with horizontal. A particle is projected at an angle α from horizontal.



x-axis

$$u_x = u \cos (\alpha - \theta)$$

$$a_x = -g \sin \theta$$

vel. at any time t

$$v_x = u \cos (\alpha - \theta) - g \sin \theta t$$

Time of flight

Displacement in y direction $s_y = 0$

$$0 = u \sin (\alpha - \theta) T - \frac{1}{2} g \cos \theta T^2$$

$$T = \frac{2 u \sin (\alpha - \theta)}{g \cos \theta}$$

Maximum distance of particle from inclined plane

Point where $v_y = 0$ is max.-distance from incline plane

$$(0)^2 = u^2 \sin^2 (\alpha - \theta) - 2 g \cos \theta H$$

$$H = \frac{u^2 \sin^2 (\alpha - \theta)}{2 g \cos \theta}$$

Range along the inclined plane

$$s_x = u_x T + \frac{1}{2} a_x T^2$$

$$R = \frac{u \cos (\alpha - \theta) \times 2 u \sin (\alpha - \theta)}{g \cos \theta} - \frac{g \sin \theta \times 2 \times 2 u^2 \sin^2 (\alpha - \theta)}{2 g^2 \cos^2 \theta}$$

$$= \frac{2 u^2 \sin (\alpha - \theta) [\cos (\alpha - \theta) \cos \theta - \sin \theta \sin (\alpha - \theta)]}{g \cos^2 \theta}$$

$$R = \frac{2u^2 \sin(\alpha - \theta) \cos \alpha}{g \cos^2 \theta}$$

$$R = \frac{u^2 [\sin(2\alpha - \theta) - \sin \theta]}{g \cos^2 \theta}$$

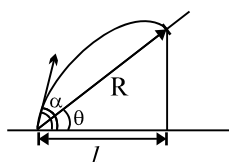
Alternate Method

$$l = u \cos \alpha T$$

$$R = \frac{l}{\cos \theta}$$

$$R = \frac{u \cos \alpha}{\cos \theta} \times \frac{2u \sin(\alpha - \theta)}{g \cos \theta}$$

$$R = \frac{2u^2 \sin(\alpha - \theta) \cos \alpha}{g \cos^2 \theta}$$



Note : Presence of incline plane does not affect the path of projectile in any way.

Maximum Range :

$$R = \frac{u^2 [\sin(2\alpha - \theta) - \sin \theta]}{g \cos^2 \theta}$$

$$\text{For max. range } 2\alpha - \theta = \frac{\pi}{2} \Rightarrow \alpha = \frac{\pi}{4} + \frac{\theta}{2}$$

$$\text{so } R_{\max} = \frac{u^2}{g(1 + \sin \theta)}$$

Projection from top of incline plane:

Incline plane is at an angle θ with horizontal and a particle is projected at an angle α from horizontal.

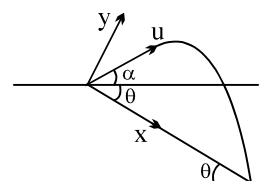
In all formulae replace θ with $-\theta$

$$H = \frac{u^2 \sin^2(\alpha + \theta)}{2g \cos \theta}$$

$$T = \frac{2u \sin(\alpha + \theta)}{g \cos \theta}$$

$$R = \frac{2u^2 \sin(\alpha + \theta) \cos \alpha}{g \cos^2 \theta}$$

$$R_{\max} = \frac{u^2}{g(1 - \sin \theta)} \text{ and } \alpha = \frac{\pi}{4} - \frac{\theta}{2}$$



Note : If a particle strikes the incline plane $\perp$ then its comp. of velocity along incline must be zero.

Ex. A particle is projected horizontally with a speed u from the top of a plane inclined at an angle θ with the horizontal. How far along the plane, from the point of projection will particle strike the plane?

Sol. **x-axis**

$$u_x = u$$

$$a_x = 0$$

$$x = ut$$

$$\Rightarrow y = \frac{gx^2}{2u^2}$$

$$\text{also } \frac{y}{x} = \tan \theta \Rightarrow x \tan \theta = \frac{gx^2}{2u^2}$$

$$x = 0, \frac{2u^2 \tan \theta}{g}$$

$$x = \frac{2u^2 \tan \theta}{g}$$

$$\Rightarrow y = \frac{2u^2 \tan^2 \theta}{g}$$

$$\text{dist. } l = \sqrt{x^2 + y^2}$$

$$l = \frac{2u^2 \tan \theta \sec \theta}{g}$$

Alternate Method.

$$R = \frac{2u^2 \sin(\alpha + \theta) \cos \alpha}{g \cos^2 \theta} \text{ as } \alpha = 0$$

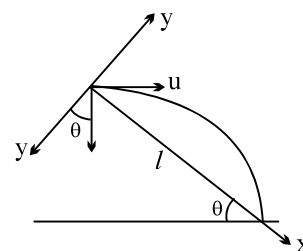
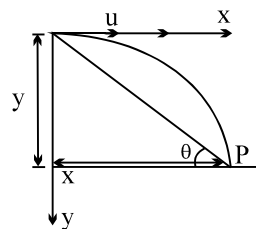
$$R = 2u^2 \tan \theta \sec \theta$$

y-axis

$$u_y = 0$$

$$a_y = g$$

$$y = \frac{gt^2}{2}$$



Ex. A particle is projected up an inclined plane. Plane is inclined at an angle θ with horizontal and particle is projected at an angle α with horizontal. If particle strikes the plane horizontally prove that $\tan \alpha = 2 \tan \theta$

Sol. We know time of flight

$$T = \frac{2u \sin(\alpha + \theta)}{g \cos \theta}$$

considering vertical motion

$$u = v \sin \alpha$$

$$a = -g$$

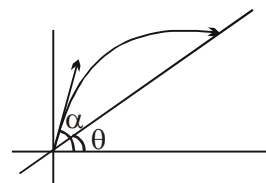
$$v = 0$$

$$\therefore T = \frac{u \sin \alpha}{g} = \frac{2u \sin(\alpha - \theta)}{g \cos \theta}$$

$$\sin \alpha \cos \theta = 2 \sin \alpha \cos \theta - 2 \cos \alpha \sin \theta$$

$$2 \cos \alpha \sin \theta = \sin \alpha \cos \theta$$

$$2 \tan \theta = \tan \alpha$$

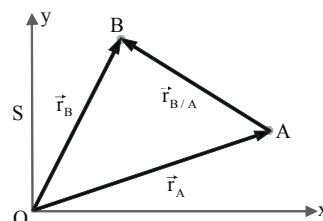


Relative Motion

Motion of a body can only be observed, when it changes its position with respect to some other body. In this sense, motion is a relative concept. To analyze motion of a body say A, therefore we have to fix our reference frame to some other body say B. The result obtained is motion of body A relative to body B.

Relative position, Relative Velocity and Relative Acceleration

Let two bodies represented by particles A and B at positions defined by position vectors $\vec{r}_A$ and $\vec{r}_B$, moving with velocities $\vec{v}_A$ and $\vec{v}_B$ and accelerations $\vec{a}_A$ and $\vec{a}_B$ with respect to a reference frame S. For analyzing motion of terrestrial bodies the reference frame S is fixed with the ground.



The vectors $\vec{r}_{B/A}$ denotes position vector of B relative to A. Following triangle law of vector addition, we have

$$\vec{r}_B = \vec{r}_A + \vec{r}_{B/A} \quad \dots(i)$$

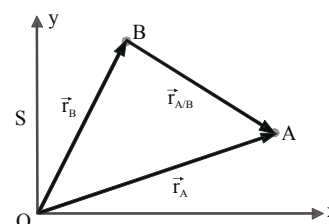
First derivatives of $\vec{r}_A$ and $\vec{r}_B$ with respect to time equals to velocity of particle A and velocity of particle B relative to frame S and first derivative of $\vec{r}_{B/A}$ with respect to time defines velocity of B relative to A.

$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A} \quad \dots(ii)$$

Second derivatives of $\vec{r}_A$ and $\vec{r}_B$ with respect to time equals to acceleration of particle A and acceleration of particle B relative to frame S and second derivative of $\vec{r}_{B/A}$ with respect to time defines acceleration of B relative to A.

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A} \quad \dots(iii)$$

In similar fashion motion of particle A relative to particle B can be analyzed with the help of adjoining figure. You can observe in the figure that position vector of A relative to B is directed from B to A and therefore



$$\vec{r}_{B/A} = -\vec{r}_{A/B}, \quad \vec{v}_{B/A} = -\vec{v}_{A/B} \quad \text{and} \quad \vec{a}_{B/A} = -\vec{a}_{A/B}.$$

The above equations elucidate that how a body A appears moving to another body B is opposite to how body B appears moving to body A.

Ex. A man when standstill observes the rain falling vertically and when he walks at 4 km/h he has to hold his umbrella at an angle of 53° from the vertical. Find velocity of the raindrops.

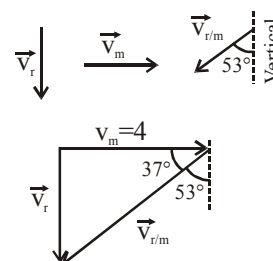
Sol. Assigning usual symbols $\vec{v}_m$, $\vec{v}_r$ and $\vec{v}_{r/m}$ to velocity of man, velocity of rain and velocity of rain

relative to man, we can express their relationship by the following eq. $\vec{v}_r = \vec{v}_m + \vec{v}_{r/m}$

The above equation suggests that a standstill man observes velocity $\vec{v}_r$ of rain relative to the ground and while he is moving with velocity $\vec{v}_m$, he observes velocity of rain relative to himself $\vec{v}_{r/m}$. It is a common intuitive fact that umbrella must be held against $\vec{v}_{r/m}$ for optimum protection from rain. According to these facts, directions of the velocity vectors are shown in the adjoining figure.

The addition of velocity vectors is represented according to the above equation. From the figure we have

$$v_r = v_m \tan 37^\circ = 3 \text{ km/h} \quad \text{Ans.}$$

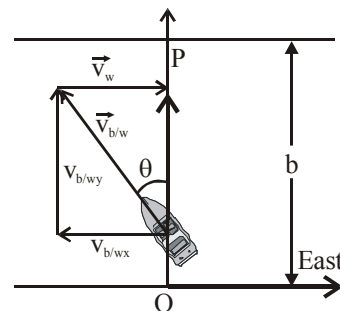


- Ex.** A boat can be rowed at 5 m/s on still water. It is used to cross a 200 m wide river from south bank to the north bank. The river current has uniform velocity of 3 m/s due east.
- (a) In which direction must it be steered to cross the river perpendicular to current?
- (b) How long will it take to cross the river in a direction perpendicular to the river flow?
- (c) In which direction must the boat be steered to cross the river in minimum time? How far will it drift?

- Sol.** (a) Velocity of a boat on still water is its capacity to move on water surface and equals to its velocity relative to water.

$\vec{v}_{b/w}$ = Velocity of boat relative to water = Velocity of boat on still water

On flowing water, the water carries the boat along with it. Thus velocity $\vec{v}_b$ of the boat relative to the ground equals to vector sum of $\vec{v}_{b/w}$ and $\vec{v}_w$. The boat crosses the river with the velocity $\vec{v}_b$.



$$\vec{v}_b = \vec{v}_{b/w} + \vec{v}_w$$

- (b) To cross the river perpendicular to current the boat must be steered in a direction so that one of the components of its velocity ($\vec{v}_{b/w}$) relative to water becomes equal and opposite to water flow velocity $\vec{v}_w$ to neutralize its effect. It is possible only when velocity of boat relative to water is greater than water flow velocity. In the adjoining figure it is shown that the boat starts from the point O and moves along the line OP (y-axis) due north relative to ground with velocity $\vec{v}_b$. To achieve this it is steered at an angle θ with the y-axis.

$$v_{b/w} \sin \theta = v_w \rightarrow 5 \sin \theta = 3 \Rightarrow \theta = 37^\circ \text{ Ans.}$$

- (c) The boat will cover river width b with velocity

$$v_b = v_{b/wy} = v_{b/w} \sin 37^\circ = 4 \text{ m/s in time } t, \text{ which is given by}$$

$$t = b / v_b \rightarrow t = 50 \text{ s Ans.}$$

- (d) To cross the river in minimum time, the component perpendicular to current of its velocity relative to ground must be kept to maximum value. It is achieved by steering the boat always perpendicular to current as shown in the adjoining figure. The boat starts from O at the south bank and reaches point P on the north bank. Time t taken by the boat is given by

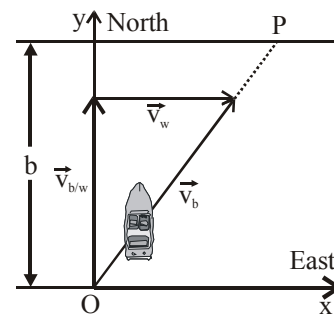
$$t = b / v_{b/w} \rightarrow t = 40 \text{ s Ans.}$$

Drift is the displacement along the river current measured from the starting point. Thus, it is given by the following equation. We denote it by x_d .

$$x_d = v_{bx} t$$

Substituting $v_{bx} = v_w = 3 \text{ m/s}$, from the figure, we have

$$x_d = 120 \text{ m Ans.}$$



EXERCISE (S-1)

General 2-D motion

- The vertical height y and horizontal distance x of a projectile on a certain planet are given by $x = (3t)m$, $y = (4t - 6t^2)m$ where t is in seconds. Find the speed of projection (in m/s).

KM0108

- The position of a particle is given by

$$\mathbf{r} = 3.0t \hat{i} - 2.0t^2 \hat{j} + 4.0 \text{ k m}$$

where t is in seconds and the coefficients have the proper units for $\mathbf{r}$ to be in metres.

(a) Find the $\mathbf{v}$ and $\mathbf{a}$ of the particle?

(b) What is the magnitude and direction of velocity of the particle at $t = 2.0 \text{ s}$?

KM0109

- A particle moves in xy plane such that $v_x = 50 - 16t$ and $y = 100 - 4t^2$ where v_x is in m/s and y is in m. It is also known that $x = 0$ when $t = 0$. Determine (i) Acceleration of particle (ii) Velocity of particle when $y = 0$.

KM0110

- The position of a particle is given by $x = 7 + 3t^3 \text{ m}$ and $y = 13 + 5t - 9t^2 \text{ m}$, where x and y are the position coordinates, and t is the time in s. Find the speed (magnitude of the velocity) when the x component of the acceleration is 36 m/s^2 .

KM0111

Projectile motion

- A particle is projected with a speed of 10 m/s at an angle 37° with the vertical. Find (i) time of flight (ii) maximum height above ground (iii) horizontal range.

KM0112

- A particle is thrown with a speed 60 ms^{-1} at an angle 60° to the horizontal. When the particle makes an angle 30° with the horizontal in downward direction, its speed at that instant is v . What is the value of v^2 in SI units?

KM0113

- A cricketer can throw a ball to a maximum horizontal distance of 100 m . How much high above the ground can the cricketer throw the same ball?

KM0114

- A particle is projected upwards with a velocity of 100 m/s at an angle of 60° with the vertical. Find the time when the particle will move perpendicular to its initial direction, taking $g = 10 \text{ m/s}^2$.

KM0115

9. A particle is projected in x - y plane with y -axis along vertical, the point of projection is origin. The equation of a path is $y = \sqrt{3}x - \frac{gx^2}{2}$. Find angle of projection and speed of projection.

KM0116

10. (a) Show that for a projectile the angle between the velocity and the x -axis as a function of time is given by

$$\theta(t) = \tan^{-1} \left(\frac{v_{0y} - gt}{v_{0x}} \right)$$

- (b) Show that the projection angle θ_0 of a projectile launched from the origin is given by

$$\theta_0 = \tan^{-1} \left(\frac{4h_m}{R} \right)$$

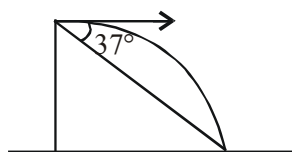
where the symbols have their usual meaning.

KM0117

11. A particle is projected in the x - y plane with y -axis along vertical. Two second after projection the velocity of the particle makes an angle 45° with the X -axis. Four second after projection, it moves horizontally. Find the velocity of projection.

KM0118

12. A ball is thrown horizontally from a cliff such that it strikes ground after 5 s. The line of sight from the point of projection to the point of landing makes an angle of 37° with the horizontal. What is the initial velocity of projection?



KM0119

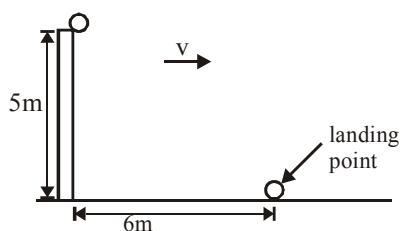
13. A Bomber flying upward at an angle of 53° with the vertical releases a bomb at an altitude of 800 m. The bomb strikes the ground 20 s after its release. Find: [Given $\sin 53^\circ = 0.8$; $g = 10 \text{ m/s}^2$]
 (i) The velocity of the bomber at the time of release of the bomb .
 (ii) The maximum height attained by the bomb .
 (iii) The horizontal distance travelled by the bomb before it strikes the ground
 (iv) The velocity (magnitude & direction) of the bomb just when it strikes the ground.

KM0120

14. A ball is projected at an angle of 30° above the horizontal from the top of a tower and strikes the ground in 5 s at an angle of 45° with the horizontal. Find the height of the tower and the speed with which it was projected.

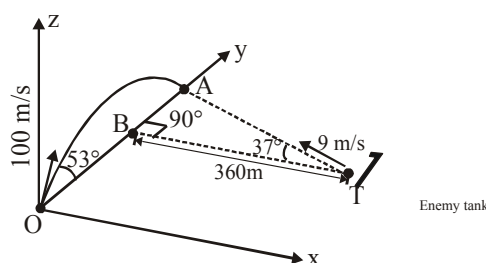
KM0121

15. A ball is dropped from rest from a tower of height 5m. As a result of the wind it lands at a distance 6m from the bottom of the tower as shown. Assuming no air resistance but that the wind gives the ball a constant horizontal velocity v . Find value of v in m/s.



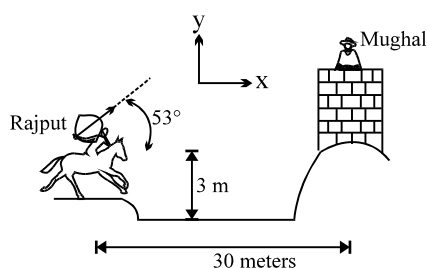
KM0122

16. A tank is initially at a perpendicular distance $BT = 360$ m from the plane of firing as shown. The enemy tank is moving with a speed of 9 m/s in direction TA as shown in figure. A gun can fire shell in y - z plane only with a speed 100 m/s at an angle of 53° such that the shell lands at points A . If tank started at $t = 0$ then time interval (in sec) after which shell is to be fired to hit the tank is



KM0123

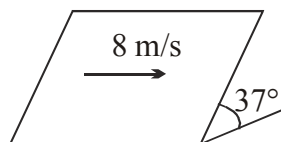
17. A Rajput soldier, sits on a horse next to a river. Across the river there is a hill and atop the hill is a fortress. He sees a Mughal, sitting on the fortress's top wall. There is a full moon, so he angrily shoots an arrow at an angle 53° relative to the horizontal. The arrow hits Mughal after a 2 second flight. The horizontal distance from Rajput to Mughal is 30 meters. The arrow is 3 meters above the river when Rajput shoots it.



- What is the original velocity, $\vec{V}$, of the arrow when Rajput shoots it?
- What is Mughal elevation above the river?
- What is the flight direction of the arrow the instant before it strikes Mughal, i.e. what is the angle, θ , between its direction and the horizontal when it skins into Mughal's tender flesh?

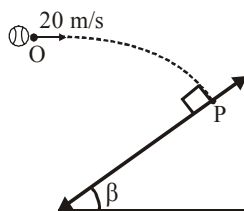
KM0124

18. A ball is projected on smooth inclined plane in direction perpendicular to line of greatest slope with velocity of 8 m/s. Find its speed after 1 s.



KM0125

19. A ball is thrown horizontally from a point O with speed 20 m/s as shown. Ball strikes the incline plane along the normal to it after two seconds. Find value of x , if $\beta = \pi/x$ (where β is the angle of incline in degree).



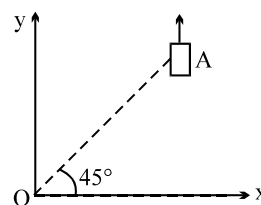
KM0126

Relative motion

20. A person decided to walk on an escalator which is moving at constant rate (speed). When he moves at the rate 1 step/sec, then he reaches top in 20 steps. Next day he goes 2 steps/sec. and reaches top in 32 steps. If speed of escalator is n steps/sec. Find the value of n .

KM0127

21. On a frictionless horizontal surface, assumed to be the x - y plane, a small trolley. A is moving along a straight line parallel to the y -axis (see figure) with a constant velocity of $(\sqrt{3} - 1)$ m/s. At a particular instant, when the line OA makes an angle of 45° with the x -axis, a ball is thrown along the surface from the origin O. Its velocity makes an angle ϕ with the x -axis and it hits the trolley.

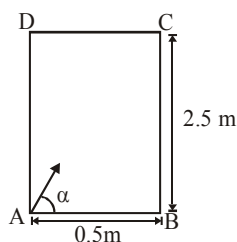


- (a) The motion of the ball is observed from the frame of trolley. Calculate the angle θ made by the velocity vector of the ball with the x -axis in this frame.

- (b) Find the speed of the ball with respect to the surface, if $\phi = \frac{4\theta}{3}$.

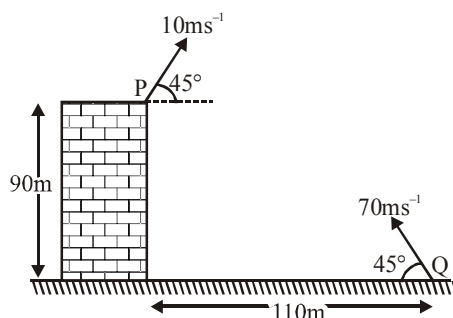
KM0128

22. A cuboidal elevator cabin is shown in the figure. A ball is thrown from point A on the floor of cabin when the elevator is falling under gravity. The plane of motion is ABCD and the angle of projection of the ball with AB, relative to elevator, if the ball collides with point C, is α . Then find the value of $\tan \alpha$.



KM0129

23. Two particles P and Q are launched simultaneously as shown in figure. Find the minimum distance between particles in meters.



KM0130

24. A man crosses a river by a boat. If he crosses the river in minimum time he takes 10 minutes with a drift 120 m. If he crosses the river taking shortest path, he takes 12.5 minutes. Assuming $v_{b/r} > v_r$, find (i) width of the river, (ii) velocity of the boat with respect to water ($v_{b/r}$) (iii) speed of the current (v_r)

KM0131

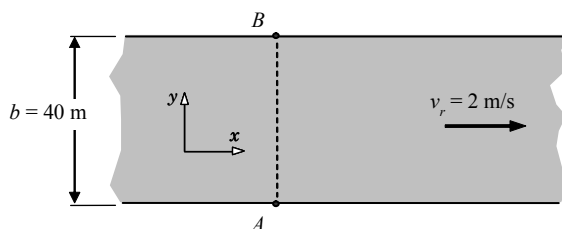
25. Rain is falling vertically with a speed of 20 m/s relative to air. A person is running in the rain with a velocity of 5 m/s and a wind is also blowing with a speed of 15 m/s (both towards east). Find the angle with the vertical at which the person should hold his umbrella for best protection from rain.

KM0132

26. A glass wind-screen of adjustable inclination is mounted on a car. The car moves horizontally with a speed of 6 m/s. At what angle α with the vertical should the wind screen be adjusted so that the rain drops falling vertically with 2 m/s strike the wind screen perpendicularly?

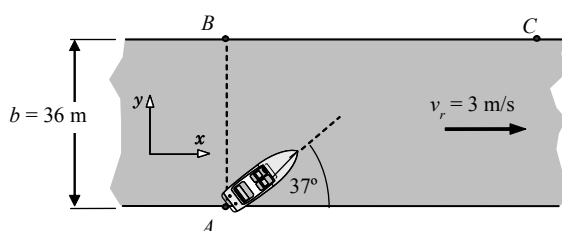
KM0133

27. Boat moves with velocity 5m/s on still water. It is steered perpendicular to the river current.
- Will it reach point B or somewhere else on the other bank ?
 - How long will it take to cross the river ?
 - How far down stream, will it reach the other bank ?
 - Does it take minimum time in this way ?



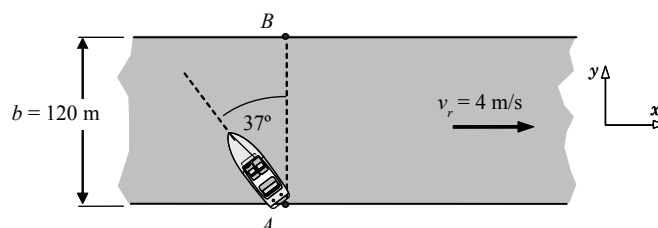
KM0134

28. Velocity of the boat with respect to river is 10 m/s. From point A it is steered in the direction shown to reach point C. Find the time of the trip and distance between B and C.



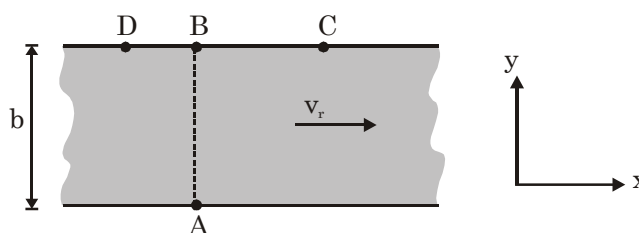
KM0135

29. Velocity of the boat with respect to river is 10 m/s. From point A it is steered in the direction shown. Where will it reach on the opposite bank?



KM0136

30. Drift is distance along a river a boat covers in crossing the river. If the boat reaches point C, distance BC is called downstream drift and if the boat reaches point D, distance BD is called upstream drift. To cross a river without drift, what should be relation between v_{br} and v_r . If a boat crosses a river without drift, in which direction must it be steered.



KM0137

EXERCISE (S-2)

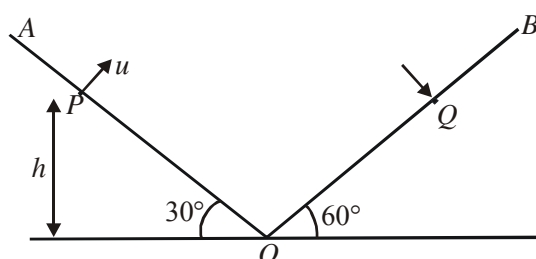
- A particle travels so that its acceleration is given by $\vec{a} = 5 \cos t \hat{i} - 3 \sin t \hat{j}$. If the particle is located at $(-3, 2)$ at time $t = 0$ and is moving with a velocity given by $(-3\hat{i} + 2\hat{j})$. Find
 - the velocity at time t and
 - the position vector of the particle at time $(t > 0)$.

KM0138

- A man can throw a stone with initial speed of 10 m/s. Find the maximum horizontal distance to which he can throw the stone in a room of height h for :
 - $h = 2$ m &
 - $h = 4$ m

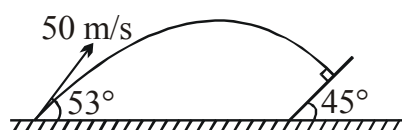
KM0139

- Two inclined planes OA and OB having inclinations 30° and 60° respectively, intersect each other at O as shown in figure. A particle is projected from point P with velocity $u = 10\sqrt{3}$ m/s along a direction perpendicular to plane OA . If the particle strikes the plane OB perpendicularly at Q . Calculate
 - Time of flight.
 - Velocity with which particle strikes the plane OB .
 - Vertical height of P from O .
 - Maximum height from O attained by the particle.
 - Distance PQ .



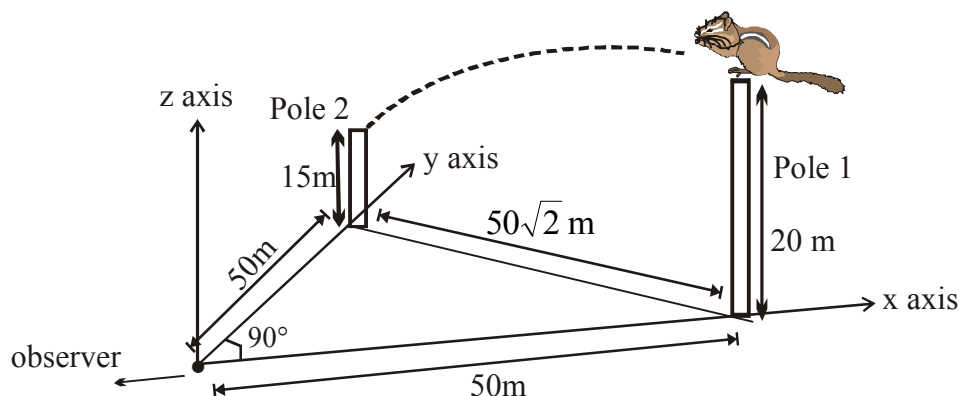
KM0140

- A projectile is thrown with velocity of 50 m/s towards an inclined plane from ground such that it strikes the inclined plane perpendicularly. The angle of projection of the projectile is 53° with the horizontal and the inclined plane is inclined at an angle of 45° to the horizontal.
 - Find the time of flight.
 - Find the distance between the point of projection and the foot of inclined plane.



KM0141

5. A small squirrel jumps from pole 1 to pole 2 in horizontal direction. Squirrels is observed by a very small observer at origin. What is average velocity vector of squirrel ? If average velocity vector is expressed as $v_x \hat{i} + v_y \hat{j} + v_z \hat{k}$, express your answer as sum of magnitudes of its components $|v_x| + |v_y| + |v_z|$ in unit m/s.



KM0142

6. Two particles start simultaneously from a point and move along line OP and OQ , one with a uniform velocity 10 m/s and other from rest with a constant acceleration 5 m/s² respectively. Line OP makes an angle 37° with the line OQ . Find the time at which they appear to each other moving at minimum speed?

KM0143

7. Two guns, situated at the top of a hill of height 10 m, fire one shot each with the same speed $5\sqrt{3}$ m/s at some interval of time. One gun fires horizontally and other fires at an angle of 60° up the horizontal. The shots collide in air at a point P . Find
(i) the time interval between the firings, and (ii) the coordinates of the point P .
(Take origin of the coordinates system at the foot of the hill vertically below the muzzle and trajectories in X - Y plane)

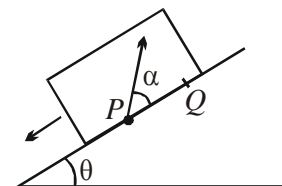
KM0144

8. A hunter is riding an elephant of height 4 m moving in straight line with uniform velocity of 2 m/s. A deer starts running with uniform velocity from a point $4\sqrt{5}$ m away in front of the elephant along a line perpendicular to velocity of the elephant. If hunter can throw his spear with a speed of 10 m/s, relative to the elephant, at what angle θ to it's direction of motion must he throw his spear horizontally for a successful hit. Find also the speed of the deer.

KM0145

9. A large heavy box is sliding without friction down a smooth plane of inclination θ . From a point P on the bottom of the box, a particle is projected inside the box. The initial speed of the particle with respect to box is u and the direction of projection makes an angle α with the bottom as shown in figure.

[JEE]



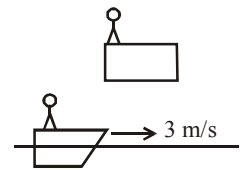
- (i) Find the distance along the bottom of the box between the point of projection P and the point Q where the particle lands. (Assume that the particle does not hit any other surface of the box. Neglect air resistance).
- (ii) If the horizontal displacement of the particle as seen by an observer on the ground is zero, find the speed of the box with respect to the ground at the instant when the particle was projected.

KM0146

10. A particle is projected from ground towards a vertical wall 80 m away at an angle of 37° with horizontal with initial velocity of 50 m/s. After its collision with wall & then once with ground find at what distance in meter from wall will it strike the ground. The component of velocity normal to the surface becomes half after collision with each surface.

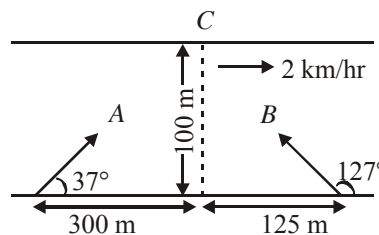
KM0147

11. You are standing on the Chambal Bridge watching the boats in the river. You see a motorboat pass directly below you, traveling perpendicular to the bridge at a speed of 3 m/s. A person on the boat throws a baseball at an initial speed of v_0 and at an angle of 37° from the vertical (Note: both v_0 and the angle are with respect to the boat). Find the value of v_0 (in m/s) necessary for the ball to travel straight up towards you.



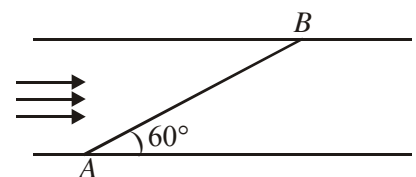
KM0148

12. Two swimmers start a race. One who reaches the point C first on the other bank wins the race. Boy A makes his strokes in a direction of 37° to the river flow with velocity 5 km/hr relative to water. Boy B makes his strokes in a direction of 127° to the river flow with same relative velocity. River is flowing with speed of 2 km/hr and is 100 m wide. Who will win the race? Compute the time taken by A and B to reach the point C if the speeds of A and B on the ground are 8 km/hr and 6 km/hr respectively.



KM0149

13. A swimmer starts to swim from point A to cross a river. He wants to reach point B on the opposite side of the river. The line AB makes an angle 60° with the river flow as shown. The velocity of the swimmer in still water is same as that of the water
- (i) In what direction should he try to direct his velocity? Calculate angle between his velocity and river velocity.
- (ii) Find the ratio of the time taken to cross the river in this situation to the minimum time in which he can cross this river.



KM0150

14. Hailstones falling vertically with speed of 10 m/s hit with respect to himself the windscreen of a moving car and rebound elastically. Find the velocity of the car if the driver find the hailstones rebound vertically after striking. Windscreen makes an angle 30° with the horizontal.

KM0151

15. A motor boat set out at 11 a.m. from a position $-6\mathbf{i} - 2\mathbf{j}$ and travels at a steady speed of magnitude $\sqrt{53}$ on a direct course to intercept a ship. The ship maintains a steady velocity vector $3\mathbf{i} + 4\mathbf{j}$ and at 12 noon is at a position $3\mathbf{i} - \mathbf{j}$. Find
- (a) the velocity vector of the motor boat,
 - (b) the time of interception and
 - (c) the position vector of point of interception. If distances are measured in kilometres and speeds in kilometres per hour.

KM0152

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

General 2-D motion

1. A particle is projected from a horizontal plane (x - z plane) such that its velocity vector at time t is given by $\vec{V} = a\hat{i} + (b - ct)\hat{j}$. Its range on the horizontal plane is given by :-

(A) $\frac{ba}{c}$ (B) $\frac{2ba}{c}$ (C) $\frac{3ba}{c}$ (D) None

KM0153

2. A body of mass 5 kg starts from the origin with an initial velocity $\vec{u} = (30\hat{i} + 40\hat{j})\text{ms}^{-1}$. If a constant force $(-6\hat{i} - 5\hat{j})\text{N}$ acts on the body, the time in which the y component of the velocity becomes zero, is :-

(A) 5s (B) 20 s (C) 40 s (D) 80 s

KM0154

Projectile motion

3. A boy throws a ball from shoulder height at an initial velocity of 30 m/s. Spending 4.8 s in air, the ball is caught by another boy as the same shoulder-height level. What is the angle of projection?

(A) 37° (B) 30° (C) 53° (D) 60°

KM0155

4. A particle is projected from the ground with velocity u at angle θ with horizontal. The horizontal range, maximum height and time of flight are R , H and T respectively. They are given by

$$R = \frac{u^2 \sin 2\theta}{g} \quad H = \frac{u^2 \sin^2 \theta}{2g} \quad \text{and} \quad T = \frac{2u \sin \theta}{g}$$

Now keeping u fixed, θ is varied from 30° to 60° , then

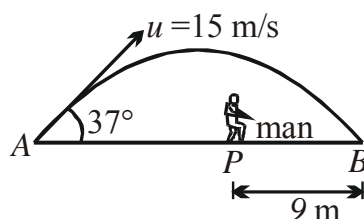
- (A) R will first increase then decrease, H will increase and T will decrease
(B) R will first increase then decrease while H and T both will increase
(C) R will decrease while H and T both will increase
(D) R will increase while H and T both will also increase

KM0156

5. Suppose a player hits several baseballs. Which baseball will be in the air for the longest time?
- (A) The one with the farthest range.
- (B) The one which reaches maximum height.
- (C) The one with the greatest initial velocity.
- (D) The one leaving the bat at 45° with respect to the ground.

KM0157

6. A ball is hit by a batsman at an angle of 37° as shown in figure. The man standing at P should run at what minimum velocity so that he catches the ball before it strikes the ground. Assume that height of man is negligible in comparison to maximum height of projectile.



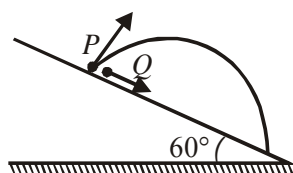
- (A) 3 m/s (B) 5 m/s (C) 9 m/s (D) 12 m/s

KM0158

7. A ball is projected horizontally. After 3 s from projection its velocity becomes 1.25 times of the velocity of projection. Its velocity of projection is :-
(A) 10 m/s (B) 20 m/s (C) 30 m/s (D) 40 m/s

KM0159

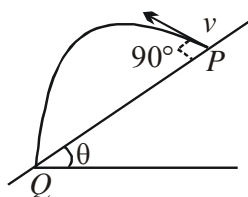
8. A particle P is projected from a point on the surface of smooth inclined plane (see figure). Simultaneously another particle Q is released on the smooth inclined plane from the same position. P and Q collide after $t = 4$ s. The speed of projection of P is :-



- (A) 5 m/s (B) 10 m/s (C) 15 m/s (D) 20 m/s

KM0160

9. In the given figure, if time taken by the projectile to reach Q is T , then $PQ =$



- (A) $T_v \sin \theta$ (B) $T_v \cos \theta$ (C) $T_v \sec \theta$ (D) $T_v \tan \theta$

KM0161

10. A particle is projected up the incline such that its component of velocity along the incline is 10 m/s . Time of flight is 2 second and maximum perpendicular distance during the motion from the incline is 5 m. Then velocity of projection will be :-

(A) 10 m/s (B) $10\sqrt{2}\text{ m/s}$ (C) $5\sqrt{5}\text{ m/s}$ (D) none of these

KM0162

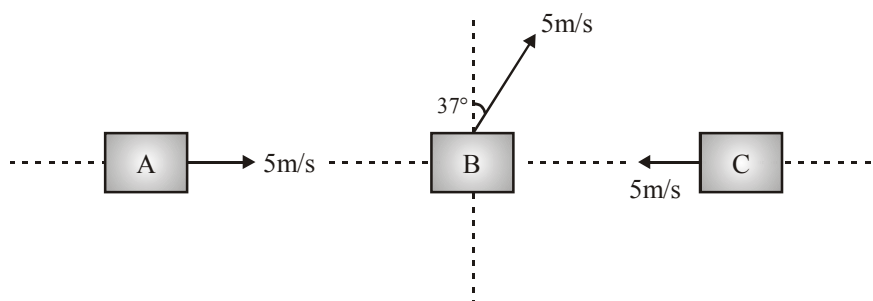
11. A particle A is projected with speed v_A from a point making an angle 60° with the horizontal. At the same instant, a second particle B is thrown vertically upward from a point directly below the maximum height point of parabolic path of A with velocity v_B . If the two particles collide then the ratio of v_A/v_B should be :-

(A) 1 (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{\sqrt{3}}{2}$ (D) $\sqrt{3}$

KM0163

Relative motion

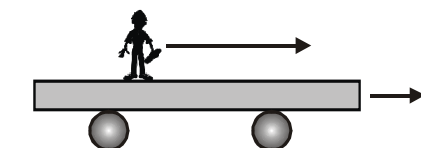
12. Consider the motion of three bodies as shown for an observer on B, what is the magnitude of relative velocity of A with respect to C ?



(A) 10 m/s (B) 0 (C) $\sqrt{20}\text{ m/s}$ (D) 8 m/s

KM0164

13. A trolley is moving horizontally with a constant velocity of v with respect to earth. A man starts running from one end of the trolley with a velocity $1.5v$ with respect to trolley. After reaching the opposite end, the man returns back and continues running with a velocity of $1.5v$ w.r.t. the trolley in the backward direction. If the length of the trolley is L then the displacement of the man with respect to earth during the process will be :-



(A) $2.5L$ (B) $1.5L$ (C) $\frac{5L}{3}$ (D) $\frac{4L}{3}$

KM0165

14. An elevator car (lift) is moving upward with uniform acceleration of 2 m/s^2 . At the instant, when its velocity is 2 m/s upwards a ball is thrown upward from its floor. The ball strikes back the floor 2 s after its projection. Find the velocity of projection of the ball relative to the lift.
 (A) $10 \text{ m/s} \uparrow$ (B) $10 \text{ m/s} \downarrow$ (C) $12 \text{ m/s} \uparrow$ (D) $12 \text{ m/s} \downarrow$

KM0166

15. A flag is mounted on a car moving due North with velocity of 20 km/hr . Strong winds are blowing due East with velocity of 20 km/hr . The flag will point in direction :-
 (A) East (B) North-East (C) South-East (D) South-West

KM0167

16. Three ships A , B & C are in motion. Ship A moves relative to B is with speed v towards North-East. Ship B moves relative to C with speed v towards the North-West. Then relative to A , C will be moving towards :-
 (A) North (B) South (C) East (D) West

KM0168

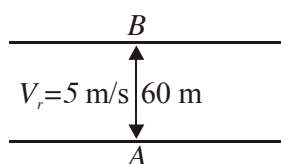
17. Wind is blowing in the north direction at speed of 2 m/s which causes the rain to fall at some angle with the vertical. With what velocity should a cyclist drive so that the rain appears vertical to him :
 (A) 2 m/s south (B) 2 m/s north (C) 4 m/s west (D) 4 m/s south

KM0169

18. A boat having a speed of 5 km/hr in still water, crosses a river of width 1 km along the shortest possible path in 15 minutes. The speed of the river in Km/hr .
 (A) 1 (B) 3 (C) 4 (D) $\sqrt{41}$

KM0170

19. A man is crossing a river flowing with velocity of 5 m/s . He reaches a point directly across the river at a distance of 60 m in 5 sec . His velocity in still water should be :-



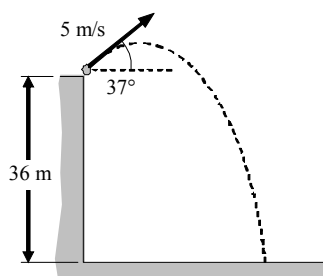
- (A) 12 m/s (B) 13 m/s (C) 5 m/s (D) 10 m/s

KM0171

20. A motor boat is to reach at a point 30° upstream (w.r.t. normal) on other side of a river flowing with velocity 5 m/s . The angle 30° is measured from a direction perpendicular to river flow. Velocity of motorboat with respect to water is $5\sqrt{3} \text{ m/s}$. The driver should steer the boat at an angle
 (A) 120° with respect to stream direction.
 (B) 30° with respect to the perpendicular to the bank.
 (C) 30° with respect to the line of destination from starting point.
 (D) None of these.

KM0172

21. A ball is thrown from the top of 36 m high tower with velocity 5 m/s at an angle 37° above the horizontal as shown. Its horizontal range on the ground is closest to [$g = 10 \text{ m/s}^2$]



- (A) 12 m (B) 18 m (C) 24 m (D) 30 m

KM0173

MULTIPLE CORRECT TYPE QUESTIONS

22. A particle moves in the xy -plane and at time t is at the point $(t^2, t^3 - 2t)$. Then

- (A) At $t = 2/3$ s, directions of velocity and acceleration are perpendicular
(B) At $t = 0$, directions of velocity and acceleration are perpendicular
(C) At $t = \sqrt{2/3}$ s, particle is moving parallel to x -axis

- (D) Acceleration of the particle when it is at point $(4, 4)$ is $(2\hat{i} + 12\hat{j}) \text{ m/s}$

KM0174

23. A particle moves in the x - y plane with a constant acceleration g in the negative y -direction. Its equation of motion is $y = ax - bx^2$, where a and b are constants. Which of the following is/are correct?

- (A) The x -component of its velocity is constant.

- (B) At the origin, the y -component of its velocity is a $\sqrt{\frac{g}{2b}}$.

- (C) At the origin, its velocity makes an angle $\tan^{-1}(a)$ with the x -axis.

- (D) The particle moves exactly like a projectile.

KM0175

24. Two particles A and B projected along different directions from the same point P on the ground with the same velocity of 70 m/s in the same vertical plane. They hit the ground at the same point Q such that $PQ = 480$ m. Then [$g = 9.8 \text{ m/s}^2$]

- (A) Ratio of their times of flight is 4 : 5

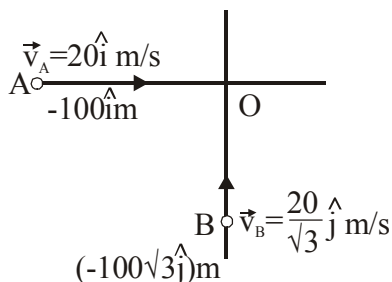
- (B) Ratio of their maximum heights is 9 : 16

- (C) Ratio of their minimum speeds during flights is 4 : 3

- (D) The bisector of the angle between their directions of projection makes 45° with horizontal

KM0176

25. Positions of two vehicles A and B with reference to origin O and their velocities are as shown.



- (A) they will collide
(B) distance of closest approach is 100 m.
(C) their relative speed is $\frac{40}{\sqrt{3}}$ m/s
(D) their relative velocity is $\frac{20}{\sqrt{3}}$ m/s

KM0177

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 26 to 28

Two projectiles are thrown simultaneously in the same vertical plane from the same point. If their velocities of projection are v_1 and v_2 at angles θ_1 and θ_2 respectively from the horizontal, then answer the following questions

26. The trajectory of particle 1 with respect to particle 2 will be
(A) a parabola
(B) a straight line
(C) a vertical straight line
(D) a horizontal straight line

KM0178

27. If $v_1 \cos \theta_1 = v_2 \cos \theta_2$, then choose the **incorrect** statement
(A) One particle will remain exactly below or above the other particle
(B) The trajectory of one with respect to other during the flight will be a vertical straight line
(C) Both will have the same range
(D) Both will attain same maximum height

KM0178

28. If $v_1 \sin \theta_1 = v_2 \sin \theta_2$, then choose the **correct** statement
(A) The time of flight of both the particles will be same
(B) The maximum height attained by the particles will be same
(C) The trajectory of one with respect to another during the flight will be a horizontal straight line
(D) None of these

KM0178

Paragraph for Question Nos. 29 to 31

A particle leaves the origin with initial velocity $\vec{v}_0 = 11\hat{i} + 14\hat{j}$ m/s. It undergoes a constant acceleration

given by $\vec{a} = -\frac{22}{5}\hat{i} + \frac{2}{15}\hat{j}$ m/s².

29. When does the particle cross the y axis ?

(A) 2 sec (B) 4 sec (C) 5 sec (D) 7 sec

KM0179

30. At the instant when particle crosses y-axis, direction in which particle is moving is :-

(A) At angle 37° from +x-axis towards +y-axis
(B) At angle 37° from -x-axis towards +y-axis
(C) At angle 53° from +x-axis towards +y-axis
(D) At angle 53° from -x-axis towards +y-axis

KM0179

31. How far is it from the origin, at that time ?

(A) 70 m (B) 71.67 m (C) 125 m (D) 15 m

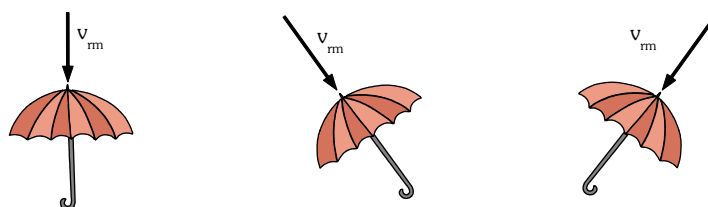
KM0179

Paragraph for Question no. 32 to 35: Rain and man

By the term velocity of rain, we mean velocity with which raindrops fall relative to the ground. In absence of wind, raindrops fall vertically and in presence of wind raindrops fall obliquely. Moreover raindrops acquire a constant terminal velocity due air resistance very quickly as they fall toward the earth. A moving man relative to himself observes an altered velocity of raindrops, which is known as velocity of rain relative to the man. It is given by the following equation.

$$\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$$

A standstill man relative to himself observes rain falling with velocity, which is equal to velocity of the raindrops relative to the ground. To protect himself a man should hold his umbrella against velocity of raindrops relative to himself as shown in the following figure.



32. Rain is falling vertically with velocity 80 cm/s.

(a) How should you hold your umbrella?
(b) You start walking towards the east with velocity 60 cm/s. How should you hold your umbrella?
(c) You are walking towards the west with velocity 60 cm/s. How should you hold your umbrella?
(d) You are walking towards the north with velocity 60 cm/s. How should you hold your umbrella?
(e) You are walking towards the south with velocity 80 cm/s. How should you hold your umbrella?

KM0180

33. When you are standstill in rain, you have to hold your umbrella vertically to protect yourself.
(a) When you walk with velocity 90 cm/s , you have to hold your umbrella at 53° above the horizontal.
What is velocity of the raindrops relative to the ground and relative to you?
(b) If you walk with speed 160 cm/s , how should you hold your umbrella?

KM0180

34. A man walks in rain at 72 cm/s due east and observes the rain falling vertically. When he stops, rain appears to strike his back at 37° from the vertical. Find velocity of raindrops relative to the ground.

KM0180

35. When you walk in rain at 75 cm/s , you have to hold your umbrella vertically and when you double your speed in the same direction, you have to hold your umbrella at 53° above the horizontal. What is the rain velocity?

Paragraph for Question no. 36 to 38 : Flag in wind

When you are standstill holding a flag, the flag flutters in the direction of wind. When you start running the direction of fluttering of the flag changes in to the direction of the wind relative to you. In all case a flag flutters in the direction of the wind relative to the flag.

36. When you are standstill holding a flag the flag flutters in the north and when you run at 8 m/s due east, the flag flutters in direction 37° north of west. Find the wind velocity.

KM0181

37. Wind is blowing uniformly due north everywhere with velocity 12 m/s . A car mounted with a flag starts running towards east. After 9 s from start the flag flutters in 53° north of west and after 16 s from the start the flag flutters in 37° north of west.

- (a) Find velocity of the car 9 s after it starts.
(b) Find velocity of the car 16 s after it starts.
(c) If the car maintains uniform acceleration, find acceleration of the car.

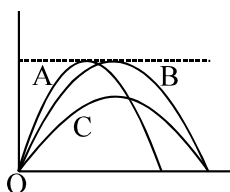
KM0181

38. Holding a flag, when you run at 8 m/s due east, the flag flutters in the north and when you run at 2 m/s due south, the flag flutters in the northeast. If the wind velocity is uniform and remain constant, find the wind velocity.

KM0181

MATRIX MATCH TYPE QUESTION

39. Trajectories are shown in figure for three kicked footballs. Initial vertical & horizontal velocity components are u_y and u_x respectively. Ignoring air resistance, choose the correct statement from **Column-II** for the value of variable in **Column-I**.



Column-I	Column-II
(A) time of flight	(P) greatest for A only
(B) u_y/u_x	(Q) greatest for C only
(C) u_x	(R) equal for A and B
(D) $u_x u_y$	(S) equal for B and C

KM0182

40. Trajectory of particle in a projectile motion is given as $y = x - \frac{x^2}{80}$. Here, x and y are in meters. For

this projectile motion, match the following with $g = 10 \text{ m/s}^2$.

Column-I	Column-II
(A) Angle of projection (in degrees)	(P) 20
(B) Angle of velocity with horizontal after 4s (in degrees)	(Q) 80
(C) Maximum height (in metres)	(R) 45
(D) Horizontal range (in metres)	(S) 30
	(T) 60

KM0183

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. A particle is moving in x-y plane. At certain instant, the components of its velocity and acceleration are as follows ; $V_x = 3\text{ m/s}$, $V_y = 4\text{ m/s}$, $a_x = 2\text{ m/s}^2$ and $a_y = 1\text{ m/s}^2$. The rate of change of speed at this moment is :-
 (A) $\sqrt{10}\text{ m/s}^2$ (B) 4 m/s^2 (C) 10 m/s^2 (D) 2 m/s^2
KM0184
2. A particle leaves the origin with an initial velocity $\vec{v} = (3\hat{i} + 4\hat{j})\text{ ms}^{-1}$ and a constant acceleration $\vec{a} = (-\hat{i} - 0.5\hat{j})\text{ ms}^{-2}$. When the particle reaches its maximum x-coordinate, what is the y-coordinate?
 (A) $\frac{27}{4}\text{ m}$ (B) $\frac{37}{4}\text{ m}$ (C) $\frac{29}{4}\text{ m}$ (D) $\frac{39}{4}\text{ m}$
KM0185
3. The position vector of a particle is determined by $\vec{r} = 3t^2\hat{i} + 4t^2\hat{j} + 7\hat{k}$. The distance travelled in first 10 sec is :-
 (A) 100 m (B) 150 m (C) 500 m (D) 300 m
KM0186
4. A point moves in x-y plane according to the law $x = 4\sin 6t$ and $y = 4(1 - \cos 6t)$. The distance traversed by the particle in 4 seconds is (x and y are in metres)
 (A) 96 m (B) 48 m (C) 24 m (D) 108 m
KM0187
5. A particle moves in the x-y plane. It x and y coordinates vary with time t according to equations $x = t^2 + 2t$ and $y = 2t$. Possible shape of path followed by the particle is
 (A) Straight line (B) Circle
 (C) Parabola (D) More information is required to decide.
KM0188
6. Particle is dropped from the height of 20 m from horizontal ground. A constant force acts on the particle in horizontal direction due to which horizontal acceleration of the particle becomes 6 m/s^2 . Find the horizontal displacement of the particle till it reaches ground.
 (A) 6 m (B) 10 m (C) 12 m (D) 24 m
KM0189
7. A projectile is fired with a velocity u making an angle θ with the horizontal. What is the magnitude of change in velocity when it is at the highest point ?
 (A) $u \cos \theta$ (B) u (C) $u \sin \theta$ (D) $u \cos \theta - u$
KM0190

8. A particle is projected at an angle of 45° from a point lying 2 m from the foot of a wall. It just touches the top of the wall and falls on the ground 4m from it. The height of the wall is
 (A) $3/4$ m (B) $2/3$ m (C) $4/3$ m (D) $1/3$ m

KM0191

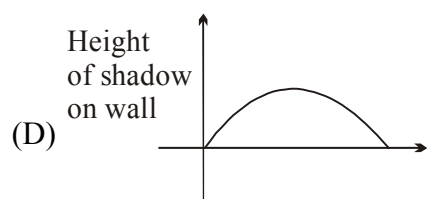
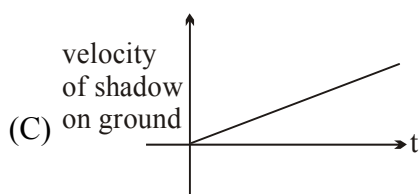
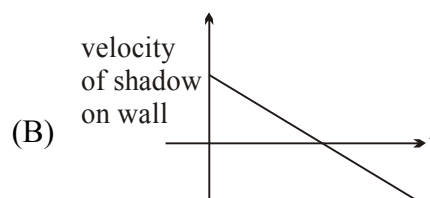
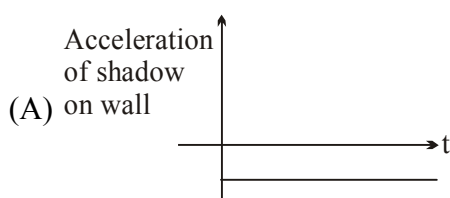
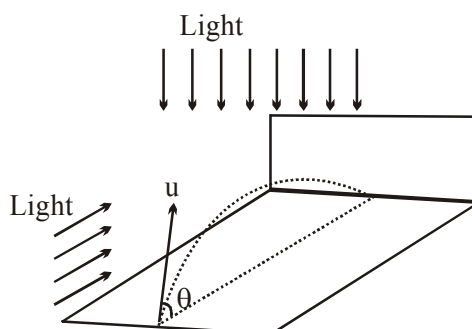
9. A ball was thrown by a boy A at angle 60° with horizontal at height 1m from ground. Boy B is running in the plane of motion of ball and catches the ball at height 1m from ground. He finds the ball falling vertically. If the boy is running at a speed 20 km/hr. Then the velocity of projection of ball is-
 (A) 20 km/hr (B) 30 km/hr (C) 40 km/hr (D) 50 km/hr

KM0192

10. A light body is projected with a velocity $(10\hat{i} + 20\hat{j} + 20\hat{k}) \text{ ms}^{-1}$. Wind blows along X-axis with an acceleration of 2.5 ms^{-2} . If Y-axis is vertical then the speed of particle after 2 second will be ($g = 10 \text{ ms}^{-2}$)
 (A) 25 ms^{-1} (B) $10\sqrt{5} \text{ ms}^{-1}$ (C) 30 ms^{-1} (D) None of these

KM0193

11. A projectile is projected as shown in figure. A proper light arrangement makes a shadow on the wall as well as on the floor ? Which of the following graphs is incorrect.



KM0194

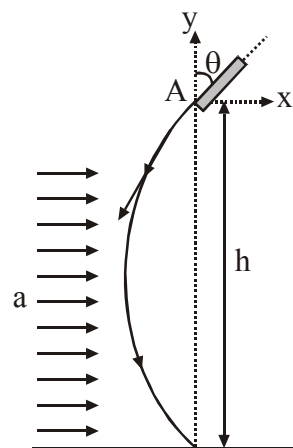
12. A particle is ejected from the tube at A with a velocity v at an angle θ with the vertical y -axis. A strong horizontal wind gives the particle a constant horizontal acceleration a in the x -directions. If the particle strikes the ground at a point directly under its released position and the downward y -acceleration is taken as g then

(A) $h = \frac{2v^2 \sin \theta \cos \theta}{a}$

(B) $h = \frac{2v^2 \sin \theta \cos \theta}{g}$

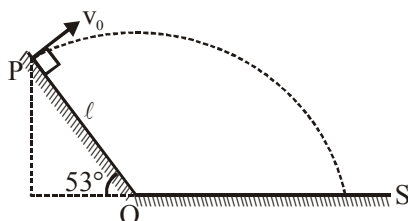
(C) $h = \frac{2v^2}{g} \sin \theta \left(\cos \theta + \frac{a}{g} \sin \theta \right)$

(D) $h = \frac{2v^2}{a} \sin \theta \left(\cos \theta + \frac{g}{a} \sin \theta \right)$



KM0195

13. A stone is projected from point P on the inclined plane with velocity $v_0 = 10$ m/s directed perpendicular to the plane. The time taken by the stone to strike the horizontal ground S is
(Given $PO = \ell = 10$ meter)



(A) 1.5 sec

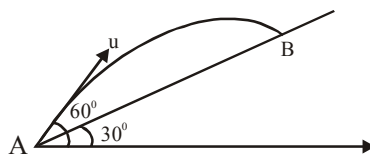
(B) 1.4 sec

(C) 2 sec

(D) 2.3 sec

KM0196

14. Time taken by the projectile to reach from A to B is t . Then the distance AB is equal to :-



(A) $\frac{ut}{\sqrt{3}}$

(B) $\frac{\sqrt{3}ut}{2}$

(C) $\sqrt{3} ut$

(D) $2 ut$

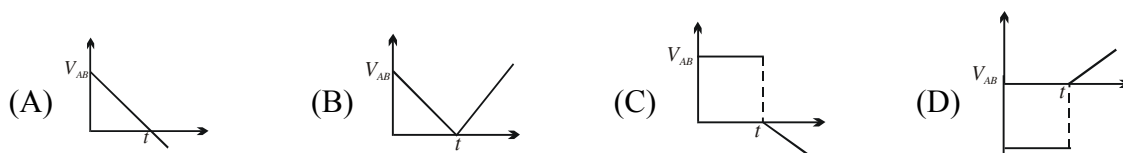
KM0197

15. A particle is projected from a point P (2 m, 0 m, 0 m) with a velocity 10 m/s making an angle 45° with the horizontal. The plane of projectile motion passes through a horizontal line PQ which makes an angle of 37° with positive x -axis and xy plane is horizontal. The coordinates of the point where the particle will strike the line PQ is ($g = 10 \text{ m/s}^2$)

(A) (10 m, 6 m, 0 m) (B) (8 m, 6 m, 0 m) (C) (10 m, 8 m, 0 m) (D) (6 m, 10 m, 0 m)

KM0198

16. A body A is thrown vertically upwards with such a velocity that it reaches a maximum height of h . Simultaneously another body B is dropped from height h . It strikes the ground and does not rebound. The velocity of A relative to B v/s time graph is best represented by (upward direction is positive)



KM0199

17. An object moves to the East across a frictionless surface with constant speed. A person then applies a constant force to the North on the object. What is the resulting path that the object takes?

(A) A straight line path partly Eastward, partly Northward
(B) A straight line path totally to the North
(C) A parabolic path opening toward the North
(D) A parabolic path opening toward the East

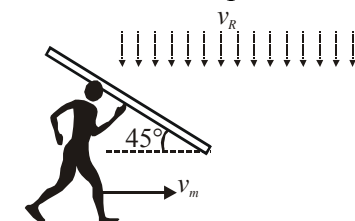
KM0200

18. A particle is thrown from a stationary platform with velocity v at an angle of 60° with the horizontal. The range obtained is R . If the platform moves horizontally in the direction of target with velocity v , the range will increase to :

(A) $\frac{3R}{2}$ (B) $\frac{5R}{2}$ (C) $2R$ (D) $3R$

KM0201

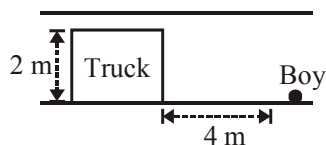
19. On a particular day rain drops are falling vertically at a speed of 5 m/s. A man holding a plastic board is running to escape from rain as shown. The lower end of board is at a height half that of man and the board makes 45° with horizontal. The maximum speed of man so that his feet does not get wet, is



(A) 5 m/s (B) $5\sqrt{2}$ m/s (C) $5/\sqrt{2}$ m/s (D) zero

KM0202

20. A 2 m wide truck is moving with a uniform speed of 8 m/s along a straight horizontal road. A pedestrian starts crossing the road at an instant when the truck is 4 m away from him. The minimum constant velocity with which he should run to avoid an accident is :-



- (A) $1.6\sqrt{5}$ m/s (B) $1.2\sqrt{5}$ m/s (C) $1.2\sqrt{7}$ m/s (D) $1.6\sqrt{7}$ m/s

KM0203

21. Two trucks are moving on parallel tracks. A person on one truck projects a ball vertically upward then path of the ball as seen by four observers: from the ground, from the second truck moving with same velocity as that first truck, from the second truck moving with speed greater than first one in same direction and from the second truck moving with speed less than the first truck in same direction are:
- (A) Parabola, Parabola, Parabola and Parabola
 (B) Straight line, Straight line, Parabola and Parabola
 (C) Parabola, Straight line, Parabola and Parabola
 (D) None of these

KM0204

22. Man A sitting in a car moving at 54 km/hr observes a man B in front of the car crossing perpendicularly the road of width 15 m in three seconds. Then the velocity of man B will be
- (A) $5\sqrt{10}$ towards the car (B) $5\sqrt{10}$ away from the car
 (C) 5 m/s perpendicular to the road (D) None

KM0205

23. A swimmer swims in still water at a speed = 5 km/hr. He enters a 200 m wide river, having river flow speed = 4 km/hr at point A and proceeds to swim at an angle of 127° with the river flow direction. Another point B is located directly across A on the other side. The swimmer lands on the other bank at a point C, from which he walks the distance CB with a speed = 3 km/hr. The total time in which he reaches from A to B is
- (A) 5 minutes (B) 4 minutes (C) 3 minutes (D) None

KM0206

24. A man wishes to swim across a river 400 m wide flowing with a speed of 3 m/s, such that he reaches the point just in front on the other bank in time not greater than 100s. The angle made by the direction he swims and river flow direction is :-
- (A) 90° (B) 127° (C) 150° (D) 143°

KM0207

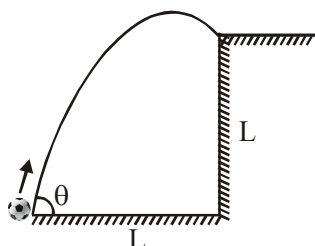
25. An observer on ground sees a boat cross a river of width 800 m perpendicular to its stream in 200 seconds. He also finds a man on a raft floating at speed of 3 m/s with river. The distance travelled by boat as seen by man on the raft in crossing the river is-
- (A) 800 m (B) 1000m (C) 1200m (D) 1600m

KM0208

26. A boatman moves his boat with a velocity 'v' (relative to water) in river and finds to his surprise that velocity of river 'u' (with respect to ground) is more than 'v'. He has to reach a point directly opposite to the starting point on another bank by travelling minimum possible distance. Then
- (A) he must steer the boat (with velocity v) at certain angle with river flow so that he can reach the opposite point on other bank directly.
- (B) his velocity 'v' must be towards directly opposite point, So, that he can travel rest of distance by walking on other bank to reach the directly opposite point.
- (C) boatman should maintain velocity v of boat at certain angle greater than 90° with direction of river flow to minimize drifting and then walk rest of distance on other bank.
- (D) boat velocity 'v' should be at an angle less than 90° with direction of river flow to minimize the drift and then walk to the point.

KM0209

27. A ball is thrown at an angle θ up to the top of a cliff of height L, from a point at a distance L from the base, as shown in figure. Assuming that one of the following quantities is the initial speed required to make the ball hit right at the edge of the cliff, which one is it :-



- (A) $\sqrt{\frac{gL}{2(\tan \theta - 1)}}$ (B) $\frac{1}{\cos \theta} \sqrt{\frac{gL}{2(\tan \theta - 1)}}$ (C) $\frac{1}{\cos \theta} \sqrt{\frac{gL}{2(\tan \theta + 1)}}$ (D) $\sqrt{\frac{gL \tan \theta}{2(\tan \theta + 1)}}$

KM0210

28. A particle is projected with a velocity of $\sqrt{20}$ m/s such that it strikes on the same level as the point of projection at a distance of $\sqrt{3}$ m. Which of the following options is/are incorrect (mass = 1 kg) :
- (A) The maximum height reached by the projectile can be 0.25 m.
- (B) The minimum velocity during its motion can be $\sqrt{5}$ m/s
- (C) The time taken for the flight can be $\sqrt{3/5}$ s.
- (D) Minimum kinetic energy during its motion can be 6 J.

KM0211

MULTIPLE CORRECT TYPE QUESTION

29. A particle is moving with a position vector, $\vec{r} = [a_0 \sin(2\pi t)\hat{i} + a_0 \cos(2\pi t)\hat{j}]$. Then-
- (A) Magnitude of displacement of the particle between time $t = 4$ sec and $t = 6$ sec is zero
 (B) Distance travelled by the particle in 1 sec is $2\pi a_0$
 (C) The speed of particle in the whole motion is constant and equal to $2\pi a_0$.
 (D) None of these

KM0212

30. A point mass is moving in the x-y plane. Its acceleration is a constant vector perpendicular to the x-axis. Which of the following do/does not change with time?
- (A) only y-component of its velocity vector
 (B) only x-component of its velocity vector
 (C) only y-component of its acceleration vector
 (D) only x-component of its acceleration vector

KM0213

31. A ball is thrown from ground such that it just crosses two poles of equal height kept 80 m apart. The maximum height attained by the ball is 80 m. When the ball passes the first pole, its velocity makes 45° with horizontal. The correct alternatives is/are :- ($g = 10 \text{ m/s}^2$)
- (A) Time interval between the two poles is 4 s.
 (B) Height of the pole is 60 m.
 (C) Range of the ball is 160 m.
 (D) Angle of projection is $\tan^{-1}(2)$ with horizontal.

KM0214

32. Position vector of a particle is expressed as function of time by equation $\vec{r} = 2t^2 + (3t - 1)\hat{j} + 5\hat{k}$, where r is in meters and t is in seconds.
- (A) It always moves in a plane that is parallel to the x-y plane.
 (B) At the instant $t = 0$ s, it is observed at point (0 m, -1 m, 5 m), moving with velocity 3 m/s in the positive y-direction.
 (C) Its acceleration vector is uniform.
 (D) It is an example of three dimensional motion.

KM0215

33. A projectile is thrown with speed u into air from a point on the horizontal ground at an angle θ with horizontal. If the air exerts a constant horizontal resistive force on the projectile then select correct alternative(s).
- (A) At the farthest point, the velocity is horizontal.
 (B) The time for ascent equals the time for descent.
 (C) The path of the projectile may be parabolic.
 (D) The path of the projectile may be a straight line.

KM0216

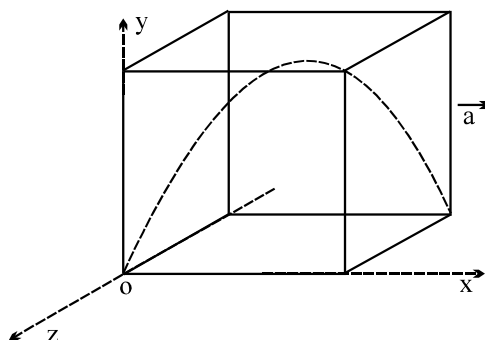
34. A block is thrown horizontally with a velocity of 2 m/s (relative to ground) on a belt, which is moving with velocity 4 m/s in opposite direction of the initial velocity of block. If the block stops slipping on the belt after 4 s it was dropped then choose the correct statement(s) :-
- (A) Displacement with respect to ground is zero after 2.66 s and magnitude of displacement with respect to ground is 12 m after 4 s.
- (B) Magnitude of displacement with respect to ground in 4 s is 4 m.
- (C) Magnitude of displacement with respect to belt in 4 s is 12 m.
- (D) Displacement with respect to ground is zero in 8/3 s.

KM0217

35. A man on a rectilinearly moving cart, facing the direction of motion, throws a ball straight up with respect to himself
- (A) The ball will always return to him.
- (B) The ball will never return to him.
- (C) The ball will return to him if the cart moves with constant velocity.
- (D) The ball will fall behind him if the cart moves with some positive acceleration.

KM0218

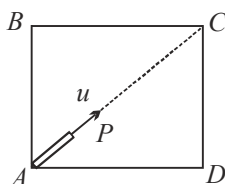
36. A cubical box dimension $L = 5/4$ metre starts moving with an acceleration $\vec{a} = 0.5 \text{ m/s}^2 \hat{i}$ from the state of rest. At the same time, a stone is thrown from the origin with velocity $\vec{V} = v_1 \hat{i} + v_2 \hat{j} - v_3 \hat{k}$ with respect to earth. Acceleration due to gravity $\vec{g} = 10 \text{ m/s}^2 (-\hat{j})$. The stone just touches the roof of box and finally falls at the diagonally opposite point then :



- (A) $v_1 = \frac{3}{2}$ (B) $v_2 = 5$ (C) $v_3 = \frac{5}{4}$ (D) $v_3 = \frac{5}{2}$

KM0219

37. A large rectangular box moves vertically downward with an acceleration a . A toy gun fixed at A and aimed towards C fires a particle P .



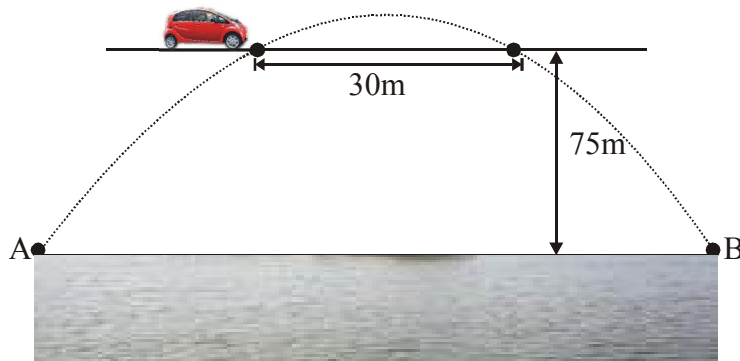
- (A) P will hit C if $a = g$
- (B) P will hit the roof BC , if $a > g$
- (C) P will hit the wall CD if $a < g$
- (D) May be either (A), (B) or (C), depending on the speed of projection of P

KM0220

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 38 to 40

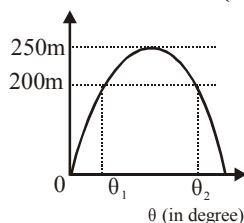
In the figure shown there is a long horizontal bridge over a river 75 m high from water surface. A strong man throws a stone in the parallel plane of the bridge. A observer in a car travelling on the bridge finds the stone going pass by the car while ascending and also while descending between two points on the road 30 m away. The car is travelling at a speed of 15 m/s. The stone is thrown from the bank of river just at the same level of water.



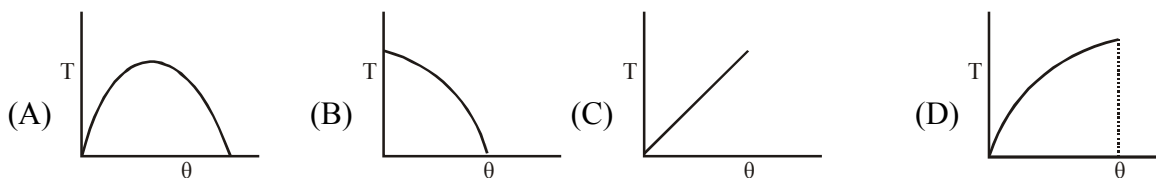
38. What is the angle the velocity makes with the bridge when it goes past the car while ascending ?
 (A) 30° (B) 45° (C) $\tan^{-1}\left(\frac{2}{3}\right)$ (D) $\tan^{-1}\left(\frac{3}{2}\right)$ **KM0221**
39. Horizontal distance AB travelled by stone is :-
 (A) 0 m (B) 75 m (C) 120 m (D) 240 m **KM0221**
40. What is the distance between car and stone at the instant when particle reaches at point B ?
 (A) 0 m (B) 75 m (C) 120 m (D) 240 m **KM0221**

Paragraph for Question 41 & 42

From the ground level, a ball is to be shot with a certain speed. Graph shows the range (R) of the particle versus the angle of projection from horizontal (θ).



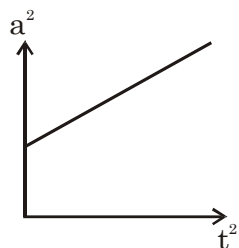
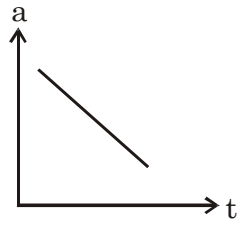
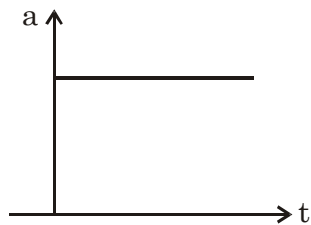
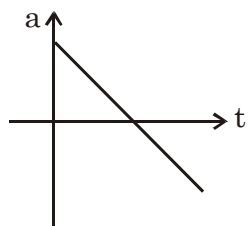
41. Values of θ_1 and θ_2 are
 (A) 53° and 37° (B) 26.5° and 63.5° (C) 18.5° and 71.5° (D) 15° and 75° **KM0222**
42. The corresponding time of flight vs θ graph is :-

**KM0222**

**MATCHING LIST TYPE ($4 \times 4 \times 4$) SINGLE OPTION CORRECT
(THREE COLUMNS AND FOUR ROWS)**

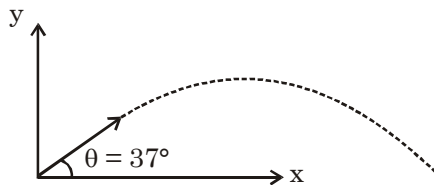
Answer Q.43, Q.44 and Q.45 by appropriately matching the information given in the three columns of the following table.

Match the following

Column-I Time of flight (in sec)	Column-II Range (in m) Along the ground/ along the inclined plane	Column-III Graph
(I) 2	(i) 3840	(P) 
(II) 1	(ii) $20\sqrt{2}$	(Q) 
(III) 10	(iii) 7	(R) 
(IV) 32	(iv) 3600	(S) 

43. A particle is projected with initial speed $u = \frac{25}{3}$ m/s as shown, here acceleration vector is given as

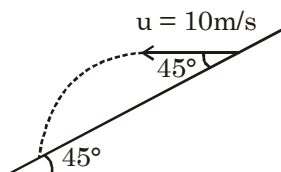
$$a_x = 2t\hat{i} \text{ m/s}^2; a_y = -10\hat{j} \text{ m/s}^2$$



- (A) (II) (iv) (P) (B) (II) (iii) (P) (C) (III) (iii) (S) (D) (II) (iii) (S)

KM0223

44. A particle is projected from a large-fixed incline plane as shown. Here $\vec{a} = g$ (Vertically downward) take $g = 10 \text{ m/s}^2$.



- (A) (II) (iv) (P) (B) (IV) (ii) (S) (C) (I) (ii) (R) (D) (IV) (ii) (P)

KM0223

45. In ground to ground projection a particle is projected at 53° from horizontal. At $t = 25$ sec after projection, its velocity vector becomes perpendicular to its initial velocity vector.

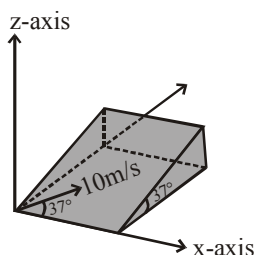
(Given $\vec{a} = g \downarrow = 10 \text{ m/s}^2$)

- (A) (IV) (i) (R) (B) (IV) (ii) (S) (C) (II) (ii) (P) (D) (I) (ii) (Q)

KM0223

MATRIX MATCH TYPE QUESTION

46. A small ball is projected along the surface of a smooth inclined plane with speed 10 m/s along the direction shown at $t = 0$. The point of projection is origin, z-axis is along vertical. The acceleration due to gravity is 10 m/s^2 . Column-I lists values of certain parameters related to motion of ball and column-II lists different time instants. Match appropriately.



Column-I

- (A) Distance from x-axis is 2.25m
(B) Speed is minimum
(C) Velocity makes angle 37° with x-axis

Column-II

- (P) 0.5 s
(Q) 1.0 s
(R) 1.5 s
(S) 2.0 s

KM0224

47.

Column-I

Column-II

(A) Time for a boat to cross a river of width ℓ by the shortest

distance ($\vec{v}$ -velocity of boat with respect to water;

$\vec{u}$ -velocity of water) $|\vec{v}| > |\vec{u}|$

(P) $\frac{\ell}{|\vec{v} + \vec{u}|}$

(B) Time for two particles moving with velocities $\vec{v}$ and $\vec{u}$

in opposite directions to meet each other.

(initial separation of particles is ℓ)

(Q) $\frac{\ell}{\sqrt{v^2 - u^2}}$

(C) Time for a boat to cross a river of width ℓ in the shortest

time ($\vec{v}$ -velocity of boat with respect to water;

$\vec{u}$ -velocity of water)

(R) $\frac{\ell}{|\vec{v}| + |\vec{u}|}$

(D) Time for a boat to travel a distance ℓ downstream

($\vec{v}$ -velocity of boat with respect to water;

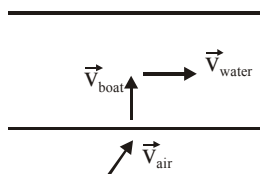
$\vec{u}$ -velocity of water)

(S) $\frac{\ell}{|\vec{v}|}$

(T) $\frac{\ell}{\sqrt{u^2 + v^2}}$

KM0225

48. A boat is being rowed in a river. Air is also blowing. Direction of velocity vectors of boat, water and air in ground frame are as shown in diagram.



Column-I

Column-II

(possible directions)

(A) Direction in which boat is being steered

(B) Direction in which a flag on the boat may flutter

(C) Direction of velocity of water relative to boat

(D) Direction of velocity of air relative to a piece of wood floating on river.

(P)

(Q)

(R)

(S)

KM0226

EXERCISE (JM)

1. A particle is moving with velocity $\vec{v} = K(y\hat{i} + x\hat{j})$, where K is a constant. The general equation for its path is : [AIEEE - 2010]

(1) $y^2 = x^2 + \text{constant}$ (2) $y = x^2 + \text{constant}$ (3) $y^2 = x + \text{constant}$ (4) $xy = \text{constant}$

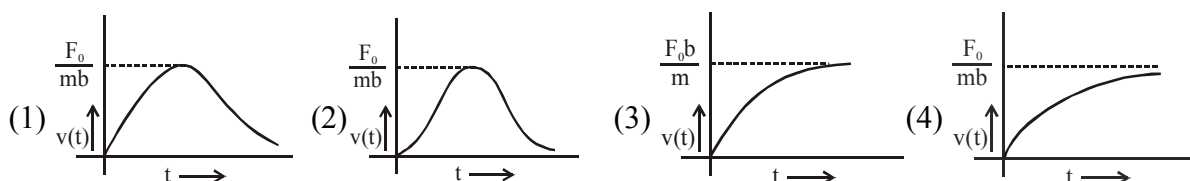
KM0227

2. A water fountain on the ground sprinkles water all around it. If the speed of water coming out of the fountain is v , the total area around the fountain that gets wet is :- [AIEEE - 2011]

(1) $\frac{\pi v^4}{2g^2}$ (2) $\pi \frac{v^2}{g^2}$ (3) $\pi \frac{v^2}{g}$ (4) $\pi \frac{v^4}{g^2}$

KM0228

3. A particle of mass m is at rest at the origin at time $t = 0$. It is subjected to a force $F(t) = F_0 e^{-bt}$ in the x direction. Its speed $v(t)$ is depicted by which of the following curves ? [AIEEE - 2012]



KM0229

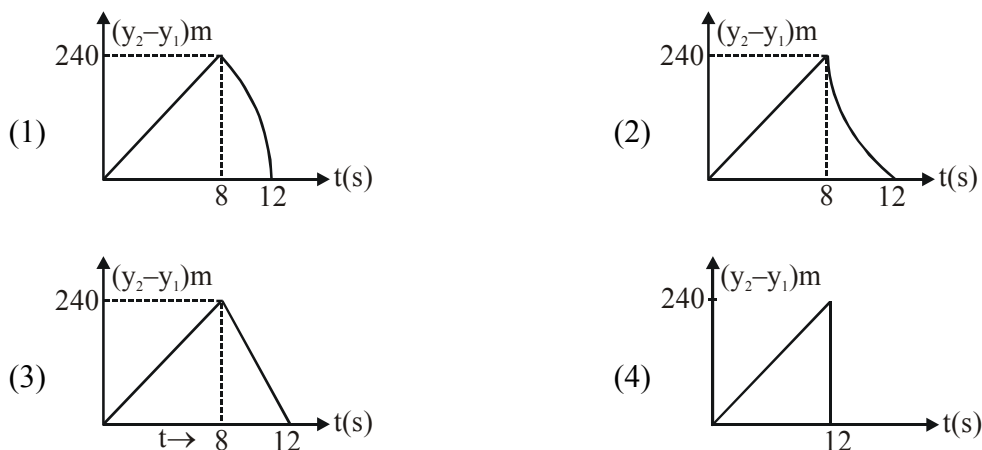
4. A projectile is given an initial velocity of $(\hat{i} + 2\hat{j})$ m/s, where $\hat{i}$ is along the ground and $\hat{j}$ is along the vertical. If $g = 10 \text{ m/s}^2$, the equation of its trajectory is : [AIEEE - 2013]

(1) $y = x - 5x^2$ (2) $y = 2x - 5x^2$ (3) $4y = 2x - 5x^2$ (4) $4y = 2x - 25x^2$

KM0230

5. Two stones are thrown up simultaneously from the edge of a cliff 240 m high with initial speed of 10 m/s and 40 m/s respectively. Which of the following graph best represents the time variation of relative position of the second stone with respect to the first ?

(Assume stones do not rebound after hitting the ground and neglect air resistance, take $g = 10 \text{ m/s}^2$) (The figure are schematic and not drawn to scale) [JEE Main-2015]



KM0231

EXERCISE (JA)

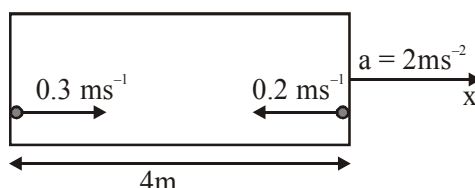
1. A train is moving along a straight line with a constant acceleration 'a'. A boy standing in the train throws a ball forward with a speed of 10 m/s, at an angle of 60° to the horizontal. The boy has to move forward by 1.15 m inside the train to catch the ball back at the initial height. The acceleration of the train, in m/s^2 , is

[IIT-JEE 2011]

KM0232

2. A rocket is moving in a gravity free space with a constant acceleration of 2 ms^{-2} along + x direction (see figure). The length of a chamber inside the rocket is 4m. A ball is thrown from the left end of the chamber in + x direction with a speed of 0.3 ms^{-1} relative to the rocket. At the same time, another ball is thrown in -x direction with a speed of 0.2 ms^{-1} from its right end relative to the rocket. The time in seconds when the two balls hit each other is

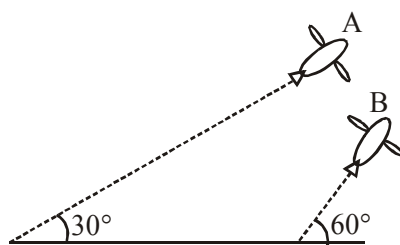
[JEE Advanced 2014]



KM0233

3. Airplanes A and B are flying with constant velocity in the same vertical plane at angles 30° and 60° with respect to the horizontal respectively as shown in figure. The speed of A is $100\sqrt{3} \text{ ms}^{-1}$. At time $t = 0$ s, an observer in A finds B at a distance of 500 m. This observer sees B moving with a constant velocity perpendicular to the line of motion of A. If at $t = t_0$, A just escapes being hit by B, t_0 in seconds is

[JEE Advanced-2014]



KM0234

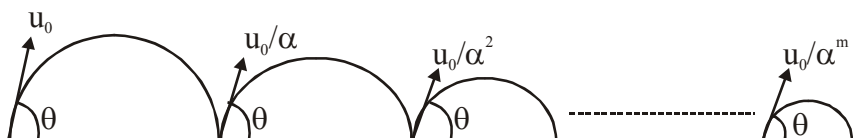
4. A ball is projected from the ground at an angle of 45° with the horizontal surface. It reaches a maximum height of 120 m and returns to the ground. Upon hitting the ground for the first time, it loses half of its kinetic energy. Immediately after the bounce, the velocity of the ball makes an angle of 30° with the horizontal surface. The maximum height it reaches after the bounce, in metres, is.....

[JEE Advanced-2018]

KM0235

5. A ball is thrown from ground at an angle θ with horizontal and with an initial speed u_0 . For the resulting projectile motion, the magnitude of average velocity of the ball up to the point when it hits the ground for the first time is V_1 . After hitting the ground, ball rebounds at the same angle θ but with a reduced speed of u_0/α . Its motion continues for a long time as shown in figure. If the magnitude of average velocity of the ball for entire duration of motion is $0.8 V_1$, the value of α is _____

[JEE Advanced-2019]



KM0236

6. Starting at time $t = 0$ from the origin with speed 1 ms^{-1} , a particle follows a two-dimensional trajectory in the x - y plane so that its coordinates are related by the equation $y = \frac{x^2}{2}$. The x and y components of its acceleration are denoted by a_x and a_y , respectively. Then
- (A) $a_x = 1 \text{ ms}^{-2}$ implies that when the particle is at the origin, $a_y = 1 \text{ ms}^{-2}$
- (B) $a_x = 0$ implies $a_y = 1 \text{ ms}^{-2}$ at all times
- (C) at $t = 0$, the particle's velocity points in the x -direction
- (D) $a_x = 0$ implies that at $t = 1 \text{ s}$, the angle between the particle's velocity and the x axis is 45°

[JEE Advanced-2020]

KM0254

ANSWER KEY

EXERCISE (S-1)

1. Ans. 5 2. Ans. (a) $\vec{v}(t) = (3.0\hat{i} - 4.0t\hat{j})$ $\hat{a}(t) = -4.0\hat{j}$ (b) 8.54 ms^{-1} , 70° with x-axis
3. Ans. (i) $\vec{a} = (-16\hat{i} - 8\hat{j}) \text{ m/s}^2$ (ii) $\vec{v} = (-30\hat{i} - 40\hat{j}) \text{ m/s}$ 4. Ans. $\sqrt{2257} \text{ m/s}$
5. Ans. (i) 1.6 sec (ii) 3.2 m (iii) 9.6 m 6. Ans. 1200 7. Ans. 50 m
8. Ans. 20 s 9. Ans. 60° , 2 m/s. 11. Ans. $20\sqrt{5} \text{ m/s}$ 12. Ans. $\frac{100}{3} \text{ m/s}$
13. Ans. (i) 100 m/s (ii) 980 m (iii) 1600 m (iv) $(80\hat{i} - 140\hat{j})$
14. Ans. $u = 50(\sqrt{3} - 1) \text{ m/s}$, $H = 125(2 - \sqrt{3}) \text{ m}$ 15. Ans. 6 16. Ans. 34
17. Ans. (a) With $\hat{i}$ to right and $\hat{j}$ up $\vec{V} = (15\hat{i} + 20\hat{j}) \text{ m/s}$; (b) 23 meters; (c) It is horizontal. $\theta = 0^\circ$
18. Ans. 10 m/s 19. Ans. 4 20. Ans. 3
21. Ans. (a) 45° , (b) 2 m/sec 22. Ans. 5 23. Ans. 6
24. Ans. 200 m, 20 m/min, 12 m/min 25. Ans. $\tan^{-1}(1/2)$ 26. Ans. $\tan^{-1}(1/3)$
27. Ans. (a) Somewhere down stream (b) 8 s (c) 16 m (d) Yes 28. Ans. 6 s, 66 m
29. Ans. 30 m upstream 30. Ans. $v_{br} > v_r$, $\theta = \sin^{-1}\left(\frac{v_r}{v_{br}}\right)$ upstream of line AB

EXERCISE (S-2)

1. Ans. (a) $\vec{v} = (5 \sin t - 3)\hat{i} + (3 \cos t - 1)\hat{j}$, (b) $(2 - 5 \cos t - 3t)\hat{i} + (2 + 3 \sin t - t)\hat{j}$
2. Ans. (i) $4\sqrt{6} \text{ m}$, (ii) 10 m 3. Ans. (i) 2 s (ii) 10 m/s (iii) 5 m (iv) 16.25 m (v) 20 m
4. Ans. (i) 7 s, (ii) 175 m 5. Ans. 0105 6. Ans. $t = 1.6 \text{ s}$
7. Ans. (i) 1 s, (ii) $(5\sqrt{3} \text{ m}, 5 \text{ m})$ 8. Ans. $\theta = 37^\circ$, $v = 6 \text{ m/s}$
9. Ans. (i) $\frac{u^2 \sin 2\alpha}{g \cos \theta}$ (ii) $v = \frac{u \cos(\alpha + \theta)}{\cos \theta}$ 10. Ans. 140 11. Ans. 5
12. Ans. B, $t_A = 165 \text{ s}$, $t_B = 150 \text{ s}$ 13. Ans. (i) 120° (ii) $2/\sqrt{3}$
14. Ans. $10\sqrt{3} \text{ m/s}$
15. Ans. (a) $7\hat{i} + 2\hat{j}$; (b) 12.30 p.m., (c) $9/2\hat{i} + \hat{j}$

EXERCISE (O-1)

1. Ans. (B) 2. Ans. (C) 3. Ans. (C) 4. Ans. (B) 5. Ans. (B) 6. Ans. (B)
 7. Ans. (D) 8. Ans. (B) 9. Ans. (D) 10. Ans. (B) 11. Ans. (B) 12. Ans. (A)
 13. Ans. (D) 14. Ans. (C) 15. Ans. (C) 16. Ans. (B) 17. Ans. (B) 18. Ans. (B)
 19. Ans. (B) 20. Ans. (C) 21. Ans. (A) 22. Ans. (A,B,C,D)
 23. Ans. (A,B,C,D) 24. Ans. (B,C,D) 25. Ans. (B,C) 26. Ans. (B) 27. Ans. (C,D)
 28. Ans. (A,B,C) 29. Ans. (C) 30. Ans. (D) 31. Ans. (B)
 32. Ans. (a) Vertically (b) 53° above east (c) 53° above west (d) 53° above north (e) 45° above south
 33. Ans. (a) 120 cm/s vertically 150 cm/s 53° above horizontal (b) 37° above the horizontal.
 34. Ans. 120 cm/s 35. Ans. 125 cm/s at 37° from the vertical 36. Ans. 6 m/s due north
 37. Ans. (a) 9 m/s (b) 16 m/s (c) 1 m/s^2 38. Ans. 10 m/s, 37° north of east
 39. Ans. (A) R; (B) P; (C) Q; (D) S 40. Ans. (A) $\rightarrow$ (R); (B) $\rightarrow$ (R); (C) $\rightarrow$ (P); (D) $\rightarrow$ (Q)

EXERCISE (O-2)

1. Ans. (D) 2. Ans. (D) 3. Ans. (C) 4. Ans. (A) 5. Ans. (C) 6. Ans. (C)
 7. Ans. (C) 8. Ans. (C) 9. Ans. (C) 10. Ans. (A) 11. Ans. (C) 12. Ans. (D)
 13. Ans. (C) 14. Ans. (A) 15. Ans. (A) 16. Ans. (C) 17. Ans. (C) 18. Ans. (D)
 19. Ans. (A) 20. Ans. (A) 21. Ans. (C) 22. Ans. (B) 23. Ans. (B) 24. Ans. (B)
 25. Ans. (B) 26. Ans. (C) 27. Ans. (B) 28. Ans. (D) 29. Ans. (A,B,C)
 30. Ans. (B,C,D) 31. Ans. (A, B,C,D) 32. Ans. (A,B,C)
 33. Ans. (B,C,D) 34. Ans. (B,C,D) 35. Ans. (C,D) 36. Ans. (A,B,C)
 37. Ans. (A,B) 38. Ans. (C) 39. Ans. (C) 40. Ans. (B) 41. Ans. (B)
 42. Ans. (D) 43. Ans. (B) 44. Ans. (C) 45. Ans. (A)
 46. Ans. (A) - (P,R) ; (B) - (Q) ; (C) - (S)
 47. Ans. (A) $\rightarrow$ (P,Q); (B) $\rightarrow$ (R); (C) $\rightarrow$ (S); (D) $\rightarrow$ (P,R)
 48. Ans. (A)-P; (B)-Q, S; (C)-S; (D)-P,R

EXERCISE (JM)

1. Ans. (1) 2. Ans. (4) 3. Ans. (4) 4. Ans. (2) 5. Ans. (1)

EXERCISE (JA)

1. Ans. 5 2. Ans. 8 or 2 3. Ans. 5 4. Ans. 30 [29.60, 30.40] 5. Ans. 4.00
 6. Ans. (A,B,C,D)

S. No.	CHAPTER NAME	Pg.No.
1	CIRCULAR MOTION	01-38
2	WORK, POWER & ENERGY	39-86
3	CENTER OF MASS, MOMENTUM & COLLISION	87-146
4	ROTATIONAL DYNAMICS	147-237



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Important Notes

[illegible]

CIRCULAR MOTION

KEY CONCEPT

Cross product (Vector Product) of two vectors :

Vector product of two vectors $\vec{A}$ and $\vec{B}$, also called the cross product, is denoted by $\vec{A} \times \vec{B}$. As the name suggests, the vector product is itself a vector. We will use this product to describe torque and angular momentum and extensively to describe magnetic field forces.

Vector product of two vectors $\vec{A}$ and $\vec{B}$ which are at an angle ϕ is defined as

$$\vec{C} = \vec{A} \times \vec{B},$$

then $C = AB \sin \phi \hat{n}$

where $\hat{n}$ is a unit vector perpendicular to plane containing vector $\vec{A}$ and $\vec{B}$. Direction of $\hat{n}$ is given by right hand rule as described below.

Note: the vector product of two parallel or antiparallel vectors is always zero. In particular, the vector product of any vector with itself is zero.

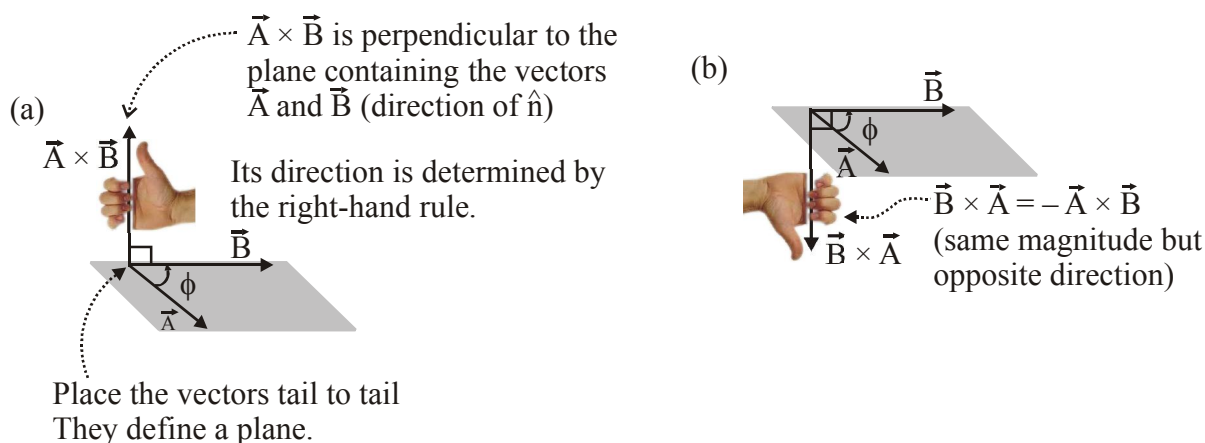


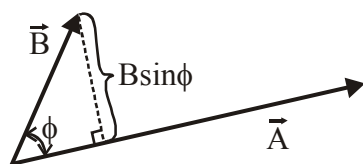
Figure : (a) The vector product $\vec{A} \times \vec{B}$, determined by the right-hand rule. (b) $\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$

Note: The vector product is not commutative! In fact, for any two vectors $\vec{A}$ and $\vec{B}$,

$$\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}$$

(Magnitude of $\vec{A} \times \vec{B}$) equals $A(B \sin \phi)$.

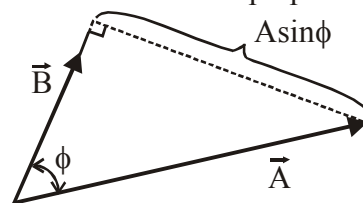
(Magnitude of $\vec{A}$) times (Component of $\vec{B}$ perpendicular to $\vec{A}$)



(a)

(Magnitude of $\vec{A} \times \vec{B}$) also equals $B(A \sin \phi)$.

(Magnitude of $\vec{B}$) times (Component of $\vec{A}$ perpendicular to $\vec{B}$)



(b)

Figure: Calculating the magnitude $AB \sin \phi$ of the vector product of two vector, $\vec{A} \times \vec{B}$.

Calculating the Vector Product Using Components

$$\hat{i} \times \hat{i} = \hat{j} \times \hat{j} = \hat{k} \times \hat{k} = 0$$

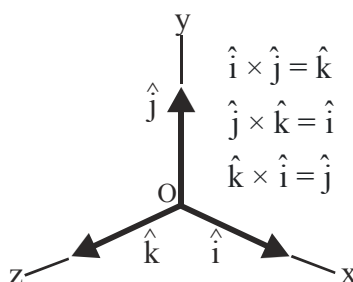
Using the right-hand rule, we find

$$\hat{i} \times \hat{j} = -\hat{j} \times \hat{i} = \hat{k};$$

$$\hat{j} \times \hat{k} = -\hat{k} \times \hat{j} = \hat{i};$$

$$\hat{k} \times \hat{i} = -\hat{i} \times \hat{k} = \hat{j}$$

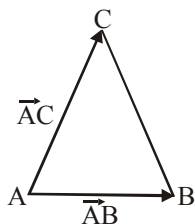
A right-handed coordinate system



Ex. Find the area of the triangle formed by the points having position vectors

$$\vec{A} = \hat{i} - \hat{j} - 3\hat{k}, \quad \vec{B} = 4\hat{i} - 3\hat{j} + \hat{k} \quad \text{and} \quad \vec{C} = 3\hat{i} - \hat{j} + 2\hat{k}$$

Sol.



Let ABC is the triangle formed by the points A, B and C. Then

$$\vec{AB} = \vec{B} - \vec{A} = (4\hat{i} - 3\hat{j} + \hat{k}) - (\hat{i} - \hat{j} - 3\hat{k}) = 3\hat{i} - 2\hat{j} + 4\hat{k}$$

$$\vec{AC} = \vec{C} - \vec{A} = (3\hat{i} - \hat{j} + 2\hat{k}) - (\hat{i} - \hat{j} - 3\hat{k}) = 2\hat{i} + 5\hat{k}$$

$$\text{Now, } \vec{AB} \times \vec{AC} = (3\hat{i} - 2\hat{j} + 4\hat{k}) \times (2\hat{i} + 5\hat{k})$$

$$= \hat{i}(-10 - 0) + \hat{j}(8 - 15) + \hat{k}(0 + 4) = -10\hat{i} - 7\hat{j} + 4\hat{k}$$

Ex. Find unit vector perpendicular to the plane determined by the points P(1, -1, 2), Q(2, 0, -1) and R(0, 2, 1).

Sol. $\vec{PQ} = (\text{Position vector of Q}) - (\text{Position vector of P})$

$$= (2\hat{i} - \hat{k}) - (\hat{i} - \hat{j} + 2\hat{k}) = \hat{i} + \hat{j} - 3\hat{k}$$

$$\text{Similarly, } \vec{PR} = (2\hat{j} + \hat{k}) - (\hat{i} - \hat{j} + 2\hat{k}) = -\hat{i} + 3\hat{j} - \hat{k}$$

$$\therefore |\vec{PQ} \times \vec{PR}| = \sqrt{8^2 + 4^2 + 4^2} = 4\sqrt{6}$$

$\therefore$ Required unit vectors

$$= \pm \frac{1}{4\sqrt{6}} (8\hat{i} + 4\hat{j} + 4\hat{k}) = \pm \frac{1}{\sqrt{6}} (2\hat{i} + \hat{j} + \hat{k})$$

Definition of Circular Motion

When a particle moves in a plane such that its distance from a fixed (or moving) point remains constant then its motion is called as circular motion with respect to that fixed point.

That fixed point is called centre and the distance is called radius of circular path.

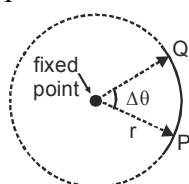
Radius Vector

The vector joining the centre of the circle and the center of the particle performing circular motion is called radius vector. It has constant magnitude and variable direction.

KINEMATICS OF CIRCULAR MOTION

Angular Displacement

- Angle traced by position vector of a particle moving w.r.t. some fixed point is called angular displacement.



$\Delta\theta = \text{angular displacement}$

$$\text{Angle} = \frac{\text{Arc}}{\text{Radius}} \text{ or } \Delta\theta = \frac{\text{Arc PQ}}{r}$$

- Small Angular displacement $d\vec{\theta}$ is a vector quantity, but large angular displacement θ is scalar quantity.

$$d\vec{\theta}_1 + d\vec{\theta}_2 = d\vec{\theta}_2 + d\vec{\theta}_1 \text{ But } \vec{\theta}_1 + \vec{\theta}_2 \neq \vec{\theta}_2 + \vec{\theta}_1$$

- Its direction is perpendicular to plane of rotation and given by right hand screw rule.
- It is dimensionless and has S.I. unit "Radian" while other units are degree or revolution.

$$2\pi \text{ radian} = 360^\circ = 1 \text{ revolution}$$

Frequency (n) : Number of revolutions describes by particle per second is its frequency. Its unit is revolutions per second (r.p.s.) or revolutions per minute (r.p.m.)

Time Period (T) : It is time taken by particle to complete one revolution. $T = \frac{1}{n}$

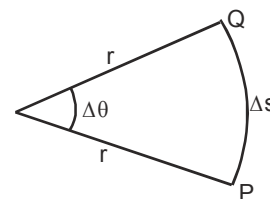
Angular Velocity (ω) : It is defined as the rate of change of angular displacement of moving particle

$$\omega = \frac{\text{Angle traced}}{\text{Time taken}} = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$$

Relation between linear and Angular velocity

$$\text{Angle} = \frac{\text{Arc}}{\text{Radius}} \text{ or } \frac{\Delta s}{r}$$

$$\Delta\theta = \frac{\Delta s}{r} \text{ or } \Delta s = r\Delta\theta$$



$$\therefore \frac{\Delta s}{\Delta t} = \frac{r\Delta\theta}{\Delta t} \text{ if } \Delta t \rightarrow 0 \text{ then } \frac{ds}{dt} = r \frac{d\theta}{dt} \quad \boxed{v = \omega r}$$

$$\boxed{\vec{v} = \vec{\omega} \times \vec{r}} \text{ (direction of } \vec{v} \text{ is according to right hand thumb rule)}$$

Average Angular Velocity (ω_{av})

$$\omega_{av} = \frac{\text{total angle of rotation}}{\text{total time taken}} = \frac{\theta_2 - \theta_1}{t_2 - t_1} = \frac{\Delta\theta}{\Delta t} = \frac{2\pi}{T} = 2\pi n$$

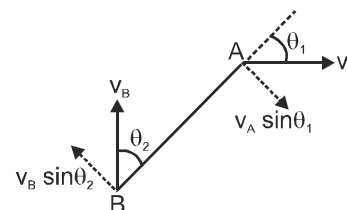
where θ_1 and θ_2 are angular position of the particle at instant t_1 and t_2 .

Instantaneous Angular Velocity

The angular velocity at some particular instant $\omega = \lim_{\Delta t \rightarrow 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt}$ or $\vec{\omega} = \frac{d\vec{\theta}}{dt}$

Relative Angular Velocity

- Relative angular velocity of a particle 'A' w.r.t. other moving particle B is the angular velocity of the position vector of A w.r.t. B. That means it is the rate at which position vector of 'A' w.r.t. B rotates at that instant



$$\omega_{AB} = \frac{(v_{AB})_{\perp}}{r_{AB}} = \frac{\text{Relative velocity of A w.r.t. B perpendicular to line AB}}{\text{separation between A and B}}$$

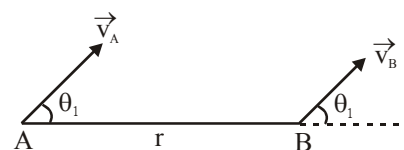
$$\text{here } (v_{AB})_{\perp} = v_A \sin \theta_1 + v_B \sin \theta_2 \quad \therefore \omega_{AB} = \frac{v_A \sin \theta_1 + v_B \sin \theta_2}{r}$$

- It is an axial vector quantity.
- Its direction is same as that of angular displacement i.e. perpendicular to the plane of rotation and along the axis according to right hand screw rule.
- Its unit is radian/second.

If particles A and B are moving with a velocity $\vec{v}_A$ and $\vec{v}_B$

and separated by a distance r at a given instant then

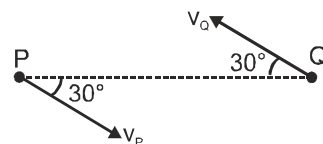
$$(i) \frac{dr}{dt} = v_B \cos \theta_2 - v_A \cos \theta_1 \quad (ii) \frac{d\theta_{AB}}{dt} = \omega_{AB} = \frac{v_B \sin \theta_2 - v_A \sin \theta_1}{r}$$



Ex. A particle revolving in a circular path completes first one third of circumference in 2 s, while next one third in 1 s. Calculate the average angular velocity of particle.

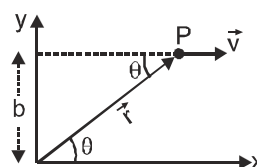
Sol. $\theta_1 = \frac{2\pi}{3}$ and $\theta_2 = \frac{2\pi}{3}$ total time $T = 2 + 1 = 3$ s $\therefore \langle \omega_{av} \rangle = \frac{\theta_1 + \theta_2}{T} = \frac{\frac{2\pi}{3} + \frac{2\pi}{3}}{3} = \frac{\frac{4\pi}{3}}{3} = \frac{4\pi}{9}$ rad/s

Ex. Two moving particles P and Q are 10 m apart at any instant. Velocity of P is 8 m/s at 30° , from line joining the P and Q and velocity of Q is 6 m/s at 30° . Calculate the angular velocity of P w.r.t. Q

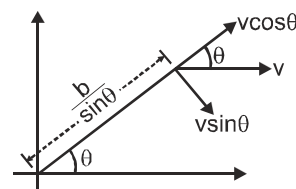


Sol. $\omega_{PQ} = \frac{8 \sin 30^\circ - (-6 \sin 30^\circ)}{10} = 0.7$ rad/s.

Ex. A particle moving parallel to x-axis as shown in fig. such that at all instant the y-axis component of its position vector is constant and is equal to 'b'. Find the angular velocity of the particle about the origin when its radius vector makes angle θ from the x-axis.



Sol. $\therefore \omega_{PO} = \frac{v \sin \theta}{\frac{b}{\sin \theta}} = \frac{v}{b} \sin^2 \theta$



Ex. The angular velocity of a particle is given by $\omega = 1.5t - 3t^2 + 2$, Find the time when its angular acceleration becomes zero.

Sol. $\alpha = \frac{d\omega}{dt} = 1.5 - 6t = 0 \Rightarrow t = 0.25$ s.

Ex. A disc starts from rest and on the application of a torque, it gains an angular acceleration given by $\alpha = 3t - t^2$. Calculate the angular velocity after 2 s.

Sol. $\frac{d\omega}{dt} = 3t - t^2 \Rightarrow \int_0^\omega d\omega = \int_0^t (3t - t^2) dt \Rightarrow \omega = \frac{3t^2}{2} - \frac{t^3}{3} \Rightarrow$ at $t = 2$ s, $\omega = \frac{10}{3}$ rad/s

Angular Acceleration (α)

- Rate of change of angular velocity is called angular acceleration. $\alpha = \lim_{\Delta t \rightarrow 0} \frac{\Delta \omega}{\Delta t} = \frac{d\omega}{dt}$ or $\vec{\alpha} = \frac{d\vec{\omega}}{dt}$
- Its an axial vector quantity. Its direction is along the axis according to right hand screw rule.
- Unit $\rightarrow$ rad/s²

Relation between Angular and Linear Acceleration

$\vec{v} = \vec{\omega} \times \vec{r}$ ($\vec{v}$ is a tangential vector, $\vec{\omega}$ is a axial vector and $\vec{r}$ is a radial vector.)

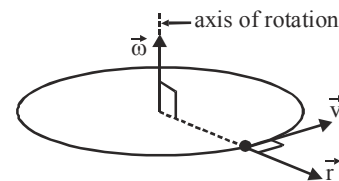
These three vectors are mutually perpendicular. but $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d}{dt}(\vec{\omega} \times \vec{r}) = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt}$

$$\text{or } \vec{a} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v} \quad \left(\frac{d\vec{\omega}}{dt} = \vec{\alpha} \text{ and } \frac{d\vec{r}}{dt} = \vec{v} \right) \Rightarrow \vec{a} = \vec{a}_T + \vec{a}_C$$

($\vec{a}_T = \vec{\alpha} \times \vec{r}$ is tangential acceleration &

$\vec{a}_C = \vec{\omega} \times \vec{v}$ is centripetal acceleration)

$\vec{a} = \vec{a}_T + \vec{a}_C$ ($\vec{a}_T$ and $\vec{a}_C$ are two component of net linear acceleration)



Tangential Acceleration

$\vec{a}_T = \vec{\alpha} \times \vec{r}$, its direction is parallel to velocity. $\vec{v} = \vec{\omega} \times \vec{r}$ and $\vec{a}_T = \vec{\alpha} \times \vec{r}$ as $\vec{\omega}$ and $\vec{\alpha}$ both are parallel and along the axis so that $\vec{v}$ and $\vec{a}_T$ are also parallel and along the tangential direction.

Magnitude of tangential acceleration in case of circular motion.

$$a_T = \alpha r \sin 90^\circ = \alpha r \quad (\vec{\alpha} \text{ is axial, } \vec{r} \text{ is radial so that } \vec{\alpha} \perp \vec{r})$$

As $\vec{a}_T$ is along the direction of motion (in the direction of $\vec{v}$) so that $\vec{a}_T$ is responsible for change in speed of the particle. Its magnitude is rate of change of speed of the particle. If particle is moving on a circular path with constant speed then tangential acceleration is zero.

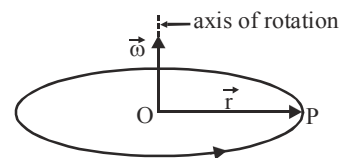
Centripetal acceleration

$$\vec{a}_C = \vec{\omega} \times \vec{v} \Rightarrow \vec{a}_C = \vec{\omega} \times (\vec{\omega} \times \vec{r}) \quad (\because \vec{v} = \vec{\omega} \times \vec{r})$$

Let $\vec{r}$ is in $\hat{i}$ direction and $\vec{\omega}$ is in $\hat{j}$ direction then direction of

$$\vec{a}_C \text{ is along } \hat{j} \times (\hat{j} \times \hat{i}) = \hat{j} \times (-\hat{k}) = -\hat{i}$$

opposite direction of $\vec{r}$ i.e., from P to O and it is



centripetal direction. Magnitude of centripetal acceleration, $a_C = \omega v = \frac{v^2}{r} = \omega^2 r$, $\vec{a}_C = \frac{v^2}{r}(-\hat{r})$

- Centripetal acceleration is always perpendicular to the velocity or displacement at each point.

Net Linear Acceleration :

$$\vec{a} = \vec{a}_T + \vec{a}_C \text{ and } \vec{a}_T \perp \vec{a}_C \text{ so that } |\vec{a}| = \sqrt{a_T^2 + a_C^2}$$

About uniform circular motion :—

- Position vector ($\vec{r}$) is always perpendicular to the velocity vector ($\vec{v}$) i.e. $\vec{r} \cdot \vec{v} = 0$
- velocity vector is always perpendicular to the acceleration. $\vec{v} \cdot \vec{a} = 0$
- $\therefore |\vec{v}| = \text{constant}$ so tangential acc. $a_t = 0$ $\therefore f_t = 0$
- Important difference between the projectile motion and uniform circular motion :
In projectile motion, both the magnitude and the direction of acceleration (g) remain constant, while in uniform circular motion the magnitude remains constant but the direction continuously changes.

DYNAMICS OF CIRCULAR MOTION

If a particle moves with constant speed in a circle, motion is called uniform circular. In uniform circular motion a resultant non-zero force acts on the particle. The acceleration is due to the change in direction of the velocity vector. In uniform circular motion tangential acceleration (a_t) is zero. The

acceleration of the particle is towards the centre and its magnitude is $\frac{v^2}{r}$. Here, v is the speed of the particle and r the radius of the circle. The direction of the resultant force F is therefore towards centre

and its magnitude is $F = \frac{mv^2}{r} = mr\omega^2$ (as $v = r\omega$)

Here, ω is the angular speed of the particle. This force F is called the centripetal force. Thus, a

centripetal force of magnitude $\frac{mv^2}{r}$ is needed to keep the particle moving in a circle with constant speed. This force is provided by some external source such as friction, magnetic force, coulomb force, gravitation, tension, etc.

Circular Turning of Roads

When vehicles go through turnings, they travel along a nearly circular arc. There must be some force which will produce the required centripetal acceleration. If the vehicles travel in a horizontal circular path, this resultant force is also horizontal. The necessary centripetal force is being provided to the vehicles by following three ways:

- By friction only.
- By banking of roads only.
- By friction and banking of roads both.

In real life the necessary centripetal force is provided by friction and banking of roads both.

• By Friction only

Suppose a car of mass m is moving at a speed v in a horizontal circular arc of radius r . In this case, the necessary centripetal force to the car will be provided by force of friction f acting towards centre.

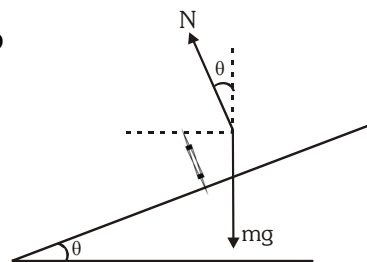
$$\text{Thus, } f = \frac{mv^2}{r} \quad \therefore f_{\max} = \mu N = \mu mg$$

Therefore, for a safe turn without sliding $\frac{mv^2}{r} \leq f_{\max} \Rightarrow \frac{mv^2}{r} \leq \mu mg \Rightarrow v \leq \sqrt{\mu rg}$

• By Banking of Roads only

Friction is not always reliable at circular turns if high speeds and sharp turns are involved. To avoid dependence on friction, the roads are banked at the turn so that the outer part of the road is some what lifted compared to the inner part.

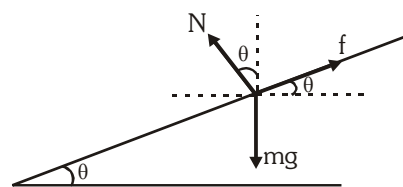
$$N \sin \theta = \frac{mv^2}{r} \text{ and } N \cos \theta = mg \Rightarrow \tan \theta = \frac{v^2}{rg} \Rightarrow v = \sqrt{rg \tan \theta}$$



- **Friction and Banking of Road both**

If a vehicle is moving on a circular road which is rough and banked also, then three forces may act on the vehicle, of these the first force, i.e., weight (mg) is fixed both in magnitude and direction.

The direction of second force, i.e., normal reaction N is also fixed (perpendicular to road) while the direction of the third force, i.e., friction f can be either inwards or outwards while its magnitude can be varied upto a maximum limit ($f_{\max} = \mu N$). So, the magnitude of normal reaction N and direction plus magnitude of friction f are so adjusted that the resultant of the



three forces mentioned above is $\frac{mv^2}{r}$ towards the centre.

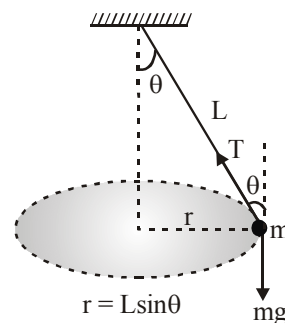
Conical Pendulum

If a small particle of mass m tied to a string is whirled in a horizontal circle, as shown in figure. The arrangement is called the 'conical pendulum'. In case of conical pendulum the vertical component of tension balances the weight while its horizontal component provides the necessary centripetal force. Thus,,

$$T \sin \theta = \frac{mv^2}{r} \quad \text{and} \quad T \cos \theta = mg \Rightarrow v = \sqrt{rg \tan \theta}$$

$$\therefore \text{Angular speed } \omega = \frac{v}{r} = \sqrt{\frac{g \tan \theta}{r}}$$

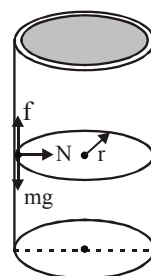
So, the time period of pendulum is $T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{r}{g \tan \theta}} = 2\pi\sqrt{\frac{L \cos \theta}{g}}$



'Death Well' or Rotor

In case of 'death well' a person drives a motorcycle on a vertical surface of a large wooden well while in case of a rotor at a certain angular speed of rotor a person hangs resting against the wall without any support from the bottom. In death well walls are at rest and person revolves while in case of rotor person is at rest and the walls rotates.

In both cases friction balances the weight of person while reaction provides the centripetal force for circular motion,



Death well



Rotor

$$\text{i.e., } f = mg \text{ and } N = \frac{mv^2}{r} = mr\omega^2$$

Ex. Find the maximum speed at which a car can turn round a curve of 30 m radius on a level road if the coefficient of friction between the tyres and the road is 0.4 [acceleration due to gravity = 10 m/s^2]

Sol. Here centripetal force is provided by friction so

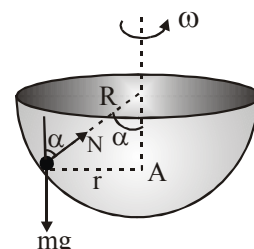
$$\frac{mv^2}{r} \leq \mu mg \Rightarrow v_{\max} = \sqrt{\mu rg} = \sqrt{120} \approx 11 \text{ ms}^{-1}$$

Ex. For traffic moving at 60 km/hr, if the radius of the curve is 0.1 km, what is the correct angle of banking of the road ? ($g = 10 \text{ m/s}^2$)

Sol. In case of banking $\tan \theta = \frac{v^2}{rg}$ Here $v = 60 \text{ km/hr} = 60 \times \frac{5}{18} \text{ ms}^{-1} = \frac{50}{3} \text{ ms}^{-1}$ $r = 0.1 \text{ km} = 100 \text{ m}$

$$\text{So } \tan \theta = \frac{50/3 \times 50/3}{100 \times 10} = \frac{5}{18} \Rightarrow \theta = \tan^{-1} \left(\frac{5}{18} \right)$$

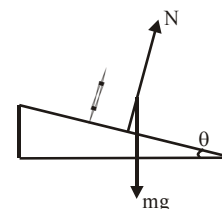
Ex. A hemispherical bowl of radius R is rotating about its axis of symmetry which is kept vertical. A small ball kept in the bowl rotates with the bowl without slipping on its surface. If the surface of the bowl is smooth and the angle made by the radius through the ball with the vertical is α . Find the angular speed at which the bowl is rotating.



Sol. $N \cos \alpha = mg$ and $N \sin \alpha = m r \omega^2$ but $r = R \sin \alpha$
 $\Rightarrow N \sin \alpha = m R \sin \alpha \omega^2 \Rightarrow N = m R \omega^2$

$$\Rightarrow (m R \omega^2) \cos \alpha = mg \Rightarrow \omega = \sqrt{\frac{g}{R \cos \alpha}}$$

Ex. A car is moving along a banked road laid out as a circle of radius r . (a) What should be the banking angle θ so that the car travelling at speed v needs no frictional force from the tyres to negotiate the turn? (b) The coefficients of friction between tyres and road are $\mu_s = 0.9$ and $\mu_k = 0.8$. At what maximum speed can a car enter the curve without sliding towards the top edge of the banked turn?



Sol. (a) $N \sin \theta = \frac{mv^2}{r}$ and $N \cos \theta = mg \Rightarrow \tan \theta = \frac{v^2}{rg}$

Note : In above case friction does not play any role in negotiating the turn.

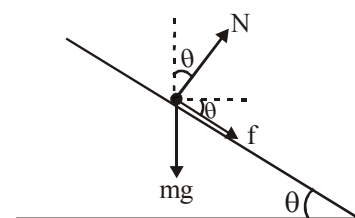
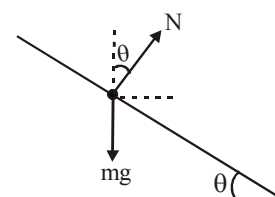
(b) If the driver moves faster than the speed mentioned above, a friction force must act parallel to the road, inward towards centre of the turn.

$$\Rightarrow F \cos \theta + N \sin \theta = \frac{mv^2}{r} \text{ and } N \cos \theta = mg + f \sin \theta$$

For maximum speed of $f = \mu N$

$$\Rightarrow N (\mu \cos \theta + \sin \theta) = \frac{mv^2}{r} \text{ and } N (\cos \theta - \mu \sin \theta) = mg$$

$$\Rightarrow \frac{v^2}{rg} = \frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} \Rightarrow v = \sqrt{\left(\frac{\sin \theta + \mu \cos \theta}{\cos \theta - \mu \sin \theta} \right) rg}$$



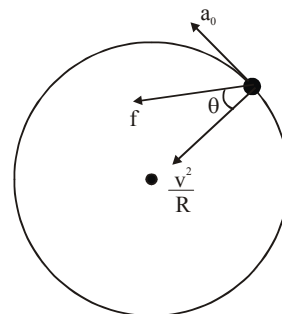
Ex. A car starts from rest with a constant tangential acceleration a_0 in a circular path of radius r . At time t , the car skids, find the value of coefficient of friction.

Sol. The tangential and centripetal acceleration is provided only by the frictional force.

$$\text{Thus, } f \sin \theta = ma_0, f \cos \theta = \frac{mv^2}{r} = \frac{m(a_0 t)^2}{r}$$

$$\Rightarrow f = m \sqrt{a_0^2 + \frac{(a_0 t)^4}{r^2}} = ma_0 \sqrt{1 + \frac{a_0^2 t^4}{r^2}} = f_{\max}$$

$$\mu mg = ma_0 \sqrt{1 + \frac{a_0^2 t^4}{r^2}} \Rightarrow \mu = \frac{a_0}{g} \sqrt{1 + \frac{a_0^2 t^4}{r^2}}$$



Centrifugal force :-

Centrifugal force is a pseudo force which an observer needs to consider while making observations in a rotating frame. This force is non physical and arise from kinematics and not due to physical interactions. Centrifugal force is directed away from axis of rotation of rotating frame and its value is $m\omega^2 r$, where ω is angular speed of rotating frame where observer has kept himself fixed and r is distance of object of mass m from axis of rotation.

EXERCISE (S-1)

Kinematics of circular motion

1. A stone tied to the end of a string 80 cm long is whirled in a horizontal circle with a constant speed. If the stone makes 14 revolutions in 25 s, what is the magnitude and direction of acceleration of the stone ?

CR0001

2. A particle is revolving in a circle of radius 1m with an angular speed of 12 rad/s. At $t = 0$, it was subjected to a constant angular acceleration α and its angular speed increased to $(480/\pi)$ rotation per minute (rpm) in 2 sec. Particle then continues to move with attained speed. Calculate
 - (i) angular acceleration of the particle,
 - (ii) tangential velocity of the particle as a function of time.
 - (iii) acceleration of the particle at $t = 0.5$ second and at $t = 3$ second
 - (iv) angular displacement at $t = 3$ second.

CR0002

3. A particle moves in a circle of radius 1.0 cm at a speed given by $v = 2.0 t$ where v is in cm/s and t in seconds.
 - (a) Find the radial acceleration of the particle at $t = 1$ s.
 - (b) Find the tangential acceleration at $t = 1$ s.
 - (c) Find the magnitude of the acceleration at $t = 1$ s.

CR0003

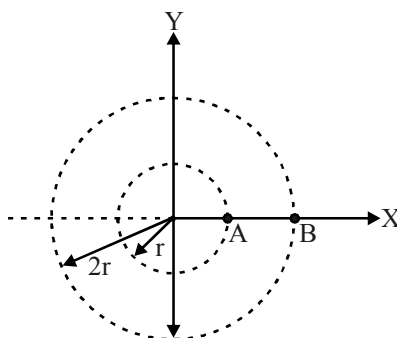
4. A particle is travelling in a circular path of radius 4m. At a certain instant the particle is moving at 20m/s and its acceleration is at an angle of 37° from the direction to the centre of the circle as seen from the particle
 - (i) At what rate is the speed of the particle increasing?
 - (ii) What is the magnitude of the acceleration?

CR0004

5. A disk rotates about its central axis starting from rest and accelerates with constant angular acceleration. At one instant it is rotating at 12 rad/s and after 80 radian of more angular displacement, its angular speed becomes 28 rad/s. How much time (seconds) does the disk takes to complete the mentioned angular displacement of 80 radians.

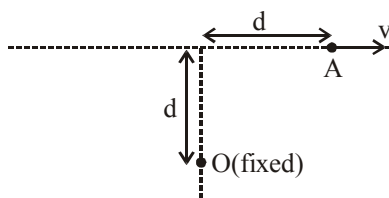
CR0005

6. Two particles A and B are moving in a horizontal plane anticlockwise on two different concentric circles with different constant angular velocities 2ω and ω respectively. Find the relative velocity (in m/s) of B w.r.t. A after time $t = \pi/\omega$. They both start at the position as shown in figure (Take $\omega = 3\text{ rad/sec}$, $r = 2\text{ m}$)



CR0006

7. Find angular velocity of A with respect to O at the instant shown in the figure.



CR0007

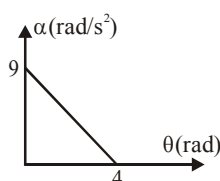
8. A stone is thrown horizontally with the velocity 15 m/s. Determine the tangential and normal accelerations of the stone in 1 second after it begins to move.

CR0008

9. A particle moves in the x-y plane with the velocity $\vec{v} = a\hat{i} + b t\hat{j}$. At the instant $t = a\sqrt{3}/b$ the magnitude of tangential, normal and total acceleration are _____, _____, & _____.

CR0009

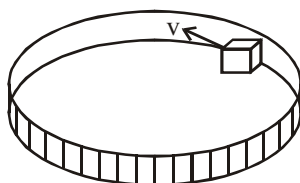
10. A particle starts moving in a non-uniform circular motion, has angular acceleration as shown in figure. The angular velocity at the end of 4 radian is given by ω rad/s then find the value of ω .



CR0010

Dynamics of circular motion

11. A block of mass m moves with speed v against a smooth, fixed vertical circular groove of radius r kept on smooth horizontal surface.



Find :

- normal reaction of the floor on the block.
- normal reaction of the vertical wall on the block.

CR0011

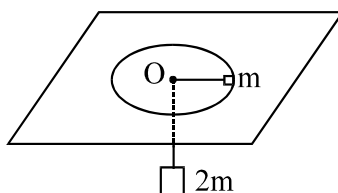
12. A cyclist speeding at 18 km/h on a level road takes a sharp circular turn of radius 3 m without reducing the speed. The coefficient of static friction between the tyres and the road is 0.1. Will the cyclist slip while taking the turn?

CR0012

13. A stone of mass 0.25 kg tied to the end of a string is whirled round in a circle of radius 1.5 m with a speed of 40 rev/min in a horizontal plane. What is the tension in the string ? What is the maximum speed with which the stone can be whirled around if the string can withstand a maximum tension of 200 N ?

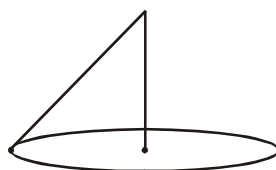
CR0013

14. A mass m rotating freely in a horizontal circle of radius 1 m on a frictionless smooth table supports a stationary mass $2m$, attached to the other end of the string passing through smooth hole O in table, hanging vertically. Find the angular velocity of rotation.



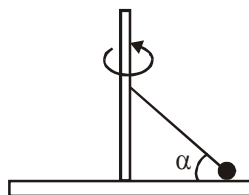
CR0014

15. Consider a conical pendulum having bob of mass m is suspended from a ceiling through a string of length L . The bob moves in a horizontal circle of radius r . Find (a) the angular speed of the bob and (b) the tension in the string.



CR0015

16. A circular platform rotates around a vertical axis with angular velocity $\omega = 10 \text{ rad/s}$. On the platform is a ball of mass 1 kg , attached to the long axis of the platform by a thin rod of length 10 cm ($\alpha = 30^\circ$). Find normal force exerted by the ball on the platform (in newton). Friction is absent.



CR0016

17. An aircraft executes a horizontal loop at a speed of 720 km/h with its wings banked at 15° . What is the radius of the loop ?

CR0017

18. A block of mass $m = 20 \text{ kg}$ is kept at a distance $R = 1 \text{ m}$ from central axis of rotation of a round turn table (A table whose surface can rotate about central axis). Table starts from rest and rotates with constant angular acceleration, $\alpha = 3 \text{ rad/sec}^2$. The friction coefficient between block and table is $\mu = 0.5$. At time

$t = \frac{x}{3} \text{ sec}$ from starting of motion (i.e. $t = 0 \text{ sec}$) the block is just about to slip. Find the value of x .

CR0018

EXERCISE (S-2)

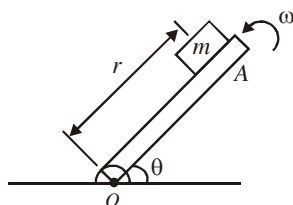
- A stone is launched upward at 45° with speed v_0 . A bee follows the trajectory of the stone at a constant speed equal to the initial speed of the stone.
 - Find the radius of curvature at the top point of the trajectory.
 - What is the acceleration of the bee at the top point of the trajectory? For the stone, neglect the air resistance.

CR0019

- A particle is moving along a circular path of radius R in such a way that at any instant magnitude of radial acceleration & tangential acceleration are equal. If at $t = 0$ velocity of particle is V_0 . Find the speed of the particle after time $t = \frac{R}{2V_0}$

CR0020

- The member OA rotates in vertical plane about a horizontal axis through O with a constant counter clockwise velocity $\omega = 3 \text{ rad/s}$. As it passes the position $\theta = 0$, a small mass m is placed upon it at a radial distance $r = 0.5 \text{ m}$. If the mass is observed to slip at $\theta = 37^\circ$, find the coefficient of friction between the mass & the member.



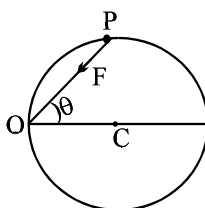
CR0021

- Two blocks of mass $m_1 = 10 \text{ kg}$ and $m_2 = 5 \text{ kg}$ connected to each other by a massless inextensible string of length 0.3 m are placed along a diameter of a turn table. The coefficient of friction between the table and m_1 is 0.5 while there is no friction between m_2 and the table. The table is rotating with an angular velocity of 10 rad/sec about a vertical axis passing through its centre. The masses are placed along the diameter of the table on either side of the centre O such that m_1 is at a distance of 0.124 m from O. The masses are observed to be at rest with respect to an observer on the turn table.
 - Calculate the frictional force on m_1
 - What should be the minimum angular speed of the turn table so that the masses will slip from this position.
 - How should the masses be placed with the string remaining taut, so that there is no frictional force acting on the mass m_1 .

CR0022

5. A particle P is moving on a circle under the action of only one force acting always towards fixed point

O on the circumference. Find ratio of $\frac{d^2\theta}{dt^2}$ & $\left(\frac{d\theta}{dt}\right)^2$.



CR0023

6. A thin circular loop of radius R rotates about its vertical diameter with an angular frequency ω . Show that a small bead on the wire loop remains at its lowermost point for $\omega \leq \sqrt{g/R}$. What is the angle made by the radius vector joining the centre to the bead with the vertical downward direction for $\omega = \sqrt{2g/R}$? Neglect friction.

CR0024

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Cross product of vectors

- The area of parallelogram represented by the vectors $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = \hat{i} + 4\hat{j}$ is :-
(A) 14 unit (B) 7.5 unit (C) 10 unit (D) 5 unit CR0025
- What is the angle between $(\vec{P} + \vec{Q})$ and $(\vec{P} \times \vec{Q})$?
(A) 0 (B) $\frac{\pi}{2}$ (C) $\frac{\pi}{4}$ (D) π CR0026
- The value of n so that vectors $2\hat{i} + 3\hat{j} - 2\hat{k}$, $5\hat{i} + n\hat{j} + \hat{k}$ and $-\hat{i} + 2\hat{j} + 3\hat{k}$ may be coplanar, will be :-
(A) 18 (B) 28 (C) 9 (D) 36 CR0027
- If $\vec{a}$ and $\vec{b}$ are two vectors then the value of $(\vec{a} + \vec{b}) \times (\vec{a} - \vec{b})$ is :-
(A) $2(\vec{b} \times \vec{a})$ (B) $-2(\vec{b} \times \vec{a})$ (C) $\vec{b} \times \vec{a}$ (D) $\vec{a} \times \vec{b}$ CR0028
- If $|\vec{a} \cdot \vec{b}| = \sqrt{3} |\vec{a} \times \vec{b}|$, then the angle between $\vec{a}$ and $\vec{b}$ is :-
(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$ CR0029

Kinematics of circular motion

- A body is executing circular motion in the vertical plane containing directions. If the direction of velocity ($\vec{v}$) at the top most point is towards west, what is the direction of angular velocity ($\vec{\omega}$) ?
(A) east (B) west (C) north (D) south CR0030
- If the magnitude of velocity in the previous question is decreasing with time, what is the direction of angular acceleration ($\vec{\alpha}$) ?
(A) east (B) west (C) north (D) south CR0031

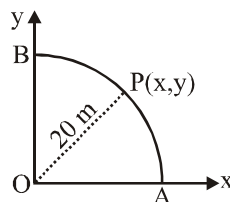
8. A mass is revolving in a circle which lies in a plane of paper. The direction of angular acceleration can be :-
 (A) perpendicular to radius and velocity (B) towards the radius
 (C) tangential (D) at right angle to angular velocity

CR0032

9. The second's hand of a watch has length 6 cm. Speed of end point and magnitude of difference of velocities at two perpendicular positions will be
 (A) 2π & 0 mm/s (B) $2\sqrt{2}\pi$ & 4.44 mm/s
 (C) $2\sqrt{2}\pi$ & 2π mm/s (D) 2π & $2\sqrt{2}\pi$ mm/s

CR0033

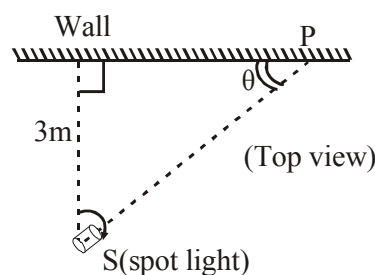
10. A point P moves in counter clockwise direction on a circular path as shown in the figure. The movement of 'P' is such that it sweeps out a length $s = t^2 + 5$, where s is in metres and t is in seconds. The radius of the path is 20 m. The acceleration of 'P' when $t = 5\sqrt{\frac{3}{10}}$ seconds is nearly :



- (A) 2 m/s^2 (B) 1.5 m/s^2 (C) 2.5 m/s^2 (D) 3 m/s^2

CR0034

11. A spot light S rotates in a horizontal plane with a constant angular velocity of 0.1 rad/s . The spot of light P moves along the wall at a distance 3m. What is the velocity of the spot P when $\theta = 45^\circ$?



- (A) 0.6 m/s (B) 0.5 m/s (C) 0.4 m/s (D) 0.3 m/s

CR0035

12. Which of the following statements is false for a particle moving in a circle with a constant angular speed ? [AIEEE - 2004]
- (A) The velocity vector is tangent to the circle
 - (B) The acceleration vector is tangent to the circle
 - (C) the acceleration vector points to the centre of the circle
 - (D) The velocity and acceleration vectors are perpendicular to each other

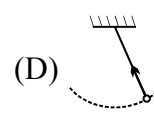
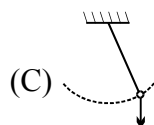
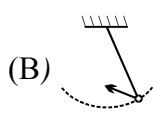
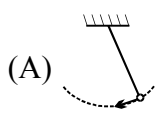
CR0036

Dynamics of circular motion

13. A particle is moving in a circle :
- (A) the resultant force on the particle must be towards the centre
 - (B) the cross product of the tangential acceleration and the angular velocity will be zero
 - (C) the direction of the angular acceleration and the angular velocity must be the same
 - (D) the resultant force may be towards the centre
14. A particle of mass m is tied to a light string and rotated with a speed v along a circular path of radius r . If T = tension in the string and mg = gravitational force on the particle then the actual forces acting on the particle are :
- (A) mg and T only
 - (B) mg , T and an additional force of $\frac{mv^2}{r}$ directed inwards.
 - (C) mg , T and an additional force of $\frac{mv^2}{r}$ directed outwards.
 - (D) only a force $\frac{mv^2}{r}$ directed outwards.

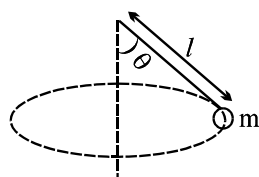
CR0038

15. Which vector in the figures best represents the acceleration of a pendulum mass at the intermediate point in its swing?



CR0039

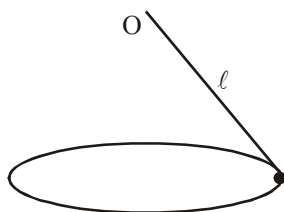
16. A conical pendulum is moving in a circle with angular velocity ω as shown. If tension in the string is T , which of following equations are correct ?



- (A) $T = m\omega^2 l$ (B) $T \sin\theta = m\omega^2 l$ (C) $T = mg \cos\theta$ (D) $T = m\omega^2 l \sin\theta$

CR0040

17. A point mass m is suspended from a light thread of length ℓ , fixed at O , is whirled in a horizontal circle at constant speed as shown. From your point of view, stationary with respect to the mass, the forces on the mass are :



- (A) (B) (C) (D)

CR0041

18. A long horizontal rod has a bead which can slide along its length and is initially placed at a distance L from one end A of the rod. The rod is set in angular motion about A with a constant angular acceleration, α . If the coefficient of friction between the rod and bead is μ , and gravity is neglected, then the time after which the bead starts slipping is :—

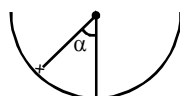
[IIT-JEE 2000]

- (A) $\sqrt{\frac{\mu}{\alpha}}$ (B) $\frac{\mu}{\sqrt{\alpha}}$ (C) $\frac{1}{\sqrt{\mu\alpha}}$ (D) infinitesimal

CR0042

19. A insect crawls up a hemispherical surface very slowly (see the figure). The coefficient of friction between the surface and the insect is $1/3$. If the line joining the centre of the hemispherical surface to the insect makes an angle α with the vertical, the maximum possible value of α is given

[IIT-JEE 2001]



- (A) $\cot \alpha = 3$ (B) $\tan \alpha = 3$ (C) $\sec \alpha = 3$ (D) $\operatorname{cosec} \alpha = 3$

CR0043

20. The maximum velocity (in ms^{-1}) with which a car driver must traverse a flat curve of radius 150 m and coefficient of friction 0.6 to avoid skidding is -

[AIEEE - 2002]

- (A) 60 (B) 30 (C) 15 (D) 25

CR0044

21. A particle is acted upon by a force of constant magnitude which is always perpendicular to the velocity of the particle. The motion of the particle takes place in a plane, it follows that-

[AIEEE - 2004]

- (A) Its velocity is constant (B) Its acceleration is constant
(C) Its kinetic energy is constant (D) It moves in a straight line

CR0045

MULTIPLE CORRECT TYPE QUESTIONS

22. A car runs around a curve of radius 10 m at a constant speed of 10 ms^{-1} . Consider the time interval for which car covers a curve of 120° arc :-

(A) Resultant change in velocity of car is $10\sqrt{3} \text{ ms}^{-1}$

(B) Instantaneous acceleration of car is 10 ms^{-2}

(C) Average acceleration of car is $\frac{5}{24} \text{ ms}^{-2}$

(D) Instantaneous and average acceleration are same for the given period of motion.

CR0046

23. A car is moving with constant speed on a rough banked road.

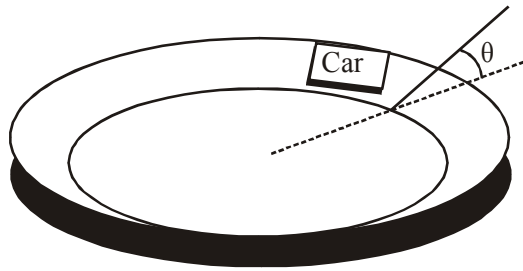
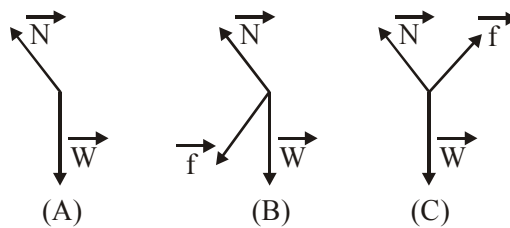


Figure shows the free body diagram of car in three situation A, B & C respectively:-



- (A) Car in A has more speed than car in C
 (B) Car in A has less speed than car in B
 (C) FBD for car in A is not possible
 (D) If $\mu > \tan\theta$ the FBD for car C is not possible

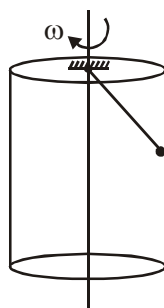
CR0047

24. A heavy particle is tied to the end A of a string of length 1.6 m. Its other end O is fixed. It revolves as a conical pendulum with the string making 60° with the vertical. Then

- (A) its period of revolution is $\frac{4\pi}{7}$ sec.
 (B) the tension in the string is double the weight of the particle
 (C) the velocity of the particle = $2.8\sqrt{3}$ m/s
 (D) the centripetal acceleration of the particle is $9.8\sqrt{3}$ m/s².

CR0048

25. In the shown figure inside a fixed hollow cylinder with vertical axis a pendulum is rotating in a conical path with its axis same as that of the cylinder with uniform angular velocity. Radius of cylinder is 30 cm, length of string is 50 cm and mass of bob is 400 gm. The bob makes contact with the inner frictionless wall of the cylinder while moving :-

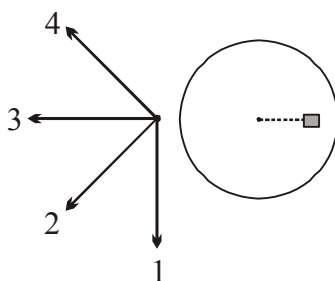


- (A) The minimum value of angular velocity of the bob so that it does not leave contact is 5 rad/s
 (B) Tension in the string is 5N for all values of angular velocity
 (C) For angular velocity of 10 rad/s the bob pushes the cylinder with a force of 9N
 (D) For angular velocity of 10 rad/s, tension in the string is 20N

CR0049

MATRIX MATCH TYPE QUESTION

26. A block is placed on a horizontal table which can rotate about its axis. The block is placed at a certain distance from centre as shown in figure. Table rotates such that particle does not slide. Select possible direction of net acceleration of block at the instant shown in figure.



Column-I

- (A) When rotation is clockwise with constant ω
 (B) When rotation is clockwise with decreasing ω
 (C) When rotation is clockwise with increasing ω
 (D) Just after clockwise rotation begins from rest

Column-II

- (P) 1
 (Q) 2
 (R) 3
 (S) 4

CR0050

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. A ring of radius r and mass per unit length m rotates with an angular velocity ω in free space. The tension in the ring is :

(A) zero (B) $\frac{1}{2}m\omega^2 r^2$ (C) $m\omega^2 r^2$ (D) $mr\omega^2$

CR0051

2. A uniform rod of mass m and length ℓ rotates in a horizontal plane with an angular velocity ω about a vertical axis passing through one end. The tension in the rod at a distance x from the axis is :

(A) $\frac{1}{2}m\omega^2 x$ (B) $\frac{1}{2}m\omega^2 \frac{x^2}{\ell}$ (C) $\frac{1}{2}m\omega^2 \ell \left(1 - \frac{x}{\ell}\right)$ (D) $\frac{1}{2} \frac{m\omega^2}{\ell} [\ell^2 - x^2]$

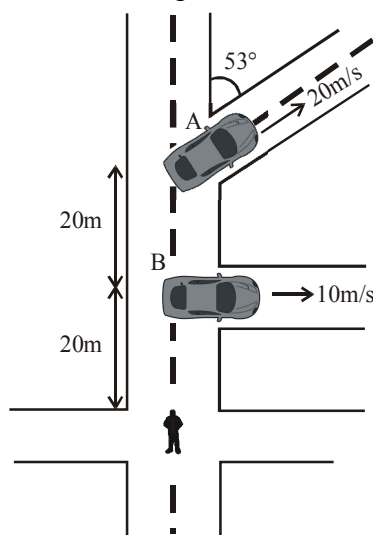
CR0052

3. The magnitude of displacement of a particle moving in a circle of radius a with constant angular speed ω varies with time t as :-

(A) $2a \sin \omega t$ (B) $2a \sin \frac{\omega t}{2}$ (C) $2a \cos \omega t$ (D) $2a \cos \frac{\omega t}{2}$

CR0053

4. A traffic policeman standing at the intersection sees 2 cars A & B turning at angles 53° & 90° respectively (as shown in the figure). Their velocities are $V_A = 20$ m/s, $V_B = 10$ m/s. Then, which car appears to be moving faster to the traffic policeman :-



(A) A (B) B (C) Both equally fast (D) Insufficient info.

CR0054

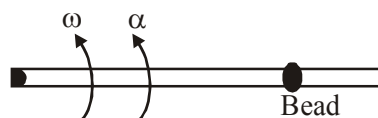
5. A bead is constrained to move on rod in gravity free space as shown in figure. The rod is rotating with angular velocity ω and angular acceleration α about its end. If μ is coefficient of friction. Mark the correct option. (Rod rotates in the plane of paper.)

(A) If $\mu = \frac{\omega^2}{\alpha}$ friction on bead is static in nature

(B) If $\mu > \frac{\omega^2}{\alpha}$ friction on bead is kinetic in nature

(C) If $\mu < \frac{\omega^2}{\alpha}$ friction is static

(D) If bead does not slide relative to rod. Friction will not exist between bead and rod.



CR0055

6. A particle is moving in a circular path. The acceleration and momentum of the particle at a certain moment are $\vec{a} = (4\hat{i} + 3\hat{j}) \text{ m/s}^2$ and $\vec{p} = (8\hat{i} - 6\hat{j}) \text{ kg-m/s}$. The motion of the particle is

(A) uniform circular motion

(B) accelerated circular motion

(C) de-accelerated circular motion

(D) we can not say anything with $\vec{a}$ and $\vec{p}$ only

CR0056

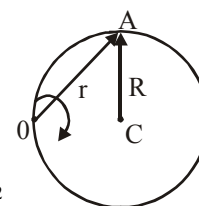
7. A particle A moves along a circle of radius $R=50 \text{ cm}$ so that its radius vector r relative to the point O (figure) rotates with the constant angular velocity $\omega=0.40 \text{ rad/s}$. Then modulus of the velocity of the particle, and the modulus of its total acceleration will be

(A) $v=0.4 \text{ m/s}$, $a=0.4 \text{ m/s}^2$

(B) $v=0.32 \text{ m/s}$, $a=0.32 \text{ m/s}^2$

(C) $v=0.32 \text{ m/s}$, $a=0.4 \text{ m/s}^2$

(D) $v=0.4 \text{ m/s}$, $a=0.32 \text{ m/s}^2$



CR0057

MULTIPLE CORRECT TYPE QUESTIONS

8. For a curved track of radius R , banked at angle θ (Take $v_0 = \sqrt{Rg \tan \theta}$)
- (A) a vehicle moving with a speed v_0 is able to negotiate the curve without calling friction into play at all
- (B) a vehicle moving with any speed $v > v_0$ is always able to negotiate the curve, with friction called into play
- (C) a vehicle moving with any speed $v < v_0$ must have the force of friction into play
- (D) the minimum value of the angle of banking for a vehicle parked on the banked road can stay there without slipping, is given by $\theta = \tan^{-1} \mu_0$ (μ_0 = coefficient of static friction)

CR0058

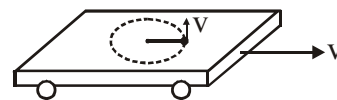
9. On a train moving along east with a constant speed v , a boy revolves a bob with string of length ℓ on smooth surface of a train, with equal constant speed v relative to train. Mark the correct option(s).

(A) Maximum speed of bob is $2v$ in ground frame.

(B) Tension in string connecting bob is $\frac{4mv^2}{\ell}$ at an instant.

(C) Tension in string is $\frac{mv^2}{\ell}$ at all the moments.

(D) Minimum speed of bob is zero in ground frame.



CR0059

10. Let $\vec{v}(t)$ be the velocity of a particle at time t . Then :

(A) $|\frac{d\vec{v}(t)}{dt}|$ and $\frac{d|\vec{v}(t)|}{dt}$ are always equal

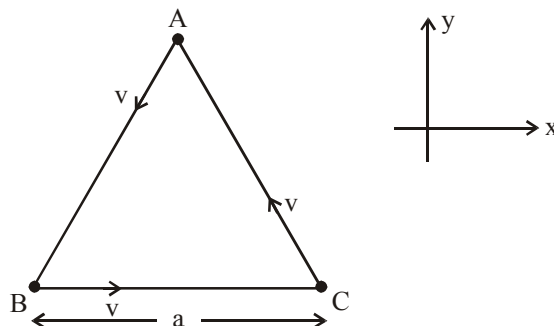
(B) $|\frac{d\vec{v}(t)}{dt}|$ and $\frac{d|\vec{v}(t)|}{dt}$ may be equal

(C) $\frac{d|\vec{v}(t)|}{dt}$ can be zero while $|\frac{d\vec{v}(t)}{dt}|$ is not zero

(D) $\frac{d|\vec{v}(t)|}{dt} \neq 0$ implies $|\frac{d\vec{v}(t)}{dt}| \neq 0$

CR0060

11. Three particles A, B, C are located at the corners of an equilateral triangle as shown in figure. Each of the particle is moving with velocity v . Then at the instant shown, the relative angular velocity of



(A) A wrt B is $\frac{v \cos 30^\circ}{a}$ in z-direction

(B) B wrt C is $\frac{v \cos 30^\circ}{a}$ in z-direction

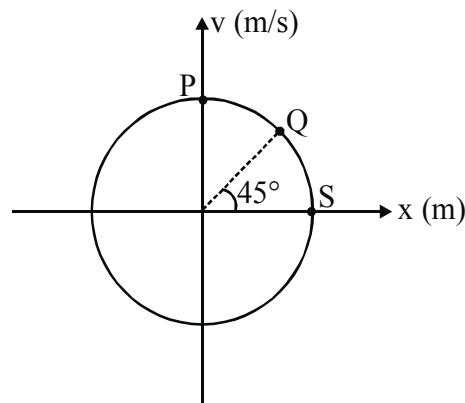
(C) A wrt C is $\frac{v \cos 30^\circ}{a}$ in z-direction

(D) B wrt A is $\frac{v \cos 30^\circ}{a}$ in $-z$ direction

CR0061

12. A particle is in motion on the x-axis. The variation of its velocity with position is as shown. The graph is circle and its equation is $x^2 + v^2 = 1$, where x is in m and v in m/s. The **CORRECT** statement(s) is/are :-

- (A) When x is positive, acceleration is negative.
 (B) When x is negative, acceleration is positive.
 (C) At Q, acceleration has magnitude $\frac{1}{\sqrt{2}} \text{ m/s}^2$
 (D) At S, acceleration is infinite.



CR0062

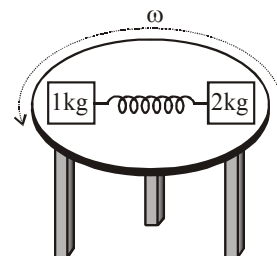
13. An ant travels along a long rod with a constant velocity $\vec{u}$ relative to the rod starting from the origin. The rod is kept initially along the positive x-axis. At $t = 0$, the rod also starts rotating with an angular velocity ω (anticlockwise) in x-y plane about origin. Then

- (A) the position of the ant at any time t is $\vec{r} = ut[\cos \omega t \hat{i} + \sin \omega t \hat{j}]$
 (B) the speed of the ant at any time t is $u\sqrt{1 + \omega^2 t^2}$
 (C) the magnitude of the tangential acceleration of the ant at any time t is $\frac{\omega^2 tu}{\sqrt{1 + \omega^2 t^2}}$
 (D) the speed of the ant at any time t is $\sqrt{1 + 2\omega^2 t^2}u$

CR0063

14. On a circular turn table rotating about its center horizontally with uniform angular velocity ω rad/s placed two blocks of mass 1 kg and 2 kg, on a diameter symmetrically about center. Their separation is 1m and friction is sufficient to avoid slipping. The spring between them as shown is stretched and applied force of 5N. If f_1 and f_2 are values of friction on 1 kg & 2kg block respectively:-

- (A) For $\omega = 2 \text{ rad/s}$, $f_1 = 3\text{N}$ & $f_2 = 1\text{N}$
 (B) For $\omega = 3 \text{ rad/s}$, $f_1 = 0.5 \text{ N}$ & $f_2 = 4\text{N}$
 (C) For $\omega = \sqrt{10} \text{ rad/s}$, $f_1 = 0$ & $f_2 = 5\text{N}$
 (D) For $\omega = \sqrt{10} \text{ rad/s}$, $f_1 = 0$ & $f_2 = 0\text{N}$



CR0064

15. Two particles move on a circular path (one just inside and the other just outside) with angular velocities ω and 5ω starting from the same point. Then

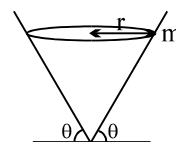
- (A) they cross each other at regular intervals of time $\frac{2\pi}{4\omega}$ when their angular velocities are oppositely directed.
- (B) they cross each other at points on the path subtending an angle of 60° at the centre if their angular velocities are oppositely directed.
- (C) they cross at intervals of time $\frac{\pi}{3\omega}$ if their angular velocities are oppositely directed.
- (D) they cross each other at points on the path subtending 90° at the centre if their angular velocities are in the same sense.

CR0065

16. A ball of mass 'm' is rotating in a circle of radius 'r' with speed v inside a smooth cone as shown in figure. Let N be the normal reaction on the ball by the cone, then choose the correct option.

(A) $N = mg \cos \theta$

(B) $g \sin \theta = \frac{v^2}{r} \cos \theta$



(C) $N \sin \theta - \frac{mv^2}{r} = 0$

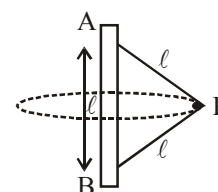
(D) none of these

CR0066

17. A particle P of mass m is attached to a vertical axis by two strings AP and BP of length l each. The separation AB=l. P rotates around the axis with an angular velocity ω . The tensions in the two strings are T_1 and T_2

(A) $T_1 = T_2$

(B) $T_1 + T_2 = m\omega^2 l$



(C) $T_1 - T_2 = 2mg$

(D) BP will remain taut only if $\omega \geq \sqrt{\frac{2g}{l}}$

CR0067

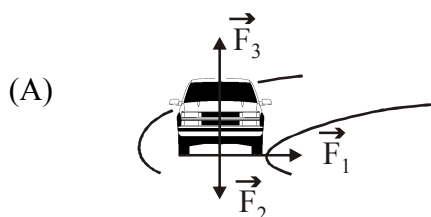
MATRIX MATCH TYPE QUESTION

18. Column-I shows certain situations and column-II shows information about forces.

Column - I

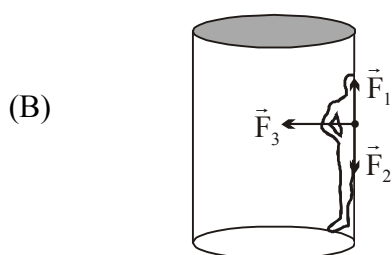
Column - II

Situation



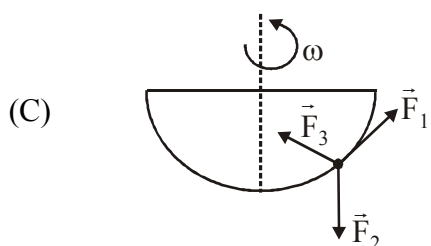
Front view of a car rounding a curve with constant speed.

(P) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3$ is centripetal force.



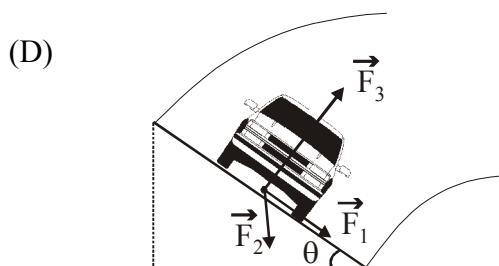
Passengers in a rotor not sliding relative to rotor wall cylindrical rotor is rotating with constant angular velocity about its symmetry axis.

(Q) $\vec{F}_1$ is static friction.



Particle kept on rough surface of a bowl, no relative motion of particle in bowl, bowl has constant angular velocity

(R) $\vec{F}_1$ can be in direction opposite to that shown in figure.



Car moving on a banked road with constant speed, no sideways skidding

(S) $\vec{F}_1 + \vec{F}_2 = \vec{0}$

(T) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$

CR0068

EXERCISE (J-M)

1. For a particle in uniform circular motion, the acceleration $\vec{a}$ at a point $P(R, \theta)$ on the circle of radius R is (Here θ is measured from the x-axis). [AIEEE - 2010]

(1) $\frac{v^2}{R} \hat{i} + \frac{v^2}{R} \hat{j}$

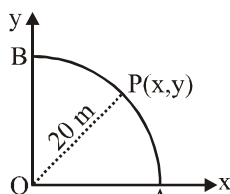
(2) $-\frac{v^2}{R} \cos \theta \hat{i} + \frac{v^2}{R} \sin \theta \hat{j}$

(3) $-\frac{v^2}{R} \sin \theta \hat{i} + \frac{v^2}{R} \cos \theta \hat{j}$

(4) $-\frac{v^2}{R} \cos \theta \hat{i} - \frac{v^2}{R} \sin \theta \hat{j}$

CR0069

2. A point P moves in counter clockwise direction on a circular path as shown in the figure. The movement of 'P' is such that it sweeps out a length $s = t^3 + 5$, where s is in metres and t is in seconds. The radius of the path is 20 m. The acceleration of 'P' when $t = 2$ s is nearly : [AIEEE - 2010]



(1) 14 m/s²

(2) 13 m/s²

(3) 12 m/s²

(4) 7.2 m/s²

CR0070

3. Two cars of masses m_1 and m_2 are moving in circles of radii r_1 and r_2 , respectively. Their speeds are such that they make complete circles in the same time t . The ratio of their centripetal acceleration is :- [AIEEE - 2012]

(1) 1 : 1

(2) $m_1 r_1 : m_2 r_2$

(3) $m_1 : m_2$

(4) $r_1 : r_2$

CR0071

4. Concrete mixture is made by mixing cement, stone and sand in a rotating cylindrical drum. If the drum rotates too fast, the ingredients remain stuck to the wall of the drum and proper mixing of ingredients does not take place. The maximum rotational speed of the drum in revolutions per minute (rpm) to ensure proper mixing is close to : (Take the radius of the drum to be 1.25 m and its axle to be horizontal): [JEE Main (Online) - 2016]

(1) 8.0

(2) 0.4

(3) 1.3

(4) 27.0

CR0072

5. A particle is moving with a uniform speed in a circular orbit of radius R in a central force inversely proportional to the n^{th} power of R . If the period of rotation of the particle is T , then, [JEE Main-2018]

(1) $T \propto R^{\frac{n}{2}+1}$

(2) $T \propto R^{(n+1)/2}$

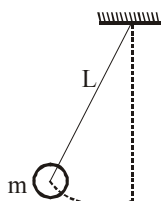
(3) $T \propto R^{n/2}$

(4) $T \propto R^{3/2}$ for any n

CR0073

EXERCISE (J-A)

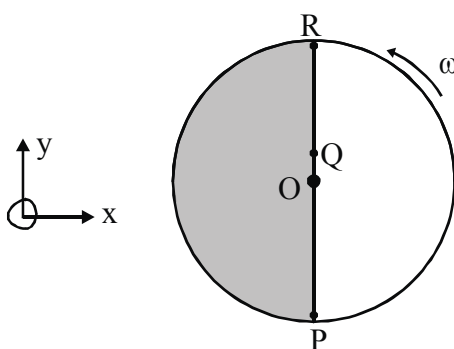
1. A ball of mass (m) 0.5 kg is attached to the end of a string having length (L) 0.5 m. The ball is rotated on a horizontal circular path about vertical axis. The maximum tension that the string can bear is 324 N. The maximum possible value of angular velocity of ball (in radian/s) is [IIT-JEE-2011]



- (A) 9 (B) 18 (C) 27 (D) 36

CR0074

2. Consider a disc rotating in the horizontal plane with a constant angular speed ω about its centre O. The disc has a shaded region on one side of the diameter and an unshaded region on the other side as shown in the figure. When the disc is in the orientation as shown, two pebbles P and Q are simultaneously projected at an angle towards R. The velocity of projection is in the y-z plane and is same for both pebbles with respect to the disc. Assume that (i) they land back on the disc before the disc has completed $\frac{1}{8}$ rotation, (ii) their range is less than half the disc radius, and (iii) ω remains constant throughout. Then [IIT-JEE-2012]

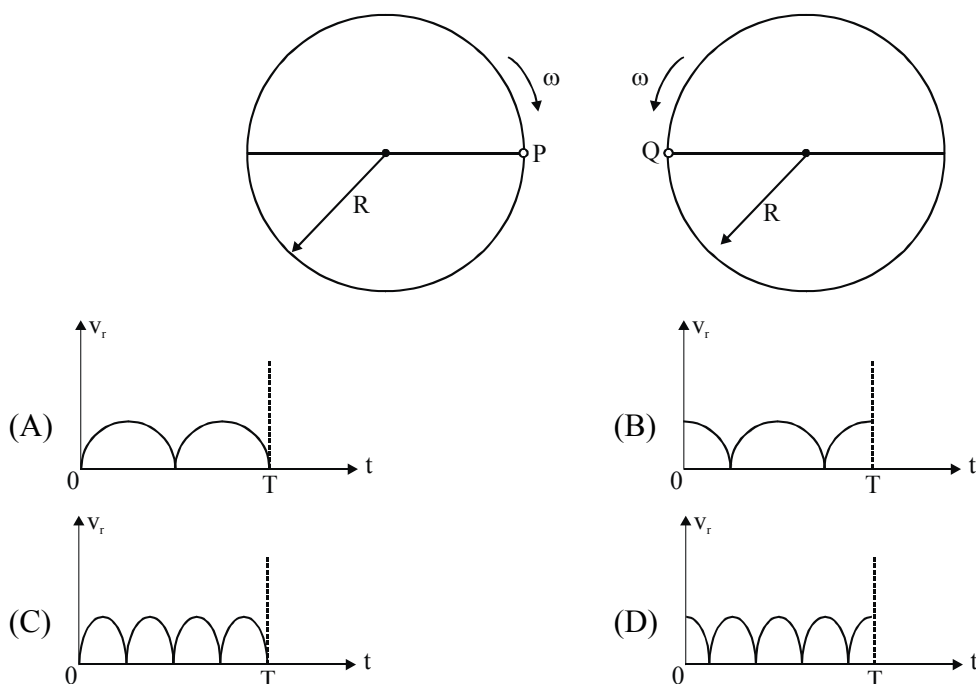


- (A) P lands in the shaded region and Q in the unshaded region.
 (B) P lands in the unshaded region and Q in the shaded region.
 (C) Both P and Q land in the unshaded region.
 (D) Both P and Q land in the shaded region.

CR0075

3. Two identical discs of same radius R are rotating about their axes in opposite directions with the same constant angular speed ω . The discs are in the same horizontal plane. At time $t=0$, the points P and Q are facing each other as shown in the figure. The relative speed between the two points P and Q is v_r . In one time period (T) of rotation of the discs, v_r as a function of time is best represented by

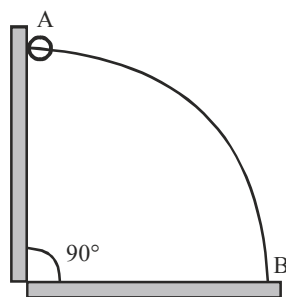
[IIT-JEE 2012]



CR0076

4. A wire, which passes through the hole in a small bead, is bent in the form of quarter of a circle. The wire is fixed vertically on ground as shown in the figure. The bead is released from near the top of the wire and it slides along the wire without friction. As the bead moves from A to B , the force it applies on the wire is :-

[IIT-JEE Advanced 2014]



- (A) Always radially outwards
 (B) Always radially inwards
 (C) Radially outwards initially and radially inwards later.
 (D) Radially inwards initially and radially outwards later.

CR0077

Paragraph for Question No. 5 and 6

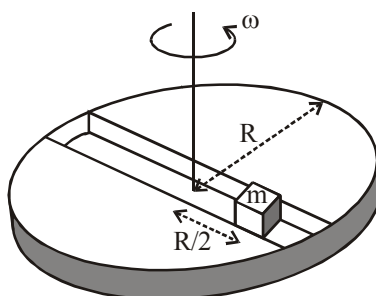
A frame of reference that is accelerated with respect to an inertial frame of reference is called a non-inertial frame of reference. A coordinate system fixed on a circular disc rotating about a fixed axis with a constant angular velocity ω is an example of a non-inertial frame of reference. The relationship between the force $\vec{F}_{\text{rot}}$ experienced by a particle of mass m moving on the rotating disc and the force $\vec{F}_{\text{in}}$ experienced by the particle in an inertial frame of reference is

$$\vec{F}_{\text{rot}} = \vec{F}_{\text{in}} + 2m(\vec{v}_{\text{rot}} \times \vec{\omega}) + m(\vec{\omega} \times \vec{r}) \times \vec{\omega},$$

where $\vec{v}_{\text{rot}}$ is the velocity of the particle in the rotating frame of reference and $\vec{r}$ is the position vector of the particle with respect to the centre of the disc.

Now consider a smooth slot along a diameter of a disc of radius R rotating counter-clockwise with a constant angular speed ω about its vertical axis through its center. We assign a coordinate system with the origin at the centre of the disc, the x -axis along the slot, the y -axis perpendicular to the slot and the z -axis along the rotation axis ($\vec{\omega} = \omega\hat{k}$). A small block of mass m is gently placed in the slot at $\vec{r} = (R/2)\hat{i}$ at $t = 0$ and is constrained to move only along the slot.

[IIT-JEE Advanced-2016]



5. The distance r of the block at time t is :

(A) $\frac{R}{4}(e^{2\omega t} + e^{-2\omega t})$ (B) $\frac{R}{2}\cos 2\omega t$ (C) $\frac{R}{2}\cos \omega t$ (D) $\frac{R}{4}(e^{\omega t} + e^{-\omega t})$

CR0078

6. The net reaction of the disc on the block is :

(A) $-\omega^2 R \cos \omega t \hat{j} - mg\hat{k}$ (B) $\omega^2 R \sin \omega t \hat{j} - mg\hat{k}$
 (C) $\frac{1}{2}\omega^2 R (e^{\omega t} - e^{-\omega t}) \hat{j} + mg\hat{k}$ (D) $\frac{1}{2}\omega^2 R (e^{2\omega t} - e^{-2\omega t}) \hat{j} + mg\hat{k}$

CR0078

ANSWER KEY

EXERCISE (S-1)

1. Ans. 9.9 ms^{-2} , in radial direction towards the centre at all points.
2. Ans. (i) 2 rad/s^2 (ii) $12+2t$ for $t \leq 2\text{s}$, 16 for $t \geq 2\text{s}$ (iii) $a = 169.01 \text{ m/s}^2$ (iv) 44 rad
3. Ans. (a) 4 cm/s^2 (b) 2 cm/s^2 (c) $\sqrt{20} \text{ cm/s}^2$ 4. Ans. (i) 75 m/s^2 , (ii) 125 m/s^2
5. Ans. 4 6. Ans. 024 7. Ans. $\frac{v}{2d}$ 8. Ans. $a_t = \frac{20}{\sqrt{13}}$, $a_n = \frac{30}{\sqrt{13}}$
9. Ans. $\sqrt{3}b/2$, $b/2$, b 10. Ans. 6 11. Ans. (i) mg , (ii) $\frac{mv^2}{r}$
12. Ans. Yes, $a_c = 8$, $\mu g = 1$ 13. Ans. $T = 6.6 \text{ N}$, $v_{\max} = 35 \text{ ms}^{-1}$
14. Ans. $\sqrt{2g} \text{ rad/s}$ 15. Ans. (a) $\sqrt{\frac{g}{\sqrt{L^2 - r^2}}}$ (b) $\frac{mgL}{\sqrt{L^2 - r^2}}$
16. Ans. 5 17. Ans. 15 km 18. Ans. 2

EXERCISE (S-2)

1. Ans. (i) $\frac{V_0^2}{2g}$, (ii) $2g$ 2. Ans. $2V_0$ 3. Ans. $\mu = \frac{3}{16}$
4. Ans. (i) 36 N , (ii) 11.66 rad/sec , (iii) 0.1 m , 0.2 m 5. Ans. $2 \tan \theta$ 6. Ans. $\theta = 60^\circ$

EXERCISE (O-1)

1. Ans. (D) 2. Ans. (B) 3. Ans. (A) 4. Ans. (A) 5. Ans. (A) 6. Ans. (D)
7. Ans. (C) 8. Ans. (A) 9. Ans. (D) 10. Ans. (C) 11. Ans. (A) 12. Ans. (B)
13. Ans. (D) 14. Ans. (A) 15. Ans. (B) 16. Ans. (A) 17. Ans. (C) 18. Ans. (A)
19. Ans. (A) 20. Ans. (B) 21. Ans. (C)
22. Ans. (A, B) 23. Ans. (A, B) 24. Ans. (A, B, C, D) 25. Ans. (A, B, C)
26. Ans. (A)-R; (B)-S; (C)-Q; (D)-P

EXERCISE (O-2)

- | | | | |
|--|--------------------|--------------------|--------------------|
| 1. Ans. (C) | 2. Ans. (D) | 3. Ans. (B) | 4. Ans. (B) |
| 5. Ans. (A) | 6. Ans. (B) | 7. Ans. (D) | |
| 8. Ans. (A, C) | 9. Ans. (A,C,D) | 10. Ans. (B,C,D) | 11. Ans. (A,B,C) |
| 12. Ans. (A, B, C) | 13. Ans. (A, B, C) | 14. Ans. (A, B, C) | 15. Ans. (B, C, D) |
| 16. Ans. (B, C) | 17. Ans. (B, C, D) | | |
| 18. Ans. (A) P,Q (B) P,Q,S (C) P,Q,R (D) P,Q,R | | | |

EXERCISE (J-M)

- | | | | |
|-------------|-------------|-------------|-------------|
| 1. Ans. (4) | 2. Ans. (1) | 3. Ans. (4) | 4. Ans. (4) |
| 5. Ans. (2) | | | |

EXERCISE (J-A)

- | | | | | |
|-------------|------------------|-------------|-------------|-------------|
| 1. Ans. (D) | 2. Ans. (C OR D) | 3. Ans. (A) | 4. Ans. (D) | 5. Ans. (D) |
| 6. Ans. (C) | | | | |

Important Notes

Chapter Contents

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Important Notes

[illegible]

WORK, POWER & ENERGY

KEY CONCEPT

WORK

- Whenever a force acting on a body, displaces it in its direction, work is said to be done by the force.
- Work done by a force is equal to scalar product of force applied and displacement of the point of application, $W = \vec{F} \cdot \vec{d}$
- Work is a scalar quantity.

Work done by a constant force :

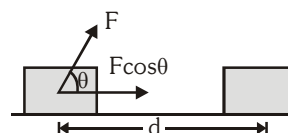
If the direction and magnitude of a force applied on a body is constant, the force is said to be constant. Work done by a constant force, $W = \text{Force} \times \text{component of displacement along force}$

$= \text{displacement} \times \text{component of force along displacement.}$

The work done will, be $W = (F \cos \theta) d$

$$= F (d \cos \theta)$$

In vector form, $W = \vec{F} \cdot \vec{d}$



Note : The force of gravity is the example of constant force, hence work done by it is the example of work done by a constant force.

Work done by a variable force

If the force applying on a body is changing its direction or magnitude or both, the force is said to be variable. Suppose a variable force causes displacement in a body from position P_1 to position P_2 . To calculate the work done by the force the path from P_1 to P_2 can be divided into infinitesimal element, each element is so small that during displacement of body through it, the force is supposed to be constant. If $d\vec{r}$ be small displacement of point of application and $\vec{F}$ be the force applied on the body, the work done by force is $dW = \vec{F} \cdot d\vec{r}$

The total work done in displacing body from P_1 to P_2 is given by $\int dW = \int_{P_1}^{P_2} \vec{F} \cdot d\vec{r} \Rightarrow W = \int_{P_1}^{P_2} \vec{F} \cdot d\vec{r}$

If $\vec{r}_1$ and $\vec{r}_2$ be the position vectors of the points P_1 and P_2 respectively, the total work done $W = \int_{r_1}^{r_2} \vec{F} \cdot d\vec{r}$

Note : When we consider a block attached to a spring, the force on the block is k times the elongation of the spring, where k is spring constant. As the elongation changes with the motion of the block, therefore the force is variable. This is an example of work done by variable force.

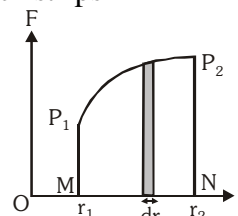
$$\text{Here : } W_s = \int_{x_i}^{x_f} -k x \, dx = -\frac{1}{2} k (x_f^2 - x_i^2)$$

Calculation of work done from force displacement graph :

Suppose a body, whose initial position is r_1 , is acted upon by a variable force $\vec{F}$ and consequently the body acquires its final position r_2 . From position r to $r + dr$ or for small displacement dr , the work done will be $\vec{F} \cdot d\vec{r}$ whose value will be the area of the shaded strip of width dr . The work done on the body in displacing it from position r_1 to r_2 will be equal to the sum of areas of all such strips

$$\text{Thus, total work done, } W = \sum_{r_1}^{r_2} dW = \sum_{r_1}^{r_2} F \cdot dr = \text{Area of } P_1 P_2 NM$$

The area of the graph between curve and displacement axis is equal to the work done.



Note : To calculate the work done by graphical method, for the sake of simplicity, here we have assumed the direction of force and displacement as same, but if they are not in same direction, the graph must be plotted between $F \cos \theta$ and r .

Nature of work done

Although work done is a scalar quantity, yet its value may be positive, negative or even zero

Negative work	Zero work	Positive work
($\theta > 90^\circ$) Work done by friction force ($\theta = 180^\circ$) Work done by gravity ($\theta = 180^\circ$)	($\theta = 90^\circ$) Motion of particle on circular path ($\theta = 90^\circ$) As $f = F$, hence $S = 0$	($\theta < 90^\circ$) Motion under gravity ($\theta = 0^\circ$) Work done by friction force on block A ($\theta = 0^\circ$)

UNITS :

SI Unit : joule (J).

joule : One joule of work is said to be done when a force of one newton displaces a body by one meter in the direction of force.

$$1 \text{ joule} = 1 \text{ newton} \times 1 \text{ meter} = 1 \text{ kgm}^2\text{s}^{-2}$$

erg : One erg of work is said to be done when a force of one dyne displaces a body by one cm. in the direction of force.

$$1 \text{ erg} = 1 \text{ dyne} \times 1 \text{ cm.} = 1 \text{ gm. cm}^2 \text{ s}^{-2}$$

Other Units : (a) 1 joule = 10^7 erg

(b) 1 erg = 10^{-7} joule

(c) 1 eV = 1.6×10^{-19} joule

(d) 1 joule = 6.25×10^{18} eV

(e) 1 MeV = 1.6×10^{-13} J

(f) 1 J = 6.25×10^{12} MeV

(g) 1 kilo watt hour (kWh) = 3.6×10^6 joule

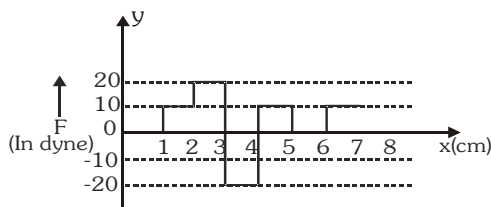
DIMENSIONS :

$$[\text{Work}] = [\text{Force}] [\text{Displacement}] = [\text{MLT}^{-2}][\text{L}] = [\text{ML}^2\text{T}^{-2}]$$

Ex. A position dependent force $F = 7 - 2x + 3x^2$ acts on a small body of mass 2kg and displaces it from $x = 0$ to $x = 5$ m. Calculate the work done in joule.

Sol.
$$W = \int_{x_1}^{x_2} F dx = \int_0^5 (7 - 2x + 3x^2) dx = \left[7x - \frac{2x^2}{2} + \frac{3x^3}{3} \right]_0^5 = 135\text{J}$$

Ex. For the force displacement diagram shown in adjoining diagram. Calculate the work done by the force in displacing the body from $x = 1$ cm to $x = 5$ cm.



Sol. Work = Area under the curve and displacement axis = $10 + 20 - 20 + 10 = 20$ erg

Ex. Calculate work done to move a body of mass 10 kg along a smooth inclined plane ($\theta = 30^\circ$) with constant velocity through a distance of 10 m.

Sol. Here the motion is not accelerated, the resultant force parallel to the plane must be zero. So $F - Mg \sin 30^\circ = 0 \Rightarrow F = Mg \sin 30^\circ$ & $d = 10\text{m}$

$$W = Fd \cos \theta = (Mg \sin 30^\circ) d \cos 0^\circ = 10 \times 10 \times \frac{1}{2} \times 10 \times 1 = 500 \text{ J}$$

Ex. Calculate work done in pulling an object with a constant force F as shown in figure. (given that the ground is rough with coefficient of friction μ)

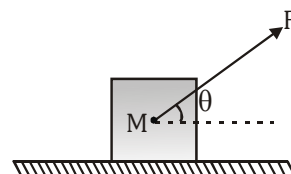
Sol. From the figure $F \sin \theta + N = Mg$

$$\therefore N = Mg - F \sin \theta$$

$$F \cos \theta = f = \mu N = \mu [Mg - F \sin \theta]$$

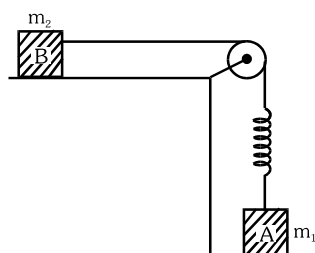
$$F (\cos \theta + \mu \sin \theta) = \mu Mg$$

$$\therefore F = \frac{\mu Mg d}{\cos \theta + \mu \sin \theta} = \text{force required to pull an object}$$



Work done in pulling an object $W = Fd = \frac{\mu Mg d}{\cos \theta + \mu \sin \theta}$

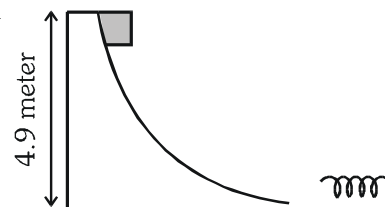
Ex. If the two blocks moves with a constant uniform speed then find coefficient of friction between the surface of the block B and the table. The spring is massless and the pulley is frictionless.



Sol. For block B : $T = f = \mu m_2 g$ and For block A : $T = m_1 g$

By solving above equations $\mu = \frac{m_1}{m_2}$

Ex. Figure shows a smooth curved track terminating in a smooth horizontal part. A spring of force constant 400 N/m is attached at one end to a wedge fixed rigidly with a horizontal part. A 40 g mass is released from rest at a height of 4.9 m on the curved track. Find the maximum compression of the spring.



Sol. From the law of conservation of mechanical energy $\Rightarrow mgh = \frac{1}{2} kx^2$

$$\Rightarrow x = \sqrt{\frac{2mgh}{k}} = \sqrt{\frac{2 \times (0.04) \times 9.8 \times 4.9}{400}} = 9.8 \text{ cm.}$$

ENERGY

- The energy of a body is defined as the capacity of doing work.
- There are various form of energy
 - (i) mechanical energy (ii) chemical energy (iii) electrical energy (iv) sound energy
 - (v) light energy etc (vi) magnetic energy (vii) nuclear energy
- Energy of an isolated system always remain constant it can neither be created nor it can be destroyed however it may be converted from one form to another

Examples

Electrical energy	Motor	Mechanical energy
Mechanical energy	Generator	Electrical energy
Light energy	Photocell	Electrical energy
Electrical energy	Heater	Heat energy
Electrical energy	Radio / Speaker	Sound energy
Nuclear energy	Nuclear Reactor	Electrical energy
Chemical energy	Cell	Electrical energy
Electrical energy	Secondary Cell Chargeable	Chemical energy

- Energy is a scalar quantity
- Unit : Its unit is same as that of work or torque. In MKS : joule, watt second ; In CGS : erg
Note : 1 eV = 1.6×10^{-19} joule; 1 kWh = 3.6×10^6 joule; 10^7 erg = 1 joule.
- Dimension $[M^1L^2T^{-2}]$
- According to Einstein's mass energy equivalence principle mass and energy are inter convertible i.e. they can be changed into each other. Energy equivalent of mass m is, $E = mc^2$ where, m : mass of the particle, c : velocity of light, E : equivalent energy corresponding to mass m .
- In mechanics we are concerned with mechanical energy only which is of two type :
 - (i) kinetic energy (ii) potential energy

Kinetic energy

- The energy possessed by a body by virtue of its motion is called kinetic energy.
- If a body of mass m is moving with velocity v , its kinetic energy $KE = \frac{1}{2}mv^2$.
- Kinetic energy is always positive.
- If linear momentum of body is p , the kinetic energy for translatory motion is $KE = \frac{p^2}{2m} = \frac{1}{2}mv^2$.

Ex. In a ballistics demonstration, a police officer fires a bullet of mass 50.0 g with speed 200 ms^{-1} on soft plywood of thickness 2.00 cm. The bullet emerges with only 10% of its initial kinetic energy. What is the emergent speed of the bullet?

Sol. Initial kinetic energy, $K_i = \frac{1}{2} \times \frac{50}{1000} \times 200 \times 200 \text{ J} = 1000 \text{ J}$

Final kinetic energy, $K_f = \frac{10}{100} \times 1000 \text{ J} = 100 \text{ J}$

If v_f is emergent speed of the bullet, then $\frac{1}{2} \times \frac{50}{1000} \times v_f^2 = 100$

$\Rightarrow v_f^2 = 4000 \Rightarrow v_f = 63.2 \text{ ms}^{-1}$.

Note that the speed is reduced by approximately 68% and not 90%.

Work Energy Theorem

Work done by all the forces (conservative or non conservative, external or internal) acting on a particle or an object is equal to the change in its kinetic energy. So work done by all the forces = change in kinetic energy

$$W = \Delta KE = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2$$

Ex. A particle of mass m moves with velocity $v = a\sqrt{x}$ where a is a constant. Find the total work done by all the forces during a displacement from $x = 0$ to $x = d$.

Sol. Work done by all forces = $W = \Delta KE = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_1^2$

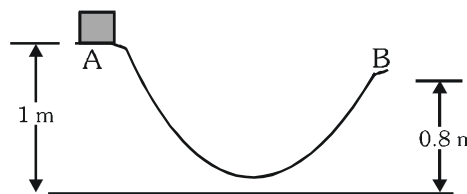
Here $v_1 = a\sqrt{0} = 0$, $v_2 = a\sqrt{d}$, So $W = \frac{1}{2}ma^2d - 0 = \frac{1}{2}ma^2d$

Ex. The displacement x of a body of mass 1 kg on horizontal smooth surface as a function of time t is given by $x = \frac{t^3}{3}$. Find the work done by the external agent for the first one second.

Sol. $\therefore x = \frac{t^3}{3} \therefore v = \frac{dx}{dt} = t^2$, Velocity at $t = 0$, $u = 0$ and at $t = 1 \text{ s}$ $v = 1 \text{ m/s}$

Using work energy theorem : $W = \frac{1}{2}mv^2 - \frac{1}{2}mu^2 = \frac{1}{2}1(1)^2 = 0.5 \text{ J}$

Ex. A block of mass 1 kg is placed at the point A of a rough track shown in figure. If it slightly pushed towards right it stops at the point B of the track. Calculate the workdone by the frictional force on the block during its transit from A to B.



Sol. $W_c + W_{nc} + W_{ext} = \Delta K$

$$mg(1 - 0.8) + W_{nc} + 0 = 0 \Rightarrow W_{nc} = -1 \times 9.8 \times 0.2 = -1.96 \text{ J}$$

POWER

When we purchase a car or jeep we are interested in the horsepower of its engine. We know that usually an engine with large horsepower is most effective in accelerating the automobile.

In many cases it is useful to know not just the total amount of work being done, but how fast the work is done. We define power as the rate at which work is being done.

$$\text{Average Power} = \frac{\text{Work done}}{\text{Time taken to do work}} = \frac{\text{Total change in kinetic energy}}{\text{Total change in time}}$$

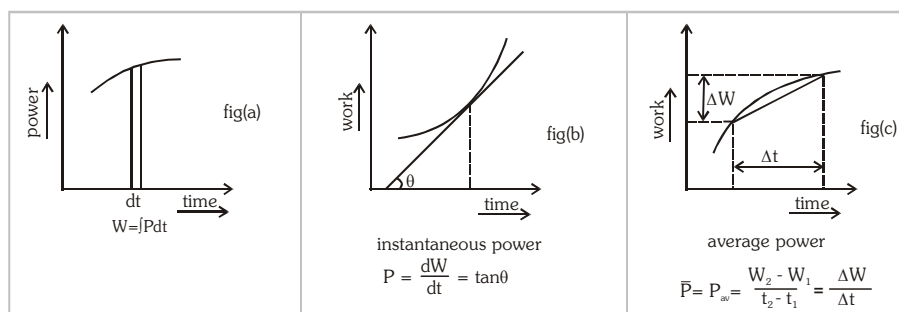
If ΔW is the amount of work done in the time interval Δt . Then $P = \frac{\Delta W}{\Delta t} = \frac{W_2 - W_1}{t_2 - t_1}$

When work is measured in joules and t is in seconds, the unit for power is the joule per second, which is called watt. For motors and engines, power is usually measured in horsepower, where horsepower is $1 \text{ hp} = 746 \text{ W}$. The definition of power is applicable to all types of work like mechanical, electrical,

thermal. Instantaneous power $P = \frac{dW}{dt} = \frac{\vec{F} \cdot d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$

Where v is the instantaneous velocity of the particle and dot product is used as only that component of force will contribute to power which is acting in the direction of instantaneous velocity.

- Power is a scalar quantity with dimension $M^1L^2T^{-3}$
- SI unit of power is J/s or watt
- 1 horsepower = 746 watt



- Area under power–time graph gives the work done. $W = \int P dt$ (See Fig. a)
- The slope of tangent at a point on work time graph gives instantaneous power (See Fig. b)
- The slope of a straight line joining two points on work time graph gives average power between two points (See Fig. c)

- For a system of varying mass $F = \frac{d}{dt}(mv) = m \frac{dv}{dt} + v \frac{dm}{dt}$
- If $v = \text{constant}$ then $F = v \frac{dm}{dt}$ then $P = \vec{F} \cdot \vec{v} = v^2 \frac{dm}{dt}$

Ex. A truck pulls a mass of 1200 kg at constant speed of 10m/s on a level road. The tension in coupling is 1000 N. What is the power spent on the mass. Find tension when truck moves up a road with inclination 1 in 6.

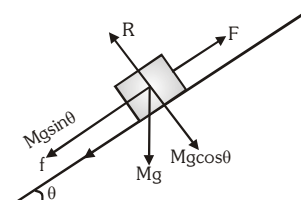
Sol. Force applied by truck $f = 1000$ N

Power spent in pulling the mass $P = fv = 1000 \times 10 = 10^4$ W

Here $\sin\theta = 1/6$, the required force for truck to move up is

$F = f + Mg \sin\theta$

$$F = 1000 \text{ N} + 1200 \times 9.8 \times \frac{1}{6} = 2960 \text{ N}$$



CONSERVATIVE FORCE

- A force is said to be conservative if the work done by or against the force is independent of path and depends only on initial and final positions
- It does not depend on the nature of path followed between the initial and final positions.

Examples of Conservative Force

All central forces are conservative like gravitational, electrostatic, elastic force, restoring force due to spring etc.

- In presence of conservative forces mechanical energy remains constant.
- Work done along a closed path or in a cyclic process is zero. i.e. $\oint \vec{F} \cdot d\vec{r} = 0$

CENTRAL FORCE

The force whose line of action always passes through a fixed point (which is known as centre of force) and magnitude of force depends only on the distance from this point is known as **central force**.

$$\vec{F} = F(r) \hat{r}$$

All forces following inverse square law are called central forces.

$$\vec{F} = \frac{k}{r^2} \hat{r} \text{ is central force like Gravitational force and Coulomb force.}$$

- All central forces are conservative forces
- Central forces are function of position

NON CONSERVATIVE FORCE

A force is said to be non-conservative if work done by or against the force in moving a body depends upon the path between the initial and final positions.

Work done in a closed path is not zero in a non-conservative force field.

The frictional forces are non-conservative forces. This is because the work done against friction depends on the length of the path along which a body is moved. It does not depend on the initial and final positions. The work done by frictional force in a round trip is not zero.

Examples of non-conservative force

The velocity-dependent forces such as air resistance, viscous force etc. are non-conservative forces.

Conservative Forces	Non-conservative Forces
• Work done does not depend upon path. • Work done in a round trip is zero. • Central forces, spring forces etc. are conservative forces • When only a conservative force acts within a system, the kinetic energy and potential energy can change. However, their sum, the mechanical energy of the system, does not change. • Work done is completely recoverable.	• Work done depends upon path. • Work done in a round trip is not zero. • Force are velocity-dependent & retarding in nature e.g. friction, viscous force etc. • Work done against a non-conservative force may be dissipated as heat energy. • Work done is not completely recoverable.

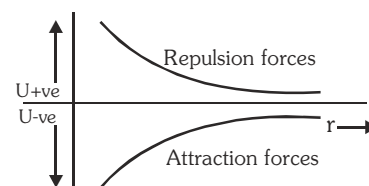
Potential energy

- The energy which a body has by virtue of its position or configuration in a conservative force field.
- Potential energy is a relative quantity.
- Potential energy is defined only for conservative force field.
- Potential energy of a body at any position in a conservative force field is defined as the work done by an external agent against the action of conservative force in order to shift it from reference point. (PE = 0) to the present position.
- Potential energy of a body in a conservative force field is equal to the work done by the body in moving from its present position to reference position.
- At reference position, the potential energy of the body is zero or the body has lost the capacity of doing work.
- Relationship between conservative force field and potential energy :

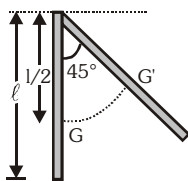
$$\vec{F} = -\nabla U = -\text{grad}(U) = -\frac{\partial U}{\partial x}\hat{i} - \frac{\partial U}{\partial y}\hat{j} - \frac{\partial U}{\partial z}\hat{k}$$

- If force varies only with one dimension (along x-axis) then $F = -\frac{dU}{dx} \Rightarrow U = -\int_{x_1}^{x_2} F dx$

- Potential energy may be positive or negative
 - i) Potential energy is positive, if force field is repulsive in nature
 - ii) Potential energy is negative, if force field is attractive in nature
- If $r \uparrow$ (separation between body and force centre), $U \uparrow$, force field is attractive or vice-versa.
- If $r \uparrow$, $U \downarrow$, force field is repulsive in nature.



Ex. A meter scale of mass m initially vertical is displaced at 45° keeping the upper end fixed. Find out the change in potential energy.



Sol. $\Delta U = mg \Delta h_{cm} = mg \frac{l}{2} (1 - \cos \theta) = mg \times \frac{1}{2} (1 - \cos 45^\circ) = \frac{mg}{2} \left(1 - \frac{1}{\sqrt{2}}\right)$

Ex. A uniform rod of length 4m and mass 20kg is lying horizontal on the ground. Calculate the work done in keeping it vertical with one of its ends touching the ground.

Sol. As the rod is kept in vertical position the shift in the centre of gravity is equal to the half the length

$$= l/2 \text{ Work done } W = mgh = mg \frac{l}{2} = (20)(9.8) \left(\frac{4}{2}\right) = 392 \text{ J}$$

POTENTIAL ENERGY CURVE AND EQUILIBRIUM

It is a curve which shows change in potential energy

with position of a particle.

Stable Equilibrium :

When a particle is slightly displaced from equilibrium position and it tends to come back towards equilibrium

then it is said to be in stable equilibrium

At point **C** : slope $\frac{dU}{dx}$ is negative so F is positive

At point **D** : slope $\frac{dU}{dx}$ is positive so F is negative

At point **A** : it is the point of stable equilibrium.

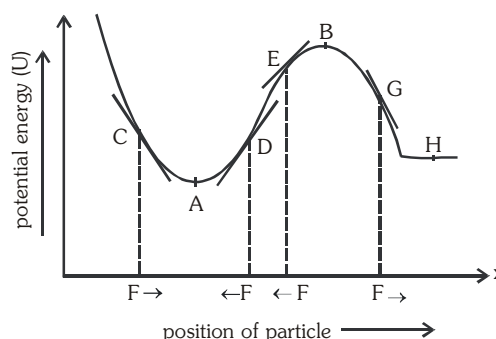
At **A** $U = U_{min}$, $\frac{dU}{dx} = 0$ and $\frac{d^2U}{dx^2} = \text{positive}$

Unstable equilibrium :

When a particle is slightly displaced from equilibrium and it tends to move away from equilibrium position then it is said to be in unstable equilibrium

At point **E** : slope $\frac{dU}{dx}$ is positive so F is negative

At point **G** : slope $\frac{dU}{dx}$ is negative so F is positive



At point **B** : it is the point of unstable equilibrium.

At **B** $U = U_{\max}$, $\frac{dU}{dx} = 0$ and $\frac{d^2U}{dx^2} = \text{negative}$

Neutral equilibrium :

When a particle is slightly displaced from equilibrium position and no force acts on it then equilibrium is said to be neutral equilibrium

Point **H** is at neutral equilibrium $\Rightarrow U = \text{constant}$; $\frac{dU}{dx} = 0$, $\frac{d^2U}{dx^2} = 0$

Ex. The potential energy for a conservative force system is given by $U = ax^2 - bx$. Where a and b are constants find out

- The expression of force
- Equilibrium position
- Potential energy at equilibrium.

Sol. (a) For conservative force $F = -\frac{dU}{dx} = -(2ax - b) = -2ax + b$

(b) At equilibrium $F = 0 \Rightarrow -2ax + b = 0 \Rightarrow x = \frac{b}{2a}$ (c) $U = a\left(\frac{b}{2a}\right)^2 - b\left(\frac{b}{2a}\right) = \frac{b^2}{4a} - \frac{b^2}{2a} = -\frac{b^2}{4a}$

Law of conservation of Mechanical energy

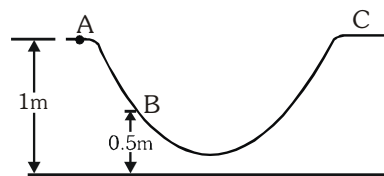
Total mechanical (kinetic + potential) energy of a system remains constant if only conservative forces are acting on the system of particles and the work done by all other forces is zero.

From work energy theorem $W = \Delta KE$

For internal conservative forces $W_{\text{int}} = -\Delta U$

So $W = W_{\text{ext}} + W_{\text{int}} = 0 + W_{\text{int}} = -\Delta U \Rightarrow -\Delta U = \Delta KE \Rightarrow (KE + U) = 0 \Rightarrow KE + U = (\text{constant})$

Ex. A particle is placed at the point A of a frictionless track ABC. It is pushed slightly towards right. Find its speed when it reaches the point B. [Take $g = 10 \text{ m/s}^2$]



Sol. $mg(1 - 0.5) = \frac{1}{2} \times m \times v^2 \Rightarrow v^2 = (2)(10)(0.5) \Rightarrow v = \sqrt{10} \text{ m/s}$

Ex. Determine the average force necessary to stop a bullet of mass 20 g and speed 250 ms^{-1} as it penetrates wood to a distance of 12 cm.

Sol. If F newton be the retarding force, then the work done by force $W = F \times s = F \times 0.12 \text{ joule}$

Loss of kinetic energy $= \frac{1}{2} \times \frac{20}{1000} \times (250)^2 = 625 \text{ joule}$

(This kinetic energy is consumed in stopping the bullet and is converted into heat energy)

Applying work-energy theorem, $F \times 0.12 = 625 \Rightarrow F = \frac{625}{0.12} \text{ N} = 5.2 \times 10^3 \text{ N}$

It is interesting to note that the retarding force is nearly 30,000 times the weight of the bullet.

Ex. A body of mass 8 kg moves under the influence of a force. The position of the body and time are related as $x = \frac{1}{2} t^2$ where x is in meter and t in sec. Find the work done by the force in first two seconds.

Sol. Work done = change in kinetic energy $= \frac{1}{2} mv^2 = \frac{1}{2} m \left(\frac{dx}{dt} \right)^2 = \frac{1}{2} m \left(\frac{2t}{2} \right)^2 = \frac{1}{2} \times 8 \times \left[\frac{2 \times 2}{2} \right]^2 = 16 \text{ J}$

Ex. A body falls on the surface of the earth from a height of 20 cm. If after colliding with the earth, its mechanical energy is lost by 75%, then determine height upto the body would reach.

Sol. $\frac{1}{4} mgh = mgh' \therefore h' = \frac{h}{4} = \frac{1}{4} \times 20 = 5 \text{ cm}$

Ex. Calculate the stopping distance for a vehicle of mass m moving with speed v along level road. (μ is the coefficient of friction between tyres and the road)

Sol. When the vehicle of mass m is moving with velocity v , the kinetic energy of the vehicle $K = \frac{1}{2} mv^2$

and if S is the stopping distance, work done by the friction

$$W = FS \cos \theta = \mu mgS \cos 180^\circ = -\mu mgS$$

$$\text{So by Work-Energy theorem, } W = \Delta K = K_f - K_i \Rightarrow -\mu mgS = 0 - \frac{1}{2} mv^2 \Rightarrow S = \frac{v^2}{2\mu g}$$

Ex. A particle of mass m is moving in a horizontal circle of radius r , under a centripetal force equal to $(-k/r^2)$, where k is constant. Calculate the total energy of the particle.

Sol. As the particle is moving in a circle, so $\frac{mv^2}{r} = \frac{k}{r^2}$. Now $K.E = \frac{1}{2} mv^2 = \frac{k}{2r}$

$$\text{As } U = -\int_{\infty}^r F \cdot dr = \int_{\infty}^r \left(\frac{k}{r^2} \right) dr = -\frac{k}{r}. \text{ So total energy} = U + K.E. = -\frac{k}{r} + \frac{k}{2r} = -\frac{k}{2r}$$

Negative energy means that particle is in bound state.

Ex. A man throws the bricks to the height of 12 m where they reach with a speed of 12 m/sec. If he throws the bricks such that they just reach this height, what percentage of energy will he save?

Sol. In first case, $W_1 = \frac{1}{2} m(v_1)^2 + mgh = \frac{1}{2} m(12)^2 + m \times 10 \times 12 = 72 \text{ m} + 120 \text{ m} = 192 \text{ m}$

and in second case, $W_2 = mgh = 120 \text{ m}$

$$\text{The percentage of energy saved} = \frac{192 \text{ m} - 120 \text{ m}}{192 \text{ m}} \times 100 = 38\%$$

CIRCULAR MOTION IN VERTICAL PLANE

Suppose a particle of mass m is attached to an inextensible light string of length R . The particle is moving in a vertical circle of radius R about a fixed point O . It is imparted a velocity u in horizontal direction at lowest point A . Let v be its velocity at point B of the circle as shown in figure. Here, $h = R(1 - \cos\theta)$... (i)

From conservation of mechanical energy

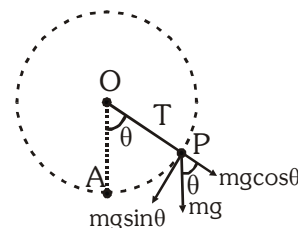
$$\frac{1}{2} m(u^2 - v^2) = mgh \Rightarrow v^2 = u^2 - 2gh \quad \dots (ii)$$

The necessary centripetal force is provided by the resultant of tension

$$T - mg \cos\theta = \frac{mv^2}{R} \quad \dots (iii)$$

Since speed of the particle decreases with height, hence tension is maximum at the bottom, where $\cos\theta = 1$ (as $\theta = 0^\circ$)

$$\Rightarrow T_{\max} = \frac{mv^2}{R} + mg; T_{\min} = \frac{mv^2}{R} - mg \text{ at the top. Here, } v' = \text{speed of the particle at the top.}$$



Condition of Looping the Loop ($u \geq \sqrt{5gR}$)

The particle will complete the circle if the string does not slack even at the highest point ($\theta = \pi$).

Thus, tension in the string should be greater than or equal to zero ($T \geq 0$) at $\theta = \pi$. In critical case

$$\text{substituting } T = 0 \text{ and } \theta = \pi \text{ in Eq. (iii), we get } mg = \frac{mv_{\min}^2}{R} \Rightarrow v_{\min} = \sqrt{gR} \text{ (at highest point)}$$

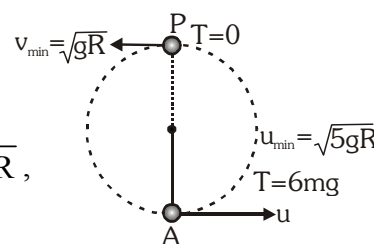
Substituting $\theta = \pi$ in Eq. (i), Therefore, from Eq. (ii)

$$u_{\min}^2 = v_{\min}^2 + 2gR = gR + 2g(2R) = 5gR \Rightarrow u_{\min} = \sqrt{5gR}$$

Thus, if $u \geq \sqrt{5gR}$, the particle will complete the circle. At $u = \sqrt{5gR}$,

velocity at highest point is $v = \sqrt{gR}$ and tension in the string is zero.

Substituting $\theta = 0^\circ$ and $v = \sqrt{5gR}$ in Eq. (iii), we get $T = 6mg$ or in the critical condition tension in the string at lowest position is $6mg$. This is shown in figure. If $u < \sqrt{5gR}$, following two cases are possible.



Condition of Leaving the Circle ($\sqrt{2gR} < u < \sqrt{5gR}$)

If $u < \sqrt{5gR}$, the tension in the string will become zero before reaching the highest point. From Eq.

$$(iii), \text{ tension in the string becomes zero } (T = 0) \text{ where, } \cos\theta = \frac{-v^2}{Rg} \Rightarrow \cos\theta = \frac{2gh - u^2}{Rg}$$

$$\text{Substituting, this value of } \cos\theta \text{ in Eq. (i), we get } \frac{2gh - u^2}{Rg} = 1 - \frac{h}{R} \Rightarrow h = \frac{u^2 + Rg}{3g} = h_1 \text{ (say)}$$

or we can say that at height h_1 tension in the string becomes zero. Further, if $u < \sqrt{5gR}$, velocity of

the particle becomes zero when $0 = u^2 - 2gh \Rightarrow h = \frac{u^2}{2g} = h_2$ (say)...(v) i.e., at height h_2 velocity of particle becomes zero.

Now, the particle will leave the circle if tension in the string becomes zero but velocity is not zero. or $T = 0$ but $v \neq 0$. This is possible only when $h_1 < h_2$

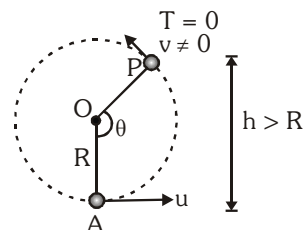
$$\Rightarrow \frac{u^2 + Rg}{3g} < \frac{u^2}{2g} \Rightarrow 2u^2 + 2Rg < 3u^2 \Rightarrow u^2 > 2Rg \Rightarrow u > \sqrt{2Rg}$$

Therefore, if $\sqrt{2gR} < u < \sqrt{5gR}$, the particle leaves the circle.

From Eq. (iv), we can see that $h > R$ if $u^2 > 2gR$. Thus, the particle, will leave the circle when $h > R$ or $90^\circ < \theta < 180^\circ$. This situation is shown in the figure

$$\sqrt{2gR} < u < \sqrt{5gR} \text{ or } 90^\circ < \theta < 180^\circ$$

Note : That after leaving the circle, the particle will follow a parabolic path.

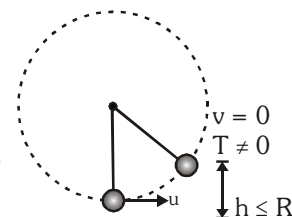


Condition of Oscillation ($0 < u \leq \sqrt{2gR}$)

The particle will oscillate if velocity of the particle becomes zero but tension in the string is not zero or $v = 0$, but $T \neq 0$. This is possible when $h_2 < h_1$

$$\Rightarrow \frac{u^2}{2g} < \frac{u^2 + Rg}{3g} \Rightarrow 3u^2 < 2u^2 + 2Rg \Rightarrow u^2 < 2Rg \Rightarrow u < \sqrt{2Rg}$$

Moreover, if $h_1 = h_2$, $u = \sqrt{2Rg}$ and tension and velocity both becomes zero simultaneously. Further, from Eq. (iv), we can see that $h \leq R$ if $u \leq \sqrt{2Rg}$.



Thus, for $0 < u \leq \sqrt{2gR}$, particle oscillates in lower half of the circle ($0^\circ < \theta \leq 90^\circ$)

This situation is shown in the figure. $0 < u \leq \sqrt{2gR}$ or $0^\circ < \theta \leq 90^\circ$

Ex. Calculate following for shown situation :-

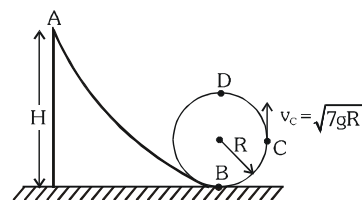
(a) Speed at D (b) Normal reaction at D (c) Height H

Sol. (a) $v_D^2 = v_C^2 - 2gR = 5gR \Rightarrow v_D = \sqrt{5gR}$

$$(b) mg + N_D = \frac{mv_D^2}{R} \Rightarrow N_D = \frac{m(5gR)}{R} - mg = 4mg$$

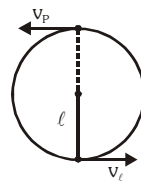
(c) by energy conservation between point A & C

$$mgH = \frac{1}{2}mv_C^2 + mgR = \frac{1}{2}m(5gR) + mg2R = \frac{9}{2}mgR \Rightarrow H = \frac{9}{2}R$$



Ex. A stone of mass 1 kg tied to a light string of length $\ell = \frac{10}{3}$ m is whirling in a circular path in vertical plane. If the ratio of the maximum to minimum tension in the string is 4, find the speed of the stone at the lowest and highest points

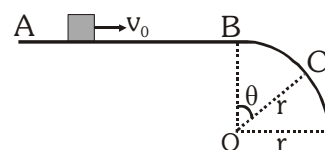
Sol. $\therefore \frac{T_{\max}}{T_{\min}} = 4 \therefore \frac{\frac{mv_{\ell}^2}{\ell} + mg}{\frac{mv_p^2}{\ell} - mg} = 4 \Rightarrow \frac{v_{\ell}^2 + g\ell}{v_p^2 - g\ell} = 4$



We know $v_{\ell}^2 = v_p^2 + 4g\ell \Rightarrow \frac{v_p^2 + 5g\ell}{v_p^2 - g\ell} = 4 \Rightarrow 3v_p^2 = 9g\ell$

$\Rightarrow v_p^2 = \sqrt{3g\ell} = \sqrt{3 \times 10 \times \frac{10}{3}} = 10 \text{ ms}^{-1} \Rightarrow v_{\ell} = \sqrt{7g\ell} = \sqrt{7 \times 10 \times \frac{10}{3}} = 15.2 \text{ ms}^{-1}$

Ex. A small block slides with velocity $0.5\sqrt{gr}$ on the horizontal frictionless surface as shown in the figure. The block leaves the surface at point C. Calculate angle θ in the figure.



Sol. As block leaves the surface at C so at C, normal reaction = 0 $\Rightarrow mg \cos \theta = \frac{mv_C^2}{r}$

By energy conservation at point B & C $\frac{1}{2} mv_C^2 - \frac{1}{2} mv_0^2 = mgr (1 - \cos \theta)$

$\Rightarrow \frac{1}{2} m (rg \cos \theta) - \frac{1}{2} m (0.5\sqrt{gr})^2 = mgr (1 - \cos \theta) \Rightarrow \cos \theta = \frac{3}{4} \Rightarrow \theta = \cos^{-1} \left(\frac{3}{4} \right)$

Ex. A particle of mass m is attached to the ceiling of a cabin with an inextensible light string of length ℓ . The cabin is moving upward with an acceleration 'a'. The particle is taken to a position such that the string makes an angle θ with vertical. When string becomes vertical, find the tension in the string.

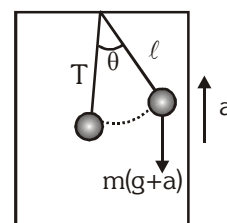
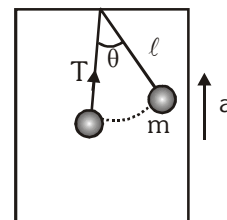
Sol. In a frame associated with cabin work done on the particle when it comes in the vertical position = $mg\ell(1 - \cos \theta) + ma(1 - \cos \theta)$

By work energy theorem,

$\frac{mv^2}{2} = (mg\ell + ma\ell)(1 - \cos \theta) \Rightarrow \frac{v^2}{2} = (g+a)\ell(1 - \cos \theta)$

At vertical position, $T = (mg + ma) = \frac{mv^2}{\ell}$

$\Rightarrow T = (mg + ma) + 2m(g+a)(1 - \cos \theta) = mg(g+a)(3 - 2 \cos \theta)$



Ex. A heavy particle hanging from a fixed point by a light inextensible string of length ℓ , is projected horizontally with speed $\sqrt{g\ell}$. Find the speed of the particle and the inclination of the string to the vertical at the instant of the motion when the tension in the string is equal to the weight of the particle.

Sol. Let tension in the string becomes equal to the weight of the particle when particle reaches the point B and deflection of the string from vertical is θ . Resolving mg along the string and perpendicular to the string, we get net radial force on the particle at B i.e.

$$F_R = T - mg \cos \theta \quad \dots(i)$$

If v_B be the speed of the particle at B, then

$$F_R = \frac{mv_B^2}{\ell} \quad \dots(ii)$$

From (i) and (ii), we get, $T - mg \cos \theta = \frac{mv_B^2}{\ell} \quad \dots(iii)$

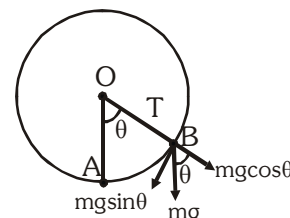
Since at B, $T = mg \Rightarrow mg(1 - \cos \theta) = \frac{mv_B^2}{\ell} \Rightarrow v_B^2 = g\ell(1 - \cos \theta) \quad \dots(iv)$

Applying conservation of mechanical energy of the particle at point A and B, we have

$$\frac{1}{2} mv_A^2 = mg\ell(1 - \cos \theta) + \frac{1}{2} mv_B^2; \text{ where } v_A = \sqrt{g\ell} \text{ and } v_B = \sqrt{g\ell(1 - \cos \theta)}$$

$$\Rightarrow g\ell = 2g\ell(1 - \cos \theta) + g\ell(1 - \cos \theta) \Rightarrow \cos \theta = \frac{2}{3} \Rightarrow \theta = \cos^{-1}\left(\frac{2}{3}\right)$$

Putting the value of $\cos \theta$ in equation (iv), we get : $v = \sqrt{\frac{g\ell}{3}}$



EXERCISE (S-1)

Work

1. The sign of work done by a force on a body is important to understand. State carefully if the following quantities are positive or negative: **(NCERT)**
- work done by a man in lifting a bucket out of a well by means of a rope tied to the bucket.
 - work done by gravitational force in the above case,
 - work done by friction on a body sliding down an inclined plane,
 - work done by an applied force on a body moving on a rough horizontal plane with uniform velocity,
 - work done by the resistive force of air on a vibrating pendulum in bringing it to rest.

WE0001

2. A body constrained to move along the z-axis of a coordinate system is subject to a constant force F given by **(NCERT)**

$$F = -\hat{i} + 2\hat{j} + 3\hat{k} \text{ N}$$

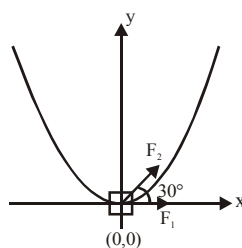
where $\hat{i}, \hat{j}, \hat{k}$ are unit vectors along the x-, y- and z-axis of the system respectively. What is the work done by this force in moving the body a distance of 4 m along the z-axis ?

WE0002

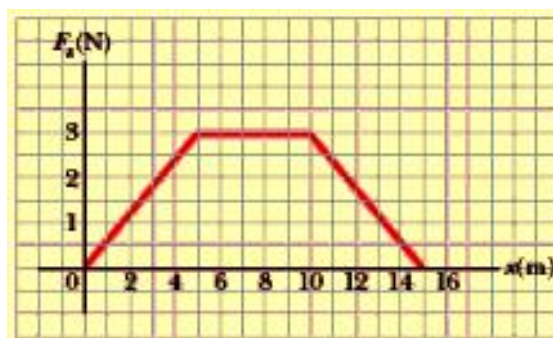
3. A point mass of 0.5 kg is moving along x-axis as $x = t^2 + 2t$, where, x is in meters and t is in seconds. Find the work done (in J) by all the forces acting on the body during the time interval [0, 2s].

WE0003

4. A sleeve of mass 2 kg at origin can move on wire of parabolic shape $x^2 = 4y$. Two forces $F_1 = 6\text{ N}$ and $F_2 = 8\text{ N}$ are applied on the sleeve. F_1 is constant and is in x-direction. F_2 is constant in direction and magnitude. Body is displaced from origin to $x = 4$, then net work done by F_1 and F_2 is

**WE0004**

5. A particle is subject to a force F_x that varies with position as in figure. Find the work done by the force on the body as it moves (a) from $x = 0$ to $x = 5.00$ m, (b) from $x = 5.00$ m to $x = 10.0$ m, and (c) from $x = 10.0$ m to $x = 15.0$ m. (d) What is the total work done by the force over the distance $x = 0$ to $x = 15.0$ m?



WE0005

6. A spring, which is initially in its unstretched condition, is first stretched by a length x and then again by a further length x . The work done in the first case is W_1 and in the second case is W_2 . Find $\frac{W_2}{W_1}$.

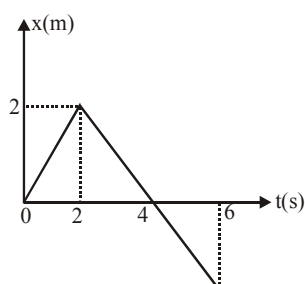
WE0006

Kinetic energy, Work energy theorem, Power

7. A body of mass 2 kg initially at rest moves under the action of an applied horizontal force of 7 N on a table with coefficient of kinetic friction = 0.1. Compute the (NCERT)
- work done by the applied force in 10 s,
 - work done by friction in 10 s,
 - work done by the net force on the body in 10 s,
 - change in kinetic energy of the body in 10 s, and interpret your results.

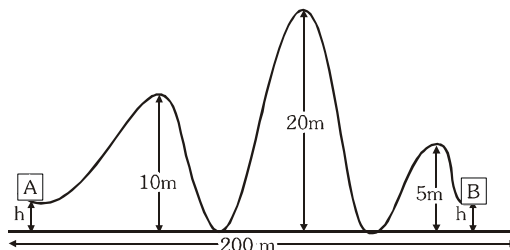
WE0007

8. Position-time graph of a particle of mass 2 kg is shown in figure. Total work done on the particle from $t = 0$ to $t = 4$ s is



WE0008

9. A point object of mass 2 kg is moved from point A to point B very slowly on a curved path by applying a tangential force on a curved path as shown in figure. Then find the work done by external force in moving the body. Given that $\mu_s = 0.3$, $\mu_k = 0.1$. [$g = 10 \text{ m/s}^2$]



WE0009

10. A 4 kg particle moves along the X-axis. Its position x varies with time according to $x(t) = t + 2t^3$, where x is in m and t is in seconds. Compute:
- The kinetic energy at time t .
 - The force acting on the particle at time t .
 - The power delivered to the particle at time t .
 - The work done on the particle from $t = 0$ to $t = 2$ seconds.

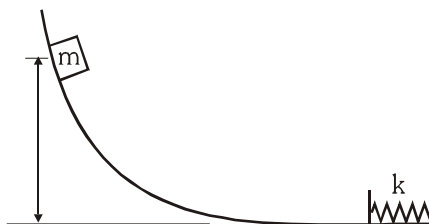
WE0010

11. A block is released from rest from top of a rough curved track as shown in figure. It comes to rest at some point on the horizontal part. If its mass is 200 gm, calculate negative of work by friction in joules.



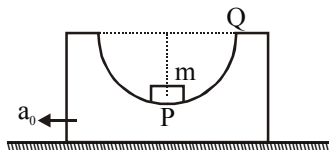
WE0011

12. A mass m slides from rest at height h down a smooth curved surface which becomes horizontal at zero height (see figure). A spring is fixed horizontally on the level part of the surface. The spring constant is $k \text{ N/m}$. When the mass encounters the spring it compresses it by an amount $x = h/10$. If $m = 1 \text{ kg}$, $h = 5 \text{ m}$ then find $k/100$.



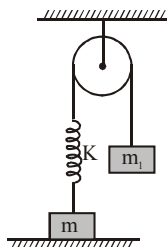
WE0012

13. A small block of mass m is lying at rest at point P of a wedge having a smooth semi circular track of radius R . The minimum value of horizontal acceleration a_0 (in m/s^2) of wedge so that mass can just reach the point Q, is



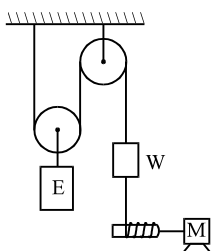
WE0013

14. Initially spring is relaxed when m_1 is released from rest. Calculate for what minimum value of m_1 the block of mass m will just leave the contact with surface? Give answer in terms of $\frac{m}{m_1}$.



WE0014

15. The elevator E has a mass of 3000 kg when fully loaded and is connected as shown to a counterweight W of mass 1000 kg. Determine the power in kilowatts delivered by the motor
- (a) when the elevator is moving down at a constant speed of 3 m/s,
- (b) when it has an upward velocity of 3 m/s and a deceleration of 0.5 m/s^2 .



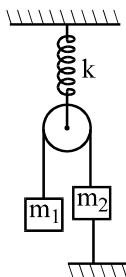
WE0015

16. Power applied to a particle varies with time as $P = (3t^2 - 2t + 1)$ watt, where t is in second. Find the change in its kinetic energy between time $t = 2$ s and $t = 4$ s.

WE0016

Conservative & non conservation forces, Potential energy, Conservation of energy

17. In the figure shown, pulley and spring are ideal. Find the potential energy stored in the spring ($m_1 > m_2$).



WE0017

18. The potential energy (in joules) function of a particle in a region of space is given as :

$$U = (2x^2 + 3y^3 + 2z)$$

Here x , y and z are in metres. Find the magnitude of x component of force (in newton) acting on the particle at point P ($1\text{m}, 2\text{m}, 3\text{m}$).

WE0018

19. The P.E. of a particle oscillating on x -axis is given as $U = 20 + (x - 2)^2$ here U is in Joules & x is in meters. Total mechanical energy of particle is 36 J

- (i) Find the mean position (ii) Find the max. K.E. of the particle

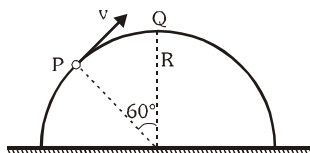
WE0019

20. The potential energy of a 2 kg particle moving along the x axis is given by $U(x) = (4.0\text{J/m}^2)x^2 + (1.0\text{J/m}^4)x^4$. When the particle is at $x = 1.0\text{m}$, find its acceleration. [only conservative forces are acting]

WE0020

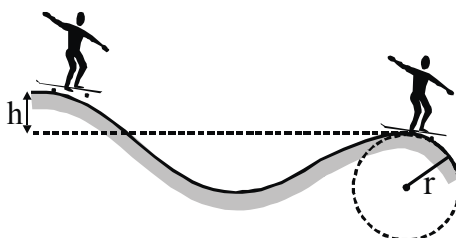
Potential energy diagram, Equilibrium, Vertical circular motion

21. A particle is given a certain velocity v at point P as shown on a hemispherical smooth surface. Find the value of v (in m/s), such that when particle reaches Q , the normal reaction of surface becomes equal to particle's weight. [$R = 1.6$ m, $g = 10$ m/s^2]



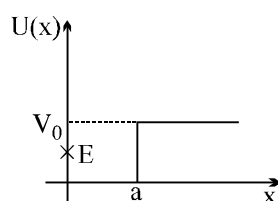
WE0021

22. A skier starts from rest at the top of a hill. The skier coasts down the hill and up a second hill, as the drawing illustrates. The crest of the second hill is circular, with a radius of $r = 36$ m. Neglect friction and air resistance. What must be the height h (in m) of the first hill so that the skier just loses contact with the snow at the crest of the second hill?

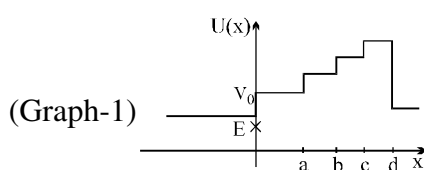


WE0022

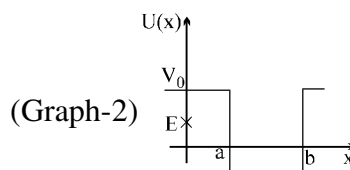
23. The given graph is a potential energy function in one dimension. The total energy of particle is indicated by cross on the ordinate axis. The graph of figure-1 is given as an example. From the figure-1, it can be interpreted that for the given total energy indicated by cross on the ordinate axis the particle cannot be found in the Region : $x > a$. Now, for the following potential functions in one dimensions, specify the regions, in which the particle cannot be found for the energy marked as E on graphs. Give your answer in the blocks shown.



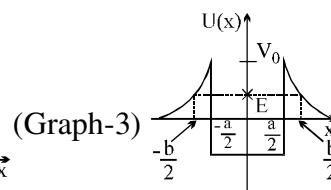
(Figure -1)



(Graph-1)



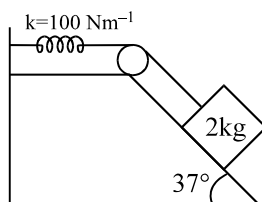
(Graph-2)



(Graph-3)

WE0023

24. (a) A 2 kg block situated on a smooth fixed incline is connected to a spring of negligible mass, with spring constant $k = 100 \text{ Nm}^{-1}$, via a frictionless pulley. The block is released from rest when the spring is unstretched. How far does the block move down the incline before coming (momentarily) to rest? What is its acceleration at its lowest point?
- (b) The experiment is repeated on a rough incline. If the block is observed to move 0.20 m down along the incline before it comes to instantaneous rest, calculate the coefficient of kinetic friction.

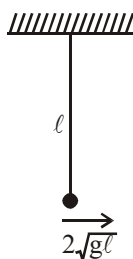


WE0024

25. A ball is attached to a horizontal cord of length L whose other end is fixed, (a) If the ball is released, what will be its speed at the lowest point of its path? (b) A peg is located a distance h directly below the point of attachment of the cord. If $h = 0.75 L$, what will be the speed of the ball when it reaches the top of its circular path about the peg?

WE0025

26. One end of a string of length $\ell = \frac{14}{9} \text{ m}$ is fixed and a mass of 1 kg is tied to the other end. The ball is given a velocity $2\sqrt{g\ell}$ at the bottom most point as shown in figure. The string is cut when the ball becomes horizontal. Find the distance (in m) travelled till it stop for the 1st time (Take $\pi = \frac{22}{7}$).



WE0026

EXERCISE (S-2)

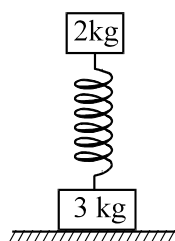
1. A ring of mass m can slide over a smooth vertical rod. The ring is connected to a spring of force constant $K = \frac{4mg}{R}$ where $2R$ is the natural length of the spring. The other end of the spring is fixed to the ground at a horizontal distance $2R$ from the base of the rod. The mass is released at a height of $1.5R$ from ground

(i) calculate the work done by the spring.

(ii) calculate the velocity of the ring as it reaches the ground.

WE0027

2. The ends of spring are attached to blocks of mass 3kg and 2kg . The 3kg block rests on a horizontal surface and the 2kg block which is vertically above it is in equilibrium producing a compression of 1cm of the spring. The 2kg mass must be compressed further by at least _____, so that when it is released, the 3kg block may be lifted off the ground.

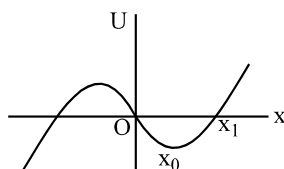


WE0028

3. A uniform rod of mass m length L is sliding along its length on a horizontal table whose top is partly smooth & rest rough with friction coefficient μ . If the rod after moving through smooth part, enters the rough with velocity v_0 .
- (i) What will be the magnitude of the friction force when its x length ($< L$) lies in the rough part during sliding.
- (ii) Determine the minimum velocity v_0 with which it must enter so that it lies completely in rough region before coming to rest.
- (iii) If the velocity is double the minimum velocity as calculated in part (a) then what distance does its front end A would have travelled in rough region before rod comes to rest.

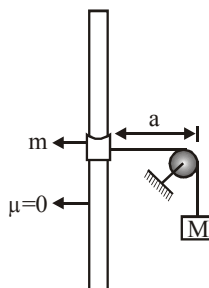
WE0029

4. A particle is confined to move along the +x axis under the action of a force $F(x)$ that is derivable from the potential $U(x) = ax^3 - bx$.
- Find the expression for $F(x)$
 - When the total energy of the particle is zero, the particle can be trapped within the interval $x = 0$ to $x = x_1$. For this case find the values of x_1 .
 - Determine the maximum kinetic energy that the trapped particle has in its motion. Express all answers in terms of a and b . At what value of x will the kinetic energy be maximum?



WE0030

5. A ring of mass m slides on a smooth vertical rod. A light string is attached to the ring and is passing over a smooth peg distant a from the rod, and at the other end of the string is a mass M ($M > m$). The ring is held on a level with the peg and released. Show that it first comes to rest after falling a distance $\frac{2mMa}{M^2 - m^2}$.



WE0031

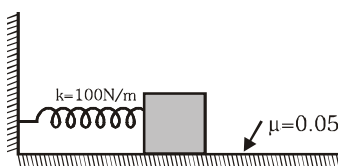
6. A 650-kg elevator starts from rest. It moves upward for 3.00 s with constant acceleration until it reaches its cruising speed of 1.75 m/s.
- What is the average power of the elevator motor during this period?
 - How does this power compare with its power when it moves at its cruising speed?

WE0032

7. A particle is moving with kinetic energy E , straight up an inclined plane with angle α , the coefficient of friction being μ . The work done against friction before the particle comes down to rest is

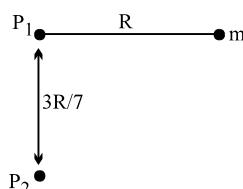
WE0033

8. A block of mass 1 kg is attached to a spring with a force constant 100 N/m and rests on a rough horizontal ground as shown in the figure. Initial displacement of block from natural length is 50 cm. The total distance covered by the block if coefficient of friction between block & ground is 0.05. [$g = 10 \text{ m/s}^2$]



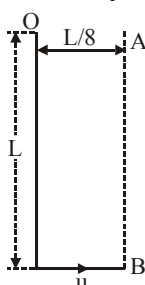
WE0034

9. A simple pendulum consists of a bob of mass m and a string of length R suspended from a peg P_1 on the wall. A second peg P_2 is fixed vertically below the first one at a distance $3R/7$ from it. The pendulum is drawn aside such that the string is horizontal and released. Calculate the maximum height (with respect to the lowest point) to which it rises



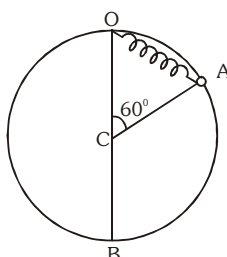
WE0035

10. A particle is suspended vertically from a point O by an inextensible massless string of length L . A vertical line AB is at a distance $\frac{L}{8}$ from O as shown in figure. The object is given a horizontal velocity u . At some point, its motion ceases to be circular and eventually the object passes through the line AB . At the instant of crossing AB , its velocity is horizontal. Find u .



WE0036

11. A particle of mass 5 kg is free to slide on a smooth ring of radius $r = 20 \text{ cm}$ fixed in a vertical plane. The particle is attached to one end of a spring whose other end is fixed to the top point O of the ring. Initially the particle is at rest at a point A of the ring such that $\angle OCA = 60^\circ$, C being the centre of the ring. The natural length of the spring is also equal to $r = 20 \text{ cm}$. After the particle is released and slides down the ring the contact force between the particle & the ring becomes zero when it reaches the lowest position B . Determine the force constant of the spring.



WE0037

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Work

1. A force of magnitude of 30 N acting along $\hat{i} + \hat{j} + \hat{k}$, displaces a particle from point (2, 4, 1) to (3, 5, 2). The work done during this displacement is :-

(A) 90 J (B) 30 J (C) $30\sqrt{3}$ J (D) $30/\sqrt{3}$ J

WE0038

2. A spring of force constant 800 N/m has an extension of 5 cm. The work done in extending it from 5 cm to 15 cm is-

[AIEEE - 2002]

(A) 16 J (B) 8 J (C) 32 J (D) 24 J

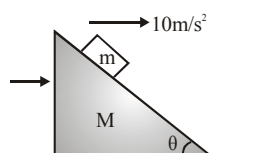
WE0039

3. A rope is used to lower vertically a block of mass M by a distance x with a constant downward acceleration $g/2$. The work done by the rope on the block is :

(A) Mgx (B) $\frac{1}{2}Mgx^2$ (C) $-\frac{1}{2}Mgx$ (D) Mgx^2

WE0040

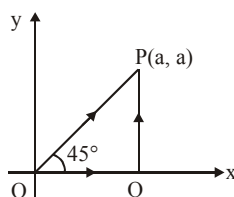
4. In the figure shown all the surfaces are frictionless, and mass of the block, $m = 1$ kg. The block and wedge are held initially at rest. Now wedge is given a horizontal acceleration of 10 m/s^2 by applying a force on the wedge, so that the block does not slip on the wedge. Then work done by the normal force in ground frame on the block in $\sqrt{3}$ seconds is :



(A) 30 J (B) 60 J (C) 150 J (D) $100\sqrt{3}$ J

WE0041

5. A particle is moved from (0, 0) to (a, a) under a force $\vec{F} = (3\hat{i} + 4\hat{j})$ from two paths. Path 1 is OP and path 2 is OQP. Let W_1 and W_2 be the work done by this force in these two paths. Then :

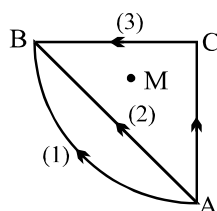


(A) $W_1 = W_2$ (B) $W_1 = 2W_2$ (C) $W_2 = 2W_1$ (D) $W_2 = 4W_1$

WE0042

6. In a region of only gravitational field of mass 'M' a particle is shifted from A to B via three different paths in the figure. The work done in different paths are W_1 , W_2 , W_3 respectively then

[IIT-JEE (Scr.)'2003]



- (A) $W_1 = W_2 = W_3$ (B) $W_1 = W_2 > W_3$ (C) $W_1 > W_2 > W_3$ (D) $W_1 < W_2 < W_3$

WE0043

Kinetic energy, Work energy theorem, Power

7. A body of mass 0.5 kg travels in a straight line with velocity $v = ax^{3/2}$ where $a = 5 \text{ m}^{-1/2}\text{s}^{-1}$. The work done by the net force during its displacement from $x = 0$ to $x = 2 \text{ m}$ is

- (A) 1.5 J (B) 50 J (C) 10 J (D) 100 J

WE0044

8. A man who is running has half the kinetic energy of the boy of half his mass. The man speeds up by 1 m/s and then has the same kinetic energy as the boy. The original speed of the man was

- (A) $\sqrt{2} \text{ m/s}$ (B) $(\sqrt{2} - 1) \text{ m/s}$ (C) 2 m/s (D) $(\sqrt{2} + 1) \text{ m/s}$

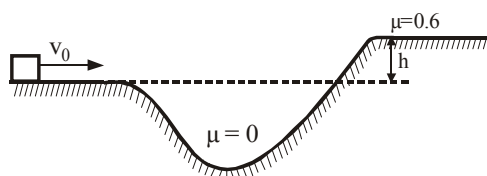
WE0045

9. A mass of 5 kg is moving along a circular path of radius 1 m. If the mass moves with 300 revolutions per minute, its kinetic energy would be

- (A) $250 \pi^2 \text{ joule}$ (B) $100 \pi^2 \text{ joule}$ (C) $5 \pi^2 \text{ joule}$ (D) 0 joule

WE0046

10. In the figure, a block slides along a track from one level to a higher level, by moving through an intermediate valley. The track is frictionless until the block reaches the higher level. There a frictional force stops the block in a distance d . The block's initial speed v_0 is 6 m/s, the height difference h is 1.1 m and the coefficient of kinetic friction μ is 0.6. The value of d is



- (A) 1.17 m (B) 1.71 m (C) 7.11 m (D) 11.7 m

WE0047

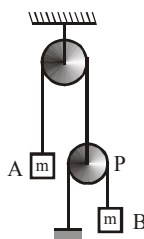
11. In a shotput event an athlete throws the shotput of mass 10 kg with an initial speed of 1 m/s at 45° from a height 1.5 m above ground. Assuming air resistance to be negligible and acceleration due to gravity to be 10 m/s^2 , the kinetic energy of the shotput when it just reaches the ground will be
 (A) 2.5 J (B) 5.0 J (C) 52.5 J (D) 155.0 J

WE0048

12. A particle moves on a rough horizontal ground with some initial velocity say v_0 . If $3/4$ of its kinetic energy is lost in friction in time t_0 then coefficient of friction between the particle and the ground is :-
 (A) $\frac{v_0}{2gt_0}$ (B) $\frac{v_0}{4gt_0}$ (C) $\frac{3v_0}{4gt_0}$ (D) $\frac{v_0}{gt_0}$

WE0049

13. In the figure shown, the system is released from rest. Find the velocity of block A when block B has fallen a distance ' ℓ '. Assume all pulleys to be massless and frictionless.



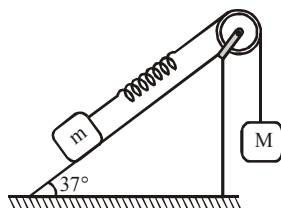
- (A) $\sqrt{\frac{g\ell}{5}}$ (B) $\sqrt{g\ell}$ (C) $\sqrt{5g\ell}$ (D) None of these

WE0050

14. A block of mass m is hung vertically from an elastic thread of force constant mg/a . Initially the thread was at its natural length and the block is allowed to fall freely. The kinetic energy of the block when it passes through the equilibrium position will be :
 (A) mga (B) $mga/2$ (C) zero (D) $2mga$

WE0051

15. A block of mass m is attached with a massless spring of force constant k . The block is placed over a rough inclined surface for which the coefficient of friction is $\mu = \frac{3}{4}$. The minimum value of M required to move the block up the plane is : (Neglect mass of string and pulley and friction in pulley)



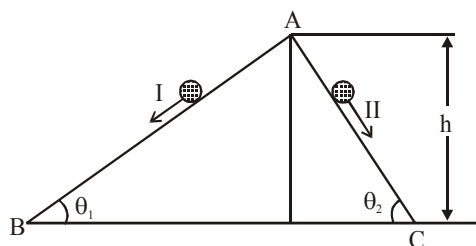
- (A) $\frac{3}{5} m$ (B) $\frac{4}{5} m$ (C) $2 m$ (D) $\frac{3}{2} m$

WE0052

16. A body is moved along a straight line by a machine delivering a constant power. The distance moved by the body in time t is proportional to- [AIEEE - 2003]
 (A) $t^{3/4}$ (B) $t^{3/2}$ (C) $t^{1/4}$ (D) $t^{1/2}$ WE0053
17. A particle moves in a straight line with retardation proportional to its displacement. Its loss of kinetic energy for any displacement x is proportional to- [AIEEE - 2004]
 (A) x^2 (B) e^x (C) x (D) $\log_e x$ WE0054
18. A body of mass m accelerates uniformly from rest to v_1 in time t_1 . The instantaneous power delivered to the body as a function of time t is- [AIEEE - 2004]
 (A) $\frac{mv_1 t}{t_1}$ (B) $\frac{mv_1^2 t}{t_1^2}$ (C) $\frac{mv_1 t^2}{t_1}$ (D) $\frac{mv_1^2 t}{t_1}$ WE0055
19. Assume the aerodynamic drag force on a car is proportional to its speed. If the power output from the engine is doubled, then the maximum speed of the car.
 (A) is unchanged (B) increases by a factor of $\sqrt{2}$
 (C) is also doubled (D) increases by a factor of four. WE0056

Conservative & non conservative forces, Potential energy, Conservation of energy

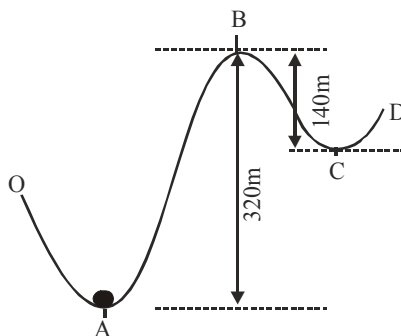
20. Two inclined frictionless tracks, one gradual and the other steep meet at A from where two stones are allowed to slide down from rest, one on each track as shown in figure. Which of the following statement is correct?



- (A) Both the stones reach the bottom at the same time but not with the same speed.
 (B) Both the stones reach the bottom with the same speed and stone I reaches the bottom earlier than stone II.
 (C) Both the stones reach the bottom with the same speed and stone II reaches the bottom earlier than stone I.
 (D) Both the stones reach the bottom at different times and with different speeds.

WE0057

21. Track OABCD (as shown in figure) is smooth and fixed in vertical plane. What minimum speed has to be given to a particle lying at point A, so that it can reach point C?

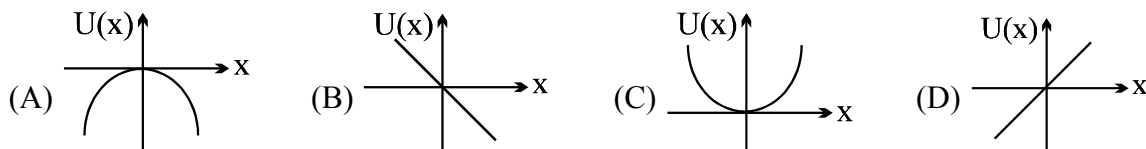


- (A) 60 m/s (B) 100 m/s (C) 70 m/s (D) 80 m/s

WE0058

22. A particle is placed at the origin and a force $F = kx$ is acting on it (where k is a positive constant). If $U(0) = 0$, the graph of $U(x)$ versus x will be (where U is the potential energy function)

[IIT-JEE' 2004(Scr)]

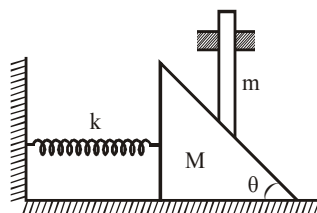


WE0059

23. When a conservative force does positive work on a body
 (A) the potential energy increases (B) the potential energy decreases
 (C) total energy increases (D) total energy decreases

WE0060

24. A wedge of mass M fitted with a spring of stiffness ' k ' is kept on a smooth horizontal surface. A rod of mass m is kept on the wedge as shown in the figure. System is in equilibrium. Assuming that all surfaces are smooth, the potential energy stored in the spring is:

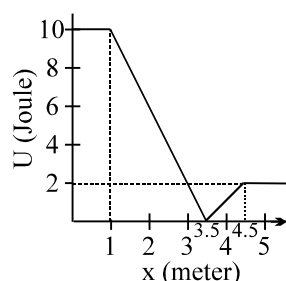


- (A) $\frac{mg^2 \tan^2 \theta}{2k}$ (B) $\frac{m^2 g \tan^2 \theta}{2k}$ (C) $\frac{m^2 g^2 \tan^2 \theta}{2k}$ (D) $\frac{m^2 g^2 \tan^2 \theta}{k}$

WE0061

Potential energy diagram, Equilibrium, Vertical circular motion

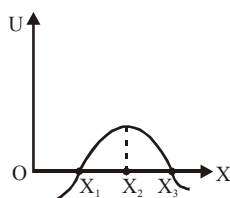
25. A body with mass 2 kg moves in one direction in the presence of a force which is described by the potential energy graph. If the body is released from rest at $x = 2\text{m}$, then its speed when it crosses $x = 5\text{ m}$ is



- (A) zero (B) 1 ms^{-1} (C) 2 ms^{-1} (D) 3 ms^{-1}

WE0062

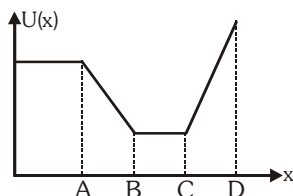
26. In the figure shown the potential energy (U) of a particle is plotted against its position ' x ' from origin. Then which of the following statement is correct. A particle at :



- (A) x_1 is in stable equilibrium (B) x_2 is in stable equilibrium
(C) x_3 is in stable equilibrium (D) None of these

WE0063

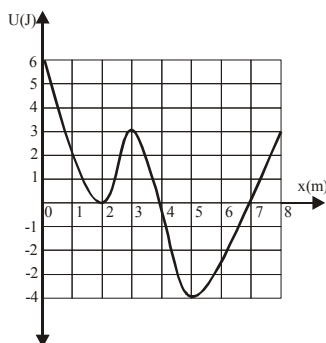
27. As a particle moves along the x -axis it is acted upon by a conservative force. The potential energy is shown below as a function of the coordinate x of the particle. Rank the labelled regions according to the magnitude of the force, least to greatest.



- (A) AB, BC, CD (B) AB, CD, BC (C) BC, CD, AB (D) BC, AB, CD

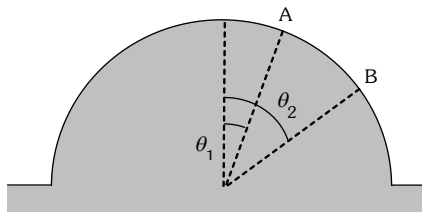
WE0064

28. Potential energy curve U of a particle as function of the position of a particle is shown. The particle has total mechanical energy E of 3.0 joules.



- (A) It can never be present at $x = 0$ m. (B) It can never be present at $x = 5$ m
(C) At $x = 2$ its kinetic energy is 0 J (D) At $x = 1$ its kinetic energy 3 J
29. A small block slides down from rest at point A on the surface of a smooth circular cylinder, as shown. At point B, the block falls off (leaves) the cylinder. The equation relating the angles θ_1 and θ_2 is given by

WE0065



- (A) $\sin \theta_2 = \frac{2}{3} \sin \theta_1$ (B) $\sin \theta_2 = \frac{3}{2} \sin \theta_1$ (C) $\cos \theta_2 = \frac{2}{3} \cos \theta_1$ (D) $\cos \theta_2 = \frac{3}{2} \cos \theta_1$

WE0066

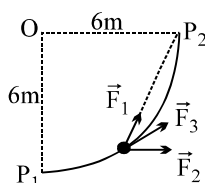
MULTIPLE CORRECT TYPE QUESTIONS

30. Which of the following statements is TRUE for a system comprising of two bodies in contact exerting frictional force on each other :
- (A) total work done by static friction on whole system is always zero.
(B) work done by static friction on a body is always zero
(C) work done by kinetic friction on a body is always negative
(D) total work done by internal kinetic friction on whole system is always negative
31. A particle of mass m is at rest in a train moving with constant velocity with respect to ground. Now the particle is accelerated by a constant force F_0 acting along the direction of motion of train for time t_0 . A girl in the train and a boy on the ground measure the work done by this force. Which of the following are **INCORRECT**?
- (A) Both will measure the same work
(B) Boy will measure higher value than the girl
(C) Girl will measure higher value than the boy
(D) Data are insufficient for the measurement of work done by the force F_0

WE0067

WE0068

32. A smooth track in the form of a quarter circle of radius 6 m lies in the vertical plane. A particle moves from P_1 to P_2 under the action of forces $\vec{F}_1$, $\vec{F}_2$ and $\vec{F}_3$. Force $\vec{F}_1$ is always toward P_2 and is always 20 N in magnitude. Force $\vec{F}_2$ always acts horizontally and is always 30 N in magnitude. Force $\vec{F}_3$ always acts tangentially to the track and is of magnitude 15 N. Select the correct alternative(s)



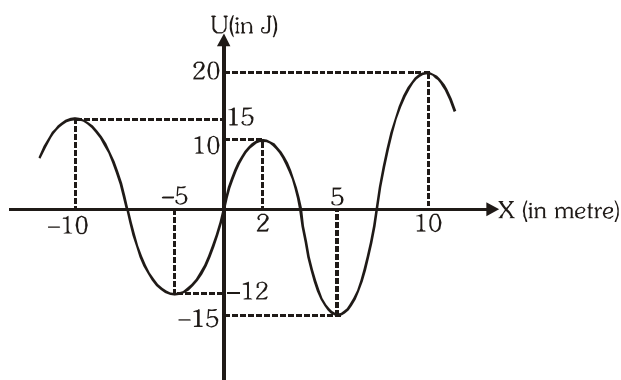
- (A) work done by $\vec{F}_1$ is 120 J
(B) work done by $\vec{F}_2$ is 180 J
(C) work done by $\vec{F}_3$ is 45π
(D) $\vec{F}_1$ is conservative in nature

WE0069

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 33 to 35

In the figure the variation of potential energy of a particle of mass $m = 2\text{ kg}$ is represented w.r.t. its x -coordinate. The particle moves under the effect of this conservative force along the x -axis.



33. If the particle is released at the origin then :
(A) It will move towards positive x -axis.
(B) It will move towards negative x -axis.
(C) It will remain stationary at the origin.
(D) Its subsequent motion cannot be decided due to lack of information.
34. If the particle is released at $x = 2 + \Delta$ where $\Delta \rightarrow 0$ (it is positive) then its maximum speed in subsequent motion will be-
(A) $\sqrt{22}$ m/s
(B) $\sqrt{25}$ m/s
(C) $\sqrt{24}$ m/s
(D) $\sqrt{23}$ m/s
35. $x = -5$ m and $x = 10$ m positions of the particle are respectively of-
(A) Neutral and stable equilibrium
(B) Neutral and unstable equilibrium
(C) Unstable and stable equilibrium
(D) Stable and unstable equilibrium.

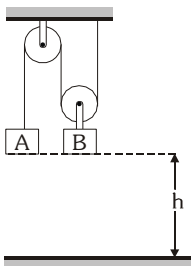
WE0070

WE0070

WE0070

MATRIX MATCH TYPE QUESTIONS

36. In the figure shown are two blocks A and B of same mass connected with pulley and string to each other. Initially both of them are at a height of $h = 0.5$ m from ground. After they are released they move in either direction and one of them strike the ground. For, the interval from releasing to when one of them strike, some physical quantities are in column I and their modulus values in SI units are in column II.



Column I

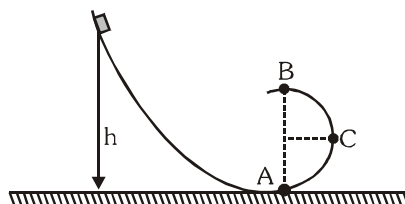
- (A) Velocity of A immediately before any one of them strike ground.
 (B) Velocity of B immediately before any one of them strike ground.
 (C) Ratio of work done by gravity on A to workdone by gravity on B.
 (D) Acceleration of block A before any one of them strike ground.

Column II

- (P) 1
 (Q) 2
 (R) 3
 (S) 4
 (T) 5

WE0071

37. A block of mass m is released from top of a smooth track as shown in the figure. The end part of the track is a circle in vertical plane of radius R . N is normal reaction of the track at any point of the track. Match the entries of column I with entries of column-II.



Column I

- (A) $h = \frac{5}{2} R$
 (B) $h = \frac{9}{2} R$
 (C) $h = R$
 (D) $h = 2R$

Column II

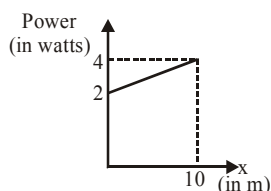
- (P) Net force on the block at C is mg
 (Q) $N_A - N_B = 6 mg$
 (R) Block leaves contact before B
 (S) Block will keep contact with the track is region between A & B.
 (T) $N_C > mg$

WE0072

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

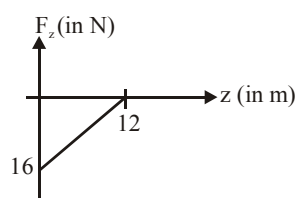
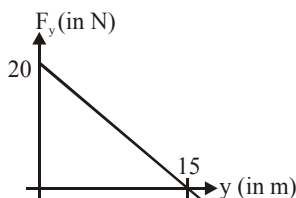
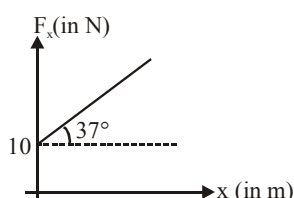
1. A particle 'A' of mass $\frac{10}{7}$ kg is moving in the positive x-direction. Its initial position is $x = 0$ & initial velocity is 1 m/s. The velocity at $x = 10$ m is : (use the graph given)



- (A) 4 m/s (B) 2 m/s (C) $3\sqrt{2}$ m/s (D) $100/3$ m/s

WE0073

2. The components of a force acting on a particle are varying according to the graphs shown. To reach at point B (8, 20, 0) from point A(0, 5, 12) the particle moves on paths parallel to x-axis then y-axis and then z-axis, then work done by this force is :-



- (A) 192 J (B) 58 J (C) 250 J (D) 125 J

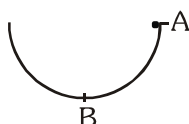
WE0074

3. A light spring of length 20 cm and force constant 2 N/cm is placed vertically on a table. A small block of mass 1 kg falls on it. The length h from the surface of the table at which the block will have the maximum velocity is :

- (A) 20 cm (B) 15 cm (C) 10 cm (D) 5 cm

WE0075

4. A small ball can move in a vertical plane along a semi-circle of radius r without friction. At what speed is the ball to launch from point A so that its acceleration is $3g$ at point B ?



- (A) $(3gr)^{1/2}$ (B) $(2gr)^{1/2}$ (C) $(gr)^{1/2}$ (D) $2(gr)^{1/2}$

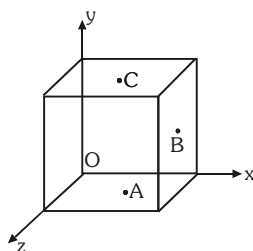
WE0076

MULTIPLE CORRECT TYPE QUESTIONS

5. A particle of mass 2 kg is projected with an initial speed $u = 10$ m/sec at an angle $\theta = 30^\circ$ with the horizontal
- (A) Total work done on the particle during the first half of the total time of flight of the particle is (-25) J.
- (B) Total work done on the particle during the total time of flight of the particle is 0 J.
- (C) Average power delivered to the particle during the first half of the flight is (-50) watt.
- (D) The radius of curvature of the trajectory of the particle at the highest point of the projectile is 7.5m.

WE0077

6. A particle is shifted from A to B and then from B to C where A, B and C are the midpoints of the corresponding faces of a cube of side 2m. If a force $\vec{F} = (3\hat{i} + 4\hat{j} - 5\hat{k})$ N is continuously acting on the particle, then select correct alternative :-



- (A) work done from A to B is 7 J
- (B) work done from B to C is 1 J
- (C) work done A to C is 8 J
- (D) $\vec{F}$ is a conservative force

WE0078

7. Which of the following is/are conservative force(s)?

- (A) $\vec{F} = 2r^3\hat{r}$
- (B) $\vec{F} = -\frac{5}{r}\hat{r}$
- (C) $\vec{F} = \frac{3(xi + yj)}{(x^2 + y^2)^{3/2}}$
- (D) $\vec{F} = \frac{3(yi + xj)}{(x^2 + y^2)^{3/2}}$

WE0079

8. If one of the forces acting on a particle is conservative then :
- (A) Work done by this force is zero when the particle moves exactly once around any closed path.
- (B) Work done by this force equals the change in the kinetic energy of the particle.
- (C) It obeys Newton's second law.
- (D) Work done by this force depends on the end points of the motion, not on the path in between.

WE0080

9. A particle of mass $m = 1$ kg lying on x-axis experiences a force given by law $F = x(3x - 2)$ Newton, where x is the x-coordinate of the particle in meters. The points on x-axis where the particle is in equilibrium are :
- (A) $x = 0$
- (B) $x = 1/3$
- (C) $x = 2/3$
- (D) $x = 1$

WE0081

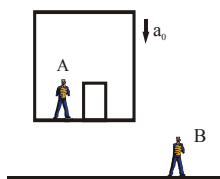
10. A particle is given a velocity at the bottom most position in a smooth spherical shell of radius $2m$. It just complete a vertical circular motion. Then
- (A) acceleration of particle when the velocity of the particle is in vertically upward direction is $g\sqrt{10}$
- (B) acceleration of particle when the velocity of the particle is in vertically downward direction is $g\sqrt{10}$
- (C) acceleration of the particle at the top most point of path is g
- (D) acceleration of the particle at the bottom most point is $5g$

WE0082

COMPREHENSION TYPE QUESTIONS

Comprehension for Question no. 11 to 14

A block of mass m is kept in an elevator which starts moving downward with an acceleration a_0 as shown in figure. The block is observed by two observers A and B for a time interval t_0 .



11. The observer B finds that the work done by gravity is
- (A) $\frac{1}{2} mg^2 t_0^2$ (B) $-\frac{1}{2} mg^2 t_0^2$ (C) $\frac{1}{2} m g a_0 t_0^2$ (D) $-\frac{1}{2} m g a_0 t_0^2$
12. The observer B finds that work done by normal reaction N is :-
- (A) zero (B) $-N a_0 t_0^2$ (C) $\frac{N a_0 t_0^2}{2}$ (D) None of these
13. According to observer B, the net work done on the block is
- (A) $-\frac{1}{2} m a_0 t_0^2$ (B) $\frac{1}{2} m a_0^2 t_0^2$ (C) $\frac{1}{2} m g a_0 t_0^2$ (D) $-\frac{1}{2} m g a_0 t_0^2$
14. According to the observer A
- (A) the work done by gravity is zero (B) the work done by normal reaction is zero
- (C) the work done by pseudo force is zero (D) all the above

WE0083

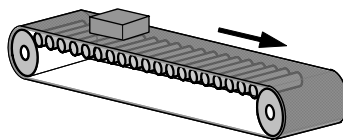
WE0083

WE0083

WE0083

Paragraph for Question 15 & 16

An object of mass M is gently placed on a horizontal conveyor belt, which is moving with uniform velocity v_0 as shown in the figure. The coefficient of static friction is μ_s , the coefficient of kinetic friction is μ_k , and the acceleration of gravity is g . Initially the object slips for a while but finally moves without slipping together with the belt.



15. How far the conveyor belt moves while the object is slipping?

(A) $\frac{v_0^2}{\mu_k g}$ (B) $\frac{v_0^2}{2\mu_k g}$ (C) $\frac{v_0^2}{\mu_s g}$ (D) $\frac{v_0^2}{2\mu_s g}$

WE0084

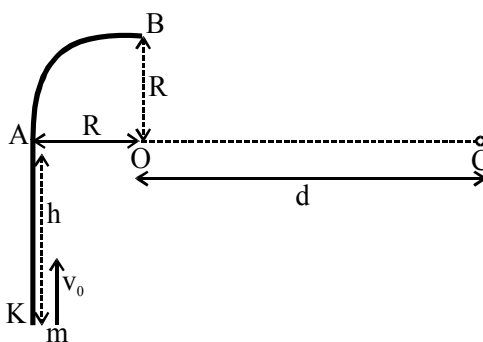
16. Work done on the object by friction force relative to the reference frame moving with the conveyor belt is

(A) $\frac{1}{2} M v_0^2$ (B) $-\frac{1}{2} M v_0^2$ (C) $M v_0^2$ (D) zero

WE0084

Paragraph for Question Nos.17 to 19

A point like object of mass m starts from point K as shown in the figure. It slides inside along the full length of the smooth track of radius R , and then moves freely and travels to point C. [The track is kept in vertical plane]



17. Determine the vertical initial velocity of the pointlike object.

(A) $v_0 = \sqrt{2g(h+R) + \frac{d^2 g}{R}}$ (B) $v_0 = \sqrt{2g(h-R) + \frac{d^2 g}{2R}}$
 (C) $v_0 = \sqrt{2g(h+R) + \frac{d^2 g}{2R}}$ (D) $v_0 = \sqrt{2g(h+R) + \frac{dg}{2R}}$

WE0085

18. What is the minimum possible distance $OC = d$, necessary for the object to slide along the entire length of the track ?

(A) $d_{\min} = R\sqrt{2}$ (B) $d_{\min} = \sqrt{3} R$ (C) $d_{\min} = \frac{R}{\sqrt{2}}$ (D) $d_{\min} = \frac{R}{\sqrt{3}}$

WE0085

19. Find the normal force exerted by the track at point A.

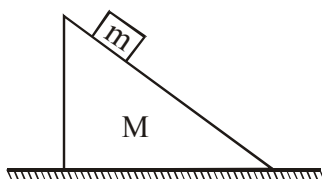
(A) $F_A = mg \left(\frac{d^2}{2R^2} - 2 \right)$ (B) $F_A = mg \left(\frac{d^2}{2R^2} + 2 \right)$

(C) $F_A = mg \left(\frac{3d^2}{R^2} + 2 \right)$ (D) $F_A = mg \left(\frac{d}{3R^2} + 2 \right)$

WE0085

MATRIX MATCH TYPE QUESTIONS

20. A block of mass m lies on wedge of mass M . The wedge in turn lies on smooth horizontal surface. Friction is absent everywhere. The wedge block system is released from rest. All situation given in column-I are to be estimated in duration the block undergoes a vertical displacement 'h' starting from rest (assume the block to be still on the wedge, g is acceleration due to gravity).



Column I	Column II
(A) Work done by normal reaction acting on the block is	(p) Positive
(B) Work done by normal reaction (exerted by block) acting on wedge is	(q) Negative
(C) The sum of work done by normal reaction on block and work done by normal reaction (exerted by block) on wedge is	(r) Zero
(D) Net work done by all forces on block is	(s) Less than mgh in magnitude

WE0086

EXERCISE (JM)

1. The potential energy function for the force between two atoms in a diatomic molecule is approximately given by $U(x) = \frac{a}{x^{12}} - \frac{b}{x^6}$, where a and b are constant and x is the distance between the atoms. if the dissociation energy of the molecule is $D = [U(x = \infty) - U_{\text{at equilibrium}}]$, D is : [AIEEE-2010]

(1) $\frac{b^2}{6a}$ (2) $\frac{b^2}{2a}$ (3) $\frac{b^2}{12a}$ (4) $\frac{b^2}{4a}$

WE0087

2. At time $t = 0$ s particle starts moving along the x -axis. If its kinetic energy increases uniformly with time ' t ', the net force acting on it must be proportional to :- [AIEEE-2011]

(1) $\sqrt{t}$ (2) constant (3) t (4) $\frac{1}{\sqrt{t}}$

WE0088

3. This question has Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements. [AIEEE-2012]

If two springs S_1 and S_2 of force constants k_1 and k_2 , respectively, are stretched by the same force, it is found that more work is done on spring S_1 than on spring S_2 .

Statement-1: If stretched by the same amount, work done on S_1 , will be more than that on S_2

Statement-2 : $k_1 < k_2$

- (1) Statement-1 is true, Statement-2 is true and Statement-2 is not the correct explanation of Statement-1.
 (2) Statement-1 is false, Statement-2 is true
 (3) Statement-1 is true, Statement-2 is false
 (4) Statement-1 is true, Statement-2 is true and Statement-2 is the correct explanation of statement-1.

WE0089

4. When a rubber-band is stretched by a distance x , it exerts a restoring force of magnitude $F = ax + bx^2$ where a and b are constants. The work done in stretching the unstretched rubber-band by L is:- [JEE-Main-2014]

(1) $\frac{aL^2}{2} + \frac{bL^3}{3}$ (2) $\frac{1}{2} \left(\frac{aL^2}{2} + \frac{bL^3}{3} \right)$ (3) $aL^2 + bL^3$ (4) $\frac{1}{2}(aL^2 + bL^3)$

WE0090

5. A person trying to lose weight by burning fat lifts a mass of 10 kg upto a height of 1 m 1000 times. Assume that the potential energy lost each time he lowers the mass is dissipated. How much fat will he use up considering the work done only when the weight is lifted up? Fat supplies 3.8×10^7 J of energy per kg which is converted to mechanical energy with a 20% efficiency rate. Take $g = 9.8 \text{ ms}^{-2}$:-

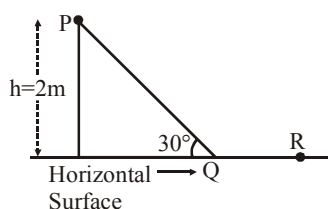
[JEE-Main-2016]

- (1) $12.89 \times 10^{-3} \text{ kg}$ (2) $2.45 \times 10^{-3} \text{ kg}$ (3) $6.45 \times 10^{-3} \text{ kg}$ (4) $9.89 \times 10^{-3} \text{ kg}$

WE0091

6. A point particle of mass, moves along the uniformly rough track PQR as shown in the figure. The coefficient of friction, between the particle and the rough track equals μ . The particle is released, from rest, from the point P and it comes to rest at a point R. The energies, lost by the ball, over the parts, PQ and PR, of the track, are equal to each other, and no energy is lost when particle changes direction from PQ to QR. The values of the coefficient of friction μ and the distance $x(=QR)$ are, respectively close to :-

[JEE-Main-2016]



- (1) 0.29 and 6.5 m (2) 0.2 and 6.5 m (3) 0.2 and 3.5 m (4) 0.29 and 3.5 m

WE0092

7. A body of mass $m = 10^{-2} \text{ kg}$ is moving in a medium and experiences a frictional force $F = -kv^2$. Its initial speed is $v_0 = 10 \text{ ms}^{-1}$. If, after 10 s, its energy is $\frac{1}{8}mv_0^2$, the value of k will be :-

[JEE-Main-2017]

- (1) $10^{-4} \text{ kg m}^{-1}$ (2) $10^{-1} \text{ kg m}^{-1} \text{ s}^{-1}$ (3) $10^{-3} \text{ kg m}^{-1}$ (4) $10^{-3} \text{ kg s}^{-1}$

WE0093

8. A time dependent force $F = 6t$ acts on a particle of mass 1 kg. If the particle starts from rest, the work done by the force during the first 1 sec. will be :

[JEE-Main-2017]

- (1) 9 J (2) 18 J (3) 4.5 J (4) 22 J

WE0094

9. A particle is moving in a circular path of radius a under the action of an attractive potential $U = -\frac{k}{2r^2}$. Its total energy is :-

[JEE-Main-2018]

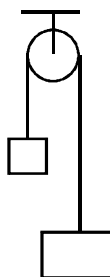
- (1) $\frac{k}{2a^2}$ (2) Zero (3) $-\frac{3}{2} \frac{k}{a^2}$ (4) $-\frac{k}{4a^2}$

WE0095

EXERCISE (JA)

1. A light inextensible string that goes over a smooth fixed pulley as shown in the figure connects two blocks of masses 0.36 kg and 0.72 kg. Taking $g = 10 \text{ m/s}^2$, find the work done (in **joules**) by the string on the block of mass 0.36 kg during the first second after the system is released from rest.

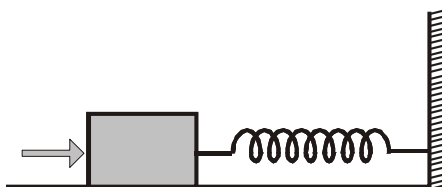
[IIT-JEE-2009]



WE0096

2. A block of mass 0.18 kg is attached to a spring of force-constant 2 N/m. The coefficient of friction between the block and the floor is 0.1. Initially the block is at rest and the spring is un-stretched. An impulse is given to the block as shown in the figure. The block slides a distance of 0.06 m and comes to rest for the first time. The initial velocity of the block in m/s is $V = N/10$. Then N is

[IIT-JEE-2011]



WE0097

3. The work done on a particle of mass m by a force, $K \left[\frac{x}{(x^2 + y^2)^{3/2}} \hat{i} + \frac{y}{(x^2 + y^2)^{3/2}} \hat{j} \right]$ (K being a constant of appropriate dimensions), when the particle is taken from the point $(a, 0)$ to the point $(0, a)$ along a circular path of radius a about the origin in the x - y plane is :-

[JEE-Advance-2013]

- (A) $\frac{2K\pi}{a}$ (B) $\frac{K\pi}{a}$ (C) $\frac{K\pi}{2a}$ (D) 0

WE0098

4. A particle of mass 0.2 kg is moving in one dimension under a force that delivers a constant power 0.5 W to the particle. If the initial speed (in ms^{-1}) of the particle is zero, the speed (in ms^{-1}) after 5 s is.

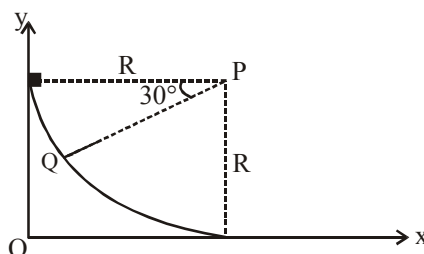
[JEE-Advance-2013]

WE0099

Paragraph for Questions 5 and 6

A small block of mass 1 kg is released from rest at the top of a rough track. The track is a circular arc of radius 40 m. The block slides along the track without toppling and a frictional force acts on it in the direction opposite to the instantaneous velocity. The work done in overcoming the friction up to the point Q, as shown in the figure below, is 150 J. (Take the acceleration due to gravity, $g = 10 \text{ m s}^{-2}$)

[JEE-Advance-2013]



5. The magnitude of the normal reaction that acts on the block at the point Q is

(A) 7.5 N (B) 8.6 N (C) 11.5 N (D) 22.5 N

WE0100

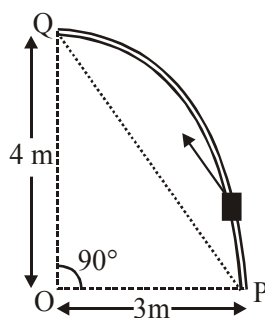
6. The speed of the block when it reaches the point Q is

(A) 5 ms^{-1} (B) 10 ms^{-1} (C) $10\sqrt{3} \text{ ms}^{-1}$ (D) 20 ms^{-1}

WE0100

7. Consider an elliptically shaped rail PQ in the vertical plane with $OP = 3 \text{ m}$ and $OQ = 4 \text{ m}$. A block of mass 1 kg is pulled along the rail from P to Q with a force of 18 N, which is always parallel to line PQ (see the figure given). Assuming no frictional losses, the kinetic energy of the block when it reaches Q is $(n \times 10)$ Joules. The value of n is (take acceleration due to gravity $= 10 \text{ ms}^{-2}$)

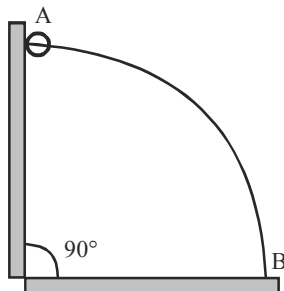
[JEE-Advance-2014]



WE0101

8. A wire, which passes through the hole in a small bead, is bent in the form of quarter of a circle. The wire is fixed vertically on ground as shown in the figure. The bead is released from near the top of the wire and it slides along the wire without friction. As the bead moves from A to B, the force it applies on the wire is :-

[JEE Advanced-2014]



- (A) Always radially outwards
 (B) Always radially inwards
 (C) Radially outwards initially and radially inwards later.
 (D) Radially inwards initially and radially outwards later.

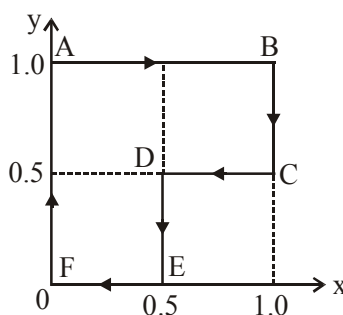
WE0102

9. A particle of mass m is initially at rest at the origin. It is subjected to a force and starts moving along the x -axis. Its kinetic energy K changes with time as $dK/dt = \gamma t$, where γ is a positive constant of appropriate dimensions. Which of the following statements is (are) true ? [JEE Advanced-2018]
- (A) The force applied on the particle is constant
 (B) The speed of the particle is proportional to time
 (C) The distance of the particle from the origin increases linearly with time
 (D) The force is conservative

WE0103

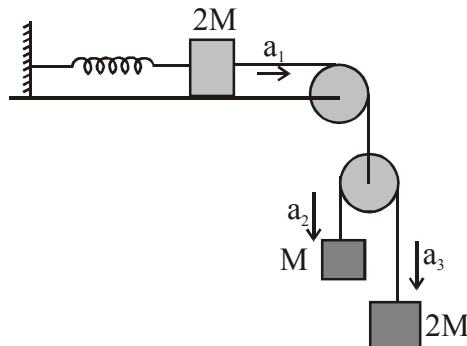
10. A particle is moved along a path AB-BC-CD-DE-EF-FA, as shown in figure, in presence of a force $\vec{F} = (\alpha y \hat{i} + 2\alpha x \hat{j}) \text{ N}$, where x and y are in meter and $\alpha = -1 \text{ N/m}^{-1}$. The work done on the particle by this force $\vec{F}$ will be ____ Joule.

[JEE Advanced-2019]



WE0104

11. A block of mass $2M$ is attached to a massless spring with spring-constant k . This block is connected to two other blocks of masses M and $2M$ using two massless pulleys and strings. The accelerations of the blocks are a_1 , a_2 and a_3 as shown in figure. The system is released from rest with the spring in its unstretched state. The maximum extension of the spring is x_0 . Which of the following option(s) is/are correct ? [g is the acceleration due to gravity. Neglect friction] **[JEE Advanced-2019]**



(1) $x_0 = \frac{4Mg}{k}$

- (2) When spring achieves an extension of $\frac{x_0}{2}$ for the first time, the speed of the block connected to

the spring is $3g\sqrt{\frac{M}{5k}}$

(3) $a_2 - a_1 = a_1 - a_3$

- (4) At an extension of $\frac{x_0}{4}$ of the spring, the magnitude of acceleration of the block connected to the

spring is $\frac{3g}{10}$

WE0105

ANSWER KEY

EXERCISE (S-1)

1. Ans. (a) +ve (b) -ve (c) -ve (d) +ve (e) -ve 2. Ans. 12 J 3. Ans. 8
 4. Ans. 67.7 J 5. Ans. (a) 7.5 J; (b) 15 J; (c) 7.5 J; (d) 30 J 6. Ans. 3
 7. Ans. (a) 875 J; (b) -250 J; (c) 625 J; (d) 625 J; 8. Ans. 0 9. Ans. 400 J
 10. Ans. (i) $2 + 24t^2 + 72t^4$ J, (ii) 48 t N, (iii) $48t + 288t^3$ W, (iv) 1248 J 11. Ans. 4
 12. Ans. 4 13. Ans. 10 14. Ans. $m_1 = m/2$
 15. Ans. (a) -30 kW, 19.5 kW 16. Ans. 46 J 18. Ans. 4
 19. Ans. (i) $x=2$, (ii) 16 J 20. Ans. -6 m/s^2 21. Ans. 4 22. Ans. 018
 23. Ans. Graph-1 : For all x, Graph-2 : $x < a$ & $x > b$, Graph-3 : $-\frac{b}{2} < x < -\frac{a}{2}$ & $\frac{a}{2} < x < \frac{b}{2}$
 24. Ans. (a) $s = 0.24 \text{ m}$, $a = 6 \text{ m/s}^2$, (b) $\mu = 1/8$ 25. Ans. $\sqrt{2gL}$, $\sqrt{gL}$ 26. Ans. 4

EXERCISE (S-2)

1. Ans. (i) $mgR/2$, (ii) $2\sqrt{gR}$ 2. Ans. 2.5cm
 3. Ans. (i) $f = \frac{\mu m}{\ell} xg$; (ii) $\sqrt{\mu g \ell}$; (iii) $\frac{5\ell}{2}$ 4. Ans. $F = -3ax^2 + b$, $x = \sqrt{\frac{b}{a}}$, $K_{\max} = \frac{2b}{3} \sqrt{\frac{b}{3a}}$
 6. Ans. (i) $6 \times 10^3 \text{ W}$ (ii) $1 \times 10^4 \text{ W}$ 7. Ans. $\frac{E\mu \cos \alpha}{\sin \alpha + \mu \cos \alpha}$ 8. Ans. 25 m
 9. Ans. 27R/28 10. Ans. $u = \sqrt{gL \left(2 + \frac{3\sqrt{3}}{2} \right)}$ 11. Ans. 500 N/m

EXERCISE (O-1)

1. Ans. (C) 2. Ans. (B) 3. Ans. (C) 4. Ans. (C) 5. Ans. (A)
 6. Ans. (A) 7. Ans. (B) 8. Ans. (D) 9. Ans. (A) 10. Ans. (A)
 11. Ans. (D) 12. Ans. (A) 13. Ans. (A) 14. Ans. (B) 15. Ans. (A)
 16. Ans. (B) 17. Ans. (A) 18. Ans. (B) 19. Ans. (B) 20. Ans. (C)
 21. Ans. (D) 22. Ans. (A) 23. Ans. (B) 24. Ans. (C) 25. Ans. (C)
 26. Ans. (D) 27. Ans. (D) 28. Ans. (A) 29. Ans. (C) 30. Ans. (A,D)
 31. Ans. (A,C) 32. Ans. (B,C,D) 33. Ans. (B) 34. Ans. (B) 35. Ans. (D)
 36. Ans. (A) Q; (B) P; (C) Q; (D) S 37. Ans. (A) Q,S,T; (B) Q,S,T; (C) P; (D) R,T

EXERCISE (O-2)

1. Ans. (A) 2. Ans. (C) 3. Ans. (B) 4. Ans. (C) 5. Ans. (A,B,C,D)
 6. Ans. (ABCD) 7. Ans. (A,B,C) 8. Ans. (A,C,D) 9. Ans. (A,C) 10. Ans. (A,B,C,D)
 11. Ans. (C) 12. Ans. (D) 13. Ans. (B) 14. Ans. (D) 15. Ans. (A)
 16. Ans. (B) 17. Ans. (C) 18. Ans. (A) 19. Ans. (B)
 20. Ans. (A) Q,S; (B) P,S; (C) R, S; (D) P,S

EXERCISE (J-M)

1. Ans. (4) 2. Ans. (4) 3. Ans. (2) 4. Ans. (1) 5. Ans. (1)
 6. Ans. (4) 7. Ans. (1) 8. Ans. (3) 9. Ans. (2)

EXERCISE (J-A)

1. Ans. 8 2. Ans. 4 3. Ans. (D) 4. Ans. 5 5. Ans. (A) 6. Ans. (B)
 7. Ans. 5 8. Ans. (D) 9. Ans. (A,B,D) 10. Ans. 0.75 11. Ans. (3)



Chapter Contents

03

CENTER OF MASS, MOMENTUM & COLLISION

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Important Notes

[illegible]

CENTER OF MASS, MOMENTUM & COLLISION

KEY CONCEPTS

CENTRE OF MASS :

For a system of particles centre of mass is that point at which its total mass is supposed to be concentrated. The centre of mass of an object is a point that represents the entire body and moves in the same way as a point mass having mass equal to that of the object, when subjected to the same external forces that act on the object.

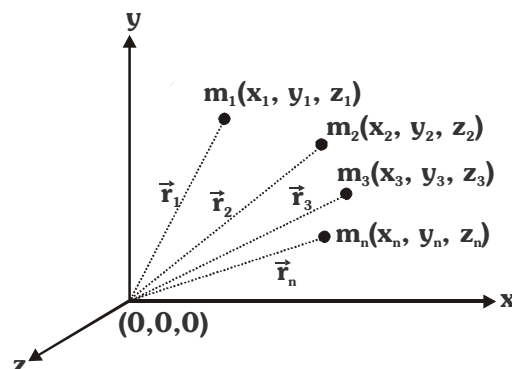
Centre of mass of system of discrete particles

Total mass of the body : $M = m_1 + m_2 + \dots + m_n$

$$\text{Then } \vec{R}_{CM} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \dots}{m_1 + m_2 + m_3 + \dots} = \frac{1}{M} \sum m_i \vec{r}_i$$

co-ordinates of centre of mass :

$$x_{cm} = \frac{1}{M} \sum m_i x_i, y_{cm} = \frac{1}{M} \sum m_i y_i \text{ and } z_{cm} = \frac{1}{M} \sum m_i z_i$$



Centre of mass of continuous distribution of particles

If the system has continuous distribution of mass, treating the mass element dm at position $\vec{r}$ as a point mass and replacing summation by integration. $\vec{R}_{CM} = \frac{1}{M} \int \vec{r} dm$.

$$\text{So that } x_{cm} = \frac{1}{M} \int x dm, y_{cm} = \frac{1}{M} \int y dm \text{ and } z_{cm} = \frac{1}{M} \int z dm$$

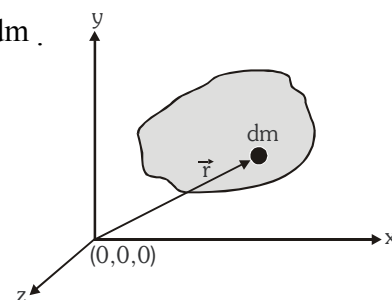
If co-ordinates of particles of mass $m_1, m_2, \dots$ are

$$(x_1, y_1, z_1), (x_2, y_2, z_2), \dots$$

then position vector of their centre of mass is

$$\begin{aligned} \vec{R}_{CM} &= x_{cm} \hat{i} + y_{cm} \hat{j} + z_{cm} \hat{k} \\ &= \frac{m_1(x_1 \hat{i} + y_1 \hat{j} + z_1 \hat{k}) + m_2(x_2 \hat{i} + y_2 \hat{j} + z_2 \hat{k}) + m_3(x_3 \hat{i} + y_3 \hat{j} + z_3 \hat{k}) + \dots}{m_1 + m_2 + m_3 + \dots} \\ &= \frac{(m_1 x_1 + m_2 x_2 + \dots) \hat{i} + (m_1 y_1 + m_2 y_2 + \dots) \hat{j} + (m_1 z_1 + m_2 z_2 + \dots) \hat{k}}{m_1 + m_2 + m_3 + \dots} \end{aligned}$$

$$\text{So, } x_{cm} = \left(\frac{m_1 x_1 + m_2 x_2 + \dots}{m_1 + m_2 + m_3 + \dots} \right), y_{cm} = \left(\frac{m_1 y_1 + m_2 y_2 + \dots}{m_1 + m_2 + \dots} \right), z_{cm} = \left(\frac{m_1 z_1 + m_2 z_2 + \dots}{m_1 + m_2 + \dots} \right)$$



The centre of mass after removal of a part of a body

If a portion of a body is taken out, the remaining portion may be considered as,

$$\text{Original mass (M)} - \text{mass of the removed part (m)} = \{\text{original mass (M)}\} + \{-\text{mass of the removed part (m)}\}$$

$$\text{The formula changes to : } x_{CM} = \frac{Mx - mx'}{M - m}; y_{CM} = \frac{My - my'}{M - m}; z_{CM} = \frac{Mz - mz'}{M - m}$$

Where x', y' and z' represent the coordinates of the centre of mass of the removed part.

MOTION OF CENTRE OF MASS

As for a system of particles, position of centre of mass is $\vec{R}_{CM} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2 + m_3 \vec{r}_3 + \dots}{m_1 + m_2 + m_3 + \dots}$

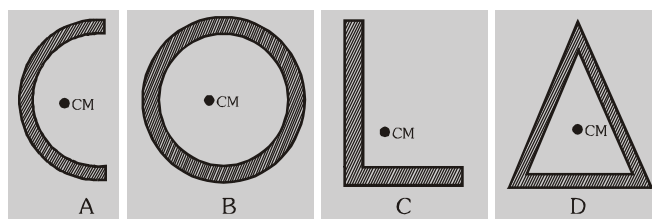
$$\text{So } \frac{d}{dt}(\vec{R}_{CM}) = \frac{m_1 \frac{d\vec{r}_1}{dt} + m_2 \frac{d\vec{r}_2}{dt} + m_3 \frac{d\vec{r}_3}{dt} + \dots}{m_1 + m_2 + m_3 + \dots} \Rightarrow$$

$$\text{Similarly acceleration } \vec{a}_{CM} = \frac{d}{dt}(\vec{v}_{CM}) = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2 + \dots}{m_1 + m_2 + \dots}$$

We can write $M\vec{v}_{CM} = m_1 \vec{v}_1 + m_2 \vec{v}_2 + \dots = \vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots$ [$\because \vec{p} = m\vec{v}$]

$$M\vec{v}_{CM} = \vec{p}_{CM} \quad [\because \Sigma \vec{p}_i = \vec{p}_{CM}]$$

IMPORTANT POINTS



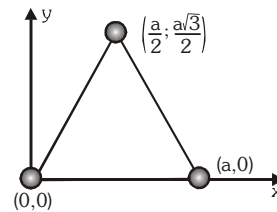
- There may or may not be any mass present physically at centre of mass (See Figure A, B, C)
- Centre of mass may be inside or outside of the body (See figure A, B, C)
- Position of centre of mass depends on the shape of the body. (See figure A, B, C)
- For a given shape it depends on the distribution of mass of within the body and is closer to massive part. (See figure A,C)
- For symmetrical bodies having homogeneous distribution of mass it coincides with centre of symmetry of geometrical centre. (See figure B,D).
- If we know the centre of mass of parts of the system and their masses, we can get the combined centre of mass by treating the parts as point particles placed at their respective centre of masses.
- It is independent of the co-ordinate system, e.g., the centre of mass of a ring is at its centre whatever be the co-ordinate system.
- If the origin of co-ordinate system is at centre of mass, i.e., $\vec{R}_{CM} = 0$, then by definition.

$$\frac{1}{M} \Sigma m_i \vec{r}_i = 0 \Rightarrow \Sigma m_i \vec{r}_i = 0$$

The sum of the moments of the masses of a system about its centre of mass is always zero.

Ex. Three bodies of equal masses are placed at $(0, 0)$, $(a, 0)$ and at $\left(\frac{a}{2}, \frac{a\sqrt{3}}{2}\right)$.

Find out the co-ordinates of centre of mass.

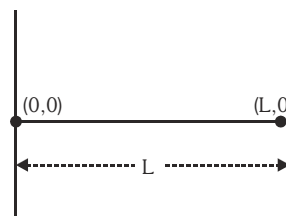


Sol.
$$x_{CM} = \frac{0 \times m + a \times m + \frac{a}{2} \times m}{m + m + m} = \frac{a}{2}, \quad y_{CM} = \frac{0 \times m + 0 \times m + \frac{a\sqrt{3}}{2} \times m}{m + m + m} = \frac{a\sqrt{3}}{6}$$

Ex. Calculate the position of the centre of mass of a system consisting of two particles of masses m_1 and m_2 separated by a distance L apart, from m_1 .

Sol. Treating the line joining the two particles as x axis

$$x_{CM} = \frac{m_1 \times 0 + m_2 \times L}{m_1 + m_2} = \frac{m_2 L}{m_1 + m_2}, \quad y_{CM} = 0 \quad z_{CM} = 0$$

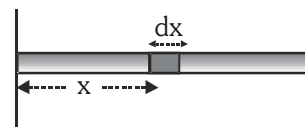


Ex. If the linear density of a rod of length L varies as $\lambda = A + Bx$, compute position of its centre of mass.

Sol. Let the x -axis be along the length of the rod and origin at one of its end as shown in figure. As rod is along x -axis, for all points on it y and z will be zero so, $y_{CM} = 0$ and $z_{CM} = 0$ i.e., centre of mass will be on the rod.

Now consider an element of rod of length dx at a distance x from the origin, mass of this element $dm = \lambda dx = (A + Bx)dx$

$$\text{so, } x_{CM} = \frac{\int_0^L x dm}{\int_0^L dm} = \frac{\int_0^L x(A + Bx)dx}{\int_0^L (A + Bx)dx} = \frac{\frac{AL^2}{2} + \frac{BL^3}{3}}{AL + \frac{BL^2}{2}} = \frac{L(3A + 2BL)}{3(2A + BL)}$$



Note : (i) If the rod is of uniform density then $\lambda = A = \text{constant}$ & $B = 0$ then $X_{CM} = L/2$

(ii) If the density of rod varies linearly with x , then $\lambda = Bx$ and $A = 0$ then $X_{CM} = 2L/3$

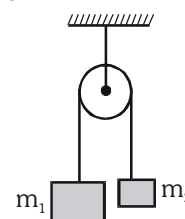
Ex. Two bodies of masses m_1 and m_2 ($< m_1$) are connected to the ends of a massless cord and allowed to move as shown in. The pulley is both massless and frictionless. Calculate the acceleration of the centre of mass.

Sol. If $\vec{a}$ is the acceleration of m_1 , then $-\vec{a}$ is the acceleration of m_2 then

$$\text{acceleration of each body } a = \frac{\text{Net force}}{\text{Net mass}} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) g$$

$$\vec{a}_{cm} = \frac{m_1 \vec{a} + m_2 (-\vec{a})}{m_1 + m_2} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{a}$$

$$\text{But } \vec{a} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) \vec{g} \text{ so } \vec{a}_{cm} = \left(\frac{m_1 - m_2}{m_1 + m_2} \right)^2 \vec{g}$$



Ex. A circle of radius R is cut from a uniform thin sheet of metal. A circular hole of radius $\frac{R}{2}$ is now cut out of the circle, with the hole tangent to the rim. Find the distance of centre of mass from the centre of the original uncut circle to the centre of mass.

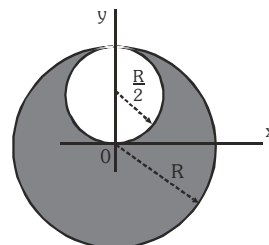
Sol. We treat the hole as a 'negative mass' object that is combined with the original uncut circle. (When the two are added together, the hole region then has zero mass). By symmetry, the CM lies along the $+y$ -axis in figure, so $x_{CM} = 0$. With the origin at the centre of the original circle whose mass is assumed to be m .

Mass of original uncut circle

$$m_1 = m \text{ \& } (0, 0)$$

Mass of hole of negative mass : $m_2 = \frac{m}{4}$; Location of CM $\left(0, \frac{R}{2}\right)$

$$\text{Thus } y_{CM} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2} = \frac{m(0) + \left(-\frac{m}{4}\right)\frac{R}{2}}{m + \left(-\frac{m}{4}\right)} = \frac{R}{6}$$



So the centre of mass is at the point $\left(0, -\frac{R}{6}\right)$

Ex. Two particles of mass 1 kg and 0.5 kg are moving in the same direction with speeds of 2m/s and 6m/s, respectively, on a smooth horizontal surface. Find the speed of centre of mass of the system.

Sol. Velocity of centre of mass of the system $\vec{v}_{cm} = \frac{m_1 \vec{v}_1 + m_2 \vec{v}_2}{m_1 + m_2}$ Since the particles m_1 and m_2 are moving

in same direction, $m_1 \vec{v}_1$ and $m_2 \vec{v}_2$ are parallel. $\Rightarrow |m_1 \vec{v}_1 + m_2 \vec{v}_2| = m_1 v_1 + m_2 v_2$

$$\text{Therefore, } v_{cm} = \frac{|m_1 \vec{v}_1 + m_2 \vec{v}_2|}{m_1 + m_2} = \frac{m_1 v_1 + m_2 v_2}{m_1 + m_2} = \frac{(1)(2) + \left(\frac{1}{2}\right)(6)}{\left(1 + \frac{1}{2}\right)} = 3.33 \text{ ms}^{-1}$$

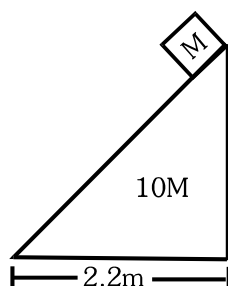
Ex. Two particles of masses 2 kg and 4 kg are approaching towards each other with acceleration 1 m/s^2 and 2 m/s^2 , respectively, on a smooth horizontal surface. Find the acceleration of centre of mass of the system.

Sol. The acceleration of centre of mass of the system $\vec{a}_{cm} = \frac{m_1 \vec{a}_1 + m_2 \vec{a}_2}{m_1 + m_2} \Rightarrow a_{cm} = \frac{|m_1 \vec{a}_1 + m_2 \vec{a}_2|}{m_1 + m_2}$

$$\text{Since } \vec{a}_1 \text{ and } \vec{a}_2 \text{ are anti-parallel, so } a_{cm} = \frac{|m_1 a_1 - m_2 a_2|}{m_1 + m_2} = \frac{|(2)(1) - (4)(2)|}{2 + 4} = 1 \text{ ms}^{-2}$$

Since $m_2 a_2 > m_1 a_1$ so the direction of acceleration of centre of mass will be directed in the direction of a_2 .

Ex. A block of mass M is placed on the top of a bigger block of mass $10M$ as shown in figure. All the surfaces are frictionless. The system is released from rest. Find the distance moved by the bigger block at the instant the smaller block reaches the ground.



Sol. If the bigger block moves towards right by a distance (X) , the smaller block will move towards left by a distance $(2.2 - X)$ (taking the two blocks together as the system). The horizontal position of CM remains same $\Rightarrow M(2.2 - X) = 10MX \Rightarrow X = 0.2 \text{ m}$.

MOMENTUM

The total quantity of motion possessed by a moving body is known as the momentum of the body. It is the product of the mass and velocity of a body i.e. momentum $\vec{p} = m\vec{v}$

IMPULSE

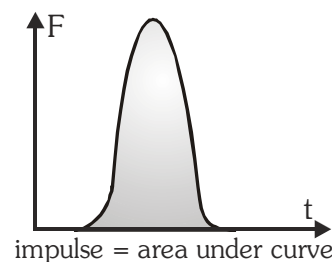
When a large force act for an extremely short duration, neither the magnitude of the force nor the time for which it acts is important. In such a case, the total effect of force is measured. The total effect of force is called impulse (measure of the action of force). This type of force is generally variable in magnitude and is sometimes called impulsive force.

If a large force act on a body or particle for a small time then

Impulse = product of force with time.

Suppose a force $\vec{F}$ acts for a short time dt then impulse = $\vec{F}dt$

For a finite interval of time from t_1 to t_2 then the impulse = $\int_{t_1}^{t_2} \vec{F}dt$

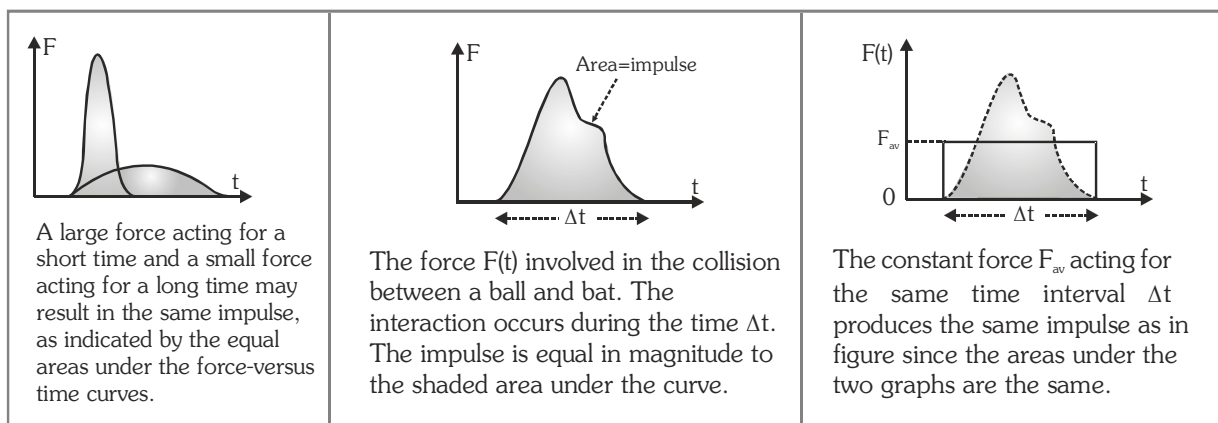


If constant force $\vec{F}$ acts for an interval Δt then Impulse = $\vec{F}\Delta t$

Impulse – Momentum theorem :

Impulse of a force is equal to the change of momentum

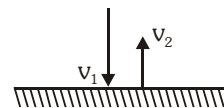
$$\vec{F}\Delta t = \Delta\vec{p}$$



Ex. A ball of mass 50 g is dropped from a height $h = 10$ m. It rebounds losing 75 percent of its total mechanical energy. If it remains in contact with the ground for 0.01 s, find the impulse of the impact force.

Sol. Impulse = change in momentum = $m(v_1 + v_2)$

Here $v_1 = \sqrt{2gh}$ and for v_2 , $\frac{1}{2}mv_2^2 = \frac{1}{2}mv_1^2 \left(1 - \frac{75}{100}\right) \Rightarrow v_2 = \frac{v_1}{2}$



So impulse = $m \left(v_1 + \frac{v_1}{2} \right) = \frac{3mv_1}{2} = \frac{3}{2} m \times \sqrt{2gh} = \frac{3}{2} \times 50 \times 10^{-3} \times \sqrt{2 \times 9.8 \times 10} = 1.05 \text{ N-s}$

LAW OF CONSERVATION OF LINEAR MOMENTUM

According to Newton's Second law of motion the rate of change of momentum is equal to the applied force.

$$\vec{F} = \frac{d\vec{p}}{dt} \quad \text{if } \vec{F} = \vec{0} \text{ then } \frac{d\vec{p}}{dt} = \vec{0} \text{ i.e. } \vec{p} = \text{constant}$$

This leads to the law of conservation of momentum which is "In the absence of external forces, the total momentum of the system is conserved."

IMPORTANT POINTS

- For an isolated system, the initial momentum of the system is equal to the final momentum of the system. If the system consists of n bodies having momentum $\vec{p}_1, \vec{p}_2, \vec{p}_3, \dots, \vec{p}_n$, then $\vec{p}_1 + \vec{p}_2 + \vec{p}_3 + \dots + \vec{p}_n = \text{constant}$
- As linear momentum depends on frame of reference. Observers in different frames would find different values of linear momentum of a given system but each would agree that his own value of linear momentum does not change with time provided. But the system should be isolated and closed, i.e., law of conservation of linear momentum is independent of frame of reference though linear momentum depends on frame of reference.
- Conservation of linear momentum is equivalent to Newton's III law of motion for a system of two particles in absence of external force by law of conservation of linear momentum.

$$\Rightarrow \vec{p}_1 + \vec{p}_2 = \text{constant} \quad \text{i.e. } m_1\vec{v}_1 + m_2\vec{v}_2 = \text{constant}$$

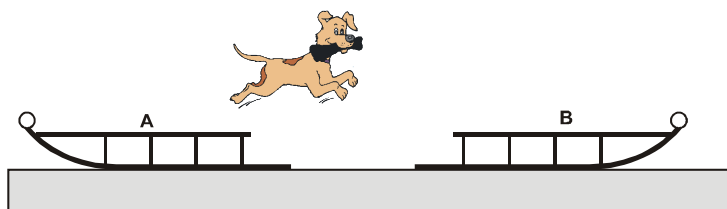
Differentiating above with respect to time $m_1 \frac{d\vec{v}_1}{dt} + m_2 \frac{d\vec{v}_2}{dt} = \vec{0}$ [as m is constant]

$$\Rightarrow m_1\vec{a}_1 + m_2\vec{a}_2 = \vec{0} \quad [\because \frac{d\vec{v}}{dt} = \vec{a}] \Rightarrow \vec{F}_1 + \vec{F}_2 = \vec{0} \quad [\because \vec{F} = m\vec{a}] \Rightarrow \vec{F}_1 = -\vec{F}_2$$

i.e., for every action there is equal and opposite reaction which is Newton's III law of motion.

- This law is universal, i.e., it applies to body macroscopic as well as microscopic systems.

Ex. Two 22.7 kg ice sleds A and B are placed a short distance apart, one directly behind the other, as shown in figure. A 3.63 kg dog, standing on one sled, jumps across to the other and immediately back to the first. Both jumps are made at a speed of 3.05 ms^{-1} relative to the ice. Find the final speeds of the two sleds.



Sol. Total momentum imparted to B $p_B = 2 \times 3.63 \times 3.05 \text{ kg ms}^{-1}$.

$$\text{Velocity of B} = \frac{p_B}{m_B} = \frac{2 \times 3.63 \times 3.05}{22.7} = 0.975 \text{ ms}^{-1}.$$

$$\text{Velocity of A when the dog jumps away from A} = \frac{p_A}{m_A} = \frac{3.63 \times 3.05}{22.7} = 0.4877 \text{ ms}^{-1}.$$

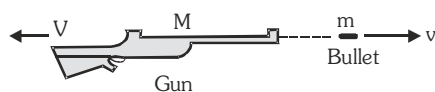
$$\text{When the dog comes back to A, Velocity of A} = \frac{22.7 \times 0.4877 + 3.63 \times 3.05}{22.7 + 3.63} = 0.841 \text{ ms}^{-1}.$$

APPLICATIONS OF CONSERVATION OF LINEAR MOMENTUM

Firing a Bullet from a Gun :

- If the bullet is the system, the force exerted by trigger will be external and so the linear momentum of the bullet will change from 0 to mv . This is not the violation of law of conservation of linear momentum as linear momentum is conserved only in absence of external force.
- If the bullet and gun is the system, the force exerted by trigger will be internal so.

Total momentum of the system $\vec{p}_s = \vec{p}_B + \vec{p}_G = \text{constant}.$



Now as initially both bullet and gun are at rest so $\vec{p}_B + \vec{p}_G = \vec{0}$ From this it is evident that :

- $\vec{p}_G = -\vec{p}_B$, i.e., if bullet acquires forward momentum, the gun will acquire equal and opposite (backward) momentum.
- As $\vec{p} = m\vec{v}$, $m\vec{v} + M\vec{V} = \vec{0}$, i.e., $\vec{V} = -\frac{m}{M}\vec{v}$ i.e, if the bullet moves forward, gun 'recoils' or 'kicks' backward. Heavier the gun lesser will be the recoil velocity V .
- Kinetic energy $K = \frac{p^2}{2m}$ and $|\vec{p}_B| = |\vec{p}_G| = p$ Kinetic energy of gun $K_G = \frac{p^2}{2M}$,

Kinetic energy of bullet $K_B = \frac{p^2}{2m}$ $\therefore \frac{K_G}{K_B} = \frac{m}{M} < 1$ ($\because M \gg m$) Thus kinetic energy of gun is smaller than bullet i.e., kinetic energy of bullet and gun will not be equal.

- Initial kinetic energy of the system is zero as both are at rest initially.
Final kinetic energy of the system $[(1/2)(mv^2 + MV^2)] > 0$.
So, here kinetic energy of the system is not constant but increases. If PE is assumed to be constant then Mechanical energy = (kinetic energy + potential energy) will also increase. However, energy is always conserved. Here chemical energy of gun powder is converted into KE.

Ex. A bullet of mass 100g is fired by a gun of 10kg with a speed 2000 m/s. Find recoil velocity of gun.

Sol. According to conservation of momentum $mv + MV = 0$.

$$\text{Velocity of gun } V = -\frac{mv}{M} = -\frac{0.1 \times 2000}{10} = -20 \text{ m/s}$$

Block Bullet System :**(a) When bullet remains in the block**

Conserving momentum of bullet and block $mv + 0 = (M+m) V$

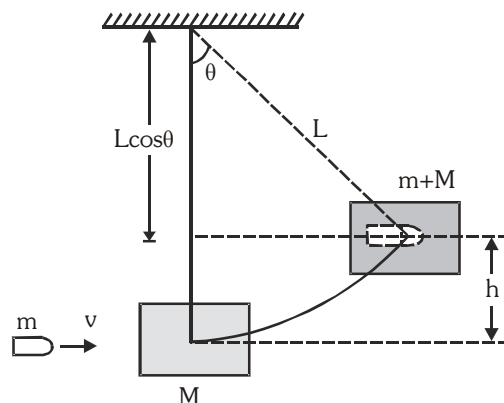
$$\text{Velocity of block } V = \frac{mv}{M+m} \dots(i)$$

By conservation of mechanical energy

$$\frac{1}{2}(M+m)V^2 = (M+m)gh \Rightarrow V = \sqrt{2gh} \dots(ii)$$

$$\text{From eqn. (i) and eqn. (ii) } \frac{mv}{M+m} = \sqrt{2gh};$$

$$\text{Speed of bullet } v = \frac{(M+m)\sqrt{2gh}}{m},$$



$$\text{Maximum height gained by block } h = \frac{V^2}{2g} = \frac{m^2 v^2}{2g(M+m)^2}$$

$$h = L - L \cos \theta \quad \therefore \cos \theta = 1 - \frac{h}{L} \Rightarrow \theta = \cos^{-1} \left(1 - \frac{h}{L} \right)$$

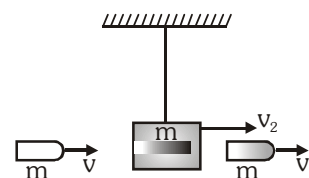
(b) If bullet moves out of the block

Conserving momentum $mv + 0 = mv_1 + Mv_2$

$$m(v - v_1) = Mv_2 \dots(i)$$

$$\text{Conserving energy } \frac{1}{2}Mv_2^2 = Mgh \Rightarrow v_2 = \sqrt{2gh} \dots(ii)$$

$$\text{From eqn. (i) \& eqn. (ii) } m(v - v_1) = M\sqrt{2gh} \Rightarrow h = \frac{m^2(v - v_1)^2}{2gM^2}$$

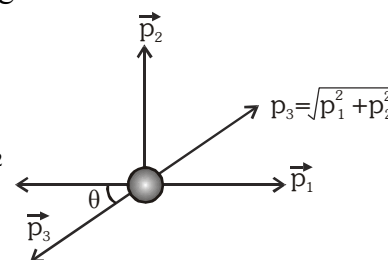
**Explosion of a Bomb at rest**

Conserving momentum

$$\vec{p}_1 + \vec{p}_2 + \vec{p}_3 = \vec{0} \Rightarrow \vec{p}_3 = -(\vec{p}_1 + \vec{p}_2) \Rightarrow p_3 = \sqrt{p_1^2 + p_2^2} \text{ as } \vec{p}_1 \perp \vec{p}_2$$

Angle made by $\vec{p}_3$ from $\vec{p}_1 = \pi + \theta$

$$\text{Angle made by } \vec{p}_3 \text{ from } \vec{p}_2 = \frac{\pi}{2} + \theta$$



$$\text{Energy released in explosion} = K_f - K_i = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + \frac{p_3^2}{2m_3} - 0 = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + \frac{p_3^2}{2m_3}$$

Motion of Two Masses Connected to a Spring

Consider two blocks, resting on a frictionless surface and connected by a massless spring as shown in figure. If the spring is stretched (or compressed) and then released from rest,

Then $F_{\text{ext}} = 0$ so $\vec{p}_s = \vec{p}_1 + \vec{p}_2 = \text{constant}$

However, initially both the blocks were at rest so, $\vec{p}_1 + \vec{p}_2 = \vec{0}$

It is clear that :



- $\vec{p}_2 = -\vec{p}_1$, i.e., at any instant the two blocks will have momentum equal in magnitude but opposite in direction (Though they have different values of momentum at different positions).
- As momentum $\vec{p} = m\vec{v}$, $m_1\vec{v}_1 + m_2\vec{v}_2 = \vec{0} \Rightarrow \vec{v}_2 = -\left(\frac{m_1}{m_2}\right)\vec{v}_1$

The two blocks always move in opposite directions with lighter block moving faster.

- Kinetic energy $KE = \frac{p^2}{2m}$ and $|\vec{p}_1| = |\vec{p}_2|$, $\frac{KE_1}{KE_2} = \frac{m_2}{m_1}$ or the kinetic energy of two blocks will not be equal but in the inverse ratio of their masses and so lighter block will have greater kinetic energy.
- Initially kinetic energy of the blocks is zero (as both are at rest) but after some time kinetic energy of the blocks is not zero (as both are in motion). So, kinetic energy is not constant but changes. Here during motion of blocks KE is converted into elastic potential energy of the spring and vice-versa but total mechanical energy of the system remain constant.

$$\text{Kinetic energy} + \text{Potential energy} = \text{Mechanical Energy} = \text{Constant}$$

Note – If $\vec{F}$ is the average of the time varying force during collision and Δt is the duration of collision then impulse $\vec{I} = \vec{F}\Delta t$.

Conservation of Linear Momentum During Impact :

If two bodies of masses m_1 and m_2 collide in air, the total external force acting on the system

of bodies ($m_1 + m_2$) is equal to $\vec{F}_1 + m_1\vec{g} + \vec{F}_2 + m_2\vec{g} \Rightarrow F_{\text{total}} = m_1\vec{g} + m_2\vec{g} + \vec{F}_1 + \vec{F}_2$

During collision the impact forces $\vec{F}_1$ and $\vec{F}_2$ are equal in magnitude and opposite in direction.

According to Newton's 3rd law of motion, $\vec{F}_1 + \vec{F}_2 = \vec{0} \Rightarrow \vec{F}_{\text{net}} = m_1\vec{g} + m_2\vec{g}$

So Impulse = $\vec{F}_{\text{net}}\Delta t = (m_1\vec{g} + m_2\vec{g})\Delta t$

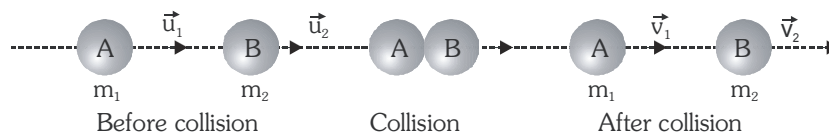
Since Δt is a very small time interval, the impulse $F(\Delta t)$ will be negligibly small. As impulse is equal to change in momentum of the system, a negligible impulse means negligible change of momentum. Let the change of momentum of 1 & 2 be $\Delta\vec{p}_1$ & $\Delta\vec{p}_2$, respectively then the total change in momentum of the system $\Rightarrow \Delta\vec{p} = \Delta\vec{p}_1 + \Delta\vec{p}_2 = \vec{F}_{\text{net}}.dt \approx \vec{0} \Rightarrow \Delta(\vec{p}_1 + \vec{p}_2) = 0 \Rightarrow \vec{p}_1 + \vec{p}_2 = \text{constant}$.

Therefore, the net or total momentum of the colliding bodies remains practically unchanged along the line of action (impact) during the collision. In other words, the momentum of the system remains constant or conserved during the period of impact. Therefore, we can conveniently equate the net momentum of the colliding bodies at the beginning and at the end of the collision (or just before and just after the impact).

Note : Remember that the impact force F is not an external force for the system of colliding bodies. If no external force acts on the system, its momentum remains constant for all the times including the time of collision. Even if some external forces like gravitation and friction (known as non-impulsive forces in general) are present, we can conserve the momentum of the system during the impact, because the finite external forces cannot change the momentum of the system significantly in very short time. Therefore, the change in position of the system during infinitesimal time of impact can also be neglected.

HEAD ON ELASTIC COLLISION

The elastic collision in which the colliding bodies move along the same straight line path before and after the collision.



Assuming initial direction of motion to be positive and $u_1 > u_2$ (so that collision may take place) and applying law of conservation of linear momentum

$$m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \Rightarrow m_1 (u_1 - v_1) = m_2 (v_2 - u_2) \quad \dots (i)$$

For elastic collision, kinetic energy before collision must be equal to kinetic energy after collision, i.e.,

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 \Rightarrow m_1 (u_1^2 - v_1^2) = m_2 (v_2^2 - u_2^2) \quad \dots (ii)$$

$$\text{Dividing equation (ii) by (i)} \quad u_1 + v_1 = v_2 + u_2 \Rightarrow (u_1 - u_2) = (v_2 - v_1) \quad \dots (iii)$$

In 1-D elastic collision 'velocity of approach' before collision is equal to the 'velocity of recession' after collision, no matter what the masses of the colliding particles be.

This law is called **Newton's law for elastic collision**

Now if we multiply equation (iii) by m_2 and subtracting it from (i)

$$(m_1 - m_2) u_1 + 2m_2 u_2 = (m_1 + m_2) v_1 \Rightarrow v_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 + \frac{2m_2}{m_1 + m_2} u_2 \quad \dots (iv)$$

Similarly multiplying equation (iii) by m_1 and adding it to equation (i)

$$2m_1 u_1 + (m_2 - m_1) u_2 = (m_2 + m_1) v_2 \Rightarrow v_2 = \frac{2m_1}{m_1 + m_2} u_1 + \frac{m_2 - m_1}{m_1 + m_2} u_2 \quad \dots (v)$$

IMPORTANT POINTS

- If the two bodies are of equal masses :** $m_1 = m_2 = m$, $v_1 = u_2$ and $v_2 = u_1$
Thus, if two bodies of equal masses undergo elastic collision in one dimension, then after the collision, the bodies will exchange their velocities.
- If two bodies are of equal masses and second body is at rest.**
 $m_1 = m_2$ and initial velocity of second body $u_2 = 0$, $v_1 = 0$, $v_2 = u_1$
When body A collides against body B of equal mass at rest, the body A comes to rest and the body B moves on with the velocity of the body A. In this case transfer of energy is hundred percent e.g., Billiard's Ball, Nuclear moderation.
- If the mass of a body is negligible as compared to other.**
If $m_1 \gg m_2$ and $u_2 = 0$ then $v_1 = u_1$, $v_2 = 2u_1$
When a heavy body A collides against a light body B at rest, the body A should keep on moving with same velocity and the body B will move with velocity double that of A.
If $m_2 \gg m_1$ and $u_2 = 0$ then $v_2 = 0$, $v_1 = -u_1$
When light body A collides against a heavy body B at rest, the body A should start moving with same velocity just in opposite direction while the body B should practically remains at rest.

Ex. Two ball of mass 5kg each is moving in opposite directions with equal speed 5m/s. collides head on with each other. Find out the final velocities of the balls if collision is elastic.

Sol. Here $m_1 = m_2 = 5\text{kg}$, $u_1 = 5\text{ m/s}$, $u_2 = -5\text{ m/s}$

In such type of condition velocity get interchange so $v_2 = u_1 = 5\text{ m/s}$ & $v_1 = u_2 = -5\text{ m/s}$

Ex. A ball of 0.1 kg makes an elastic head on collision with a ball of unknown mass that is initially at rest. If the 0.1 kg ball rebounds at one third of its original speed. What is the mass of other ball ?

Sol. Here $m_1 = 0.1$ kg, $m_2 = ?$, $u_2 = 0$, $u_1 = u$, $v_1 = -u/3$

$$\text{As } v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2} \right) u + \frac{2m_2 u_2}{m_1 + m_2} \Rightarrow -\frac{u}{3} = \left(\frac{0.1 - m_2}{0.1 + m_2} \right) u \Rightarrow m_2 = 0.2 \text{ kg}$$

HEAD ON INELASTIC COLLISION OF TWO PARTICLES

Let the coefficient of restitution for collision is e

(i) Momentum is conserved $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \dots (i)$

(ii) Kinetic energy is not conserved.

(iii) According to Newton's law $\frac{v_2 - v_1}{u_2 - u_1} = -e \dots (ii)$

By solving eq. (i) and (ii)

$$v_1 = \left(\frac{m_1 - em_2}{m_1 + m_2} \right) u_1 + \left(\frac{(1+e)m_2}{m_1 + m_2} \right) u_2, \quad v_2 = \left(\frac{m_2 - em_1}{m_1 + m_2} \right) u_2 + \left(\frac{(1+e)m_1}{m_1 + m_2} \right) u_1$$

PERFECT INELASTIC COLLISION

In case of inelastic collision, after collision two bodies move with same velocity (or stick together).

If two particles of masses m_1 and m_2 , moving with velocity u_1 and u_2 ($u_2 < u_1$) respectively along the same line collide 'head on' and after collision they have same common velocity v , then by conservation of linear momentum,

$$m_1 u_1 + m_2 u_2 = m_1 v + m_2 v \Rightarrow v = \frac{m_1 u_1 + m_2 u_2}{(m_1 + m_2)} \dots (i)$$

Kinetic energy of the system before collision is $KE_i = \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$

And after collision is $KE_f = \frac{1}{2} (m_1 + m_2) v^2$

Loss in KE during collision

$$\Delta KE = KE_i - KE_f = \left[\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 \right] - \frac{1}{2} (m_1 + m_2) v^2 \dots (ii)$$

Substituting the value of v from eq. (i),

$$\Delta KE = \frac{1}{2} \left[(m_1 u_1^2 + m_2 u_2^2) - \frac{(m_1 u_1 + m_2 u_2)^2}{(m_1 + m_2)} \right]$$

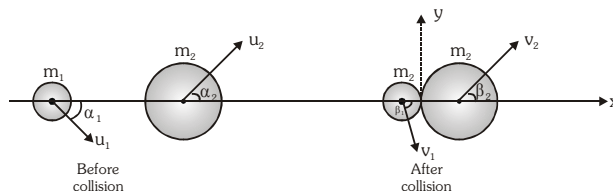
$$\Rightarrow \Delta KE = \frac{1}{2} \left[\frac{m_1 m_2 (u_1^2 + u_2^2 - 2u_1 u_2)}{(m_1 + m_2)} \right] \Rightarrow \boxed{\Delta KE = \frac{1}{2} \frac{m_1 m_2}{(m_1 + m_2)} (u_1 - u_2)^2}$$

If the target is initially at rest $u_2 = 0$ and $u_1 = u$

$$\Delta KE = \frac{m_1 m_2}{2(m_1 + m_2)} u^2, \quad \frac{\Delta KE}{KE_i} = \frac{m_2}{(m_1 + m_2)} \left[\because KE_i = \frac{1}{2} m_1 u^2 \right]$$

Now if target is massive, i.e., $m_2 \gg m_1$ then $\frac{\Delta KE}{KE_i} \approx 1$ so percentage loss in KE = 100%

i.e., if a light moving body strikes a heavy target at rest and sticks to it, practically all its KE is lost.



Change in Kinetic energy: $\Delta K.E. = \frac{1}{2} mv^2 - \frac{1}{2} mu^2$

Putting the value of v we obtain

$$\begin{aligned}\Delta KE &= \frac{1}{2} m \left[\left[\sqrt{u^2 (\sin^2 \theta + e^2 \cos^2 \theta)} \right]^2 - u^2 \right] = \frac{1}{2} mu^2 [\sin^2 \theta + e^2 \cos^2 \theta - 1] \\ &= -\frac{1}{2} mu^2 [\cos^2 \theta - e^2 \cos^2 \theta] = -\frac{1}{2} (1 - e^2) mu^2 \cos^2 \theta\end{aligned}$$

Negative sign indicates the loss of kinetic energy

IMPORTANT POINTS

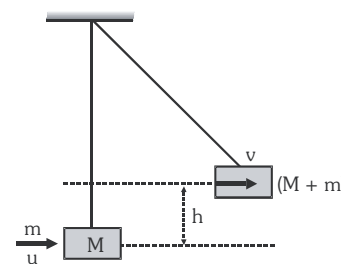
- Momentum remains conserved in all types of collisions.
- Total energy remains conserved in all types of collisions.
- Only conservative forces work in elastic collisions.
- In inelastic collisions all the forces are not conservative.

Ex. A simple pendulum of length 1m has a wooden bob of mass 1kg. It is struck by a bullet of mass 10^{-2} kg moving with a speed of 2×10^2 m/s. The bullet gets embedded into the bob. Obtain the height to which the bob rises before swinging back.

Sol. Applying principle of conservation of linear momentum

$$mu = (M + m)v \Rightarrow 10^{-2} \times (2 \times 10^2) = (1 + .01)v \Rightarrow v = \frac{2}{1.01}$$

KE_i of the block with bullet in it, is converted into P.E. as it rises through a height h



$$\frac{1}{2}(M + m)v^2 = (M + m)gh \Rightarrow v^2 = 2gh \Rightarrow h = \frac{v^2}{2g} = \left(\frac{2}{1.01} \right)^2 \times \frac{1}{2 \times 9.8} = 0.2 \text{ m}$$

Ex. A body falling on the ground from a height of 10m, rebounds to a height 2.5m calculate

(i) The percentage loss in K.E.

(ii) Ratio of the velocities of the body just before and just after the collision.

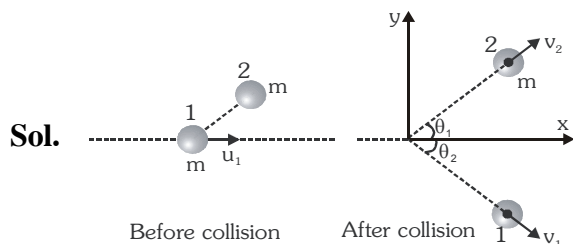
Sol. Let v_1 and v_2 be the velocity of the body just before and just after the collision

$$KE_1 = \frac{1}{2} mv_1^2 = mgh_1 \dots (i) \text{ and } KE_2 = \frac{1}{2} mv_2^2 = mgh_2 \dots (ii)$$

$$\Rightarrow \frac{v_1^2}{v_2^2} = \frac{h_1}{h_2} = \frac{10}{2.5} = 4 \Rightarrow \frac{v_1}{v_2} = 2$$

$$\text{Percentage loss in KE} = \frac{mg(h_1 - h_2)}{mgh_1} \times 100 = \frac{10 - 2.5}{10} \times 100 = 75\%$$

Ex. A body strikes obliquely with another identical stationary rest body elastically. Prove that they will move perpendicular to each other after collision.



Conservation of linear momentum in x-direction gives

$$mu_1 = mv_1 \cos \theta_1 + mv_2 \cos \theta_2 \Rightarrow u_1 = v_1 \cos \theta_1 + v_2 \cos \theta_2 \quad \dots (i)$$

Conservation of linear momentum in y-direction gives

$$0 = mv_1 \sin \theta_1 - mv_2 \sin \theta_2 \Rightarrow 0 = v_1 \sin \theta_1 - v_2 \sin \theta_2 \quad \dots (ii)$$

Conservation of kinetic energy

$$\frac{1}{2} mu_1^2 = \frac{1}{2} mv_1^2 + \frac{1}{2} mv_2^2 \Rightarrow u_1^2 = v_1^2 + v_2^2 \quad \dots (iii)$$

$$(i)^2 + (ii)^2$$

$$\Rightarrow u_1^2 + 0 = v_1^2 \cos^2 \theta_1 + v_2^2 \cos^2 \theta_2 + 2v_1 v_2 \cos \theta_1 \cos \theta_2 + v_1^2 \sin^2 \theta_1 + v_2^2 \sin^2 \theta_2 - 2v_1 v_2 \sin \theta_1 \sin \theta_2$$

$$\Rightarrow u_1^2 = v_1^2 (\cos^2 \theta_1 + \sin^2 \theta_1) + v_2^2 (\cos^2 \theta_2 + \sin^2 \theta_2) + 2v_1 v_2 (\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2)$$

$$\Rightarrow u_1^2 = v_1^2 + v_2^2 + 2v_1 v_2 \cos (\theta_1 + \theta_2) \quad \{ \because u_1^2 = v_1^2 + v_2^2 \}$$

$$\Rightarrow \cos (\theta_1 + \theta_2) = 0 \Rightarrow \theta_1 + \theta_2 = 90^\circ$$

Ex. A steel ball is dropped on a smooth horizontal plane from certain height h . Assuming coefficient of restitution of impact as e , find the average speed of the ball till it stops.

Sol. Since the ball falls through a height h , just before the first impact its speed v will be given as

$v = \sqrt{2gh}$. Let its speed be v_1 just after the first impact. Then, Newton's experimental formula yields,

$$\frac{0 - v_1}{v} = e \Rightarrow v_1 = ev$$

Similarly, its speed just before 2nd impact, $v_1 = ev = e\sqrt{2gh}$

Speed just after n^{th} impact, $v_n = e^n v = e^n \sqrt{2gh}$

The maximum height attained after 1st impact $= h_1 = \frac{v_1^2}{2g} = (e\sqrt{2gh})^2 = e^2 h$. Similarly, the maximum

height attained after 2nd impact, $h_2 = e^4 h$. Hence, the maximum height attained after n^{th} impact $= e^{2n} h$

The ball experiences infinite impacts till it becomes stationary. $\Rightarrow$ The total distance covered,

$$d = h + 2h_1 + 2h_2 + \dots = h + 2e^2 h + 2e^4 h + \dots = h[1 + 2(e^2 + e^4 + e^6 + \dots)]$$

$$= \left[1 + 2 \left(\frac{e^2}{1 - e^2} \right) \right] h = \left(\frac{1 + e^2}{1 - e^2} \right) h.$$

The total time taken by the ball till it stops bouncing $T = \sqrt{\frac{2h}{g}} + 2\sqrt{\frac{2h_1}{g}} + 2\sqrt{\frac{2h_2}{g}} + \dots$

Putting $h_1 = e^2h$, $h_2 = e^4h$, $T = \sqrt{\frac{2h}{g}} + 2\sqrt{\frac{2e^2h}{g}} + 2\sqrt{\frac{2e^4h}{g}} + \dots$

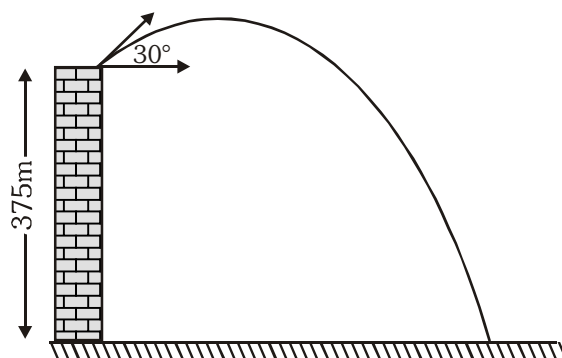
$$\Rightarrow T = \sqrt{\frac{2h}{g}} [1 + 2(e + e^2 + \dots)] = \sqrt{\frac{2h}{g}} \left[1 + \frac{2e}{1-e}\right] = \frac{1+e}{1-e} \sqrt{\frac{2h}{g}}$$

Therefore, average speed of the ball for its total time of motion, $\vec{v} = \frac{\text{total distance}}{\text{total time}} = \frac{d}{T}$

Putting the values of d and T , we obtain $\vec{v} = \frac{1+e^2}{(1+e)^2} \sqrt{\frac{gh}{2}}$

Ex. A particle of mass 1 kg is projected from a tower of height 375m with initial velocity 100 ms^{-1} at an angle 30° with the horizontal. Find out its kinetic energy in joule just after collision with ground if

collision is inelastic with $e = \frac{1}{2}$ ($g = 10 \text{ ms}^{-2}$)



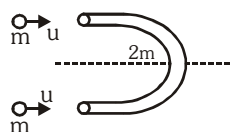
Sol. $v_y^2 = u_y^2 + 2gh \Rightarrow v_y = \sqrt{(50)^2 + 2 \times 10 \times 375} = 100 \text{ ms}^{-1}$

Horizontal velocity just after collision = $50\sqrt{3} \text{ ms}^{-1}$

Vertical velocity just after collision = $100 \times \frac{1}{2} = 50 \text{ ms}^{-1}$

Kinetic energy just after collision = $\frac{1}{2} \times 1 \times [(50\sqrt{3})^2 + (50)^2] = 5000 \text{ J}$

Ex A U shaped tube of mass $2m$ is placed on a horizontal surface. Two spheres each of diameter d (just less than the inner diameter of tube) and mass m enter into the tube with a velocity u as shown in figure. Taking all collisions to be elastic and all surfaces smooth. Match the following-



Column-I

- (A) The speed of the tube with respect to ground, when spheres are just about to collide inside the tube.
- (B) The speed of spheres when spheres are just about to collide.
- (C) The speed of the spheres when they come out the tube.
- (D) The speed of the tube when spheres come out the

Column-II

- (p) u
- (q) $u/2$
- (r) $\frac{\sqrt{3}}{2}u$
- (s) zero

Sol. For (A) From conservation of linear momentum $2mu = (m + m)v \Rightarrow v = \frac{u}{2}$

For (B) Let v_1 be the velocity of spheres w.r.t. tube when they are just about to collide then by using

conservation of kinetic energy $\frac{1}{2} (2m)u^2 = \frac{1}{2} (4m) \left(\frac{u}{2}\right)^2 + 2 \frac{1}{2} mv_1^2$

$$\Rightarrow v_1 = \frac{u}{\sqrt{2}}$$

$$\Rightarrow \text{Required speed of spheres} = \sqrt{\left(\frac{u}{2}\right)^2 + \left(\frac{u}{\sqrt{2}}\right)^2} = \sqrt{\frac{u^2}{4} + \frac{u^2}{2}} = \frac{\sqrt{3}u}{2}$$

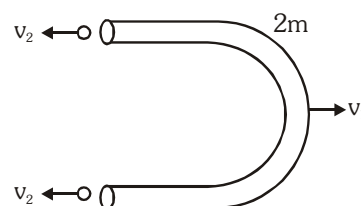
For (C) $2mu = 2mv_1 - 2mv_2$

$$2 \times \frac{1}{2} mu^2 = 2 \times \frac{1}{2} mv_1^2 + \frac{1}{2} (2m) v_1^2$$

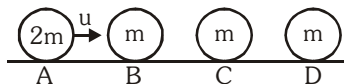
$$\Rightarrow u^2 = v_1^2 + v_2^2 - 2v_1v_2 \text{ \& } u^2 = v_1^2 + v_2^2$$

$$\Rightarrow v_1v_2 = 0 \text{ but } v_1 \neq 0 \text{ so } v_2 = 0$$

For (D) Speed of tube $v_1 = u$



Ex. Four balls A, B, C and D are kept on a smooth horizontal surface as shown in figure. Ball A is given velocity u towards B-

**Column-I**

(A) Total impulse of all collisions on A

(B) Total impulse of all collisions on B

(C) Total impulse of all collision on C

(D) Total impulse of all collisions on D

Column-II

(p) $\frac{4mu}{9}$

(q) $\frac{4mu}{27}$

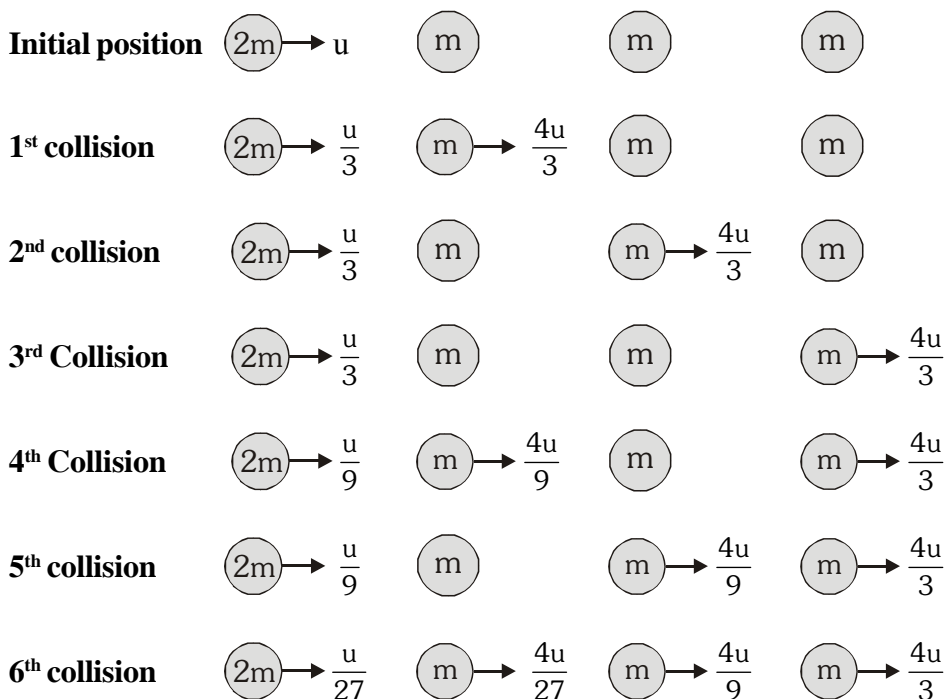
(r) $\frac{4mu}{3}$

(s) $\frac{52}{27}mu$

Sol. In 1st collision between A & B

$$2mu = 2mv_A + 2mv_B \text{ \& } e = 1 = \frac{v_B - v_A}{u} \Rightarrow v_A = \frac{u}{3}, v_B = \frac{4u}{3}$$

Situation of all collisions is shown in figure.



For (A) Total impulse on A = $2m \left(u - \frac{u}{27} \right) = \frac{52}{27}mu$

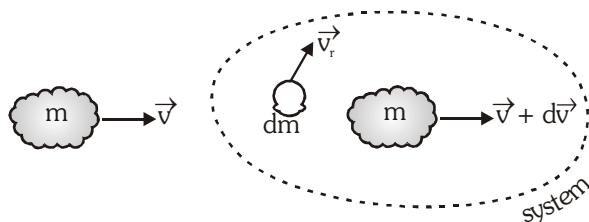
For (B) Total impulse on B = $m \left(\frac{4u}{27} - 0 \right) = \frac{4}{27} mu$

For (C) Total impulse on C = $m \left(\frac{4u}{9} - 0 \right) = \frac{4}{9} mu$

For (D) Total impulse on D = $m \left(\frac{4u}{3} - 0 \right) = \frac{4}{3} mu$

Variable mass system:

In previous discussion of the conservation of linear momentum, we assume that system's mass remains constant. Now we consider those system whose mass is variable i.e. those in which mass enters or leaves the system. Suppose at some moment $t = t$ mass of a body is m and its velocity is $\vec{v}$. After some time at $t = t + dt$ its mass becomes $(m - dm)$ and velocity becomes $\vec{v} + d\vec{v}$. The mass dm is ejected with relative velocity $\vec{v}_r$.



If no forces are acting on the system then the linear momentum of the system will remain conserved.

$$\Rightarrow \vec{F}_{\text{ex}} dt = (m - dm)(\vec{v} + d\vec{v}) + dm(\vec{v}_r + \vec{v} + d\vec{v}) - m\vec{v}$$

$$\therefore F_{\text{ex}} = 0 \Rightarrow m d\vec{v} = -\vec{v}_r dm \Rightarrow m \left(\frac{d\vec{v}}{dt} \right) = \vec{v}_r \left(-\frac{dm}{dt} \right)$$

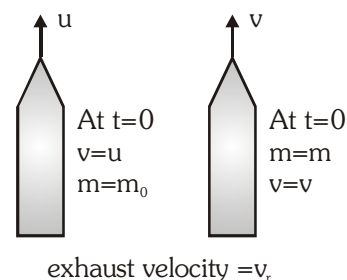
Rocket propulsion :

Thrust force on the rocket = $v_r \left(-\frac{dm}{dt} \right)$

So for motion of rocket $m \frac{dv}{dt} = v_r \left(-\frac{dm}{dt} \right) - mg$

$$\Rightarrow dv = v_r \left(-\frac{dm}{m} \right) - g dt \Rightarrow \int_{u_0}^v dv = -v_r \int_{m_0}^m \frac{dm}{m} - g \int_0^t dt$$

$$\Rightarrow v - u = v_r \ln \left(\frac{m_0}{m} \right) - gt \Rightarrow v = u - gt + v_r \ln \left(\frac{m_0}{m} \right)$$



Ex An open topped rail road car of mass M has an initial velocity v_0 along a straight horizontal frictionless track. It suddenly starts raining at time $t=0$. The rain drops fall vertically with velocity u and add a mass m kg/sec of water. Find the velocity of car after t second (assuming that it is not completely filled with water).

Sol. According to law of conservation of momentum, $Mv_0 = (M + m \times t) v$. Where m is the mass of water added per second and v is the velocity of the car after t second. $\therefore v = \frac{Mv_0}{M + mt}$

Ex. A uniform chain of mass m and length ℓ hangs on a thread and touches the surface of a table by its lower end. Find the force exerted by the chain on the table when half of its length has fallen on the table. The fallen part does not form heap.

Sol. At given condition force exerted by the chain on the table consists of two parts

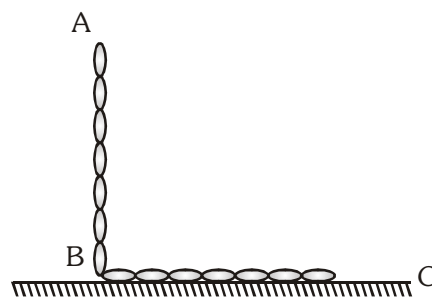
(i) Weight of portion BC = $\frac{mg}{2}$

(ii) Thrust force = $v_r \left(-\frac{dm}{dt} \right) = v \left(\frac{m}{\ell} v \right) = \frac{m}{\ell} v^2$

but $v = \sqrt{2g \left(\frac{\ell}{2} \right)} = \sqrt{g\ell}$

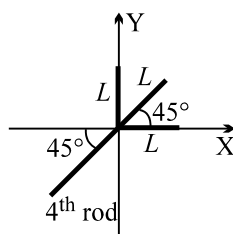
$\Rightarrow$ Thrust force = $\frac{m}{\ell} (g\ell) = mg$

$\therefore$ Net force exerted by falling chain = $\frac{mg}{2} + mg = \frac{3mg}{2}$



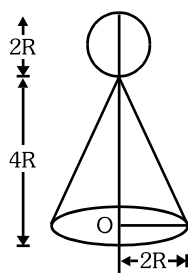
EXERCISE (S-1)

- Four particles of mass 5, 3, 2, 4 kg are at the points (1, 6), (-1, 5), (2, -3), (-1, -4). Find the coordinates of their centre of mass. **CM0001**
- A rigid body consists of a 3 kg mass connected to a 2 kg mass by a massless rod. The 3kg mass is located at $\vec{r}_1 = (2\hat{i} + 5\hat{j})$ m and the 2 kg mass at $\vec{r}_2 = (4\hat{i} + 2\hat{j})$ m. Find the length of rod and the coordinates of the centre of mass. **CM0002**
- Three identical uniform rods of the same mass M and length L are arranged in xy plane as shown in the figure. A fourth uniform rod of mass $3M$ has been placed as shown in the xy plane. What should be the value of the length of the fourth rod such that the center of mass of all the four rods lie at the origin?



CM0003

- From a circle of radius a , an isosceles right angled triangle with the hypotenuse as the diameter of the circle is removed. The distance of the centre of gravity of the remaining position from the centre of the circle is **CM0004**
- A man has constructed a toy as shown in figure. If density of the material of the sphere is 12 times of the cone compute the position of the centre of mass. [Centre of mass of a cone of height h is at height of $h/4$ from its base.]



CM0005

6. The figure shows the positions and velocities of two particles. If the particles move under the mutual attraction of each other, then find the position of centre of mass at $t = 1$ s.

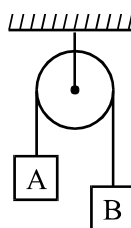


CM0006

7. Mass centers of a system of three particles of masses 1, 2, 3 kg is at the point (1 m, 2 m, 3 m) and mass center of another group of two particles of masses 2 kg and 3 kg is at point (-1 m, 3 m, -2 m). Where a 5 kg particle should be placed, so that mass center of the system of all these six particles shifts to mass center of the first system?

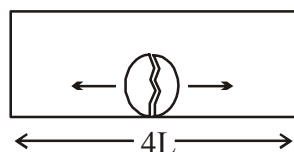
CM0007

8. In the arrangement shown in the figure, $m_A = 2$ kg and $m_B = 1$ kg. String is light and inextensible. Find the acceleration of centre of mass of both the blocks. Neglect friction everywhere.



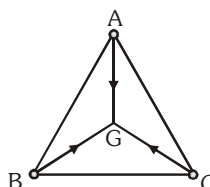
CM0008

9. A bomb of mass $3m$ is kept inside a closed box of mass $3m$ and length $4L$ at its centre. It explodes in two parts of mass m & $2m$. The two parts move in opposite direction and stick to the opposite side of the walls of box. Box is kept on a smooth horizontal surface. What is the distance moved by the box during this time interval.



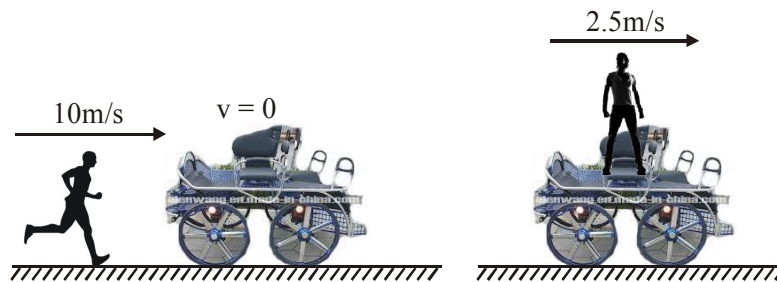
CM0009

10. Three particles A, B and C of equal mass move with equal speed v along the medians of an equilateral triangle as shown in fig. They collide at the centroid G of the triangle. After the collision, A comes to rest, B retraces its path with the speed v . What is the velocity of C?



CM0010

11. A 50 kg boy runs at a speed of 10 m/s and jumps onto a cart as shown in the figure. The cart is initially at rest. If the speed of the cart with the boy on it is 2.50 m/s, what is the mass of the cart ?
(Assuming friction is absent between cart and ground)



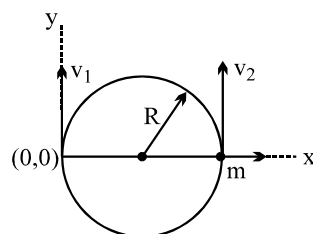
CM0011

12. Two cars initially at rest are free to move in the x direction. Car A has mass 4 kg and car B has mass 2 kg. They are tied together, compressing a spring in between them. When the spring holding them together is burned, car A moves off with a speed of 2 m/s.
(i) With what speed does car B leave.
(ii) How much energy was stored in the spring before it was burned.
13. A 24 kg projectile is fired at an angle of 53° above the horizontal with an initial speed of 50 m/s. At the highest point in its trajectory, the projectile explodes into two fragments of equal mass, the first of which falls vertically with zero initial speed.
(i) How far from the point of firing does the second fragment strike the ground? (Assume the ground is level.)
(ii) How much energy was released during the explosion?
14. A particle of mass m , moving in a circular path of radius R with a constant speed v_2 is located at point $(2R, 0)$ at time $t = 0$ and a man starts moving with a velocity v_1 along the +ve y-axis from origin at time $t = 0$. Calculate the linear momentum of the particle w.r.t. the man as a function of time.

CM0012

CM0013

[IIT-JEE' 2003]

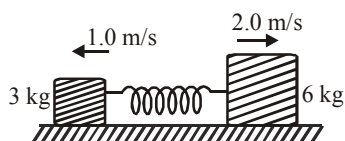


CM0014

15. A spaceship is moving with constant speed v_0 in gravity free space along +Y-axis suddenly shoots out one third of its part with speed $2v_0$ along + X-axis. Find the speed of the remaining part.

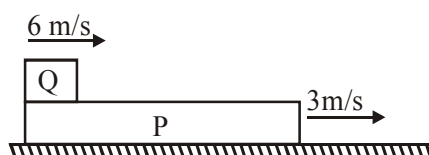
CM0015

16. Two blocks of mass 3 kg and 6 kg respectively are placed on a smooth horizontal surface. They are connected by a light spring of force constant $k = 200 \text{ N/m}$. Initially the spring is unstretched. The indicated velocities are imparted to the blocks. The maximum extension of the spring will be :-



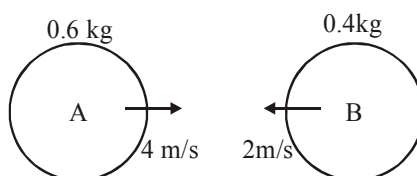
CM0016

17. A plank P and block Q are arranged as shown on a smooth table top. They are given velocities 3 m/s and 6 m/s respectively. The length of plank is 1 m and block is of negligible size. After some time when the block has reached the other end of plank it stops slipping on plank. Find the coefficient of friction between plank P and block Q if mass of plank is double of block).



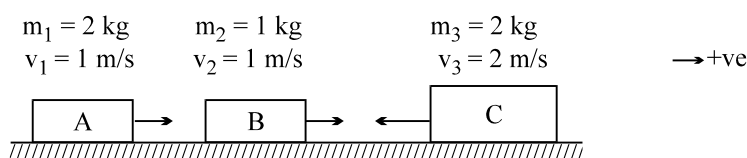
CM0017

18. A bullet of mass m strikes an obstruction and deviates off at 60° to its original direction. If its speed is also changed from u to v , find the magnitude of the impulse acting on the bullet. CM0018
19. The velocities of two steel balls before impact are shown. If after head on impact the velocity of ball B is observed to be 3 m/s to the right, the coefficient of restitution is



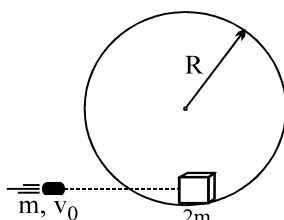
CM0019

20. Three carts move on a frictionless track with inertias and velocities as shown. The carts collide and stick together after successive collisions.
- (i) Find loss of mechanical energy when B & C stick together.
- (ii) Find magnitude of impulse experienced by A when it sticks to combined mass (B & C).



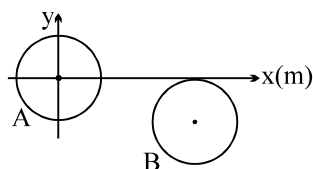
CM0020

21. A small block of mass $2m$ initially rests at the bottom of a fixed circular, vertical track, which has a radius of R . The contact surface between the mass and the loop is frictionless. A bullet of mass m strikes the block horizontally with initial speed v_0 and remain embedded in the block as the block and the bullet circle the loop. Determine each of the following in terms of m , v_0 , R and g .
- The speed of the masses immediately after the impact.
 - The minimum initial speed of the bullet if the block and the bullet are to successfully execute a complete ride on the loop



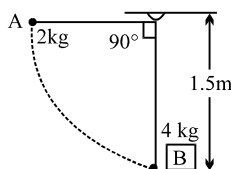
CM0021

22. Two smooth balls A and B, each of mass m and radius R , have their centres at $(0,0,R)$ and at $(5R,-R,R)$ respectively, in a coordinate system as shown. Ball A, moving along positive x axis, collides with ball B. Just before the collision, speed of ball A is 4 m/s and ball B is stationary. The collision between the balls is elastic. Find Velocity of the ball A just after the collision and impulse of the force exerted by A on B during the collision.



CM0022

23. A sphere A is released from rest in the position shown and strikes the block B which is at rest. If $e = 0.75$ between A and B and $\mu_k = 0.5$ between B and the support, determine
- the velocity of A just after the impact
 - the maximum displacement of B after the impact.



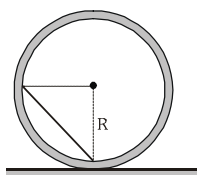
CM0023

24. Bullets of mass 10 g each are fired from a machine gun at rate of 60 bullets/minute. The muzzle velocity of bullets is 100 m/s. The thrust force due to firing bullets experienced by the person holding the gun stationary is _____.

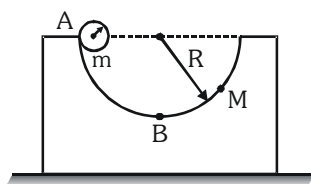
CM0024

EXERCISE (S-2)

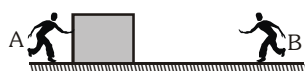
- The linear mass density of a ladder of length ℓ increases uniformly from one end A to the other end B, (i) Form an expression for linear mass density as a function of distance x from end A where linear mass density λ_0 . The density at one end being twice that of the other end. (ii) Find the position of the centre of mass from end A. **CM0025**
- Inside a hollow uniform sphere of inner radius R a uniform rod of length $R\sqrt{2}$ is released from the state of rest as shown. The mass of the rod is same as that of the sphere. Assume friction to be absent everywhere. Horizontal displacement of sphere with respect to earth in the time in which the rod becomes horizontal, is

**CM0026**

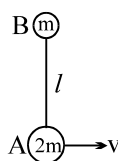
- A block of mass M with a semicircular track of radius R , rests on a horizontal frictionless surface. A uniform cylinder of radius r and mass m is released from rest at the top point A (see Fig). The cylinder slips on the semicircular frictionless track. How far has the block moved when the cylinder reaches the bottom (point B) of the track? How fast is the block moving when the cylinder reaches the bottom of the track?

**CM0027**

- Two persons A and B each of mass 100 kg are on a frictionless horizontal surface. Person A at rest is holding a block of mass 50 kg, suddenly pushes the block with some velocity (v) towards the person B approaching at a velocity of 5 m/s. B catches the block & slows down. Now the separation between A and B becomes constant. Find the speed v (in m/s).

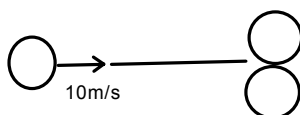
**CM0028**

5. Two masses A and B connected with an inextensible string of length ℓ lie on a smooth horizontal plane. A is given a velocity of v m/s along the ground perpendicular to line AB as shown in figure. Find the tension in string during their subsequent motion.



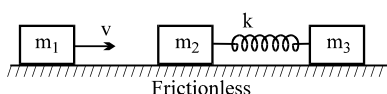
CM0029

6. A ball with initial speed of 10 m/s collides elastically with two other identical ball whose centres are on a line perpendicular to the initial velocity and which are initially in contact with each other. All the three ball are lying on a smooth horizontal table. The first ball is aimed directly at the contact point of the other two balls. All the balls are smooth. Find the velocities of the three balls after the collision.



CM0030

7. Mass m_1 hits & sticks with m_2 while sliding horizontally with velocity v along the common line of centres of the three equal masses ($m_1 = m_2 = m_3 = m$). Initially masses m_2 and m_3 are stationary and the spring is unstretched. Find the
- velocities of m_1 , m_2 and m_3 immediately after impact.
 - maximum kinetic energy of m_3 .
 - minimum kinetic energy of m_2 .
 - maximum compression of the spring.



CM0031

8. A sphere of mass m is moving with a velocity $4\hat{i} - \hat{j}$ when it hits a smooth wall and rebounds with velocity $\hat{i} + 3\hat{j}$. Find the impulse it receives. Find also the coefficient of restitution between the sphere and the wall.

CM0032

9. Two particles A and B of mass $2m$ and m respectively are attached to the ends of a light inextensible string of length $4a$ which passes over a small smooth peg at a height $3a$ from an inelastic table. The system is released from rest with each particle at a height a from the table. Find
- The speed of B when A strikes the table.
 - The time that elapses before A first hits the table.
 - The time for which A is resting on the table after the first collision & before it is first jerked off.

CM0033

10. Two particles, each of mass m , are connected by a light inextensible string of length 2ℓ . Initially they lie on a smooth horizontal table at points A and B distant ℓ apart. The particle at A is projected across the table with velocity u . Find the speed with which the second particle begins to move if the direction of u is :-
- (i) along BA.
 - (ii) at an angle of 120° with AB.
 - (iii) perpendicular to AB.

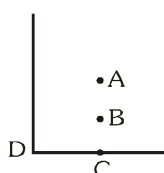
In each case calculate (in terms of m & u) the impulsive tension in the string.

CM0034

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

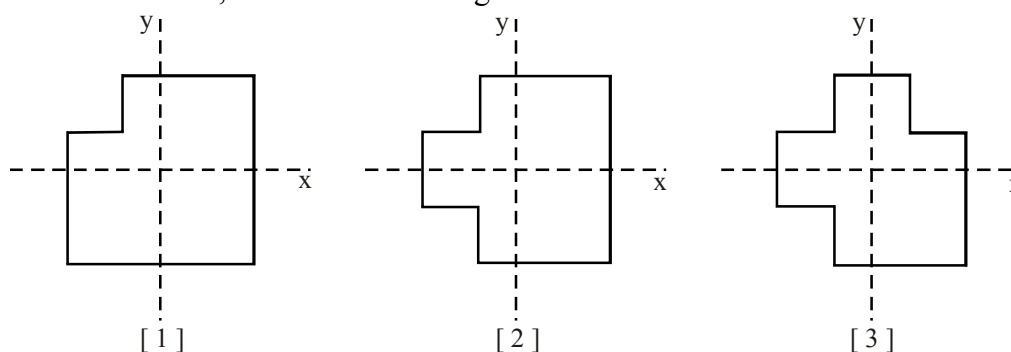
1. A thick uniform wire is bent into the shape of the letter "U" as shown. Which point indicates the location of the center of mass of this wire? A is the midpoint of the line joining mid points of two parallel sides of 'U' shaped wire.



- (A) D (B) A (C) B (D) C

CM0035

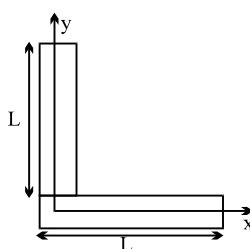
2. A machinist starts with three identical square plates but cuts one corner from one of them, two corners from the second, and three corners from the third. Rank the three plates according to the x-coordinate of their centers of mass, from smallest to largest.



- (A) 3, 1, 2 (B) 1, 3, 2 (C) 3, 2, 1 (D) 1 and 3 tie, then 2

CM0036

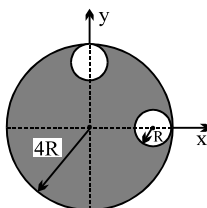
3. Centre of mass of two thin uniform rods of same length but made up of different materials & kept as shown, can be, if the meeting point is the origin of co-ordinates



- (A) $(L/2, L/2)$ (B) $(2L/3, L/2)$ (C) $(L/3, L/3)$ (D) $(L/3, L/6)$

CM0037

4. From the circular disc of radius $4R$ two small disc of radius R are cut off. The centre of mass of the new structure will be :



- (A) $i\frac{R}{5} + j\frac{R}{5}$ (B) $-i\frac{R}{5} + j\frac{R}{5}$ (C) $\frac{-3R}{14}(\hat{i} + \hat{j})$ (D) None of these

CM0038

5. Seven identical birds are flying south together at constant velocity. A hunter shoots one of them, which immediately dies and falls to the ground. The other six continue flying south at the original velocity. After the one bird has hit the ground, the centre of mass of all seven birds
- (A) continues south at the original speed, but is now located some distance behind the flying birds
 (B) continues south, but at $6/7$ the original velocity
 (C) continues south, but at $1/7$ the original velocity
 (D) stops with the dead bird

CM0039

6. A man weighing 80 kg is standing at the centre of a flat boat and he is 20 m from the shore. He walks 8 m on the boat towards the shore and then halts. The boat weight 200 kg . How far is he from the shore at the end of this time ?
- (A) 11.2 m (B) 13.8 m (C) 14.3 m (D) 15.4 m

CM0040

7. There are some passengers inside a stationary railway compartment. The track is frictionless. The centre of mass of the compartment itself (without the passengers) is C_1 , while the centre of mass of the 'compartment plus passengers' system is C_2 . If the passengers move about inside the compartment along the track.
- (A) both C_1 and C_2 will move with respect to the ground.
 (B) neither C_1 nor C_2 will move with respect to the ground.
 (C) C_1 will move but C_2 will be stationary with respect to the ground.
 (D) C_2 will move but C_1 will be stationary with respect to the ground.

CM0041

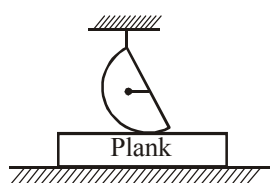
8. A non-zero external force acts on a system of particles. The velocity and acceleration of the centre of mass are found to be v_0 and a_c respectively at any instant t . It is possible that
- (i) $v_0 = 0, a_c = 0$ (ii) $v_0 \neq 0, a_c = 0$ (iii) $v_0 = 0, a_c \neq 0$ (iv) $v_0 \neq 0, a_c \neq 0$

Then

- (A) (iii) and (iv) are true. (B) (i) and (ii) are true.
(C) (i) and (iii) are true. (D) (ii), (iii) and (iv) are true.

CM0042

9. Lower surface of a plank is rough and lying at rest on a rough horizontal surface. Upper surface of the plank is smooth and has a smooth hemisphere placed over it through a light string as shown in the figure. After the string is burnt, trajectory of centre of mass of the sphere is :-



- (A) a circle (B) an ellipse
(C) a straight line (D) a parabola

CM0043

10. Three interacting particles of masses 100 g, 200 g and 400 g each have a velocity of 20 m/s magnitude along the positive direction of x-axis, y-axis and z-axis. Due to force of interaction the third particle stops moving. The velocity of the second particle is $(10\hat{j} + 5\hat{k})$. What is the velocity of the first particle?

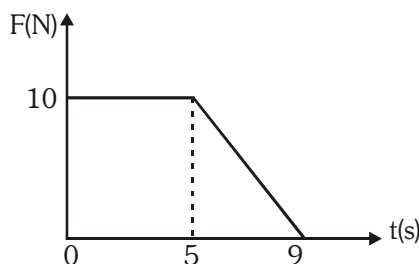
- (A) $20\hat{i} + 20\hat{j} + 70\hat{k}$ (B) $10\hat{i} + 20\hat{j} + 8\hat{k}$
(C) $30\hat{i} + 10\hat{j} + 7\hat{k}$ (D) $15\hat{i} + 5\hat{j} + 60\hat{k}$

CM0044

11. A system of N particles is free from any external forces.
- (i) Which of the following is true for the magnitude of the total momentum of the system?
- (A) It must be zero
(B) It could be non-zero, but it must be constant
(C) It could be non-zero, and it might not be constant
(D) The answer depends on the nature of the internal forces in the system
- (ii) Which of the following must be true for the sum of the magnitudes of the momenta of the individual particles in the system?
- (A) It must be zero
(B) It could be non-zero, but it must be constant
(C) It could be non-zero, and it might not be constant
(D) It could be zero, even if the magnitude of the total momentum is not zero

CM0045

12. A body of mass 4 kg is acted on by a force which varies as shown in the graph below. The momentum acquired is

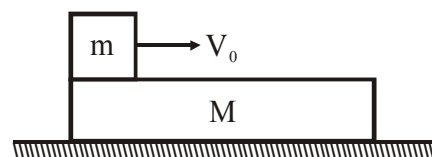


- (A) 280 N-s (B) 140 N-s (C) 70 N-s (D) 210 N-s

CM0046

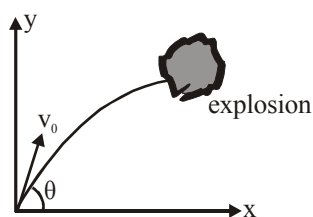
13. The coefficient of friction between the block and plank is μ and ground is smooth. The value of μ is such that block becomes stationary with respect to plank before it reaches the other end. Then which of the following statement is **incorrect**.

- (A) The work done by friction on the block is negative.
 (B) The work done by friction on the plank is positive.
 (C) The net work done by friction is negative.
 (D) Net work done by the friction is zero.



CM0047

14. A projectile is projected in x-y plane with velocity v_0 . At top most point of its trajectory projectile explodes into two identical fragments. Both the fragments land simultaneously on ground and stick there. Taking point of projection as origin and R as range of projectile if explosion had not taken place. Which of the following can not be position vectors of two pieces, when they land on ground.



- (A) $\frac{R}{2}\hat{i}, \frac{3R}{2}\hat{i}$ (B) $0\hat{i}, 2R\hat{i}$ (C) $R\hat{i} - R\hat{k}, R\hat{i} + R\hat{k}$ (D) $2R\hat{i} + \frac{R}{2}\hat{k}, R\hat{i} - \frac{R}{2}\hat{k}$

CM0048

15. A boy hits a baseball with a bat and imparts an impulse J to the ball. The boy hits the ball again with the same force, except that the ball and the bat are in contact for twice the amount of time as in the first hit. The new impulse equals:
- (A) half the original impulse (B) the original impulse
 (C) twice the original impulse (D) four times the original impulse

CM0049

16. Two balls of same mass are dropped from the same height h , on to the floor. The first ball bounces to a height $h/4$, after the collision & the second ball to a height $h/16$. The impulse applied by the first & second ball on the floor are I_1 and I_2 respectively. Then

(A) $5I_1 = 6I_2$ (B) $6I_1 = 5I_2$ (C) $I_1 = 2I_2$ (D) $2I_1 = I_2$

CM0050

17. Ball A of mass 5.0 kilograms moving at 20 m/s collides with ball B of unknown mass moving at 10m/s in the same direction. After the collision, ball A moves at 10 m/s and ball B at 15 m/s, both still in the same direction. What is the mass of ball B?

(A) 6.0 kg (B) 10. kg (C) 2.0 kg (D) 12 kg

CM0051

18. A smooth small spherical ball of mass m , moving with velocity u collides head on with another small spherical ball of mass $3m$, which was initially at rest. Two-third of the initial kinetic energy of the system is lost. The coefficient of restitution between the spheres is

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{2}$ (D) zero

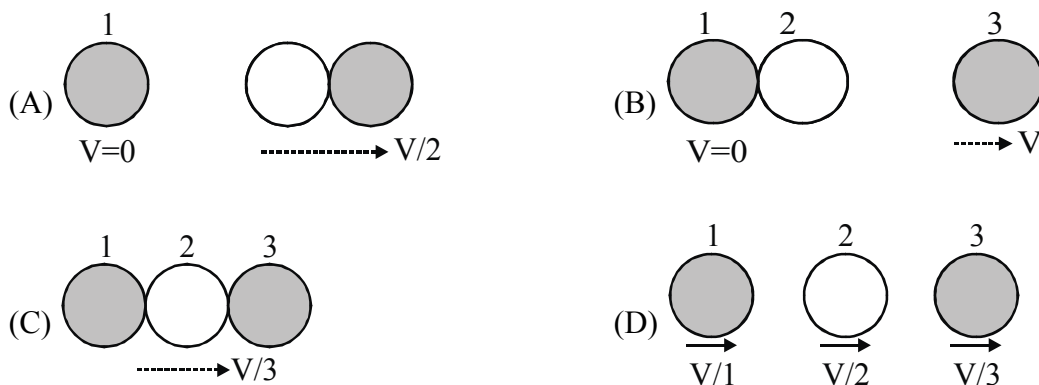
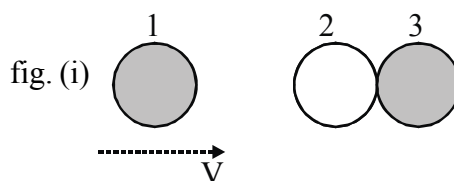
CM0052

19. A ball strikes a smooth horizontal ground at an angle of 45° with the vertical. What cannot be the possible angle of its velocity with the vertical after the collision. (Assume $e \leq 1$).

(A) 45° (B) 30° (C) 53° (D) 60°

CM0053

20. Two identical ball bearings in contact with each other and resting on a frictionless table are hit head-on by another ball bearing of the same mass moving initially with a speed V as shown in figure (i). If the collision is elastic, which of the following is a possible result after collision?



CM0054

21. A ball is projected from ground with a velocity V at an angle θ to the vertical. On its path it makes an elastic collision with a vertical wall and returns to ground. The total time of flight of the ball is

(A) $\frac{2v \sin \theta}{g}$ (B) $\frac{2v \cos \theta}{g}$ (C) $\frac{v \sin 2\theta}{g}$ (D) $\frac{v \cos \theta}{g}$

CM0055

22. A ball is thrown downwards with initial speed = 6 m/s, from a point at height = 3.2 m above a horizontal floor. If the ball rebounds back to the same height then coefficient of restitution equals to
- (A) 1/2 (B) 0.75 (C) 0.8 (D) None

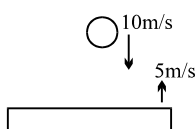
CM0056

23. A particle is projected from a smooth horizontal surface with velocity v at an angle θ from horizontal. Coefficient of restitution between the surface and ball is e . The distance of the point where ball strikes the surface second time from the point of projection is

(A) $\frac{v^2 \sin 2\theta(1+e^2)}{g}$ (B) $\frac{v^2 \sin 2\theta(1+e^4)}{g}$
 (C) $\frac{v^2 \sin 2\theta(1+e^3)}{g}$ (D) $\frac{v^2 \sin 2\theta(1+e)}{g}$

CM0057

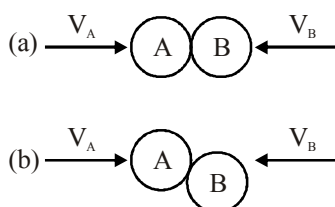
24. A ball of mass 1 kg strikes a heavy platform, elastically, moving upwards with a velocity of 5 m/s. The speed of the ball just before the collision is 10 m/s downwards. Then the impulse imparted by the platform on the ball is :-



(A) 15 N-s (B) 10 N-s (C) 20 N-s (D) 30 N-s

CM0058

25. Two bodies, A and B, collide as shown in figures a and b below. Circle the true statement :



- (A) They exert equal and opposite forces on each other in (a) but not in (b)
 (B) They exert equal and opposite force on each other in (b) but not in (a)
 (C) They exert equal and opposite force on each other in both (a) and (b)
 (D) The forces are equal and opposite to each other in (a), but only the components of the forces parallel to the velocities are equal in (b).

CM0059

26. A mass 'm' moves with a velocity 'v' and collides inelastically with another identical mass at rest.

After collision the 1st mass moves with velocity $\frac{v}{\sqrt{3}}$ in a direction perpendicular to the initial direction of motion. Find the speed of the 2nd mass after collision :-

(A) $\frac{2v}{\sqrt{3}}$

(B) $\frac{v}{\sqrt{3}}$

(C) $v\sqrt{\frac{2}{3}}$

(D) the situation of the problem is not possible without external impulse

CM0060

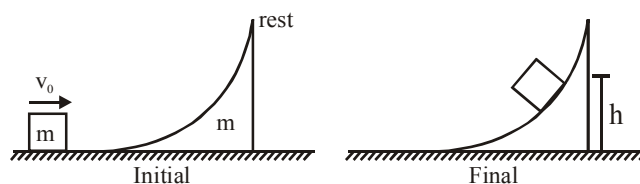
MULTIPLE CORRECT TYPE QUESTIONS

27. Two charges moving under their only own mutual attraction separated by large distance initially. Then choose the correct statement(s)

- (A) If both are free, mechanical energy is conserved.
 (B) If one is fixed and other is free, mechanical energy is conserved.
 (C) If one is fixed and other is free, momentum is conserved.
 (D) If both are free momentum is conserved.

CM0061

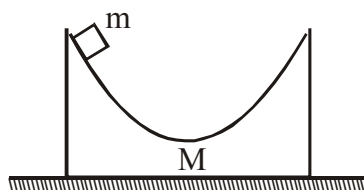
28. In the arrangement shown, horizontal surface is smooth, but friction is present between the block and the surface of the wedge. Block is given velocity v_0 at $t = 0$. After achieving height 'h' on the wedge, block comes to rest with respect to wedge at $t = t_0$. Then from $t = 0$ to $t = t_0$:-



- (A) Work done by friction on the block is negative
 (B) Work done by friction on the wedge is negative
 (C) Work done by block on the wedge is positive
 (D) Work done by wedge on the block is positive

CM0062

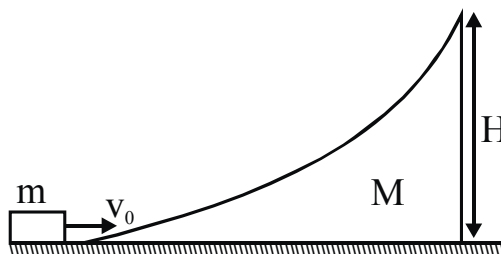
29. Figure shows a wedge on which a small block is released from rest. All the surfaces are smooth system comprises of wedge and blocks. Mark the correct statement(s) regarding motion of block on wedge till block attains maximum height on wedge.



- (A) Acceleration of centre of mass of system is initially vertically down then vertically up.
 (B) Initially centre of mass moves down and then up.
 (C) At the maximum height block and wedge move with common velocity.
 (D) Centre of mass of wedge moves towards left then right

CM0063

30. Figure shows a block of mass m projected with velocity v_0 towards a wedge. Consider all the surfaces to be smooth. Block does not have sufficient energy to negotiate (over come) wedge. Mark the correct option(s)



- (A) when block is at the maximum height on wedge, block and wedge have velocity equal to velocity of centre of mass of block wedge system
 (B) wedge acquires maximum speed with respect to ground when block returns to lowest point on wedge.
 (C) momentum of wedge and block is conserved at all times
 (D) centre of mass of wedge and block remains stationary

CM0064

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 31 and 32

A projectile of mass "m" is projected from ground with a speed of 50 m/s at an angle of 53° with the horizontal. It breaks up into two equal parts at the highest point of the trajectory. One particle coming to rest immediately after the explosion.

31. The ratio of the radii of curvatures of the moving particle just before and just after the explosion are:
 (A) 1 : 4 (B) 1 : 3 (C) 2 : 3 (D) 4 : 9

CM0065

32. The distance between the pieces of the projectile when they reach the ground are:
 (A) 240 (B) 360 (C) 120 (D) none

CM0065

Paragraph for Question 33 to 35

2 kg and 3 kg blocks are placed on a smooth horizontal surface and connected by spring which is unstretched initially. The blocks are imparted velocities as shown in the figure.



33. The maximum energy stored in the spring in the subsequent motion will be
 (A) $5v_0^2$ (B) $15v_0^2$ (C) zero (D) $10v_0^2$

CM0066

34. Maximum speed of 3 kg block in the subsequent motion will be
 (A) v_0 (B) $2v_0$ (C) $3v_0$ (D) $4v_0$

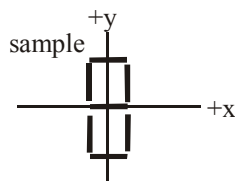
CM0066

35. Maximum speed of 2 kg block in the subsequent motion will be
 (A) v_0 (B) $2v_0$ (C) $3v_0$ (D) $4v_0$

CM0066

MATRIX MATCH TYPE QUESTIONS

36. On the left are statements about the location of the center of mass of the objects depicted on the right. The objects on the right are symbols constructed out of sticks of equal length and mass. The location of the center of mass is described using the coordinate system depicted in the sample.



The centre of mass lies at $x = 0, y = 0$

Column I

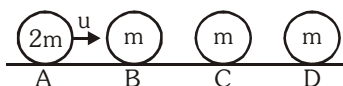
- (A) The center of mass is at $x > 0$ and $y = 0$
- (B) The center of mass is at $x = 0$ and $y > 0$
- (C) The center of mass is at $x > 0$ and $y > 0$
- (D) The center of mass is at $x = 0$ and $y = 0$

Column II

- (P)
- (Q)
- (R)
- (S)
- (T)

CM0067

37. Four balls A, B, C and D are kept on a smooth horizontal surface as shown in figure. Ball A is given velocity u towards B- (Assume each collision to be elastic)



Column-I

Column-II

- | | |
|--|-----------------------|
| (A) Total impulse of all collisions on A | (p) $\frac{4mu}{9}$ |
| (B) Total impulse of all collisions on B | (q) $\frac{4mu}{27}$ |
| (C) Total impulse of all collision on C | (r) $\frac{4mu}{3}$ |
| (D) Total impulse of all collisions on D | (s) $\frac{52}{27}mu$ |

CM0068

38. In Column-I, 4 situations are depicted and in column-II, 4 possible kinds of collision are listed. Match the situation with type of collision.

Column-I

Column-II

Before

After

- | | |
|--|-------------------------------|
| (A) $\begin{array}{cc} \xrightarrow{3\text{m/s}} & \xrightarrow{1.5\text{m/s}} \\ \boxed{2\text{kg}} & \boxed{8\text{kg}} \end{array}$ | (P) Elastic |
| (B) $\begin{array}{cc} \xleftarrow{2\text{m/s}} & \xrightarrow{1\text{m/s}} \\ \boxed{2\text{kg}} & \boxed{8\text{kg}} \end{array}$ | (Q) Perfectly Inelastic |
| (C) $\begin{array}{cc} \xrightarrow{3\text{m/s}} & \xleftarrow{1\text{m/s}} \\ \boxed{2\text{kg}} & \boxed{8\text{kg}} \end{array}$ | (R) Partially elastic |
| | (S) Collision is not possible |

CM0069

39. A particle of mass m , kinetic energy K and momentum p collides head on elastically with another particle of mass $2m$ at rest. After collision :

Column I

Column II

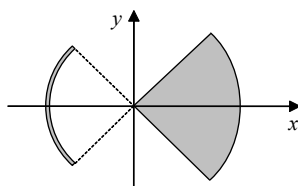
- | | |
|---------------------------------------|--------------------|
| (A) Momentum of first particle | (P) $\frac{3}{4}p$ |
| (B) Momentum of second particle | (Q) $-K/9$ |
| (C) Kinetic energy of first particle | (R) $-p/3$ |
| (D) Kinetic energy of second particle | (S) $\frac{8K}{9}$ |
| | (T) None |

CM0070

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

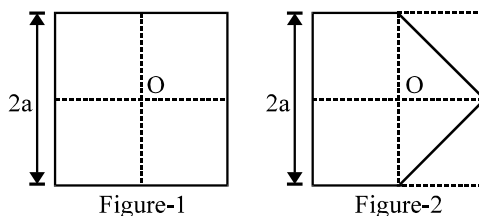
1. A sector cut from a uniform disk of radius 12 cm and a uniform rod of the same mass bent into shape of an arc are arranged facing each other as shown in the figure. If center of mass of the combination is at the origin, what is the radius of the arc?



- (A) 8 cm (B) 9 cm (C) 12 cm (D) 18 cm

CM0071

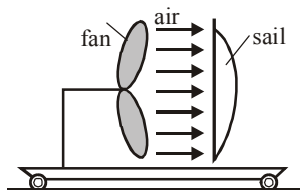
2. A piece of paper (shown in figure-1) is in form of a square. Two corners of this square are folded to make it appear like figure-2. Both corners are put together at centre of square 'O'. If O is taken to be (0, 0), the centre of mass of new system will be at



- (A) $\left(\frac{-a}{8}, 0\right)$ (B) $\left(\frac{-a}{6}, 0\right)$ (C) $\left(\frac{a}{12}, 0\right)$ (D) $\left(\frac{-a}{12}, 0\right)$

CM0072

3. A fan and a sail are mounted vertically on a cart that is initially at rest on a horizontal table as shown in the diagram. When the fan is turned on, an air stream is blown towards the right and is incident on the sail. The cart is free to move with negligible resistance forces. After the fan has been turned on the cart will



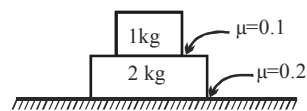
- (A) move to the right and then to the left (B) remain at rest
(C) move towards the right (D) move towards the left

CM0073

4. Two identical carts constrained to move on a straight line, on which sit two twins of same mass, are moving with same velocity. At some time snow begins to drop uniformly vertically downward. Ram, sitting on one of the trolleys, throws off the falling snow sideways with respect to himself and in the second cart shyam is asleep. (Assume that friction is absent)
- (A) Cart carrying Ram will speed up while cart carrying shyam will slow down
 (B) Cart carrying Ram will remain at the same speed while cart carrying shyam will slow down
 (C) Cart carrying Ram will speed up while cart carrying shyam will remain at the same speed
 (D) Cart carrying Ram as well as shyam will slow down

CM0074

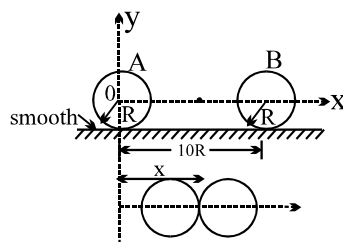
5. If both the blocks as shown in the given arrangement are given together a horizontal velocity towards right. If a_{cm} be the subsequent acceleration of the centre of mass of the system of blocks then a_{cm} equals



- (A) 0 m/s^2 (B) $\frac{5}{3} \text{ m/s}^2$
 (C) $\frac{7}{3} \text{ m/s}^2$ (D) 2 m/s^2

CM0075

6. Two uniform non conducting balls A & B have identical size having radius R but made of different density material (density of A = 2 density of B). The ball A is +vely charged & ball B is -vely charged. The balls are released on the horizontal smooth surface at the separation $10R$ as shown in figure. Because of mutual attraction the balls start moving towards each other. They will collide at a point.



- (A) $x = \frac{10R}{3}$ (B) $x = \frac{11R}{3}$
 (C) $x = 5R$ (D) $x = \frac{7R}{5}$

CM0076

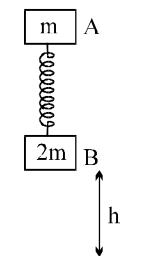
7. In adjacent figure a boy, on a horizontal platform A, kept on a smooth horizontal surface, holds a rope attached to a box B. Boy pulls the rope with a constant force of 50N. The coefficient of friction between boy and platform is 0.5. (Mass of boy = 80 kg, mass of platform = 120kg and mass of box = 100 kg)



- (A) Velocity of platform relative to box after 4 sec. is 2m/s
 (B) Velocity of boy relative to platform after 4sec is 2m/s
 (C) Friction force between boy and platform is 30N
 (D) Friction force between boy and platform is 50N

CM0077

8. From what minimum height h must the system be released when spring is unstretched so that after perfectly inelastic collision ($e = 0$) with ground, B may be lifted off the ground (Spring constant = k).



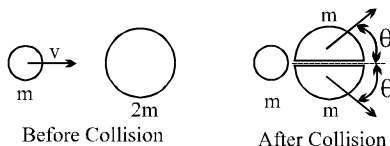
- (A) $mg/(4k)$ (B) $4mg/k$ (C) $mg/(2k)$ (D) none

CM0078

9. An isolated particle of mass m is moving in horizontal plane (x - y), along the x -axis, at a certain height above the ground. It suddenly explodes into two fragment of masses $\frac{m}{4}$ and $\frac{3m}{4}$. An instant later, the smaller fragment is at $y = +15$ cm. The larger fragment at this instant is at :—
 (A) $y = -5$ cm (B) $y = +20$ cm (C) $y = +5$ cm (D) $y = -20$ cm

CM0079

10. A particle of mass m is moving along the x -axis with speed v when it collides with a particle of mass $2m$ initially at rest. After the collisions, the first particle has come to rest, and the second particle has split into two equal-mass pieces that move at equal angles $\theta > 0$ with the x -axis, as shown in the figure. Which of the following statements correctly describes the speeds of the two pieces?



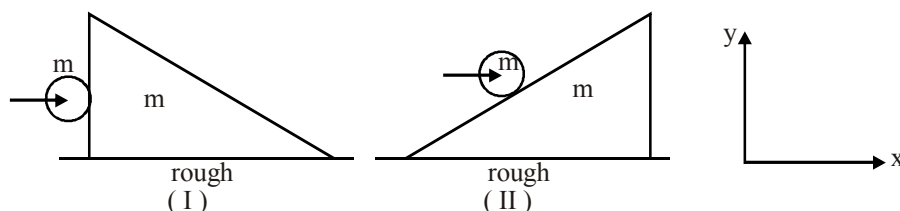
- (A) Each piece moves with speed v
 (B) One of the pieces moves with speed v , the other moves with speed less than v
 (C) Each piece moves with speed $v/2$
 (D) Each piece moves with speed greater than $v/2$

CM0080

11. A ball of mass m collides horizontally with a stationary wedge on a rough horizontal surface, in the two orientations as shown. Neglect friction between ball and wedge. Two student comment on system of ball and wedge in these situations

Saurav : Momentum of system in x -direction will change by significant amount in both cases.

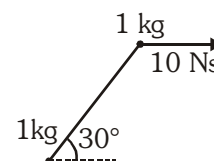
Rahul : There are no impulsive external forces in y -direction in both cases hence the total momentum of system in y -direction can be treated as conserved in both cases.



- (A) **Saurav** is wrong and **Rahul** is correct
(B) **Saurav** is correct and **Rahul** is wrong
(C) Both are correct
(D) Both are wrong

CM0081

12. Two balls of masses 1 kg each are connected by an inextensible massless string. The system is resting on a smooth horizontal surface. An impulse of 10 Ns is applied to one of the balls at an angle 30° with the line joining two balls in horizontal direction as shown in the figure. Assuming that the string remains taut after the impulse, the magnitude of impulse of tension is :-



- (A) 6Ns
(B) $\frac{5}{2}\sqrt{3}$ Ns
(C) 5 Ns
(D) $\frac{5}{\sqrt{3}}$ Ns

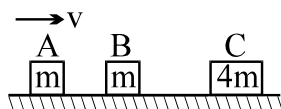
CM0082

13. A force exerts an impulse I on a particle changing its speed from u to $2u$. The applied force and the initial velocity are oppositely directed along the same line. The work done by the force is

- (A) $\frac{3}{2} I u$
(B) $\frac{1}{2} I u$
(C) $I u$
(D) $2 I u$

CM0083

14. Three blocks are initially placed as shown in the figure. Block A has mass m and initial velocity v to the right. Block B with mass m and block C with mass $4m$ are both initially at rest. Neglect friction. All collisions are elastic. The final velocity of block A is



- (A) $0.6v$ to the left
(B) $1.4v$ to the left
(C) v to the left
(D) $0.4v$ to the right

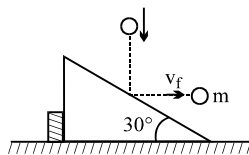
CM0084

15. Two billiard balls undergo a head-on collision. Ball 1 is twice as heavy as ball 2. Initially, ball 1 moves with a speed v towards ball 2 which is at rest. Immediately after the collision, ball 1 travels at a speed of $v/3$ in the same direction. What type of collision has occurred?

- (A) inelastic
(B) elastic
(C) completely inelastic
(D) cannot be determined from the information given

CM0085

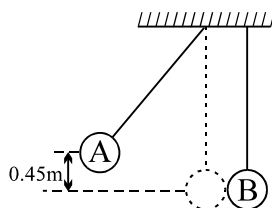
16. As shown in the figure a body of mass m moving vertically with speed 3 m/s hits a smooth fixed inclined plane and rebounds with a velocity v_f in the horizontal direction. If $\angle$ of inclined is 30° , the velocity v_f will be



- (A) 3 m/s (B) $\sqrt{3} \text{ m/s}$ (C) $1/\sqrt{3} \text{ m/s}$ (D) this is not possible

CM0086

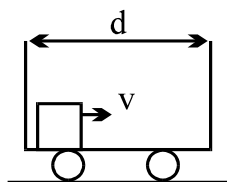
17. Two massless string of length 5 m hang from the ceiling very near to each other as shown in the figure. Two balls A and B of masses 0.25 kg and 0.5 kg are attached to the string. The ball A is released from rest at a height 0.45 m as shown in the figure. The collision between two balls is completely elastic. Immediately after the collision, the kinetic energy of ball B is 1 J . The velocity of ball A just after the collision is



- (A) 5 ms^{-1} to the right (B) 5 ms^{-1} to the left (C) 1 ms^{-1} to the right (D) 1 ms^{-1} to the left

CM0087

18. In a smooth stationary cart of length d , a small block is projected along its length with velocity v towards front. Coefficient of restitution for each collision is e . The cart rests on a smooth ground and can move freely. The time taken by block to come to rest w.r.t. cart is



- (A) $\frac{ed}{(1-e)v}$ (B) $\frac{ed}{(1+e)v}$ (C) $\frac{d}{e}$ (D) infinite

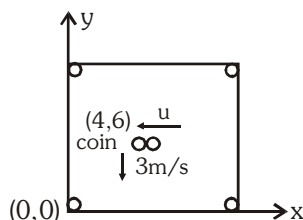
CM0088

19. A smooth sphere is moving on a horizontal surface with a velocity vector $(2\hat{i} + 2\hat{j}) \text{ m/s}$ immediately before it hit a vertical wall. The wall is parallel to vector $\hat{j}$ and coefficient of restitution between the sphere and the wall is $e = 1/2$. The velocity of the sphere after it hits the wall is

- (A) $\hat{i} - \hat{j}$ (B) $-\hat{i} + 2\hat{j}$ (C) $-\hat{i} - \hat{j}$ (D) $2\hat{i} - \hat{j}$

CM0089

20. On a smooth carom board, a coin moving in negative y -direction with a speed of 3 m/s is being hit at the point $(4, 6)$ by a striker moving along negative x -axis. The line joining centres of the coin and the striker just before the collision is parallel to x -axis. After collision the coin goes into the hole located at the origin. Masses of the striker and the coin are equal. Considering the collision to be elastic, the initial and final speeds of the striker in m/s will be



- (A) $(1.2, 0)$ (B) $(2, 0)$ (C) $(3, 0)$ (D) None of these

CM0090

21. Figure shows a block A of mass 5 kg kept at rest on a horizontal smooth surface. A spring ($K = 200 \text{ N/m}$) which is compressed by 10 cm and tied with the help of a string to maintain the compression is attached to block A as shown in figure. Block B also of mass 5 kg moving with 2 m/s collides with A, as shown. During the collision the string breaks and after the collision the spring is in its natural state. Assume the bodies to be elastic and let the velocities of A and B be v_1 and v_2 respectively assuming positive direction towards right, after collision. Then



- (A) $v_1 + v_2 > 2$
 (B) Initial kinetic energy of system = final kinetic energy of system
 (C) $v_1^2 + v_2^2 = 4.4 \text{ (m/s)}^2$
 (D) $v_1 - v_2 = 2$

CM0091

22. An open water tight railway wagon of mass $5 \times 10^3 \text{ kg}$ coasts at an initial velocity 1.2 m/s without friction on a railway track. Rain drops fall vertically downwards into the wagon. The velocity of the wagon after it has collected 10^3 kg of water will be
- (A) 0.5 m/s (B) 2 m/s (C) 1 m/s (D) 1.5 m/s

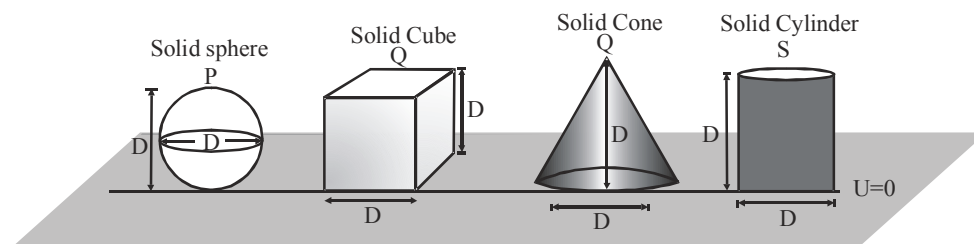
CM0092

23. A rocket of mass 4000 kg is set for vertical firing. How much gas must be ejected per second so that the rocket may have initial upwards acceleration of magnitude 19.6 m/s^2 . [Exhaust speed of fuel = 980 m/s .]
- (A) 240 kg s^{-1} (B) 60 kg s^{-1} (C) 120 kg s^{-1} (D) None

CM0093

MULTIPLE CORRECT TYPE QUESTIONS

24. Assuming potential energy 'U' at ground level to be zero.



All objects are made up of same material.

U_P = Potential energy of solid sphere

U_Q = Potential energy of solid cube

U_R = Potential energy of solid cone

U_S = Potential energy of solid cylinder

- (A) $U_S > U_P$ (B) $U_Q > U_S$ (C) $U_P > U_Q$ (D) $U_S > U_R$

CM0094

25. A blast breaks a body initially at rest of mass 0.5 kg into three pieces, two smaller pieces of equal mass and the third double the mass of either of small piece. After the blast the two smaller masses move at right angles to one another with equal speed. Find the statements that is/are true for this case assuming that the energy of blast is totally transferred to masses.

- (A) All the three pieces share the energy of blast equally
 (B) The speed of bigger mass is $\sqrt{2}$ times the speed of either of the smaller mass
 (C) The direction of motion of bigger mass makes an angle of 135° with the direction of smaller pieces
 (D) The bigger piece carries double the energy of either piece.

CM0095

26. A particle moving with kinetic energy = 3 joule makes an elastic head on collision with a stationary particle which has twice its mass during the impact.

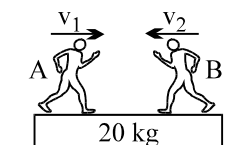
- (A) The minimum kinetic energy of the system is 1 joule.
 (B) The maximum elastic potential energy of the system is 2 joule.
 (C) Momentum and total kinetic energy of the system are conserved at every instant.
 (D) The ratio of kinetic energy to potential energy of the system first decreases and then increases.

CM0096

27. In a one dimensional collision between two identical particles A and B, B is stationary and A has momentum p before impact. During impact, B gives impulse J to A.
- (A) The total momentum of the 'A plus B' system is p before and after the impact, and $(p-J)$ during the impact.
- (B) During the impact A gives impulse of magnitude J to B
- (C) The coefficient of restitution is $\frac{2J}{p} - 1$
- (D) The coefficient of restitution is $\frac{J}{p} + 1$

CM0097

28. In the figure shown the system is at rest initially. Two persons 'A' and 'B' of masses 40 kg each move with speeds v_1 and v_2 respectively towards each other on a plank lying on a smooth horizontal surface as shown in figure. Plank travels a distance of 20 m towards right direction in 5 sec. (Here v_1 and v_2 are given with respect to the plank). Then the possible condition(s) can be



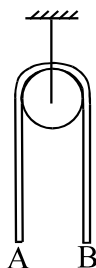
- (A) $v_1 = 0$ m/s, $v_2 = 10$ m/s
- (B) $v_1 = 5$ m/s, $v_2 = 15$ m/s
- (C) $v_1 = 10$ m/s, $v_2 = 20$ m/s
- (D) $v_1 = 2$ m/s, $v_2 = 12$ m/s

CM0098

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 29 and 30

A uniform chain of length $2L$ is hanging in equilibrium position, if end B is given a slightly downward displacement the imbalance causes an acceleration. Here pulley is small and smooth & string is inextensible



29. The acceleration of end B when it has been displaced by distance x , is

- (A) $\frac{x}{L} g$
- (B) $\frac{2x}{L} g$
- (C) $\frac{x}{2} g$
- (D) g

CM0099

30. The velocity v of the string when it slips out of the pulley (height of pulley from floor $> 2L$)

- (A) $\sqrt{\frac{gL}{2}}$ (B) $\sqrt{2gL}$ (C) $\sqrt{gL}$ (D) none of these

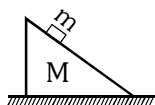
CM0099

MATRIX MATCH TYPE QUESTION

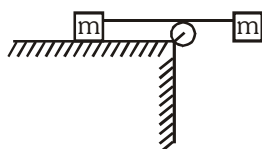
31. In each situation of column-I, a system involving two bodies is given. All strings and pulleys are light and friction is absent everywhere. Initially each body of every system is at rest. Consider the system in all situation of column I from rest till any collision occurs. Then match the statements in column-I with the corresponding results in column-II

Column I

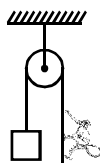
- (A) The block plus wedge system is placed over smooth horizontal surface. After the system is released from rest, the centre of mass of system



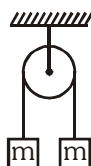
- (B) The string connecting both the blocks of mass m is horizontal. Left block is placed over smooth horizontal table as shown. After the two block system is released from rest, the centre of mass of system



- (C) The block and monkey have same mass. The monkey starts climbing up the rope. After the monkey starts climbing up, the centre of mass of monkey+block system



- (D) Both block of mass m are initially at rest. The left block is given initial velocity u downwards. Then, the centre of mass of two block system afterwards



Column II

- (P) Shifts towards right

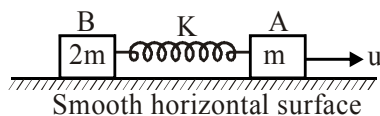
- (Q) Shifts downwards

- (R) Shifts upwards

- (S) Does not shift

CM0100

32. Two blocks A and B of mass m and $2m$ respectively are connected by a massless spring of spring constant K . This system lies over a smooth horizontal surface. At $t = 0$ the block A has velocity u towards right as shown while the speed of block B is zero, and the length of spring is equal to its natural length at that instant.



Column-I

- (A) The velocity of block A
- (B) The velocity of block B
- (C) The kinetic energy of system of two block
- (D) The potential energy of spring

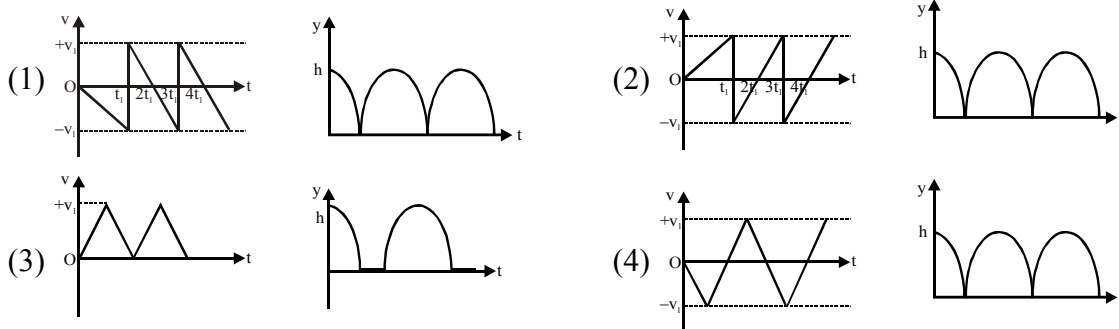
Column-II

- (P) can never be zero
- (Q) may be zero at certain instants of time
- (R) is minimum at maximum compression of spring
- (S) is maximum at maximum extension of spring

CM0101

EXERCISE (J-M)

1. Consider a rubber ball freely falling from a height $h = 4.9$ m onto a horizontal elastic plate. Assume that the duration of collision is negligible and the collision with the plate is totally elastic. Then the velocity as a function of time and the height as a function of time will be :- **[AIEEE - 2009]**

**CM0102**

Directions : Question number 4 contain Statement-1 and Statement-2. Of the four choices given after the statements, choose the one that best describes the two statements.

- 2. Statement-1 :** Two particles moving in the same direction do not lose all their energy in a completely inelastic collision.

Statement-2 : Principle of conservation of momentum holds true for all kinds of collisions.

[AIEEE - 2010]

- (1) Statement-1 is true, Statement-2 is false
- (2) Statement-1 is true, Statement-2 is true; Statement-2 is the correct explanation of Statement-1
- (3) Statement-1 is true, Statement-2 is true; Statement-2 is not the correct explanation of Statement-1
- (4) Statement-1 is false, Statement-2 is true

CM0103

3. This question has Statement I and Statement II. Of the four choices given after the Statements, choose the one that best describes the two Statements. **[JEE Main-2013]**

Statement - I : A point particle of mass m moving with speed v collides with stationary point particle

of mass M. If the maximum energy loss possible is given as $f\left(\frac{1}{2}mv^2\right)$ then $f = \left(\frac{m}{M+m}\right)$.

Statement - II : Maximum energy loss occurs when the particles get stuck together as a result of the collision.

- (1) Statement-I is true, Statement-II is true, Statement-II is a correct explanation of Statement-I.
- (2) Statement-I is true, Statement-II is true, Statement-II is a not correct explanation of Statement-I.
- (3) Statement-I is true, Statement-II is false.
- (4) Statement-I is false, Statement-II is true

CM0104

4. A particle of mass m moving in the x direction with speed $2v$ is hit by another particle of mass $2m$ moving in the y direction with speed v . If the collisions perfectly inelastic, the percentage loss in the energy during the collision is close to : [JEE Main-2015]

(1) 56 % (2) 62% (3) 44% (4) 50%

CM0105

5. Distance of the centre of mass of a solid uniform cone from its vertex is z_0 . If the radius of its base is R and its height is h then z_0 is equal to :- [JEE Main-2015]

(1) $\frac{5h}{8}$ (2) $\frac{3h^2}{8R}$ (3) $\frac{h^2}{4R}$ (4) $\frac{3h}{4}$

CM0106

6. It is found that if a neutron suffers an elastic collinear collision with deuterium at rest, fractional loss of its energy is p_d ; while for its similar collision with carbon nucleus at rest, fractional loss of energy is p_c . The values of p_d and p_c are respectively : [JEE Main-2018]

(1) (-28, -89) (2) (0, 0) (3) (0, 1) (4) (-89, -28)

CM0107

7. In a collinear collision, a particle with an initial speed v_0 strikes a stationary particle of the same mass. If the final total kinetic energy is 50 % greater than the original kinetic energy, the magnitude of the relative velocity between the two particles, after collision, is : [JEE Main-2018]

(1) $\sqrt{2} v_0$ (2) $\frac{v_0}{2}$ (3) $\frac{v_0}{\sqrt{2}}$ (4) $\frac{v_0}{4}$

CM0108

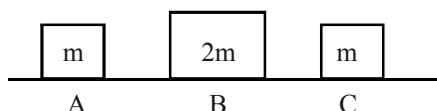
8. The mass of a hydrogen molecule is 3.32×10^{-27} kg. If 10^{23} hydrogen molecules strike, per second, a fixed wall of area 2 cm^2 at an angle of 45° to the normal, and rebound elastically with a speed of 10^3 m/s , then the pressure on the wall is nearly : [JEE Main-2018]

(1) $4.70 \times 10^3 \text{ N/m}^2$ (2) $2.35 \times 10^2 \text{ N/m}^2$ (3) $4.70 \times 10^2 \text{ N/m}^2$ (4) $2.35 \times 10^3 \text{ N/m}^2$

CM0109

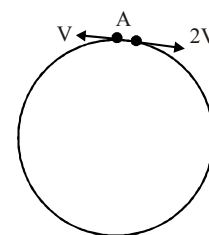
EXERCISE (J-A)

1. Three objects A, B and C are kept in a straight line on a frictionless horizontal surface. These have masses m , $2m$ and m , respectively. The object A moves towards B with a speed 9 m/s and makes an elastic collision with it. Thereafter, B makes completely inelastic collision with C. All motions occur on the same straight line. Find the final speed (in m/s) of the object C. [IIT-JEE-2009]



CM0110

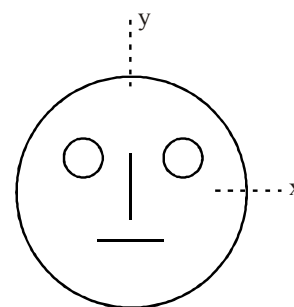
2. Two small particles of equal masses start moving in opposite directions from a point A in a horizontal circular orbit. Their tangential velocities are v and $2v$, respectively, as shown in the figure. Between collisions, the particles move with constant speeds. After making how many elastic collisions, other than that at A, these two particles will again reach the point A? [IIT-JEE-2009]



- (A) 4 (B) 3 (C) 2 (D) 1

CM0111

3. Look at the drawing given in the figure which has been drawn with ink of uniform line-thickness. The mass of ink used to draw each of the two inner circles, and each of the two line segments is m . The mass of the ink used to draw the outer circle is $6m$. The coordinates of the centres of the different parts are: outer circle $(0, 0)$, left inner circle $(-a, a)$, right inner circle (a, a) , vertical line $(0, 0)$ and horizontal line $(0, -a)$. The y -coordinate of the centre of mass of the ink in this drawing is [IIT-JEE-2009]



- (A) $\frac{a}{10}$ (B) $\frac{a}{8}$ (C) $\frac{a}{12}$ (D) $\frac{a}{3}$

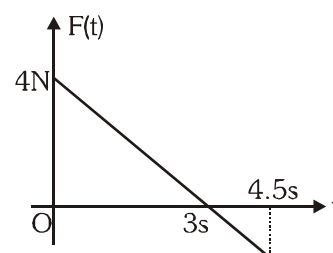
CM0112

4. A point mass of 1 kg collides elastically with a stationary point mass of 5 kg . After their collision, the 1 kg mass reverses its direction and moves with a speed of 2 m/s . Which of the following statement(s) is (are) correct for the system of these two masses? [IIT-JEE 2010]

- (A) Total momentum of the system is 3 kg m/s .
 (B) Momentum of 5 kg mass after collision is 4 kg m/s .
 (C) Kinetic energy of the centre of mass is 0.75 J .
 (D) Total kinetic energy of the system is 4 J .

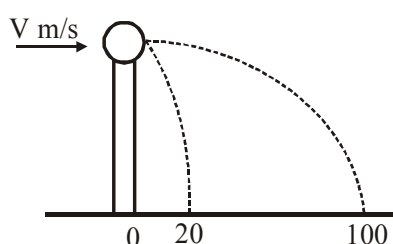
CM0113

5. A block of mass 2 kg is free to move along the x-axis. It is at rest and from $t=0$ onwards it is subjected to a time-dependent force $F(t)$ in the x-direction. The force $F(t)$ varies with t as shown in the figure. The kinetic energy of the block after 4.5 second is [IIT-JEE-2010]
 (A) 4.50 J (B) 7.50 J
 (C) 5.06 J (D) 14.06 J



CM0114

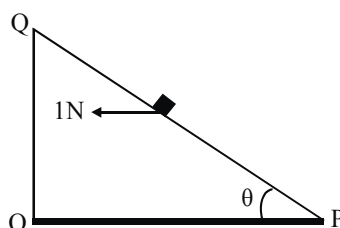
6. A ball of mass 0.2 kg rests on a vertical post of height 5m. A bullet of mass 0.01 kg, traveling with a velocity V m/s in a horizontal direction, hits the centre of the ball. After the collision, the ball and bullet travel independently. The ball hits the ground at a distance of 20 m and the bullet at a distance of 100 m from the foot of the post. The initial velocity V of the bullet is + [IIT-JEE 2011]



- (A) 250 m/s (B) $250\sqrt{2}$ m/s (C) 400 m/s (D) 500 m/s

CM0115

7. A small block of mass of 0.1 kg lies on a fixed inclined plane PQ which makes an angle θ with the horizontal. A horizontal force of 1 N acts on the block through its center of mass as shown in the figure. The block remains stationary if (take $g = 10 \text{ m/s}^2$) [IIT-JEE 2012]



- (A) $\theta = 45^\circ$
 (B) $\theta > 45^\circ$ and a frictional force acts on the block towards P
 (C) $\theta > 45^\circ$ and a frictional force acts on the block towards Q
 (D) $\theta < 45^\circ$ and a frictional force acts on the block towards Q

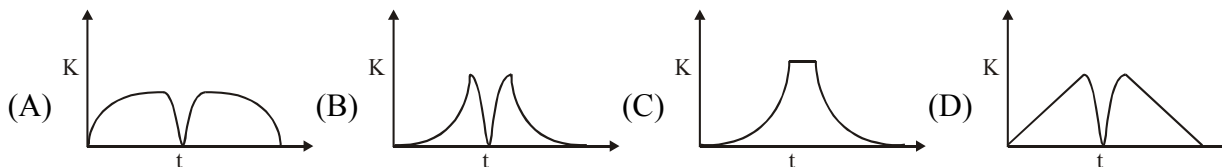
CM0116

8. A bob of mass m , suspended by a string of length ℓ_1 is given a minimum velocity required to complete a full circle in the vertical plane. At the highest point, it collides elastically with another bob of mass m suspended by a string of length ℓ_2 , which is initially at rest. Both the strings are mass-less and inextensible. If the second bob, after collision acquires the minimum speed required to complete a full circle in the vertical plane, the ratio $\frac{\ell_1}{\ell_2}$ is. [JEE Advanced-2013]

CM0117

9. A tennis ball is dropped on a horizontal smooth surface. It bounces back to its original position after hitting the surface. The force on the ball during the collision is proportional to the length of compression of the ball. Which one of the following sketches describes the variation of its kinetic energy K with time t most appropriately? The figures are only illustrative and not to the scale.

[JEE Advanced-2014]



CM0118

10. A block of mass M has a circular cut with a frictionless surface as shown. The block rests on the horizontal frictionless surface of a fixed table. Initially the right edge of the block is at $x = 0$, in a co-ordinate system fixed to the table. A point mass m is released from rest at the topmost point of the path as shown and it slides down. When the mass loses contact with the block, its position is x and the velocity is v . At that instant, which of the following options is/are correct?

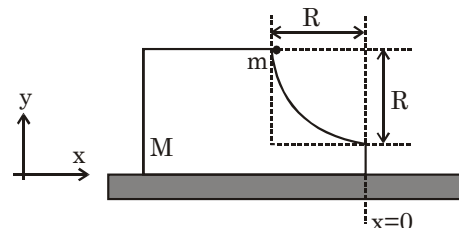
[JEE Advanced-2017]

(A) The x component of displacement of the centre of mass of the block M is : $-\frac{mR}{M+m}$

(B) The position of the point mass is : $x = -\sqrt{2} \frac{mR}{M+m}$

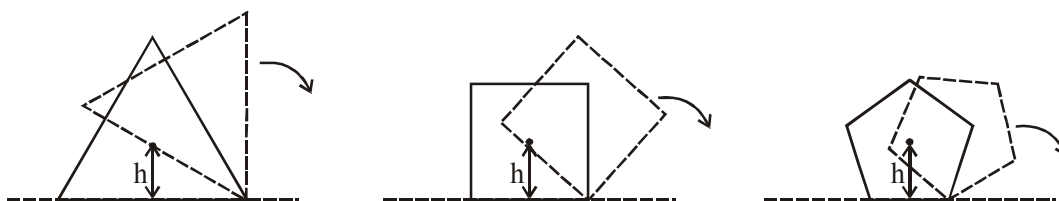
(C) The velocity of the point mass m is : $v = \sqrt{\frac{2gR}{1 + \frac{m}{M}}}$

(D) The velocity of the block M is : $v = -\frac{m}{M} \sqrt{2gR}$



CM0119

11. Consider regular polygons with number of sides $n = 3, 4, 5, \dots$ as shown in the figure. The center of mass of all the polygons is at height h from the ground. They roll on a horizontal surface about the leading vertex without slipping and sliding as depicted. The maximum increase in height of the locus of the center of mass for each polygon is Δ . Then Δ depends on n and h as : [JEE Advanced-2017]



(A) $\Delta = h \sin^2\left(\frac{\pi}{n}\right)$

(B) $\Delta = h \sin\left(\frac{2\pi}{n}\right)$

(C) $\Delta = h \left(\frac{1}{\cos\left(\frac{\pi}{n}\right)} - 1 \right)$

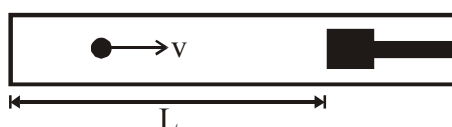
(D) $\Delta = h \tan^2\left(\frac{\pi}{2n}\right)$

CM0120

12. A solid horizontal surface is covered with a thin layer of oil. A rectangular block of mass $m = 0.4\text{ kg}$ is at rest on this surface. An impulse of 1.0 N s is applied to the block at time $t = 0$ so that it starts moving along the x -axis with a velocity $v(t) = v_0 e^{-t/\tau}$, where v_0 is a constant and $\tau = 4\text{ s}$. The displacement of the block, in metres, at $t = \tau$ is..... Take $e^{-1} = 0.37$? [JEE Advanced-2018]

CM0121

13. A small particle of mass m moving inside a heavy, hollow and straight tube along the tube axis undergoes elastic collision at two ends. The tube has no friction and it is closed at one end by a flat surface while the other end is fitted with a heavy movable flat piston as shown in figure. When the distance of the piston from closed end is $L = L_0$ the particle speed is $v = v_0$. The piston is moved inward at a very low speed V such that $V \ll \frac{dL}{L} v_0$, where dL is the infinitesimal displacement of the piston. Which of the following statement(s) is/are correct ? [JEE Advanced-2019]



- (1) The rate at which the particle strikes the piston is v/L
- (2) After each collision with the piston, the particle speed increases by $2V$
- (3) The particle's kinetic energy increases by a factor of 4 when the piston is moved inward from L_0 to $\frac{1}{2}L_0$
- (4) If the piston moves inward by dL , the particle speed increases by $2v \frac{dL}{L}$

CM0122

ANSWER KEY

EXERCISE (S-1)

1. Ans. $(1/7, 23/14)$ 2. Ans. $\sqrt{13} \text{ m}, \left(\frac{14}{5}, \frac{19}{5}\right)$ 3. Ans. $L(\sqrt{2} + 1)/3$

4. Ans. $\frac{a}{3(\pi - 1)}$ 5. Ans. $4R$ from O 6. Ans. $x = 6\text{m}$

7. Ans. $(3 \text{ m}, 1 \text{ m}, 8 \text{ m})$

$$\frac{6(\hat{i} + 2\hat{j} + 3\hat{k}) + 5(-\hat{i} + 3\hat{j} - 2\hat{k}) + 5\vec{r}}{16} = (\hat{i} + 2\hat{j} + 3\hat{k})$$

$$\vec{r} = (3\hat{i} + \hat{j} + 8\hat{k})$$

8. Ans. $g/9$ downwards 9. Ans. $\frac{L}{3}$ 10. Ans. $\vec{v}_C = -\vec{v}_B$ 11. Ans. 150 kg

12. Ans. (i) 4 m/s , (ii) 24 J 13. Ans. (i) 360 m , (ii) 10800 J

14. Ans. $\vec{P}_{PM} = m\vec{v}_{PM}$
 $= -mv_2 \sin \omega t \hat{i} + m(v_2 \cos \omega t - v_1) \hat{j}$

15. Ans. $\frac{\sqrt{13}}{2} v_0$ 16. Ans. 30 cm 17. Ans. 0.3 18. Ans. $m \times \sqrt{u^2 - uv + v^2}$

19. Ans. $\frac{7}{18}$ 20. Ans. (i) 3 J , (ii) $\frac{12}{5} \text{ N-s}$

21. Ans. (i) $v_0/3$, (ii) $3\sqrt{5gR}$ 22. Ans. $(\hat{i} + \sqrt{3}\hat{j}) \text{ m/s}$, $(3m\hat{i} - \sqrt{3}m\hat{j}) \text{ kg-m/s}$

23. Ans. (i) $v_A = \sqrt{g/12} \text{ m/s}$, (ii) $S_{\max} = 49/48 \text{ m}$ 24. Ans. 1 N

EXERCISE (S-2)

1. Ans. (i) $\lambda(x) = \lambda_0 + \frac{\lambda_0 x}{\ell}$, (ii) $\frac{5}{9} \ell$

2. Ans. $\frac{R}{4}$

3. Ans. $\frac{m(R-r)}{M+m}$, $m\sqrt{\frac{2g(R-r)}{M(M+m)}}$

4. Ans. 4

5. Ans. $\frac{2mv^2}{3\ell}$

6. Ans. -2m/s , $6.93 \text{ m/s} \angle 30^\circ$

7. Ans. (i) $v/2, v/2, 0$; (ii) $2mv^2/9$; (iii) $mv^2/72$; (d) $x = \sqrt{m/6k} v$

8. Ans. $m(-3\hat{i} + 4\hat{j})$, $e = \frac{9}{16}$

9. Ans. (i) $\sqrt{\frac{2ag}{3}}$ (ii) $\frac{3v}{g}$ (iii) $\frac{2v}{g}$

10. Ans. (i) $\frac{u}{2}, \frac{mu}{2}$ (ii) $\frac{u\sqrt{13}}{8}, \frac{mu\sqrt{13}}{8}$ (iii) $\frac{u\sqrt{3}}{4}, \frac{mu\sqrt{3}}{4}$

EXERCISE (O-1)

- | | | | | | |
|-------------------------------------|--------------|---------------|--|---------------------------|-----------------|
| 1. Ans. (C) | 2. Ans. (B) | 3. Ans. (D) | 4. Ans. (C) | 5. Ans. (B) | 6. Ans. (C) |
| 7. Ans. (C) | 8. Ans. (A) | 9. Ans. (C) | 10. Ans. (A) | 11. Ans. (i) (B) (ii) (C) | |
| 12. Ans. (C) | 13. Ans. (D) | 14. Ans. (D) | 15. Ans. (C) | 16. Ans. (A) | 17. Ans. (B) |
| 18. Ans. (A) | 19. Ans. (B) | 20. Ans. (B) | 21. Ans. (B) | 22. Ans. (C) | 23. Ans. (D) |
| 24. Ans. (D) | 25. Ans. (C) | 26. Ans. (D) | 27. Ans. (A, B,D) | | 28. Ans. (A, C) |
| 29. Ans. (B,C) | | 30. Ans. (AB) | 31. Ans. (A) | 32. Ans. (A) | 33. Ans. (B) |
| 34. Ans. (C) | 35. Ans. (D) | | 36. Ans. (A)-P; (B)-S; (C)-Q, R; (D)-T | | |
| 37. Ans. A-(s), B-(q), C-(p), D-(r) | | | 38. Ans. (A)-Q (B)-S (C)-P | | |
| 39. Ans. (A) R, (B) T, (C) T, (D)S | | | | | |

EXERCISE (O-2)

- | | | | | | |
|--|--------------|------------------|--|----------------|------------------|
| 1. Ans. (A) | 2. Ans. (D) | 3. Ans. (B) | 4. Ans. (D) | 5. Ans. (D) | 6. Ans. (B) |
| 7. Ans. (C) | 8. Ans. (B) | 9. Ans. (A) | 10. Ans. (D) | 11. Ans. (D) | 12. Ans. (B) |
| 13. Ans. (B) | 14. Ans. (A) | 15. Ans. (B) | 16. Ans. (B) | 17. Ans. (D) | 18. Ans. (D) |
| 19. Ans. (B) | 20. Ans. (B) | 21. Ans. (C) | 22. Ans. (C) | 23. Ans. (C) | 24. Ans. (A,B,D) |
| 25. Ans. (A,C) | | 26. Ans. (A,B,D) | | 27. Ans. (B,C) | |
| 28. Ans. (A,B,C,D) | | 29. Ans. (A) | 30. Ans. (C) | | |
| 31. Ans. (A)-Q; (B)-P, Q; (C)-R; (D) S | | | 32. Ans. (A)-Q; (B)-Q; (C)-P,R,, (D)-Q,S | | |

EXERCISE (J-M)

- | | | | | | |
|-------------|-------------|-------------|-------------|-------------|-------------|
| 1. Ans. (1) | 2. Ans. (2) | 3. Ans. (4) | 4. Ans. (1) | 5. Ans. (4) | 6. Ans. (4) |
| 7. Ans. (1) | 8. Ans. (4) | | | | |

EXERCISE (J-A)

- | | | | | | |
|---------------------------|-------------|-----------------|---------------|----------------|--------------|
| 1. Ans. 4m/s | 2. Ans. (C) | 3. Ans. (A) | 4. Ans. (A,C) | 5. Ans. (C) | 6. Ans. (D) |
| 7. Ans. (A,C) | | 8. Ans. 5 | 9. Ans. (B) | 10. Ans. (A,C) | 11. Ans. (C) |
| 12. Ans. 6.3 [6.29, 6.31] | | 13. Ans. (2, 3) | | | |

Important Notes



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Important Notes

[illegible]

KINEMATICS OF ROTATION MOTION

Rigid Body

A rigid body is an assemblage of a large number of material particles, which do not change their mutual distances under any circumstance or in other words, they are not deformed under any circumstance.

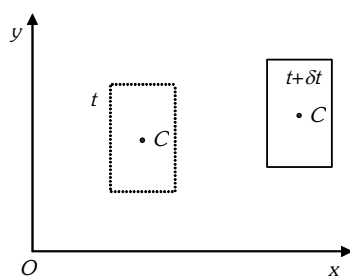
Actual material bodies are never perfectly rigid and are deformed under action of external forces. When these deformations are small enough to be considered during their course of motion, the body is assumed a rigid body. Hence, all solid objects such as stone, ball, vehicles etc are considered as rigid bodies while analyzing their translation as well as rotation motion.

To analyze rotation of a body relative motion between its particles cannot be neglected and size of the body becomes a considerable factor. This is why study of rotation motion is also known as *mechanics of rigid bodies*.

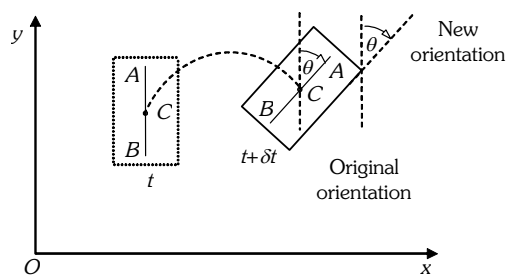
Rotation Motion of a Rigid Body

Any kind of motion of a body is identified by change in position or change in orientation or change in both. If a body changes its orientation during its motion it said to be in rotation motion.

In the following figures, a rectangular plate is shown moving in the x - y plane. The point C is its mass center. In the first case it does not changes orientation, therefore is in pure translation motion. In the second case it changes its orientation by during its motion. It is a combination of translation and rotation motion.



Pure Translation

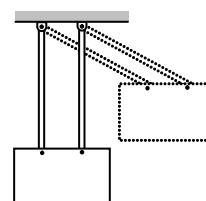


Combination of translation and rotation

Rotation i.e. change in orientation is identified by the angle through which a linear dimension or a straight line drawn on the body turns. In the figure this angle is shown by θ .

Ex. Identify Translation and rotation motion

A rectangular plate is suspended from the ceiling by two parallel rods each pivoted at one end on the plate and at the other end on the ceiling. The plate is given a side-push to oscillate in the vertical plane containing the plate. Identify motion of the plate and the rods.



Sol.

Neither of the linear dimensions of the plate turns during the motion. Therefore, the plate does not change its orientation. Here edges of the body easily fulfill our purpose to measure orientation; therefore, no line is drawn on it.

The plate is in *curvilinear translation motion* and the rods are in rotation motion.

Types of Motions involving Rotation

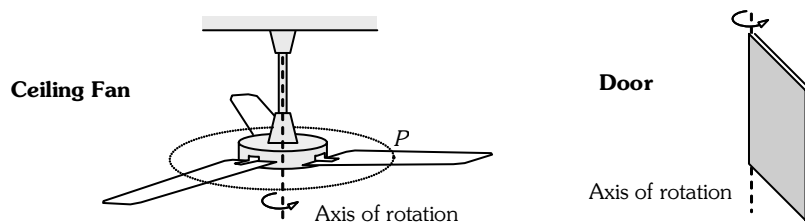
Motion of body involving rotation can be classified into following three categories.

- I Rotation about a fixed axis.
- II Rotation about an axis in translation.
- III Rotation about an axis in rotation

Rotation about a fixed axis

Rotation of ceiling fan, potter's wheel, opening and closing of doors and needles of a wall clock etc. come into this category.

When a ceiling fan rotates, the vertical rod supporting it remains stationary and all the particles on the fan move on circular paths. Circular path of a particle P on one of its blades is shown by dotted circle. Centers of circular paths followed by every particle are on the central line through the rod. This central line is known as *axis of rotation* and is shown by a dashed line. All the particles on the axis of rotation are at rest, therefore the axis is stationary and the fan is in rotation about this fixed axis.

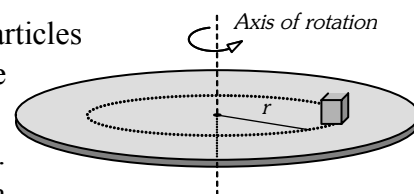


A door rotates about a vertical line that passes through its hinges. This vertical line is the axis of rotation. In the figure, the axis of rotation is shown by dashed line.

Axis of rotation

An imaginary line perpendicular to plane of circular paths of particles of a rigid body in rotation and containing the centers of all these circular paths is known as *axis of rotation*.

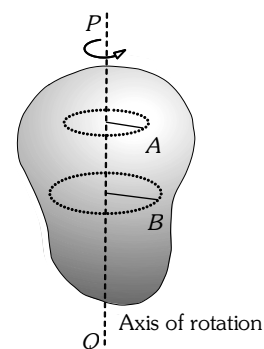
It is not necessary that the axis of rotation pass through the body. Consider system shown in the figure, where a block is fixed on a rotating disk. The axis of rotation passes through the center of the disk but not through the block.



Important observations

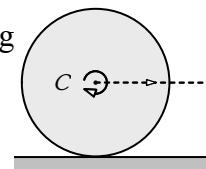
Let us consider a rigid body of arbitrary shape rotating about a fixed axis PQ passing through the body. Two of its particles A and B are shown moving on their circular paths.

- ⇒ All of its particles, not on the axis of rotation, move on circular paths with centers on the axis of rotation. All these circular paths are in parallel planes that are perpendicular to the axis of rotation.
- ⇒ All the particles of the body cover same angular displacement in the same time interval, therefore all of them move with the same angular velocity and angular acceleration.
- ⇒ Particles moving on circular paths of different radii move with different speeds and different magnitudes of linear acceleration. Furthermore, no two particles in the same plane perpendicular to the axis of rotation have same velocity and acceleration vectors.
- ⇒ All the particles on a line parallel to the axis of rotation move circular paths of the same radius therefore have same velocity and acceleration vectors.
- ⇒ Consider two particles in a plane perpendicular to the rotational axis. Every such particle on a rigid body in rotation motion moves on circular path relative to another one. Radius of the circular path equals to the distance between the particles. In addition, angular velocity and angular acceleration equals to that of rotation motion of the body.



Rotation about an axis in translation

Rotation about an axis in translation includes a broad category of motions. Rolling is an example of this kind of motion. A rod lying on table when pushed from its one in its perpendicular direction also executes this kind of motion. To understand more let us discuss few examples.



Consider rolling of wheels of a vehicle, moving on straight level road. Relative to a reference frame, moving with the vehicle wheel appears rotating about its stationary axel. The rotation of the wheel from this frame is rotation about fixed axis. Relative to a reference frame fixed with the ground, the wheel appears rotating about the moving axel, therefore, rolling of a wheel is superposition two simultaneous but distinct motions – rotation about the axel fixed with the vehicle and translation of the axel together with the vehicle.

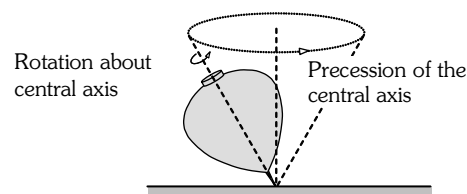
Important observations

- ⇒ Every particle of the body always remains in a plane perpendicular to the rotational axis. Therefore, this kind of motion is also known as *general plane motion*.
- ⇒ Relative to every particle another particle in a plane perpendicular to axis of rotation moves on circular path. Radius of the circular path equals to the distance between the particles and angular velocity and angular acceleration equals to that of rotation motion of the body.
- ⇒ Rotation about axis in translation is superposition of pure rotation about the axis and simultaneous translation motion of the axis.

Rotation about an axis in rotation.

In this kind of motion, the body rotates about an axis that also rotates about some other axis. Analysis of rotation about rotating axes is not in the scope of JEE, therefore we will discuss it to have an elementary idea only.

As an example consider a rotating top. The top rotates about its central axis of symmetry and this axis sweeps a cone about a vertical axis. The central axis continuously changes its orientation, therefore is in rotation motion. This type of rotation in which the axis of rotation also rotates and sweeps out a cone is known as *precession*.



Another example of rotation about axis in rotation is a table-fan swinging while rotating. Table-fan rotates about its horizontal shaft along which axis of rotation passes. When the rotating table-fan swings, its shaft rotates about a vertical axis.

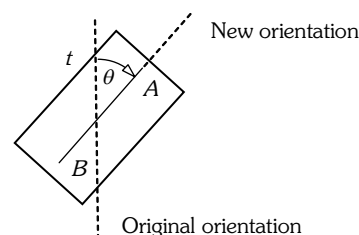
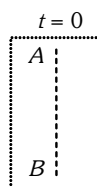
Angular displacement, angular velocity and angular acceleration

Rotation motion is the change in orientation of a rigid body with time. It is measured by turning of a linear dimension or a straight line drawn on the body.

In the figure is shown at two different instants $t = 0$ and t a rectangular plate moving in its own plane. Change in orientation during time t equals to the angle θ through which all the linear dimensions of the plate or a line AB turns.

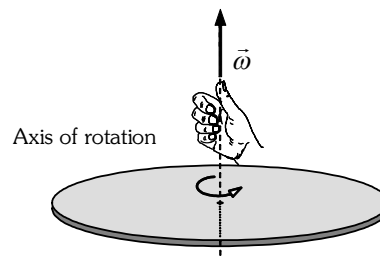
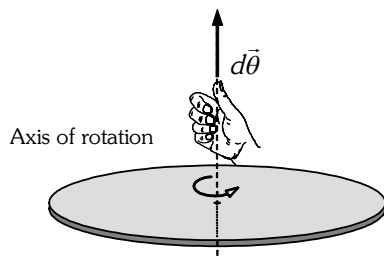
If the angle θ continuously changes with time t , instantaneous angular velocity ω and angular acceleration α for rotation of the body are defined by the following equations.

$$\omega = \frac{d\theta}{dt} \quad [1] \quad \alpha = \frac{d^2\theta}{dt^2} = \frac{d\omega}{dt} = \omega \frac{d\omega}{d\theta} \quad [2]$$

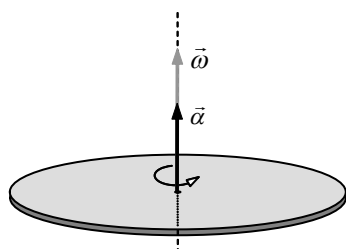


Direction of angular motion quantities

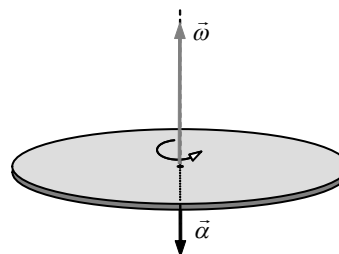
Angular displacement, angular velocity and angular acceleration are known as angular motion quantities. Infinitesimally small angular displacement, instantaneous angular velocity and angular acceleration are vector quantities. Direction of infinitesimally small angular displacement and instantaneous angular velocity is given by the right hand rule. For a disk rotating as shown in the figure, the angular velocity points upwards along the axis of rotation.



The direction of angular acceleration depends, whether angular velocity increases or decreases with time. For increasing angular velocity, the angular acceleration vector points in the direction of angular velocity vector and for decreasing angular velocity, the angular acceleration vector points opposite to the angular velocity vector.



Angular acceleration: Increasing angular speed



Angular acceleration: Decreasing angular speed

In rotation about fixed axis and rotation about axis in translation, the axis of rotation does not rotate and angular velocity and acceleration always point along the axis of rotation. Therefore, in dealing these kinds of motions, the angular motion quantities can be used in scalar notations by assigning them positive sign for one direction and negative sign for the opposite direction.

These quantities have similar mathematical relations as position coordinate, velocity, acceleration and time have in rectilinear motion.

⇒ A body rotating with constant angular velocity ω and hence zero angular acceleration is said to be uniform rotation. Angular position θ is given by equation

$$\theta = \theta_o + \omega t \quad [3]$$

⇒ Thus for a body rotating with uniform angular acceleration α , the angular position θ and angular velocity ω can be expressed by the following equation.

$$\omega = \omega_o + \alpha t \quad [4]$$

$$\theta = \theta_o + \omega_o t + \frac{1}{2} \alpha t^2 = \theta_o + \frac{1}{2} (\omega_o + \omega) t \quad [5]$$

$$\omega^2 = \omega_o^2 + 2\alpha(\theta - \theta_o) \quad [6]$$

Ex. A disk rotates about a fixed axis. Its angular velocity ω varies with time according to equation $\omega = at + b$. At the instant $t = 0$ its angular velocity is 1.0 rad/s at angular position is 2 rad and at the instant $t = 2$ s, angular velocity is 5.0 rad/s. Determine angular position θ and angular acceleration α when $t = 4$ s.

Sol. The given equation $\omega = at + b$ has form similar to eq.[4], therefore motion is rotation with uniform angular acceleration. Initial angular velocity $= \omega_o = b = 1.0 \text{ rad/s}$, Angular acceleration $\alpha = a$,

$$\theta = \frac{1}{2}at^2 + t + c$$

Since at $t = 0$, $\omega = 1.0 \text{ rad/s}$, we obtain the constant c .

$$\text{Initial angular position} = \theta_o = c = 2.0 \text{ rad}$$

Since at $t = 2.0 \text{ s}$ angular velocity is 5.0 rad/s , from given expression of angular velocity, we have

$$\omega = at + b \rightarrow \text{Substituting } b = 1.0 \text{ rad/s, } t = 2.0 \text{ s and } \omega = 5.0 \text{ rad/s, we have } a = 2.0 \text{ rad/s}^2$$

Now we can write expressions for angular position, angular velocity and angular acceleration.

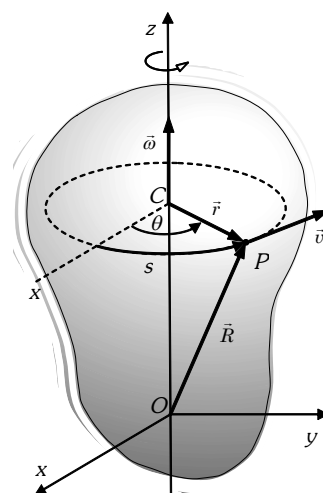
$$\theta = t^2 + t + 2.0 \quad (1) \quad \omega = 2.0t + 1.0 \quad (2)$$

From the above equations, we can calculate angular position, angular velocity and angular acceleration at $t = 4.0 \text{ s}$

$$\theta_4 = 22 \text{ rad, } \omega_4 = 9.0 \text{ rad/s, } \alpha = 2.0 \text{ rad/s}^2 \quad \text{Ans.}$$

Kinematics of rotation about fixed axis

In figure is shown a rigid body of arbitrary shape rotating about the z -axis. In the selected frame (here the coordinate system) all the three axes are at rest, therefore the z -axis that is the axis of rotation is at rest and the body is in fixed axis rotation. All of its particles other than those on the z -axis move on circular paths with their centers on the z -axis. All these circular paths are parallel to the x - y plane. In the figure, one of its particles P is shown moving with velocity $\vec{v}$ on a circular path of radius r and center C . Its position vector is $\vec{R}$. It were at the line Cx at $t = 0$ and at the position shown at the instant t . During time interval t , it covers the circular arc of length s and its radius vector turns through angle θ .



In an infinitesimally small time interval dt let, the particle covers infinitesimally small distance ds along its circular path.

$$d\vec{s} = d\vec{\theta} \times \vec{r} = d\vec{\theta} \times \vec{R} \quad [7]$$

$$\vec{v} = \frac{d\vec{s}}{dt} = \frac{d\vec{r}}{dt} = \frac{d\vec{R}}{dt} \quad [8]$$

From eq. [7] and [8] we have

$$\vec{v} = \vec{\omega} \times \vec{r} = \vec{\omega} \times \vec{R} \quad [9]$$

The above equation tells us the relation between the linear and angular velocity. Now we explore relation between the linear and angular accelerations. For the purpose, differentiate the above equation with respect to time.

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d\vec{\omega}}{dt} \times \vec{r} + \vec{\omega} \times \frac{d\vec{r}}{dt} = \vec{\alpha} \times \vec{r} + \vec{\omega} \times \vec{v} \quad [10]$$

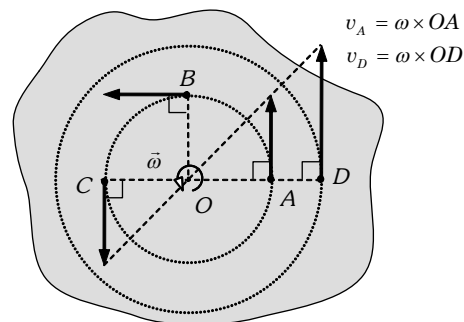
The first term on the RHS points along the tangent in the direction of the velocity vector and it is known as tangential acceleration $\vec{a}_t$ same as we have in circular motion. In addition, the second term point towards the center C . It is known as centripetal acceleration of normal component $\vec{a}_n$ of acceleration same as in circular motion. Now we have

$$\text{Tangential acceleration} \quad \vec{a}_t = \vec{\alpha} \times \vec{r} \quad [11]$$

$$\text{Normal acceleration} \quad \vec{a}_n = \vec{\omega} \times \vec{v} = -\omega^2 \vec{r} \quad [12]$$

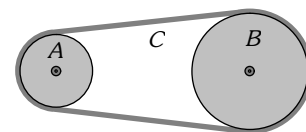
How to Locate Axis of Rotation

Every particle in a plane perpendicular to the axis of rotation move with different velocities and accelerations, moreover, they all have the same angular velocity and angular acceleration. Such a section of a body in rotation is shown here. The particles A , B and C at equal distance from the axis of rotation move with equal speeds v_A and the particle D moves with speed v_D on concentric circular paths. The location of rotational axis can be determined by any of the two graphical techniques.



- ⇒ Lines perpendicular to velocity vectors and passing through the particles, whose velocity vectors are neither parallel nor antiparallel intersect at the axis of rotation. See pairs of particles A and B , B and C and B and D .
- ⇒ Lines perpendicular to velocity vectors and passing through the particles, whose velocity vectors are either parallel or antiparallel, coincide and intersect the line joining tips of their velocity vectors at the axis of rotation. Refer pairs of particles A and C , A and D and C and D .

Ex. A belt moves over two pulleys A and B as shown in the figure. The pulleys are mounted on two fixed horizontal axels. Radii of the pulleys A and B are 50 cm and 80 cm respectively. Pulley A is driven at constant angular acceleration 0.8 rad/s^2 until the pulley B acquires an angular velocity of 10 rad/s . The belt does not slide on either of the pulleys.



- (a) Find acceleration of a point C on the belt and angular acceleration of the pulley B .
- (b) How long after the pulley B achieve angular velocity of 10 rad/s .

Sol. Since the belt does not slide on the pulleys, magnitude of velocity and acceleration of any point on the belt are same as velocity tangential acceleration of any point on periphery of either of the pulleys. Using the above fact with eq.[11], we have

$$\vec{a}_T = \vec{\alpha} \times \vec{r} \rightarrow$$

$$a_C = \alpha_A r_A = \alpha_B r_B$$

Substituting $r_A = 0.5 \text{ m}$, $r_B = 0.8 \text{ m}$ and $\alpha = 0.8 \text{ rad/s}^2$, we have

$$a_C = 5 \text{ m/s}^2 \text{ and } \alpha_B = \frac{a_C}{r_B} = \frac{\alpha_A r_A}{r_B} = 0.5 \text{ rad/s}^2 \quad \text{Ans.}$$

From eq. [4], we have

$$\omega = \omega_o + \alpha t \rightarrow$$

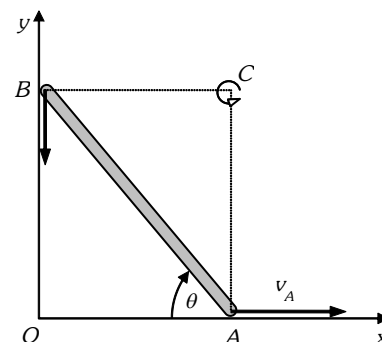
$$t = \frac{\omega_B - \omega_{Bo}}{\alpha_B}$$

Substituting $\omega_{Bo} = 0$, $\omega_B = 10 \text{ rad/s}$ and $\alpha_B = 0.5 \text{ rad/s}^2$, we have $t = 20 \text{ s}$ **Ans.**

Kinematics of rotation about axis in translation

In this kind of motion, the body rotates about an axis and the axis moves without rotation. Rolling is a very common example of this kind of motion.

As an example consider a rod whose ends A and B are sliding on the x and y -axis as shown in the figure. Change in its orientation measured by change in angle θ indicates that the rod is in rotation. Perpendiculars drawn to velocity vector of its end points intersect at the axis of rotation, which is continuously changing its position.

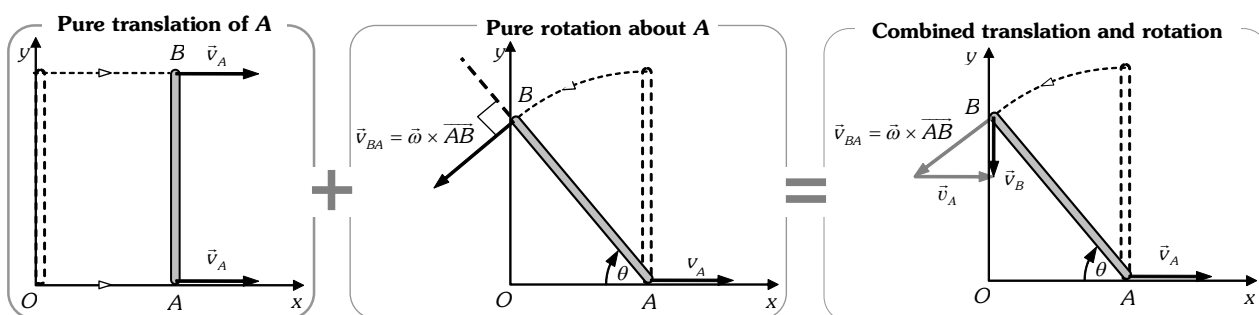


Instantaneous Axis of Rotation (IAR)

It is a mathematical line about that a body in combined translation and rotation can be conceived in pure rotation at an instant. It continuously changes its location.

Now we explore how the combined translation and rotational motion of the rod is superposition of translation motion of any of its particle and pure rotation about an axis through that particle.

Consider motion of the rod from beginning when it was parallel to the y-axis. In the following figure translation motion of point A is superimposed with pure rotation about A.



The motion of the rod can be conceived as superposition of translation of point A and simultaneous rotation about an axis through A.

The same experiment can be repeated to demonstrate that motion of the rod can be conceived as superposition of translation of any of its particle and simultaneous rotation about an axis through that particle.

Considering translation of A and rotation about A this fact can be expressed by the following equation.

Combined Motion = Translation of point A + Pure rotation about point A

$$\vec{v}_B = \vec{v}_A + \vec{v}_{B/A} \quad [13]$$

Since point B moves relative A moving on circular path its velocity relative to A is given by the equation

$$\vec{v}_{BA} = \vec{\omega} \times \vec{AB}.$$

Now we have

$$\vec{v}_B = \vec{v}_A + \vec{\omega} \times \vec{AB} \quad [14]$$

The above fact is true for any rigid body in combined translation and rotation motion.

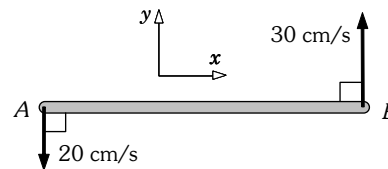
Rotation about an axis in translation of a rigid body can be conceived as well as analyzed as superposition of translation motion of any of its particle and simultaneous rotation about an axis passing through that particle provided that the axis is parallel to the actual one.

Similar to eq.[13], we can write equation for acceleration.

$$\left. \begin{aligned} \vec{a}_A &= \vec{a}_B + \vec{a}_{BA} \\ \vec{a}_A &= \vec{a}_B + \overbrace{\vec{a}_{BAT} + \vec{a}_{BA_n}} \\ \vec{a}_A &= \vec{a}_B + \vec{\alpha} \times \vec{AB} + -\omega^2 \vec{AB} \end{aligned} \right\} \quad [15]$$

Ex. A 100 cm rod is moving on a horizontal surface. At an instant, when it is parallel to the x-axis its ends A and B have velocities 30 cm/s and 20 cm/s as shown in the figure.

- (a) Find its angular velocity and velocity of its center.
(b) Locate its instantaneous axis of rotation.



Sol.

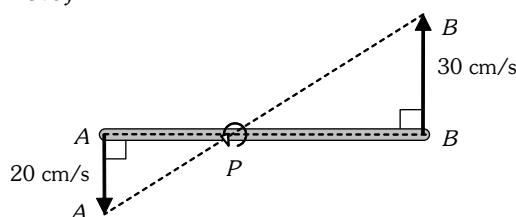
Let the rod is rotating anticlockwise, therefore its angular velocity is given by $\vec{\omega} = \omega \hat{k}$. Velocity vectors of all the points on the rod and its angular velocity must satisfy the relative motion eq.[14].

- (a) Substituting velocities $\vec{v}_A = -20\hat{j}$ cm/s and $\vec{v}_B = 30\hat{j}$ cm/s and angular velocity $\vec{\omega}$ in eq.[14], we have $\vec{v}_B = \vec{v}_A + \vec{\omega} \times \vec{AB} \rightarrow \omega = 0.5 \text{ rad/s}$ **Ans.**

Velocity vector of the center C of the rod also satisfy the following equation.

$$\vec{v}_C = \vec{v}_A + \vec{\omega} \times \vec{AC} \rightarrow \vec{v}_C = -20\hat{j} + 0.5\hat{k} \times 50\hat{i} = 5.0\hat{j} \text{ cm/s} \quad \text{Ans.}$$

- (b) Here velocity vectors of the particles A and B are antiparallel, therefore the instantaneous axis of rotation passes through intersection of the common perpendicular to their velocity vectors and a line joining tips of the velocity vectors. The required geometrical construction is shown in the following figure.



Since triangles AA'P and BB'P are similar and $AB = 100 \text{ cm}$, we have $AP = 40 \text{ cm}$.

The instantaneous axis of rotation passes through the point P, which is 40 cm from A. **Ans.**

Analytical Approach.

The instantaneous center of rotation is at instantaneous rest. Using this fact in eq.[14], we have

$$\vec{v}_P = \vec{v}_A + \vec{\omega} \times \vec{AP} \rightarrow \vec{0} = -20\hat{j} + 0.5\hat{k} \times (AP)\hat{j} \Rightarrow AP = 40 \text{ cm} \quad \text{Ans.}$$

Concept of Rotational Inertia (Moment of inertia)

Total mass of a body in translation motion is the measure of its inertia to translation motion. Similarly if a point mass m is rotating about an axis at a distance r from the axis, then term mr^2 provides suitable measure of its inertia to rotation motion. *The inertia to rotation motion is known as rotational inertia or more commonly moment of inertia.*

Moment of inertia of a rigid body

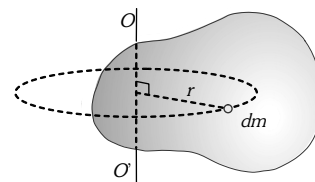
A rigid body is continuous distribution of mass and can be assumed consisting of infinitely large number of point particles. If one of the point particle of infinitely small mass dm is at a distance r from the axis of rotation OO' , the moment of inertia of this point particle is given by

$$dI_o = r^2 dm$$

The moment of inertia of the whole body about the axis OO' can now be obtained by integrating term of the above equation over the limits to cover whole of the body.

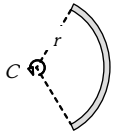
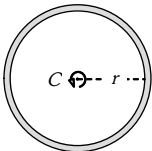
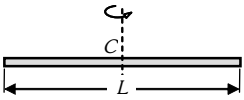
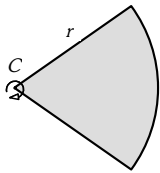
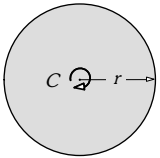
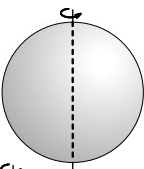
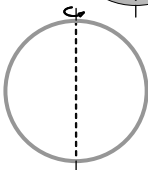
$$I_o = \int dI_o = \int r^2 dm$$

Expression for moment of inertia contains product of two terms. One of them is the mass of the body and the other is a characteristic dimension, which depends on the manner how mass of the body is distributed relative to the axis of rotation. Therefore moment of inertia of a rigid body depends on the mass of the body and distribution of the mass relative to the axis of rotation. Obviously for uniform bodies expression of moment of inertia depends on their shape and location and orientation of the axis of rotation. Based on these facts we can conclude



1. If mass distribution is similar for two bodies about an axis, expressions of their moment of inertia must be of the same form about that axis.
2. If the whole body or any of its portions is shifted parallel to the axis of rotation, moment of inertia remains unchanged.

Moment of Inertia for some commonly used bodies

Body	Axis	Moment of Inertia
Uniform thin rod bent into shape of an arc of mass m		Passing through center and perpendicular to the plane containing the arc $I_C = mr^2$
Uniform ring of mass m		Passing through center and perpendicular to the plane containing the arc or the centroidal axis. $I_C = mr^2$
Straight uniform rod		Passing through center and perpendicular to the rod or the centroidal axis. $I_C = \frac{mL^2}{12}$
Sector of a uniform disk of mass m		Passing through center and perpendicular to the plane containing the sector. $I_C = \frac{mr^2}{2}$
Uniform disk of mass m		Passing through center and perpendicular to the plane containing the disk or the centroidal axis. $I_C = \frac{mr^2}{2}$
Homogeneous cylinder of mass m		Axis of the cylinder or the centroidal axis. $I_C = \frac{mr^2}{2}$
Homogeneous sphere of mass m		Diameter or the centroidal axis $I_C = \frac{2}{5}mR^2$
Spherical shell of mass m		Diameter or the centroidal axis $I = \frac{2}{3}mR^2$

Theorems on Moment of Inertia

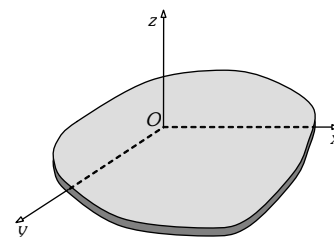
Moment of inertias of a rigid body about different axes may be different. There are two theorems known as *theorem of perpendicular axes* and *theorem of parallel axes*, which greatly simplify calculation of moment of inertia about an axis if moment of inertia of a body about another suitable axis is known.

Theorem of Perpendicular Axes

This theorem is applicable for a rigid body that lies entirely within a plane i.e. a laminar body or a rod bent into shape of a plane curve. The moment of inertia I_x , I_y and I_z of the body about the x , y and z -axis can be expressed by the following equations.

$$I_z = I_y + I_x$$

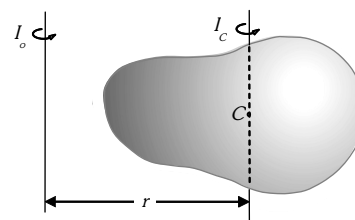
For a planar body, the moment of inertia about an axis perpendicular to the plane of the body is the sum of the moment of inertias about two perpendicular axes in the plane of the object provided that all the three axes are concurrent.

**Theorem of Parallel Axes**

This theorem also known as Steiner's theorem can be used to determine the moment of inertia of a rigid body about any axis, if the moment of inertia of the body about a parallel axis passing through mass center of the body and perpendicular distance between both the axes is known.

Consider a body of arbitrary shape and mass m shown in the figure. Its moment of inertia I_o and I_c are defined about two parallel axes. The axis about which moment of inertia I_c is defined passes through the mass center C . Separation between the axes is r . These two moment of inertias are related by the following equation.

$$I_o = I_c + Mx_c^2$$



The above equation is known as the theorem of parallel axes or Steiner's theorem.

- ⇒ The moment of inertia about any axis parallel to an axis through the mass center is given by sum of moment of inertia about the axis through the mass center and product term of mass of the body and square of the distance between the axes.
- ⇒ Among all the parallel axes the moment of inertia of a rigid body about the axis through the mass center is the minimum moment of inertia.

The second term added to the moment of inertia I_c about the centroidal axis in the above equation can be recognized as the moment of inertia of a particle of mass equal to that of the body and located at its mass center. It again reveals that the plane motion of a rigid body is superposition of pure rotation about the mass center or centroidal rotation and translation of its mass center.

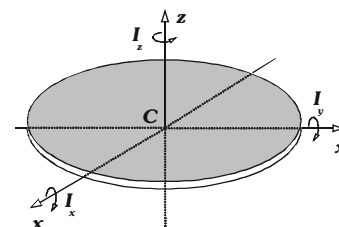
Ex. Find moment of inertia of a uniform disk of mass m and radius r about one of its diameter.

Sol. In the adjoining figure a disk is shown with two of its diameter perpendicular to each other. These diameters are along the x and the y -axis of a coordinate system. The x -axis is perpendicular to the plane of the disk and passes through its center is also shown. Since the disk is symmetric about both the diameters, moment of inertias about both the diameters must be equal. Thus substituting this in the theorem of perpendicular axes, we have

$$I_z = I_y + I_x \rightarrow I_z = 2I_x = 2I_y$$

Moment of inertia of the disk about the z -axis is $I_z = \frac{1}{2}mr^2$. Substituting it in the above equation, we have

$$I_x = I_y = \frac{1}{2}I_z = \frac{1}{4}mr^2 \quad \text{Ans.}$$



Radius of Gyration

It is the radial distance from a rotation axis at which the mass of an object could be concentrated without altering the moment of inertia of the body about that axis.

If the mass m of the body were actually concentrated at a distance k from the axis, the moment of inertia about that axis would be mk^2 .

$$k = \sqrt{\frac{I}{m}}$$

The radius of gyration has dimensions of length and is measured in appropriate units of length such as meters.

DYNAMICS OF RIGID BODY

Torque: Moment of a force

Torque is rotational analogue of force and expresses tendency of a force applied to an object to cause the object to rotate about a given point.

To investigate further let us discuss an experiment. Consider a rod pivoted at the point O . A force $\vec{F}$ is applied on it at the point P .

The component $F \cos \theta$ of the force along the rod is counterbalanced by the reaction force of the pivot and cannot contribute in rotating the rod. It is the component $F \sin \theta$ of the force perpendicular to the rod, which is responsible for rotation of the rod. Moreover, farther is the point P from O , where the force is applied easier is to rotate the rod. This is why handle on a door is attached as far away as possible from the hinges.

Magnitude of torque of a force is proportional to the product of distance of point of application of the force from the pivot and magnitude of the perpendicular component $F \sin \theta$ of the force. Denoting torque by symbol τ , the distance of point of application of force from the pivot by r , we can write

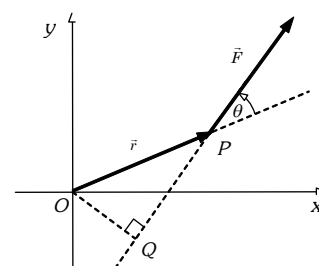
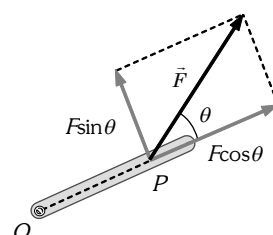
$$\tau_o \propto rF \sin \theta$$

Since rotation has sense of direction, torque should also be a vector. Its direction is given by right hand rule. Now we can express torque by the cross product of $\vec{r}$ and $\vec{F}$.

$$\vec{\tau}_o = \vec{r} \times \vec{F}$$

Here constant of proportionality has been assumed a dimensionless number unity because a unit of torque has been chosen as product of unit of force and unit of length.

The geometrical construction shown in figure suggests a simple way to calculate torque. The line OQ ($= r \sin \theta$) known as *moment arm*, is the length of perpendicular drawn from O on the line of action of the force. The magnitude of the torque equals to the product of OQ and magnitude of the force $\vec{F}$.



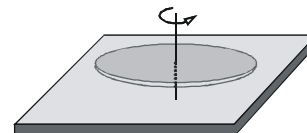
Torque about a Point and Torque about an Axis

We have defined *torque of a force about a point* as the moment of the force about that point. In dealing with rotation about a fixed axis we need to know torque about the axis rotation.

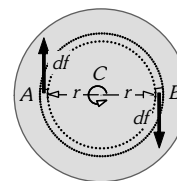
When a body is in plane motion the net torque of all the forces including the forces necessary to restrain rotation of the axis is along the axis of rotation. It is known as torque about the axis. Torque of a force about an axis of rotation equals to the moment of force about the point where plane of motion of the point of application of the force intersects the axis.

In analyzing plane motion we always consider torque about an axis under consideration and in rest of the book by the term torque of force we mean torque about an axis.

Ex. A uniform disk of mass M and radius R rotating about a vertical axis passing through its center and perpendicular to its plane is placed gently on a rough horizontal ground, where coefficient of friction is μ . Calculate torque of the frictional forces.



Sol. When the disk rotates on the ground, kinetic friction acts at every contact point. Since the gravity acts uniformly everywhere and the disk is also uniform, the normal reaction from the ground is uniformly distributed over the entire contact area. Consider two diametrically opposite identical portions A and B of the disk each of mass dm at distance r from the center as shown in the adjacent figure. The normal reaction from the ground on each of these portions equals to their weights and hence frictional forces are $df = \mu dm g$.



Consider a ring of radius r and width dr shown by dashed lines. Net torque $d\tau_c$ of friction force on this ring can easily be expressed by the following equation.

$$d\tau_c = r\mu(\text{mass of the ring})g = r\mu\left(\frac{\text{Mass of the disk}}{\text{Area of the disk}} \times \text{Area of the ring}\right)g = r\mu\left(\frac{2Mrdr}{R^2}\right)g$$

Integrating both sides of the above equation, we have

$$\tau_c = \frac{2\mu Mg}{R^2} \int_0^R r^2 dr = \frac{2}{3} \mu MgR \quad \text{Ans.}$$

Rotational equilibrium

A rigid body is said to be state of rotational equilibrium if its angular acceleration is zero. Therefore a body in rotational equilibrium must either be in rest or rotation with constant angular velocity.

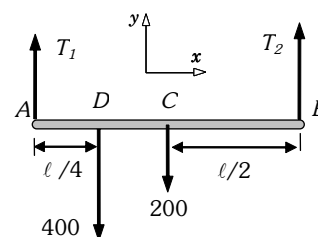
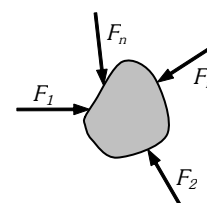
Since scope of JEE syllabus is confined only to rotation about a fixed axis or rotation about an axis in translation motion, the discussion regarding rotational equilibrium is limited here to situations involving only coplanar forces. Under these circumstances the necessary and sufficient condition for rotational equilibrium is

If a rigid body is in rotational equilibrium under the action of several coplanar forces, the resultant torque of all the forces about any axis perpendicular to the plane containing the forces must be zero.

In the figure a body is shown under the action of several external coplanar forces $F_1, F_2, \dots, F_i$ and F_n .

$$\sum \bar{\tau}_P = 0$$

Here P is a point in the plane of the forces about which we calculate torque of all the external forces acting on the body. The flexibility available in selection of the point P provides us with advantages that we can select such a point about which torques of several unknown forces will become zero or we can make as many number of equations as desired by selecting several different points. The first situation yields to a simpler equation to be solved and second situation though does not give independent equation, which can be used to determine additional unknowns yet may be used to check the solution.



The above condition reveals that a body cannot be in rotational equilibrium under the action of a single force unless the line of action passes through the mass center of the body.

A case of particular interest arises where only three coplanar forces are involved and the body is in rotational equilibrium. It can be shown that *if a body is in rotational equilibrium under the action of three forces, the lines of action of the three forces must be either concurrent or parallel*. This condition provides us with a graphical technique to analyze rotational equilibrium.

Equilibrium of Rigid Bodies

A rigid body is said to be in equilibrium, if it is in translational as well as rotational equilibrium both. To analyze such problems conditions for both the equilibriums must be applied.

Ex. A uniform rod of 20 kg is hanging in horizontal position with the help of two threads. It also supports a 40 kg mass as shown in the figure. Find the tension developed in each thread.

Sol. Free body diagram of the rod is shown in the figure.

Translational equilibrium

$$\Sigma F_y = 0 \rightarrow T_1 + T_2 = 400 + 200 = 600 \text{ N} \quad (1)$$

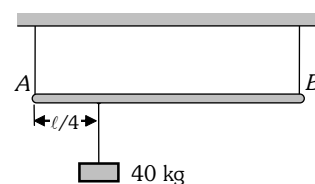
Rotational equilibrium: Applying the condition about A, we get T_2 .

$$\Sigma \tau_A = 0 \rightarrow 400(l/4) + 200(l/2) - T_2 l = 0$$

$$T_2 = 200 \text{ N} \quad \text{Ans.}$$

Similarly writing torque equation about B, we have

$$\Sigma \tau_B = 0 \rightarrow T_1 = 400 \text{ N. Ans.}$$



Force and Torque equations in General Plane Motion

Motion of a rigid body either pure rotation or rotation about axis in translation can be thought and analyzed as superposition of translation of any of its particle and simultaneous rotation about an axis passing through that particle provided that the axis remain parallel to the original one. As far as kinematics is concerned this particle may or may not be the mass center. Whereas in dealing with kinetics, general plane motion is conceived as superposition of translation motion of the mass center and simultaneous centroidal rotation.

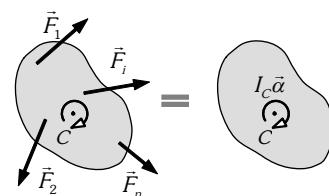
To make use the above idea and equations developed in the previous section we classify pure rotation i.e. rotation about fixed axis into two categories and deal with general plane motion as the third category.

Pure centroidal rotation: Rotation about fixed axis through mass centre

In this kind of rotation motion the axis of rotation passes through the mass center and remain fixed in space. Rotation of ceiling fan is a common example of this category. It is a subcategory of pure rotation. The axis of rotation passes through the mass center and remains fixed. In this kind of rotation the mass center of the body does not move.

In the figure, free body diagram and kinetic diagram of a body rotating about a fixed axis passing through its mass center C is shown. The mass center of the body does not accelerate; therefore we only need to write the torque equation.

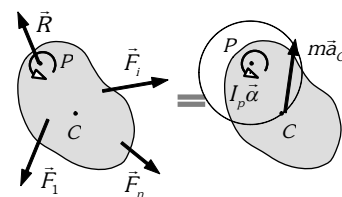
$$\Sigma \tau_C = I_C \alpha$$



Rotation about fixed axis not passing through mass center

In this kind of rotation the axis of rotation remains fixed and does not pass through the mass center. Rotation of door is a common example of this category. Doors are hinged about their edges; therefore their axis of rotation does not pass through the mass center. In this kind of rotation motion the mass center executes circular motion about the axis of rotation.

In the figure, free body diagram and kinetic diagram of a body rotating about a fixed axis through point P is shown. It is easy to conceive that as the body rotates its mass center moves on a circular path of radius $\vec{r}_{P/C}$. The mass center of the body is in translation motion with acceleration $\vec{a}_C$ on circular path of radius $r_{P/C}$. To deal with this kind of motion, we have to make use of both the force and the torque equations.



Translation of mass center

$$\Sigma \vec{F}_i = M\vec{a}_C = M\vec{\alpha} \times \vec{r}_{C/P} - M\omega^2 \vec{r}_{C/P}$$

Centroidal Rotation

$$\Sigma \vec{\tau}_C = I_C \vec{\alpha}$$

Making use of parallel axis theorem

$$(I_P = Mr_{P/C}^2 + I_C) \text{ and } \vec{a}_{C/P} = \vec{\alpha} \times \vec{r}_{C/P} - \omega^2 \vec{r}_{C/P}$$

we can write the following equation also.

Pure Rotation about P

$$\Sigma \vec{\tau}_P = I_P \vec{\alpha}$$

General Plane Motion: Rotation about axis in translation motion

Rotation of bodies about an axis in translation motion can be dealt with either as superposition of translation of mass center and centroidal rotation or assuming pure rotation about the instantaneous axis of rotation. In the figure is shown the free body diagram and kinetic diagram of a body in general plane motion.

Translation of mass center

$$\sum_{i=1}^n \vec{F}_i = M\vec{a}_C$$

Centroidal Rotation

$$\sum_{i=1}^n \vec{\tau}_C = I_C \vec{\alpha}$$

This kind of situation can also be dealt with considering it rotation about IAR. It gives sometimes quick solutions, especially when IAR is known and forces if acting at the IAR are not required to be found.

Example

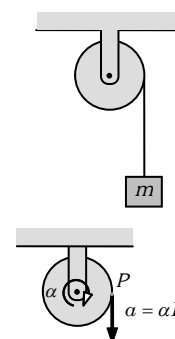
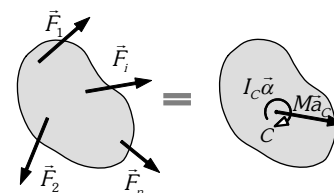
A block of mass m is suspended with the help of a light cord wrapped over a cylindrical pulley of mass M and radius R as shown in the figure. The system is released from rest. Find the angular acceleration of the pulley and the acceleration of the block.

Solution.

After the system is released, the block is in translation motion and the pulley in rotation about an axis passing through its mass center i.e. in pure rotation.

Let the block moves vertically down with acceleration a pulling the cord down and causing the pulley to rotate clockwise. Since the cord is inextensible every point on its vertical portion and point of contact P of the pulley move down with acceleration a as shown in the adjacent figure. It is the tangential acceleration of point P so the angular acceleration α of the pulley rotating in clockwise sense is given by

$$a = \alpha R \quad (1)$$



The forces acting on the pulley and on the block are shown in their free-body diagrams along with the effective torque $I_C \alpha$ of the pulley and effective force ma of the block. Here T is the tension in the string, R is the reaction by the axel of the pulley, Mg is weight of the pulley and mg is weight of the block.

The pulley is in rotation about fixed axis through its mass center so we use eq. .

$$\sum \vec{\tau}_C = I_C \vec{\alpha} \rightarrow TR = I_C \alpha$$

After substituting $I_C = \frac{1}{2}MR^2$ and α from eq. (1), we have

$$T = \frac{1}{2}Ma \quad (2)$$

The block is in translation motion, so we use Newton's second law

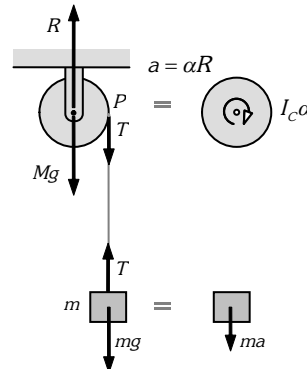
$$\sum \vec{F} = m\vec{a} \rightarrow mg - T = ma \quad (3)$$

From equation (2) and (3), we have

$$\text{Acceleration of the block} \quad a = \frac{2mg}{M+2m} \quad \text{Ans.}$$

From eq. (1) and the above, we have

$$\alpha = \frac{2mg}{R(M+2m)} \quad \text{Ans.}$$



Ex. A thread is wrapped around a uniform disk of radius r and mass m . One end of the thread is attached to a fixed support on the ceiling and the disk is held stationary in vertical plane below the fixed support as shown in the figure. When the disk is set free, it rolls down due to gravity. Find the acceleration of the center of the disk and tension in the thread.

Sol. The point P , where the thread leaves the disk is always at instantaneous rest; therefore the disk can be assumed rolling without slipping with ICR at point P . Acceleration of the mass center $\vec{a}_C$ and angular acceleration of the disk are shown in the adjacent figure. Applying condition for rolling on stationary surface, we have

$$\vec{a}_C = \vec{\alpha} \times \vec{r}_{C/P} \rightarrow a_C = \alpha r \quad (1)$$

The disk rolls down on the vertical stationary thread. Its motion can either be analyzed as superposition of translation of the mass center and simultaneous centroidal rotation or a pure rotation about ICR. Since tension, which acts at the ICR is asked; we prefer superposition of translation of the mass center and simultaneous centroidal rotation.

Forces acting on the disk are tension T applied by the thread at point P and weight of the disk. These forces and the effective force ma_C and effective torque $I_C \alpha$ are shown in the adjacent figure.

Applying Newton's second law for translation of mass center, we have

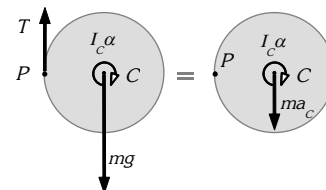
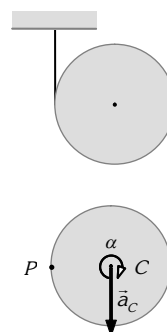
$$\sum \vec{F}_i = M\vec{a}_C \rightarrow mg - T = ma_C \quad (2)$$

Applying torque equation for centroidal rotation, we have

$$\sum \vec{\tau}_C = I_C \vec{\alpha} \rightarrow Tr = I_C \alpha$$

Substituting $\frac{1}{2}mr^2$ for I_C and α from eq. (1), we have

$$T = \frac{1}{2}ma_C \quad (3)$$



From eq. (2) and (3), we have

Acceleration of the mass center $a_C = \frac{2}{3}g$ **Ans.**

Tension in the string $T = \frac{1}{3}mg$ **Ans.**

Energy Methods

Newton's laws of motion tell us what is happening at an instant, while method of work and energy equips us to analyze what happens when a body moves from one place to other or a system changes its configuration. In this section, we introduce how to use methods of work and energy to analyze motion of rigid bodies.

Concept of Work in rotation motion

Work of a force is defined as the scalar product of the force vector and displacement vector of the point of application of the force. If during the action of a force $\vec{F}$ its point of application moves from position $\vec{r}_1$ to $\vec{r}_2$, the work $W_{1 \rightarrow 2}$ done by the force is expressed by the following equation.

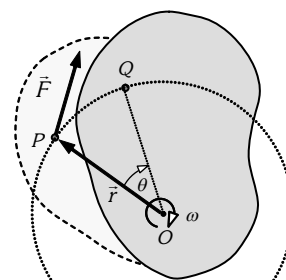
$$W_{1 \rightarrow 2} = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}$$

Either we can use of this idea to calculate work of a force or its modified

version in terms of torque and angular displacement.

The work done by a torque during a finite rotation of the rigid body from initial value θ_i of the angle θ to final value θ_f can be obtained by integrating both the sides of the equation given

$$W_{i \rightarrow f} = \int_{\theta_i}^{\theta_f} \vec{\tau}_o \cdot d\vec{\theta}$$



Kinetic Energy of a rigid body in rotation motion

A rigid body can be represented as a system of large number of particles, which keep their mutual distances unchanged in all circumstances. Kinetic energy of the whole body must be sum of kinetic energies of all of its particles. In this section we develop expressions for kinetic energy of a rigid body.

Kinetic Energy of a rigid body in plane motion

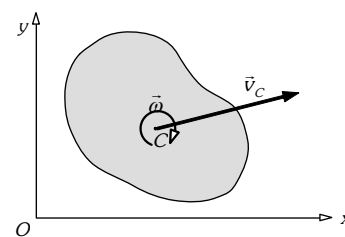
In the figure is shown a body in plane motion. Its mass center at an instant is moving with velocity $\vec{v}_C$ and rotating with angular velocity $\vec{\omega}$. Both these motions are shown superimposed in the given figure.

Kinetic energy too can be written as sum of kinetic energy $(\frac{1}{2}Mv_C^2)$ due to translation motion of the mass center and kinetic energy $(\frac{1}{2}I_C\omega^2)$ due to centroidal rotation.

$$K = \frac{1}{2}Mv_C^2 + \frac{1}{2}I_C\omega^2$$

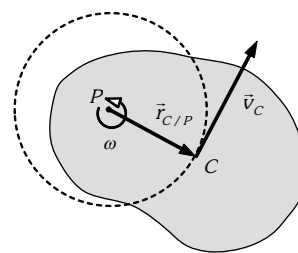
If location of the instantaneous axis of rotation (IAR) is known, making use of the parallel axis theorem we can write kinetic energy by the following equation also.

$$K = \frac{1}{2}I_{IAR}\omega^2$$



Kinetic Energy of a rigid body in rotation about fixed axis not passing through the mass centre

In this kind of motion the mass center is in circular motion about the axis of rotation. In the figure is shown a body rotation with angular velocity ω about a fixed axis through point P and perpendicular to plane of the paper. Mass center moves with speed $v_C = \omega r$. Kinetic energy of the body can now be expressed by the following equation.



$$K = \frac{1}{2} M v_C^2 + \frac{1}{2} I_C \omega^2$$

Making use of the parallel axis theorem ($I_P = M r_{P/C}^2 + I_C$) we can write kinetic energy by the following equation also.

$$K = \frac{1}{2} I_P \omega^2$$

Kinetic Energy of a rigid body in pure centroidal rotation

In pure centroidal rotation the mass center remain at rest; therefore kinetic energy due to translation of mass center vanishes.

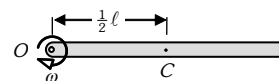
$$K = \frac{1}{2} I_C \omega^2$$

Ex. A rod of mass m and length ℓ is pivoted to a fixed support at one of its ends O . It is rotating with constant angular velocity ω . Write expression for its kinetic energy.



Sol. If the point C is the mass center of the rod, from theorem of parallel axes, the moment of inertia I_O of the rod about the fixed axis is

$$I_O = I_C + m(OC)^2 \rightarrow I_O = I_C + \frac{1}{4} m \ell^2$$



Substituting $\frac{1}{12} m \ell^2$ for I_C , we have

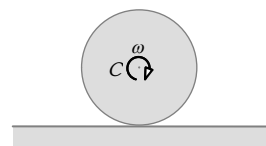
$$I_O = \frac{1}{3} m \ell^2$$

Kinetic energy of the rod equals to kinetic energy due to rotation about the fixed axis.

$$K = \frac{1}{2} I_O \omega^2 \rightarrow \text{Using above expression for } I_O, \text{ we have}$$

$$K = \frac{1}{6} m \ell^2 \omega^2 \quad \text{Ans.}$$

Ex. A uniform rigid body of mass m and round section of radius r is rolling on horizontal ground with angular velocity ω . Its radius of gyration about the centroidal axis is k .



(a) Write expression of its kinetic energy.

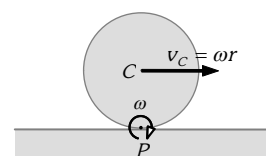
(b) Also express the kinetic energy as sum of kinetic energy due to translation of mass center and kinetic energy due to simultaneous centroidal rotation.

Sol. (a) The point of contact with ground of a body rolling on the ground is its ICR. Let the point P is the ICR as shown in the adjacent figure. The geometrical center C of a uniform body and the mass center coincide. Therefore moment of inertia I_P of the body about the ICR can be written by using the theorem of parallel axes.

$$I_P = I_C + m(PC)^2 \rightarrow I_P = I_C + m r^2$$

Substituting $I_C = m k^2$, we have

$$I_P = m(k^2 + r^2) \quad (1)$$



Kinetic energy of a rigid body equals to kinetic energy due to rotation about the ICR.

$$K = \frac{1}{2} I_P \omega^2 \rightarrow \text{Substituting } I_P \text{ from eq. (1), we have}$$

$$K = \frac{1}{2} m(k^2 + r^2) \omega^2 \quad \text{Ans.}$$

- (b) Kinetic energy of the body also equals to sum of kinetic energy due to translation of its mass center and kinetic energy due to simultaneous centroidal rotation.

$$K = \frac{1}{2} m v_C^2 + \frac{1}{2} I_C \omega^2 \rightarrow \text{Substituting condition for rolling } v_C = \omega r \text{ and } I_C = m k^2, \text{ we have}$$

$$K = \frac{1}{2} m(\omega r)^2 + \frac{1}{2} m k^2 \omega^2 = \frac{1}{2} m(r^2 + k^2) \omega^2 \quad \text{Ans.}$$

Rolling as rotation about an axis in translation

If the point of contact of the of the rolling body does not slide it is known as *rolling without slipping* or *pure rolling* or simply *rolling* and if the point of contact slides it is known as *rolling with slipping*. All kind of rolling motion is examples of rotation about an axis in translation.

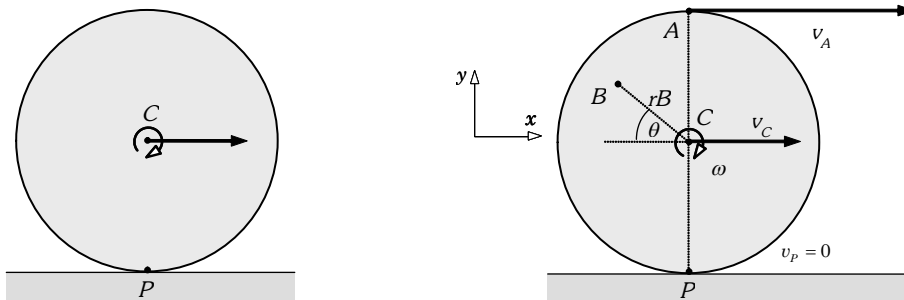
Rolling without slipping on stationary surface.

We first discuss velocity relations and thereafter accelerations relations of two points of a body of round section rolling on a stationary surface. For the purpose, we can use any of the following methods.

- | | | |
|----|-----------------------|---|
| I | Analytical Method: | By using relative motion equations. |
| II | Superposition Method: | By superimposing translation of a point and pure rotation about that point. |

Velocity relations by Analytical Method

Its point of contact P does not slide on the surface, therefore velocity of the point of contact relative to the surface is zero. In the next figure, velocity vectors of its center C and top point A are shown.



Velocity of the center C can be obtained with the help of relative motion equation.

$$\vec{v}_C = \vec{v}_P + \vec{\omega} \times \vec{PC} \rightarrow \vec{v}_C = \vec{0} + (-\omega \hat{k}) \times R \hat{j}$$

$$\vec{v}_C = \omega R \hat{i}$$

[16]

The above equation is used as condition of rolling without slipping on stationary surface.

Velocity of the top point A can be obtained by relative motion equation.

$$\vec{v}_A = \vec{v}_P + \vec{\omega} \times \vec{PA} \rightarrow \vec{v}_A = \vec{0} + (-\omega \hat{k}) \times (2R \hat{j})$$

$$\vec{v}_A = 2\omega R \hat{i} = 2\vec{v}_C$$

[17]

Once velocity of the center is obtained, we can use relative motion between A and C as well.

$$\vec{v}_A = \vec{v}_C + \vec{\omega} \times \vec{CA} \rightarrow \vec{v}_A = \omega R \hat{i} + (-\omega \hat{k}) \times (2R \hat{j})$$

$$\vec{v}_A = 2\omega R \hat{i} = 2\vec{v}_C$$

[18]

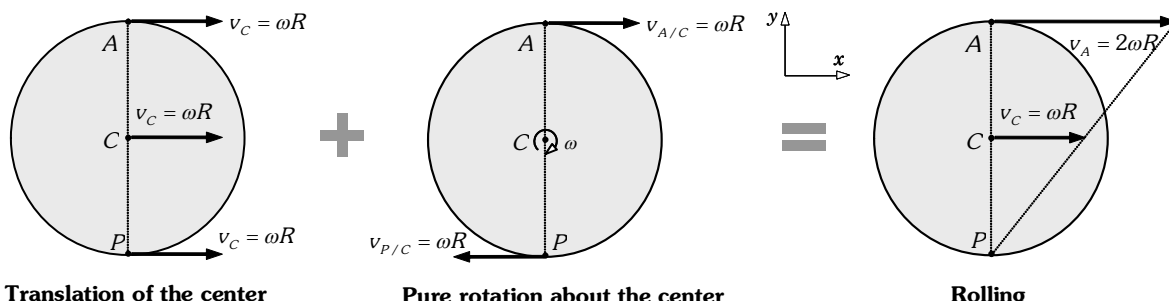
In similar fashion, velocity vector of an arbitrarily chosen point B .

$$\vec{v}_B = \vec{v}_C + \vec{\omega} \times \vec{CB} \rightarrow \vec{v}_B = v_C \hat{i} + (-\omega \hat{k}) \times (-r \cos \theta \hat{i} + r \sin \theta \hat{j})$$

$$\vec{v}_B = (v_C + \omega r \sin \theta) \hat{i} + \omega r \cos \theta \hat{j} \quad [19]$$

Velocity relations by Superposition Method

Now we will see that the above velocity relation can also be obtained by assuming rolling of the wheel as superposition of translation of its center and simultaneous rotation about the center.

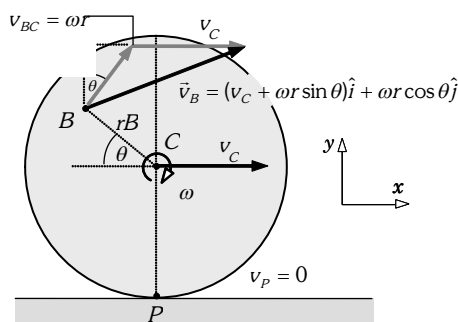


Translation of the center

Pure rotation about the center

Rolling

Velocity of an arbitrary point B as superposition of translation the center and rotation about the center.



Ex. A cylinder of radius 5 m rolls on a horizontal surface. Velocity of its center is 25 m/s. Find its angular velocity and velocity of the point A .

Sol. In rolling the angular velocity $\vec{\omega}$ and velocity of the center of a round section body satisfy condition described in the relative motion eq. [14]. So we have

$$\vec{v}_C = \vec{\omega} \times \vec{r}_{C/P} \rightarrow 25\hat{i} = \omega \hat{k} \times 5\hat{j} \Rightarrow \vec{\omega} = -5\hat{k} \text{ rad/s}$$

Angular velocity vector points in the negative z -axis so the cylinder rotates in clockwise sense.

Velocity of the point A can be calculated by either analytical method, superposition method.

Analytical Method

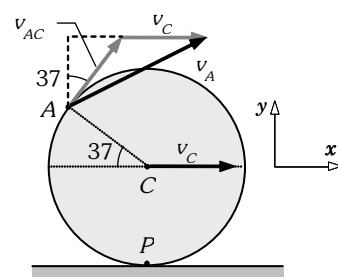
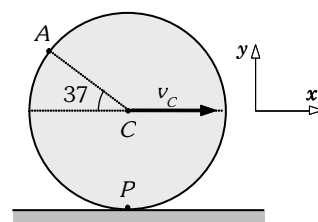
$$\vec{v}_A = \vec{v}_C + \vec{\omega} \times \vec{AC} \rightarrow \vec{v}_A = 25\hat{i} + (-5\hat{k}) \times (-5 \cos 37^\circ \hat{i} + 5 \sin 37^\circ \hat{j})$$

$$\vec{v}_A = (40\hat{i} + 20\hat{j}) \text{ m/s}$$

Superposition Method

In rolling $v_C = v_{AC} = \omega R = 25 \text{ m/s}$. The superposition i.e. vector addition of the terms of equation $\vec{v}_A = \vec{v}_C + \vec{v}_{AC}$ are shown in the following figure. Resolving $v_{AC} = \omega R = 25 \text{ m/s}$ into its Cartesian components and adding to $\vec{v}_C$, we obtain

$$\vec{v}_A = \vec{v}_C + \vec{v}_{A/C} \rightarrow \vec{v}_A = 25\hat{i} + 15\hat{i} + 20\hat{j} = (40\hat{i} + 20\hat{j}) \text{ m/s}$$



Acceleration relations by Analytical Method

The point of contact P does not slide on the surface, therefore component of its acceleration parallel to the surface must be zero. However, it has an acceleration component towards the center. The center always moves parallel to the horizontal surface and does not change direction of its velocity; therefore, its acceleration can only be parallel to the surface.

Relation between acceleration of acceleration vector of the center C and point of contact P can be obtained with the help of relative motion [15] equation together with the above fact.

$$\vec{a}_C = \vec{a}_P + \vec{\alpha} \times \vec{PC} - \omega^2 \vec{PC} \rightarrow a_C \hat{i} = a_{Py} \hat{j} + (-\alpha \hat{k}) \times R \hat{j} - \omega^2 R \hat{j} = a_{Py} \hat{j} + \alpha R \hat{i} - \omega^2 R \hat{j}$$

Equating coefficients of x and y -components on both the sides of the above equation, we have

$$\vec{a}_C = \alpha R \hat{i} \quad [20]$$

$$\vec{a}_P = \omega^2 R \hat{j} \quad [21]$$

The eq. [20] is used as condition for rolling without slipping together with eq. [16]

In the given figure, acceleration vectors the point of contact; center and the top point are shown. Now we will see how these accelerations can be calculated by using relative motion equation.

Once velocity of the center is obtained, we can use relative motion between A and C as well. Now we calculate acceleration of the top point A .

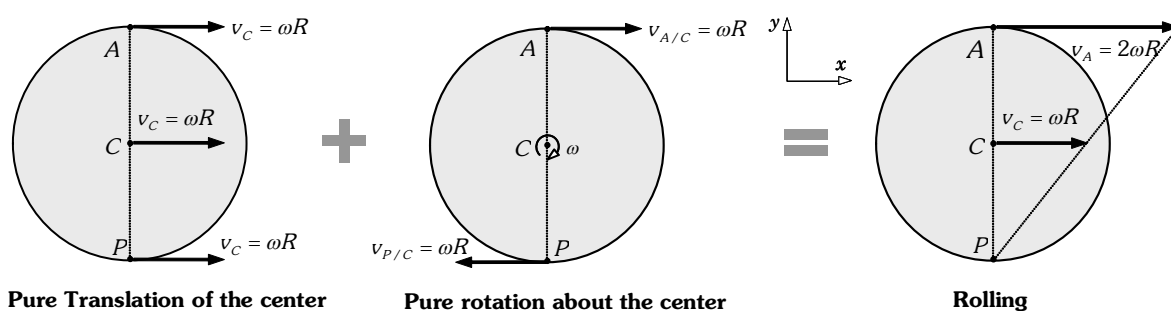
$$\vec{a}_A = \vec{a}_C + \vec{\alpha} \times \vec{CA} - \omega^2 \vec{CA} \rightarrow \vec{a}_A = \alpha R \hat{i} + (-\alpha \hat{k}) \times R \hat{j} - \omega^2 R \hat{j}$$

$$\vec{a}_A = 2\alpha R \hat{i} - \omega^2 R \hat{j} \quad [22]$$

Acceleration vector of point A and its components are shown in the given figure.

Acceleration relations by Superposition Method

Now we see how acceleration relations are expressed for a rolling wheel by assuming its rolling as superposition of its translation with the velocity of center and simultaneous rotation about the centre.

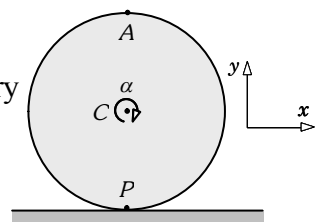
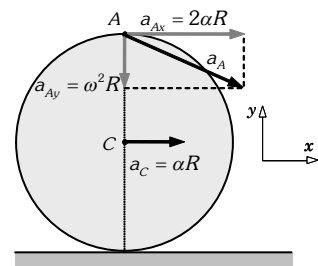
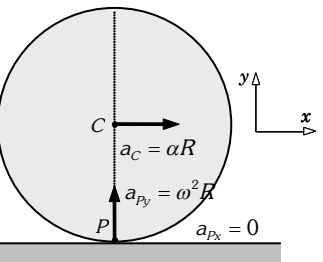


Ex. A body of round section of radius 10 cm starts rolling on a horizontal stationary surface with uniform angular acceleration 2 rad/s^2 .

- Find initial acceleration of the center C and top point A .
- Find expression for acceleration of the top point A as function of time.

Sol. Initially when the body starts, it has no angular velocity; therefore, the last term in relative motion equation [15] for acceleration vanishes and for a pair of two points A and B the equation reduces to

$$\vec{a}_A = \vec{a}_B + \vec{\alpha} \times \vec{BA}$$



The angular acceleration vector is $\vec{\alpha} = -2\hat{k} \text{ rad/s}^2$.

- (a) Acceleration of the center C is obtained by using condition for rolling without slipping.

$$\vec{a}_C = \vec{\alpha} \times \vec{PC} \rightarrow \vec{a}_C = -2\hat{k} \times 10\hat{j} = 20\hat{i} \text{ cm/s}^2 \text{ Ans.}$$

Acceleration of the point A can be obtained either by analytical method, superposition method or by use of ICR. These methods for calculation of acceleration of the top point are already described; therefore, we use the result.

$$\vec{a}_A = 2\alpha R\hat{i} \rightarrow \vec{a}_A = 40\hat{i} \text{ cm/s}^2 \text{ Ans.}$$

- (b) Initially at the instant $t = 0$, when the body starts, its angular velocity is zero. At latter time it acquires angular velocity $\vec{\omega}$, therefore acceleration of any point on the body, other than its center, has an additional component of acceleration. This additional component is accounted for by the last term in the relative motion equation [15].

Angular velocity acquired by the body at time t is obtained by eq.[4] used for a body rotating with constant angular acceleration.

$$\vec{\omega} = \vec{\omega}_0 + \vec{\alpha}t \rightarrow \text{Substituting } \omega_0 = 0, \text{ we have}$$

$$\vec{\omega} = -2t\hat{k}$$

Analytical Method

Using the relative motion equation for the pair of points C and A , we have

$$\vec{a}_A = \vec{a}_C + \vec{\alpha} \times \vec{CA} - \omega^2 \vec{CA} \rightarrow \vec{a}_A = \alpha R\hat{i} + (-\alpha\hat{k}) \times R\hat{j} - \omega^2 R\hat{j} = 2\alpha R\hat{i} - \omega^2 R\hat{j}$$

Substituting the known values

$$\vec{\alpha} = -2\hat{k} \text{ rad/s}^2, \vec{\omega} = -2t\hat{k} \text{ rad/s and } R = 10 \text{ cm}$$

$$\text{we have } \vec{a}_A = 40\hat{i} - 40t^2\hat{j} \text{ cm/s}^2 \text{ Ans.}$$

Superposition Method

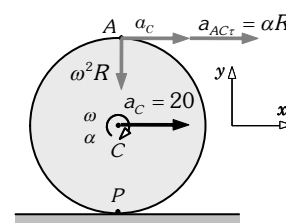
We superimpose translation motion of the center and rotation motion about the center. In fact it is vector addition of terms of above equation used in analytical method.

From the above figure, we have

$$\vec{a}_A = (a_C + \alpha R)\hat{i} - \omega^2 R\hat{j}$$

$$\text{Substituting known values } \vec{\alpha} = -2\hat{k} \text{ rad/s}^2, \vec{\omega} = -2t\hat{k} \text{ rad/s and } R = 10 \text{ cm,}$$

$$\text{we have } \vec{a}_A = 40\hat{i} - 40t^2\hat{j} \text{ cm/s}^2 \text{ Ans.}$$



Methods of Impulse and Momentum

Methods of impulse and momentum describe what happens over a time interval. When motion of a body involves rotation we have to consider angular impulse as well as angular momentum. In this section we discuss concept of angular impulse, angular momentum of rigid body, angular impulse momentum principle and conservation of angular momentum.

Angular Impulse

Like impulse of a force angular impulse of a constant torque equals to product of the torque and concerned time interval and if the torque is not constant it must be integrated with time over the concerned time interval.

If torque $\vec{\tau}_o$ about an axis passing through O is constant, its angular impulse during a time interval from t_1 to t_2 denoted by $\vec{J}_{o,1 \rightarrow 2}$ is given by the following equation.

$$\vec{J}_{o,1 \rightarrow 2} = \vec{\tau}_o (t_2 - t_1)$$

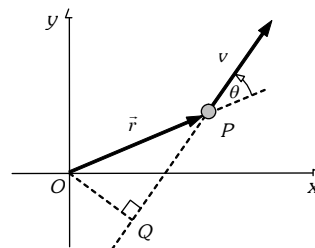
If torque $\vec{\tau}_o$ about an axis passing through O is time varying, its angular impulse during a time interval from t_1 to t_2 denoted by $\vec{J}_{o,1 \rightarrow 2}$ is given by the following equation.

$$\vec{J}_{o,1 \rightarrow 2} = \int_{t_1}^{t_2} \vec{\tau}_o dt$$

Angular momentum of a particle

Angular momentum $\vec{L}_o$ about the origin O of a particle of mass m moving with velocity $\vec{v}$ is defined as the moment of its linear momentum $\vec{p} = m\vec{v}$ about the point O .

$$\vec{L}_o = \vec{r} \times (m\vec{v})$$



Angular Momentum of a Rigid Body

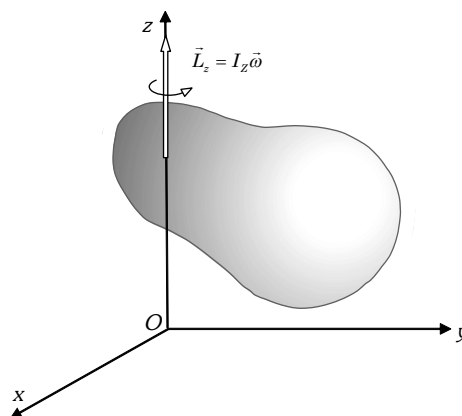
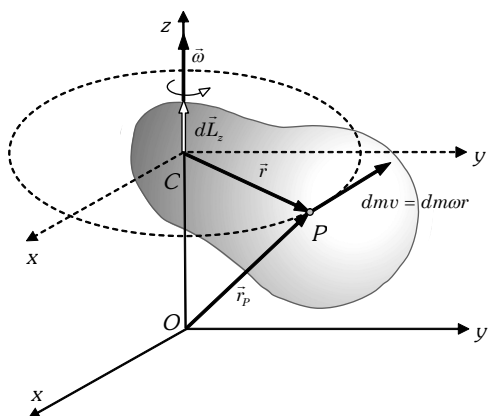
Angular momentum is quantity of rotation motion in a body. The angular momentum of a system of particles is the sum of angular momentum all the particles within the system. A rigid body is an assemblage of large number of particles maintaining their mutual distances intact under all circumstances, therefore angular momentum of a rigid body must be sum of angular momenta of all of its particles.

Angular Momentum about a point and about an axis

Angular momentum of a particle is not defined about an axis instead it is defined about a point. Therefore above idea of summing up angular momenta of all the particles about a point gives angular momentum of the rigid body about a point. But while dealing with fixed axis rotation or rotation about axis in translation we need angular momentum about an axis.

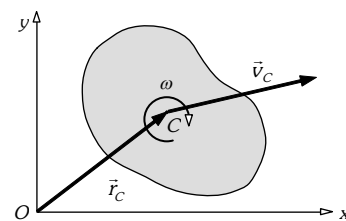
Angular momentum about an axis is calculated similar to torque about an axis. To calculate angular momentum of a particle of rigid body about an axis we take moment of momentum of the particle about the point where plane of motion of the point of application of the force intersects the axis.

In the following figure is shown angular momentum $d\vec{L}_z = \vec{r} \times (dm\vec{v}) = r^2 dm\omega$ of a particle P of a rigid body rotating about the z -axis. It is along the z -axis i.e. axis of rotation. In the next figure total angular momentum $\vec{L}_z = \int d\vec{L}_z = I_z \vec{\omega}$ about the axis of rotation is shown. It is also along the axis of rotation.



Angular Momentum in general plane motion

Angular momentum of a body in plane motion can also be written similar to torque equation or kinetic energy as sum of angular momentum about the axis due to translation of mass center and angular momentum of centroidal rotation about centroidal axis parallel to the original axis.



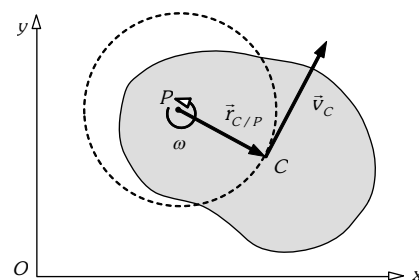
Consider a rigid body of mass M in plane motion. At the instant shown its mass center has velocity $\vec{v}$ and it is rotating with angular velocity $\vec{\omega}$ about an axis perpendicular to the plane of the figure. Its angular momentum $\vec{L}_O$ about an axis passing through the origin and parallel to the original is expressed by the following equation.

$$\vec{L}_c = \vec{r}_c \times (M\vec{v}_c) + I_c \vec{\omega}$$

The first term of the above equation represent angular momentum due to translation of the mass center and the second term represents angular momentum in centroidal rotation.

Angular momentum in rotation about fixed axis

Consider a body of mass M rotating with angular velocity ω about a fixed axis perpendicular to plane of the figure passing through point P . Making use of the parallel axis theorem $I_P = Mr_{C/P}^2 + I_C$ and equation $\vec{v}_C = \vec{\omega} \times \vec{r}_{C/P}$ we can express the angular momentum $\vec{L}_P$ of the body about the fixed rotational axis.



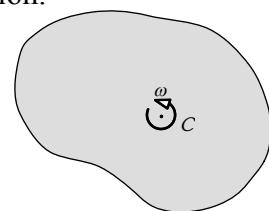
$$\vec{L}_D = I_D \vec{\omega}$$

The above equation reveals that the angular momentum of a rigid body in plane motion can also be expressed in a single term due to rotation about the instantaneous axis of rotation.

Angular momentum in pure centroidal rotation

In pure centroidal rotation, mass center remains at rest, therefore angular momentum due to translation of the mass center vanishes.

$$\vec{L}_C = I_C \vec{\omega}$$



Rotational Equivalent of the Newton's Laws of Motion

Differentiating terms on both the sides of equation $\vec{L}_o = \vec{r}_C \times (M\vec{v}_C) + I_C\vec{\omega}$ with respect to time, and making substitution of $\vec{v}_C = d\vec{r}_C/dt$, $\vec{a}_C = d\vec{v}_C/dt$ and $\vec{\alpha} = d\vec{\omega}/dt$ we have

$$\frac{d\vec{L}_o}{dt} = \vec{v}_C \times (M\vec{v}_C) + \vec{r}_C \times M\vec{a}_C + I_C \vec{\alpha}$$

The first on the right hand side vanishes, so we can write

$$\frac{d\vec{L}_o}{dt} = \vec{r}_C \times M\vec{a}_C + I_C \vec{\alpha}$$

Now comparing the above equation with torque equation $\Sigma \vec{\tau}_o = \vec{r}_C \times M\vec{a}_C + I_C\vec{\alpha}$, we have

$$\sum \vec{\tau}_o = \frac{d\vec{L}_o}{dt}$$

The above equation though developed for plane motion only yet is valid for rotation about an axis in rotation also. It states that the net torque about the origin of an inertial frame equals to the time rate of change in angular momentum about the origin and can be treated as a parallel to Newton's second law which states that net external force on a body equals to time rate of change in its linear momentum.

Angular Impulse Momentum Principle

Rearranging the terms and integrating both the sides obtained from previous equation, we can write

$$\Sigma \int_{t_1}^{t_2} \vec{\tau}_o dt = \vec{L}_{o2} - \vec{L}_{o1}$$

The left hand side of the above equation is the angular impulse of torque of all the external forces in the time interval in the time interval t_1 to t_2 .

$$\Sigma \vec{J}_{o,1 \rightarrow 2} = \vec{L}_{o2} - \vec{L}_{o1}$$

The idea expressed by the above equation is known as angular impulse momentum principle and states that increment in the angular momentum of a body about a point in a time interval equals to the net angular impulse of all the external forces acting on it during the concerned time interval.

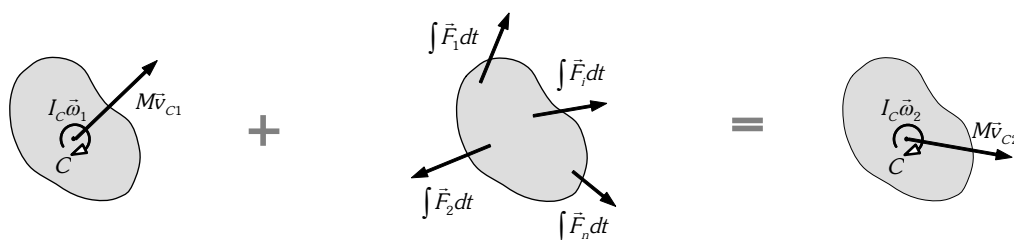
For the ease of application the above equation is rearranged as

$$\vec{L}_{o1} + \Sigma \vec{J}_{o,1 \rightarrow 2} = \vec{L}_{o2}$$

Like linear impulse momentum principle, the angular impulse momentum principle provides us solution of problems concerned with change in angular velocity in a time interval or change in angular velocity during very short interval interactions.

Method of Impulse Momentum Principle for Plane motion of a Rigid Body

Linear momentum and angular momentum serve as measures of amount of translation and rotation motion respectively. The external forces acting on a rigid body can change its state of translation as well as rotation motion which is reflected by change in linear as well as angular momentum according to the principles of linear impulse and momentum and angular impulse and momentum.



Linear and angular momenta at the instant t_1

Impulse of all the forces during time interval t_1 to t_2

Linear and angular momenta at the instant t_2

In the above figure is shown strategy to apply method of impulse and momentum. Consider a rigid body of mass M in plane motion. Its moment of inertia about the centroidal axis perpendicular to plane of motion is I_c . Let $\vec{v}_{c1}$ and $\vec{\omega}_1$ represent velocity of its mass center and its angular velocity at the beginning of a time interval t_1 to t_2 . Under the action of several forces $\vec{F}_1, \vec{F}_2, \dots, \vec{F}_i, \dots, \vec{F}_n$ during the time interval its mass center velocity and angular velocity become $\vec{v}_{c2}$ and $\vec{\omega}_2$ respectively. The adjacent figure shows strategy representing how to write equations for linear and angular impulse momentum principles.

While applying the principle it becomes simpler to consider translation of the mass center and centroidal rotation separately. Thus in an alternative way we apply linear impulse momentum principle for translation of the mass center and angular impulse momentum principle for centroidal rotation.

Translation of mass center: Linear impulse momentum principle.

$$\vec{p}_1 + \Sigma \vec{I}_{mp1 \rightarrow 2} = \vec{p}_2$$

Here $\vec{p}_1 = M\vec{v}_{c1}$ and $\vec{p}_2 = M\vec{v}_{c2}$ represent linear momentums at the beginning and end of the time interval and $\Sigma \vec{I}_{mp1 \rightarrow 2}$ stands for impulse of all the external forces during the time interval.

Centroidal rotation: Angular impulse momentum principle.

$$\vec{L}_{c1} + \Sigma \vec{J}_{c,1 \rightarrow 2} = \vec{L}_{c2}$$

Here $\vec{L}_{C1} = I_C \vec{\omega}_1$ and $\vec{L}_{C2} = I_C \vec{\omega}_2$ represent angular momentums about the centroidal axis at the beginning and end of the time interval and $\sum \vec{J}_{C,1 \rightarrow 2}$ stands for angular impulse of all the external forces about the centroidal axis during the time interval.

Conservation of Angular Momentum

If angular impulse of all the external forces about an axis in time interval vanishes, the angular momentum of the system about the same axis in that time interval remain unchanged.

$$\text{If } \sum \int_{t_1}^{t_2} \vec{\tau}_o dt = 0, \text{ we have } \vec{L}_{o1} = \vec{L}_{o2}$$

The condition of zero net angular impulse required for conservation of angular momentum can be fulfilled in the following cases.

- If no external force acts, the angular impulse about all axes will be zero and hence angular momentum remains conserved about all axes.
- If net torque of all the external forces or torques of each individual force about an axis vanishes the angular momentum about that axes will be conserved.
- If all the external forces are finite in magnitude and the concerned time interval is infinitely small, the angular momentum remain conserved.
- If a system of rigid bodies changes its moment of inertia by changing its configuration due to internal forces only its angular momentum about any axes remains conserved. If we denote the moment of inertias in two configurations by I_1 and I_2 and angular velocities by ω_1 and ω_2 , we can write

$$I_1 \vec{\omega}_1 = I_2 \vec{\omega}_2$$

The principle of conservation of angular momentum governs a wide range of physical processes from subatomic to celestial world. The following examples explicate some of these applications.

Student on rotating turntable

The student, the turntable and dumbbells make an isolated system on which no external torque acts, if we ignore friction in the bearing of the turntable and air friction. Initially the student has his arm stretched on rotating turntable. When he pulls dumbbells close to his body, angular velocity increases due to conservation of angular momentum.

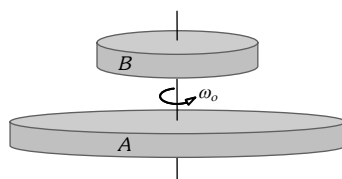


Larger moment of inertia and smaller angular velocity



Smaller moment of inertia and larger angular velocity

Ex. Consider the disk A of moment of inertia I_1 rotating freely in horizontal plane about its axis of symmetry with angular velocity ω_0 . Another disk B of moment of inertia I_2 held at rest above the disk A . The axis of symmetry of the disk B coincides with that of the disk A as shown in the figure. The disk B is released to land on the disk A . When sliding stops, what will be the angular velocity of both the disks?



Sol. Both the disks are symmetric about the axis of rotation therefore does not require any external torque to keep the axis stationary. When the disk B lands on A slipping starts. The force of friction provides an internal torque to system of both the disk. It slows down rotation rate of A and increases that of B till both acquire same angular velocity ω .

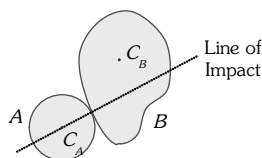
Since there is no external torques on the system of both the disks about the axis of rotation, the total angular momentum of the system remains conserved. The total angular momentum of the system is the sum of angular momentum of both disks. Denoting the angular momentum of the disk A before B lands on it and long after slipping between them stops by symbols $\vec{L}_{A1}$, $\vec{L}_{B1}$, $\vec{L}_{A2}$ and $\vec{L}_{B2}$ respectively, we can express conservation of angular momentum by the following equation.

$$\vec{L}_{A1} + \vec{L}_{B1} = \vec{L}_{A2} + \vec{L}_{B2} \rightarrow I_1\omega_0 + 0 = I_1\omega + I_2\omega \Rightarrow \omega = \frac{I_1\omega_0}{I_1 + I_2} \quad \text{Ans.}$$

Eccentric Impact

In eccentric impact the line of impact which is the common normal drawn at the point of impact does not pass through mass center of at least one of the colliding bodies. It involves change in state of rotation motion of either or both the bodies.

Consider impact of two A and B such that the mass center C_B of B does not lie on the line of impact as shown in figure. If we assume bodies to be frictionless their mutual forces must act along the line of impact. The reaction force of A on B does not pass through the mass center of B as a result state of rotation motion of B changes during the impact.



Problems of Eccentric Impact

Problems of eccentric impact can be divided into two categories. In one category both the bodies under going eccentric impact are free to move. No external force act on either of them. Their mutual forces are responsible for change in their momentum and angular momentum. In another category either or both of the bodies are hinged.

Eccentric Impact of bodies free to move

Since no external force acts on the two body system, we can use principle of conservation of linear momentum, principle of conservation of angular momentum about any point and concept of coefficient of restitution.

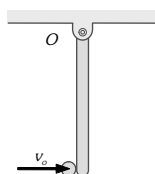
The coefficient of restitution is defined for components of velocities of points of contacts of the bodies along the line of impact.

While applying principle of conservation of angular momentum care must be taken in selecting the point about which we write the equation. The point about which we write angular momentum must be at rest relative to the selected inertial reference frame and as far as possible its location should be selected on line of velocity of the mass center in order to make zero the first term involving moment of momentum of mass center.

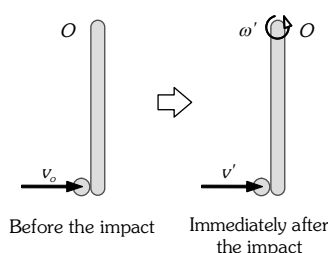
Eccentric Impact of hinged bodies

When either or both of the bodies are hinged the reaction of the hinge during the impact act as external force on the two body system, therefore linear momentum no longer remain conserved and we cannot apply principle of conservation of linear momentum. When both the bodies are hinged we cannot also apply conservation of angular momentum, and we have to use impulse momentum principle on both the bodies separately in addition to making use of coefficient of restitution. But when one of the bodies is hinged and other one is free to move, we can apply conservation of angular momentum about the hinge.

- Ex.** A uniform rod of mass m and length ℓ is suspended from a fixed support and can rotate freely in the vertical plane. A small ball of mass m moving horizontally with velocity v_o strikes elastically the lower end of the rod as shown in the figure. Find the angular velocity of the rod and velocity of the ball immediately after the impact.



- Sol.** The rod is hinged and the ball is free to move. External forces acting on the rod ball system are their weights and reaction from the hinge. Weight of the ball as well as the rod are finite and contribute negligible impulse during the impact, but impulse of reaction of the hinge during impact is considerable and cannot be neglected. Obviously linear momentum of the system is not conserved. The angular impulse of the reaction of hinge about the hinge is zero. Therefore angular momentum of the system about the hinge is conserved. Let velocity of the ball after the impact becomes v'_B and angular velocity of the rod becomes ω' .



We denote angular momentum of the ball and the rod about the hinge before the impact by L_{B1} and L_{R1} and after the impact by L_{B2} and L_{R2} .

Applying conservation of angular momentum about the hinge, we have

$$\vec{L}_{B1} + \vec{L}_{R1} = \vec{L}_{B2} + \vec{L}_{R2} \rightarrow mv_o \ell + 0 = mv'_B \ell + I_o \omega'$$

Substituting $\frac{1}{3}M\ell^2$ for I_o , we have

$$3mv'_B + M\ell\omega' = 3mv_o \quad (1)$$

The velocity of the lower end of the rod before the impact was zero and immediately after the impact it becomes $\ell\omega'$ towards right. Employing these facts we can express the coefficient of restitution according to eq.

$$e = \frac{v'_{Qn} - v'_{Pn}}{v_{pn} - v_{Qn}} \rightarrow \ell\omega' - v'_B = ev_o \quad (2)$$

From eq. (1) and (2), we have

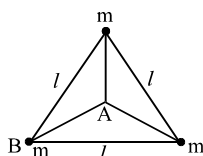
$$\text{Velocity of the ball immediately after the impact } v'_B = \frac{(3m - eM)v_o}{3m + M} \quad \text{Ans.}$$

$$\text{Angular velocity of the rod immediately after the impact } \omega' = \frac{3(1+e)mv_o}{(3m+M)\ell} \quad \text{Ans.}$$

EXERCISE (S-1)

Moment of inertia

- Three equal masses m are rigidly connected to each other by massless rods of length l forming an equilateral triangle, as shown in the figure. What is the ratio of the moment of inertia of the assembly for an axis through B compared with that for an axis through A (centroid). Both the axis are perpendicular to the plane of triangle.



RD0001

- Two rods of equal mass m and length l lie along the x axis and y axis with their centres at origin. What is the moment of inertia of the system about the line $x = y$:

RD0002

Torque and its calculation

- A force $\vec{F} = 2\hat{i} + 3\hat{j}$ N is applied to an object that is pivoted about a fixed axle aligned along the z coordinate axis. If the force is applied at the point $\vec{r} = 4\hat{i} + 5\hat{j}$ m, find (a) the magnitude of the net torque about the z axis and (b) the direction of the torque vector τ .

RD0003

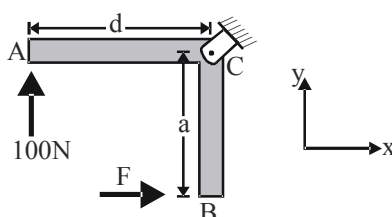
Equilibrium

- In an experiment with a massless beam balance an unknown mass m is balanced by two known masses of 16kg and 4 kg as shown in figure. Find the value of the unknown mass m .



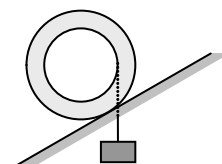
RD0004

- Find the force F required to keep the system in equilibrium. The dimensions of the system are $d = 0.3$ m and $a = 0.2$ m. Assume the rods to be massless.



RD0005

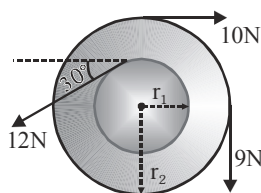
6. A 3.0 kg bobbin consists of a central cylinder of radius 5.0 cm and two end plates each of radius 6.0 cm. It is placed on a slotted incline, where friction is sufficient to prevent sliding. A block of mass 4.5 kg is suspended from a cord wound around the bobbin and passing through the slot under the incline. If the bobbin is in static equilibrium, what is the angle of tilt of the incline?



RD0006

$\tau = I\alpha$ and its calculation

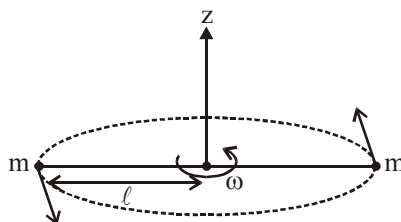
7. In the following figure r_1 and r_2 are 5 cm and 30 cm respectively. If the moment of inertia of the wheel is 5100 kg-m^2 about the axis passing through its centre and perpendicular to the plane of wheel, then what will be its angular acceleration?



RD0007

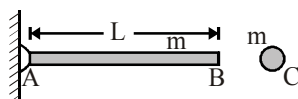
8. A dumbbell consists of two identical particles of mass m connected by a rigid light rod of length 2ℓ . The dumbbell is set spinning with angular speed ω_0 on a surface with a small friction coefficient μ_k . If

dumbbell stops in time $t = \frac{K\omega_0\ell}{2\mu g}$ where K is a constant, then find the value of K .



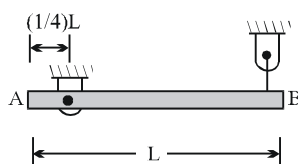
RD0008

9. A uniform bar AB of mass m and a ball of the same mass are released from rest from the same horizontal position. The bar is hinged at end A. There is gravity downwards. What is the distance of the point from point B that has the same acceleration as that of ball, immediately after release?



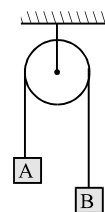
RD0009

10. A uniform beam of length L and mass m is supported as shown. If the cable at B suddenly breaks, determine; (a) the acceleration of end B. (b) the reaction at the pin support.



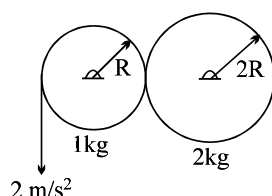
RD0010

11. In the figure, A & B are two blocks of mass 4 kg & 2 kg respectively attached to the two ends of a light string passing over a disc C of mass 40 kg and radius 0.1 m. The disc is free to rotate about a fixed horizontal axes, coinciding with its own axis. The system is released from rest and the string does not slip over the disc. Find:
- the linear acceleration of mass B.
 - the number of revolutions made by the disc at the end of 10 sec. from the start.
 - the tension in the string segment supporting the block A.



RD0011

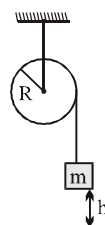
12. Two discs A and B touch each other as shown in figure. A rope tightly wound on A is pulled down at 2 m/s^2 . Find the friction force between A and B if slipping is absent



RD0012

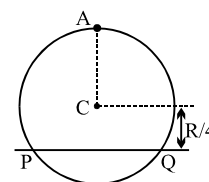
Kinetic energy in pure rotation

13. A mass m is attached to a pulley through a cord as shown in the figure. The pulley is a solid disk with radius R . The cord does not slip on the disk. The mass is released from rest at a height h from the ground and at the instant the mass reaches the ground, the disk is rotating with angular velocity ω . Find the mass of the disk.



RD0013

14. A uniform circular disc has radius R and mass m . A particle also of mass m is fixed at a point A on the edge of the disc as in figure. The disc can rotate freely about a fixed horizontal chord PQ that is at a distance $R/4$ from the centre C of the disc. The line AC is perpendicular to PQ. Initially the disc is held vertical with the point A at its highest position. It is then allowed to fall so that it starts rotating about PQ. Find the linear speed of the particle at it reaches its lowest position.



RD0014

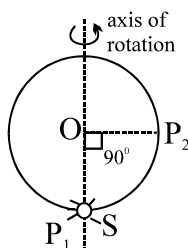
15. A rod of mass m & length ℓ is hinged at its upper end. It can rotate in vertical plane. It is given angular velocity ω so that it can complete vertical circle. Find (a) ω (b) Tension at centre of rod at initial moment.



RD0015

Angular momentum and its conservation

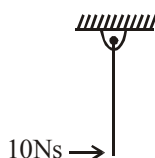
16. A uniform ring is rotating about vertical axis with angular velocity ω initially. A point insect (S) having the same mass as that of the ring starts walking from the lowest point P_1 and finally reaches the point P_2 (as shown in figure). What is the final angular velocity of the ring?

**RD0016**

17. Two men, each of mass 75 kg, stand on the rim of a horizontal large disc, diametrically opposite to each other. The disc has a mass 450 kg and is free to rotate about its axis. Each man simultaneously start along the rim clockwise with the same speed and reaches their original starting points on the disc. Find the angle turned by the disc with respect to the ground in this duration.

RD0017

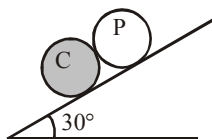
18. A thin uniform straight rod of mass 2 kg and length 1 m is free to rotate about its upper end when at rest. It receives an impulsive blow of 10 Ns at its lowest point, normal to its length as shown in figure. Find the kinetic energy of rod just after impact.

**RD0018****Combined rotation and translation**

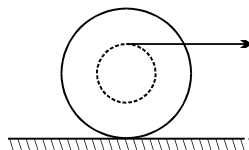
19. A solid uniform disk of mass m rolls down a fixed inclined plane without slipping with an acceleration a . Find the frictional force on the disk due to surface of the plane :

RD0019

20. A solid cylinder C and a hollow pipe P of same diameter are in contact when they are released from rest as shown in the figure on a long incline plane. Cylinder C and pipe P roll without slipping. Determine the clear gap (in m) between them after 6 seconds.

**RD0020**

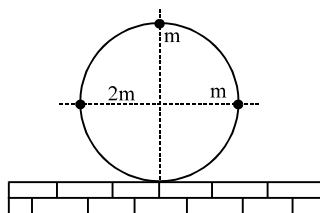
21. A spool of inner radius R and outer radius $3R$ has a moment of inertia $= MR^2$ about an axis passing through its geometric centre, where M is the mass of the spool. A thread wound on the inner surface of the spool is pulled horizontally with a constant force $= Mg$. Find the acceleration of the point on the thread which is being pulled assuming that the spool rolls purely on the floor.



RD0021

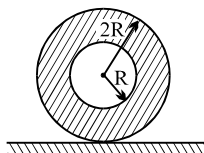
Kinetic energy in rolling

22. A ring of mass m and radius R has three particles attached to the ring as shown in the figure. The centre of the ring has a speed v_0 . Find the kinetic energy of the system. (Slipping is absent)



RD0022

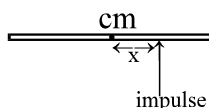
23. A hollow cylinder with inner radius R , outer radius $2R$ and mass M is rolling with speed of its axis v . Its kinetic energy is :-



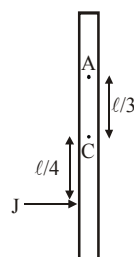
RD0023

Angular impulse

24. A uniform rod of length l is given an impulse at right angle to its length as shown. Find the distance of instantaneous centre of rotation from the centre of the rod.



25. A uniform rod of mass m and length ℓ is placed in gravity free space and linear impulse J is given to the rod at a distance $x = \ell/4$ from centre 'C' and perpendicular to the rod. Point A is at a distance $\ell/3$ from centre as shown in the figure. Then find
- Speed of centre of rod
 - Speed of point A
 - Speed of upper end of rod
 - Speed of lower end of rod



RD0024

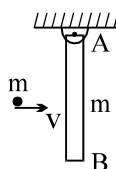
RD0025

26. A solid sphere of mass m and radius R is placed on a smooth horizontal surface. A sudden blow is given horizontally to the sphere at a height $h = 4R/5$ above the centre line. If I is the impulse of the blow then find
- the minimum time after which the highest point B will touch the ground
 - the displacement of the centre of mass during this interval.

RD0026

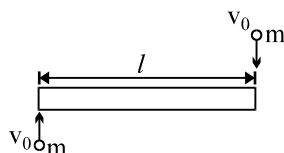
Eccentric collision

27. A uniform rod AB of length L and mass m is suspended freely at A and hangs vertically at rest when a particle of same mass m is fired horizontally with speed v to strike the rod at its mid point. If the particle comes to rest after the impact, then find the impulsive reaction at A.



RD0027

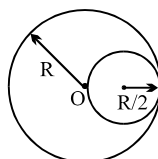
28. On a smooth table two particles of mass m each, travelling with a velocity v_0 in opposite directions, strike the ends of a rigid massless rod of length l , kept perpendicular to their velocity. The particles stick to the rod after the collision. Find the tension in rod during subsequent motion.



RD0028

EXERCISE (S-2)

1. A hole of radius $R/2$ is cut from a solid sphere of radius R . If the mass of the remaining part of sphere is M , then find moment of inertia of the body about an axis through O .

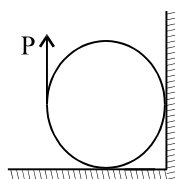


RD0029

2. Find the MI of a rod about an axis through its centre of mass and perpendicular to the length whose linear density varies as $\lambda = ax$ where a is a constant and x is the position of an element of the rod relative to its left end. The length of the rod is ℓ .

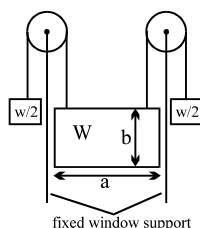
RD0030

3. Figure shows a vertical force applied tangentially to a uniform cylinder of weight F_g . The coefficient of static friction between the cylinder and both surfaces is 0.500 . In terms of F_g , find the maximum force P that can be applied that does not cause the cylinder to rotate.



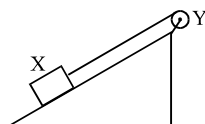
RD0031

4. A slight loosely fit window is balanced by two strings which are connected to weights $w/2$ each. The strings pass over the frictionless pulleys as shown in the figure. The strings are tied almost at the corner of the window. The string on the right is cut and then the window accelerates downwards. If the coefficients of friction between the window and the side supports is μ then calculate the acceleration of the window in terms of μ , a , b and g , where a is width and b is the length of the window.



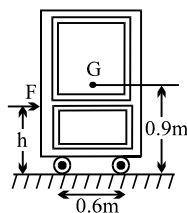
RD0032

5. A block X of mass 0.5kg is held by a long massless string on a frictionless fixed inclined plane of inclination 30° to the horizontal. The string is wound on a uniform solid cylindrical drum Y of mass 2kg and of radius 0.2m as shown in the fig. The drum is given an initial angular velocity such that the block X starts moving up the plane. ($g = 9.8\text{ m/s}^2$)
(i) Find the tension in the string during the motion
(ii) At a certain instant of time the magnitude of the angular velocity of Y is 10 rad/sec . Calculate the distance travelled by X from that instant of time until it comes to rest.



RD0033

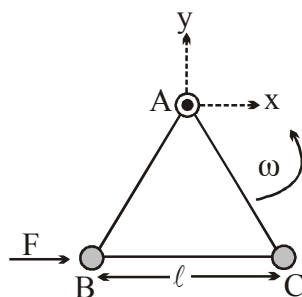
6. A 20 kg cabinet is mounted on small casters that allow it to move freely ($\mu = 0$) on the floor. If a 100 N force is applied as shown, determine
- the acceleration of the cabinet,
 - the range of values of h for which the cabinet will not tip.



RD0034

7. Three particles A, B, and C each of mass m , are connected to each other by three massless rigid rods to form a rigid, equilateral triangular body of side ℓ . This body is placed on a horizontal frictionless table (x - y plane) and is hinged to it at the point A, so that it can move without friction about the vertical axis through A (see figure). The body is set into rotational motion on the table about A with a constant angular velocity ω .
- Find the magnitude of the horizontal force exerted by the hinge on the body.
 - At time T , when the side BC is parallel to the x -axis, a force F is applied on B along BC (as shown). Obtain the x -component and the y -component of the force exerted by the hinge on the body, immediately after time T .

[IIT-JEE' 2001]

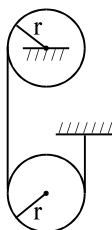


RD0035

8. The door of an automobile is open and perpendicular to the body. The automobile starts with an acceleration of 2 ft/sec^2 , and the width of the door is 30 inches. Treat the door as a uniform rectangle, and neglect friction to find the speed of its outside edge as seen by the driver when the door closes.

RD0036

9. Two uniform cylinders, each of mass $m = 10 \text{ kg}$ and radius $r = 150 \text{ mm}$, are connected by a rough belt as shown. If the system is released from rest, determine
- the velocity of the centre of cylinder A after it has moved through 1.2 m &
 - the tension in the portion of the belt connecting the two cylinders.



RD0037

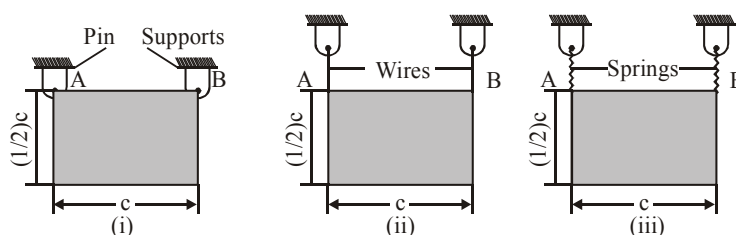
- 10.** A circular disc of mass 300 gm and radius 20 cm can rotate freely about a vertical axis passing through its centre O. A small insect of mass 100 gm is initially at a point A on the disc (which is initially stationary) the insect starts walking from rest along the rim of the disc with such a time varying relative velocity that the disc rotates in the opposite direction with a constant angular acceleration $= 2\pi \text{ rad/s}^2$. After some time T, the insect is back at the point A. By what angle θ has the disc rotated till now ; as seen by a stationary earth observer? Also find the time T.

RD0038

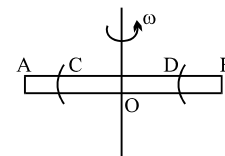
- 11.** A uniform plate of mass m is suspended in each of the ways shown. For each case determine immediately after the connection at B has been released ;

(a) the angular acceleration of the plate .

(b) the acceleration of its mass center .

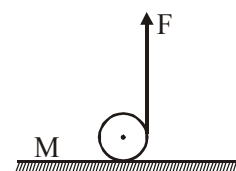


- 12.** A rigid horizontal smooth rod AB of mass 0.75 kg and length 40 cm can rotate freely about a fixed vertical axis through its mid point O. Two rings each of mass 1 kg are initially at rest at a distance of 10 cm from O on either side of the rod. The rod is set in rotation with an angular velocity of 30 radians per second. Find the velocity of each ring along the length of the rod in m/s when they reach the ends of the rod



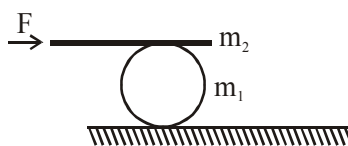
RD0039

- 13.** The free end of the string wound on the surface of a solid cylinder of mass M & radius r is pulled up by a force F as shown. If there is sufficient friction between cylinder & floor so that the cylinder is able to roll without slipping, find the maximum angular acceleration that the cylinder can have (without cylinder leaving contact with surface).



RD0040

- 14.** A man pushes a cylinder of mass m_1 with help of a plank of mass m_2 as shown. There is no slipping at any contact. the horizontal component of the force applied by the man is F . Find:



- (a) the accelerations of the plank and the centre of mass of the cylinder, and
(b) the magnitudes and directions of frictional forces at contact points. **[JEE '99, 6 + 4]**

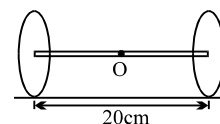
RD0042

15. A 160 mm diameter pipe of mass 6 kg rests on a 1.5 kg plate. The pipe and plate are initially at rest when a force P of magnitude 25 N is applied for 0.75 s. Knowing that $\mu_s = 0.25$ & $\mu_k = 0.20$ between the plate and both the pipe and the floor, determine;
- whether the pipe slides with respect to the plate.
 - the resulting velocities of the pipe and of the plate.



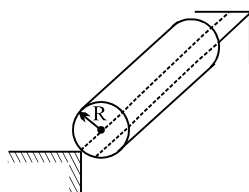
RD0043

16. Two thin circular disks of mass 2 kg and radius 10 cm each are joined by a rigid massless rod of length 20 cm. The axis of the rod is along the perpendicular to the planes of the disks through their centre. The object is kept on a truck in such a way that the axis of the object is horizontal and perpendicular to the direction of motion of the truck. Its friction with the floor of the truck is large enough so that the object can roll on the truck without slipping. Take x -axis as the direction of motion of the truck and z -axis as the vertically upwards direction. If the truck has an acceleration of 9 m/s^2 calculate :
- the force of friction on each disk
 - The magnitude and the direction of the frictional torque acting on each disk about the centre of mass O of the object. Express the torque in the vector form of unit vectors in the x - y and z directions.



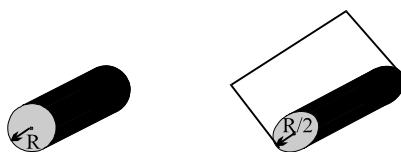
RD0044

17. A rectangular rigid fixed block has a long horizontal edge. A solid homogeneous cylinder of radius R is placed horizontally at rest with its length parallel to the edge such that the axis of the cylinder and the edge of the block are in the same vertical plane as shown in figure. there is sufficient friction present at the edge so that a very small displacement cause the cylinder to roll of the edge without slipping. Determine
- the angle θ_c through which the cylinder rotates before it leaves contact with the edge,
 - the speed of the centre of mass of the cylinder before leaving contact with the edge, and
 - the ratio of the translational to rotational kinetic energies of the cylinder when its centre of mass is in horizontal line with the edge.



RD0045

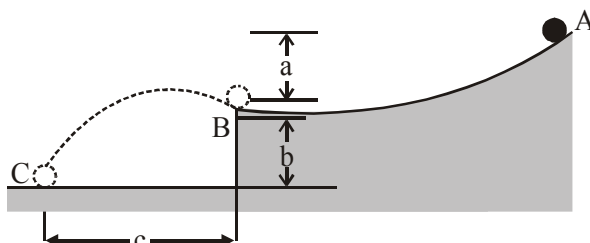
18. A carpet of mass ' M ' made of inextensible material is rolled along its length in the form of a cylinder of radius ' R ' and is kept on a rough floor. The carpet starts unrolling without sliding on the floor when a negligibly small push is given to it. The horizontal velocity of the axis of the cylindrical part of the carpet when its radius reduces to $R/2$ will be:



RD0046

19. A small sphere of mass m and radius r is released from rest at A and rolls without sliding on the curved surface to point B where it leaves the surface with a horizontal velocity. Knowing that $a = 1.5 \text{ m}$ & $b = 1.2 \text{ m}$, determine;

- (a) the speed of the sphere as it strikes the ground at C .
(b) the corresponding distance c .



RD0047

20. A uniform disk of mass m and radius R is projected horizontally with velocity v_0 on a rough horizontal floor so that it starts off with a purely sliding motion at $t = 0$. After t_0 seconds it acquires a purely rolling motion as shown in figure.

- (i) Calculate the velocity of the centre of mass of the disk at t_0 .
(ii) Assuming the coefficient of friction to be μ calculate t_0 . Also calculate the work done by the frictional force as a function of time and the total work done by it over a time t much longer than t_0 .



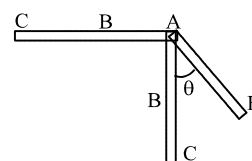
RD0048

21. A uniform disc of mass m and radius R rotates about a fixed vertical axis passing through its centre with angular velocity ω . A particle of same mass m and having velocity $2\omega R$ towards centre of the disc collides with the disc moving horizontally and sticks to its rim. Find

- (a) the angular velocity of the disc.
(b) the impulse on the particle due to disc.
(c) the impulse on the disc due to hinge.

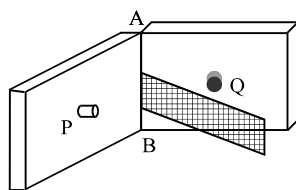
RD0049

22. A rod hinged at one end is released from the horizontal position as shown in the figure. When it becomes vertical its lower half separates without exerting any reaction at the breaking point. Then find the maximum angle ' θ ' made by the hinged upper half with the vertical.



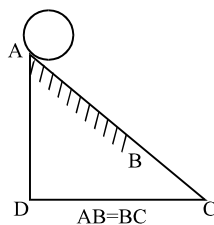
RD0050

23. Two heavy metallic plates are joined together at 90° to each other. A laminar sheet of mass 30 Kg is hinged at the line AB joining the two heavy metallic plates. The hinges are frictionless. The moment of inertia of the laminar sheet about an axis parallel to AB and passing through its centre of mass is 1.2 Kg-m^2 . Two rubber obstacles P and Q are fixed, one on each metallic plate at a distance 0.5 m from the line AB. This distance is chosen so that the reaction due to the hinges on the laminar sheet is zero during the impact. Initially the laminar sheet hits one of the obstacles with an angular velocity 1 rad/s and turns back. If the impulse on the sheet due to each obstacle is 6 N-s . [IIT-JEE' 2001]
- (a) Find the location of the centre of mass of the laminar sheet from AB.
- (b) At what angular velocity does the laminar sheet come back after the first impact ?
- (c) After how many impacts, does the laminar sheet come to rest ?



RD0051

24. Portion AB of the wedge shown in figure is rough and BC is smooth. A solid cylinder rolls without slipping from A to B. Find the ratio of translational kinetic energy to rotational kinetic energy, when the cylinder reaches point C.



RD0052

25. A rod AB of mass M and length L is lying on a horizontal frictionless surface. A particle of mass m travelling along the surface hits the end 'A' of the rod with a velocity v_0 in the direction perpendicular to AB. The collision is completely elastic. After the collision the particle comes to rest.
- (a) Find the ratio m/M . [IIT-JEE' 2000]
- (b) A point P on the rod is at rest immediately after the collision. Find the distance AP.
- (c) Find the linear speed of the point P at a time $\pi L/(3v_0)$ after the collision.

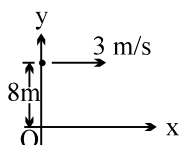
RD0053

EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

Concept of ω , α , v , a

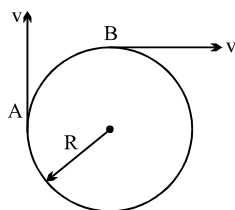
1. A particle starts from the point (0m, 8m) and moves with uniform velocity of $3\hat{i}$ m/s. After 5 seconds, the angular velocity of the particle about the origin will be :



- (A) $\frac{8}{289}$ rad/s (B) $\frac{3}{8}$ rad/s (C) $\frac{24}{289}$ rad/s (D) $\frac{8}{17}$ rad/s

RD0054

2. Two points of a rigid body are moving as shown. The angular velocity of the body is:



- (A) $\frac{v}{2R}$ (B) $\frac{v}{R}$ (C) $\frac{2v}{R}$ (D) $\frac{2v}{3R}$

RD0055

Moment of inertia

3. Three bodies have equal mass m . Body A is solid cylinder of radius R , body B is a square lamina of side R , and body C is a solid sphere of radius R . Which body has the smallest moment of inertia about an axis passing through their centre of mass and perpendicular to the plane (in case of lamina)
- (A) A (B) B (C) C (D) A and C both

RD0056

4. For the same total mass which of the following will have the largest moment of inertia about an axis passing through its centre of mass and perpendicular to the plane of the body
- (A) a disc of radius a (B) a ring of radius a
 (C) a square lamina of side $2a$ (D) four rods forming a square of side $2a$

RD0057

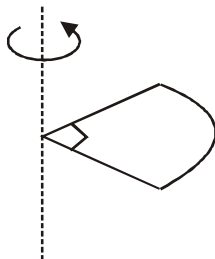
5. Let I be the moment of inertia of a uniform square plate about an axis AB that passes through its centre and is parallel to two of its sides. CD is a line in the plane of the plate that passes through the centre of the plate and makes an angle θ with AB . The moment of inertia of the plate about the axis CD is then equal to

(A) I (B) $I \sin^2 \theta$ (C) $I \cos^2 \theta$ (D) $I \cos^2(\theta/2)$

RD0058

6. One quarter sector is cut from a uniform circular disc of radius R . This sector has mass M . It is made to rotate about a line perpendicular to its plane and passing through the center of the original disc. Its moment of inertia about the axis of rotation is

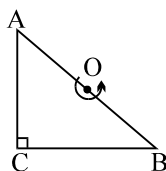
[IIT-JEE 2000]



(A) $\frac{1}{2} MR^2$ (B) $\frac{1}{4} MR^2$ (C) $\frac{1}{8} MR^2$ (D) $\sqrt{2} MR^2$

RD0059

7. Find the moment of inertia of a plate cut in shape of a right angled triangle of mass M , about an axis perpendicular to the plane of the plate and passing through the mid point of side AB . (Side $AC = BC = a$)



(A) $\frac{Ma^2}{12}$ (B) $\frac{Ma^2}{6}$ (C) $\frac{Ma^2}{3}$ (D) $\frac{2Ma^2}{3}$

RD0060

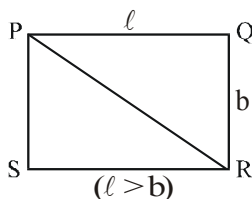
8. A circular disc X of radius R is made from an iron plate of thickness t and another disc Y of radius $4R$ is made from an iron plate of thickness $t/4$. Then the relation between the moment of inertia I_X and I_Y is-

[AIEEE - 2003]

(A) $I_Y = 32 I_X$ (B) $I_Y = 16 I_X$ (C) $I_Y = I_X$ (D) $I_Y = 64 I_X$

RD0061

9. Moment of inertia of a rectangular plate about an axis passing through P and perpendicular to the plate is I . Then moment of PQR about an axis perpendicular to the plane of the plate:



(A) about $P = I/2$ (B) about $R = I/2$ (C) about $P > I/2$ (D) about $R > I/2$

RD0062

10. A thin uniform rod of mass M and length L has its moment of inertia I_1 about its perpendicular bisector. The rod is bend in the form of a semicircular arc. Now its moment of inertia through the centre of the semi circular arc and perpendicular to its plane is I_2 . The ratio of $I_1 : I_2$ will be _____
 (A) < 1 (B) > 1 (C) $= 1$ (D) can't be said

RD0063

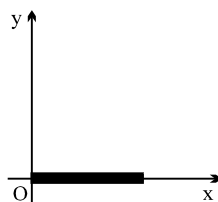
11. One solid sphere A and another hollow sphere B are of same mass and same outer radii. Their moment of inertia about their diameters are respectively I_A and I_B such that- [AIEEE - 2004]

- (A) $I_A = I_B$ (B) $I_A > I_B$ (C) $I_A < I_B$ (D) $\frac{I_A}{I_B} = \frac{d_A}{d_B}$

where d_A and d_B are their densities.

RD0064

12. The figure shows a uniform rod lying along the x-axis. The locus of all the points lying on the xy-plane, about which the moment of inertia of the rod is same as that about O is :-



- (A) an ellipse (B) a circle (C) a parabola (D) a straight line

RD0065

13. A rigid body can be hinged about any point on the x-axis. When it is hinged such that the hinge is at x , the moment of inertia is given by $I = 2x^2 - 12x + 27$. The x-coordinate of centre of mass is
 (A) $x = 2$ (B) $x = 0$ (C) $x = 1$ (D) $x = 3$

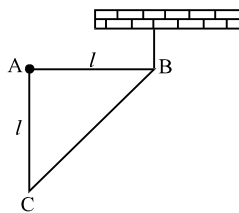
RD0066

Equilibrium

14. A weightless rod is acted on by upward parallel forces of 2N and 4N at ends A and B respectively. The total length of the rod is $AB = 3\text{m}$. To keep the rod in equilibrium a force of 6N should act in the following manner:
 (A) Downwards at any point between A and B.
 (B) Downwards at mid point of AB.
 (C) Downwards at a point C such that $AC = 1\text{m}$.
 (D) Downwards at a point D such that $BD = 1\text{m}$.

RD0067

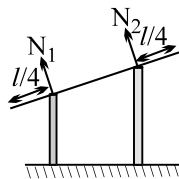
15. A right triangular plate ABC of mass m is free to rotate in the vertical plane about a fixed horizontal axis through A. It is supported by a string such that the side AB is horizontal. The reaction at the support A in equilibrium is:



- (A) $\frac{mg}{3}$ (B) $\frac{2mg}{3}$ (C) $\frac{mg}{2}$ (D) mg

RD0068

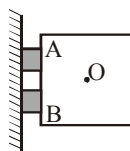
16. A uniform rod of length l is placed symmetrically on two walls as shown in figure. The rod is in equilibrium. If N_1 and N_2 are the normal forces exerted by the walls on the rod then :-



- (A) $N_1 > N_2$ (B) $N_1 < N_2$
(C) $N_1 = N_2$ (D) N_1 and N_2 would be in the vertical directions

RD0069

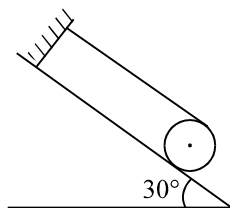
17. A vertical rectangular door with its centre of gravity at O (see figure) is fixed on two hinges A and B along one vertical length side of the door. The entire weight of the door is supported by the hinge A. Then the free body force diagram for the door (the arrows indicate the direction of the forces) is :-



- (A) (B)
- (C) (D)

RD0070

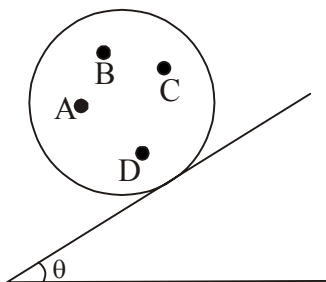
18. A thin hoop of weight 500 N and radius 1 m rests on a rough inclined plane as shown in the figure. The minimum coefficient of friction needed for this configuration to be in equilibrium is:



- (A) $\frac{1}{3\sqrt{3}}$ (B) $\frac{1}{\sqrt{3}}$ (C) $\frac{1}{2}$ (D) $\frac{1}{2\sqrt{3}}$

RD0071

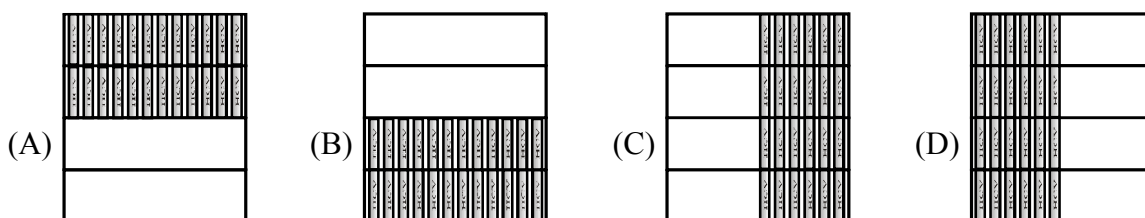
19. A non uniform sphere can be kept on a rough inclined plane so that it is in equilibrium. In the figure below the dots represents location of centre of mass. In which one of the positions can sphere be in equilibrium?



RD0072

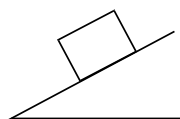
Toppling

20. Same number of books are placed in four book cases as shown. Which bookcase is most likely to topple forward if pulled a little at the top towards right :



RD0073

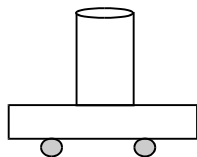
21. A homogeneous cubical brick lies motionless on a rough inclined surface. The half of the brick which applies greater pressure on the plane is :



- (A) left half (B) right half
(C) both applies equal pressure (D) the answer depend upon coefficient of friction

RD0074

22. A uniform 2 kg cylinder rests on a laboratory cart as shown. The coefficient of static friction between the cylinder and the cart is 0.5. If the cylinder is 4 cm in diameter and 10 cm in height, which of the following is closest to the maximum acceleration of the cart such that cylinder neither slips nor tips over?

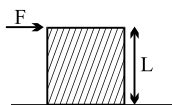


- (A) 2 m/s^2 (B) 4 m/s^2 (C) 5 m/s^2 (D) 6 m/s^2

RD0075

23. A cubical block of side L rests on a rough horizontal surface with coefficient of friction μ . A horizontal force F is applied on the block as shown. If the coefficient of friction is sufficiently high so that the block does not slide before toppling, the minimum force required to topple the block is:

[IIT-JEE'(Scr)'2000]

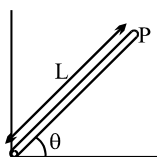


- (A) infinitesimal (B) $mg/4$ (C) $mg/2$ (D) $mg(1-\mu)$

RD0076

$$\tau = I\alpha$$

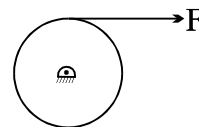
24. A uniform flag pole of length L and mass M is pivoted on the ground with a frictionless hinge. The flag pole makes an angle θ with the horizontal. The moment of inertia of the flag pole about one end is $(1/3)ML^2$. If it starts falling from the position shown in the accompanying figure, the linear acceleration of the free end of the flag pole (labeled P) immediately after it starts falling off would be:



- (A) $(2/3)g\cos\theta$ (B) $(2/3)g$ (C) g (D) $(3/2)g\cos\theta$

RD0077

25. A pulley is hinged at the centre and a massless thread is wrapped around it. The thread is pulled with a constant force F starting from rest. As time increases,
- (A) its angular velocity increases, but force on hinge remains constant
 (B) its angular velocity remains same, but force on hinge increases
 (C) its angular velocity increases and force on hinge increases
 (D) its angular velocity remains same and force on hinge is constant



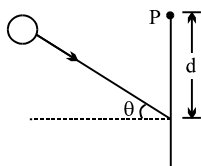
RD0078

Angular momentum and its conservation

26. A particle of mass 2 kg located at the position $(\hat{i} + \hat{j})$ m has a velocity $2(+\hat{i} - \hat{j} + \hat{k})$ m/s. Its angular momentum about z-axis in $\text{kg-m}^2/\text{s}$ is:
 (A) zero (B) +8 (C) 12 (D) -8

RD0079

27. A ball of mass m moving with velocity v , collide with the wall elastically as shown in the figure. After impact the change in angular momentum about P is:



- (A) $2 mvd$ (B) $2 mvd \cos \theta$ (C) $2 mvd \sin \theta$ (D) zero

RD0080

28. Two uniform spheres of mass M have radii R and $2R$. Each sphere is rotating about a fixed axis through a diameter. The rotational kinetic energies of the spheres are identical. What is the ratio of the magnitude of the angular momenta of these spheres? That is, $\frac{L_{2R}}{L_R} =$

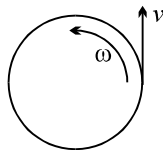
- (A) 4 (B) $2\sqrt{2}$ (C) 2 (D) $\sqrt{2}$

RD0081

29. A spinning ice skater can increase his rate of rotation by bringing his arms and free leg closer to his body. How does this procedure affect the skater's angular momentum and kinetic energy and what is the work done by the skater?
 (A) angular momentum remains the same while kinetic energy increases and work done is positive.
 (B) angular momentum remains the same while kinetic energy decreases and work done is negative.
 (C) both angular momentum and kinetic energy remain the same and work done is zero.
 (D) angular momentum increases while kinetic energy remains the same and work done may be positive or negative.

RD0082

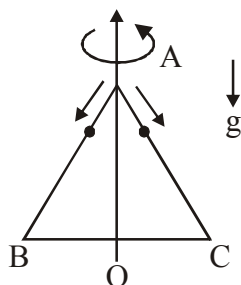
30. A child with mass m is standing at the edge of a disc with moment of inertia I , radius R , and initial angular velocity ω . See figure given below. The child jumps off the edge of the disc with tangential velocity v with respect to the ground. The new angular velocity of the disc is



- (A) $\sqrt{\frac{I\omega^2 - mv^2}{I}}$ (B) $\sqrt{\frac{(I + mR^2)\omega^2 - mv^2}{I}}$
 (C) $\frac{I\omega - mvR}{I}$ (D) $\frac{(I + mR^2)\omega - mvR}{I}$

RD0083

31. An equilateral triangle ABC formed from a uniform wire has two small identical beads initially located at A. The triangle is set rotating about the vertical axis AO. Then the beads are released from rest simultaneously and allowed to slide down, one along AB and the other along AC as shown. Neglecting frictional effects, the quantities that are conserved as the beads slide down, are



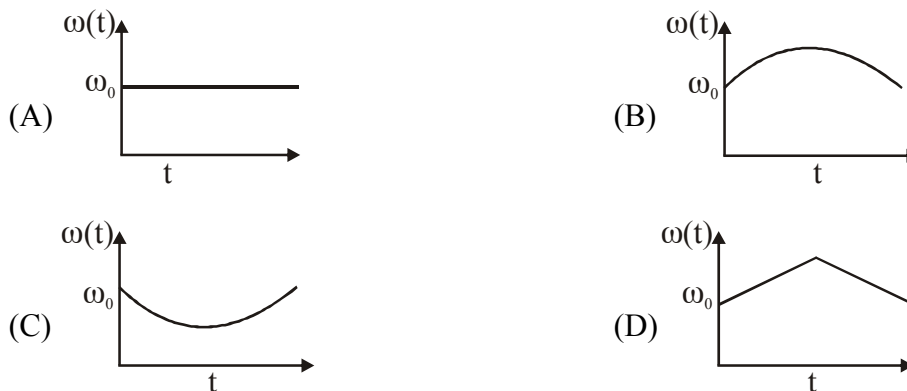
- (A) Angular velocity and total energy (kinetic and potential)
 (B) Total angular momentum and total energy
 (C) Angular velocity and moment of inertia about the axis of rotation.
 (D) Total angular momentum and moment of inertia about the axis of rotation.

[IIT-JEE 2000]

RD0084

32. A circular platform is free to rotate in a horizontal plane about a vertical axis passing through its centre. A tortoise is sitting at the edge of the platform. Now, the platform is given an angular velocity ω_0 . When the tortoise move along a chord of the platform with a constant velocity (with respect to the platform), the angular velocity of the platform $\omega(t)$ will vary with time t as

[IIT-JEE 2002]

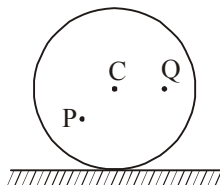


RD0085

Combined rotation and translation

Kinematics

33. A disc is rolling without slipping with angular velocity ω . P and Q are two points equidistant from the centre C. The order of magnitude of velocity is :- [IIT-JEE 2004]

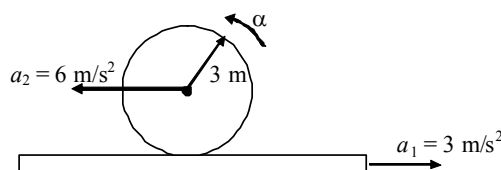


- (A) $v_Q > v_C > v_P$ (B) $v_P > v_C > v_Q$
(C) $v_P = v_C, v_Q = v_C/2$ (D) $v_P < v_C > v_Q$

RD0086

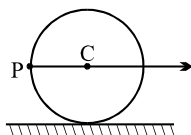
34. In the following figure, a sphere of radius 3 m rolls on a plank. The accelerations of the sphere and the plank are indicated. The value of α is

- (A) 3 rad/s^2
(B) 6 rad/s^2
(C) 3 rad/s^2 (opposite to the direction shown in figure)
(D) 1 rad/s^2



RD0087

35. A disc of radius R is rolling purely on a flat horizontal surface, with a constant angular velocity. The angle between the velocity and acceleration vectors of point P is

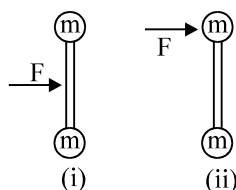


- (A) zero (B) 45° (C) 135° (D) $\tan^{-1}(1/2)$

RD0088

Dynamics

36. A force F is applied to a dumbbell for a time interval, t, first as in (i) and then as in (ii). In which case does the dumbbell acquire the greater centre-of-mass speed?



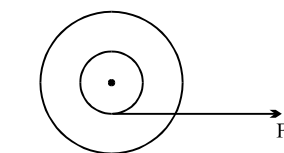
- (A) (i)
(B) (ii)
(C) there is no difference
(D) the answer depends on the rotational inertia of the dumbbell

RD0089

37. A cylinder rolls up an inclined plane, reaches some height, and then rolls down (without slipping throughout these motions). The directions of the frictional force acting on the cylinder are
 (A) Up the incline while ascending and down the incline while descending [IIT-JEE 2002]
 (B) Up the incline while ascending as well as descending
 (C) down the incline while ascending and up the incline while descending
 (D) down the incline while ascending as well as descending

RD0090

38. Inner and outer radii of a spool are r and R respectively. A thread is wound over its inner surface and placed over a rough horizontal surface. Thread is pulled by a force F as shown in fig. then in case of pure rolling



- (A) Thread unwinds, spool rotates anticlockwise and friction act leftwards
 (B) Thread winds, spool rotates clockwise and friction acts leftwards
 (C) Thread winds, spool moves to the right and friction act rightwards
 (D) Thread winds, spool moves to the right and friction does not come into existence.

RD0091

39. A body kept on a smooth horizontal surface is pulled by a constant horizontal force applied at the top point of the body. If the body rolls purely on the surface, its shape can be :
 (A) thin pipe (B) uniform cylinder
 (C) uniform sphere (D) thin spherical shell

RD0092

40. A solid sphere with a velocity (of centre of mass) v and angular velocity ω is gently placed on a rough horizontal surface. The frictional force on the sphere:
 (A) must be forward (in direction of v) (B) must be backward (opposite to v)
 (C) cannot be zero (D) none of the above

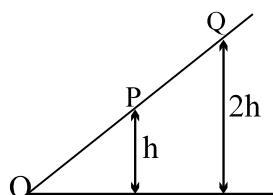
RD0093

Kinetic energy

41. A hoop and a solid cylinder have the same mass and radius. They both roll, without slipping, on a horizontal surface. If their kinetic energies are equal
 (A) the hoop has a greater translational speed than the cylinder
 (B) the cylinder has a greater translational speed than the hoop
 (C) the hoop and the cylinder have the same translational speed
 (D) the hoop has a greater rotational speed than the cylinder

RD0094

42. A ball rolls down an inclined plane as shown in figure. The ball is first released from rest from P and then later from Q. Which of the following statement is/ are correct?
- (i) The ball takes twice as much time to roll from Q to O as it does to roll from P to O.
 - (ii) The acceleration of the ball at Q is twice as large as the acceleration at P.
 - (iii) The ball has twice as much K.E. at O when rolling from Q as it does when rolling from P.



- (A) i, ii only (B) ii, iii only (C) i only (D) iii only

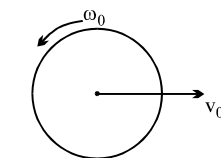
RD0095

43. The moment of inertia of a solid cylinder about its axis is given by $(1/2)MR^2$. If this cylinder rolls without slipping, the ratio of its rotational kinetic energy to its translational kinetic energy is
- (A) 1 : 1 (B) 2 : 2 (C) 1 : 2 (D) 1 : 3

RD0096

Angular momentum

44. A uniform circular disc placed on a rough horizontal surface has initially a velocity v_0 and an angular velocity ω_0 as shown in the figure. The disc comes to rest after moving some distance in the direction of motion. Then $\frac{v_0}{r\omega_0}$ is



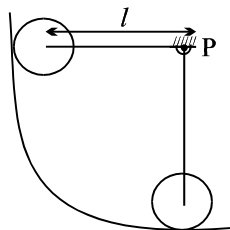
- (A) $\frac{1}{2}$ (B) 1 (C) $\frac{3}{2}$ (D) 2

RD0097

45. A hollow sphere of radius R and mass m is fully filled with non viscous liquid of mass m . It is rolled down a horizontal plane such that its centre of mass moves with a velocity v . If it purely rolls
- (A) Kinetic energy of the sphere is $\frac{5}{6}mv^2$
 - (B) Kinetic energy of the sphere is $\frac{4}{5}mv^2$
 - (C) Angular momentum of the sphere about a fixed point on ground is $\frac{8}{3}m v R$
 - (D) Angular momentum of the sphere about a fixed point on ground is $\frac{14}{5}m v R$

RD0098

46. A sphere of mass M and radius R is attached by a light rod of length ℓ to a point P . The sphere rolls without slipping on a circular track as shown. It is released from the horizontal position. The angular momentum of the system about P when the rod becomes vertical is :

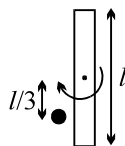


- (A) $M\sqrt{\frac{10}{7}gl} [l + R]$ (B) $M\sqrt{\frac{10}{7}gl} \left[l + \frac{2}{5}R\right]$
 (C) $M\sqrt{\frac{10}{7}gl} \left[l + \frac{7}{5}R\right]$ (D) $M\sqrt{\frac{10}{7}gl} \left[l - \frac{2}{5}R\right]$

RD0099

Eccentric collision

47. A uniform rod of length l and mass M rotating about a fixed vertical axis on a smooth horizontal table. It elastically strikes a particle placed at a distance $l/3$ from its axis and stops. Mass of the particle is



- (A) $3M$ (B) $\frac{3M}{4}$ (C) $\frac{3M}{2}$ (D) $\frac{4M}{3}$

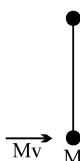
RD0100

48. A mass m is moving at speed v perpendicular to a rod of length d and mass $M = 6m$ which pivots around a frictionless axle running through its centre. It strikes and sticks to the end of the rod. The moment of inertia of the rod about its centre is $Md^2/12$. Then the angular speed of the system right after the collision is

- (A) $2v/d$ (B) $2v/(3d)$ (C) v/d (D) $3v/(2d)$

RD0101

49. Two particles each of mass M are connected by a massless rod of length l . The rod is lying on the smooth surface. If one of the particle is given an impulse Mv as shown in the figure then angular velocity of the rod would be
 [IIT-JEE'(Scr)2003]

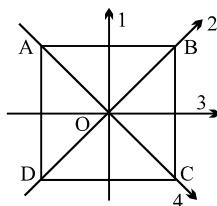


- (A) v/l (B) $2v/l$ (C) $v/2l$ (D) None

RD0102

MULTIPLE CORRECT TYPE QUESTIONS

50. ABCD is a square plate with centre O. The moments of inertia of the plate about the perpendicular axis through O is I and about the axes 1, 2, 3 & 4 are I_1, I_2, I_3 & I_4 respectively. It follows that :



- (A) $I_2 = I_3$ (B) $I = I_1 + I_4$ (C) $I = I_2 + I_4$ (D) $I_1 = I_3$

RD0103

51. A rod of weight w is supported by two parallel knives at end points A and B and is in equilibrium in a horizontal position. The knives are at a distance ' d ' from each other. The centre of mass of the rod is at a distance ' x ' from A.

- (A) the normal reaction at A is $\frac{wx}{d}$ (B) the normal reaction at A is $\frac{w(d-x)}{d}$
(C) the normal reaction at B is $\frac{wx}{d}$ (D) the normal reaction at B is $\frac{w(d-x)}{d}$

RD0104

52. A block with a square base measuring ' a ' $\times$ ' a ' and height h , is placed on an inclined plane. The coefficient of friction is μ . The angle of inclination (θ) of the plane is gradually increased. The block will

- (A) topple before sliding if $\mu > \frac{a}{h}$ (B) topple before sliding if $\mu < \frac{a}{h}$
(C) slide before toppling if $\mu > \frac{a}{h}$ (D) slide before toppling if $\mu < \frac{a}{h}$

RD0105

53. A particle falls freely near the surface of the earth. Consider a fixed point O (not vertically below the particle) on the ground.

- (A) Angular momentum of the particle about O is increasing .
(B) Torque of the gravitational force on the particle about O is decreasing.
(C) The moment of inertia of the particle about O is decreasing .
(D) The angular velocity of the particle about O is increasing.

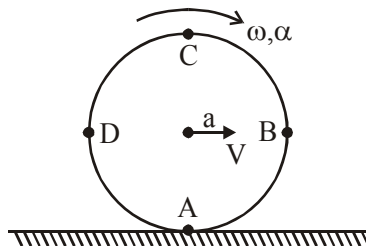
RD0106

54. A man spinning in free space changes the shape of his body, eg. by spreading his arms or curling up. By doing this, he can change his

- (A) moment of inertia (B) angular momentum
(C) angular velocity (D) rotational kinetic energy

RD0107

55. A circular disc of radius R rolls without slipping on a rough horizontal surface. At the instant shown its linear velocity is V , linear acceleration a , angular velocity ω and angular acceleration α . Four points A, B, C and D lie on its circumference such that the diameter AC is vertical & BD horizontal then choose the **CORRECT** option(s).



(A) $V_B = \sqrt{V^2 + (R\omega)^2}$

(B) $V_C = V + R\omega$

(C) $a_A = \sqrt{(a - R\alpha)^2 + (\omega^2 R)^2}$

(D) $a_D = \sqrt{(a + \omega^2 R)^2 + (R\alpha)^2}$

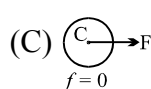
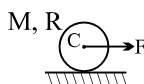
RD0108

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 56 to 58

In the following problems, indicate the correct direction of friction force acting on the cylinder, which is pulled on a rough surface by a constant force F .

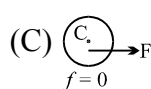
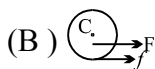
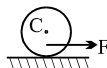
56. A cylinder of mass M and radius R is pulled horizontally by a force F . The friction force can be given by which of the following diagrams :-



(D) cannot be interpreted

RD0109

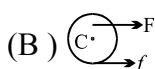
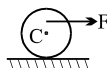
57. A cylinder is pulled horizontally by a force F acting at a point below the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams :-



(D) cannot be interpreted

RD0109

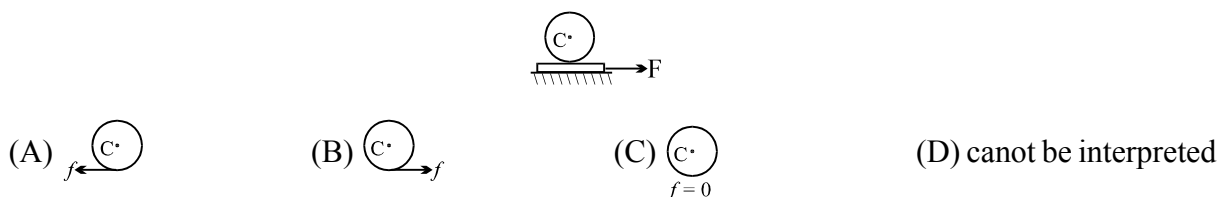
58. A cylinder is pulled horizontally by a force F acting at a point above the centre of mass of the cylinder, as shown in figure. The friction force can be given by which of the following diagrams



(D) cannot be interpreted

RD0109

59. A cylinder is placed on a rough plank which in turn is placed on a smooth surface. The plank is pulled with a constant force F . The friction force can be given by which of the following diagrams



RD0109

Paragraph for Question No. 60 to 62

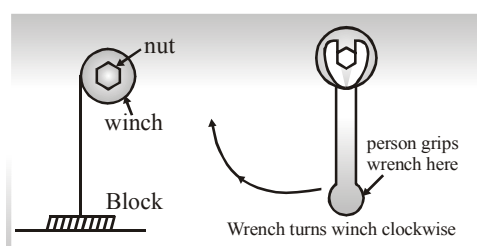
In figure, the winch is mounted on an axle, and the 6-sided nut is welded to the winch. By turning the nut with a wrench, a person can rotate the winch. For instance, turning the nut clockwise lifts the block off the ground, because more and more rope gets wrapped around the winch.

Three students agree that using a longer wrench makes it easier to turn the winch. But they disagree about why. All three students are talking about the case where the winch is used, over a 10 s time interval, to lift the block one metre off the ground.

Student 1 : By using a longer wrench, the person decreases the average force he must exert on the wrench, in order to lift the block one metre in 10 s.

Student 2 : Using a longer wrench reduces the work done by the person as he uses the winch to lift the block 1m in 10s.

Student 3 : Using a longer wrench reduces the power that the person must exert to lift the block 1m in 10s.



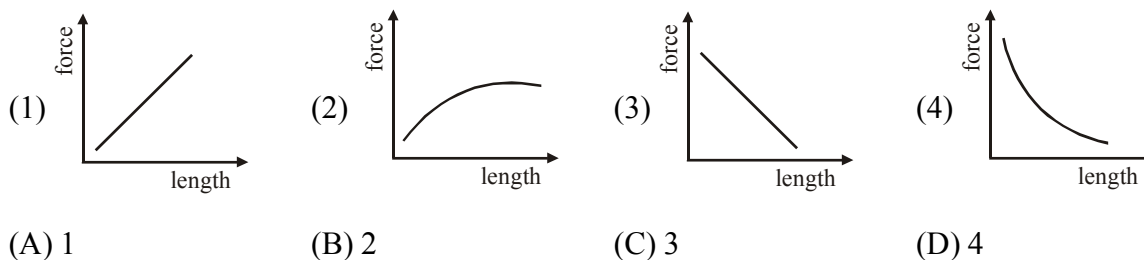
60. Student 1 is :-
- (A) correct, because the torque that the wrench must exert to lift the block doesn't depend on the wrench's length
- (B) correct, because using a longer wrench decreases the torque it must exert on the winch
- (C) incorrect, because the torque that the wrench must exert to lift the block doesn't depend on the wrench's length
- (D) Incorrect, because using a longer wrench decreases the torque it must exert on the winch.

RD0110

61. Which of the following is true about student 2 and 3 :-
- (A) Student 2 and 3 are both correct
- (B) Student 2 is correct, but student 3 is incorrect
- (C) Student 3 is correct, but student 2 is incorrect
- (D) Student 2 and 3 are both incorrect

RD0110

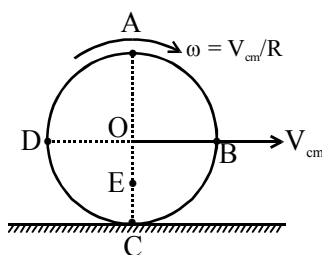
62. If several wrenches all apply the same torque to a nut, which graph best expresses the relationship between the force the person must apply to the wrench, and the length of the wrench :-



RD0110

MATRIX MATCH TYPE QUESTION

63. Consider a body rolling on a horizontal surface as shown in figure (Symbols have their usual meaning)



Column-I

- (A) Velocity is zero
(B) Speed is maximum
(C) $0 < \text{speed} < v_{\text{cm}}$
(D) $1.3v_{\text{cm}} < \text{speed} < 2v_{\text{cm}}$

Column-II

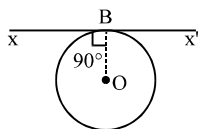
- (P) Point A
(Q) Point B
(R) Point C
(S) Point D
(T) Point E

RD0111

EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

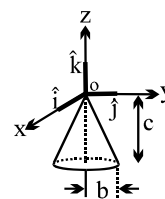
1. A thin wire of length L and uniform linear mass density ρ is bent into a circular loop with centre at O as shown. The moment of inertia of the loop about the axis XX' is : [IIT-JEE'(Scr)'2000]



- (A) $\rho L^3/8\pi^2$ (B) $\rho L^3/16\pi^2$ (C) $5\rho L^3/16\pi^2$ (D) $3\rho L^3/8\pi^2$

RD0112

2. A solid cone hangs from a frictionless pivot at the origin O , as shown. If $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors, and a , b , and c are positive constants, which of the following forces F applied to the rim of the cone at a point P results in a torque τ on the cone with a negative component τ_z ?



- (A) $F = a\hat{k}$, P is $(0, b, -c)$ (B) $F = -a\hat{k}$, P is $(0, -b, -c)$
(C) $F = a\hat{j}$, P is $(-b, 0, -c)$ (D) None

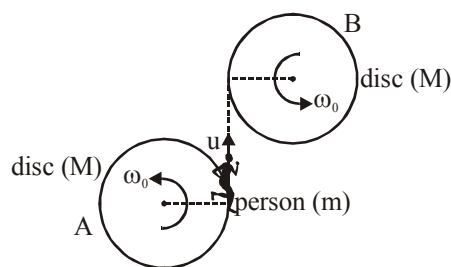
RD0113

3. A heavy seesaw (i.e., not massless) is out of balance. A light girl sits on the end that is tilted downward, and a heavy body sits on the other side so that the seesaw now balances. If they both move forward so that they are one-half their original distance from the pivot point (the fulcrum) what will happen to the seesaw ?
- (A) The side the body is sitting on will tilt downward
(B) The side the girl is sitting on will once again tilt downward
(C) Nothing ; the seesaw will still be balanced
(D) It is impossible to say without knowing the masses and the distances

RD0114

4. Two identical discs each of mass $M (= 4m)$ and radius R are rotating in opposite sense with equal angular speed ω_0 about vertical axes passing through their centres, (as shown in figure). A person of mass m sitting on circumference of disc A jumps with a tangential relative velocity u (after jumping) w.r.t one rotating disc (A) and lands on other disc (B) also tangential. Now the second disc (B) comes to a stop. Find the relative velocity u .

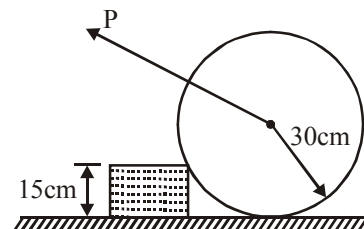
- (A) $u = \frac{\omega_0 R}{2}$
(B) $u = \omega_0 R$
(C) $u = \frac{3\omega_0 R}{2}$
(D) $u = 2\omega_0 R$



RD0115

5. In the given figure a uniform wheel of radius 30cm rests against a rigid rectangular block 15cm high. The wheel weighs 1000 N. The minimum pull P through the center which will turn the wheel over the block is :-

(A) $500\sqrt{3}$ N (B) $1000\sqrt{3}$ N
(C) 1000 N (D) $400\sqrt{3}$ N



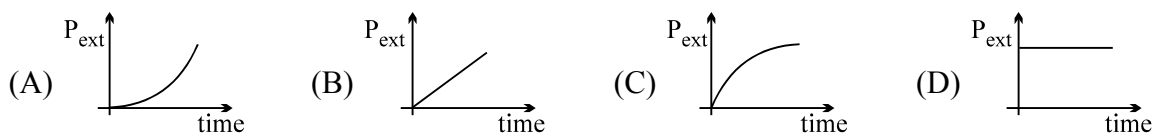
RD0116

6. A uniform rod AB of mass m and length l is at rest on a smooth horizontal surface. An impulse J is applied to the end B, perpendicular to the rod in the horizontal direction. Speed of particle P at a distance $\frac{l}{6}$ from the centre towards A of the rod after time $t = \frac{\pi m l}{12J}$ is

(A) $2\frac{J}{m}$ (B) $\frac{J}{\sqrt{2}m}$ (C) $\frac{J}{m}$ (D) $\sqrt{2}\frac{J}{m}$

RD0117

7. A rod is hinged at its centre and rotated by applying a constant torque starting from rest. The power developed by the external torque as a function of time is :



RD0118

8. A straight rod of length L is released on a frictionless horizontal floor in a vertical position. As it falls + slips, the distance of a point on the rod from the lower end, which follows a quarter circular locus is :-

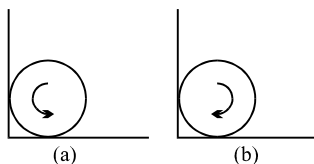
(A) $L/2$ (B) $L/4$ (C) $L/8$ (D) None

RD0119

9. A sphere is placed rotating with its centre initially at rest in a corner as shown in figure (a) & (b).

Coefficient of friction between all surfaces and the sphere is $1/3$. Find the ratio of the frictional force $\frac{f_a}{f_b}$

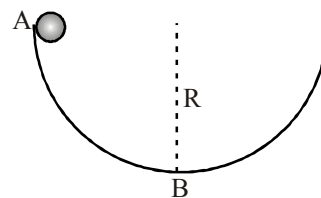
by ground in situations (a) & (b).



(A) 1 (B) $\frac{9}{10}$ (C) $\frac{10}{9}$ (D) none

RD0120

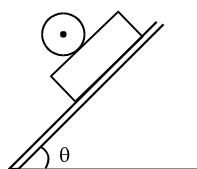
10. A small sphere A of mass m and radius r rolls without slipping inside a large fixed hemispherical bowl of radius R ($\gg r$) as shown in figure. If the sphere starts from rest at the top point of the hemisphere find the normal force exerted by the small sphere on the hemisphere when it is at the bottom B of the hemisphere.



- (A) $\frac{10}{7} mg$ (B) $\frac{17}{7} mg$ (C) $\frac{5}{7} mg$ (D) $\frac{7}{5} mg$

RD0121

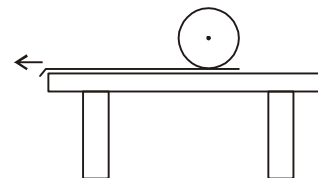
11. A plank of mass M is placed over smooth inclined plane and a sphere is also placed over the plank. Friction is sufficient between sphere and plank. If plank and sphere are released from rest, the frictional force on sphere is:



- (A) up the plane (B) down the plane (C) horizontal (D) zero

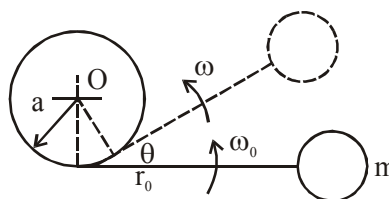
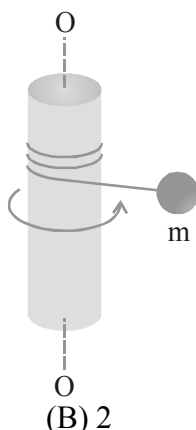
RD0122

12. A thin table cloth covers a horizontal table and a uniform body of round shape lies on top of it. The table cloth is pulled from under the body, and friction causes the body to slide and rotate. What is the body's final motion on the table? (Assume that the table is so large that the body does not fall off it.)
- (A) Body will finally roll towards left
(B) Body will finally roll towards right
(C) Body will finally come to rest
(D) Any of the above is possible depending on shape of body



RD0123

13. The small particle of mass m is given an initial high velocity in the horizontal plane and winds its cord around the fixed vertical shaft of radius $1m$. All motion occurs essentially in the horizontal plane. If the angular velocity of the cord is 0.8 rad/s when the distance from the particle to the tangential point is $5m$, determine the angular velocity ω (in rad/s) of the cord after it has turned through an angle 1 rad .

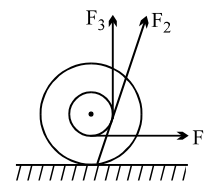


- (A) 1 (B) 2 (C) 4 (D) 6

RD0124

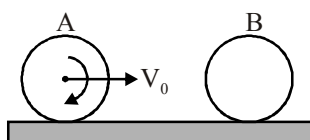
14. A yo-yo is resting on a perfectly rough horizontal table. Forces F_1 , F_2 and F_3 are applied separately as shown. The correct statement is :-

- (A) when F_3 is applied the centre of mass will move to the right.
 (B) when F_2 is applied the centre of mass will move to the left.
 (C) when F_1 is applied the centre of mass will move to the right.
 (D) when F_2 is applied the centre of mass will move to the right.



RD0125

15. A hollow smooth uniform sphere A of mass 'm' rolls without sliding on a smooth horizontal surface. It collides head on elastically with another stationary smooth solid sphere B of the same mass m and same radius. The ratio of kinetic energy of 'B' to that of 'A' just after the collision is :



- (A) 1 : 1 (B) 2 : 3 (C) 3 : 2 (D) None

RD0126

16. A wheel of radius r rolling on a straight line, the velocity of its centre being v . At a certain instant the point of contact of the wheel with the grounds is M and N is the highest point on the wheel (diametrically opposite to M). The incorrect statement is:

- (A) The velocity of any point P of the wheel is proportional to MP.
 (B) Points of the wheel moving with velocity greater than v form a larger area of the wheel than points moving with velocity less than v .
 (C) The point of contact M is instantaneously at rest.
 (D) The velocities of any two parts of the wheel which are equidistant from centre are equal.

RD0127

17. A ladder of length L is slipping with its ends against a vertical wall and a horizontal floor. At a certain moment, the speed of the end in contact with the horizontal floor is v and the ladder makes an angle $\alpha = 30^\circ$ with the horizontal. Then the speed of the ladder's center must be :-

- (A) $2v/\sqrt{3}$ (B) $v/2$ (C) v (D) None

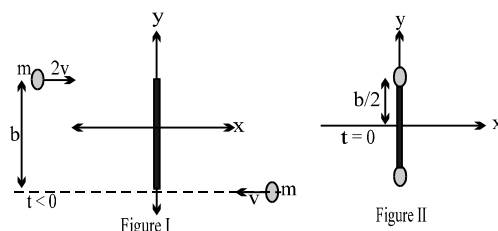
RD0128

18. In the previous question, if $dv/dt = 0$, then the angular acceleration of the ladder when $\alpha = 45^\circ$ is :-

- (A) $2v^2/L^2$ (B) $v^2/2L^2$ (C) $\sqrt{2}[v^2/L^2]$ (D) None

RD0129

19. One ice skater of mass m moves with speed $2v$ to the right, while another of the same mass m moves with speed v toward the left, as shown in figure I. Their paths are separated by a distance b . At $t = 0$, when they are both at $x = 0$, they grasp a pole of length b and negligible mass. For $t > 0$, consider the system as a rigid body of two masses m separated by distance b , as shown in figure II. Which of the following is the correct formula for the motion after $t = 0$ of the skater initially at $y = b/2$?



- (A) $x = 2vt, y = b/2$
 (B) $x = vt + 0.5b \sin(3vt/b), y = 0.5b \cos(3vt/b)$
 (C) $x = 0.5vt + 0.5b \sin(3vt/b), y = 0.5b \cos(3vt/b)$
 (D) $x = 0.5vt + 0.5b \sin(6vt/b), y = 0.5b \cos(6vt/b)$

RD0130

MULTIPLE CORRECT TYPE QUESTIONS

20. A rigid object is rotating in a counterclockwise sense around a fixed axis. If the rigid object rotates through more than 180° but less than 360° , which of the following pairs of quantities can represent an initial angular position and a final angular position of the rigid object. Which of the sets can only occur.

- (A) 3 rad, 6 rad (B) -1 rad, 1 rad (C) 1 rad, 5 rad (D) -1 rad, 2.5 rad

RD0131

21. A body is in equilibrium under the influence of a number of forces. Each force has a different line of action. The minimum number of forces required is

- (A) 2, if their lines of action pass through the centre of mass of the body.
 (B) 3, if their lines of action are not parallel.
 (C) 3, if their lines of action are parallel.
 (D) 4, if their lines of action are parallel and all the forces have the same magnitude.

RD0132

22. The torque $\vec{\tau}$ on a body about a given point is found to be equal to $\vec{A} \times \vec{L}$ where $\vec{A}$ is a constant vector and $\vec{L}$ is the angular momentum of the body about that point. From this it follows that

- (A) $d\vec{L}/dt$ is perpendicular to $\vec{L}$ at all instants of time
 (B) the components of $\vec{L}$ in the direction of $\vec{A}$ does not change with time
 (C) the magnitude of $\vec{L}$ does not change with time
 (D) $\vec{L}$ does not change with time

RD0133

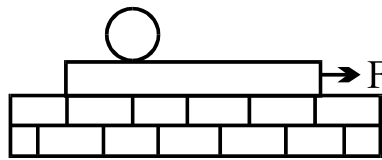
23. A block of mass m moves on a horizontal rough surface with initial velocity v . The height of the centre of mass of the block is h from the surface. Consider a point A on the surface.
- (A) angular momentum about A is mvh initially
 (B) the velocity of the block decreases as time passes.
 (C) torque of the forces acting on block is zero about A
 (D) angular momentum is not conserved about A.

RD0134

24. If a cylinder is rolling down the incline with sliding.
- (A) after some time it may start pure rolling
 (B) after sometime it will start pure rolling
 (C) it may be possible that it will never start pure rolling
 (D) none of these

RD0135

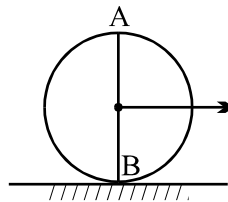
25. A plank with a uniform sphere placed on it rests on a smooth horizontal plane. Plank is pulled to right by a constant force F . If sphere does not slip over the plank. Which of the following is correct.



- (A) Acceleration of the centre of sphere is less than that of the plank.
 (B) Work done by friction acting on the sphere is equal to its total kinetic energy.
 (C) Total kinetic energy of the system is equal to work done by the force F
 (D) None of the above

RD0136

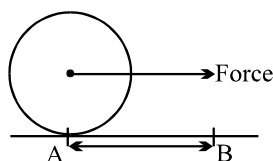
26. A uniform disc is rolling on a horizontal surface. At a certain instant B is the point of contact and A is at height $2R$ from ground, where R is radius of disc.



- (A) The magnitude of the angular momentum of the disc about B is thrice that about A.
 (B) The angular momentum of the disc about A is anticlockwise.
 (C) The angular momentum of the disc about B is clockwise
 (D) The angular momentum of the disc about A is equal to that about B.

RD0137

27. A disc of circumference s is at rest at a point A on a horizontal surface when a constant horizontal force begins to act on its centre. Between A and B there is sufficient friction to prevent slipping, and the surface is smooth to the right of B. $AB = s$. The disc moves from A to B in time T . To the right of B,



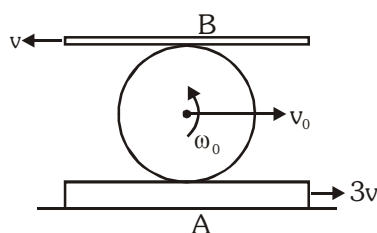
- (A) the angular acceleration of the disc will disappear, linear acceleration will remain unchanged
 (B) linear acceleration of the disc will increase
 (C) the disc will make one rotation in time $T/2$
 (D) the disc will cover a distance greater than s in further time T .

RD0138

28. A ring rolls without slipping on the ground. Its centre C moves with a constant speed u . P is any point on the ring. The speed of P with respect to the ground is v .
 (A) $0 \leq v \leq 2u$
 (B) $v = u$, if CP is horizontal
 (C) $v = u$, if CP makes an angle of 30° with the horizontal and P is below the horizontal level of C.
 (D) $v = \sqrt{2}u$, if CP is horizontal

RD0139

29. The disc of radius r is confined to roll without slipping at A and B. If the plates have the velocities shown, then



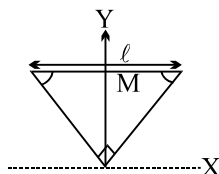
- (A) linear velocity $v_0 = v$
 (B) angular velocity of disc is $\frac{3v}{2r}$
 (C) angular velocity of disc is $\frac{2v}{r}$
 (D) None of these

RD0140

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 32 to 35

The figure shows an isosceles triangular plate of mass M and base L . The angle at the apex is 90° . The apex lies at the origin and the base is parallel to X -axis



32. The moment of inertia of the plate about the z -axis is :-

- (A) $\frac{ML^2}{12}$ (B) $\frac{ML^2}{24}$ (C) $\frac{ML^2}{6}$ (D) none of these

RD0143

33. The moment of inertia of the plate about the x -axis is :-

- (A) $\frac{ML^2}{8}$ (B) $\frac{ML^2}{32}$ (C) $\frac{ML^2}{24}$ (D) $\frac{ML^2}{6}$

RD0144

34. The moment of inertia of the plate about its base parallel to the x -axis is :-

- (A) $\frac{ML^2}{18}$ (B) $\frac{ML^2}{36}$ (C) $\frac{ML^2}{24}$ (D) none of these

RD0143

35. The moment of inertia of the plate about the y -axis is :-

- (A) $\frac{ML^2}{6}$ (B) $\frac{ML^2}{8}$ (C) $\frac{ML^2}{24}$ (D) none of these

RD0143

Paragraph for Question on 36 and 37

In the treatment of moments of inertia, introductory textbooks often present two theorems, generally called the parallel axis theorem and the perpendicular axis theorem. There is another theorem of this same genre, which is not usually included, but which is interesting and useful. It is

$$I_x + I_y + I_z = 2 \sum_i m_i r_i^2$$

Here, I_x , I_y and I_z are the moments of inertia about three mutually perpendicular intersecting axes, m_i is the mass of the i^{th} particle and r_i is the distance from the intersection. The proof is simple: Taking the three axes as coordinate axes, we have :

$$\begin{aligned} & I_x + I_y + I_z \\ &= \sum_i m_i (y_i^2 + z_i^2) + \sum_i m_i (z_i^2 + x_i^2) + \sum_i m_i (x_i^2 + y_i^2) \\ &= 2 \sum_i m_i (x_i^2 + y_i^2 + z_i^2) = 2 \sum_i m_i r_i^2 \end{aligned}$$

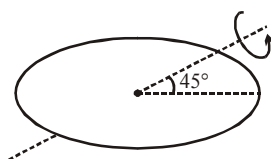
One important application is the calculation of the moment of inertia I_d of a uniform thin-walled spherical shell, of mass M and radius R , about a diameter. Taking the centre as the origin of coordinates, we have $I_x = I_y = I_z = I_d$, and $r_i = R$. The theorem gives $3I_d = 2\sum_i m_i R^2 = 2(\sum_i m_i)R^2 = 2MR^2$, where $I_d = 2MR^2/3$.

36. Consider a solid cube of mass m and side L . What will be the value of $\sum m_i r_i^2$ for this body when the point of intersection of axes is the centre of the cube :

(A) $\frac{ML^2}{2}$ (B) $\frac{ML^2}{4}$ (C) $\frac{ML^2}{3}$ (D) $\frac{ML^2}{6}$

RD0145

37. Find the moment of inertia of ring of mass m and radius R about an axis passing through its centre and making an angle 45° with its plane :

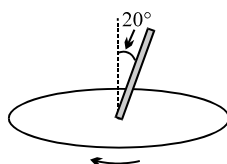


(A) $\frac{MR^2}{4}$ (B) $\frac{MR^2}{2}$ (C) $\frac{3}{4}MR^2$ (D) MR^2

RD0145

Paragraph for Question No. 38 and 39

A uniform rod is fixed to a rotating turntable so that its lower end is on the axis of the turntable and it makes an angle of 20° to the vertical. (The rod is thus rotating with uniform angular velocity about a vertical axis passing through one end.) If the turntable is rotating clockwise as seen from above.



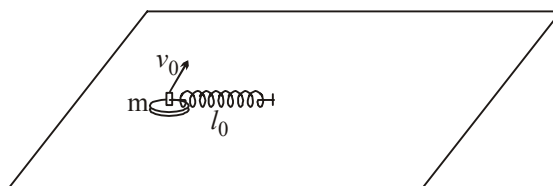
38. What is the direction of the rod's angular momentum vector (calculated about its lower end)?
 (A) vertically downwards
 (B) down at 20° to the horizontal
 (C) up at 20° to the horizontal
 (D) vertically upwards
39. Is there a torque acting on it, and if so in what direction?
 (A) yes, vertically (B) yes, horizontally
 (C) yes at 20° to the horizontal (D) no

RD0146

RD0146

Paragraph for Question No. 40 to 43

A spring having initial unstretched length ℓ_0 is lying on a smooth table. Its one end is fixed and the other one is fastened to a small particle of mass m . The particle is imparted an initial speed v_0 horizontally in a direction perpendicular to the spring. In the course of the motion in horizontal plane, the maximum elongation of the spring is $\Delta\ell = \ell_0/10$. (Given : $m = 0.1$ kg, $\ell_0 = 1$ m, $v_0 = 11$ m/s).



40. In the course of motion, which of the quantities relating to spring block system are conserved ?
 (A) kinetic energy (B) momentum (C) angular momentum (D) potential energy

RD0147

41. Which of the following is correct about initial situation and situation at maximum elongation ?
 (A) the orientation of spring in both positions should be perpendicular to each other.
 (B) the velocity at maximum extension should be zero
 (C) the velocity at maximum extension as well as at initial position should be perpendicular to spring.
 (D) Acceleration should be zero at the maximum extension as well as at initial position.

RD0147

42. **Student-A** : at maximum extension $mv'(\ell_0 + \Delta\ell) = mv_0\ell_0$.

Student-B : at maximum extension $k\Delta\ell = \frac{mv^2}{(\ell_0 + \Delta\ell)}$. **Student-C** : $\frac{1}{2}mv_0^2 = \frac{1}{2}mv'^2 + \frac{1}{2}k\Delta\ell^2$.

where v' is velocity at instant of maximum extension :

- (A) Only Student-A and B are correct (B) Only Student-A and C are correct
 (C) Only Student-B and C are correct (D) All are correct

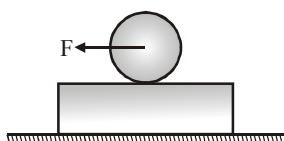
RD0147

43. The value of spring constant (in N/m) is :
 (A) 100 (B) 210 (C) 420 (D) 105

RD0147

Paragraph for Question No. 44 and 45

A disc of mass m and radius R is placed over a plank of same mass m . There is sufficient friction between disc and plank to prevent slipping. A force F is applied at the centre of the disc.



44. Acceleration of the plank is :-

- (A) $\frac{F}{2m}$ (B) $\frac{3F}{4m}$ (C) $\frac{F}{4m}$ (D) $\frac{3F}{2m}$

RD0148

45. Force of friction between the disc and the plank is :-

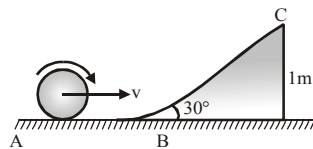
- (A) $\frac{F}{2}$ (B) $\frac{F}{4}$ (C) $\frac{F}{3}$ (D) $\frac{2F}{3}$

RD0148

Paragraph for Question No. 46 and 47

A small sphere of mass 1 kg is rolling without slipping on a rough stationary base with linear speed

$$v = \sqrt{\frac{200}{7}} \text{ m/s. It leaves the inclined plane at point C.}$$



46. Find its linear speed at point C :-

- (A) $\sqrt{\frac{100}{7}} \text{ m/s}$ (B) $\sqrt{\frac{50}{7}} \text{ m/s}$ (C) $\sqrt{\frac{100}{35}} \text{ m/s}$ (D) $\sqrt{\frac{200}{35}} \text{ m/s}$

RD0149

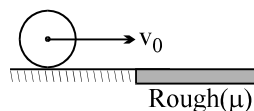
47. Find ratio of rotational and translational kinetic energy of the sphere when it strikes the ground after leaving from point C :-

- (A) $\frac{2}{5}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) $\frac{1}{2}$

RD0149

Paragraph for Question No. 48 to 51

A ring of mass M and radius R sliding with a velocity v_0 suddenly enters into rough surface where the coefficient of friction is μ , as shown in figure.



48. Choose the correct statement(s)

- (A) As the ring enters on the rough surface, the limiting friction force acts on it
 (B) The direction of friction is opposite to the direction of motion
 (C) The friction force accelerates the ring in the clockwise sense about its centre of mass
 (D) As the ring enters on the rough surface it starts rolling

RD0150

49. Choose the correct statement(s)

- (A) The momentum of the ring is conserved
- (B) The angular momentum of the ring is conserved about its centre of mass
- (C) The angular momentum of the ring conserved about any point on the horizontal surface in line of friction.
- (D) The mechanical energy of the ring is conserved

RD0150

50. Choose the correct statement(s)

- (A) The ring starts its rolling motion when the centre of mass stationary
- (B) The ring starts rolling motion when the point of contact becomes stationary
- (C) The time after which the ring starts rolling is $\frac{v_0}{2\mu g}$
- (D) The rolling velocity is $\frac{v_0}{2}$

RD0150

51. Choose the correct alternative(s) :-

- (A) The linear distance moved by the centre of mass before the ring starts rolling is $\frac{3v_0^2}{8\mu g}$
- (B) The net work done by friction force is $-\frac{3}{8}mv_0^2$
- (C) The loss in kinetic energy of the ring is $\frac{mv_0^2}{4}$
- (D) The gain in rotational kinetic energy is $+\frac{mv_0^2}{8}$

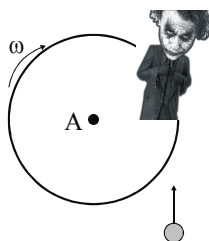
RD0150

MATRIX MATCH TYPE QUESTION

52. Column-I depicts various situations where some sudden events are taking place. Column-II describes changes in various parameters of systems immediately after the events taking place in column-I.

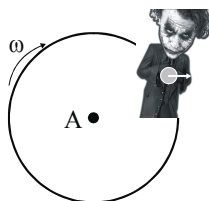
Column-I

- (A) Joker is standing on revolving platform and batman throws the ball and joker catches the ball while it was moving horizontally.



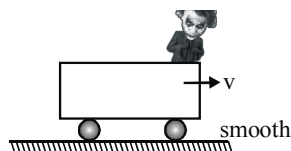
Joker, ball and platform is system.

- (B) Joker throws the ball horizontally and perpendicular to his motion while standing on the revolving platform.



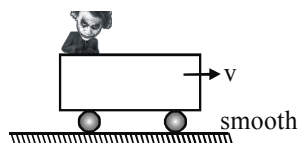
Joker, ball and platform is system.

- (C) Joker jumps horizontally towards right from the cart which is moving at speed v on smooth horizontal floor.



Joker and cart is the system

- (D) Joker drops himself vertically from the moving cart with no horizontal velocity relative to cart.



Joker and cart is the system

Column-II

- (P) Linear momentum remains conserved.

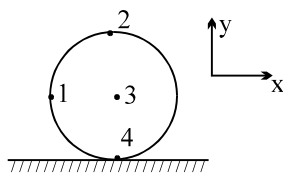
- (Q) Mechanical energy is conserved

- (R) Mechanical energy increases.

- (S) Mechanical energy decreases.

- (T) v or ω changes

53. A rigid cylinder is kept on a smooth horizontal surface as shown. If **Column-I** indicates velocities of various points (3-centre of cylinder, 2- top point, 4-bottom point, 1- on the level of 3 at the rim) on it shown, choose correct state of motion from **Column-II**.



Column-I

- (A) $\vec{v}_1 = \hat{i} + \hat{j}, \vec{v}_2 = 2\hat{i}$
 (B) $\vec{v}_1 = \hat{i} + \hat{j}, \vec{v}_3 = -\hat{i}$
 (C) $\vec{v}_2 = \hat{i}, \vec{v}_3 = 0$
 (D) $\vec{v}_4 = 0, \vec{v}_1 = -\hat{i} - \hat{j}$

Column-II

- (P) Pure rotation about centre
 (Q) Rolling without slipping to left
 (R) Rolling without slipping to right
 (S) Not possible

RD0152

EXERCISE (JM)

1. A thin uniform rod of length l and mass m is swinging freely about a horizontal axis passing through its end. Its maximum angular speed is ω . Its centre of mass rises to a maximum height of:-

[AIEEE - 2009]

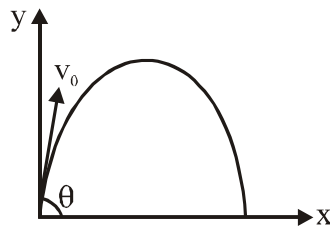
- (1) $\frac{1}{2} \frac{l^2 \omega^2}{g}$ (2) $\frac{1}{6} \frac{l^2 \omega^2}{g}$ (3) $\frac{1}{3} \frac{l^2 \omega^2}{g}$ (4) $\frac{1}{6} \frac{l \omega}{g}$

RD0153

2. A small particle of mass m is projected at an angle θ with the x -axis with an initial velocity v_0 in the x - y plane as shown in the figure. At a time $t < \frac{v_0 \sin \theta}{g}$, the angular momentum of the particle is:

Where $\hat{i}$, $\hat{j}$ and $\hat{k}$ are unit vectors along x , y and z -axis respectively.

[AIEEE-2010]



- (1) $\frac{1}{2} mg v_0 t^2 \cos \theta \hat{i}$ (2) $- mg v_0 t^2 \cos \theta \hat{j}$
 (3) $mg v_0 t \cos \theta \hat{k}$ (4) $-\frac{1}{2} mg v_0 t^2 \cos \theta \hat{k}$

RD0154

3. A pulley of radius 2 m is rotated about its axis by a force $F = (20t - 5t^2)$ newton (where t is measured in seconds) applied tangentially. If the moment of inertia of the pulley about its axis of rotation is 10 kg m^2 , the number of rotations made by the pulley before its direction of motion it reversed, is :-

- (1) more than 6 but less than 9 (2) more than 9
 (3) less than 3 (4) more than 3 but less than 6

[AIEEE-2011]

RD0155

4. A thin horizontal circular disc is rotating about a vertical axis passing through its centre. An insect is at rest at a point near the rim of the disc. The insect now moves along a diameter of the disc to reach its other end. During the journey of the insect, then angular speed of the disc :-

[AIEEE-2011]

- (1) continuously increases (2) first increases and then decreases
 (3) remains unchanged (4) continuously decreases

RD0156

5. A particle of mass 'm' is projected with a velocity v making an angle of 30° with the horizontal. The magnitude of angular momentum of the projectile about the point of projection when the particle is at its maximum height 'h' is :- [AIEEE-2011]

(1) $\frac{\sqrt{3}}{2} \frac{mv^2}{g}$ (2) zero (3) $\frac{mv^3}{\sqrt{2}g}$ (4) $\frac{\sqrt{3}}{16} \frac{mv^3}{g}$

RD0157

6. A hoop of radius r and mass m rotating with an angular velocity ω_0 is placed on a rough horizontal surface. The initial velocity of the centre of the hoop is zero. What will be the velocity of the centre of the hoop when it ceases to slip? [JEE Mains-2013]

(1) $\frac{r\omega_0}{4}$ (2) $\frac{r\omega_0}{3}$ (3) $\frac{r\omega_0}{2}$ (4) $r\omega_0$

RD0158

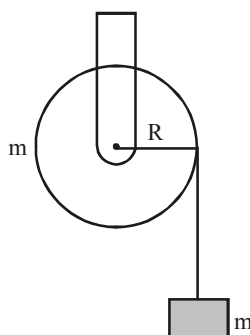
7. A bob of mass m attached to an inextensible string of length ℓ is suspended from a vertical support. The bob rotates in a horizontal circle with an angular speed ω rad/s about the vertical. About the point of suspension :

[JEE Mains-2014]

- (1) Angular momentum changes in direction but not in magnitude
(2) Angular momentum changes both in direction and magnitude
(3) Angular momentum is conserved
(4) Angular momentum changes in magnitude but not in direction.

RD0159

8. A mass 'm' is supported by a massless string wound around a uniform hollow cylinder of mass m and radius R . If the string does not slip on the cylinder, with what acceleration will the mass fall on release? [JEE Mains-2014]



(1) $\frac{5g}{6}$ (2) g (3) $\frac{2g}{3}$ (4) $\frac{g}{2}$

RD0160

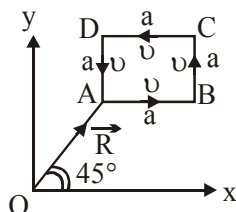
9. From a solid sphere of mass M and radius R a cube of maximum possible volume is cut. Moment of inertia of cube about an axis passing through its centre and perpendicular to one of its faces is:-

[JEE Mains-2015]

(1) $\frac{4MR^2}{9\sqrt{3}\pi}$ (2) $\frac{4MR^2}{3\sqrt{3}\pi}$ (3) $\frac{MR^2}{32\sqrt{2}\pi}$ (4) $\frac{MR^2}{16\sqrt{2}\pi}$

RD0161

10. A particle of mass m is moving along the side of a square of side ' a ', with a uniform speed v in the x - y plane as shown in the figure : [JEE Mains-2016]



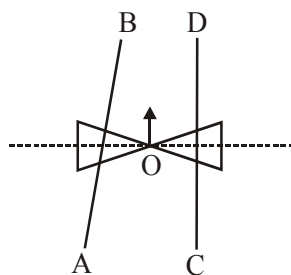
Which of the following statement is false for the angular momentum $\vec{L}$ about the origin ?

- (1) $\vec{L} = \frac{mv}{\sqrt{2}} R \hat{k}$ when the particle is moving from D to A
- (2) $\vec{L} = -\frac{mv}{\sqrt{2}} R \hat{k}$ when the particle is moving from A to B
- (3) $\vec{L} = mv \left[\frac{R}{\sqrt{2}} - a \right] \hat{k}$ when the particle is moving from C to D
- (4) $\vec{L} = mv \left[\frac{R}{\sqrt{2}} + a \right] \hat{k}$ when the particle is moving from B to C

RD0162

11. A roller is made by joining together two cones at their vertices O. It is kept on two rails AB and CD which are placed asymmetrically (see figure), with its axis perpendicular to CD and its centre O at the centre of line joining AB and CD (see figure). It is given a light push so that it starts rolling with its centre O moving parallel to CD in the direction shown. As it moves, the roller will tend to :-

[JEE Mains-2016]



- (1) turn left and right alternately.
- (2) turn left.
- (3) turn right.
- (4) go straight.

RD0163

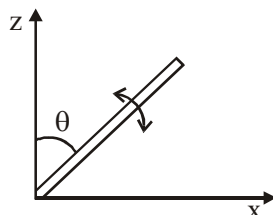
12. The moment of inertia of a uniform cylinder of length ℓ and radius R about its perpendicular bisector is I . What is the ratio ℓ/R such that the moment of inertia is minimum ? [JEE Main-2017]

- (1) 1
- (2) $\frac{3}{\sqrt{2}}$
- (3) $\sqrt{\frac{3}{2}}$
- (4) $\frac{\sqrt{3}}{2}$

RD0164

13. A slender uniform rod of mass M and length ℓ is pivoted at one end so that it can rotate in a vertical plane (see figure). There is negligible friction at the pivot. The free end is held vertically above the pivot and then released. The angular acceleration of the rod when it makes an angle θ with the vertical is :

[JEE Main-2017]

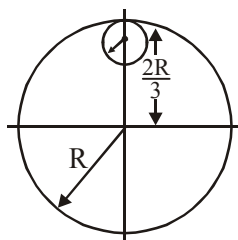


- (1) $\frac{3g}{2\ell} \cos \theta$ (2) $\frac{2g}{3\ell} \cos \theta$ (3) $\frac{3g}{2\ell} \sin \theta$ (4) $\frac{2g}{3\ell} \sin \theta$

RD0165

14. From a uniform circular disc of radius R and mass $9M$, a small disc of radius $\frac{R}{3}$ is removed as shown in the figure. The moment of inertia of the remaining disc about an axis perpendicular to the plane of the disc and passing through centre of disc is :

[JEE Main-2018]

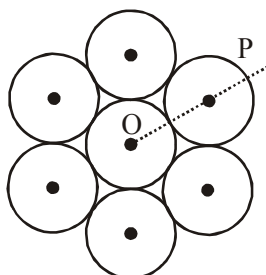


- (1) $\frac{40}{9} MR^2$ (2) $10 MR^2$ (3) $\frac{37}{9} MR^2$ (4) $4 MR^2$

RD0166

15. Seven identical circular planar disks, each of mass M and radius R are welded symmetrically as shown. The moment of inertia of the arrangement about the axis normal to the plane and passing through the point P is :

[JEE Main-2018]



- (1) $\frac{55}{2} MR^2$ (2) $\frac{73}{2} MR^2$ (3) $\frac{181}{2} MR^2$ (4) $\frac{19}{2} MR^2$

RD0167

EXERCISE (JA)

1. A block of base $10 \text{ cm} \times 10 \text{ cm}$ and height 15 cm is kept on an inclined plane. The coefficient of friction between them is $\sqrt{3}$. The inclination θ of this inclined plane from the horizontal plane is gradually increased from 0° . Then :- [IIT-JEE 2009]

- (A) at $\theta = 30^\circ$, the block will start sliding down the plane
 (B) the block will remain at rest on the plane up to certain θ and then it will topple
 (C) at $\theta = 60^\circ$, the block will start sliding down the plane and continue to do so at higher angles
 (D) at $\theta = 60^\circ$, the block will start sliding down the plane and on further increasing θ , it will topple at certain θ

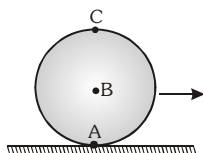
RD0168

2. If the resultant of the external forces acting on a system of particles is zero, then from an inertial frame, one can surely say that [IIT-JEE 2009]

- (A) linear momentum of the system does not change in time
 (B) kinetic energy of the system does not change in time
 (C) angular momentum of the system does not change in time
 (D) potential energy of the system does not change in time

RD0169

3. A sphere is rolling without slipping on a fixed horizontal plane surface. In the figure, A is the point of contact, B is the centre of the sphere and C is its topmost point. Then [IIT-JEE 2009]

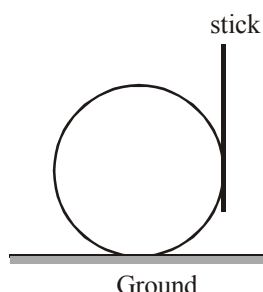


- (A) $\vec{v}_C - \vec{v}_A = 2(\vec{v}_B - \vec{v}_C)$
 (B) $\vec{v}_C - \vec{v}_B = \vec{v}_B - \vec{v}_A$
 (C) $|\vec{v}_C - \vec{v}_A| = 2|\vec{v}_B - \vec{v}_C|$
 (D) $|\vec{v}_C - \vec{v}_A| = 4|\vec{v}_B|$

RD0170

4. A boy is pushing a ring of mass 2 kg and radius 0.5 m with a stick as shown in the figure. The stick applies a force of 2N on the ring and rolls it without slipping with an acceleration of 0.3 m/s^2 . The coefficient of friction between the ground and the ring is large enough that rolling always occurs and the coefficient of friction between the stick and the ring is $(P/10)$. The value of P is

[IIT-JEE 2011]



RD0171

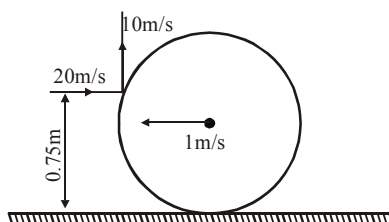
5. Four solid spheres each of diameter $\sqrt{5} \text{ cm}$ and mass 0.5 kg are placed with their centers at the corners of a square of side 4 cm. The moment of inertia of the system about the diagonal of the square is $N \times 10^{-4} \text{ kg-m}^2$, then N is

[IIT-JEE 2011]

RD0172

6. A thin ring of mass 2 kg and radius 0.5 m is rolling without slipping on a horizontal plane with velocity 1 m/s. A small ball of mass 0.1 kg, moving with velocity 20 m/s in the opposite direction, hits the ring at a height of 0.75 m and goes vertically up with velocity 10 m/s. Immediately after the collision

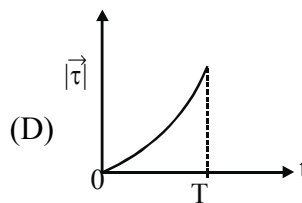
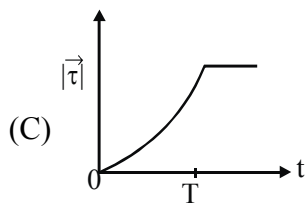
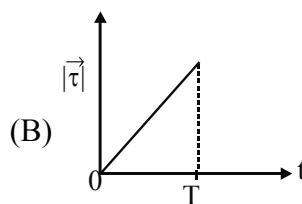
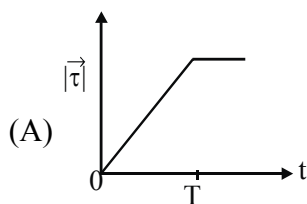
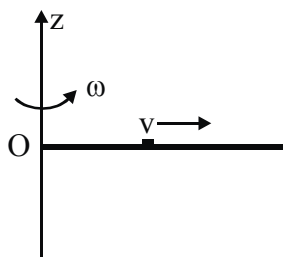
[IIT-JEE 2011]



- (A) the ring has pure rotation about its stationary CM
 (B) the ring comes to a complete stop
 (C) friction between the ring and the ground is to the left
 (D) there is no friction between the ring and the ground

RD0173

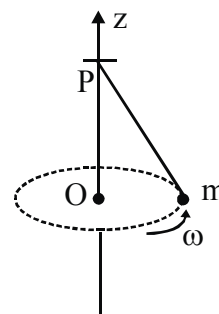
7. A thin uniform rod, pivoted at O , is rotating in the horizontal plane with constant angular speed ω , as shown in the figure. At time $t = 0$, a small insect starts from O and moves with constant speed v with respect to the rod towards the other end. It reaches the end of the rod at $t = T$ and stops. The angular speed of the system remains ω throughout. The magnitude of the torque ($|\tau|$) on the system about O , as a function of time is best represented by which plot? [IIT-JEE 2012]



8. A small mass m is attached to a massless string whose other end is fixed at P as shown in the figure. The mass is undergoing circular motion in the x - y plane with centre at O and constant angular speed ω . If the angular momentum of the system, calculated about O and P are denoted by $\vec{L}_O$ and $\vec{L}_P$ respectively, then

[IIT-JEE 2012]

- (A) $\vec{L}_O$ and $\vec{L}_P$ do not vary with time
 (B) $\vec{L}_O$ varies with time while $\vec{L}_P$ remains constant
 (C) $\vec{L}_O$ remains constant while $\vec{L}_P$ varies with time
 (D) $|\vec{L}_O|$ and $|\vec{L}_P|$ both do not vary with time



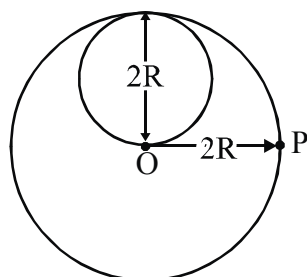
RD0174

RD0175

9. A lamina is made by removing a small disc of diameter $2R$ from a bigger disc of uniform mass density and radius $2R$, as shown in the figure. The moment of inertia of this lamina about axes passing through O and P is I_0 and I_P respectively. Both these axes are perpendicular to the plane of the lamina.

The ratio $\frac{I_P}{I_0}$ to the nearest integer is

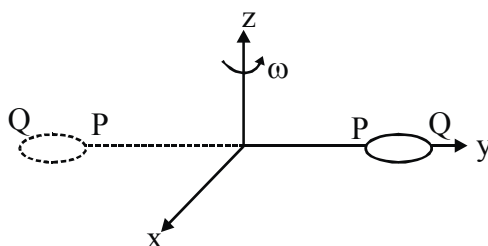
[IIT-JEE 2012]



RD0176

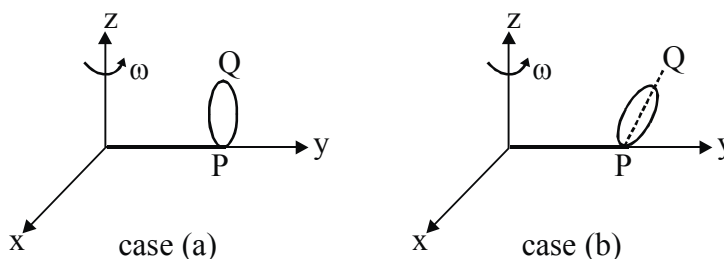
Paragraph for Questions 10 and 11

The general motion of a rigid body can be considered to be a combination of (i) a motion of its centre of mass about an axis, and (ii) its motion about an instantaneous axis passing through the centre of mass. These axes need not be stationary. Consider, for example, a thin uniform disc welded (rigidly fixed) horizontally at its rim to a massless stick, as shown in the figure. When the disc-stick system is rotated about the origin on a horizontal frictionless plane with angular speed ω , the motion at any instant can be taken as a combination of (i) a rotation of the centre of mass of the disc about the z -axis, and (ii) a rotation of the disc through an instantaneous vertical axis passing through its centre of mass (as is seen from the changed orientation of points P and Q). Both these motions have the same angular speed ω in this case.



Now consider two similar systems as shown in the figure : case (A) the disc with its face vertical and parallel to x - z plane; Case (B) the disc with its face making an angle of 45° with x - y plane and its horizontal diameter parallel to x -axis. In both the cases, the disc is welded at point P , and the systems are rotated with constant angular speed ω about the z -axis.

[IIT-JEE 2012]



10. Which of the following statements regarding the angular speed about the instantaneous axis (passing through the centre of mass) is correct?

(A) It is $\sqrt{2\omega}$ for both the cases. (B) It is ω for case (a); and $\frac{\omega}{\sqrt{2}}$ for case (b).
 (C) It is ω for case (a); and $\sqrt{2\omega}$ for case (b). (D) It is ω for both the cases.

RD0177

11. Which of the following statements about the instantaneous axis (passing through the centre of mass) is correct?

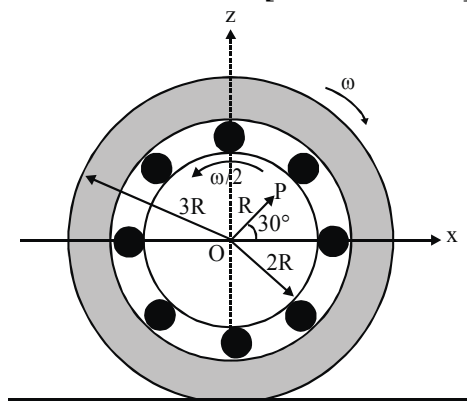
(A) It is vertical for both the cases (a) and (b).
 (B) It is vertical for case (a); and is at 45° to the x-z plane and lies in the plane of the disc for case (b).
 (C) It is horizontal for case (a); and is at 45° to the x-z plane and is normal to the plane of the disc for case (b).
 (D) It is vertical for case (a); and is at 45° to the x-z plane and is normal to the plane of the disc for case (b).

RD0177

12. The figure shows a system consisting of (i) a ring of outer radius $3R$ rolling clockwise without slipping on a horizontal surface with angular speed ω and (ii) an inner disc of radius $2R$ rotating anti-clockwise with angular speed $\omega/2$. The ring and disc are separated by frictionless ball bearing. The system is in the x-z plane. The point P on the inner disc is at a distance R from the origin, where OP makes an angle of 30° with the horizontal. Then with respect to the horizontal surface,

[IIT-JEE 2012]

- (A) the point O has a linear velocity $3R\omega\hat{i}$
 (B) the point P has a linear velocity $\frac{11}{4}R\omega\hat{i} + \frac{\sqrt{3}}{4}R\omega\hat{k}$
 (C) the point P has a linear velocity $\frac{13}{4}R\omega\hat{i} - \frac{\sqrt{3}}{4}R\omega\hat{k}$
 (D) the point P has a linear velocity $\left(3 - \frac{\sqrt{3}}{4}\right)R\omega\hat{i} + \frac{1}{4}R\omega\hat{k}$



RD0178

13. Two solid cylinders P and Q of same mass and same radius start rolling down a fixed inclined plane from the same height at the same time. Cylinder P has most of its mass concentrated near its surface, while Q has most of its mass concentrated near the axis. Which statement(s) is(are) correct?

(A) Both cylinders P and Q reach the ground at the same time.
 (B) Cylinder P has larger acceleration than cylinder Q.
 (C) Both cylinders reach the ground with same translational kinetic energy.
 (D) Cylinder Q reaches the ground with larger angular speed.

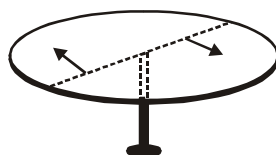
[IIT-JEE 2012]

RD0179

14. A uniform circular disc of mass 50 kg and radius 0.4 m is rotating with an angular velocity of 10 rad s^{-1} about its own axis, which is vertical. Two uniform circular rings, each of mass 6.25 kg and radius 0.2 m, are gently placed symmetrically on the disc in such a manner that they are touching each other along the axis of the disc and are horizontal. Assume that the friction is large enough such that the rings are at rest relative to the disc and the system rotates about the original axis. The new angular velocity (in rad s^{-1}) of the system is [IIT-JEE 2013]

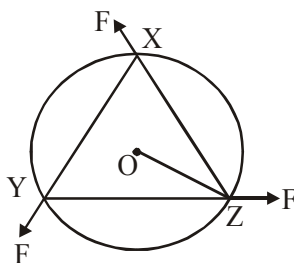
RD0180

15. A horizontal circular platform of radius 0.5 m and mass 0.45 kg is free to rotate about its axis. Two massless spring toy-guns, each carrying a steel ball of mass 0.05 kg are attached to the platform at a distance 0.25 m from the centre on its either sides along its diameter (see figure). Each gun simultaneously fires the balls horizontally and perpendicular to the diameter in opposite directions. After leaving the platform, the balls have horizontal speed of 9 ms^{-1} with respect to the ground. The rotational speed of the platform in rad s^{-1} after the balls leave the platform is [JEE Advanced-2014]



RD0181

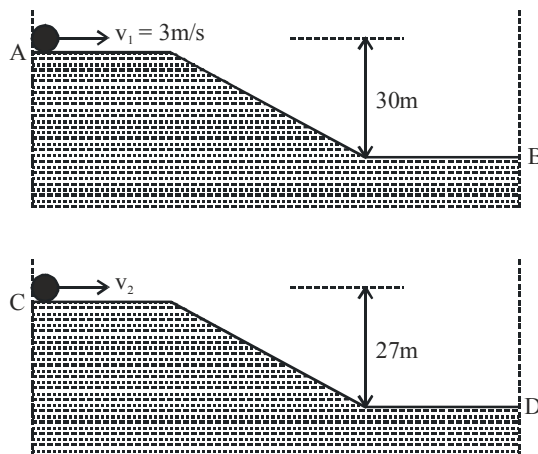
16. A uniform circular disc of mass 1.5 kg and radius 0.5 m is initially at rest on a horizontal frictionless surface. Three forces of equal magnitude $F = 0.5 \text{ N}$ are applied simultaneously along the three sides of an equilateral triangle XYZ with its vertices on the perimeter of the disc (see figure). One second after applying the forces, the angular speed of the disc in rad s^{-1} is [JEE Advanced-2014]



RD0182

17. Two identical uniform discs roll without slipping on two different surfaces AB and CD (see figure) starting at A and C with linear speeds v_1 and v_2 respectively, and always remain in contact with the surfaces. If they reach B and D with the same linear speed and $v_1 = 3 \text{ m/s}$, then v_2 in m/s is ($g = 10 \text{ m/s}^2$)

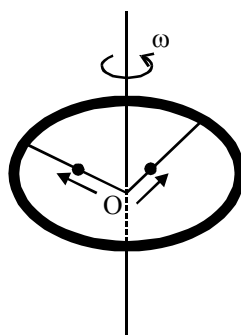
[JEE Advanced-2015]



RD0183

18. A ring of mass M and radius R is rotating with angular speed ω about a fixed vertical axis passing through its centre O with two point masses each of mass $\frac{M}{8}$ at rest at O . These masses can move radially outwards along two massless rods fixed on the ring as shown in the figure. At some instant the angular speed of the system is $\frac{8}{9}\omega$ and one of the masses is at a distance of $\frac{3}{5}R$ from O . At this instant the distance of the other mass from O is :

[JEE Advanced-2015]



(A) $\frac{2}{3}R$

(B) $\frac{1}{3}R$

(C) $\frac{3}{5}R$

(D) $\frac{4}{5}R$

RD0184

- 19.** The densities of two solid spheres A and B of the same radii R vary with radial distance r as

$\rho_A(r) = k \left(\frac{r}{R} \right)$ and $\rho_B(r) = k \left(\frac{r}{R} \right)^5$, respectively, where k is a constant. The moments of inertia of the

individual spheres about axes passing through their centres are I_A and I_B , respectively. If $\frac{I_B}{I_A} = \frac{n}{10}$, the value of n is.

[JEE Advanced-2015]

[JEE Advanced-2015]

RD0185

- 20.** A uniform wooden stick of mass 1.6 kg and length ℓ rests in an inclined manner on a smooth, vertical wall of height h ($< \ell$) such that a small portion of the stick extends beyond the wall. The reaction force of the wall on the stick is perpendicular to the stick. The stick makes an angle of 30° with the wall and the bottom of the stick is on a rough floor. The reaction of the wall on the stick is equal in magnitude to the reaction of the floor on the stick. The ratio h/ℓ and the frictional force f at the bottom of the stick are: ($g = 10 \text{ ms}^{-2}$)
- [JEE Advanced-2016]**

$$(A) \frac{h}{\ell} = \frac{\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3} N$$

(B) $\frac{h}{\ell} = \frac{3}{16}, f = \frac{16\sqrt{3}}{3} N$

$$(C) \frac{h}{\ell} = \frac{3\sqrt{3}}{16}, f = \frac{8\sqrt{3}}{3} N$$

$$(D) \frac{h}{\ell} = \frac{3\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3} N$$

RD0186

- 21.** The position vector $\vec{r}$ of a particle of mass m is given by the following equation $\vec{r}(t) = \alpha t^3 \hat{i} + \beta t^2 \hat{j}$,

where $\alpha = \frac{10}{3} \text{ ms}^{-3}$, $\beta = 5 \text{ ms}^{-2}$ and $m = 0.1 \text{ kg}$. At $t = 1 \text{ s}$, which of the following statement(s) is(are) true about the particle?

[JEE Advanced-2016]

[JEE Advanced-2016]

(A) The velocity $\vec{v}$ is given by $\vec{v} = (10\hat{i} + 10\hat{j})\text{ms}^{-1}$

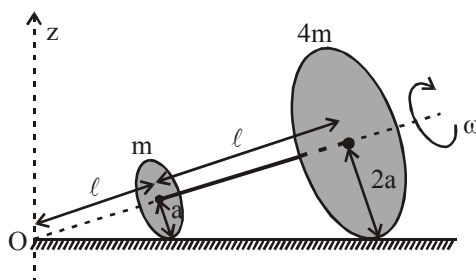
(B) The angular momentum $\vec{L}$ with respect to the origin is given by $\vec{L} = -\left(\frac{5}{3}\right) \hat{k} \text{ Nms}$

(C) The force $\vec{F}$ is given by $\vec{F} = (\hat{i} + 2\hat{j})N$

(D) The torque $\vec{\tau}$ with respect to the origin is given by $\vec{\tau} = -\left(\frac{20}{3}\right)\hat{k}$ Nm

RD0187

22. Two thin circular discs of mass m and $4m$, having radii of a and $2a$, respectively, are rigidly fixed by a massless, right rod of length $\ell = \sqrt{24}a$ through their center. This assembly is laid on a firm and flat surface, and set rolling without slipping on the surface so that the angular speed about the axis of the rod is ω . The angular momentum of the entire assembly about the point 'O' is $\vec{L}$ (see the figure). Which of the following statement(s) is(are) true ? [JEE Advanced-2016]



- (A) The magnitude of angular momentum of the assembly about its center of mass is $17ma^2\omega/2$
 (B) The magnitude of the z -component of $\vec{L}$ is $55ma^2\omega$
 (C) The magnitude of angular momentum of center of mass of the assembly about the point O is $81ma^2\omega$
 (D) The center of mass of the assembly rotates about the z -axis with an angular speed of $\omega/5$

RD0188

Paragraph for Question No. 23 and 24

A frame of reference that is accelerated with respect to an inertial frame of reference is called a non-inertial frame of reference. A coordinate system fixed on a circular disc rotating about a fixed axis with a constant angular velocity ω is an example of a non-inertial frame of reference. The relationship between the force $\vec{F}_{\text{rot}}$ experienced by a particle of mass m moving on the rotating disc and the force $\vec{F}_{\text{in}}$ experienced by the particle in an inertial frame of reference is

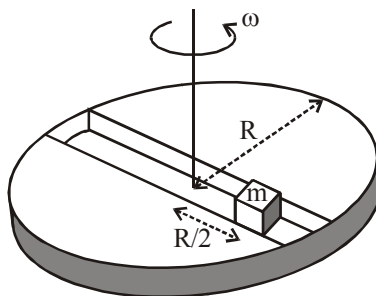
$$\vec{F}_{\text{rot}} = \vec{F}_{\text{in}} + 2m(\vec{v}_{\text{rot}} \times \vec{\omega}) + m(\vec{\omega} \times \vec{r}) \times \vec{\omega},$$

where $\vec{v}_{\text{rot}}$ is the velocity of the particle in the rotating frame of reference and $\vec{r}$ is the position vector of the particle with respect to the centre of the disc.

Now consider a smooth slot along a diameter of a disc of radius R rotating counter-clockwise with a constant angular speed ω about its vertical axis through its center. We assign a coordinate system with the origin at the centre of the disc, the x -axis along the slot, the y -axis perpendicular to the slot and the z -axis along the rotation axis ($\vec{\omega} = \omega\hat{k}$). A small block of mass m is gently placed in the slot at

$\vec{r} = (R/2)\hat{i}$ at $t = 0$ and is constrained to move only along the slot.

[JEE Advanced-2016]



23. The distance r of the block at time t is :

(A) $\frac{R}{4}(e^{2\omega t} + e^{-2\omega t})$

(B) $\frac{R}{2}\cos 2\omega t$

(C) $\frac{R}{2}\cos \omega t$

(D) $\frac{R}{4}(e^{\omega t} + e^{-\omega t})$

RD0189

24. The net reaction of the disc on the block is :

(A) $-\omega^2 R \cos \omega t \hat{j} - mg \hat{k}$

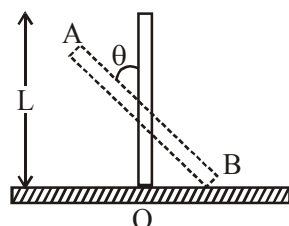
(B) $m\omega^2 R \sin \omega t \hat{j} - mg \hat{k}$

(C) $\frac{1}{2}m\omega^2 R (e^{\omega t} - e^{-\omega t}) \hat{j} + mg \hat{k}$

(D) $\frac{1}{2}m\omega^2 R (e^{2\omega t} - e^{-2\omega t}) \hat{j} + mg \hat{k}$

RD0189

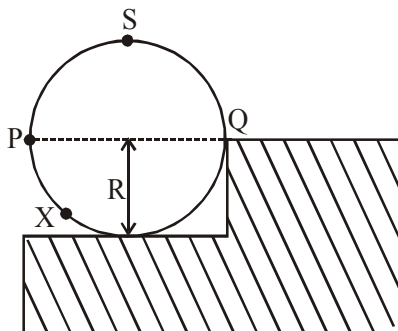
25. A rigid uniform bar AB of length L is slipping from its vertical position on a frictionless floor (as shown in the figure). At some instant of time, the angle made by the bar with the vertical is θ . Which of the following statements about its motion is/are correct ? [JEE Advanced-2017]



- (A) When the bar makes an angle θ with the vertical, the displacement of its midpoint from the initial position is proportional to $(1 - \cos\theta)$
- (B) The midpoint of the bar will fall vertically downward
- (C) Instantaneous torque about the point in contact with the floor is proportional to $\sin\theta$
- (D) The trajectory of the point A is a parabola

RD0190

26. A wheel of radius R and mass M is placed at the bottom of a fixed step of height R as shown in the figure. A constant force is continuously applied on the surface of the wheel so that it just climbs the step without slipping. Consider the torque τ about an axis normal to the plane of the paper passing through the point Q . Which of the following options is/are correct ? [JEE Advanced-2017]



- (A) If the force is applied normal to the circumference at point X then τ is constant
 (B) If the force is applied tangentially at point S then $\tau \neq 0$ but the wheel never climbs the step
 (C) If the force is applied normal to the circumference at point P then τ is zero
 (D) If the force is applied at point P tangentially then τ decreases continuously as the wheel climbs

RD0191

Paragraph for Question no. 27 & 28

One twirls a circular ring (of mass M and radius R) near the tip of one's finger as shown in Figure 1. In the process the finger never loses contact with the inner rim of the ring. The finger traces out the surface of a cone, shown by the dotted line. The radius of the path traced out by the point where the ring and the finger is in contact is r . The finger rotates with an angular velocity ω_0 . The rotating ring rolls without slipping on the outside of a smaller circle described by the point where the ring and the finger is in contact (Figure 2). The coefficient of friction between the ring and the finger is μ and the acceleration due to gravity is g . [JEE Advanced-2017]

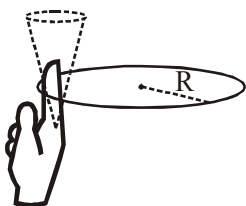


Figure 1

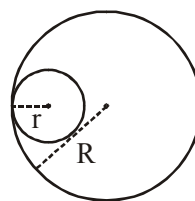


Figure 2

27. The total kinetic energy of the ring is :-

- (A) $M\omega_0^2 R^2$ (B) $M\omega_0^2 (R-r)^2$ (C) $\frac{1}{2} M\omega_0^2 (R-r)^2$ (D) $\frac{3}{2} M\omega_0^2 (R-r)^2$

RD0192

28. The minimum value of ω_0 below which the ring will drop down is :-

- (A) $\sqrt{\frac{3g}{2\mu(R-r)}}$ (B) $\sqrt{\frac{g}{\mu(R-r)}}$ (C) $\sqrt{\frac{2g}{\mu(R-r)}}$ (D) $\sqrt{\frac{2g}{2\mu(R-r)}}$

RD0192

29. The potential energy of a particle of mass m at a distance r from a fixed point O is given by $V(r) = kr^2/2$, where k is a positive constant of appropriate dimensions. This particle is moving in a circular orbit of radius R about the point O . If v is the speed of the particle and L is the magnitude of its angular momentum about O , which of the following statements is (are) true ?

[JEE Advanced-2018]

- (A) $v = \sqrt{\frac{k}{2m}}R$ (B) $v = \sqrt{\frac{k}{m}}R$ (C) $L = \sqrt{mk}R^2$ (D) $L = \sqrt{\frac{mk}{2}}R^2$

RD0193

30. Consider a body of mass 1.0 kg at rest at the origin at time $t = 0$. A force $\vec{F} = (\alpha\hat{i} + \beta\hat{j})$ is applied on the body, where $\alpha = 1.0 \text{ N s}^{-1}$ and $\beta = 1.0 \text{ N}$. The torque acting on the body about the origin at time $t = 1.0 \text{ s}$ is $\vec{\tau}$. Which of the following statements is (are) true?

[JEE Advanced-2018]

- (A) $|\vec{\tau}| = \frac{1}{3} \text{ Nm}$
 (B) The torque $\vec{\tau}$ is in the direction of the unit vector $+\hat{k}$
 (C) The velocity of the body at $t = 1 \text{ s}$ is $\vec{v} = \frac{1}{2}(\hat{i} + 2\hat{j})\text{ms}^{-1}$
 (D) The magnitude of displacement of the body at $t = 1 \text{ s}$ is $\frac{1}{6} \text{ m}$

RD0194

31. A ring and a disc are initially at rest, side by side, at the top of an inclined plane which makes an angle 60° with the horizontal. They start to roll without slipping at the same instant of time along the shortest path. If the time difference between their reaching the ground is $(2 - \sqrt{3})/\sqrt{10} \text{ s}$, then the height of the top of the inclined plane, in meters, is _____. Take $g = 10 \text{ ms}^{-2}$.

[JEE Advanced-2018]

RD0195

32. A thin and uniform rod of mass M and length L is held vertical on a floor with large friction. The rod is released from rest so that it falls by rotating about its contact-point with the floor without slipping. Which of the following statement(s) is/are correct, when the rod makes an angle 60° with vertical ? [g is the acceleration due to gravity]

[JEE Advanced-2019]

- (1) The radial acceleration of the rod's center of mass will be $\frac{3g}{4}$
 (2) The angular acceleration of the rod will be $\frac{2g}{L}$
 (3) The angular speed of the rod will be $\sqrt{\frac{3g}{2L}}$
 (4) The normal reaction force from the floor on the rod will be $\frac{Mg}{16}$

RD0196

ANSWER KEY

EXERCISE (S-1)

1. Ans. 2 2. Ans. $\frac{ml^2}{12}$ 3. Ans. 2.00 N·m, (b) $\hat{k}$
 4. Ans. 8 kg 5. Ans. $150(+\hat{i})$ 6. Ans. 30°
 7. Ans. 10^{-3} rad/s² clockwise 8. Ans. 2 9. Ans. $\frac{L}{3}$
 10. Ans. (a) $\frac{9g}{7} \downarrow$ (b) $\frac{4mg}{7} \uparrow$ 11. Ans. (i) $10/13 \text{ m/s}^2$, (ii) $5000/26\pi$, (iii) $480/13 \text{ N}$
 12. Ans. 2N 13. Ans. $M = 2m \left(\frac{2gh}{R^2 \omega^2} - 1 \right)$ 14. Ans. $\sqrt{5gR}$
 15. Ans. (a) $\omega = \sqrt{\frac{6g}{\ell}}$, (b) $T = \frac{11mg}{4}$ 16. Ans. $\frac{\omega}{3}$ 17. Ans. $\frac{4\pi}{5}$
 18. Ans. 75 J 19. Ans. $1/2 \text{ ma}$ 20. Ans. 15 21. Ans. 16 m/s^2
 22. Ans. 6 mv_0^2 23. Ans. $\frac{13}{16} \text{ Mv}^2$
 24. Ans. $\frac{l^2}{12x}$ 25. Ans. (i) $\frac{J}{m}$ (ii) zero (iii) $\frac{J}{2m}$ (iv) $\frac{5}{2} \frac{J}{m}$
 26. Ans. (a) $t = \frac{\pi R m}{2I}$; (b) $s = \frac{\pi R}{2}$ 27. Ans. $\frac{mv}{4}$ 28. Ans. $\frac{2mv_0^2}{l}$

EXERCISE (S-2)

1. Ans. $57/140 \text{ MR}^2$ 2. Ans. $\frac{a\ell^4}{36}$ 3. Ans. $(3/8) F_g$ 4. Ans. $a = \left[\frac{b - \mu a}{3b + \mu a} \right] g$
 5. Ans. 1.63 N, 1.224 m 6. Ans. (a) $5 \text{ m/s}^2 \rightarrow$, (b) $0.3 < h < 1.5 \text{ m}$
 7. Ans. (a) $\sqrt{3} \text{ ml}\omega^2$ (b) $(F_{\text{net}})_x = -\frac{F}{4}$, $(F_{\text{net}})_y = \sqrt{3} \text{ ml}\omega^2$ 8. Ans. $\sqrt{15} \text{ ft/sec}$
 9. Ans. (a) $4\sqrt{\frac{3}{7}} \text{ m/s}$, (b) $\frac{200}{7} \text{ N}$ 10. Ans. $t = 2/\sqrt{5} \text{ sec}$, $\theta = 4\pi/5 \text{ rad}$
 11. Ans. (i) (a) $\frac{1.2g}{c}$ (cw) (b) $-0.3(\hat{i} + 2\hat{j})g$ (ii) (a) $24g/17c$ (cw) (b) $12g/17 \downarrow$ (iii) (a) $2.4g/c$ (cw) (b) $0.5g \downarrow$
 12. Ans. 3 13. Ans. $\alpha_{\text{max}} = \frac{2g}{3r}$
 14. Ans. (a) $a_c = \frac{4F}{3m_1 + 8m_2}$, $a_p = 2a_c$
 (b) friction at the top of the cylinder = $3m_1 F / (3m_1 + 8m_2)$ towards right; friction at the bottom = $m_1 F / (3m_1 + 8m_2)$ towards right.
 15. Ans. (a) pipe rolls without sliding
 (b) pipe : $5/6 \text{ m/s} \rightarrow$, $125/12 \text{ rad/s}$ (anticlockwise); plate : $5/3 \text{ m/s} \rightarrow$
 16. Ans. 6N , $-0.6\hat{j} \pm 0.6\hat{k}$ 17. Ans. (a) $\theta_c = \cos^{-1}(4/7)$, (b) $v = \sqrt{4/7} gR$, (c) $K_T / K_R = 6$

18. Ans. $v = \sqrt{\frac{14gR}{3}}$

19. Ans. (a) 6.68 m/s (b) 2.27 m

20. Ans. (i) $2v_0/3$, (ii) $t = v_0/3\mu g$, $W = \frac{1}{2} [3\mu^2 m g^2 t^2 - 2\mu m g t v_0]$ ($t < t_0$), $W = -\frac{1}{6} m v_0^2$ ($t > t_0$)

21. Ans. (a) $\omega/3$, (b) $\frac{\sqrt{37}}{3} m \omega R$, (c) $\frac{\sqrt{37}}{3} m \omega R$ 22. Ans. 60°

23. Ans. (a) $\ell = 0.1\text{m}$; (b) $\omega' = 1\text{rad/s}$; (c) laminar sheet will never come to rest

24. Ans. 5 25. Ans. (a) $\frac{m}{M} = \frac{1}{4}$; (b) $x = \frac{2L}{3}$; (c) $\frac{v_0}{2\sqrt{2}}$

EXERCISE (O-1)

- | | | | | | |
|---|--------------------|----------------|----------------|------------------|--------------|
| 1. Ans. (C) | 2. Ans. (B) | 3. Ans. (B) | 4. Ans. (D) | 5. Ans. (A) | 6. Ans. (A) |
| 7. Ans. (B) | 8. Ans. (D) | 9. Ans. (C) | 10. Ans. (A) | 11. Ans. (C) | 12. Ans. (B) |
| 13. Ans. (D) | 14. Ans. (D) | 15. Ans. (B) | 16. Ans. (C) | 17. Ans. (B) | 18. Ans. (D) |
| 19. Ans. (C) | 20. Ans. (C) | 21. Ans. (A) | 22. Ans. (B) | 23. Ans. (C) | 24. Ans. (D) |
| 25. Ans. (A) | 26. Ans. (D) | 27. Ans. (B) | 28. Ans. (C) | 29. Ans. (A) | 30. Ans. (D) |
| 31. Ans. (B) | 32. Ans. (B) | 33. Ans. (A) | 34. Ans. (A) | 35. Ans. (B) | 36. Ans. (C) |
| 37. Ans. (B) | 38. Ans. (B) | 39. Ans. (A) | 40. Ans. (D) | 41. Ans. (B) | 42. Ans. (D) |
| 43. Ans. (C) | 44. Ans. (A) | 45. Ans. (C) | 46. Ans. (D) | 47. Ans. (B) | 48. Ans. (B) |
| 49. Ans. (A) | 50. Ans. (A,B,C,D) | 51. Ans. (B,C) | 52. Ans. (A,D) | 53. Ans. (A,C,D) | |
| 54. Ans. (A,C,D) | 55. Ans. (A,B,C,D) | 56. Ans. (A) | 57. Ans. (A) | | |
| 58. Ans. (D) | 59. Ans. (B) | 60. Ans. (A) | 61. Ans. (D) | 62. Ans. (D) | |
| 63. Ans. (A)→(R); (B)→(P); (C)→(T); (D)→(Q,S) | | | | | |

EXERCISE (O-2)

- | | | | | | |
|--------------------------------------|-------------------------------------|------------------|--------------------|------------------|--------------|
| 1. Ans. (D) | 2. Ans. (C) | 3. Ans. (B) | 4. Ans. (C) | 5. Ans. (A) | 6. Ans. (D) |
| 7. Ans. (B) | 8. Ans. (B) | 9. Ans. (B) | 10. Ans. (B) | 11. Ans. (D) | 12. Ans. (C) |
| 13. Ans. (A) | 14. Ans. (C) | 15. Ans. (C) | 16. Ans. (D) | 17. Ans. (C) | 18. Ans. (A) |
| 19. Ans. (C) | 20. Ans. (C,D) | 21. Ans. (B,C,D) | 22. Ans. (A,B,C) | 23. Ans. (A,B,D) | |
| 24. Ans. (A,C) | 25. Ans. (A,B,C) | 26. Ans. (A,B,C) | 27. Ans. (B,C,D) | | |
| 28. Ans. (A,C,D) | 29. Ans. (A,C) | 30. Ans. (B,D) | 31. Ans. (A, B, C) | | |
| 32. Ans. (C) | 33. Ans. (A) | 34. Ans. (C) | 35. Ans. (C) | 36. Ans. (B) | 37. Ans. (C) |
| 38. Ans. (B) | 39. Ans. (B) | 40. Ans. (C) | 41. Ans. (C) | 42. Ans. (B) | 43. Ans. (B) |
| 44. Ans. (C) | 45. Ans. (B) | 46. Ans. (A) | 47. Ans. (C) | 48. Ans. (A,B,C) | |
| 49. Ans. (C) | 50. Ans. (B,C,D) | 51. Ans. (A,C,D) | | | |
| 52. Ans. (A) ST (B) R (C) PRT (D) PQ | 53. Ans. (A)-R, (B)-S, (C)-P, (D)-Q | | | | |

EXERCISE (JM)

- | | | | | | |
|--------------|--------------|--------------|-------------------|--------------|--------------|
| 1. Ans. (2) | 2. Ans. (4) | 3. Ans. (4) | 4. Ans. (2) | 5. Ans. (4) | 6. Ans. (3) |
| 7. Ans. (1) | 8. Ans. (4) | 9. Ans. (1) | 10. Ans. (1 or 3) | 11. Ans. (2) | 12. Ans. (3) |
| 13. Ans. (3) | 14. Ans. (4) | 15. Ans. (3) | | | |

EXERCISE (JA)

- | | | | | | |
|--------------|------------------------|---------------------|----------------------------|------------------|-------------------|
| 1. Ans. (B) | 2. Ans. (A) | 3. Ans. (B,C) | 4. Ans. 4 | 5. Ans. 9 | 6. Ans. (AC or C) |
| 7. Ans. (B) | 8. Ans. (C, D) | 9. Ans. 3 | 10. Ans. (D) | 11. Ans. (A) | 12. Ans. (A,B) |
| 13. Ans. (D) | 14. Ans. 8 | 15. Ans. 4 | 16. Ans. 2 | 17. Ans. 7 | 18. Ans. (C, D) |
| 19. Ans. 6 | 20. Ans. (D) | 21. Ans. (A, B, D) | 22. Ans. (D) | 23. Ans. (D) | |
| 24. Ans. (C) | 25. Ans. (A), (B), (C) | 26. Ans. (B,C or C) | 27. Ans. (Bonus) | | |
| 28. Ans. (B) | 29. Ans. (B,C) | 30. Ans. (A,C) | 31. Ans. 0.75 [0.74, 0.76] | 32. Ans. (1,3,4) | |

GRAVITATION

KEY CONCEPT

The discovery of the law of gravitation

The way the law of universal gravitation was discovered is often considered the paradigm of modern scientific technique. The major steps involved were

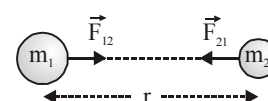
- The hypothesis about planetary motion given by **Nicolaus Copernicus (1473–1543)**.
- The careful experimental measurements of the positions of the planets and the Sun by **Tycho Brahe (1546–1601)**.
- Analysis of the data and the formulation of empirical laws by **Johannes Kepler (1571–1630)**.
- The development of a general theory by **Isaac Newton (1642–1727)**.

Newton's law of Gravitation

It states that every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and is inversely proportional to the square of the distance between them.

$$F \propto m_1 m_2 \text{ and } F \propto \frac{1}{r^2} \text{ so } F \propto \frac{m_1 m_2}{r^2}$$

$$\therefore F = -\frac{Gm_1 m_2}{r^2} \hat{r} \quad [G = \text{Universal gravitational constant}]$$



Note : This formula is only applicable for spherical symmetric masses or point masses.

Vector form of Newton's law of Gravitation :

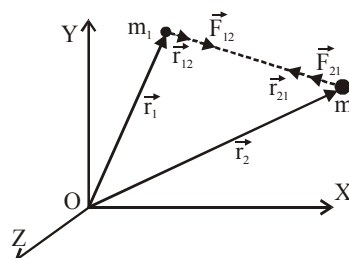
Let $\vec{r}_{12}$ = Displacement vector from m_1 to m_2

$\vec{r}_{21}$ = Displacement vector from m_2 to m_1

$\vec{F}_{21}$ = Gravitational force exerted on m_2 by m_1

$\vec{F}_{12}$ = Gravitational force exerted on m_1 by m_2

$$\vec{F}_{12} = -\frac{Gm_1 m_2}{r_{21}^2} \hat{r}_{21} = -\frac{Gm_1 m_2}{r_{21}^3} \vec{r}_{21}$$



Negative sign shows that :

- The direction of $\vec{F}_{12}$ is opposite to that $\hat{r}_{21}$
- The gravitational force is attractive in nature

$$\text{Similarly } \vec{F}_{21} = -\frac{Gm_1 m_2}{r_{12}^2} \hat{r}_{12} \text{ or } \vec{F}_{21} = -\frac{Gm_1 m_2}{r_{12}^3} \vec{r}_{12} \Rightarrow \vec{F}_{12} = -\vec{F}_{21}$$

The gravitational force between two bodies are equal in magnitude and opposite in direction.

Gravitational constant "G"

- Gravitational constant is a scalar quantity.
- Unit :** **SI :** $G = 6.67 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$
CGS : $6.67 \times 10^{-8} \text{ dyne-cm}^2/\text{g}^2$
Dimensions : $\text{M}^{-1}\text{L}^3\text{T}^{-2}$
- Its value is same throughout the universe, G does not depend on the nature and size of the bodies, it also does not depend upon nature of the medium between the bodies.
- Its value was first find out by the scientist "**Henry Cavendish**" with the help of "Torsion Balance" experiment.
- Value of G is small therefore gravitational force is weaker than electrostatic and nuclear forces.

Ex. Two particles of masses 1 kg and 2 kg are placed at a separation of 50 cm. Assuming that the only forces acting on the particles are their mutual gravitation, find the initial acceleration of heavier particle.

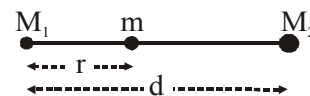
Sol. Force exerted by one particle on another $F = \frac{Gm_1m_2}{r^2} = \frac{6.67 \times 10^{-11} \times 1 \times 2}{(0.5)^2} = 5.3 \times 10^{-10} \text{ N}$

$$\text{Acceleration of heavier particle} = \frac{F}{m_2} = \frac{5.3 \times 10^{-10}}{2} = 2.65 \times 10^{-10} \text{ ms}^{-2}$$

This example shows that gravitation is very weak but only this force keep bind our solar system and also this universe, all galaxies and other interstellar system.

Ex. Two stationary particles of masses M_1 and M_2 are at a distance 'd' apart. A third particle lying on the line joining the particles, experiences no resultant gravitational forces. What is the distance of this particle from M_1 .

Sol. The force on m towards M_1 is $F_1 = \frac{GM_1m}{r^2}$



The force on m towards M_2 is $F_2 = \frac{GM_2m}{(d-r)^2}$

According to question net force on m is zero i.e. $F_1 = F_2$

$$\Rightarrow \frac{GM_1m}{r^2} = \frac{GM_2m}{(d-r)^2} \Rightarrow \left(\frac{d-r}{r} \right)^2 = \frac{M_2}{M_1}$$

$$\Rightarrow \frac{d}{r} - 1 = \frac{\sqrt{M_2}}{\sqrt{M_1}} \Rightarrow r = d \left[\frac{\sqrt{M_1}}{\sqrt{M_1} + \sqrt{M_2}} \right]$$

Ex. Three particles, each of mass m , are situated at the vertices of an equilateral triangle of side ' a '. The only forces acting on the particles are their mutual gravitational forces. It is desired that each particle moves in a circle while maintaining their original separation ' a '. Determine the initial velocity that should be given to each particle and time period of circular motion.

Sol. The resultant force on particle at A due to other two particles is

$$F_A = \sqrt{F_{AB}^2 + F_{AC}^2 + 2F_{AB}F_{AC}\cos 60^\circ} = \sqrt{3} \frac{Gm^2}{a^2} \dots (i) \quad \left[\because F_{AB} = F_{AC} = \frac{Gm^2}{a^2} \right]$$

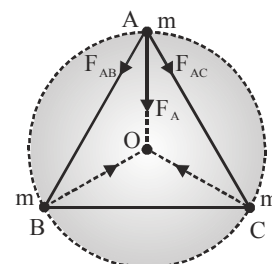
$$\text{Radius of the circle } r = \frac{2}{3} \times a \sin 60^\circ = \frac{a}{\sqrt{3}}$$

If each particle is given a tangential velocity v , so that F acts as the centripetal force,

$$\text{Now } \frac{mv^2}{r} = \sqrt{3} \frac{mv^2}{a} \dots (ii)$$

$$\text{From (i) and (ii) } \sqrt{3} \frac{mv^2}{a} = \frac{Gm^2\sqrt{3}}{a^2} \Rightarrow v = \sqrt{\frac{Gm}{a}}$$

$$\text{Time period } T = \frac{2\pi r}{v} = \frac{2\pi a}{\sqrt{3}} \sqrt{\frac{a}{Gm}} = 2\pi \sqrt{\frac{a^3}{3Gm}}$$



- Gravitational forces are central forces as they act along the line joining the centres of two bodies.
- The gravitational forces are conservative forces so work done by gravitational force does not depend upon path and therefore if any particle moves along a closed path under the action of gravitational force then the work done by this force is always zero.
- The total gravitational force on one particle due to number of particles is the resultant of forces of attraction exerted on the given particle due to individual particles i.e. $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots$ it means the principle of superposition is valid.

Gravitational Field

The gravitational field is the space around a mass or an assembly of masses over which it can exert gravitational forces on other masses.

Theoretically speaking, the gravitational field extends up to infinity. However, in actual practice, the gravitational field may become too weak to be measured beyond a particular distance.

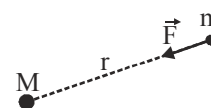
Gravitational Field Intensity [g or E_g]

Gravitational force acting per unit mass at any point in the gravitational field is called Gravitational field intensity.

$$g = \frac{GMm}{r^2} / m = \frac{GM}{r^2} \quad \text{Vector form : } \vec{g} = \frac{\vec{F}}{m} \text{ or } \vec{g} = -\frac{GM}{r^2} \hat{r}$$

Gravitational field intensity is a vector quantity having dimension $[LT^{-2}]$ and unit N/kg .

- Since the force between two point masses is having the similar expression as that of force between two point charges, we can write the gravitational field & gravitational potential in the same manner as the electric field & electric potential.



Analogy between Electrostatics & Gravitation

(1) Point Charge

$$(a) \quad E = \frac{kQ}{r^2}$$

$$(b) \quad V = \frac{kQ}{r}$$

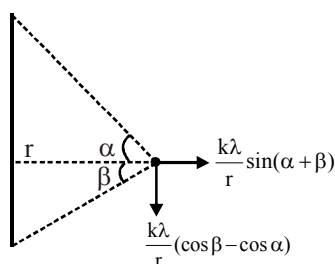
(2) Uniform charged ring

$$(a) \quad E = \frac{kQx}{(r^2 + x^2)^{3/2}} \text{ on axis}$$

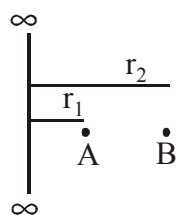
$$E \text{ is max. when } x = \frac{r}{\sqrt{2}}$$

$$(b) \quad V = \frac{kQ}{\sqrt{r^2 + x^2}} \text{ on axis, } \frac{kQ}{r} \text{ at center}$$

(3) Uniform linear charge



(4) Infinite Linear charge



$$(a) \quad E = \frac{2K\lambda}{r}$$

$$(b) \quad V_B - V_A = -2K\lambda \ln \frac{r_2}{r_1}$$

(5) Infinite Sheet of charge

$$E = \frac{\sigma}{2\epsilon_0}$$

Point Mass

$$g = \frac{GM}{r^2}$$

$$V_G = \frac{-GM}{r}$$

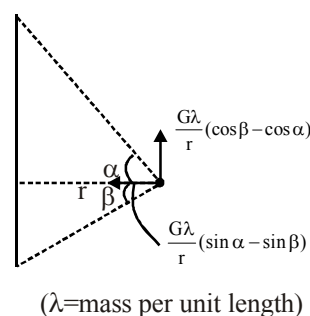
Ring of uniform mass distribution

$$g = \frac{GMx}{(r^2 + x^2)^{3/2}} \text{ on axis}$$

$$g \text{ is max. when } x = \frac{r}{\sqrt{2}}$$

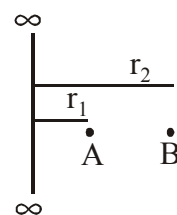
$$V_G = \frac{-GM}{\sqrt{r^2 + x^2}} \text{ on axis, } \frac{-GM}{r} \text{ at center}$$

Uniform linear mass



(λ = mass per unit length)

Infinite linear mass



$$g = \frac{2G\lambda}{r}$$

$$V_B - V_A = 2G\lambda \ln \left(\frac{r_2}{r_1} \right)$$

Infinite Sheet of mass

$$g = \frac{\sigma}{2} \times 4\pi G = 2\pi G\sigma$$

(σ = mass per unit area)

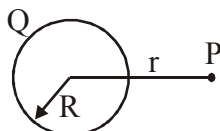
- * Notice gravitational force is always attractive and hence gravitational potential is always -ve. (for a repulsive force potential is positive). This can be explained from the sign of W_{ext} in moving the test charge from ∞ to the point under consideration.
- ** Since $\vec{g}$ points from B towards A potential increases as we move from A to B. Just like electric potential gravitational potential also increases opposite to field direction.

(6) **Uniformly charged hollow sphere**

Charge Q , radius R

distance of field point from center r

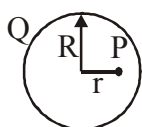
Case I $r > R$



$$E = \frac{kQ}{r^2}$$

$$V = \frac{kQ}{r}$$

Case II $r < R$



$$E = 0$$

$$V = \frac{kQ}{r}$$

(7) **Electrostatics self energy of uniformly charged thin spherical shell.**

$$U = \frac{KQ^2}{2R}$$

(8) **Uniformly charged solid sphere**

mass M , radius R

$$E = \frac{kQ}{r^2}, r > R$$

$$\frac{kQr}{R^3}, r < R$$

$$V = \frac{kQ}{r}, r > R$$

$$\frac{kQ}{2R^3} (3R^3 - r^2), r > R$$

Hollow sphere of uniform mass

Mass M , radius R

distance of field point from center r

$$g = \frac{GM}{r^2}$$

$$V_G = -\frac{GM}{r}$$

$$g = 0$$

$$V_G = -\frac{GM}{R}$$

Gravitational self energy of uniform thin spherical shell.

$$U = \frac{GM^2}{2R}$$

Uniformly solid sphere

mass M , radius R

$$g = \frac{GM}{r^2}, r > R$$

$$\frac{GM}{R^3} r, r < R$$

$$V_a = -\frac{GM}{r}, r > R$$

$$-\frac{GM}{2R^3} (3R^3 - r^2), r > R$$

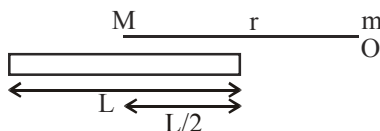
- (9) **Electrostatics self energy of uniformly charged solid sphere.**

$$U = \frac{3}{5} \frac{KQ^2}{R}$$

- Gravitational self energy of uniform solid sphere.**

$$U = \frac{3}{5} \frac{GM^2}{R}$$

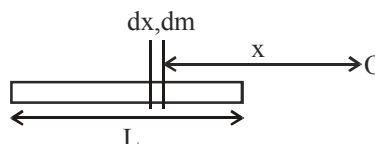
Ex. Find gravitational force between the point mass & the rod of uniform mass.



Ans.
$$\frac{GMm}{\left(r + \frac{1}{2}\right)\left(r - \frac{1}{2}\right)}$$

Sol. $dm = \frac{M}{L} dx$

$$dF = \frac{G(dM)m}{x^2}$$



$$\int_0^F dF = \frac{GMm}{L} \int_{x_1}^{x_2} \frac{dx}{x^2}$$

$$F = \frac{GMm}{L} \left[\frac{1}{x} \right]_{x_2}^{x_1} = \frac{GMm}{L} \frac{L}{\left(r - \frac{L}{2}\right)\left(r + \frac{L}{2}\right)} = \frac{GMm}{\left(r - \frac{L}{2}\right)\left(r + \frac{L}{2}\right)}$$

Ex. A thin rod of mass M and length L is bent in a semicircle as shown in figure.

- (a) What is its gravitational force (both magnitude and direction) on a particle with mass m at O, the centre of curvature?
 (b) What would be the force on m if the rod is, in the form of a complete circle?

Sol. (a) Considering an element of rod of length $d\ell$ as shown in figure and treating it as a point mass $(M/L)d\ell$ situated at a distance R from P, the gravitational force due to this element on the particle will be

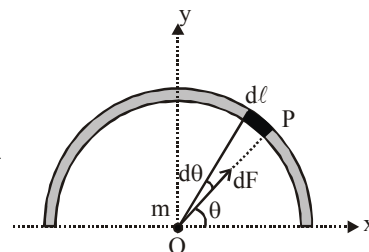
$$dF = \frac{Gm(M/L)(Rd\theta)}{R^2} \text{ along OP [as } d\ell = Rd\theta]$$

So the component of this force along x and y-axis will be

$$dF_x = dF \cos \theta = \frac{GmM}{LR} \cos \theta d\theta; dF_y = dF \sin \theta = \frac{GmM}{LR} \sin \theta d\theta$$

$$\text{So that } F_x = \frac{GmM}{LR} \int_0^\pi \cos \theta d\theta = \frac{GmM}{LR} [\sin \theta]_0^\pi = 0$$

$$\text{and } F_y = \frac{GmM}{LR} \int_0^\pi \sin \theta d\theta = \frac{GmM}{LR} [-\cos \theta]_0^\pi = \frac{2\pi GmM}{L^2} \left[\text{as } R = \frac{L}{\pi} \right]$$



So
$$F = \sqrt{F_x^2 + F_y^2} = F_y = \frac{2\pi GmM}{L^2} \text{ [as } F_x \text{ is zero]}$$

i.e., the resultant force is along the y-axis and has magnitude $(2\pi GmM/L^2)$

(b) If the rod was bent into a complete circle,

$$F_x = \frac{GmM}{LR} \int_0^{2\pi} \cos \theta d\theta = 0 \text{ and also } F_y = \frac{GmM}{LR} \int_0^{2\pi} \sin \theta d\theta = 0$$

i.e., the resultant force on m at O due to the ring is zero.

Ex. Find ratio of gravitational field on the surface of two planets which are of uniform mass density & have radius R_1 & R_2 if

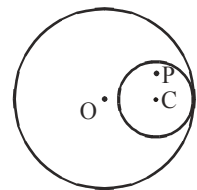
- (a) They are of same mass
(b) They are of same density

Ans. (a) $\frac{g_1}{g_2} = \frac{R_2^2}{R_1^2}$ (b) $\frac{g_1}{g_2} = \frac{R_1}{R_2}$

Sol.
$$g = \frac{GM}{R^2} = \frac{G \frac{4}{3} \pi R^3 \rho}{R^2} = \left(\frac{4\pi G R}{3} \right) \rho$$

Ex. A uniform solid sphere of density ρ and radius R has a spherical cavity of radius r inside it as shown. Find gravitation field at

- (a) O
(b) C
(c) P (prove that field inside cavity is uniform)

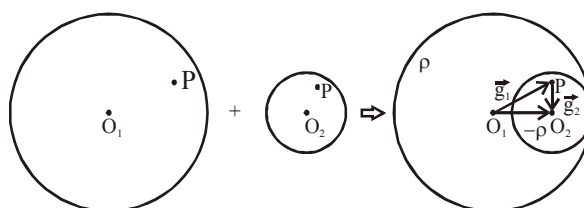


Ans. (a) $\frac{k \left(\frac{4}{3} \pi r^3 \rho \right)}{(OC)^2} \vec{CO}$ (b) $\frac{4\pi G \rho \vec{OC}}{3}$ (c) $\frac{4}{3} \pi G \rho \vec{O_1 O_2}$

Sol. $\vec{g}$ = gravitational field at any point inside sphere

$$\begin{aligned} \vec{g} &= \frac{GM}{R^3} \vec{r} \\ &= \frac{G}{R^3} \frac{4}{3} \pi R^3 \rho \vec{r} \\ \vec{g} &= \frac{4}{3} \pi G \rho \vec{r} \end{aligned}$$

Let the sphere with cavity is formed by superimposing it with a small sphere of density $(-\rho)$ as shown



Resultant field $\vec{g} = \vec{g}_1 + \vec{g}_2$

$$= \left(\frac{4}{3} \pi G \rho \right) \overrightarrow{O_1 P} + \left(\frac{4}{3} \pi G \rho \right) \overrightarrow{P O_2}$$

$$= \frac{4}{3} \pi G \rho \overrightarrow{O_1 O_2} \quad \left[\overrightarrow{O_1 O_2} = \overrightarrow{OC} \right]$$

It is independent of position of point inside cavity

At O $\vec{g} = \vec{g}_1 + \vec{g}_2$

$$= 0 + \frac{GM}{(\overline{CO})^2} (\widehat{CO})$$

$$= \frac{G \frac{4}{3} \pi r^2 \rho}{(\overline{CO})^2} \widehat{CO}$$

$$= \frac{\left(\frac{4}{3} \pi G r^2 \rho \right)}{(\overline{CO})^2} \widehat{CO}$$

Acceleration Due to Gravity (g)

Gravitational Force $F_g = ma$ if $R_e =$ Radius of Earth, $M_e =$ Mass of Earth.

$$\text{then } \frac{GM_e m}{R_e^2} = ma_g \Rightarrow a_g = g = \frac{GM_e}{R_e^2} \quad (GM_e = gR_e^2) \dots (i)$$

- In form of density $g = \frac{GM_e}{R_e^2} = \frac{G}{R_e^2} \times \frac{4}{3} \pi R_e^3 \times \rho$

$$\therefore g = \frac{4}{3} \pi G R_e \rho \quad \dots (ii)$$

If ρ is constant then $g \propto R_e$

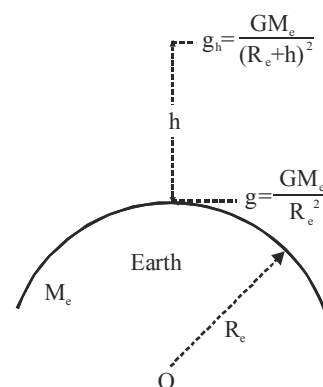
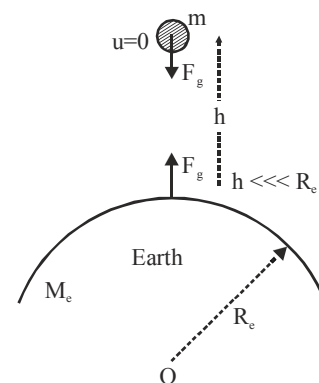
Variation in Acceleration due gravity

(a) Due to Altitude (height)

$$\frac{g_h}{g} = \frac{R_e^2}{(R_e + h)^2}$$

$$\text{By binomial expansion } \left(1 + \frac{h}{R_e} \right)^{-2} \approx \left(1 - \frac{2h}{R_e} \right)$$

$$[\text{If } h \ll R_e, \text{ then higher power terms are negligible}] \therefore g_h = g \left[1 - \frac{2h}{R_e} \right]$$



Ex. Two equal masses m and m are hung from a balance whose scale pans differ in vertical height by 'h'. Determine the error in weighing in terms of density of the Earth ρ .

Sol. $g_h = g \left[1 - \frac{2h}{R_e} \right]$, $W_2 - W_1 = mg_2 - mg_1 = 2mg \left[\frac{h_1}{R_e} - \frac{h_2}{R_e} \right] = 2m \frac{GM}{R_e^2} \times \frac{h}{R_e} \left[\because g = \frac{GM}{R_e^2} \text{ \& } h_1 - h_2 = h \right]$

$$\text{Error in weighing} = W_2 - W_1 = 2mG \frac{4}{3} \pi R_e^3 \rho \frac{h}{R_e^3} = \frac{8\pi}{3} G \rho m h$$

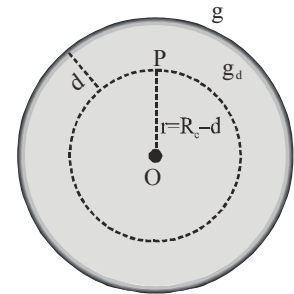
(b) Due to depth :

Assuming density of Earth remains same throughout. At depth d inside the Earth :

$$g_d = g \left[1 - \frac{d}{R_e} \right] \text{ valid for any depth}$$

$$\text{Decrement in } g \text{ with depth} = \Delta g_d = g - g_d = g - g \left[1 - \frac{d}{R_e} \right]$$

$$\therefore \frac{\Delta g_d}{g} = \frac{d}{R_e}$$



Ex. At which depth from Earth surface, acceleration due to gravity is decreased by 1%

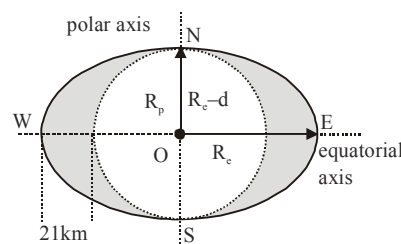
Sol. $\frac{\Delta g_d}{g} = \frac{d}{R_e} \Rightarrow \frac{1}{100} = \frac{d}{6400} \therefore d = 64 \text{ km}$

(c) Due to shape of the Earth

$$R_p < R_e$$

$$\therefore g_e < g_p$$

$$\Rightarrow \text{by putting the values } g_p - g_e = 0.02 \text{ m/s}^2$$



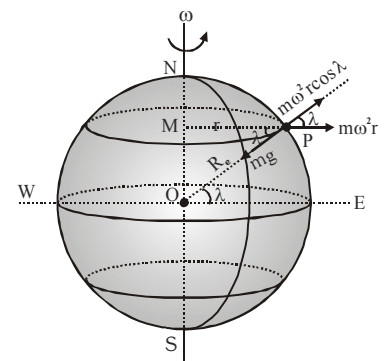
(d) Due to Rotation of the Earth

$$g' = g - R_e \omega^2 \cos^2 \lambda$$

If latitude angle $\lambda = 0$. It means at equator. $g'_{\min} = g_e = g - R_e \omega^2$

If latitude angle $\lambda = 90^\circ$. It means at poles. $g'_{\max} = g_p = g \Rightarrow g_p > g_e$

Change in "g" only due to rotation $\Delta g_{\text{rot.}} = g_p - g_e = 0.03 \text{ m/s}^2$



$$\Delta g_{\text{total}} = g_p - g_e = (0.05 \text{ m/s}^2) \rightarrow \begin{cases} 0.02 \text{ m/s}^2 & \text{(due to shape)} \\ 0.03 \text{ m/s}^2 & \text{(due to rotation)} \end{cases}$$

Weightlessness

State of the free fall $\left(\vec{a} = -\frac{GM}{r^2}\vec{r}\right)$ is called state of weightlessness. If a body is in a satellite (which does not produce its own gravity) orbiting the Earth at a height h above its surface then

$$\text{True weight} = mg_h = \frac{mGM}{(R+h)^2} = \frac{mg}{\left(1+\frac{h}{R}\right)^2}$$

$$\text{Apparent weight} = m(g_h - a)$$

$$\text{but } a = \frac{v_0^2}{r} = \frac{GM}{r^2} = \frac{GM}{(R+h)^2} = g_h \Rightarrow \text{Apparent weight} = m(g_h - g_h) = 0$$

Note : The condition of weightlessness can be overcome by creating artificial gravity by rotating the satellite in addition to its revolution.

Escape speed (v_e)

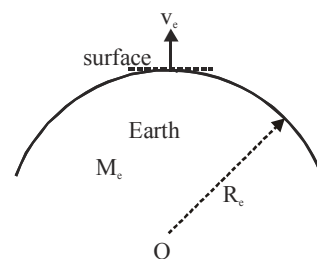
Minimum speed required for an object at Earth's surface so that it just escapes the Earth's gravitational field.

Escape energy

Minimum energy given to a particle in form of kinetic energy so that it can just escape from Earth's gravitational field.

$$\text{Escape energy} = \frac{GM_e m}{R_e} \quad (\text{-ve of PE of Earth's surface})$$

$$\text{Escape energy} = \text{Kinetic Energy} \Rightarrow \frac{GM_e m}{R_e} = \frac{1}{2}mv_e^2 \Rightarrow v_e = \sqrt{\frac{2GM_e}{R_e}}$$



- $v_e = \sqrt{\frac{2GM_e}{R_e}}$ (In form of mass) If $M = \text{constant}$ $v_e \propto \frac{1}{\sqrt{R_e}}$
- $v_e = \sqrt{2gR_e}$ (In form of g) If $g = \text{constant}$ $v_e \propto \sqrt{R_e}$
- $v_e = R_e \sqrt{\frac{8\pi G \cdot \rho}{3}}$ (In form of density) If $\rho = \text{constant}$ $v_e \propto R_e$

- Escape velocity does not depend on mass of body, angle of projection or direction of projection.
 $v_e \propto m^0$ and $v_e \propto \theta^0$
- Escape velocity at : Earth surface $v_e = 11.2 \text{ km/s}$ Moon surface $v_e = 2.31 \text{ km/s}$
- Atmosphere on Moon is absent because root mean square velocity of gas particle is greater than escape velocity. $v_{\text{rms}} > v_e$

Ex. A space-ship is launched into a circular orbit close to the Earth's surface. What additional speed should now be imparted to the spaceship so that orbit to overcome the gravitational pull of the Earth.

Sol. Let ΔK be the additional kinetic energy imparted to the spaceship to overcome the gravitation pull

then by energy conservation $-\frac{GMm}{2R} + \Delta K = 0 + 0 \Rightarrow \Delta K = \frac{GMm}{2R}$

Total kinetic energy $= \frac{GMm}{2R} + \Delta K = \frac{GMm}{2R} + \frac{GMm}{2R} = \frac{GMm}{R}$ then $\frac{1}{2}mv_2^2 = \frac{GMm}{R} \Rightarrow v_2 = \sqrt{\frac{2GM}{R}}$

But $v_1 = \sqrt{\frac{GM}{R}}$. So Additional velocity $= v_2 - v_1 = \sqrt{\frac{2GM}{R}} - \sqrt{\frac{GM}{R}} = (\sqrt{2} - 1) \sqrt{\frac{GM}{R}}$

Ex. Find the minimum speed with which an object should be projected vertically upward from earth's surface to reach a height equal to radius of earth, R_e .

Ans. $\sqrt{\frac{GM}{R_e}}$

Sol. $-\frac{GMm}{R_e} + \frac{1}{2}mv^2 = -\frac{GMm}{2R_e}$

$\therefore v = \sqrt{\frac{GM}{R_e}}$

Ex. The distance between the centres of two stars is $10a$. The masses of the stars are M and $16M$ and their radii a and $2a$ respectively. A body of mass m is fired straight from surface of the larger star towards smaller star.

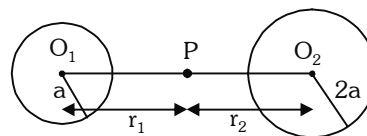
(a) Find the distance between centre of smaller star and the point of zero gravitational field strength:

Sol. P is the point where field strength is zero.

$$\frac{GM}{r_1^2} = \frac{G(16M)}{r_2^2}$$

and $r_1 + r_2 = 10a$

So, $r_1 = 2a, r_2 = 8a$



(b) The initial minimum speed of the body to reach smaller star is $K\sqrt{\frac{GM}{a}}$. Find the value of K :

Sol. From conservation of mechanical energy.

$$\frac{1}{2}mv_{\min}^2 = \text{Potential energy of body at P} - \text{Potential energy of body at larger star}$$

$$= \left[-\frac{GMm}{r_1} - \frac{16GMm}{r_2} \right] - \left[-\frac{GMm}{(10a-2a)} - \frac{16GMm}{2a} \right]$$

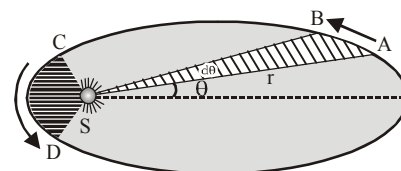
$$= \frac{3\sqrt{5}}{2} \left[\sqrt{\frac{GM}{a}} \right]$$

Kepler's Laws

Kepler found important regularities in the motion of the planets. These regularities are known as 'Kepler's three laws of planetary motion'.

- (a) **First Law (Law of Orbits)** : All planets move around the Sun in elliptical orbits, having the Sun at one focus of the orbit.
- (b) **Second Law (Law of Areas)** : A line joining any planet to the Sun sweeps out equal areas in equal times, that is, the areal speed of the planet remains constant.

According to the second law, when the planet is nearest the Sun, then its speed is maximum and when it is farthest from the Sun, then its speed is minimum. In figure if a planet moves from A to B in a given time-interval, and from C to D in the same time-interval, then the areas ASB and CSD will be equal.



$$\frac{dA}{dt} = \frac{J}{2m} \quad \dots(iii)$$

Now, the areal speed dA/dt of the planet is constant, according to Kepler's second law. Therefore, according to eq. (iii), the angular momentum J of the planet is also constant, that is, the angular momentum of the planet is conserved. Thus, Kepler's second law is equivalent to conservation of angular momentum.

- (c) **Third Law : (Law of Periods)** : The square of the period of revolution (time of one complete revolution) of any planet around the Sun is directly proportional to the cube of the semi-major axis of its elliptical orbit.

$$T^2 \propto a^3$$

So it is clear through this rule that the farthest planet from the Sun has largest period of revolution. The period of revolution of the closest planet Mercury is 88 days, while that of the farthest dwarf planet Pluto is 248 years.

Satellite motion

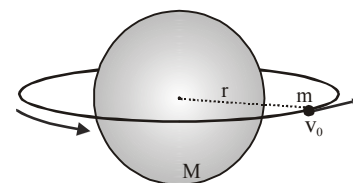
A light body revolving round a heavier body due to gravitational attraction, is called satellite. Earth is a satellite of the Sun while Moon is satellite of Earth.

Orbital velocity (v_0) : A satellite of mass m moving in an orbit of radius r with speed v_0 then required centripetal force is provided by gravitation.

$$F_{cp} = F_g \Rightarrow \frac{mv_0^2}{r} = \frac{GMm}{r^2} \Rightarrow v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{GM}{(R_e + h)}} \quad (r = R_e + h)$$

For a satellite very close to the Earth surface $h \ll R_e \therefore r = R_e$

$$v_0 = \sqrt{\frac{GM}{R_e}} = \sqrt{gR_e} = 8 \text{ km/s}$$



- If a body is taken at some height from Earth and given horizontal velocity of magnitude 8 km/sec then the body becomes satellite of Earth.
- v_0 depends upon : Mass of planet, Radius of circular orbit of satellite, g (at planet), Density of planet
- If orbital velocity of a near by satellite becomes $\sqrt{2} v_0$ (or increased by 41.4%, or K.E. is doubled) then the satellite escapes from gravitational field of Earth.

Time Period of a Satellite $T = \frac{2\pi r}{v_0} = \frac{2\pi r^{3/2}}{\sqrt{GM}} = \frac{2\pi r^{3/2}}{R\sqrt{g}} \Rightarrow T^2 = \frac{4\pi^2}{GM} r^3 \Rightarrow T^2 \propto r^3 (r = R + h)$

For Geostationary Satellite $T = 24 \text{ hr}$, $h = 36,000 \text{ km} \approx 6 R_e$ ($r \approx 7 R_e$), $v_0 = 3.1 \text{ km/s}$

For Near by satellite $v_0 = \sqrt{\frac{GM_e}{R_e}} \approx 8 \text{ km/s}$

$$T_{Ns} = 2\pi \sqrt{\frac{R_e}{g}} = 84 \text{ minute} = 1 \text{ hour } 24 \text{ minute} = 1.4 \text{ hr} = 5063 \text{ s}$$

In terms of density $T_{Ns} = \frac{2\pi(R_e)^{1/2}}{(G \times 4/3 \pi R_e \times \rho)^{1/2}} = \sqrt{\frac{3\pi}{G\rho}}$

Time period of near by satellite only depends upon density of planet.

For Moon $h_m = 380,000 \text{ km}$ and $T_m = 27 \text{ days}$

$$v_{om} = \frac{2\pi(R_e + h)}{T_m} = \frac{2\pi(386400 \times 10^3)}{27 \times 24 \times 60 \times 60} \approx 1.04 \text{ km/sec.}$$

Energies of a Satellite Kinetic energy $K.E. = \frac{1}{2}mv_0^2 = \frac{GMm}{2r} = \frac{L^2}{2mr^2}$

Potential energy $P.E. = -\frac{GMm}{r} = -mv_0^2 = -\frac{L^2}{mr^2}$

Total mechanical energy $T.E. = P.E. + K.E. = -\frac{mv_0^2}{2} = -\frac{GMm}{2r} = -\frac{L^2}{2mr^2}$

Essential Condition's for Satellite Motion

- Centre of satellite's orbit coincide with centre of Earth.
- Plane of orbit of satellite is passing through centre of Earth.

Special Points about Geo-Stationary Satellite

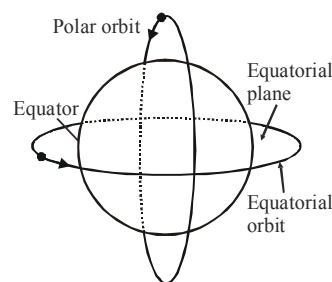
- All three essential conditions for satellite motion should be followed.
- It rotates in equatorial plane.
- Its height from Earth surface is 36000 km . ($\sim 6R_e$)
- Its angular velocity and time period should be same as that of Earth.
- Its rotating direction should be same as that of Earth (West to East).
- Its orbit is called parking orbit and its orbital velocity is 3.1 km./sec .
- Maximum latitude at which message can be recieved by geostationary satellite is

$$\theta = \cos^{-1} \left(\frac{R_e}{R_e + h} \right)$$

- The area of earth's surface covered by geostationary satellite is $S = \Omega R_e^2 = \frac{2\pi h R_e^2}{R_e + h}$

• **Polar Satellite (Sun – synchronous satellite)**

It is that satellite which revolves in polar orbit around Earth. A polar orbit is that orbit whose angle of inclination with equatorial plane of Earth is 90° and a satellite in polar orbit will pass over both the north and south geographic poles once per orbit. Polar satellites are Sun-synchronous satellites. Every location on Earth lies within the observation of polar satellite twice each day. The polar satellites are used for getting the cloud images, atmospheric data, ozone layer in the atmosphere and to detect the ozone hole over Antarctica.



Only the equatorial orbits are stable for a satellite. For any satellite to orbit around in a stable orbit, it must move in such an orbit so that the centre of Earth lies at the centre of the orbit.

Binding energy

Total mechanical energy (potential + kinetic) of a closed system is negative. The modulus of this total mechanical energy is known as the binding energy of the system. This is the energy due to which system is closed or different parts of the system are bound to each other.

Binding energy of satellite (system)

$$\text{B.E.} = -\text{T.E.} \quad \text{B.E.} = \frac{1}{2}mv_0^2 = \frac{GMm}{2r} = \frac{L^2}{2mr^2} \quad \text{Hence B.E.} = \text{K.E.} = -\text{T.E.} = \frac{-\text{P.E.}}{2}$$

Work done in Changing the Orbit of Satellite

$$W = \text{Change in mechanical energy of system but } E = \frac{-GMm}{2r} \text{ so } W = E_2 - E_1 = \frac{GMm}{2} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

Ex. A satellite moves eastwards very near the surface of the Earth in equatorial plane with speed (v_0). Another satellite moves at the same height with the same speed in the equatorial plane but westwards. If R = radius of the Earth and ω be its angular speed of the Earth about its own axis. Then find the approximate difference in the two time period as observed on the Earth.

$$\text{Sol. } T_{\text{west}} = \frac{2\pi R}{v_0 + R\omega} \text{ and } T_{\text{east}} = \frac{2\pi R}{v_0 - R\omega} \Rightarrow \Delta T = T_{\text{east}} - T_{\text{west}} = 2\pi R \left[\frac{2R\omega}{v_0^2 - R^2\omega^2} \right] = \frac{4\pi\omega R^2}{v_0^2 - R^2\omega^2}$$

Ex. An artificial satellite (mass m) of a planet (mass M) revolves in a circular orbit whose radius is n times the radius R of the planet. In the process of motion, the satellite experiences a slight resistance due to cosmic dust. Assuming the force of resistance on satellite to depend on velocity as $F = av^2$ where 'a' is a constant, calculate how long the satellite will stay in the space before it falls onto the planet's surface.

$$\text{Sol. Air resistance } F = -av^2, \text{ where orbital velocity } v = \sqrt{\frac{GM}{r}}$$

$$r = \text{the distance of the satellite from planet's centre} \Rightarrow F = -\frac{GMa}{r}$$

$$\text{The work done by the resistance force } dW = Fdx = Fvdt = \frac{GMa}{r} \sqrt{\frac{GM}{r}} dt = \frac{(GM)^{3/2}}{r^{3/2}} a dt \dots (i)$$

$$\text{The loss of energy of the satellite} = dE \therefore \frac{dE}{dr} = \frac{d}{dr} \left[-\frac{GMm}{2r} \right] = \frac{GMm}{2r^2} \Rightarrow dE = \frac{GMm}{2r^2} dr \dots (ii)$$

Since $dE = -dW$ (work energy theorem) $-\frac{GMm}{2r^2} dr = \frac{(GM)^{3/2}}{r^{3/2}} dt$

$$\Rightarrow t = -\frac{m}{2a\sqrt{GM}} \int_{nR}^R \frac{dr}{\sqrt{r}} = \frac{m\sqrt{R}(\sqrt{n}-1)}{a\sqrt{GM}} = (\sqrt{n}-1) \frac{m}{a\sqrt{gR}}$$

Ex. Two satellites S_1 and S_2 revolve round a planet in coplanar circular orbits in the same sense. Their periods of revolution are 1 h and 8h respectively. The radius of the orbit of S_1 is 10^4 km. When S_2 is closest to S_1 , find (a) the speed of S_2 relative to S_1 and (b) the angular speed of S_2 as observed by an astronaut in S_1 .

Sol. Let the mass of the planet be M , that of S_1 be m_1 and S_2 be m_2 . Let the radius of the orbit of S_1 be $R_1 (= 10^4 \text{ km})$ and of S_2 be R_2 .

Let v_1 and v_2 be the linear speeds of S_1 and S_2 with respect to the planet. Figure shows the situation. As the square of the time period is proportional to the cube of the radius.

$$\left(\frac{R_2}{R_1}\right)^3 = \left(\frac{T_2}{T_1}\right)^2 = \left(\frac{8h}{1h}\right)^2 = 64$$

or, $\frac{R_2}{R_1} = 4$

or, $R_2 = 4R_1 = 4 \times 10^4 \text{ m}$.

Now the time period of S_1 is 1 h. So,

$$\frac{2\pi R_1}{v_1} = 1h$$

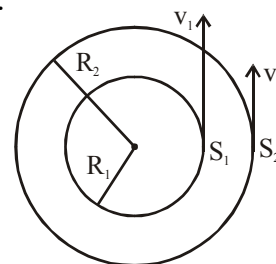
or, $v_1 = \frac{2\pi R_1}{1h} = 2\pi \times 10^4 \text{ km/h}$

similarly, $v_2 = \frac{2\pi R_2}{8h} = \pi \times 10^4 \text{ km/h}$

(a) At the closest separation, they are moving in the same direction. Hence the speed of S_2 with respect to S_1 is $|v_2 - v_1| = \pi \times 10^4 \text{ km/h}$.

(b) As seen from S_1 , the satellite S_2 is at a distance $R_2 - R_1 = 3 \times 10^4 \text{ km}$ at the closest separation. Also it is moving at $\pi \times 10^4 \text{ km/h}$ in a direction perpendicular to the line joining them. Thus, the angular speed of S_2 as observed by S_1 is

$$\omega = \frac{\pi \times 10^4 \text{ km/h}}{3 \times 10^4 \text{ km}} = \frac{\pi}{3} \text{ rad/h.}$$



Conditions for different trajectory

For a body being projected tangentially from above earth's surface, say at a distance r from earth's center, the trajectory would depend on the velocity of projection v .

Velocity

$$1. \text{ velocity, } v < \sqrt{\frac{GM}{r} \left(\frac{2R}{r+R} \right)}$$

$$2. \sqrt{\frac{GM}{r}} > v > \sqrt{\frac{GM}{r} \left(\frac{2R}{r+R} \right)}$$

3. Velocity is equal to the critical velocity

$$\text{of the orbit, } v = \sqrt{\frac{GM_e}{r}}$$

4. Velocity is between the critical and escape velocity of the orbit

$$\sqrt{\frac{2GM_e}{r}} > v > \sqrt{\frac{GM_e}{r}}$$

$$5. v = v_{\text{esc}} = \sqrt{\frac{2GM_e}{r}}$$

$$6. v > v_{\text{esc}} = \sqrt{\frac{2GM_e}{r}}$$

Orbit

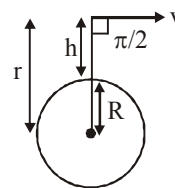
Body returns to earth

Body acquires an elliptical orbit with earth as the far-focus w.r.t. the point of projection.
Circular orbit with radius r

Body acquires an elliptical orbit with earth as the near focus w.r.t. the point of projection.

Body just escapes earth's gravity, along a parabolic path.

Body escape earth's gravity along a hyperbolic path.



Binary star system

Figure shows two particles moving due to mutually attractive gravitational force about center of mass. Since there is no external force CM of system remains fixed and time period of revolution must be same.

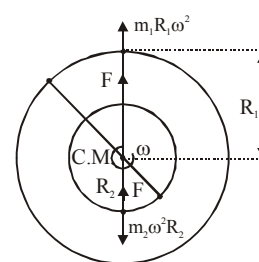
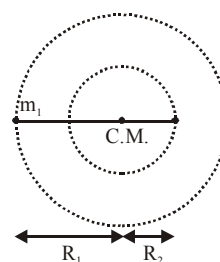
Both bodies have comparable mass and both are moving in circular orbit centre of mass as shown in diagram

$$\omega = \sqrt{\frac{G(m_1 + m_2)}{R^3}}$$

Angular momentum of the system about centre of mass.

$$L = \left(\frac{m_1 m_2}{m_1 + m_2} \right) R^2 \omega$$

$$\text{Kinetic energy} = \frac{1}{2} \left(\frac{m_1 m_2}{m_1 + m_2} \right) R^2 \omega^2$$



The time period of a simple pendulum of infinite length.

In deriving the formula $T_0 = 2\pi\sqrt{\left(\frac{L}{g}\right)}$ we have assumed that length of the pendulum L is much less than the radius of the earth R so that 'g' always remains vertical. However, if length of pendulum is comparable of the radius of earth, 'g' will not remain vertical but will be directed towards the centre of the earth.

$$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{1}{g\left[\frac{1}{L} + \frac{1}{R}\right]}} \quad (< T_0)$$

From this expression it is clear that :

(a) If $L \ll R$, $(1/L) \gg (1/R)$ so $T = 2\pi\sqrt{\left(\frac{L}{g}\right)}$ which is expected.

(b) If $L \gg R \rightarrow \infty$ $(1/L) \ll (1/R)$ so

$$T = 2\pi\sqrt{\frac{R}{g}} = 2\pi\sqrt{\frac{6.4 \times 10^6}{10}}$$

$$= 800 \times 2\pi \text{ sec} \approx 83.8 \text{ minute}$$

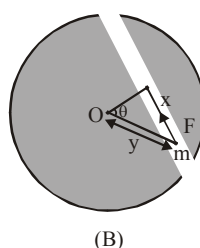
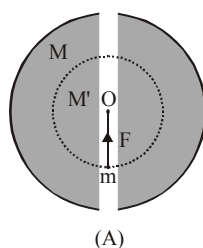
And it is also the maximum time period which an oscillating simple pendulum can have.

(c) If L is comparable to R (say $L = R$),

$$T = 2\pi\sqrt{\frac{R}{2g}} \approx 1 \text{ hour}.$$

Motion of a ball in a tunnel through the earth :

Case I : If the tunnel is along a diameter and the ball is released from the surface. The ball executes SHM.



$$\text{so that } T = 2\pi\sqrt{\frac{R^3}{GM}} ; T = 2\pi\sqrt{\left(\frac{R}{g}\right)}$$

Which is same as that of a simple pendulum of infinite length and is equal to 84.6 minute.

Case II : If the tunnel is along a chord and ball is released from the surface. The motion is SHM with the same time period.

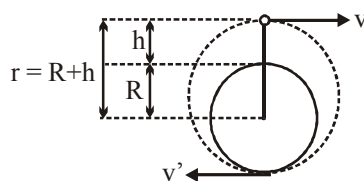
Gravitational pressure :

A uniform sphere has a mass M and radius R . The pressure p inside the sphere, caused by gravitational compression, as a function of the distance r from its centre can be found to be $p = \frac{3}{8} (1 - r^2/R^2) \gamma M^2/\pi R^4$.

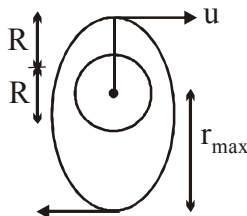
Launching of an artificial satellite around earth

Ex. A satellite is launched tangentially from a height h above earth's surface as shown.

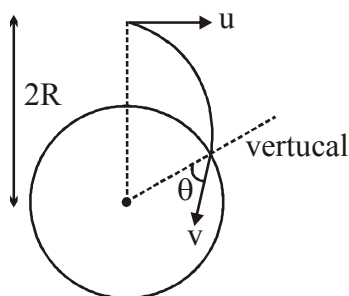
I. Find minimum launch speed ' v ' so that it just touches the earth's surface



II. If $h = R$ and satellite is launched tangentially with speed $= \sqrt{\frac{3GM}{5R}}$ find the maximum distance of satellite from earth's center



III. If $h = R$ and satellite is launched tangentially with a speed $u = \sqrt{\frac{GM}{7R}}$. Find the angle w.r.t. vertical at which the satellite will crash on earth's surface.



Sol.

(I) Angular momentum conservation

$$mv(R + h) = mv'R$$

Energy conservation

$$\frac{-GMm}{R+h} + \frac{1}{2}mv^2 = -\frac{GMm}{R} + \frac{1}{2}mv'^2$$

$$\text{Solving, } v = \sqrt{\frac{2GMR}{r(R+r)}}$$

(II) Angular momentum conservation $\mu \cdot 2R = mvr_{\max}$.

Energy conservation :

$$\frac{-GMm}{2R} + \frac{1}{2}\mu^2 = \frac{-GMm}{r_{\max}} + \frac{1}{2}mv^2$$

$$\Rightarrow GMm\left(\frac{1}{2R} - \frac{1}{r_{\max}}\right) = \frac{1}{2}\mu^2\left(1 - \left(\frac{2R}{r_{\max}}\right)^2\right)$$

$$\Rightarrow GMm\left(\frac{1}{2R} - \frac{1}{r_{\max}}\right) = \frac{3GM}{10R}\left(1 - \frac{4R^2}{r_{\max}^2}\right)$$

$$\Rightarrow 2r_{\max}^2 - 10Rr_{\max} + 12R^2 = 0$$

$$\Rightarrow (r_{\max} - 2R)(r_{\max} - 3R) = 0$$

$$\Rightarrow r_{\max} = 3R$$

(III) Energy conservation

$$\frac{1}{2}\mu^2 - \frac{GMm}{2R} = \frac{1}{2}mv^2 - \frac{GMm}{R}$$

$$\Rightarrow \frac{1}{2}m(v^2 - u^2) = \frac{GMm}{2R}$$

$$v^2 = u^2 + \frac{GM}{R} \Rightarrow v = \sqrt{\frac{8GM}{7R}}$$

Angular momentum conservation

$$\mu \cdot 2R = mvR\sin\theta$$

$$\Rightarrow \sin\theta = \frac{2u}{v}$$

$$\Rightarrow \theta = 45^\circ$$

EXERCISE (S-1)

1. A particle is fired vertically from the surface of the earth with a velocity kv_e , where v_e is the escape velocity and $k < 1$. Neglecting air resistance and assuming earth's radius as R_e . Calculate the height to which it will rise from the surface of the earth.

GR0001

2. Calculate the distance from the surface of the earth at which above and below the surface acceleration due to gravity is the same.

GR0002

3. An object is projected vertically upward from the surface of the earth of mass M with a velocity such that the maximum height reached is eight times the radius R of the earth. Calculate:
(i) the initial speed of projection
(ii) the speed at half the maximum height.

GR0003

4. A satellite close to the earth is in orbit above the equator with a period of rotation of 1.5 hours. If it is above a point P on the equator at some time, it will be above P again after time _____.

GR0004

5. A satellite is moving in a circular orbit around the earth. The total energy of the satellite is $E = -2 \times 10^5 \text{ J}$. The amount of energy to be imparted to the satellite to transfer it to a circular orbit where its potential energy is $U = -2 \times 10^5 \text{ J}$ is equal to _____.

GR0005

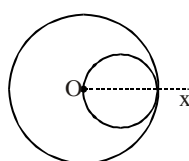
6. A satellite of mass m is orbiting the earth in a circular orbit of radius r . It starts losing energy due to small air resistance at the rate of $C \text{ J/s}$. Then the time taken for the satellite to reach the earth is _____.

GR0006

7. A pair of stars rotates about a common center of mass. One of the stars has a mass M which is twice as large as the mass m of the other. Their centres are at a distance d apart, d being large compared to the size of either star. (a) Derive an expression for the period of rotation of the stars about their common centre of mass in terms of d, m, G . (b) Compare the angular momentum of the two stars about their common centre of mass by calculating the ratio L_m/L_M . (c) Compare the kinetic energies of the two stars by calculating the ratio K_m/K_M .

GR0007

8. A sphere of radius R has its centre at the origin. It has a uniform mass density ρ_0 except that there is a spherical hole of radius $r = R/2$ whose centre is at $x = R/2$ as in fig. (a) Find gravitational field at points on the axis for $|x| > R$ (b) Show that the gravitational field inside the hole is uniform, find its magnitude and direction.



GR0008

EXERCISE (S-2)

1. A small body of mass m is projected with a velocity just sufficient to make it reach from the surface of a planet (of radius $2R$ and mass $3M$) to the surface of another planet (of radius R and mass M). The distance between the centers of the two spherical planets is $6R$. The distance of the body from the center of bigger planet is ' x ' at any moment. During the journey, find the distance x where the speed of the body is (a) maximum (b) minimum. Assume motion of body along the line joining centres of planets.

GR0009

2. A body is launched from the earth's surface at an angle $\alpha=30^\circ$ to the horizontal at a speed $v_0 = \sqrt{\frac{1.5GM}{R}}$. Neglecting air resistance and earth's rotation, find (a) the height to which the body will rise. (b) the radius of curvature of trajectory at its top point.

GR0010

3. A body moving radially away from a planet of mass M , when at distance r from planet, explodes in such a way that two of its many fragments move in mutually perpendicular circular orbits around the planet. What will be (a) then velocity in circular orbits. (b) maximum distance between the two fragments before collision and (c) magnitude of their relative velocity just before they collide.

GR0011

4. A cord of length 64 m is used to connect a 100 kg astronaut to spaceship whose mass is much larger than that of the astronaut. Estimate the value of the tension in the cord. Assume that the spaceship is orbiting near earth surface. Assume that the spaceship and the astronaut fall on a straight line from the earth centre. The radius of the earth is 6400 km.

GR0012

5. A hypothetical spherical planet of radius R and its density varies as $\rho = Kr$, where K is constant and r is the distance from the center. Determine the pressure caused by gravitational pull inside ($r < R$) the planet at a distance r measured from its center.

GR0013

6. The Earth may be regarded as a spherically shaped uniform core of density ρ_1 and radius $R/2$ surrounded by a uniform shell of thickness $R/2$ and density ρ_2 . Find the ratio of $\frac{\rho_1}{\rho_2}$ if the value of acceleration due to gravity is the same at surface as at depth $R/2$ from the surface.

GR0014

7. A binary star has a period (T) of 2 earth years while distance L between its components having masses M_1 and M_2 is four astronomical units. If $M_1 = M_S$ where M_S is the mass of sun, find the ratio $M_2/5M_S$.

GR0015

8. Two uniform spherical stars made of same material have radii R and $2R$. Mass of the smaller planet is m . They start moving from rest towards each other from a large distance under mutual force of gravity. The collision between the stars is inelastic with coefficient of restitution $1/2$.
(a) Find the kinetic energy of the system just after the collision.
(b) Find the maximum separation between their centres after their first collision.

GR0016

9. A remote sensing satellite is revolving in an orbit of radius x over the equator of earth. Find the area on earth surface in which satellite can not send message.

GR0017

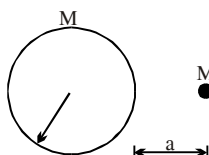
EXERCISE (O-1)

1. If the distance between the centres of Earth and Moon is D and mass of Earth is 81 times that of Moon. At what distance from the centre of Earth gravitational field will be zero?

(A) $\frac{D}{2}$ (B) $\frac{2D}{3}$ (C) $\frac{4D}{5}$ (D) $\frac{9D}{10}$

GR0018

2. A particle of mass M is at a distance a from surface of a thin spherical shell of equal mass and having radius a .



- (A) Gravitational field and potential both are zero at centre of the shell.
 (B) Gravitational field is zero not only inside the shell but at a point outside the shell also.
 (C) Inside the shell, gravitational field alone is zero.
 (D) Neither gravitational field nor gravitational potential is zero inside the shell.

GR0019

3. A hollow spherical shell is compressed to half its radius. The gravitational potential at the centre
 (A) increases (B) decreases
 (C) remains same (D) during the compression increases then returns at the previous value.

GR0020

4. Let ω be the angular velocity of the earth's rotation about its axis. Assume that the acceleration due to gravity on the earth's surface has the same value at the equator and the poles in absence of rotation of earth. An object weighed at the equator gives the same reading as a reading taken at a depth d below earth's surface at a pole ($d \ll R$) The value of d is

(A) $\frac{\omega^2 R^2}{g}$ (B) $\frac{\omega^2 R^2}{2g}$ (C) $\frac{2\omega^2 R^2}{g}$ (D) $\frac{\sqrt{Rg}}{g}$

GR0021

5. If the radius of the earth be increased by a factor of 5, by what factor its density be changed to keep the value of g the same?

(A) $1/25$ (B) $1/5$ (C) $1/\sqrt{5}$ (D) 5

GR0022

6. The mass and diameter of a planet are twice those of earth. What will be the period of oscillation of a pendulum on this planet if it is a seconds pendulum on earth?

(A) $\sqrt{2}$ second (B) $2\sqrt{2}$ seconds (C) $\frac{1}{\sqrt{2}}$ second (D) $\frac{1}{2\sqrt{2}}$ second

GR0023

7. Two identical satellites are at the heights R and $7R$ from the Earth's surface. Then which of the following statement is incorrect. (R = radius of the Earth)

(A) Ratio of total energy of both is 5
(B) Ratio of kinetic energy of both is 4
(C) Ratio of potential energy of both 4
(D) Ratio of total energy of both is 4 and ratio of magnitude of potential to kinetic energy is 2

GR0024

8. A spherical uniform planet is rotating about its axis. The velocity of a point on its equator is V . Due to the rotation of planet about its axis the acceleration due to gravity g at equator is $1/2$ of g at poles. The escape velocity of a particle on the pole of planet in terms of V is

(A) $V_e = 2V$ (B) $V_e = V$ (C) $V_e = V/2$ (D) $V_e = \sqrt{3} V$

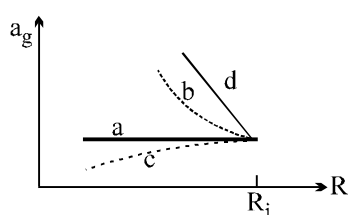
GR0025

9. The escape velocity for a planet is v_e . A tunnel is dug along a diameter of the planet and a small body is dropped into it at the surface. When the body reaches the centre of the planet, its speed will be

(A) v_e (B) $\frac{v_e}{\sqrt{2}}$ (C) $\frac{v_e}{2}$ (D) 0

GR0026

10. A (nonrotating) star collapses onto itself from an initial radius R_i with its mass remaining unchanged. Which curve in figure best gives the gravitational acceleration a_g on the surface of the star as a function of the radius of the star during the collapse?



(A) a (B) b (C) c (D) d

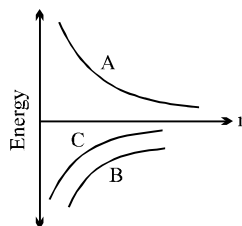
GR0027

11. A satellite of mass m , initially at rest on the earth, is launched into a circular orbit at a height equal to the radius of the earth. The minimum energy required is

(A) $\frac{\sqrt{3}}{4} mgR$ (B) $\frac{1}{2} mgR$
(C) $\frac{1}{4} mgR$ (D) $\frac{3}{4} mgR$

GR0028

12. The figure shows the variation of energy with the orbit radius of a body in circular planetary motion. Find the correct statement about the curves A, B and C



- (A) A shows the kinetic energy, B the total energy and C the potential energy of the system.
 (B) C shows the total energy, B the kinetic energy and A the potential energy of the system.
 (C) C and A are kinetic and potential energies respectively and B is the total energy of the system.
 (D) A and B are kinetic and potential energies and C is the total energy of the system.

GR0029

13. A satellite of mass $5M$ orbits the earth in a circular orbit. At one point in its orbit, the satellite explodes into two pieces, one of mass M and the other of mass $4M$. After the explosion the mass M ends up travelling in the same circular orbit, but in opposite direction. After explosion the mass $4M$ is :-
 (A) In a circular orbit
 (B) unbound
 (C) elliptical orbit
 (D) data is insufficient to determine the nature of the orbit.

GR0030

14. A satellite can be in a geostationary orbit around earth at a distance r from the centre. If the angular velocity of earth about its axis doubles, a satellite can now be in a geostationary orbit around earth if its distance from the centre is :-

- (A) $\frac{r}{2}$ (B) $\frac{r}{2\sqrt{2}}$ (C) $\frac{r}{(4)^{1/3}}$ (D) $\frac{r}{(2)^{1/3}}$

GR0031

15. An earth satellite is moved from one stable circular orbit to another larger and stable circular orbit. The following quantities increase for the satellite as a result of this change:-
 (A) gravitational potential energy (B) angular velocity
 (C) linear orbital velocity (D) centripetal acceleration

GR0032

16. Satellites A and B are orbiting around the earth in orbits of ratio R and $4R$ respectively. The ratio of their areal velocities is :
 (A) $1 : 2$ (B) $1 : 4$ (C) $1 : 8$ (D) $1 : 16$

GR0033

17. The fractional change in the value of free-fall acceleration g for a particle when it is lifted from the surface to an elevation h ($h \ll R$) is

- (A) $\frac{h}{R}$ (B) $\frac{2h}{R}$ (C) $-\frac{2h}{R}$ (D) $-\frac{h}{R}$

GR0034

18. If suddenly the gravitational force of attraction between earth and a satellite revolving around it becomes zero, then the satellite will- [AIEEE-2002]

(A) continue to move in its orbit with same velocity
(B) move tangentially to the original orbit with same velocity
(C) become stationary in its orbit
(D) move towards the earth

GR0035

19. The time period of a satellite of earth is 5 hours. If the separation between the centre of earth and the satellite is increased to 4 times the previous value, the new time period will become- [AIEEE-2003]

(A) 10 h (B) 80 h (C) 40 h (D) 20 h

GR0036

20. A communications Earth satellite

(A) goes round the earth from east to west
(B) can be in the equatorial plane only
(C) can be vertically above any place on the earth
(D) goes round the earth from west to east

GR0037

21. If a satellite orbits as close to the earth's surface as possible,

(A) its speed is maximum
(B) time period of its rotation is minimum
(C) the total energy of the 'earth plus satellite' system is minimum
(D) the total energy of the 'earth plus satellite' system is maximum

GR0038

22. For a satellite to orbit around the earth, which of the following must be true?

(A) It must be above the equator at some time
(B) It cannot pass over the poles at any time
(C) Its height above the surface cannot exceed 36,000 km
(D) Its period of rotation must be $> 2\pi\sqrt{R/g}$ where R is radius of earth

GR0039

23. In elliptical orbit of a planet, as the planet moves from apogee position to perigee position,

Column-I

(A) Speed of planet
(B) Distance of planet from centre of Sun
(C) Potential energy
(D) Angular momentum about centre of Sun

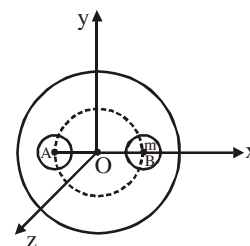
Column-II

(P) Remains same
(Q) Decreases
(R) Increases
(S) Can not say

GR0040

EXERCISE (O-2)

1. A solid sphere of uniform density and radius 4 units is located with its centre at the origin O of coordinates. Two spheres of equal radii 1 unit, with their centres at A (-2, 0, 0) and B (2, 0, 0) respectively, are taken out of the solid leaving behind spherical cavities as shown in figure. Then
- (A) The gravitational field due to this object at the origin is zero
 (B) The gravitational field at the point B (2, 0, 0) is zero
 (C) The gravitational potential is the same at all points of circle $y^2 + z^2 = 36$
 (D) The gravitational potential is the same at all points on the circle $y^2 + z^2 = 4$



GR0041

2. A planet of mass m is in an elliptical orbit about the sun ($m \ll M_{\text{sun}}$) with an orbital period T . If A be the area of orbit, then its angular momentum would be :

- (A) $\frac{2mA}{T}$ (B) mAT (C) $\frac{mA}{2T}$ (D) $2mAT$

GR0042

3. A planet revolves about the sun in elliptical orbit. The areal velocity $\left(\frac{dA}{dt}\right)$ of the planet is $4.0 \times 10^{16} \text{ m}^2/\text{s}$. The least distance between planet and the sun is $2 \times 10^{12} \text{ m}$. Then the maximum speed of the planet in km/s is :

- (A) 10 (B) 20 (C) 40 (D) None of these

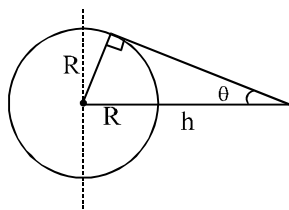
GR0043

4. The Sun travels in approximately circular orbit of radius R around the center of the galaxy and completes one revolution in time T . The Earth also revolves around the Sun in time t . Assume orbit of the Earth to be a circle of radius r ($r \ll R$) and whole mass of the galaxy centered on its center. By using only these given informations, find an expression for the ratio of the mass of the galaxy to that of the Sun.

- (A) $\left(\frac{R}{r}\right)^3 \left(\frac{t}{T}\right)^2$ (B) $\left(\frac{R}{r}\right)^3 \left(\frac{T}{t}\right)^2$ (C) $\left(\frac{R}{r}\right)^2 \left(\frac{t}{T}\right)^3$ (D) $\left(\frac{R}{r}\right)^2 \left(\frac{T}{t}\right)^3$

GR0044

5. A geostationary satellite is at a height h above the surface of earth. If earth radius is R



- (A) The minimum colatitude on earth upto which the satellite can be used for communication is $\sin^{-1}(R/R+h)$.
 (B) The maximum latitudes on earth upto which the satellite can be used for communication is $\cos^{-1}(R/R+h)$.
 (C) The area on earth escaped from this satellite is given as $2\pi R^2 (1 + \sin\theta)$
 (D) The area on earth escaped from this satellite is given as $2\pi R^2 (1 + \cos\theta)$

GR0045

6. When a satellite in a circular orbit around the earth enters the atmospheric region, it encounters small air resistance to its motion. Then
- (A) its kinetic energy increases
 - (B) its kinetic energy decreases
 - (C) its angular momentum about the earth decreases
 - (D) its period of revolution around the earth increases

GR0046

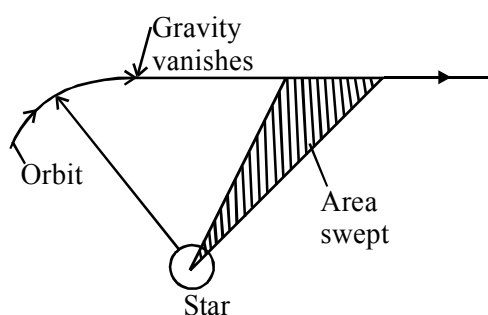
7. Two satellites s_1 & s_2 of equal masses revolve in the same sense around a heavy planet in coplanar circular orbit of radii R & $4R$
- (A) the ratio of period of revolution of s_1 & s_2 is $1 : 8$.
 - (B) their velocities are in the ratio $2 : 1$
 - (C) their angular momentum about the planet are in the ratio $2 : 1$
 - (D) the ratio of angular velocities of s_2 w.r.t. s_1 when all three are in the same line is $9 : 5$.

GR0047

8. A double star is a system of two stars of masses m and $2m$, rotating about their centre of mass only under their mutual gravitational attraction. If r is the separation between these two stars then their time period of rotation about their centre of mass will be proportional to :
- (A) $r^{3/2}$
 - (B) r
 - (C) $m^{1/2}$
 - (D) $m^{-1/2}$

GR0048

9. A planet is orbiting a star when for no apparent reason the star's gravity suddenly vanishes. After which planet moves in a straight line. Mark the **CORRECT** statement(s) :

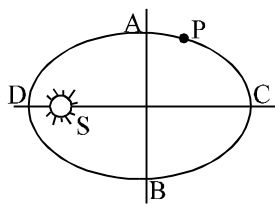


- (A) Newton's first law is obeyed on planet after gravity vanishes
- (B) Kepler's law of areas is obeyed only till the planet is in gravity of star
- (C) Kepler's law of areas is obeyed even after gravity vanishes
- (D) Angular momentum of planet about centre of star is conserved through out its motion

GR0049

Paragraph for Question No. 10 and 11

Figure shows the orbit of a planet P around the sun S. AB and CD are the minor and major axes of the ellipse.



10. If t_1 is the time taken by the planet to travel along ACB and t_2 the time along BDA, then
 (A) $t_1 = t_2$ (B) $t_1 > t_2$
 (C) $t_1 < t_2$ (D) nothing can be concluded

GR0050

11. If U is the potential energy and K kinetic energy then $|U| > |K|$ at
 (A) Only D (B) Only C (C) both D & C (D) neither D nor C

GR0050

EXERCISE (JM)

1. Two bodies of masses m and $4m$ are placed at a distance r . The gravitational potential at a point on the line joining them where the gravitational field is zero is :- [AIEEE - 2011]

(1) $-\frac{6Gm}{r}$ (2) $-\frac{9Gm}{r}$ (3) zero (4) $-\frac{4Gm}{r}$

GR0051

2. Two particles of equal mass ' m ' go around a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle with respect to their centre of mass is:- [AIEEE-2011]

(1) $\sqrt{\frac{Gm}{R}}$ (2) $\sqrt{\frac{Gm}{4R}}$ (3) $\sqrt{\frac{Gm}{3R}}$ (4) $\sqrt{\frac{Gm}{2R}}$

GR0052

3. The mass of a spaceship is 1000 kg. It is to be launched from the earth's surface out into free space. The value of ' g ' and ' R ' (radius of earth) are 10 m/s^2 and 6400 km respectively. The required energy for this work will be :- [AIEEE-2012]

(1) 6.4×10^{10} Joules (2) 6.4×10^{11} Joules (3) 6.4×10^8 Joules (4) 6.4×10^9 Joules

GR0053

4. What is the minimum energy required to launch a satellite of mass m from the surface of a planet of mass M and radius R in a circular orbit at an altitude of $2R$? [JEE-Main 2013]

(1) $\frac{5GmM}{6R}$ (2) $\frac{2GmM}{3R}$ (3) $\frac{GmM}{2R}$ (4) $\frac{GmM}{3R}$

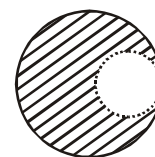
GR0054

5. Four particles, each of mass M and equidistant from each other, move along a circle of radius R under the action of their mutual gravitational attraction. The speed of each particle is : [JEE-Main 2014]

(1) $\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$ (2) $\frac{1}{2}\sqrt{\frac{GM}{R}(1+2\sqrt{2})}$ (3) $\sqrt{\frac{GM}{R}}$ (4) $\sqrt{2\sqrt{2}\frac{GM}{R}}$

GR0055

6. From a solid sphere of mass M and radius R , a spherical portion of radius $\frac{R}{2}$ is removed, as shown in the figure. Taking gravitational potential $V = 0$ at $r = \infty$, the potential at the centre of the cavity thus formed is : ($G = \text{gravitational constant}$)



(1) $\frac{-2GM}{3R}$ (2) $\frac{-2GM}{R}$ (3) $\frac{-GM}{2R}$ (4) $\frac{-GM}{R}$

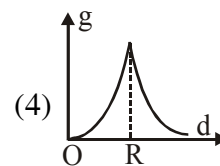
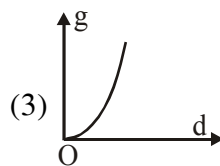
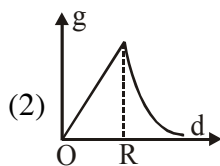
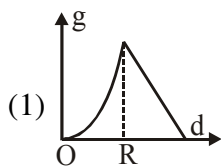
GR0056

7. A satellite is revolving in a circular orbit at a height 'h' from the earth's surface (radius of earth R ; $h \ll R$). The minimum increase in its orbital velocity required, so that the satellite could escape from the earth's gravitational field, is close to : (Neglect the effect of atmosphere). [JEE-Main 2016]

(1) $\sqrt{gR}(\sqrt{2} - 1)$ (2) $\sqrt{2gR}$ (3) $\sqrt{gR}$ (4) $\sqrt{gR/2}$

GR0057

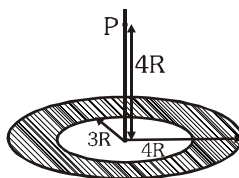
8. The variation of acceleration due to gravity g with distance d from centre of the earth is best represented by (R = Earth's radius) :- [JEE-Main 2017]



GR0058

EXERCISE (JA)

- Gravitational acceleration on the surface of a planet is $\frac{\sqrt{6}}{11}g$, where g is the gravitational acceleration on the surface of the earth. The average mass density of the planet is $\frac{2}{3}$ times that of the Earth. If the escape speed on the surface of the earth is taken to be 11 kms^{-1} , the escape speed on the surface of the planet in kms^{-1} will be
[IIT-JEE 2010]
GR0059
- A binary star consists of two stars A (mass $2.2 M_s$) and B (mass $11 M_s$), where M_s is the mass of the sun. They are separated by distance d and are rotating about their centre of mass, which is stationary. The ratio of the total angular momentum of the binary star to the angular momentum of star B about the centre of mass is :-
[IIT-JEE 2010]
GR0060
- A thin uniform annular disc (see figure) of mass M has outer radius $4R$ and inner radius $3R$. The work required to take a unit mass from point P on its axis to infinity is
[IIT-JEE 2010]



- (A) $\frac{2GM}{7R}(4\sqrt{2}-5)$ (B) $-\frac{2GM}{7R}(4\sqrt{2}-5)$ (C) $\frac{GM}{4R}$ (D) $\frac{2GM}{5R}(\sqrt{2}-1)$

GR0061

- A satellite is moving with a constant speed V in a circular orbit about the earth. An object of mass ' m ' is ejected from the satellite such that it just escapes from the gravitational pull of the earth. At the time of its ejection, the kinetic energy of the object is :-
[IIT-JEE 2011]

- (A) $\frac{1}{2}mV^2$ (B) mV^2 (C) $\frac{3}{2}mV^2$ (D) $2mV^2$

GR0062

5. Two spherical planets P and Q have the same uniform density ρ , masses M_P and M_Q , and surface areas A and 4A, respectively. A spherical planet R also has uniform density ρ and its mass is $(M_P + M_Q)$. The escape velocities from the planets P, Q and R, are V_P , V_Q and V_R , respectively. Then

[IIT-JEE 2012]

- (A) $V_Q > V_R > V_P$ (B) $V_R > V_Q > V_P$ (C) $V_R/V_P = 3$ (D) $V_P/V_Q = 1/2$

GR0063

6. A planet of radius $R = \frac{1}{10} \times$ (radius of Earth) has the same mass density as Earth. Scientists dig a well

of depth $\frac{R}{5}$ on it and lower a wire of the same length and of linear mass density $10^{-3} \text{ kg m}^{-1}$ into it. If

the wire is not touching anywhere, the force applied at the top of the wire by a person holding it in place is (take the radius of Earth $= 6 \times 10^6 \text{ m}$ and the acceleration due to gravity on Earth is 10 ms^{-2})

[JEE-Advance 2014]

- (A) 96 N (B) 108 N (C) 120 N (D) 150 N

GR0064

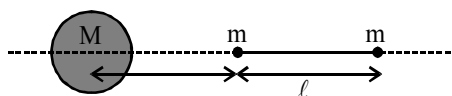
7. A bullet is fired vertically upwards with velocity v from the surface of a spherical planet. When it reaches its maximum height, its acceleration due to the planet's gravity is $1/4^{\text{th}}$ of its value at the surface of the planet. If the escape velocity from the planet is $v_{\text{esc}} = v\sqrt{N}$, then the value of N is (ignore energy loss due to atmosphere)

[JEE-Advance 2015]

GR0065

8. A large spherical mass M is fixed at one position and two identical point masses m are kept on a line passing through the centre of M (see figure). The point masses are connected by a rigid massless rod of length ℓ and this assembly is free to move along the line connecting them. All three masses interact only through their mutual gravitational interaction. When the point mass nearer to M is at a distance $r = 3\ell$ from M, the tension in the rod is zero for $m = k\left(\frac{M}{288}\right)$. The value of k is :

[JEE-Advance 2015]



GR0066

9. A spherical body of radius R consists of a fluid of constant density and is in equilibrium under its own gravity. If $P(r)$ is the pressure at $r (r < R)$, then the correct option(s) is(are) :- [JEE-Advance 2015]

- (A) $P(r = 0) = 0$ (B) $\frac{P(r = 3R/4)}{P(r = 2R/3)} = \frac{63}{80}$
 (C) $\frac{P(r = 3R/5)}{P(r = 2R/5)} = \frac{16}{21}$ (D) $\frac{P(r = R/2)}{P(r = R/3)} = \frac{20}{27}$

GR0067

10. A rocket is launched normal to the surface of the Earth, away from the Sun, along the line joining the sun and the Earth. The Sun is 3×10^5 times heavier than the Earth and is at a distance 2.5×10^4 times larger than the radius of the Earth. The escape velocity from Earth's gravitational field is $v_e = 11.2 \text{ km s}^{-1}$. The minimum initial velocity (v_s) required for the rocket to be able to leave the Sun-Earth system is closest to (Ignore the rotation and revolution of the Earth and the presence of any other planet) [JEE-Advance 2017]

(A) $v_s = 22 \text{ km s}^{-1}$ (B) $v_s = 72 \text{ km s}^{-1}$ (C) $v_s = 42 \text{ km s}^{-1}$ (D) $v_s = 62 \text{ km s}^{-1}$

GR0068

11. A planet of mass M , has two natural satellites with masses m_1 and m_2 . The radii of their circular orbits are R_1 and R_2 respectively. Ignore the gravitational force between the satellites. Define v_1 , L_1 , K_1 and T_1 to be, respectively, the orbital speed, angular momentum, kinetic energy and time period of revolution of satellite 1 ; and v_2 , L_2 , K_2 and T_2 to be the corresponding quantities of satellite 2. Given $m_1/m_2 = 2$ and $R_1/R_2 = 1/4$, match the ratios in List-I to the numbers in List-II.

[JEE-Advance 2018]

List-I	List-II
P. $\frac{v_1}{v_2}$	1. $\frac{1}{8}$
Q. $\frac{L_1}{L_2}$	2. 1
R. $\frac{K_1}{K_2}$	3. 2
S. $\frac{T_1}{T_2}$	4. 8
(A) $P \rightarrow 4$; $Q \rightarrow 2$; $R \rightarrow 1$; $S \rightarrow 3$	(B) $P \rightarrow 3$; $Q \rightarrow 2$; $R \rightarrow 4$; $S \rightarrow 1$
(C) $P \rightarrow 2$; $Q \rightarrow 3$; $R \rightarrow 1$; $S \rightarrow 4$	(D) $P \rightarrow 2$; $Q \rightarrow 3$; $R \rightarrow 4$; $S \rightarrow 1$

GR0069

12. Consider a spherical gaseous cloud of mass density $\rho(r)$ in free space where r is the radial distance from its center. The gaseous cloud is made of particles of equal mass m moving in circular orbits about the common center with the same kinetic energy K . The force acting on the particles is their mutual gravitational force. If $\rho(r)$ is constant in time, the particle number density $n(r) = \rho(r)/m$ is : [G is universal gravitational constant] [JEE-Advance 2019]

(A) $\frac{K}{\pi r^2 m^2 G}$ (B) $\frac{K}{6\pi r^2 m^2 G}$ (C) $\frac{3K}{\pi r^2 m^2 G}$ (D) $\frac{K}{2\pi r^2 m^2 G}$

GR0070

ANSWER KEY

EXERCISE (S-1)

1. Ans. $\frac{R_e k^2}{1-k^2}$ 2. Ans. $h = \frac{\sqrt{5}-1}{2} R$ 3. Ans. (i) $\frac{4}{3} \sqrt{\frac{Gm}{R}}$, (ii) $\frac{2}{3} \sqrt{\frac{2Gm}{5R}}$
4. Ans. 1.6 hours if it is rotating from west to east, 24/17 hours if it is rotating from east to west
5. Ans. $1 \times 10^5 J$ 6. Ans. $t = \frac{GMm}{2C} \left(\frac{1}{R_e} - \frac{1}{r} \right)$ 7. Ans. (a) $T = \frac{2\pi d^{3/2}}{\sqrt{3Gm}}$, (b) 2, (c) 2
8. Ans. $\vec{g} = +\frac{\pi G \rho_0 R^3}{6} \left[\frac{1}{(x-(R/2))^2} - \frac{8}{x^2} \right] \hat{i}$, $\vec{g} = -\frac{2\pi G \rho_0 R}{3} \hat{i}$

EXERCISE (S-2)

1. Ans. $2R, 3R[3-\sqrt{3}]$ 2. Ans. (a) $h = \left(\frac{\sqrt{7}}{2} + 1 \right) R$, (b) $1.125 R$ 3. Ans. (a) $\sqrt{\frac{GM}{r}}$; (b) $r\sqrt{2}$; (c) $\sqrt{\frac{2GM}{r}}$
4. Ans. $T = 3 \times 10^{-2} N$ 5. Ans. $\frac{1}{4} K^2 G \pi (R^4 - r^4)$ 6. Ans. $7/3$ 7. Ans. 3
8. Ans. (a) $\frac{2Gm^2}{3R}$ (b) $4R$ 9. Ans. $\left(1 - \frac{\sqrt{x^2 - R^2}}{x} \right) 4\pi R^2$

EXERCISE (O-1)

1. Ans. (D) 2. Ans. (D) 3. Ans. (B) 4. Ans. (A)
5. Ans. (B) 6. Ans. (B) 7. Ans. (A) 8. Ans. (A)
9. Ans. (B) 10. Ans. (B) 11. Ans. (D) 12. Ans. (D)
13. Ans. (B) 14. Ans. (C) 15. Ans. (A) 16. Ans. (A)
17. Ans. (C) 18. Ans. (B) 19. Ans. (C) 20. Ans. (B,D)
21. Ans. (A,B,C) 22. Ans. (A, D) 23. Ans. (A)-R, (B)-Q, (C)-Q (D)-P

EXERCISE O-2

1. Ans. (A,C,D) 2. Ans. (A) 3. Ans. (C) 4. Ans. (A)
5. Ans. (A,B,C) 6. Ans. (A,C) 7. Ans. (A,B,D) 8. Ans. (A,D)
9. Ans. (A,C,D) 10. Ans. (B) 11. Ans. (C)

EXERCISE - (JM)

1. Ans. (2) 2. Ans. (2) 3. Ans. (1) 4. Ans. (1)
5. Ans. (2) 6. Ans. (4) 7. Ans. (1) 8. Ans. (2)

EXERCISE-(JA)

1. Ans. 3 2. Ans. 6 3. Ans. (A) 4. Ans. (B)
5. Ans. (B,D) 6. Ans. (B) 7. Ans. 2 8. Ans. 7
9. Ans. (B,C) 10. Ans. (C) 11. Ans. (B) 12. Ans. (D)

ELECTROSTATICS

KEY CONCEPT

ELECTRIC CHARGE

Charge is the property associated with matter due to which it produces and experiences electrical and magnetic effects. The excess or deficiency of electrons in a body gives the concept of charge.

SI unit of charge : ampere $\times$ second i.e. Coulomb Dimension : $[A T]$

Practical units of charge are ampere $\times$ hour (=3600 C) and faraday (= 96500 C)

- Millikan calculated quanta of charge by 'Highest common factor' (H.C.F.) method and it is equal to charge of electron.
- $1 \text{ C} = 3 \times 10^9 \text{ state coulomb}$, $1 \text{ absolute - coulomb} = 10 \text{ C}$, $1 \text{ Faraday} = 96500 \text{ C}$.

SPECIFIC PROPERTIES OF CHARGE

- **Charge is a scalar quantity :** It represents excess or deficiency of electrons.
- **Charge is transferable :** If a charged body is put in contact with an another body, then charge can be transferred to another body.
- **Charge is always associated with mass**

Charge cannot exist without mass though mass can exist without charge.

- So the presence of charge itself is a convincing proof of existence of mass.
- In charging, the mass of a body changes.
- When body is given positive charge, its mass decreases.
- When body is given negative charge, its mass increases.
- **Charge is quantized**

The quantization of electric charge is the property by virtue of which all free charges are integral multiple of a basic unit of charge represented by e . Thus charge q of a body is always given by

$$q = ne$$

$n =$ positive integer or negative integer

The quantum of charge is the charge that an electron or proton carries.

Note : Charge on a proton = (-) charge on an electron = $1.6 \times 10^{-19} \text{ C}$

- **Charge is conserved**

In an isolated system, total charge does not change with time, though individual charge may change i.e. charge can neither be created nor destroyed. Conservation of charge is also found to hold good in all types of reactions either chemical (atomic) or nuclear. No exceptions to the rule have ever been found.

- **Charge is invariant**

Charge is independent of frame of reference. i.e. charge on a body does not change whatever be its speed.

- **Attraction – Repulsion**

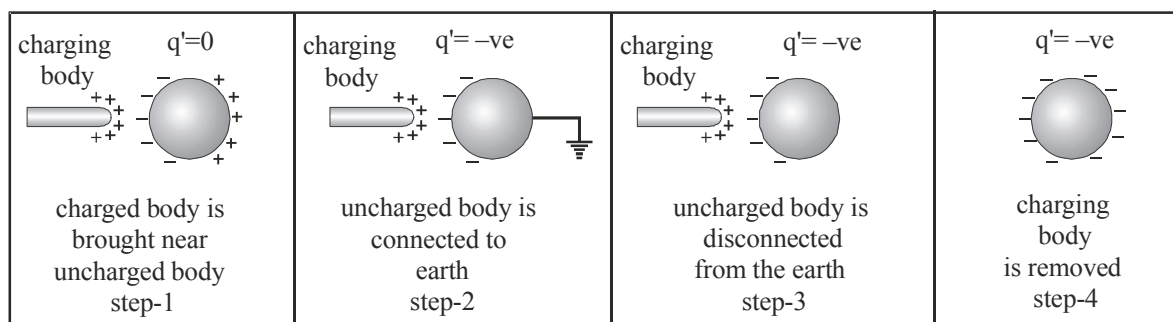
Similar charges repel each other while dissimilar attract

METHODS OF CHARGING

- **Friction:** If we rub one body with other body, electrons are transferred from one body to the other.
- **Electrostatic induction**

If a charged body is brought near a metallic neutral body, the charged body will attract opposite charge and repel similar charge present in the neutral body. As a result of this one side of the neutral body becomes negative while the other positive, this process is called 'electrostatic induction'.

- Charging a body by induction (in four successive steps)



Some important facts associated with induction-

- Inducing body neither gains nor loses charge
- The nature of induced charge is always opposite to that of inducing charge
- Induction takes place only in bodies (either conducting or non conducting) and not in particles.

- **Conduction**

The process of transfer of charge by contact of two bodies is known as conduction. If a charged body is put in contact with uncharged body, the uncharged body becomes charged due to transfer of electrons from one body to the other.

- The charged body loses some of its charge (which is equal to the charge gained by the uncharged body)
- The charge gained by the uncharged body is always lesser than initial charge present on the charged body.

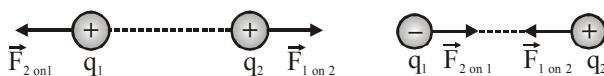
COULOMB'S LAW

The electrostatic force of interaction between two static point electric charges is directly proportional to the product of the charges, inversely proportional to the square of the distance between them and acts along the straight line joining the two charges.

Force of electrostatic interaction depends on the nature of medium between the charges. If two points charges q_1 and q_2 separated by a distance r . Let F be the electrostatic force between these two charges. According to Coulomb's law.

$$F \propto q_1 q_2 \text{ and } F \propto \frac{1}{r^2}$$

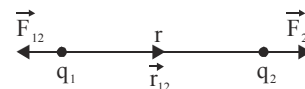
$$F_e = \frac{kq_1 q_2}{r^2}$$



where $\left[k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \frac{\text{Nm}^2}{\text{C}^2} \right]$ k = coulomb's constant or electrostatic force constant

Coulomb's law in vector form

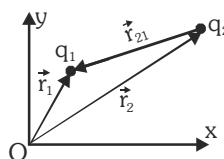
$$\vec{F}_{12} = \text{force on } q_1 \text{ due to } q_2 = \frac{kq_1q_2}{r^2} \hat{r}_{21}$$



$$\vec{F}_{21} = \frac{kq_1q_2}{r^2} \hat{r}_{12} \text{ (here } \hat{r}_{12} \text{ is unit vector from } q_1 \text{ to } q_2 \text{)}$$

Coulomb's law in terms of position vector

$$\vec{F}_{12} = \frac{kq_1q_2}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_1 - \vec{r}_2)$$



Principle of superposition

When a number of charges are interacting, the total force on a given charge is vector sum of the forces exerted on it by all other charges individually

$$F = \frac{kq_0q_1}{r_1^2} + \frac{kq_0q_2}{r_2^2} + \dots + \frac{kq_0q_i}{r_i^2} + \dots + \frac{kq_0q_n}{r_n^2}$$

$$\text{in vector form } \vec{F} = kq_0 \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i$$

Some important points regarding Coulomb's law and electric force

- The law is based on physical observations and is not logically derivable from any other concept. Experiments till today reveal its universal nature.
- The force is a two body interaction, i.e., electrical force between two point charges is independent of presence or absence of other charges and so the principle of superposition is valid, i.e., force on a charged particle due to number of point charges is the resultant of forces due to individual point charges, i.e., $\vec{F}_1 = \vec{F}_{12} + \vec{F}_{13} + \dots$
- The net Coulomb's force on two charged particles in free space and in a medium filled upto infinity are

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1q_2}{r^2} \text{ and } F' = \frac{1}{4\pi\epsilon} \frac{q_1q_2}{r^2}. \text{ So } \frac{F}{F'} = \frac{\epsilon}{\epsilon_0} = K,$$

- Dielectric constant (K) of a medium is numerically equal to the ratio of the force on two point charges in free space to that in the medium filled upto infinity.
- The law expresses the force between two point charges at rest. In applying it to the case of extended bodies of finite size care should be taken in assuming the whole charge of a body to be concentrated at its 'centre' as this is true only for spherical charged body, that too for external point.

Although net electric force on both particles change in the presence of dielectric but force due to one charge particle on another charge particle does not depend on the medium between them.

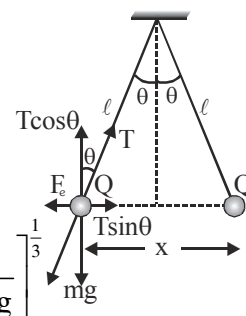
□ Equilibrium of suspended point charge system

For equilibrium position

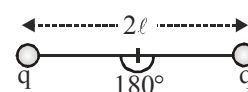
$$T \cos \theta = mg \text{ and } T \sin \theta = F_e = \frac{kQ^2}{x^2} \Rightarrow \tan \theta = \frac{F_e}{mg} = \frac{kQ^2}{x^2 mg}$$

- If θ is small then

$$\tan \theta \approx \sin \theta = \frac{x}{2\ell} \Rightarrow \frac{x}{2\ell} = \frac{kQ^2}{x^2 mg} \Rightarrow x^3 = \frac{2kQ^2 \ell}{mg} \Rightarrow x = \left[\frac{Q^2 \ell}{2\pi \epsilon_0 mg} \right]^{\frac{1}{3}}$$



- If whole set up is taken into an artificial satellite ($g_{\text{eff}} \approx 0$) $T = F_e = \frac{kq^2}{4\ell^2}$

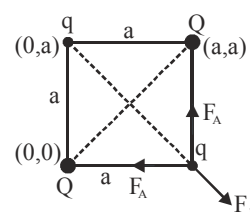


Ex. For the system shown in figure find Q for which resultant force on q is zero.

Sol. For force on q to be zero, charges q and Q must be of opposite of nature.

Net attraction force on q due to charges Q = Repulsion force on q due to q

$$\sqrt{2} F_A = F_R \Rightarrow \sqrt{2} \frac{kQq}{a^2} = \frac{kq^2}{(\sqrt{2}a)^2} \Rightarrow q = 2\sqrt{2} Q \text{ Hence } q = -2\sqrt{2} Q$$



Ex. Given a cube with point charges q on each of its vertices. Calculate the force exerted on any of the charges due to rest of the 7 charges.

Sol. The net force on particle A can be given by vector sum of force experienced by this particle due to all the other charges on vertices of the cube. For this we use vector form of coulomb's law

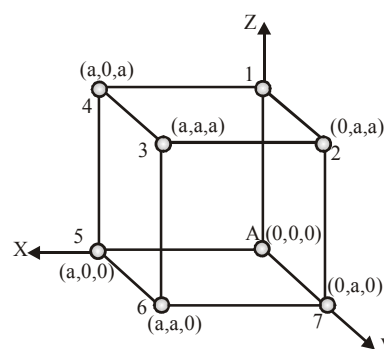
$$\vec{F} = \frac{kq_1 q_2}{|\vec{r}_1 - \vec{r}_2|^3} (\vec{r}_1 - \vec{r}_2)$$

From the figure the different forces acting on A are given as

$$\vec{F}_{A_1} = \frac{kq^2(-a\hat{k})}{a^3}$$

$$\vec{F}_{A_2} = \frac{kq^2(-a\hat{j} - a\hat{k})}{(\sqrt{2}a)^3}, \quad \vec{F}_{A_3} = \frac{kq^2(-a\hat{i} - a\hat{j} - a\hat{k})}{(\sqrt{3}a)^3},$$

$$\vec{F}_{A_4} = \frac{kq^2(-a\hat{j} - a\hat{k})}{(\sqrt{2}a)^3}, \quad \vec{F}_{A_5} = \frac{kq^2(-a\hat{j})}{a^3}, \quad \vec{F}_{A_6} = \frac{kq^2(-a\hat{i} - a\hat{j})}{(\sqrt{2}a)^3}, \quad \vec{F}_{A_7} = \frac{kq^2(-a\hat{j})}{a^3}$$



The net force experienced by A can be given as

$$\vec{F}_{\text{net}} = \vec{F}_{A_1} + \vec{F}_{A_2} + \vec{F}_{A_3} + \vec{F}_{A_4} + \vec{F}_{A_5} + \vec{F}_{A_6} + \vec{F}_{A_7} = \frac{-kq^2}{a^2} \left[\left(\frac{1}{3\sqrt{3}} + \frac{1}{\sqrt{2}} + 1 \right) (\hat{i} + \hat{j} + \hat{k}) \right]$$

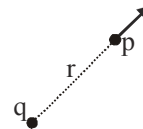
ELECTRIC FIELD

In order to explain ‘action at a distance’, i.e., ‘force without contact’ between charges it is assumed that a charge or charge distribution produces a field in space surrounding it. So the region surrounding a charge (or charge distribution) in which its electrical effects are perceptible is called the electric field of the given charge. Electric field at a point is characterized either by a vector function of position $\vec{E}$ called ‘electric intensity’ or by a scalar function of position V called ‘electric potential’. The electric field in a certain space is also visualized graphically in terms of ‘lines of force.’ So electric intensity, potential and lines of force are different ways of describing the same field.

Intensity of electric field due to point charge

Electric field intensity is defined as force on unit test charge.

$$\vec{E} = \lim_{q_0 \rightarrow 0} \frac{\vec{F}}{q_0} = \frac{kq}{r^2} \hat{r} = \frac{kq}{r^3} \vec{r}$$



Note : Test charge (q_0) is a fictitious charge that exerts no force on nearby charges but experiences forces due to them.

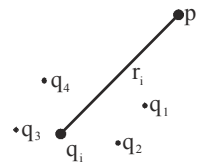
Unit : N/C, V/m **Dimensions :** $[MLT^{-3}A^{-1}]$

Due to discrete distribution of charge

Field produced by a charge distribution for discrete distribution:—

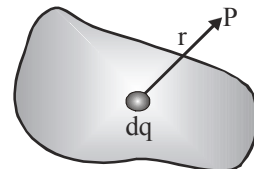
By principle of superposition intensity of electric field due to i^{th} charge $\vec{E}_{ip} = \frac{kq}{r_i^3} \vec{r}_i$

$$\therefore \text{Net electric field due to whole distribution of charge } \vec{E}_p = \sum_{i=1} \vec{E}_i$$



Continuous distribution of charge

Treating a small element as particle $\vec{E} = \frac{1}{4\pi\epsilon_0} \int \frac{dq}{r^3} \vec{r}$



Due to linear charge distribution $E = \int_{\ell} \frac{\lambda dl}{r^2}$ [λ = charge per unit length]

Due to surface charge distribution $E = k \int_s \frac{\sigma ds}{r^2}$ [σ = charge per unit area]

Due to volume charge distribution $E = k \int \frac{\rho dv}{r^2}$ [ρ = charge per unit volume]

Key points

- Charged particle in an electric field always experiences a force either it is at rest or in motion.
- In presence of a dielectric , electric field decreases and becomes $\frac{1}{\epsilon_r}$ times of its value in free space.
- Test charge is always a unit (+ ve) charge. $\vec{E} = \frac{\vec{F}_{\text{test}}}{\text{test charge}}$
- If identical charges are placed on each vertices of a regular polygon, then $\vec{E}$ at centre = zero.

Electric field strength at a general point due to a uniformly charged rod

As shown in figure, if P is any general point in the surrounding of rod, to find electric field strength at P, we consider an element on rod of length dx at a distance x from point O as shown in figure. Now if dE be the electric field at P due to the element, then

$$dE = \frac{k dq}{(x^2 + r^2)} \quad \text{Here } dq = \frac{Q}{L} dx$$

Electric field strength in x-direction due to dq at P is

$$dE_x = dE \sin \theta = \left[\frac{k dq}{(x^2 + r^2)} \right] \sin \theta = \frac{k Q \sin \theta}{L (x^2 + r^2)} dx$$

Here we have $x = r \tan \theta$ and $dx = r \sec^2 \theta d\theta$

$$\text{Thus } dE_x = \frac{kQ}{L} \frac{r \sec^2 \theta d\theta}{r^2 \sec^2 \theta} \sin \theta, \text{ Strength} = \frac{kQ}{Lr} \sin \theta d\theta$$

Net electric field strength due to dq at point P in x-direction is

$$E_x = \int dE_x = \frac{kQ}{Lr} \int_{-\theta_2}^{+\theta_1} \sin \theta d\theta = \frac{kQ}{Lr} [-\cos \theta]_{-\theta_2}^{+\theta_1} = \frac{kQ}{Lr} [\cos \theta_2 - \cos \theta_1]$$

Similarly, electric field strength at point P due to dq in y-direction is

$$dE_y = dE \cos \theta = \frac{kQ dx}{L(r^2 + x^2)} \times \cos \theta$$

Again we have $x = r \tan \theta$ and $dx = r \sec^2 \theta d\theta$.

$$\text{Thus we have } dE_y = \frac{kQ}{L} \cos \theta \times \frac{r \sec^2 \theta}{r^2 \sec^2 \theta} d\theta = \frac{kQ}{Lr} \cos \theta d\theta$$

Net electric field strength at P due to dq in y-direction is

$$E_y = \int dE_y = \frac{kQ}{Lr} \int_{-\theta_2}^{+\theta_1} \cos \theta d\theta = \frac{kQ}{Lr} [+ \sin \theta]_{-\theta_2}^{+\theta_1} = \frac{kQ}{Lr} [\sin \theta_1 + \sin \theta_2]$$

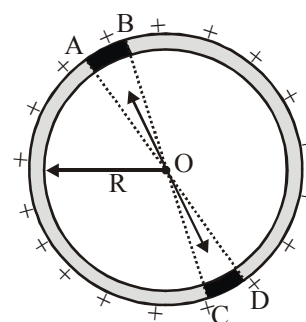
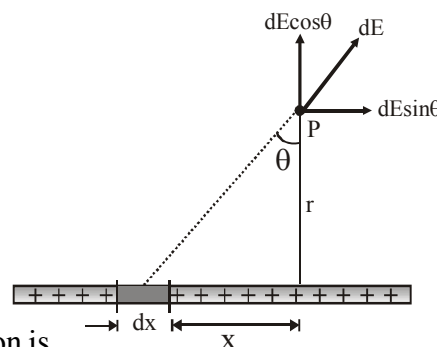
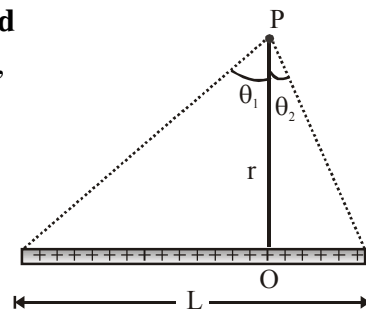
Thus electric field at a general point in the surrounding of a uniformly charged rod which subtend angles θ_1 and θ_2 at the two corners of rod can be given as

$$\text{in x-direction: } E_x = \frac{kQ}{Lr} (\cos \theta_2 - \cos \theta_1) \text{ and in y-direction } E_y = \frac{kQ}{Lr} (\sin \theta_1 + \sin \theta_2)$$

Electric field due to a uniformly charged ring

Case – I : At its centre

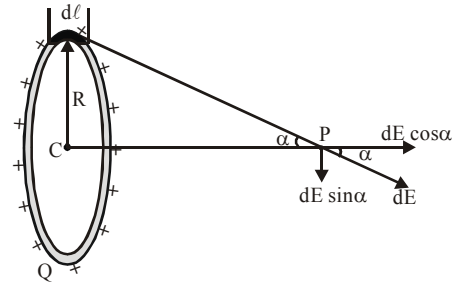
Here by symmetry we can say that electric field strength at centre due to every small segment on ring is cancelled by the electric field at centre due to the segment exactly opposite to it. The electric field strength at centre due to segment AB is cancelled by that due to segment CD. This net electric field strength at the centre of a uniformly charged ring is $E_0 = 0$



Case II : At a point on the axis of Ring

Here we'll find the electric field strength at point P due to the ring which is situated at a distance x from the ring centre. For this we consider a small section of length $d\ell$ on ring as shown. The charge on this elemental section is

$$dq = \frac{Q}{2\pi R} d\ell \quad [Q = \text{total charge of ring}]$$



Due to the element $d\ell$, electric field strength dE at point P can be given as $dE = \frac{Kdq}{(R^2 + x^2)}$

The component of this field strength $dE \sin \alpha$ which is normal to the axis of ring will be cancelled out due to the ring section opposite to $d\ell$. The component of electric field strength along the axis of ring $dE \cos \alpha$ due to all the sections will be added up. Hence total electric field strength at point P due to the ring is

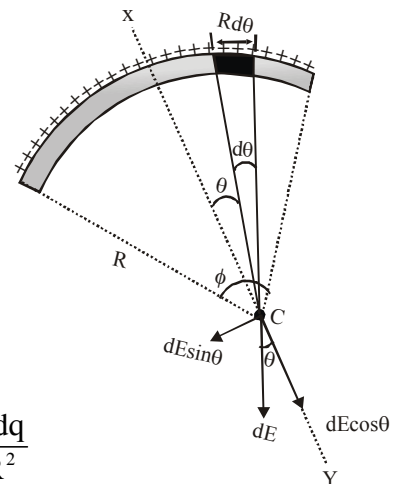
$$E_p = \int dE \cos \alpha = \int_0^{2\pi R} \frac{k dq}{(R^2 + x^2)} \times \frac{x}{\sqrt{R^2 + x^2}} = \int_0^{2\pi R} \frac{k Q x}{2\pi R (R^2 + x^2)} d\ell = \frac{k Q x}{2\pi R (R^2 + x^2)^{3/2}} \int_0^{2\pi R} d\ell$$

$$= \frac{k Q_x}{2\pi R (R^2 + x^2)^{3/2}} [2\pi R] = \frac{k Q x}{(R^2 + x^2)^{3/2}}$$

Electric field strength due to a charged circular arc at its centre :

Figure shows a circular arc of radius R which subtend an angle ϕ at its centre. To find electric field strength at C , we consider a polar segment on arc of angular width $d\theta$ at an angle θ from the angular bisector XY as shown. The length of elemental

segment is $Rd\theta$, the charge on this element $d\ell$ is $dq = \left(\frac{Q}{\phi}\right).d\theta$



Due to this dq, electric field at centre of arc C is given as $dE = \frac{k dq}{R^2}$

Now electric field component due to this segment $dE \sin \theta$ which is perpendicular to the angular bisector gets cancelled out in integration and net electric field at centre will be along angular bisector

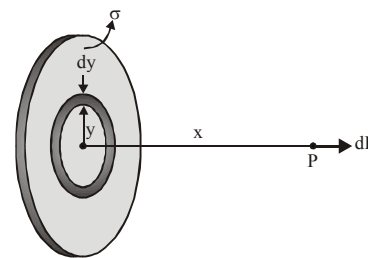
which can be calculated by integrating $dE \cos \theta$ within limits from $-\frac{\phi}{2}$ to $\frac{\phi}{2}$.

Hence net electric field strength at centre C is $E_c = \int dE \cos \theta$

$$= \int_{-\phi/2}^{+\phi/2} \frac{kQ}{\phi R^2} \cos \theta d\theta = \frac{kQ}{\phi R^2} \int_{-\phi/2}^{+\phi/2} \cos \theta d\theta = \frac{kQ}{\phi R^2} [\sin \theta]_{-\phi/2}^{+\phi/2} = \frac{kQ}{\phi R^2} \left[\sin \frac{\phi}{2} + \sin \frac{\phi}{2} \right] = \frac{2kQ \sin \left(\frac{\phi}{2} \right)}{\phi R^2}$$

Electric field strength due to a uniformly surface charged disc :

If there is a disc of radius R , charged on its surface with surface charge density σ , we wish to find electric field strength due to this disc at a distance x from the centre of disc on its axis at point P shown in figure.



To find electric field at point P due to this disc, we consider an elemental ring of radius y and width dy in the disc as shown in figure. The charge on this elemental ring $dq = \sigma \cdot 2\pi y dy$ [Area of elemental ring $ds = 2\pi y dy$]

Now we know that electric field strength due to a ring of radius R , charge Q , at a distance x from its

centre on its axis can be given as
$$E = \frac{kQx}{(x^2 + R^2)^{3/2}}$$

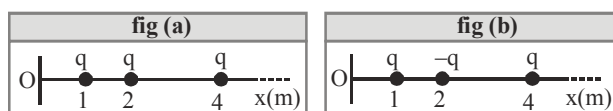
Here due to the elemental ring electric field strength dE at point P can be given as

$$dE = \frac{k dq x}{(x^2 + y^2)^{3/2}} = \frac{k \sigma 2\pi y dy x}{(x^2 + y^2)^{3/2}}$$

Net electric field at point P due to this disc is given by integrating above expression from 0 to R as

$$E = \int dE = \int_0^R \frac{k \sigma 2\pi x y dy}{(x^2 + y^2)^{3/2}} = k \sigma \pi x \int_0^R \frac{2y dy}{(x^2 + y^2)^{3/2}} = 2k \sigma \pi x \left[-\frac{1}{\sqrt{x^2 + y^2}} \right]_0^R = \frac{\sigma}{2\epsilon_0} \left[1 - \frac{x}{\sqrt{x^2 + R^2}} \right]$$

Ex. Calculate the electric field at origin due to infinite number of charges as shown in figures below.



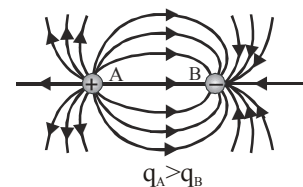
Sol. (a) $E_0 = kq \left[\frac{1}{1} + \frac{1}{4} + \frac{1}{16} + \dots \right] = \frac{kq \cdot 1}{(1 - 1/4)} = \frac{4kq}{3} \quad [\because S_\infty = \frac{a}{1-r}, a = 1 \text{ and } r = \frac{1}{4}]$

(b) $E_0 = kq \left[\frac{1}{1} - \frac{1}{4} + \frac{1}{16} - \dots \right] = \frac{kq \cdot 1}{(1 - (-1/4))} = \frac{4kq}{5}$

ELECTRIC LINES OF FORCE

Electric lines of electrostatic field have following properties

- Imaginary
- Can never cross each other
- Can never be closed loops
- The number of lines originating or terminating on a charge is proportional to the magnitude of charge. In rationalised MKS system ($1/\epsilon_0$) electric lines are associated with unit charge, so if a body encloses a charge q , total lines of force associated with it (called flux) will be q/ϵ_0 .
- If there is no electric field there will be no lines of force.
- Lines of force per unit area normal to the area at a point represents magnitude of intensity, crowded lines represent strong field while distant lines weak field.



(vii) Tangent to the line of force at a point in an electric field gives the direction of intensity. So a positive charge free to move follow the line of force.

- Lines of force starts from (+ve) charge and ends on (–ve) charge.

Electric flux (ϕ)

The word "flux" comes from a Latin word meaning "to flow" and you can consider the flux of a vector field to be a measure of the flow through an imaginary fixed element of surface in the field.

Electric flux is defined as $\phi_E = \int \vec{E} \cdot d\vec{A}$

This surface integral indicates that the surface in question is to be divided into infinitesimal elements of area $d\vec{A}$ and the scalar quantity $\vec{E} \cdot d\vec{A}$ is to be evaluated for each element and summed over the entire surface.

Electric flux :

(i) It is a scalar quantity

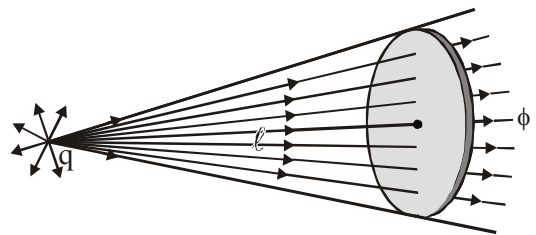
(ii) **Units** (V–m) and N – m²/C

Dimensions : [ML³T⁻³A⁻¹]

(iii) The value of ϕ does not depend upon the distribution of charges and the distance between them inside the closed surface.

Electric Flux through a circular Disc :

Figure shows a point charge q placed at a distance ℓ from a disc of radius R . Here we wish to find the electric flux through the disc surface due to the point charge q . We know a point charge q originates electric flux in radially outward direction. The flux is originated in cone shown in figure passes through the disc surface.



To calculate this flux, we consider on elemental ring an disc surface of radius x and width dx as shown. Area of this ring (strip) is $dS = 2\pi x \, dx$. The electric field due to q at this elemental ring is

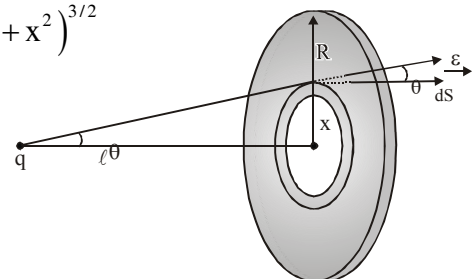
$$\text{given as } E = \frac{kq}{(x^2 + \ell^2)}$$

If $d\phi$ is the flux passing through this elemental ring, then

$$d\phi = E dS \cos \theta = \frac{kq}{(x^2 + \ell^2)} \times 2\pi x \, dx \times \frac{\ell}{\sqrt{x^2 + \ell^2}} = \frac{2\pi kq \ell x \, dx}{(\ell^2 + x^2)^{3/2}}$$

$$\phi = \int d\phi = \int_0^R \frac{q\ell}{2\epsilon_0} \frac{x \, dx}{(\ell^2 + x^2)^{3/2}} = \frac{q\ell}{2\epsilon_0} \int_0^R \frac{x \, dx}{(\ell^2 + x^2)^{3/2}}$$

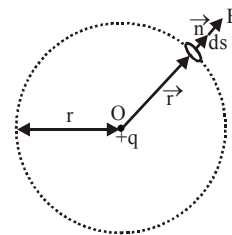
$$= \frac{q\ell}{2\epsilon_0} \left[-\frac{1}{\sqrt{\ell^2 + x^2}} \right]_0^R = \frac{q\ell}{2\epsilon_0} \left[\frac{1}{\ell} - \frac{1}{\sqrt{\ell^2 + R^2}} \right]$$



The above result can be obtained in a much simpler way by using the concept of solid angle and Gauss's law.

GAUSS'S LAW

It relates with the total flux of an electric field through a closed surface to the net charge enclosed by that surface and according to it, the total flux linked with a closed surface is $(1/\epsilon_0)$ times the charge enclosed by the closed



surface i.e., $\int_s \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$

Regarding Gauss's law it is worth noting that :

- (1) Gauss's law and Coulomb's law are equivalent, i.e., if we assume Coulomb's law we can prove Gauss's law and vice-versa. To prove Gauss's law from Coulomb's law consider a hypothetical spherical surface [called Gaussian-surface] of radius r with point charge q at its centre as shown in

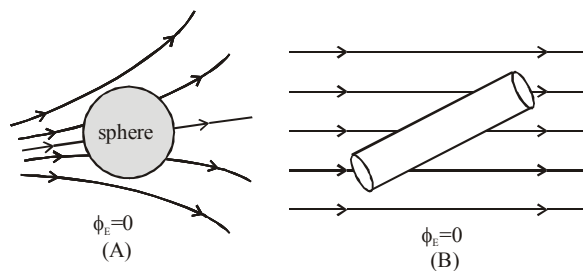
figure. By Coulomb's law intensity at a point P on the surface will be, $\vec{E} = \frac{1}{4\pi\epsilon_0 r^3} \vec{r}$ And hence

electric flux linked with area $\vec{ds} \Rightarrow \vec{E} \cdot \vec{ds} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} \vec{r} \cdot \vec{ds}$

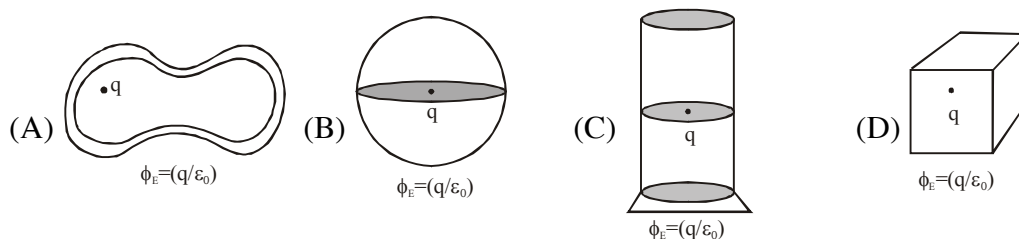
Here direction of $\vec{r}$ and $\vec{ds}$ are same, i.e., $\int_s \vec{E} \cdot \vec{ds} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \oint_s ds = \frac{1}{4\pi\epsilon_0} \frac{1}{r^2} (4\pi r^2) \oint_s \vec{E} \cdot \vec{ds} = \frac{q}{\epsilon_0}$

Which is the required result. Though here in proving it we have assumed the surface to be spherical, it is true for any arbitrary surface provided the surface is closed.

- (2) (a) If a closed body (not enclosing any charge) is placed in an electric field (either uniform or non-uniform) total flux linked with it will be zero.



(b) If a closed body encloses a charge q , then total flux linked with the body will be $\oint_s \vec{E} \cdot d\vec{s} = \frac{q}{\epsilon_0}$



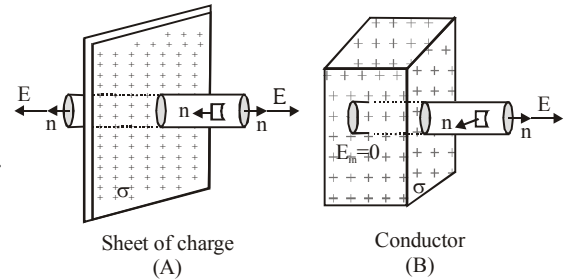
From this expression it is clear that the flux linked with a closed body is independent of the shape and size of the body and position of charge inside it.[figure]

Note : So in case of closed symmetrical body with charge at its centre, flux linked with each half will be

$$\frac{1}{2}(\phi_E) = \left(\frac{q}{2\epsilon_0} \right) \text{ and the symmetrical closed body has } n \text{ identical faces with point charge at its centre,}$$

$$\text{flux linked with each face will be } \left(\frac{\phi_E}{n} \right) = \left(\frac{q}{n\epsilon_0} \right)$$

- (3) Gauss's law is a powerful tool for calculating electric intensity in case of symmetrical charge distribution by choosing a Gaussian-surface in such a way that $\vec{E}$ is either parallel or perpendicular to its various faces. As an example, consider the case of a plane sheet of charge having charge density σ . To calculate E at a point P close to it consider a Gaussian surface in the form of a 'pill box' of cross-section S as shown in figure.



The charge enclosed by the Gaussian-surface = σS and the flux linked with the pill box = $ES + 0 + ES = 2ES$ (as E is parallel to curved surface and perpendicular to plane faces)

$$\text{So from Gauss's law, } \phi_E = \frac{1}{\epsilon_0}(q), 2ES = \frac{1}{\epsilon_0}(\sigma S) \Rightarrow E = \frac{\sigma}{2\epsilon_0}$$

- (4) If $\vec{E} = 0$, $\phi = \oint \vec{E} \cdot d\vec{s} = 0$, so $q = 0$ but if $q = 0$, $\oint \vec{E} \cdot d\vec{s} = 0$ So $\vec{E}$ may or may not be zero.

If a dipole is enclosed by a closed surface then, $q = 0$, so $\oint \vec{E} \cdot d\vec{s} = 0$, but $\vec{E} \neq 0$

Note : If instead of plane sheet of charge, we have a charged conductor, then as shown in figure (B)

$E_{in} = 0$. So $\phi_E = ES$ and hence in this case $E = \frac{\sigma}{\epsilon_0}$. This result can be verified from the fact that

intensity at the surface of a charged spherical conductor of radius R is, $E = \frac{1}{4\pi\epsilon_0} \frac{q}{R^2}$ with $q = 4\pi R^2 \sigma$

$$\text{So for a point close to the surface of conductor, } E = \frac{1}{4\pi\epsilon_0 R^2} \times (4\pi R^2 \sigma) = \frac{\sigma}{\epsilon_0}$$

Ex. If a point charge q is placed at the centre of a cube.

What is the flux linked (a) with the cube? (b) with each face of the cube?

Sol. (a) According to Gauss's law flux linked with a closed body is $(1/\epsilon_0)$ times the charge enclosed

and here the closed body cube is enclosing a charge q so, $\phi_T = \frac{1}{\epsilon_0}(q)$

(b) Now as cube is a symmetrical body with 6-faces and the point charge is at its centre, so electric

flux linked with each face will be $\phi_F = \frac{1}{6}(\phi_T) = \frac{q}{6\epsilon_0}$

Note: (i) Here flux linked with cube or one of its faces is independent of the side of cube.

(ii) If charge is not at the centre of cube (but anywhere inside it), total flux will not change, but the flux linked with different faces will be different.

Ex. Consider $\vec{E} = 3 \times 10^3 \hat{i}$ (N/C) then what is the flux through the square of 10 cm side, if the normal of its plane makes 60° angle with the X axis.

Sol. $\phi = E \cos \theta = 3 \times 10^3 \times [10 \times 10^{-2}]^2 \times \cos 60^\circ = 3 \times 10^3 \times 10^{-2} \times \frac{1}{2} = 15 \text{ Nm}^2/\text{C}$

Ex. Find the electric field due to an infinitely long cylindrical charge distribution of radius R and having linear charge density λ at a distance half of the radius from its axis.

Sol. $r = \frac{R}{2}$ point will be inside so $E = \frac{2k\lambda r}{R^2} = \frac{2k\lambda \left(\frac{R}{2}\right)}{R^2} = \frac{\lambda}{4\pi\epsilon_0 R}$

ELECTRIC FIELD DUE TO SOLID CONDUCTING OR HOLLOW SPHERE

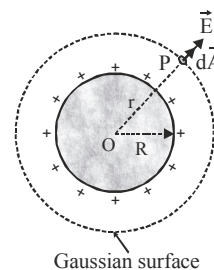
- **For outside point ($r > R$)**

Using Gauss's theorem $\oint \vec{E} \cdot d\vec{s} = \frac{\Sigma q}{\epsilon_0}$

$\therefore$ At every point on the Gaussian surface $\vec{E} \parallel d\vec{s}$;

$$\Rightarrow \vec{E} \cdot d\vec{s} = E ds \cos 0^\circ = E ds$$

$$\therefore \oint E \cdot ds = \frac{\Sigma q}{\epsilon_0} \quad [E \text{ is constant over the gaussian surface}] \Rightarrow E \times 4\pi r^2 = \frac{q}{\epsilon_0} \Rightarrow E_p = \frac{q}{4\pi\epsilon_0 r^2}$$



- **For surface point $r = R$:** $E_s = \frac{q}{4\pi\epsilon_0 R^2}$

- **For Inside point ($r < R$) :** Because charge inside the conducting sphere or hollow is zero.

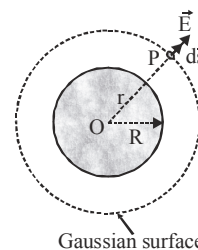
$$(i.e. \Sigma q = 0) \text{ So } \oint \vec{E} \cdot d\vec{s} = \frac{\Sigma q}{\epsilon_0} = 0 \Rightarrow E_{in} = 0$$

ELECTRIC FIELD DUE TO SOLID NON CONDUCTING SPHERE

- **Outside ($r > R$)**

From Gauss's theorem

$$\oint_s \vec{E} \cdot d\vec{s} = \frac{\Sigma q}{\epsilon_0} \Rightarrow E \times 4\pi r^2 = \frac{q}{\epsilon_0} \Rightarrow E_p = \frac{q}{4\pi\epsilon_0 r^2}$$



- **At surface ($r = R$)**

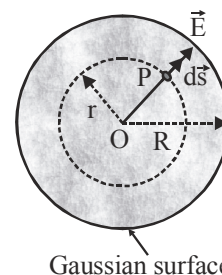
$$E_s = \frac{q}{4\pi\epsilon_0 R^2} \text{ Put } r = R$$

- **Inside ($r < R$) :**

$$\text{From Gauss's theorem } \oint_s \vec{E} \cdot d\vec{s} = \frac{\Sigma q}{\epsilon_0}$$

Where Σq charge contained within Gaussian surface of radius r

$$E(4\pi r^2) = \frac{\Sigma q}{\epsilon_0} \Rightarrow E = \frac{\Sigma q}{4\pi r^2 \epsilon_0} \dots (i)$$



As the sphere is uniformly charged, the volume charge density (charge/volume) ρ is constant throughout

the sphere $\rho = \frac{q}{\frac{4}{3}\pi R^3} \Rightarrow$ charge enclosed in gaussian surface

$$= \rho \left(\frac{4}{3} \pi r^3 \right) = \left(\frac{q}{(4/3) \pi R^3} \right) \left(\frac{4}{3} \pi r^3 \right) \Rightarrow \sum q = \frac{qr^3}{R^3}$$

put this value in equation (i) $E_{in} = \frac{1}{4\pi \epsilon_0} \frac{qr}{R^3}$

ELECTRIC FIELD DUE TO AN INFINITE LINE DISTRIBUTION OF CHARGE

Let a wire of infinite length is uniformly charged having a constant linear charge density λ .

P is the point where electric field is to be calculated.

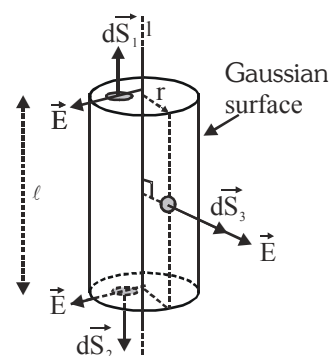
Let us draw a coaxial Gaussian cylindrical surfaces of length ℓ .

From Gauss's theorem

$$q \int_{S_1} \vec{E} \cdot d\vec{S}_1 + \int_{S_2} \vec{E} \cdot d\vec{S}_2 + \int_{S_3} \vec{E} \cdot d\vec{S}_3 = \frac{q}{\epsilon_0}$$

$$\Sigma \vec{E} \perp d\vec{S}_1 \text{ so } \vec{E} \cdot d\vec{S}_1 = 0 \text{ and } \vec{E} \perp d\vec{S}_2 \text{ so } \vec{E} \cdot d\vec{S}_2 = 0$$

$$\mathbf{E} \times 2\pi r \ell = \frac{q}{\epsilon_0} [\because \vec{E} \parallel d\vec{S}_3]$$



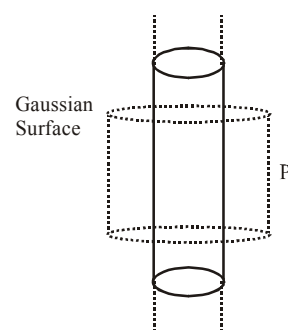
Charge enclosed in the Gaussian surface $q = \lambda \ell$.

$$\text{So } E \times 2\pi r \ell = \frac{\lambda \ell}{\epsilon_0} \Rightarrow E = \frac{\lambda}{2\pi \epsilon_0 r} \text{ or } E = \frac{2k\lambda}{r} \text{ where } k = \frac{1}{4\pi \epsilon_0}$$

Charged cylindrical nonconductor of infinite length

Electric field at outside point $\vec{E}_A = \frac{2k\lambda}{r} \hat{r} \quad r > R$

$$\text{Electric field at inside point } \vec{E}_B = \frac{\lambda \vec{r}}{2\pi \epsilon_0 R^2} \quad r < R$$



KEY POINTS

- Electric field inside a solid conductor is always zero.
- Electric field inside a hollow conductor may or may not be zero ($E \neq 0$ if non zero charge is inside the sphere).
- The electric field due to a circular loop of charge and a point charge are identical provided the distance of the observation point from the circular loop is quite large as compared to its radius i.e. $x \gg R$.

Ex. A charged particle is kept in equilibrium in the electric field between the plates of millikan oil drop experiment. If the direction of the electric field between the plates is reversed, then calculate acceleration of the charged particle.

Sol. Let mass of the particle = m ,

charge on particle = q

Intensity of electric field in between plates = E initially $mg = qE$

After reversing the field $ma = mg + qE \Rightarrow ma = 2mg$

$\therefore$ Acceleration of particle $\Rightarrow a = 2g$

ELECTROSTATIC POTENTIAL ENERGY

Potential energy of a system of particles is defined only in conservative fields. As electric field is also conservative, we define potential energy in it. Potential energy of a system of particles we define as the work done in assembling the system in a given configuration against the interaction forces of particles. Electrostatic potential energy is defined in two ways.

(i) Interaction energy of charged particles of a system

(ii) Self energy of a charged object

INFINITE SHEET CHARGE

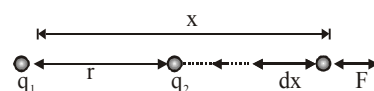
• Electrostatic Interaction Energy

Electrostatic interaction energy of a system of charged particles is defined as the external work required to assemble the particles from infinity to the given configuration. When some charged particles are at infinite separation, their potential energy is taken zero as no interaction is there between them. When these charges are brought close to a given configuration, external work is required if the force between these particles is repulsive and energy is supplied to the system, hence final potential energy of system will be positive. If the force between the particle is attractive, work will be done by the system and final potential energy of system will be negative.

• Interaction Energy of a system of two charged particles

Figure shows two +ve charges q_1 and q_2 separated by a distance r . The electrostatic interaction energy of this system can be given as work done in bringing q_2 from infinity to the given separation from q_1 . It can be calculated as

$$W = \int_{\infty}^r \vec{F} \cdot d\vec{x} = - \int_{\infty}^r \frac{kq_1q_2}{x^2} dx \quad [-ve \text{ sign shows that } x \text{ is decreasing}]$$



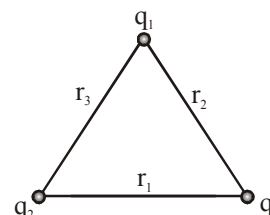
$$W = \frac{kq_1q_2}{r} = U \quad [\text{interaction energy}]$$

If the two charges here are of opposite sign, the potential energy will be negative as $U = - \frac{kq_1q_2}{r}$

• Interaction Energy for a system of charged particles

When more than two charged particles are there in a system, the interaction energy can be given by sum of interaction energies of all the pairs of particles. For example if a system of three particles having charges q_1 , q_2 and q_3 is given as shown in figure. The total interaction energy of this system can be

$$\text{given as } U = \frac{kq_1q_2}{r_3} + \frac{kq_1q_3}{r_2} + \frac{kq_2q_3}{r_1}$$



ELECTRIC POTENTIAL

Electric potential is a scalar property of every point in the region of electric field. At a point in electric field potential is defined as the interaction energy of a unit positive charge. If at a point in electric field a charge q_0 has potential energy U , then electric potential at that point can be given as

$$V = \frac{U}{q_0} \text{ joule/coulomb}$$

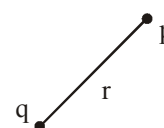
Potential energy of a charge in electric field is defined as work done in bringing the charge from infinity to the given point in electric field. Similarly we can define electric potential as "work done in bringing a unit positive charge from infinity to the given point against the electric forces."

• Electric Potential due to a point charge in its surrounding :

The potential at a point P at a distance r from the charge q , $V_p = \frac{U}{q_0}$

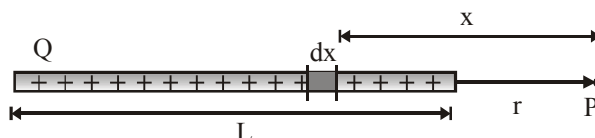
Where U is the potential energy of charge q_0 at point p, $U = \frac{kqq_0}{r}$

Thus potential at point P is $V_p = \frac{kq}{r}$



Electric Potential due to a charge Rod :

Figure shows a rod of length L , uniformly charged with a charge Q . Due to this we'll find electric potential at a point P at a distance r from one end of the rod as shown in figure.



For this we consider an element of width dx at a distance x from the point P.

Charge on this element is $dQ = \frac{Q}{L} dx$

The potential dV due to this element at point P can be given by using the result of a point charge as

$$dV = \frac{k dq}{x} = \frac{kQ}{Lx} dx$$

$$\text{Net electric potential at point P : } V = \int dV = \int_r^{r+L} \frac{kQ}{Lx} dx = \frac{kQ}{L} \ln \left(\frac{r+L}{r} \right)$$

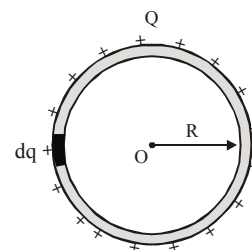
Electric potential due to a charged ring

Case-I : At its centre

To find potential at the centre C of the ring, we first find potential dV at

centre due to an elemental charge dq on ring which is given as $dV = \frac{k dq}{R}$

Total potential at C is $V = \int dV = \int \frac{k dq}{R} = \frac{kQ}{R}$.



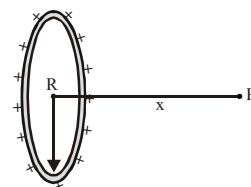
As all dq 's of the ring are situated at same distance R from the ring centre C , simply the potential due to all dq 's is added as being a scalar quantity, we can directly say that the total electric potential at ring centre is $\frac{kQ}{R}$. Here we can also state that even if charge Q is non-uniformly distributed on ring, the

electric potential C will remain same.

Case II : At a point on axis of ring

We find the electric potential at a point P on the axis of ring as shown, we can directly state the result as here also all points of ring are at same distance $\sqrt{x^2 + R^2}$ from the point P , thus the potential at P can be given

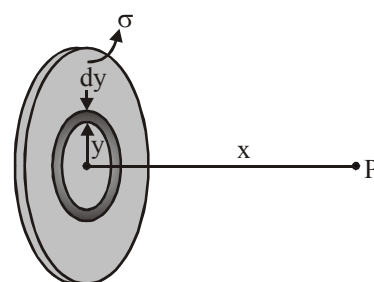
$$\text{as } V_P = \frac{kQ}{\sqrt{R^2 + x^2}}$$



Electric potential due to a uniformly charged disc :

Figure shows a uniformly disc of radius R with surface charge density ρ coul/m². To find electric potential at point P we consider an elemental ring of radius y and width dy , charge on this elemental ring is $dq = \sigma 2\pi y dy$. Due to this ring, the electric potential at point P can be given as

$$dV = \frac{k dq}{\sqrt{x^2 + y^2}} = \frac{k \cdot \sigma \cdot 2\pi y dy}{\sqrt{x^2 + y^2}}$$



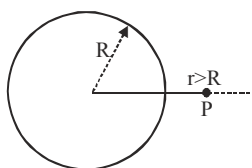
Net electric potential at Point P due to whole disc can be given as

$$V = \int dV = \int_0^R \frac{\sigma}{2\epsilon_0} \cdot \frac{y dy}{\sqrt{x^2 + y^2}} = \frac{\sigma}{2\epsilon_0} \left[\sqrt{x^2 + y^2} \right]_0^R = \frac{\sigma}{2\epsilon_0} \left[\sqrt{x^2 + R^2} - x \right]$$

ELECTRIC POTENTIAL DUE TO HOLLOW OR CONDUCTING SPHERE

• At outside sphere

According to definition of electric potential, electric potential at point P



$$V = -\int_{\infty}^r \vec{E} \cdot d\vec{r} = -\int_{\infty}^r \frac{q}{4\pi\epsilon_0 r^2} dr \left[\because E_{\text{out}} = \frac{q}{4\pi\epsilon_0 r^2} \right]; V = -\frac{q}{4\pi\epsilon_0} \int_{\infty}^r \frac{1}{r^2} dr = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_{\infty}^r = \frac{q}{4\pi\epsilon_0 r}$$

• At surface

$$= V = -\int_{\infty}^R \vec{E} \cdot d\vec{r} = -\int_{\infty}^R \frac{q}{4\pi\epsilon_0 r^2} dr \left[\because E_{\text{out}} = \frac{q}{4\pi\epsilon_0 r^2} \right]; V = -\frac{q}{4\pi\epsilon_0} \int_{\infty}^R \left(\frac{1}{r^2} \right) dr = \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_{\infty}^R \Rightarrow V = \frac{q}{4\pi\epsilon_0 R}$$

• Inside the surface :

$$\therefore \text{ Inside the surface } E = 0 \left[E = -\frac{dV}{dr} \right] \frac{dV}{dr} = 0 \text{ or } V = \text{constant so } V = \frac{q}{4\pi\epsilon_0 R}$$

ELECTRIC POTENTIAL DUE TO SOLID NON CONDUCTING SPHERE

- **At outside sphere** Same as conducting sphere.
- **At Surface** Same as conducting sphere.
- **Inside the sphere**

$$V = - \int_{\infty}^r \vec{E} \cdot d\vec{r} \Rightarrow V = - \left[\int_{\infty}^R E_1 dr + \int_R^r E_2 dr \right]$$

$$V = - \left[\int_{\infty}^R \left(\frac{kq}{r^2} \right) dr + \int_R^r \left(\frac{kqr}{R^3} \right) dr \right] \Rightarrow V = - \left[kq \left(-\frac{1}{r} \right)_{\infty}^R + \frac{kq}{R^3} \left(\frac{r^2}{2} \right)_R^r \right]$$

$$V = -kq \left[-\frac{1}{R} + \frac{r^2}{2R^3} - \frac{R^2}{2R^3} \right] \Rightarrow V = \frac{kq}{2R^3} [3R^2 - r^2]$$

Potential Difference Between Two points in electric field

Potential difference between two points in electric field can be defined as work done in displacing a unit positive charge from one point to another against the electric forces.

If a unit +ve charge is displaced from a point A to B as shown work required can be given as

$$V_B - V_A = - \int_A^B \vec{E} \cdot d\vec{x}$$

If a charge q is shifted from point A to B, work done against electric forces can be given as $W = q(V_B - V_A)$

If in a situation work done by electric forces is asked, we use $W = q(V_A - V_B)$

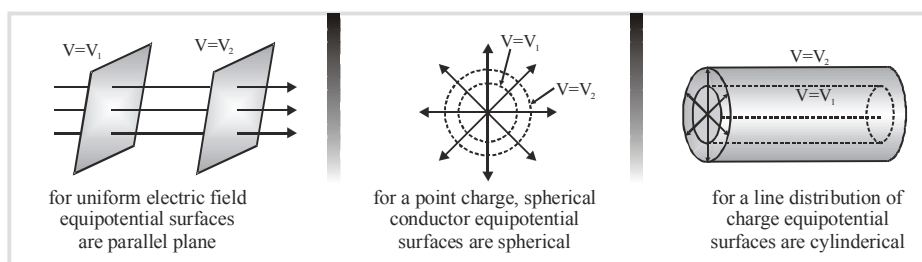
If $V_B < V_A$, then charges must have tendency to move toward B (low potential point) it implies that electric forces carry the charge from high potential to low potential points. Hence we can say that in the direction of electric field always electric potential decreases.



Equipotential surfaces

For a given charge distribution, locus of all points having same potential is called 'equipotential surface'.

- Equipotential surfaces can never cross each other (otherwise potential at a point will have two values which is absurd)
- Equipotential surfaces are always perpendicular to direction of electric field.
- If a charge is moved from one point to the other over an equipotential surface then work done $W_{AB} = -U_{AB} = q(V_B - V_A) = 0$ [$\because V_B = V_A$]
- Shapes of equipotential surfaces



- The intensity of electric field along an equipotential surface is always zero.

Electric Potential Gradient

The maximum rate of change of potential at right angles to an equipotential surface in an electric field is defined as potential gradient. $\vec{E} = -\vec{\nabla}V = -\text{grad } V$

Note : Potential is a scalar quantity but the gradient of potential is a vector quantity

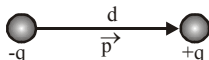
$$\text{In cartesian co-ordinates } \vec{\nabla}V = \left[\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right]$$

Ex. If $V = -5x + 3y + \sqrt{15}z$ then $E(x, y, z) = ?$

Sol. $\vec{E} = -\left[\frac{\partial V}{\partial x} \hat{i} + \frac{\partial V}{\partial y} \hat{j} + \frac{\partial V}{\partial z} \hat{k} \right] = -(-5\hat{i} + 3\hat{j} + \sqrt{15}\hat{k}) \Rightarrow |\vec{E}| = \sqrt{25 + 9 + 15} = \sqrt{49} = 7 \text{ unit}$

ELECTRIC DIPOLE

A system of two equal and opposite charges separated by a small distance is called electric dipole, shown in figure. Every dipole has a characteristic property called dipole moment. It is defined as the product of magnitude of either charge and the separation between the charges, given as

$$\vec{p} = q\vec{d}$$


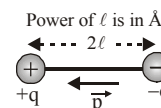
In some molecules, the centres of positive and negative charges do not coincide. This results in the formation of electric dipole. Atom is non-polar because in it the centres of positive and negative charges coincide. Polarity can be induced in an atom by the application of electric field. Hence it can be called as induced dipole.

- Dipole Moment :** Dipole moment $\vec{p} = q\vec{d}$

(i) Vector quantity, directed from negative to positive charge

(ii) **Dimension :** [LTA], **Units :** coulomb $\times$ metre (or Cm)

(iii) Practical unit is "debye" $\equiv$ Two equal and opposite point charges each having charge 10^{-10} frankline ($\approx e$) and separation of $1\text{\AA}$ then the value of dipole moment ($\vec{p}$) is 1 debye.



$$1 \text{ Debye} = 10^{-10} \times 10^{-10} \text{ Fr} \times \text{m} = 10^{-20} \times \frac{\text{C} \times \text{m}}{3 \times 10^9} \approx 3.3 \times 10^{-30} \text{ C} \times \text{m}$$

- Dipole Placed in uniform Electric Field**

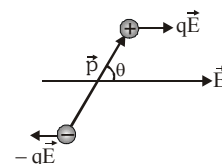
Figure shows a dipole of dipole moment $\vec{p}$ placed at an angle θ to the direction of electric field. Here the charges of dipole experience forces

$$qE \text{ in opposite direction as shown. } \vec{F}_{\text{net}} = [q\vec{E} + (-q)\vec{E}] = \vec{0}$$

Thus we can state that when a dipole is placed in a uniform electric field, net force on the dipole is zero. But as equal and opposite forces act with a separation in their line of action, they produce a couple which tend to align the dipole along the direction of electric field. The torque due to this couple can be given as

$$\tau = \text{Force} \times \text{separation between lines of actions of forces} = qE \times d \sin \theta = pE \sin \theta$$

$$\vec{\tau} = \vec{r} \times \vec{F} = \vec{d} \times q\vec{E} = q\vec{d} \times \vec{E} = \vec{p} \times \vec{E}$$



Work done in Rotation of a Dipole in Electric field

When a dipole is placed in an electric field at an angle θ , the torque on it due to electric field is

$$\tau = pE \sin \theta$$

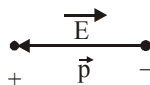
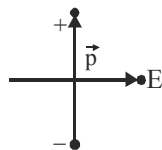
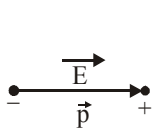
Work done in rotating an electric dipole from θ_1 to θ_2 [uniform field]

$$dW = \tau d\theta \text{ so } W = \int dW = \int \tau d\theta \text{ and } W_{\theta_1 \rightarrow \theta_2} = W = \int_{\theta_1}^{\theta_2} pE \sin\theta d\theta = pE (\cos\theta_1 - \cos\theta_2)$$

e.g. $W_{0 \rightarrow 180} = pE [1 - (-1)] = 2 pE$

$$W_{0 \rightarrow 90} = pE (1-0) = pE$$

If a dipole is rotated from field direction ($\theta = 0^\circ$) to θ then $W = pE (1 - \cos\theta)$



$$\theta = 0$$

$$\theta = 90^\circ$$

$$\theta = 180^\circ$$

$$\tau = \text{minimum} = 0$$

$$\tau = \text{maximum} = pE$$

$$\tau = \text{minimum} = 0$$

$$W = \text{minimum} = 0$$

$$W = pE$$

$$W = \text{maximum} = 2pE$$

Electrostatic potential energy :

Electrostatic potential energy of a dipole placed in a uniform field is defined as work done in rotating a dipole from a direction perpendicular to the field to the given direction i.e.,

$$W_{90^\circ \rightarrow \theta} = \int_{90^\circ}^{\theta} pE \sin \theta d\theta = -pE \cos \theta = -\vec{p} \cdot \vec{E}$$

$\vec{E}$ is a conservative field so whatever work is done in rotating a dipole from θ_1 to θ_2 is just equal to change in electrostatic potential energy $W_{\theta_1 \rightarrow \theta_2} = U_{\theta_2} - U_{\theta_1} = pE (\cos \theta_1 - \cos \theta_2)$

Work done in rotating an electric dipole in an electric field

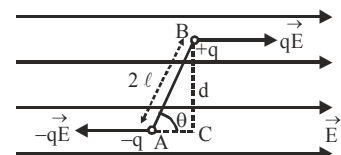
Suppose at any instant, the dipole makes an angle θ with the electric field.

The torque acting on dipole. $\tau = qEd = (q 2\ell \sin\theta)E = pE \sin\theta$

The work done in rotating dipole from θ_1 to θ_2

$$W = \int_{\theta_1}^{\theta_2} \tau d\theta = \int_{\theta_1}^{\theta_2} pE \sin \theta \, d\theta$$

$$W = pE (\cos\theta_1 - \cos\theta_2) = U_2 - U_1 \quad (\because U = -pE \cos\theta)$$



Force on an electric dipole in Non-uniform electric field :

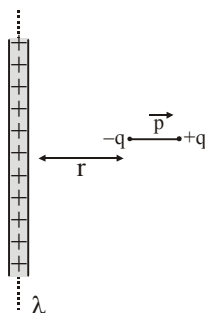
If in a non-uniform electric field dipole is placed at a point where electric field is E , the interaction

energy of dipole at this point $U = -\vec{p} \cdot \vec{E}$. Now the force on dipole due to electric field $F = -\frac{\Delta U}{\Delta x}$

For unidirectional variation in electric field, $F = -\frac{d}{dx}(\vec{p} \cdot \vec{E})$

If dipole is placed in the direction of electric field then $F = -p \frac{dE}{dx}$

Ex. Calculate force on a dipole in the surrounding of a long charged wire as shown in the figure.



Sol. In the situation shown in figure, the electric field strength due to the wire, at the position of dipole as

$$E = \frac{2k\lambda}{r}$$

$$\text{Thus force on dipole is } F = -p \cdot \frac{dE}{dr} = -p \left[-\frac{2k\lambda}{r^2} \right] = \frac{2kp\lambda}{r^2}$$

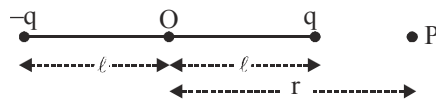
Here -ve charge of dipole is close to wire hence net force on dipole due to wire will be attractive.

ELECTRIC POTENTIAL DUE TO DIPOLE

• At axial point

$$\text{Electric potential due to } +q \text{ charge } V_1 = \frac{kq}{(r-\ell)}$$

$$\text{Electric potential due to } -q \text{ charge } V_2 = \frac{-kq}{(r+\ell)}$$



$$\text{Net electric potential } V = V_1 + V_2 = \frac{kq}{(r-\ell)} + \frac{-kq}{(r+\ell)} = \frac{kq \times 2\ell}{(r^2 - \ell^2)} = \frac{kp}{r^2 - \ell^2}$$

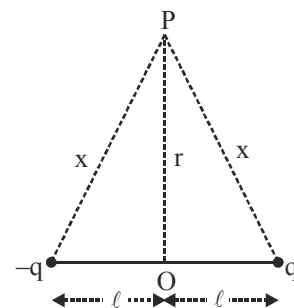
$$\text{If } r \gg \ell \Rightarrow V = \frac{kp}{r^2}$$

• At equatorial point

$$\text{Electric potential of P due to } +q \text{ charge } V_1 = \frac{kq}{x}$$

$$\text{Electric potential of P due to } -q \text{ charge } V_2 = -\frac{kq}{x}$$

$$\text{Net potential } V = V_1 + V_2 = \frac{kq}{x} - \frac{kq}{x} = 0 \therefore V = 0$$

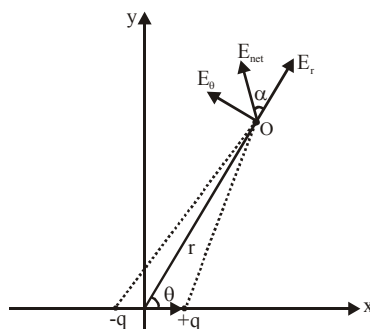


• At general point

$$V = \frac{p \cos \theta}{4\pi\epsilon_0 r^2} = \frac{\vec{p} \cdot \vec{r}}{4\pi\epsilon_0 r^3} \quad \vec{p} = q\vec{a} \text{ electric dipole moment}$$

Electric field due to an electric dipole

Figure shows an electric dipole placed on x-axis at origin. Here we wish to find the electric field and potential at a point O having coordinates (r, θ). Due to the positive charge of dipole electric field at O is in radially outward direction and due to the negative charge it is radially inward as shown in figure.



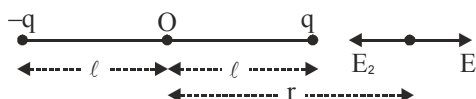
$$E_r = -\frac{\partial V}{\partial r} = \frac{2kp \cos \theta}{r^3} \text{ and } E_\theta = -\frac{1}{r} \frac{\partial V}{\partial \theta} = \frac{kp \sin \theta}{r^3}$$

Thus net electric field at point O, $E_{\text{net}} = \sqrt{E_r^2 + E_\theta^2} = \frac{kp}{r^3} \sqrt{1 + 3 \cos^2 \theta}$

If the direction of E_{net} is at an angle α from radial direction, then $\alpha = \tan^{-1} \frac{E_\theta}{E_r} = \left(\frac{1}{2} \tan \theta \right)$

Thus the inclination of net electric field at point O is $(\theta + \alpha)$

• At a point on the axis of a dipole :



Electric field due to +q charge $E_1 = \frac{kq}{(r - \ell)^2}$

Electric field due to -q charge $E_2 = \frac{kq}{(r + \ell)^2}$

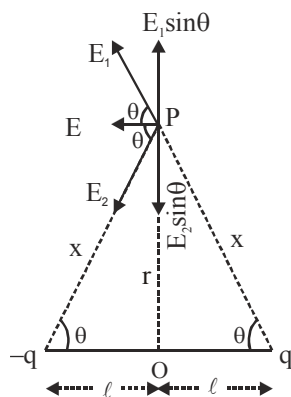
Net electric field $E = E_1 - E_2 = \frac{kq}{(r - \ell)^2} - \frac{kq}{(r + \ell)^2} = \frac{kq \times 4r\ell}{(r^2 - \ell^2)^2} \quad [\because p = q \times 2\ell = \text{Dipole moment}]$

$E = \frac{2kpr}{(r^2 - \ell^2)^2}$ If $r \gg \ell$ then $E = \frac{2kp}{r^3}$

- At a point on equatorial line of dipole :

Electric field due to $+q$ charge $E_1 = \frac{kq}{x^2}$; Electric field due to $-q$ charge $E_2 = \frac{kq}{x^2}$

Vertical component of E_1 and E_2 will cancel each other and horizontal components will be added



So net electric field at P

$$E = E_1 \cos \theta + E_2 \cos \theta \quad [\because E_1 = E_2]$$

$$E = 2E_1 \cos \theta = \frac{2kq}{x^2} \cos \theta \quad \because \cos \theta = \frac{\ell}{x} \text{ and } x = \sqrt{r^2 + \ell^2}$$

$$E = \frac{2kq\ell}{x^3} = \frac{2kq\ell}{(r^2 + \ell^2)^{3/2}} = \frac{kp}{(r^2 + \ell^2)^{3/2}} \quad \text{If } r \gg \ell$$

$$\text{then } E = \frac{kp}{r^3} \text{ or } \vec{E} = \frac{-k\vec{p}}{r^3}$$

- Ex.** A short electric dipole is situated at the origin of coordinate axis with its axis along x-axis and equator along y-axis. It is found that the magnitudes of the electric intensity and electric potential due to the dipole are equal at a point distant $r = \sqrt{5}$ m from origin. Find the position vector of the point.

Sol. $\because |E_P| = |V_P| \therefore \frac{kp}{r^3} \sqrt{1 + 3 \cos^2 \theta} = \frac{kp \sin \theta}{r^2} \Rightarrow 1 + 3 \cos^2 \theta = 5 \cos^2 \theta \Rightarrow \cos \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$

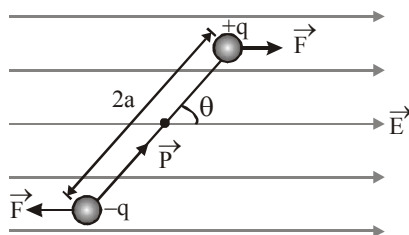
Position vector $\vec{r}$ of point P is $\vec{r} = \frac{\sqrt{5}}{2} (\hat{i} + \hat{j})$

- Ex.** Prove that the frequency of oscillation of an electric dipole of moment p and rotational inertia I for

small amplitudes about its equilibrium position in a uniform electric field strength E is $\frac{1}{2\pi} \sqrt{\left(\frac{pE}{I}\right)}$

- Sol.** Let an electric dipole (charge q and $-q$ at a distance $2a$ apart) placed in a uniform external electric field of strength E .

Restoring torque on dipole



$$\tau = -pE \sin \theta = -pE \theta \quad (\text{as } \theta \text{ is small})$$

Here -ve sign shows the restoring tendency of torque.

$$\therefore \tau = I\alpha \therefore \text{angular acceleration} = \alpha = \frac{\tau}{I} = \frac{pE}{I} \theta$$

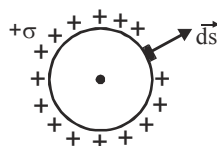
$$\text{For SHM } \alpha = \omega^2 \theta \text{ comparing we get } \omega = \sqrt{\frac{pE}{I}}$$

$$\text{Thus frequency of oscillations of dipole } n = \frac{\omega}{2\pi} = \frac{1}{2\pi} \sqrt{\left(\frac{pE}{I}\right)}$$

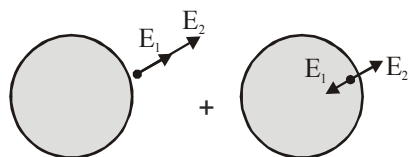
ELECTROSTATIC PRESSURE

Force due to electrostatic pressure is directed normally outwards to the surface. Force on small element ds of charged conductor

$$dF = (\text{Charge on } ds) \times \text{Electric field} = (\sigma ds) \frac{\sigma}{2\epsilon_0} = \frac{\sigma^2}{2\epsilon_0} ds$$



$$\text{Inside } E_1 - E_2 = 0 \Rightarrow E_1 = E_2$$



$$\text{Just outside } E = E_1 + E_2 = 2E_2 \Rightarrow E_2 = \frac{\sigma}{2\epsilon_0}$$

(E_1 is field due to point charge on the surface and E_2 is field due to rest of the sphere).

The electric force acting per unit area of charged surface is defined as electrostatic pressure.

$$P_{\text{electrostatic}} = \frac{dF}{dS} = \frac{\sigma^2}{2\epsilon_0}$$

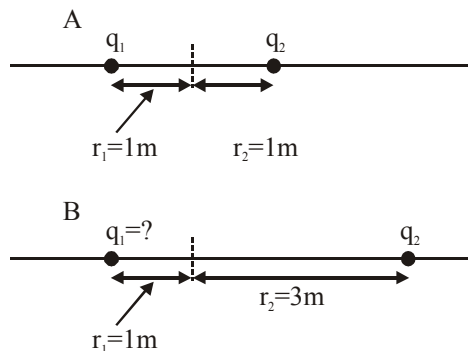
Electric field

6. Two point charge $-4Q$ and $9Q$ are placed at a distance $2m$ from each other. The position at which net electric field is zero from the charge $-4Q$ is x (in m). What is the value of x .

ES0006

7. Two charges are separated as shown in the diagram A below. Each charge has a magnitude of 50 nC in diagram A. If the charge on the right is moved further to the right as depicted in diagram B, additional charge that must be removed from the left in order to maintain the initial electric field

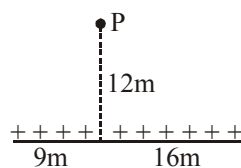
magnitude at the center (in nC) is given by $\frac{\alpha}{9}$ nC. Fill the value of α in OMR sheet.

**ES0007**

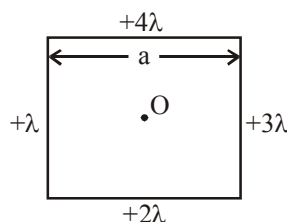
8. A clock face has negative charges $-q, -2q, -3q, \dots, -12q$ fixed at the positions of the corresponding numerals on the dial. Assume that the clock hands do not disturb the net field due to point charges. At what time does the hour hand point in the same direction as the electric field at the centre of the dial.

ES0008

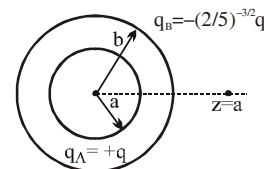
9. Find the magnitude of electric field (in N/C) due to a line charge of $\lambda = (2\sqrt{2}) \text{ nC/m}$ at a point P as shown.

**ES0009**

10. Four uniformly charged wires of length a are arranged to form a square. Linear charge density of each wire is as shown. Electric field intensity at centre of square is $\frac{nk\lambda}{a}$ then value of n

**ES0010**

11. Two concentric rings, one of radius 'a' and the other of radius 'b' have the charges $+q$ and $-(2/5)^{3/2} q$ respectively as shown in the figure. Find the ratio b/a if a charge particle placed on the axis at $z = a$ is in equilibrium.

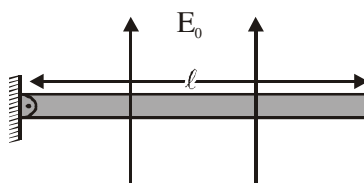


ES0011

12. The intensity of electric field is required to exert a force on a proton equal to its weight at sea level is given by $\alpha \times 10^{-7}$ V/m. Fill the value of α (Given $g = 10 \text{ m/s}^2$, mass of proton $= 1.6 \times 10^{-27} \text{ kg}$ and charge on proton $= 1.6 \times 10^{-19} \text{ C}$).

ES0012

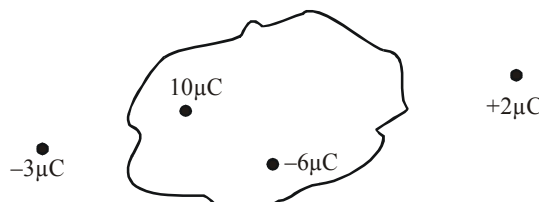
13. A thin insulating uniformly charged (linearly charged density λ) rod is hinged about one of its ends. It can rotate in vertical plane. If rod is in equilibrium by applying vertical electric field E as shown in figure. Find the value of E (in N/C). (Given that mass of rod 2 kg , $\lambda = 10 \text{ C/m}$, $\ell = 1 \text{ m}$, $g = 10 \text{ m/s}^2$)



ES0013

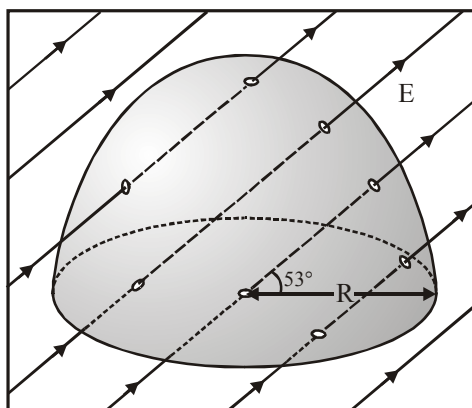
Gauss' law

14. In a particular system, if number of electric field lines associated by 1 C charge is 10^9 . Then net number of electric field lines passing through the given closed surface is $n \times 10^3$ find n .



ES0014

15. A uniform electric field $E = 500 \text{ N/C}$ passes through a hemispherical surface of radius $R = 1.2 \text{ m}$ as shown in figure. The net electric flux (in S.I. units) through the hemispherical surface only is $N\pi$. Then find the value of N .



ES0015

16. A non-uniform electric field in x-direction is increasing uniformly from 2 N/C at $x = 1\text{ m}$ to 8 N/C at $x = 7\text{ m}$. The center of cube is at $x = 2\text{ m}$. If the charge enclosed inside a small cube of side length 10 cm is $8.85 \times 10^{-5n}\text{ C}$. The value of n will be.

ES0016

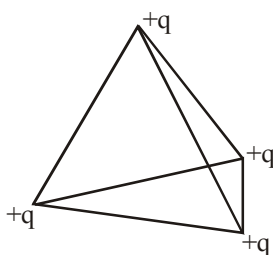
17. $20\text{ }\mu\text{C}$ charge is placed inside a closed surface then flux related to surface is ϕ . If $80\text{ }\mu\text{C}$ charge is added inside the surface then change in flux is given by $\alpha\phi$. Fill the value of α .

ES0017

18. A sphere of radius R has charge density given by $\rho = \rho_0 \left(1 - \frac{nr}{3R}\right)$, where ρ_0 is a constant, r is distance from centre of sphere. For a spherical gaussian surface of radius R centered at the centre of sphere, the flux is zero. Find 'n'.

ES0018

19. Four point charges are kept at the vertices of a regular tetrahedron of side R . Total electrostatic energy of the configuration is $\frac{\alpha kq^2}{R}$. The value of α is



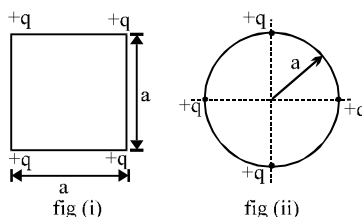
ES0019

Electric potential energy and electric potential

20. Four identical free point positive charges q ($=1\text{ }\mu\text{C}$) each are located at the four corners of a square of side 1 m . A negative charge is placed at the centre of the square to obtain equilibrium of all the charges. What is the total potential energy (in milli Joules) of the system assuming the reference point at infinity?

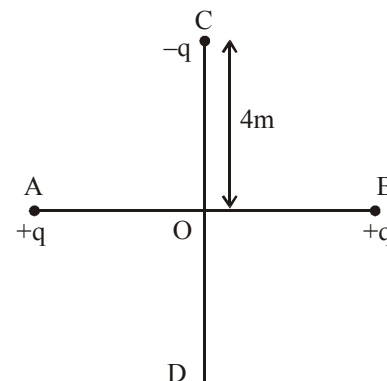
ES0020

21. Consider the configuration of a system of four charges each of value $+q$. Find the work done by external agent in changing the configuration of the system from figure (i) to fig (ii).



ES0021

22. Two fixed, equal positive charges, each $+q$ are located at point A & B separated by a distance of 6 m. A particle of mass m having equal and opposite charge $-q$ moves towards them along the perpendicular bisector line COD where O is the centre of line joining A and B. If $-q$ charge is released from rest from point C, then speed of charge $-q$ at O is given by



$$v = q \sqrt{\frac{x}{15\pi \epsilon_0 m}} \text{ find value of } x. \text{ (Neglect gravity)}$$

ES0022

23. An electric field $(-30\hat{i} + 20\hat{j}) \text{ Vm}^{-1}$ exists in the space. If potential at the origin is zero then find the potential at $(5\text{m}, 3\text{m})$ in volts.

ES0023

24. A circular ring of radius a with uniform charge density λ is in the xy plane with centre at origin. A particle of mass m and charge q is projected from $P(0, 0, a\sqrt{3})$ on $+z$ -axis towards origin with initial

velocity u . The minimum value of the velocity so that the particle does not return to P is $\sqrt{\frac{\lambda q}{x\epsilon_0 m}}$.

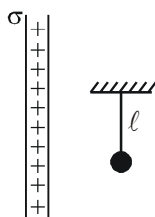
Find 'x' (neglect gravity).

ES0024

25. A particle of mass m carrying charge ' q ' is projected with velocity (v) from point P towards an infinite line charge from a distance ' a '. Its speed reduces to zero momentarily at point Q which is at a distance $a/2$ from the line charge. If another particle with mass m and charge $-q$ is projected with the same velocity v from point P towards the line charge. Its speed is found to be $\frac{Nv}{\sqrt{2}}$ at point 'Q'. Find the value of N .

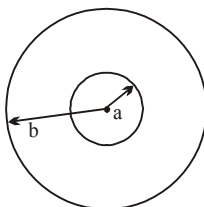
ES0025

26. A simple pendulum of length ℓ and bob mass m is hanging in front of a large nonconducting sheet of surface charge density σ . If suddenly a charge $+q$ is given to the bob in the position shown in figure. Find the maximum angle through which the string is deflected from vertical.



ES0026

27. Outer cylinder of the coaxial nonconductor of radius 'b' is given a positive potential V relative to the inner cylinder of radius 'a' as shown in the figure (charge distribution is uniform). A charge q (mass = m) is set free with negligible velocity at the surface of the inner cylinder. Find the velocity (in m/s), when it hits the outer cylinder. [consider $V = 10$, $q = -20$, $m = 1$ all in S.I. Units]



ES0027

28. A positive charge Q is uniformly distributed throughout the volume of a nonconducting sphere of radius R . A point mass having charge $+q$ and mass m is fired towards the centre of the sphere with velocity v from a point at distance r ($r > R$) from the centre of the sphere. Find the minimum velocity v so that it can penetrate $R/2$ distance of the sphere. Neglect any resistance other than electric interaction. Charge on the small mass remains constant throughout the motion.

ES0028

Electric dipole

29. A plastic rod of length 1.0 m carries uniform positive charge $+4.0 \mu\text{C}$ on half of its length and uniform negative charge $-4.0 \mu\text{C}$ on the remaining half of its length. Find magnitude of its net dipole moment in $\mu\text{C}\cdot\text{m}$.

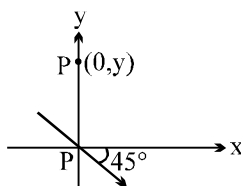


ES0029

30. A dipole of dipole moment $\vec{p} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ is placed at point $A(2, -3, 1)$. The electric potential due to this dipole at the point $B(4, -1, 0)$ is $(ab) \times 10^9$ volt here 'a' represents sign (for negative answer select 0 for positive answer select 1). Write the value of $(a+b)^2$. The parameters specified here are in S.I. units.

ES0030

31. A dipole is placed at origin of coordinate system as shown in figure, find the electric field at point $P(0, y)$.



ES0031

32. A small rigid object carries positive and negative charge of magnitude 4 coulomb each. It is oriented so that the positive charge has coordinates $(-1.2\text{mm}, 1.1\text{mm})$ and negative charge has coordinates $(1.4\text{mm}, -1.3\text{mm})$. The object is kept in an electric field of $(2500\hat{i} - 5000\hat{j})\text{N/C}$. Find the magnitude of torque (in N-m) acting on the dipole.

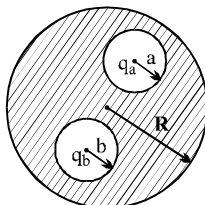
ES0032

33. A charge $+Q$ is fixed at the origin of the coordinate system while a small electric dipole of dipole-moment $\vec{p}$ pointing away from the charge along the x -axis is set free from a point far away from the origin.
- (a) calculate the K.E. of the dipole when it reaches to a point $(d, 0)$ [IIT-JEE 2003]
- (b) calculate the force on the charge $+Q$ due to the dipole at this moment.

ES0033

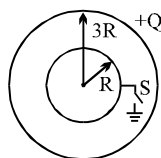
Conductors

34. A conducting sphere of radius R has two spherical cavities of radius a and b . The cavities have charges q_a and q_b respectively at their centres. Find:
- (a) The electric field and electric potential at a distance r
- (i) r (distance from O , the centre of sphere $> R$)
- (ii) r (distance from B , the centre of cavity $b < b$)
- (b) Surface charge densities on the surface of radius R , radius a and radius b .
- (c) What is the force on q_a and q_b ?



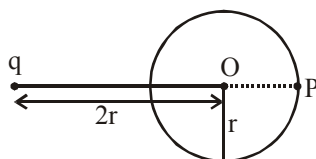
ES0034

35. Two thin conducting shells of radii R and $3R$ are shown in figure. The outer shell carries a charge $+Q$ and the inner shell is neutral. The inner shell is earthed with the help of switch S . Find the charge attained by the inner shell.



ES0035

36. A point charge q is placed at a distance $2r$ from the centre O of a conducting neutral sphere of radius r . Due to the induced charge on the sphere, the electric potential at point P on the surface of sphere is x volt. Then find the value of x . (If $\frac{kq}{r} = 18$ volt)

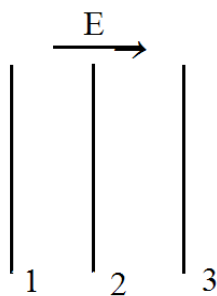


ES0036

37. A conducting liquid bubble of radius a and thickness t ($t \ll a$) is charged to potential V . If the bubble collapses to a droplet, find the potential on the droplet. **[IIT-JEE 2005]**

ES0037

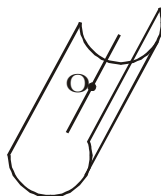
38. Three uncharge conducting large plates are placed parallel to each other in a uniform electric field. Find the induced charge density on each surface of each plate.



ES0038

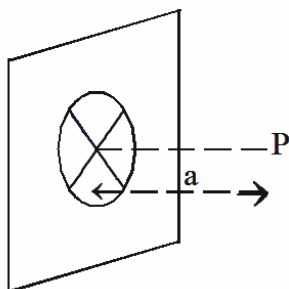
EXERCISE (S-2)

1. A thin long strip whose cross-section is a semicircle carries a uniform surface charge of density σ on its inner surface. Find the electric field at a point O located midway on its axis.



ES0039

2. An infinitely long non-conducting plane of charge density σ has circular aperture of certain radius carved out from it. The electric field at a point which is at a distance 'a' from the centre of the aperture is $\sigma/2\sqrt{2}\epsilon_0$. Find the radius of aperture.

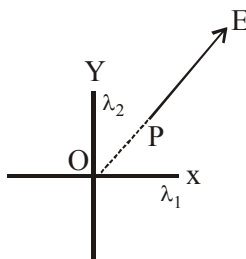


ES0040

3. A disc of radius R is kept such that its axis coincide with the x-axis and its centre is at (d,0,0). The thickness of disc is t and it carries a uniform volume charge density ρ . The external electric field in the space is given by $\vec{E} = K \vec{r}$ where $K = \text{Constant}$ and $\vec{r}$ is position vector of any point in space with respect to the origin of the coordinate system. Find the electric force on the disc.

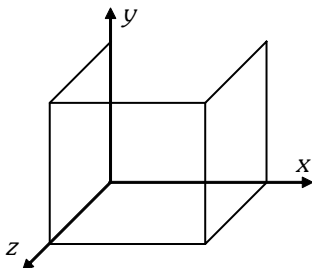
ES0041

4. Two mutually perpendicular infinite wires along x-axis and y-axis carry charge densities λ_1 and λ_2 . The electric line of force at P is along the line $y = \frac{1}{\sqrt{3}}x$, where P is also a point lying on the same line then find λ_2/λ_1 .



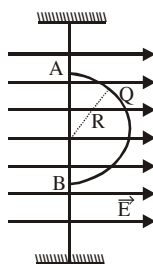
ES0042

5. An electric field given by $\vec{E} = 4\hat{i} - 3(y^2 + 2)\hat{j}$ pierces Gaussian's cube of side 1m placed at origin such that its three sides represents x, y and z axes. The net charge enclosed within the cube is given by $-n\epsilon_0$. Find the value of n.



ES0043

6. The charge $Q = \pi C$ is distributed on a thin semicircular ring of radius $R = 2m$. There is a uniform electrostatic field $|\vec{E}| = 2N/C$ directed horizontally. The semicircular ring can rotate freely about a fixed vertical axis AB. Initially the ring is in static equilibrium as shown in figure. If we want to rotate it about the fixed axis by 90° then minimum work required on the ring is x J. Find the value of x.



ES0044

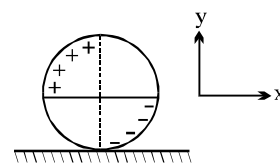
7. Four point charges $+8\mu C$, $-1\mu C$, $-1\mu C$ and $+8\mu C$, are fixed at the points, $-\sqrt{\frac{27}{2}}$ m, $-\sqrt{\frac{3}{2}}$ m, $+\sqrt{\frac{3}{2}}$ m and $+\sqrt{\frac{27}{2}}$ m respectively on the y-axis. A particle of mass 6×10^{-4} kg and of charge $+0.1 \mu C$ moves along the $-x$ direction. Its speed at $x = +\infty$ is v_0 . Find the least value of v_0 for which the particle will cross the origin. Find also the kinetic energy of the particle at the origin. Assume that space is gravity free. (Given : $1/(4\pi\epsilon_0) = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$)

ES0045

8. The electric potential in a region is given by $V(x, y, z) = ax^2 + ay^2 + abz^2$. 'a' is a positive constant of appropriate dimensions and b, a positive constant such that V is volts when x, y, z are in m. Let $b = 2$. The work done by the electric field when a point charge $+4\mu C$ moves from the point $(0, 0, 0.1\text{m})$ to the origin is $50 \mu J$. The radius of the circle of the equipotential curve corresponding to $V = 6250$ volts and $z = \sqrt{2}$ m is α m. Fill α^2 in OMR sheet.

ES0046

9. A nonconducting ring of mass m and radius R is charged as shown. The charged density i.e. charge per unit length is λ . It is then placed on a rough nonconducting horizontal surface plane. At time $t = 0$, a uniform electric field $\vec{E} = E_0 \hat{i}$ is switched on and the ring starts rolling without sliding. Determine the friction force (magnitude and direction) acting on the ring, when it starts moving.

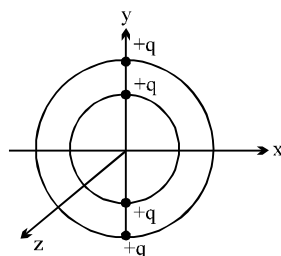


ES0047

10. Small identical balls with equal charges are fixed at vertices of regular 2008-gon with side a . At a certain instant, one of the balls is released & a sufficiently long time interval later, the ball adjacent to the first released ball is freed. The kinetic energies of the released balls are found to differ by K at a sufficiently long distance from the polygon. Determine the charge q of each ball.

ES0048

11. Two concentric rings of radii r and $2r$ are placed with centre at origin. Two charges $+q$ each are fixed at the diametrically opposite points of the rings as shown in figure. Smaller ring is now rotated by an angle 90° about Z-axis then it is again rotated by 90° about Y-axis. Find the work done by electrostatic forces in each step. If finally larger ring is rotated by 90° about X-axis, find the total work required to perform all three steps.



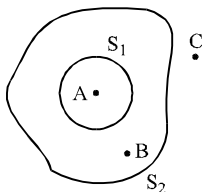
ES0049

12. A non-conducting disc of radius a and uniform positive surface charge density σ is placed on the ground, with its axis vertical. A particle of mass m & positive charge q is dropped, along the axis of the disc, from a height H with zero initial velocity. The particle has $\frac{q}{m} = \frac{4\epsilon_0 g}{\sigma}$.

- (a) Find the value of H if the particle just reaches the disc.
 (b) Sketch the potential energy of the particle as a function of its height and find its equilibrium position.

ES0050

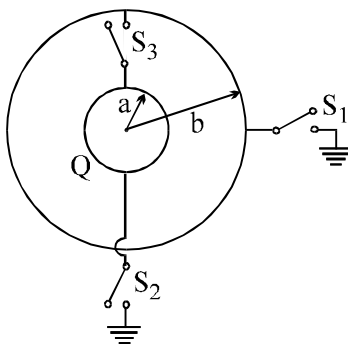
- 13.** S_1 and S_2 are two conducting surfaces. Between S_1 and S_2 and inside S_1 is air. S_1 is spherical with A its centre. S_1 has total charge Q . S_2 is uncharged. Find (if possible) :



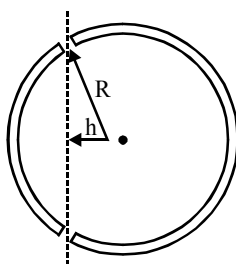
- (i) Charges induced on inner and outer surface of S_2 .
 - (ii) Total electric field at A, B.
 - (iii) Electric field at B due to induced charges on S_2 .
 - (iv) Electric field at C due to induced charges on inner surface of S_2 .
 - (v) Electric field produced by induced charges on outer surface of S_2 inside the body of S_2 .
 - (vi) Can you find electric field at C easily ?
- (take the required distances from A). Which charge will produce electric field here.

ES0051

14. The figure shows a conducting sphere 'A' of radius 'a' which is surrounded by a neutral conducting spherical shell B of radius 'b' ($b > a$). Initially switches S_1 , S_2 and S_3 are open and sphere 'A' carries a charge Q . First the switch ' S_1 ' is closed to connect the shell B with the ground and then opened. Now the switch ' S_2 ' is closed so that the sphere 'A' is grounded and then S_2 is opened. Finally, the switch ' S_3 ' is closed to connect the spheres together. Find the heat (in joule) which is produced after closing the switch S_3 . [Consider $b = 4$ cm, $a = 2$ cm and $Q = 80 \mu\text{C}$]

**ES0052**

- 15.** Consider a metal sphere, of radius R that is cut in two along a plane whose minimum distance from the sphere's centre is h . Sphere is uniformly charged by a total electric charge Q . What force is necessary to hold the two parts of the sphere together ?

**ES0053**

- 16.** Two fixed charges $-2Q$ and Q are located at the points with co-ordinates $(-3a, 0)$ and $(3a, 0)$ respectively in the x - y plane. (a) Show that all the points in the x - y plane where the electric potential due to the two charges is zero lie on a circle. Find its radius and the location of its centre. (b) Give the expression for the potential $V_{(x)}$ at a general point on the x -axis and sketch the function $V_{(x)}$ on the whole x -axis. (c) If a particle of charge $+q$ starts from rest at the centre of the circle, show by a short qualitative argument that the particle eventually crosses the circle. Find its speed when it does so.

ES0054

- 17.** The electric field strength depends only on the x , y and z coordinates according to the law

E = $\frac{a(\hat{x}i + \hat{y}j + \hat{z}k)}{(x^2 + y^2 + z^2)^{3/2}}$, where a = 122.5 SI unit and is a constant. Find the potential difference (in volt)

between $(3, 2, 6)$ and $(0, 3, 4)$.

ES0055

EXERCISE (O-1)

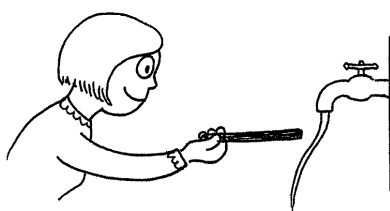
SINGLE CORRECT TYPE QUESTIONS

Properties of charges and coulomb's law

- If an object made of substance A rubs an object made of substance B, then A becomes positively charged and B becomes negatively charged. If, however, an object made of substance A is rubbed against an object made of substance C, then A becomes negatively charged. What will happen if an object made of substance B is rubbed against an object made of substance C?
 (A) B becomes positively charged and C becomes positively charged.
 (B) B becomes positively charged and C becomes negatively charged.
 (C) B becomes negatively charged and C becomes positively charged.
 (D) B becomes negatively charged and C becomes negatively charged.

ES0056

- In normal cases thin stream of water bends toward a negatively charged rod. When a positively charged rod is placed near the stream, it will bend in the



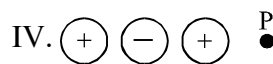
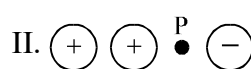
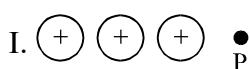
- (A) Opposite direction. (B) Same direction.
 (C) It won't bend at all. (D) Can't be predicted.

ES0057

- Two charged bodies A and B exert repulsive forces on each other. If charge on A is more than that on B, which of the following statement is true.
 (A) Body A experiences more Colombian force than B.
 (B) Body A experiences less Colombian force than B.
 (C) Both of them experience Colombian forces of equal magnitude.
 (D) It depends whether the bodies can be treated as point like charges or not.

ES0058

- Given are four arrangements of three fixed electric charges. In each arrangement, a point labeled P is also identified — test charge, $+q$, is placed at point P. All of the charges are of the same magnitude, Q , but they can be either positive or negative as indicated. The charges and point P all lie on a straight line. The distances between adjacent items, either between two charges or between a charge and point P, are all the same. Correct order of choices in a decreasing order of magnitude of force on P is



- (A) $II > I > III > IV$ (B) $I > II > III > IV$ (C) $II > I > IV > III$ (D) $III > IV > I > II$

ES0059

5. Two point charge of $100 \mu\text{C}$ and $-4 \mu\text{C}$ are positioned at points $(-2\sqrt{3}, 3\sqrt{3}, -4)$ and $(4\sqrt{3}, -5\sqrt{3}, 6)$ respectively of a Cartesian coordinate system. Find the force vector on the $-4 \mu\text{C}$ charge? All the coordinates are in meters.

- (A) $9 \times 10^{-4} (3\sqrt{3}\hat{i} - 4\sqrt{3}\hat{j} + 5\hat{k})$
 (B) $9 \times 10^{-4} (-3\sqrt{3}\hat{i} + 4\sqrt{3}\hat{j} - 5\hat{k})$
 (C) $2.25 \times 10^{-4} (-3\sqrt{3}\hat{i} + 4\sqrt{3}\hat{j} - 5\hat{k})$
 (D) $2.25 \times 10^{-4} (3\sqrt{3}\hat{i} - 4\sqrt{3}\hat{j} + 5\hat{k})$

ES0060

Electric field

6. Five positive equal charges are placed at vertices of a regular hexagon and net electric field at the centre is E_1 . A negative charge having equal magnitude is placed sixth vertex and then net electric

field is E_2 . Find $\frac{E_2}{E_1}$.

- (A) 2 (B) 1 (C) 3 (D) None of these

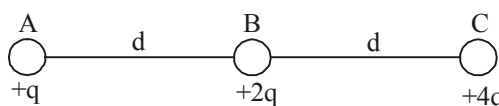
ES0061

7. There are two point charges q_1 and q_2 lying on a circle of unit radius. Electric field intensity at the center of circle due to these charges is $\vec{E}$. Find the position vector of the center with respect to q_2 if the position vector of the center with respect to q_1 is $\vec{r}_1$.

- (A) $\frac{\vec{E} + kq_1\vec{r}_1}{kq_2}$ (B) $-\left(\frac{\vec{E} + kq_1\vec{r}_1}{kq_2}\right)$
 (C) $\frac{kq_1\vec{r}_1 - \vec{E}}{kq_2}$ (D) $\frac{\vec{E} - kq_1\vec{r}_1}{kq_2}$

ES0062

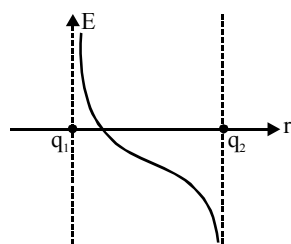
8. Three charges $+q$, $+2q$ and $+4q$ are connected by strings as shown in the figure. What is ratio of tensions in the strings AB and BC.



- (A) 1 : 2 (B) 1 : 3 (C) 2 : 1 (D) 3 : 1

ES0063

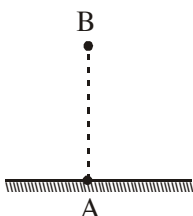
9. The variation of electric field between the two charges q_1 and q_2 along the line joining the charges is plotted against distance from q_1 (taking rightward direction of electric field as positive) as shown in the figure. Then the correct statement is :-



- (A) q_1 and q_2 are positive and $q_1 < q_2$ (B) q_1 and q_2 are positive and $q_1 > q_2$
(C) q_1 is positive and q_2 is negative and $q_1 < q_2$ (D) q_1 and q_2 are negative and $q_1 < q_2$

ES0064

10. Particle B of charge Q and mass m is in equilibrium under weight and electrostatics force applied by a fixed charged A , which is directly beneath the particle B as shown in figure. When particle B is disturbed along vertical, the equilibrium is



- (A) stable (B) unstable
(C) neutral (D) there can not be in equilibrium

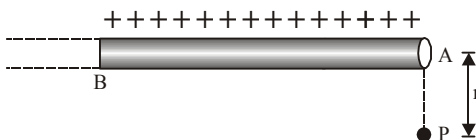
ES0065

11. A charge q is placed at the centroid of an equilateral triangle. Three equal charges Q are placed at the vertices of the triangle. The system of four charges will be in equilibrium if q is equal to :-

- (A) $\frac{-Q}{\sqrt{3}}$ (B) $\frac{-Q}{3}$ (C) $-Q\sqrt{3}$ (D) $\frac{Q}{\sqrt{3}}$

ES0066

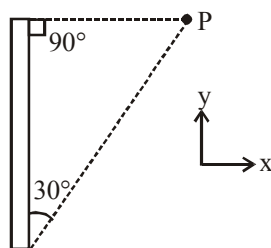
12. A semi-infinite insulating rod has linear charge density λ . The electric field at the point P shown in figure is :-



- (A) $\frac{2\lambda^2}{(4\pi\epsilon_0 r)^2}$ at 45° with AB (B) $\frac{\sqrt{2}\lambda}{4\pi\epsilon_0 r^2}$ at 45° with AB
(C) $\frac{\sqrt{2}\lambda}{4\pi\epsilon_0 r}$ at 45° with AB (D) $\frac{\sqrt{2}\lambda}{4\pi\epsilon_0 r}$ at perpendicular to AB

ES0067

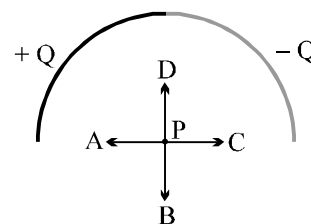
13. The direction (θ) of $\vec{E}$ at point P due to uniformly charged finite rod will be :-



- (A) at angle 30° from x-axis
(B) 45° from x - axis
(C) 60° from x-axis
(D) none of these

ES0068

14. As shown in the figure, an insulating rod is set into the shape of a semicircle. The left half of the rod has a charge of $+Q$ uniformly distributed along its length, and the right half of the rod has a charge of $-Q$ uniformly distributed along its length. What vector shows the correct direction of the electric field at point P, the centre of the semicircle ?



- (A) A
(B) B
(C) C
(D) D

ES0069

15. A nonconducting ring of radius R has uniformly distributed positive charge Q . A small part of the ring, of length d , is removed ($d \ll R$). The electric field at the centre of the ring will now be
(A) directed towards the gap, inversely proportional to R^3 .
(B) directed towards the gap, inversely proportional to R^2 .
(C) directed away from the gap, inversely proportional to R^3 .
(D) directed away from the gap, inversely proportional to R^2 .

ES0070

16. The maximum electric field at a point on the axis a uniformly charged ring is E_0 . At how many points on the axis will the magnitude of electric field be $E_0/2$:-
(A) 1
(B) 2
(C) 3
(D) 4

ES0071

17. A particle of mass m , charge $-Q$ is constrained to move along the axis of a ring of radius a . The ring carries a uniform charge density $+\lambda$ along its circumference. Initially, the particle lies in the plane of the ring at a point where no net force acts on it. The period of oscillation of the particle when it is displaced slightly from its equilibrium position is

- (A) $T = 4\pi \sqrt{\frac{\epsilon_0 m a^2}{\lambda Q}}$
(B) $T = 2\pi \sqrt{\frac{2\epsilon_0 m a^2}{\lambda Q}}$
(C) $T = 2\pi \sqrt{\frac{4\epsilon_0 m a^2}{\lambda Q}}$
(D) $T = 2\pi \sqrt{\frac{\epsilon_0 m a^2}{2\lambda Q}}$

ES0072

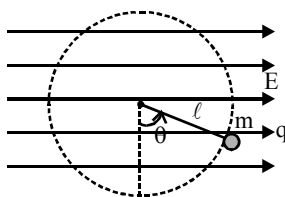
18. The surface charge density of a thin charged disc of radius R is σ . The value of the electric field at the centre of the disc is $\frac{\sigma}{2\epsilon_0}$. With respect to the field at the centre, the electric field along the axis at a

distance $\sqrt{3}R$ from the centre of the disc :

- (A) reduces by 70.7% (B) reduces by 29.3% (C) reduces by 86.6% (D) reduces by 13.4%

ES0073

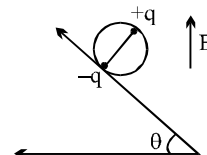
19. A small ball of mass m and charge $+q$ tied with a string of length ℓ , is rotating in a vertical circle under gravity and a uniform horizontal electric field E as shown. The tension in the string will be minimum for:-



- (A) $\theta = \tan^{-1}\left(\frac{qE}{mg}\right)$ (B) $\theta = \pi$ (C) $\theta = \pi - \tan^{-1}\left(\frac{qE}{mg}\right)$ (D) $\theta = \pi + \tan^{-1}\left(\frac{qE}{mg}\right)$

ES0074

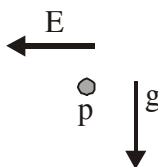
20. A wheel having mass m has charges $+q$ and $-q$ on diametrically opposite points. It remains in equilibrium on a rough inclined plane in the presence of uniform vertical electric field $E =$

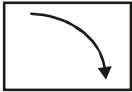
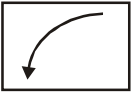
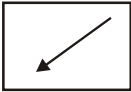
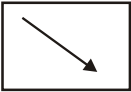


- (A) $\frac{mg}{q}$ (B) $\frac{mg}{2q}$ (C) $\frac{mg \tan \theta}{2q}$ (D) none

ES0075

21. A negatively charged particle p is placed, initially at rest, in a constant, uniform gravitational field and a constant, uniform electric field as shown in the diagram. What qualitatively, is the shape of the trajectory of the electron ?



- (A)  (B)  (C)  (D) 

ES0076

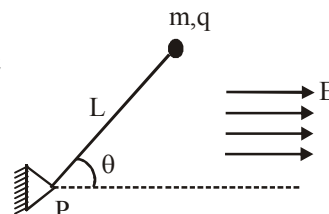
22. A particle of mass m and charge q is attached to a light rod of length L . The rod can rotate freely in the plane of paper about the other end, which is hinged at P . The entire assembly lies in a uniform electric field E also acting in the plane of paper as shown. The rod is released from rest when it makes an angle θ with the electric field direction. Determine the speed of the particle when the rod is parallel to the electric field.

(A) $\left(\frac{2qEL(1 - \cos \theta)}{m}\right)^{1/2}$

(B) $\left(\frac{2qEL(1 - \sin \theta)}{m}\right)^{1/2}$

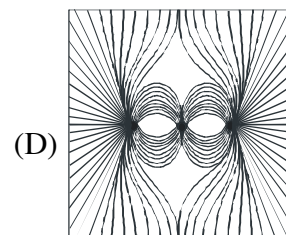
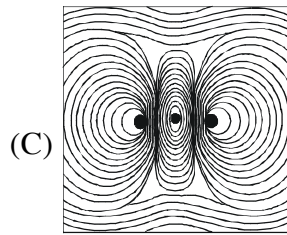
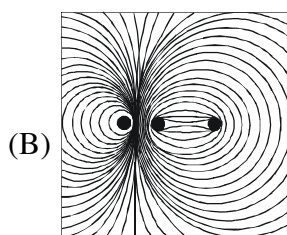
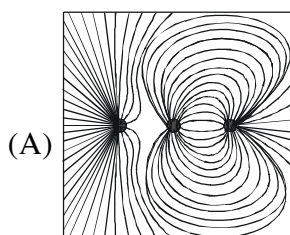
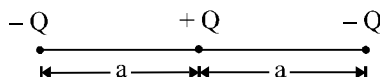
(C) $\left(\frac{qEL(1 - \cos \theta)}{2m}\right)^{1/2}$

(D) $\left(\frac{2qEL \cos \theta}{m}\right)^{1/2}$



ES0077

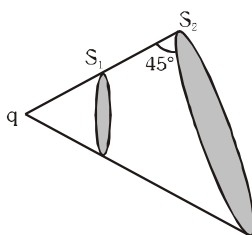
23. The fig. shows the distribution of three charges $-Q$, $+Q$ and $-Q$ on the X -axis. Which of the following figures shows the possible electric field lines for the distribution?



ES0078

Gauss' law

24. In the given figure flux through surface S_1 is ϕ_1 & through S_2 is ϕ_2 . Which is correct ?



(A) $\phi_1 = \phi_2$

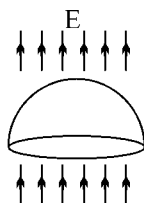
(B) $\phi_1 > \phi_2$

(C) $\phi_1 < \phi_2$

(D) None of these

ES0079

- 25.** A hemispherical surface (half of a spherical surface) of radius R is located in a uniform electric field E that is parallel to the axis of the hemisphere. What is the magnitude of the electric flux through the hemisphere surface?

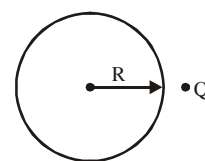


- (A) 0 (B) $4\pi R^2 E/3$ (C) $2\pi R^2 E$ (D) $\pi R^2 E$

ES0080

- 26. Statement 1:** A charge is outside the Gaussian sphere of radius R . Then electric field on the surface of sphere is zero.

and

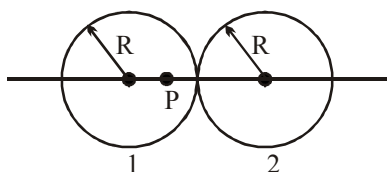


Statement 2 : As $\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$, for the sphere q_{in} is zero, so $\oint \vec{E} \cdot d\vec{s} = 0$.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false.
(D) Statement-1 is false, statement-2 is true.

ES0081

27. Figure shows, in cross section, two solid spheres with uniformly distributed charge throughout their volumes. Each has radius R . Point P lies on a line connecting the centres of the spheres, at radial distance $R/2$ from the center of sphere 1. If the net electric field at point P is zero and Q_1 is $64 \mu\text{C}$, what is Q_2 (in μC).



- (A) 64 (B) 36 (C) 32 (D) 72

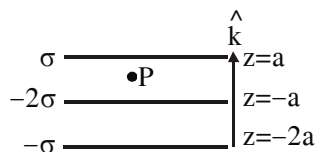
ES0082

- 28.** A sphere of radius R carries charge density proportional to the square of the distance from the center: $\rho = Ar^2$, where A is a positive constant. At a distance of R/2 from the center, the magnitude of the electric field is :-

- (A) $A/4\pi\epsilon_0$ (B) $AR^3/40\epsilon_0$ (C) $AR^3/24\epsilon_0$ (D) $AR^3/5\epsilon_0$

ES0083

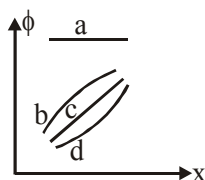
29. Three large parallel plates have uniform surface charge densities as shown in the figure. What is the electric field at P ? [IIT-JEE 2005 (Scr)]



- (A) $-\frac{4\sigma}{\epsilon_0} \hat{k}$ (B) $\frac{4\sigma}{\epsilon_0} \hat{k}$ (C) $-\frac{2\sigma}{\epsilon_0} \hat{k}$ (D) $\frac{2\sigma}{\epsilon_0} \hat{k}$

ES0084

30. A line of charge extends along a X-axis whose linear charge density varies directly as x . Imagine a spherical volume with its centre located on X-axis and is moving gradually along it. Which of the graphs shown in figure correspond to the flux ϕ with the x coordinate of the centre of the volume?



- (A) a (B) b (C) c (D) d

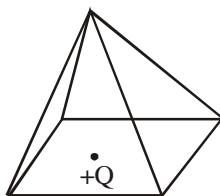
ES0085

31. The electric field in a region is given by $\vec{E} = 200\hat{i}$ N/C for $x > 0$ and $-200\hat{i}$ N/C for $x < 0$. A closed cylinder of length 2m and cross-section area 10^2 m^2 is kept in such a way that the axis of cylinder is along X-axis and its centre coincides with origin. The total charge inside the cylinder is [Take : $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2\text{m}^2.\text{N}$]

- (A) 0 (B) $1.86 \times 10^{-5} \text{ C}$ (C) $1.77 \times 10^{-11} \text{ C}$ (D) $35.4 \times 10^{-8} \text{ C}$

ES0086

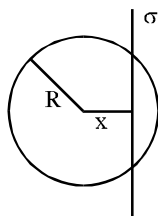
32. A point charge $+Q$ is positioned at the center of the base of a square pyramid as shown. The flux through one of the four identical upper faces of the pyramid is :-



- (A) $\frac{Q}{16\epsilon_0}$ (B) $\frac{Q}{4\epsilon_0}$ (C) $\frac{Q}{8\epsilon_0}$ (D) None of these

ES0087

- 33.** An infinite, uniformly charged sheet with surface charge density σ cuts through a spherical Gaussian surface of radius R at a distance x from its center, as shown in the figure. The electric flux Φ through the Gaussian surface is :-



- (A) $\frac{\pi R^2 \sigma}{\epsilon_0}$ (B) $\frac{2\pi(R^2 - x^2) \sigma}{\epsilon_0}$ (C) $\frac{\pi(R - x)^2 \sigma}{\epsilon_0}$ (D) $\frac{\pi(R^2 - x^2) \sigma}{\epsilon_0}$

ES0088

- 34.** Consider a circle of radius R . A point charge lies at a distance ' a ' from its center and on its axis such that $R = a\sqrt{3}$. If electric flux passing through the circle is ϕ then the magnitude of the point charge is:-

- (A) $\sqrt{3}\varepsilon_0\phi$ (B) $2\varepsilon_0\phi$ (C) $4\varepsilon_0\phi/\sqrt{3}$ (D) $4\varepsilon_0\phi$

ES0089

Electric potential energy and electric potential

- 35.** Two particles X and Y, of equal mass and with unequal positive charges, are free to move and are initially far away from each other. With Y at rest, X begins to move towards it with initial velocity u . After a long time, finally :-
- (A) X will stop, Y will move with velocity u .
(B) X and Y will both move with velocities $u/2$ each.
(C) X will stop, Y will move with velocity $< u$.
(D) both will move with velocities $< u/2$.

ES0090

- 36.** Two positively charged particles X and Y are initially far away from each other and at rest. X begins to move towards Y with some initial velocity. The total momentum and energy of the system are p and E .
- (A) If Y is fixed, both p and E are conserved.
- (B) If Y is fixed, E is conserved, but not p .
- (C) If both are free to move, p is conserved but not E .
- (D) If both are free, E is conserved, but not p .

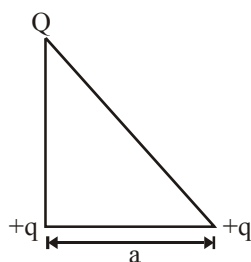
ES0091

37. Potential energy of a system comprising of point charges is U_1 . When a charge q is added in the system without disturbing other charges, the potential energy becomes U_2 . The potential of the point where the charge q is placed in the system is

(A) $\frac{U_2 - U_1}{q}$ (B) $\frac{U_1 - U_2}{q}$ (C) $\frac{U_1 + U_2}{2q}$ (D) $\frac{U_2 - U_1}{2q}$

ES0092

38. Three charges Q , $+q$ and $+q$ are placed at the vertices of a right-angled isosceles triangle as shown. The net electrostatic energy of the configuration is zero if Q is equal to : [JEE 2000(Scr) 1 + 1]



(A) $\frac{-q}{1+\sqrt{2}}$ (B) $\frac{-2q}{2+\sqrt{2}}$ (C) $-2q$ (D) $+q$

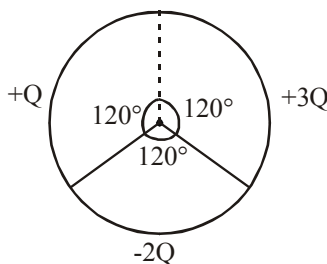
ES0093

39. Two fixed charges A and B of $5 \mu\text{C}$ each are separated by a distance of 6m . C is the mid point of the line joining A and B. A charge 'Q' of $-5 \mu\text{C}$ is shot perpendicular to the line joining A and B through C with a kinetic energy of 0.06J . The charge 'Q' comes to rest at a point D. The distance CD is:-

(A) 3 m (B) $\sqrt{3}\text{ m}$ (C) $3\sqrt{3}\text{ m}$ (D) 4 m

ES0094

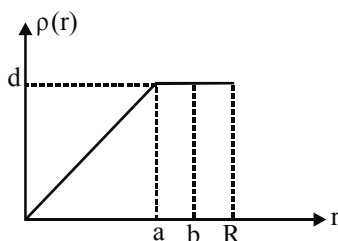
40. Figure shows three circular arcs, each of radius R and total charge as indicated. The net electric potential at the centre of curvature is :-



(A) $\frac{Q}{4\pi\epsilon_0 R}$ (B) $\frac{Q}{2\pi\epsilon_0 R}$
 (C) $\frac{2Q}{\pi\epsilon_0 R}$ (D) $\frac{Q}{\pi\epsilon_0 R}$

ES0095

41. The nuclear charge (Ze) is non-uniformly distributed within a nucleus of radius R . The charge density $\rho(r)$ [charge per unit volume] is dependent on the radial distance r from the centre of the nucleus as shown in figure. Select correct alternative/s.



- (A) Electric field at $r = R$ is independent of b
 (B) Electric potential at $r = R$ is proportional to b
 (C) Electric field at $r = R$ is proportional to a
 (D) Electric potential at $r = R$ is proportional to a

ES0096

42. A solid sphere of radius R is charged uniformly. At what distance from its surface is the electrostatic potential half of the potential at the centre?

- (A) R (B) $R/2$ (C) $R/3$ (D) $2R$

ES0097

43. When a negative charge is released and moves in electric field, it moves toward a position of

- (A) lower electric potential and lower potential energy
 (B) lower electric potential and higher potential energy
 (C) higher electric potential and lower potential energy
 (D) higher electric potential and higher potential energy

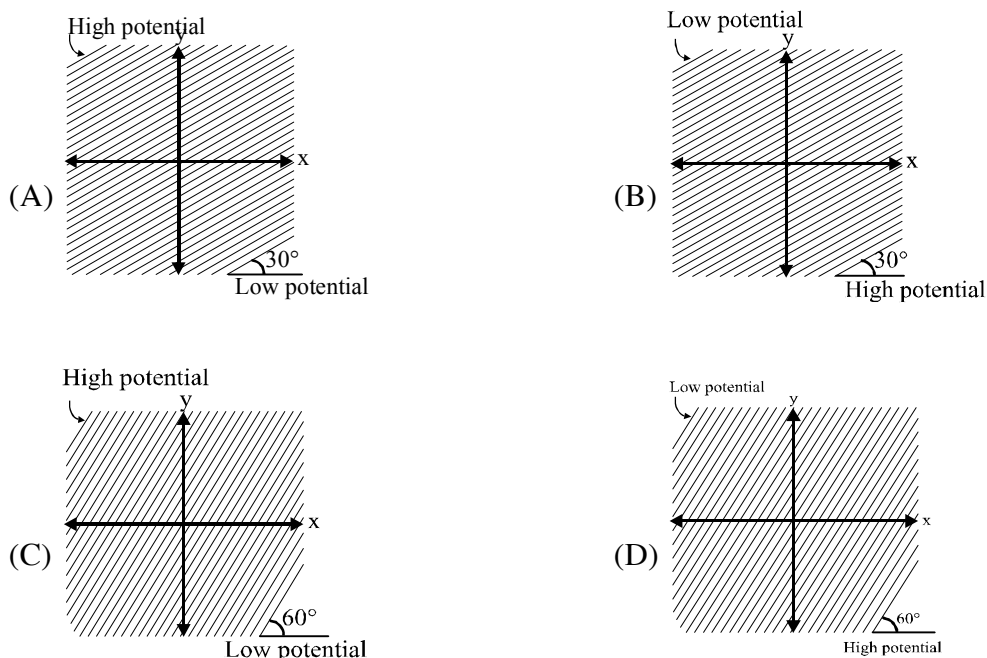
ES0098

44. A positively charged particle is released from rest in an uniform electric field. The electric potential energy of the charge

- (A) remains a constant because the electric field is uniform.
 (B) increases because the charge moves along the electric field.
 (C) decreases because the charge moves along the electric field.
 (D) decreases because the charge moves opposite to the electric field.

ES0099

45. The electric field intensity at all points in space is given by $\vec{E} = \sqrt{3}\hat{i} - \hat{j}$ volts/metre. The nature of equipotential lines in x-y plane is given by :-

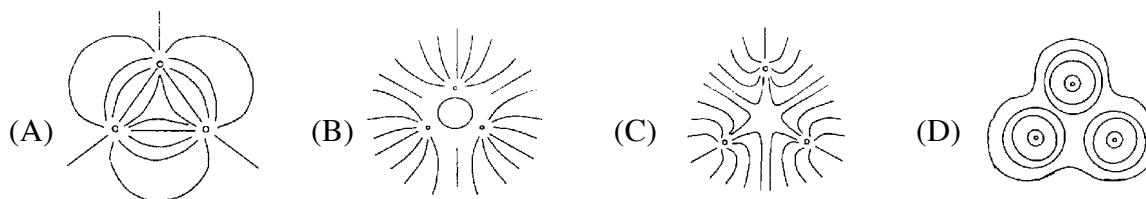


ES0100

46. Equipotential at a great distance from a collection of charges whose total sum is not zero are approximately
 (A) spheres. (B) planes.
 (C) paraboloids (D) ellipsoids.

ES0101

47. Three positive charges of equal value q are placed at the vertices of an equilateral triangle. The resulting lines of force should be sketched as in [JEE 2001 (Scr)]



ES0102

48. In an electric field the potential at a point is given by the following relation $V = \frac{343}{r}$. The electric field at $\vec{r} = 3\hat{i} + 2\hat{j} + 6\hat{k}$ is :

(A) $21\hat{i} + 14\hat{j} + 42\hat{k}$ (B) $3\hat{i} + 2\hat{j} + 6\hat{k}$ (C) $\frac{1}{7}(3\hat{i} + 2\hat{j} + 6\hat{k})$ (D) $-(3\hat{i} + 2\hat{j} + 6\hat{k})$

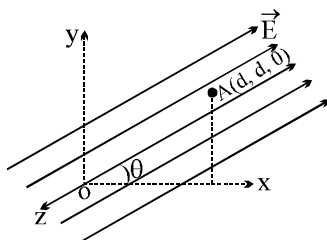
ES0103

49. From a point if we move in a direction making an angle θ measured from +ve x-axis, the potential gradient is given as $\frac{dv}{dr} = 2 \cos \theta$. Find the direction and magnitude of electric field at that point.

(A) $2\hat{i}$ (B) $-2\hat{i}$ (C) $\hat{i} + \hat{j}$ (D) $-\hat{i} + \hat{j}$

ES0104

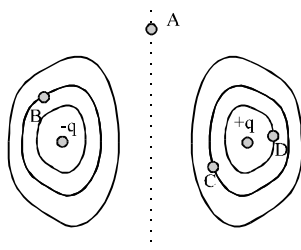
50. A uniform electric field having strength $\vec{E}$ is existing in x-y plane as shown in figure. Find the p.d. between origin O & A(d, d, 0)



(A) $Ed(\cos\theta + \sin\theta)$ (B) $-Ed(\sin\theta - \cos\theta)$ (C) $\sqrt{2} Ed$ (D) none of these

ES0105

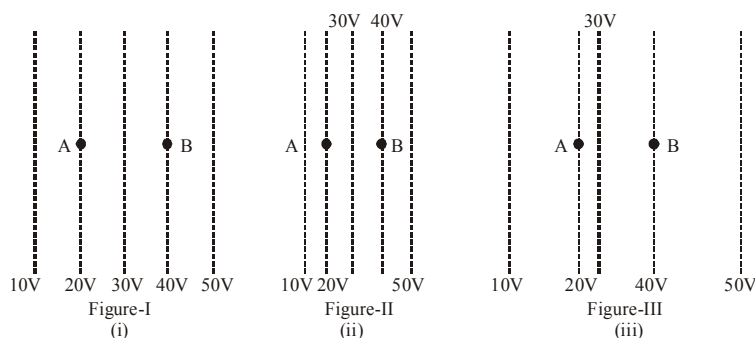
51. Figure shows equi-potential surfaces for a two charges system. At which of the labeled points point will an electron have the highest potential energy?



(A) Point A (B) Point B (C) Point C (D) Point D

ES0106

52. Figure shows some equipotential lines distributed in space. A charged object is moved from point A to point B.



- (A) The work done in Fig. (i) is the greatest.
 (B) The work done in Fig. (ii) is least.
 (C) The work done is the same in Fig. (i), Fig. (ii) and Fig. (iii).
 (D) The work done in Fig. (iii) is greater than Fig. (ii) but equal to that in Fig. (i).

ES0107

53. Consider the following conclusions regarding the components of an electric field at a certain point in space given by

$$E_x = -Ky \quad E_y = Kx$$

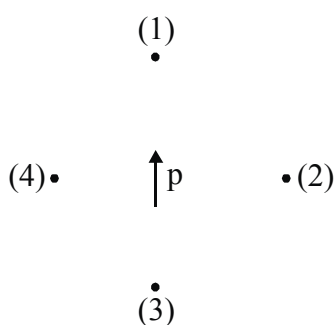
$$E_z = 0$$

- (I) The field is conservative. (II) The field is non-conservative.
 (III) The lines of force are straight lines (IV) The lines of force are circles.
 Of these conclusions
 (A) II and IV are valid (B) I and III are valid
 (C) I and IV are valid (D) II and III are valid

ES0108

Electric dipole

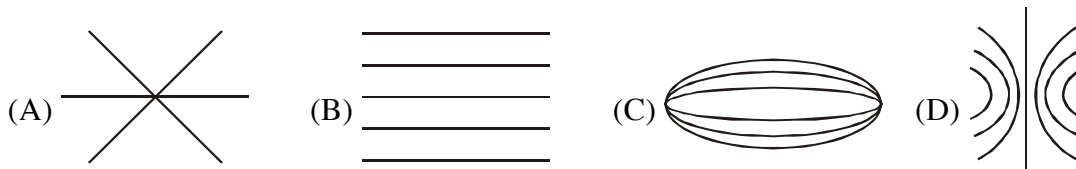
54. The drawing shows four points surrounding an electric dipole. Which one of the following expressions best ranks the electric potential at these four locations?



- (A) $1 > 2 = 4 > 3$ (B) $3 > 2 > 4 > 1$ (C) $3 > 2 = 4 > 1$ (D) $2 = 4 > 1 = 3$

ES0109

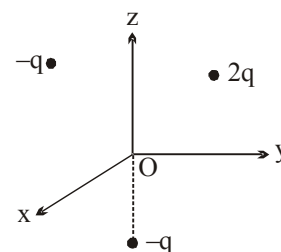
55. Which of the following represents the equipotential lines of a dipole (two equal and opposite charges placed at small separation)?



ES0110

56. Three point charges $2q$, $-q$ and $-q$ are located respectively at $(0, a, a)$, $(0, a, -a)$ and $(0, 0, -a)$ as shown. The dipole moment of this distribution is :-

- (A) $2qa$ in the y - z plane at $\tan^{-1} \left(\frac{1}{4} \right)$ with z -axis
 (B) $\sqrt{17}qa$ in the y - z plane at $\tan^{-1} \left(\frac{1}{4} \right)$ with z -axis
 (C) $\sqrt{5}qa$ in the x - y plane at $\tan^{-1} (4)$ with y -axis
 (D) $4qa$ in the x - y plane at $\tan^{-1} (4)$ with y -axis



ES0111

- ES0112**

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- ES0113**

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- A diagram showing a dipole consisting of two point charges, $-q$ and $+q$, separated by a distance p . The dipole is placed in a uniform electric field represented by five parallel horizontal lines pointing to the right. The $-q$ charge is on the left and the $+q$ charge is on the right, with an arrow labeled p indicating the separation between them.

-

- ES0115**

61. An electric dipole (dipole moment p) is placed at a radial distance $r \gg a$ (where a is dipole length) from an infinite line of charge having linear charge density $+\lambda$. Dipole moment vector is aligned along radial vector $\vec{r}$ force experienced by dipole is :-

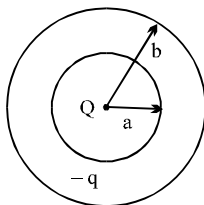
- (A) $\frac{\lambda p}{2\pi\epsilon_0 r^2}$, attractive (B) $\frac{\lambda p}{2\pi\epsilon_0 r^3}$, attractive
(C) $\frac{\lambda p}{2\pi\epsilon_0 r^2}$, repulsive (D) $\frac{\lambda p}{2\pi\epsilon_0 r^3}$, repulsive

ES0116

Conductors

62. Both question (a) and (b) refer to the system of charges as shown in the figure. A spherical shell with an inner radius 'a' and an outer radius 'b' is made of conducting material. A point charge $+Q$ is placed at the centre of the spherical shell and a total charge $-q$ is placed on the shell.

a. Charge $-q$ is distributed on the surfaces as



- (A) $-Q$ on the inner surface, $-q$ on outer surface
(B) $-Q$ on the inner surface, $-q + Q$ on the outer surface
(C) $+Q$ on the inner surface, $-q - Q$ on the outer surface
(D) The charge $-q$ is spread uniformly between the inner and outer surface.

b. Assume that the electrostatic potential is zero at an infinite distance from the spherical shell. The

electrostatic potential at a distance R ($a < R < b$) from the centre of the shell is (where $K = \frac{1}{4\pi\epsilon_0}$)

- (A) 0 (B) $\frac{KQ}{a}$ (C) $K \frac{Q-q}{R}$ (D) $K \frac{Q-q}{b}$

ES0117

63. The electrostatic potential on the surface of a charged conducting sphere is 100V. Two statements are made in this regard:

S_1 : At any point inside the sphere, electric intensity is zero.

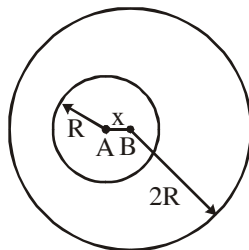
S_2 : At any point inside the sphere, the electrostatic potential is 100V.

Which of the following is a correct statement?

- (A) S_1 is true but S_2 is false.
(B) Both S_1 & S_2 are false.
(C) S_1 is true, S_2 is also true and S_1 is the cause of S_2 .
(D) S_1 is true, S_2 is also true but the statements are independent.

ES0118

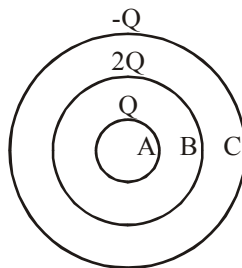
- 64.** Figure shows two shells of radii R and $2R$. The inner shell (centre at A) is nonconducting and uniformly charged with charge Q while the outer shell (centre at B) is conducting and uncharged. the potential at the point B is :-



- (A) zero (B) $\frac{KQ}{R}$ (C) $\frac{KQ}{x}$ (D) None of these

ES0119

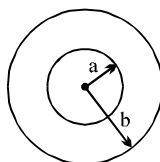
- 65.** Charge Q , $2Q$ and $-Q$ are given to three concentric conducting spherical shells A, B and C respectively. The ratio of charges on the inner and the outer surfaces of the shell 'C' will be :-



- (A) $\frac{3}{4}$ (B) $-\frac{3}{4}$ (C) $\frac{3}{2}$ (D) $-\frac{3}{2}$

ES0120

- 66.** If the electric potential of the inner metal sphere is 10 volt & that of the outer shell is 5 volt, then the potential at the centre will be :



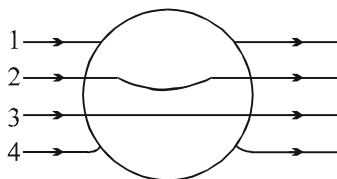
- (A) 10 volt (B) 5 volt (C) 15 volt (D) 0

ES0121

- 67.** n small drops of same size are charged to V volts each. If they coalesce to form a single large drop, then its potential will be :-
- (A) V/n (B) Vn (C) $Vn^{1/3}$ (D) $Vn^{2/3}$

ES0122

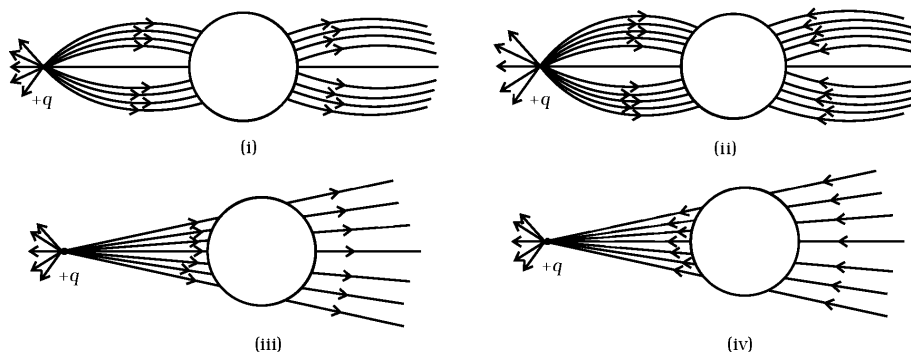
- 68.** A metallic solid sphere is placed in a uniform electric field. The lines of force follow the path (s) shown in figure as :



- (A) 1 (B) 2 (C) 3 (D) 4

ES0123

69. A point positive charge is brought near an isolated conducting sphere. The electric field is best given by



- (A) Fig (i) (B) Fig (iii) (C) Fig (ii) (D) Fig (iv)

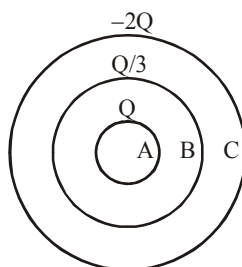
ES0124

70. There are two uncharged identical metallic spheres 1 and 2 of radius r separated by a distance d ($d \gg r$). A charged metallic sphere of same radius having charge q is touched with one of the sphere. After some time it is moved away from the system. Now the uncharged sphere is earthed. Charge on earthed sphere is

- (A) q (B) $-q$ (C) $-qr/2d$ (D) 0

ES0125

71. Three conducting concentric spherical shells of radius R , $2R$ and $3R$ have charges Q , $\frac{Q}{3}$ and $-2Q$ respectively. The intermediate shell is now grounded. Find the charge flow into the earth.

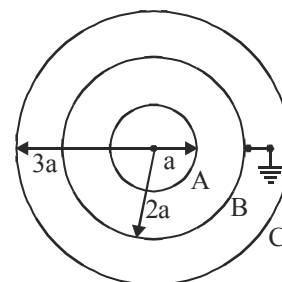


- (A) $\frac{Q}{3}$ (B) $\frac{2Q}{3}$ (C) Q (D) 0

ES0126

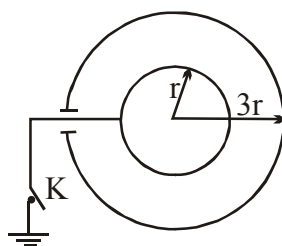
72. Figure shows a system of three concentric metal shells A, B and C with radii a , $2a$ and $3a$ respectively. Shell B is earthed and shell C is given a charge Q . Now if shell C is connected to shell A, then the final charge on the shell B, is equal to :

- (A) $-\frac{4Q}{13}$ (B) $-\frac{8Q}{11}$
(C) $-\frac{5Q}{3}$ (D) $-\frac{3Q}{7}$



ES0127

73. Figure shows two conducting thin concentric shells of radii r and $3r$. The outer shell carries charge q and inner shell is neutral. The amount of charge which flows from inner shell to the earth after the key K is closed, is equal to :-

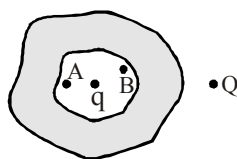


- (A) $-q/3$ (B) $q/3$ (C) $3q$ (D) $-3q$

ES0128

74. **Statement-1** : A point charge q is placed inside a cavity of conductor as shown. Another point charge Q is placed outside the conductor as shown. Now as the point charge Q is pushed away from conductor, the potential difference ($V_A - V_B$) between two points A and B within the cavity of sphere remains constant.

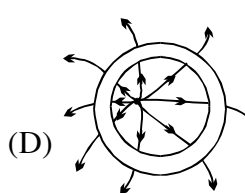
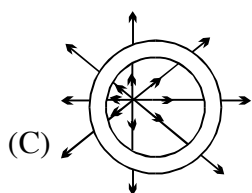
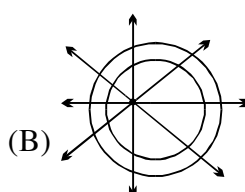
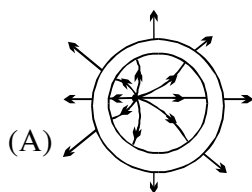
Statement-2 : The electric field due to charges on outer surface of conductor and outside the conductor is zero at all points inside the conductor.



- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

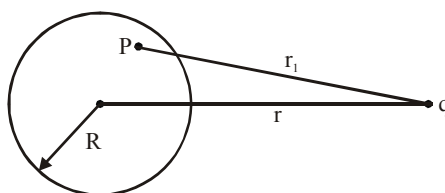
ES0129

75. A point charge q is placed at a point inside a hollow conducting sphere. Which of the following electric force pattern is correct ? [IIT-JEE'2003 (scr)]



ES0130

76. A point charge q is placed at a distance r from center of a conducting neutral sphere of radius R ($r > R$). The potential at any point P inside the sphere at a distance r_1 from point charge due to induced charge of the sphere is given by :-



- (A) $\frac{Kq}{r_1} - \frac{Kq}{R}$ (B) $\frac{Kq}{r_1} - \frac{Kq}{R}$ (C) $\frac{Kq}{r} - \frac{Kq}{r_1}$ (D) $-\frac{Kq}{r_1} + \frac{Kq}{R}$

ES0131

77. Consider a conductor with a spherical cavity in it. A point charge q_0 is placed at the centre of cavity and a point charge Q is placed outside conductor.

Statement-1 : Total charge induced on cavity wall is equal and opposite to the charge inside.

Statement-2 : If cavity is surrounded by a Gaussian surface, where all parts of Gaussian surface are located inside the conductor, $\oint \vec{E} \cdot d\vec{A} = 0$; hence $q_{\text{induced}} = -q_0$

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

ES0132

78. Two non-conducting hemispherical surfaces, which are having uniform charge density σ are placed on smooth horizontal surface as shown in figure. Assuming springs are ideal, calculate compression in each spring if both the hemispherical surface are just touching each other.

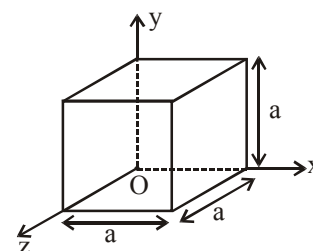


- (A) $\frac{\sigma^2 R^2}{2\epsilon_0 k}$ (B) R (C) $\frac{\sigma^2 \pi R^2}{2\epsilon_0 k}$ (D) None of these

ES0133

79. Electric field in a region is found to be $E = 3y\hat{j}$. The total energy stored in electric field inside the cube shown will be

- (A) $9a^5 \epsilon_0$ (B) $3 \epsilon_0 a^5$
 (C) $\frac{3}{2} \epsilon_0 a^5$ (D) 0



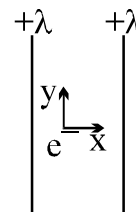
ES0134

MULTIPLE CORRECT TYPE QUESTIONS

- 80.** If a body is charged by rubbing it with another body then its weight :-
- (A) may increase slightly (B) may decrease slightly
- (C) must increase slightly (D) remains precisely constant

ES0135

- 81.** An electron is placed just in the middle between two long fixed line charges of charge density $+\lambda$ each. The wires are in the xy plane (Do not consider gravity)
- (A) The equilibrium of the electron will be unstable along x-direction
(B) The equilibrium of the electron will be neutral along y-direction
(C) The equilibrium of the electron will be stable along z-direction
(D) The equilibrium of the electron will be stable along y-direction

**ES0136**

- 82.** Which statement(s) concerning electrostatic fields is/are **CORRECT**?
- (A) Electric field lines never cross each other.
 - (B) Positive charge experiences force in the direction of electric field line.
 - (C) Electric field lines always start on negative charges and end on positive charges.
 - (D) Electric field lines that are closer indicate a stronger electric field while electric field lines that are far apart indicate a weaker electric field.

ES0137

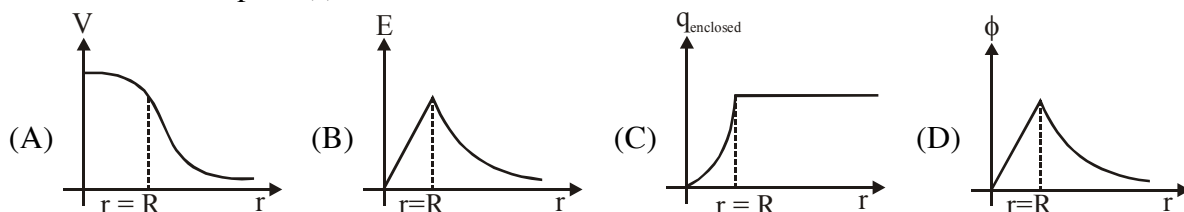
- 83.** Two point charges Q and $-\frac{Q}{4}$ are separated by distance L then



- (A) Potential is zero at a point on the axis which is at a distance $L/3$ on the right side of charge $-\frac{Q}{4}$
- (B) Potential is zero at a point on the axis which is at a distance $L/5$ on the left side of charge $-\frac{Q}{4}$
- (C) Electric field is zero at a point on the axis which is at a distance L on the right side of charge $-\frac{Q}{4}$
- (D) There exist two points on the axis, where electric field is zero

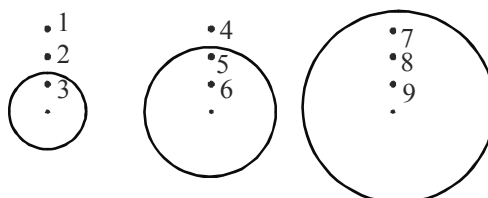
ES0138

84. Graph shows the variation of potential (V), magnitude of electric field (E), charge enclosed (q_{enclosed}) within the concentric sphere and net flux (ϕ) through a concentric spherical Gaussian surface as a function of distance (r) from centre of a uniformly positive charged solid sphere of radius R . Choose the **CORRECT** option(s):-



ES0139

85. Figure shows three spherical shells in separate situations, with each shell having the same uniform positive net charge. Points 1, 4 and 7 are at the same radial distances from the centre of the their respective shells; so are points 2, 5 and 8 ; and so are points 3, 6 and 9. With the electric potential taken equals to zero at an infinite distance, choose correct statement.



- (A) Point 3 has highest potential
(B) point 1, 4 and 7 are at same potential
(C) Point 9 has lowest potential
(D) point 5 and 8 are at same potential

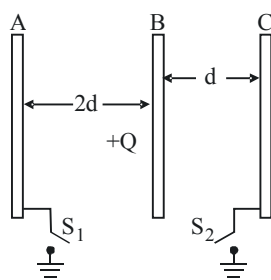
ES0140

86. Which of the following represent(s) an electrostatic field :-

- (A) $\vec{E} = y\hat{i} + x\hat{j}$
(B) $\vec{E} = y\hat{i}$
(C) $\vec{E} = y\hat{i} + x\hat{j} + z\hat{k}$
(D) $\vec{E} = \alpha\vec{r}$ (α is constant)

ES0141

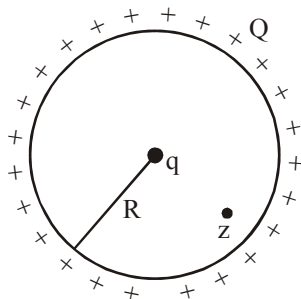
87. Three identical, parallel conducting plates A, B and C are placed as shown. Switches S_1 and S_2 are open, and can connect A and C to earth when closed. $+Q$ charge is given to B.



- (A) If S_1 is closed with S_2 open, a charge of amount Q will pass through S_1
(B) If S_2 is closed with S_1 open, a charge of amount Q will pass through S_2
(C) If S_1 and S_2 are closed together, a charge of amount $Q/3$ will pass through S_1 , and a charge of amount $2Q/3$ will pass through S_2 .
(D) All the above statements are incorrect

ES0142

88. A positive charge Q is uniformly distributed along a circular ring of radius R . A small test charge q is placed at the centre of the ring. Then



- (A) If $q > 0$ and is displaced away from the centre in the plane of the ring, it will be pushed back towards the centre.
 (B) If $q < 0$ and is displaced away from the centre in the plane of the ring, it will never return to the centre and will continue moving till it hits the ring.
 (C) If $q < 0$, it will perform SHM for small displacement along the axis.
 (D) q at the centre of the ring is in an unstable equilibrium within the plane of the ring for $q > 0$.

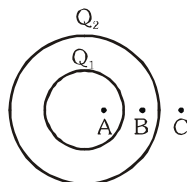
ES0143

89. If there were only one type of charge in the universe, then

- (A) $\oint_s \mathbf{E} \cdot d\mathbf{S} \neq 0$ on any surface.
 (B) $\oint_s \mathbf{E} \cdot d\mathbf{S} = 0$ if the charge is outside the surface.
 (C) $\oint_s \mathbf{E} \cdot d\mathbf{S}$ could not be defined.
 (D) $\oint_s \mathbf{E} \cdot d\mathbf{S} = \frac{q}{\epsilon_0}$ if charges of magnitude q were inside the surface.

ES0144

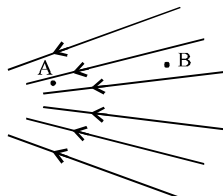
90. Figure shows two uniform charged concentric spherical shell. Both charges are positive, Select correct statement



- (A) Electric field intensity at B may be greater than electric field intensity at C.
 (B) Electric field intensity at B must be greater than electric field intensity at C.
 (C) Potential at A greater than potential at B
 (D) If a charge moves from B to C work done by electric force must be positive.

ES0145

91. Which of the following is true for the figure showing electric lines of force? (E is electrical field, V is potential)



- (A) $E_A > E_B$ (B) $E_B > E_A$ (C) $V_A > V_B$ (D) $V_B > V_A$

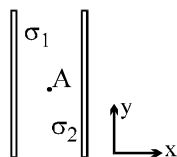
ES0146

92. Consider a uniform electric field in the $\hat{z}$ direction. The potential is a constant

- (A) in all space. (B) for any x for a given z .
(C) for any y for a given z . (D) on the x - y plane for a given z .

ES0147

93. Two large conducting sheets are kept parallel to each other as shown. In equilibrium, the charge density on facing surfaces is σ_1 and σ_2 . What is the value of electric field at A.



- (A) $\frac{\sigma_1}{\epsilon_0} \hat{i}$ (B) $-\frac{\sigma_2}{\epsilon_0} \hat{i}$ (C) $\frac{\sigma_1 + \sigma_2}{2\epsilon_0} \hat{i}$ (D) $\frac{\sigma_1 - \sigma_2}{2\epsilon_0} \hat{i}$

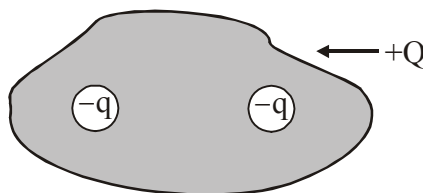
ES0148

94. A hollow closed conductor of irregular shape is given some charge. Which of the following statements are correct ?

- (A) The entire charge will appear on its outer surface.
(B) All points on the conductor will have the same potential.
(C) All points on its surface will have the same charge density.
(D) All points just outside it will have the same electric intensity.

ES0149

95. A conducting body is given charge Q and charge $-q$ has been placed in each of the cavity, which of the following statements is/are true?



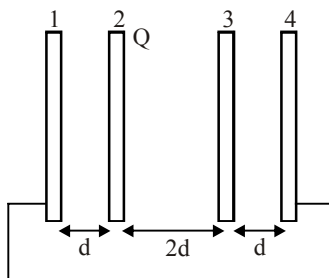
- (A) If $Q = 2q$, then conducting body will be at zero potential.
(B) If an external electric field is applied then the charge distribution on the outer surface of conductor would change.
(C) The potential of any point inside the cavity is less than that of conducting body.
(D) None of these

ES0150

COMPREHENSION TYPE QUESTIONS

Paragraph for Question No. 96 to 98

Four metallic plates are placed as shown in the figure. Plate 2 is given a charge Q whereas all other plates are uncharged. Plates 1 and 4 are joined together. The area of each plate is same.



96. The charge appearing on the right side of plate 3 is
 (A) 0 (B) $+Q/4$ (C) $-3Q/4$ (D) $Q/2$
97. The charge appearing on right side of plate 4 is
 (A) 0 (B) $-Q/4$ (C) $-3Q/4$ (D) $+Q/2$
98. The potential difference between plates 1 and 2 is
 (A) $\frac{3}{2} \frac{Qd}{\epsilon_0 A}$ (B) $\frac{Qd}{\epsilon_0 A}$ (C) $\frac{3}{4} \frac{Qd}{\epsilon_0 A}$ (D) $\frac{3Qd}{\epsilon_0 A}$

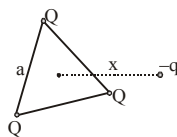
ES0151

ES0151

ES0151

Paragraph for Questions No. 99 to 101

Three charged particles each of $+Q$ are fixed at the corners of an equilateral triangle of side ' a '. A fourth particle of charge $-q$ and mass m is placed at a point on the line passing through centroid of triangle and perpendicular to the plane of triangle at a distance x from the centre of triangle.



99. Magnitude of resultant force on the fourth charged particle is
 (A) $\frac{1}{4\pi\epsilon_0} \frac{9\sqrt{3} Qqx}{(3x^2 + a^2)^{3/2}}$ (B) $\frac{1}{4\pi\epsilon_0} \frac{27\sqrt{3} Qqx}{(3x^2 + a^2)^{3/2}}$
 (C) $\frac{1}{4\pi\epsilon_0} \frac{2\sqrt{2} Qqx}{(2x^2 + a^2)^{3/2}}$ (D) $\frac{1}{4\pi\epsilon_0} \frac{4\sqrt{2} Qqx}{(2x^2 + a^2)^{3/2}}$

ES0152

100. Value of x for which the force is maximum is

- (A) $\frac{a}{\sqrt{3}}$ (B) $\frac{a}{\sqrt{2}}$ (C) $\frac{a}{\sqrt{6}}$ (D) $\frac{a}{\sqrt{5}}$

ES0152

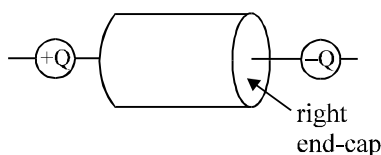
101. For small oscillation the period of oscillation of fourth particle is

- (A) $2\pi\sqrt{\frac{4\pi\epsilon_0 ma^3}{9\sqrt{3}Qq}}$ (B) $\pi\sqrt{\frac{4\pi\epsilon_0 ma^3}{9\sqrt{3}Qq}}$ (C) $2\pi\sqrt{\frac{2\pi\epsilon_0 ma^3}{27\sqrt{3}Qq}}$ (D) $2\pi\sqrt{\frac{\pi\epsilon_0 ma^3}{27\sqrt{3}Qq}}$

ES0152

Paragraph for Question No. 102 and 103

The figure applies to the following two questions. Positive and negative charges of equal magnitude lie along the symmetry axis of a cylinder. The distance from the positive charge to the left end-cap of the cylinder is the same as the distance from the negative charge to the right end -cap.



102. What is the flux of the electric field through the closed cylinder ?

- (A) 0 (B) $+Q/\epsilon_0$ (C) $+2Q/\epsilon_0$ (D) $-Q/\epsilon_0$

ES0153

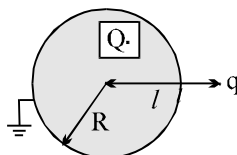
103. What is the sign of the flux through the right end-cap of the cylinder ?

- (A) Positive
(B) Negative
(C) There is no flux through the right end-cap.
(D) None of these

ES0153

Paragraph for Question No. 104 to 106

There is a cubical cavity inside a conducting sphere of radius R . A positive point charge Q is placed at the centre of the cube and another positive charge q is placed at a distance l ($>R$) from the centre of the sphere. The sphere is earthed



104. Charge induced on the inner surface of cavity is

- (A) $-Q$, uniformly distributed (B) $-Q$, non-uniformly distributed
(C) $-(Q+q)$ non-uniformly distributed (D) none

ES0154

105. Net charge on the outer surface of conducting sphere is is

(A) + Q

(B) $Q - qR/l$

(C) $-qR/l$

(D) none

ES0154

106. Potential at a point inside the cavity is

(A) zero

(B) positive

(C) negative

(D) can not be determined.

ES0154

MATRIX MATCH TYPE QUESTION

107. Column II corresponds to the graph of magnitude of electric field versus distance from centre of charge distribution in Column I.

Column-I

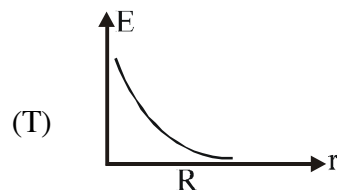
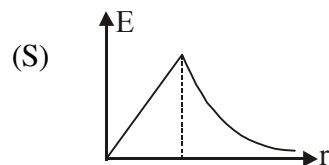
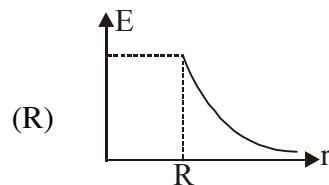
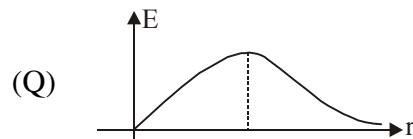
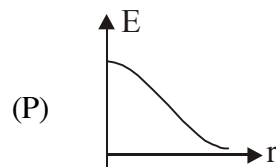
(A) Ring along its axis

(B) Uniformly charged solid sphere

(C) Uniformly charged spherical shell

(D) Combination of charge $+Q$ and $-Q$ at the perpendicular bisector

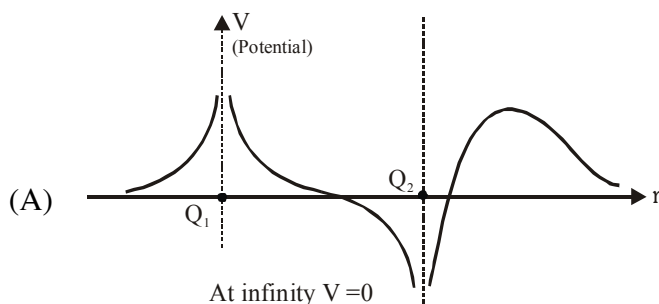
Column-II

**ES0155**

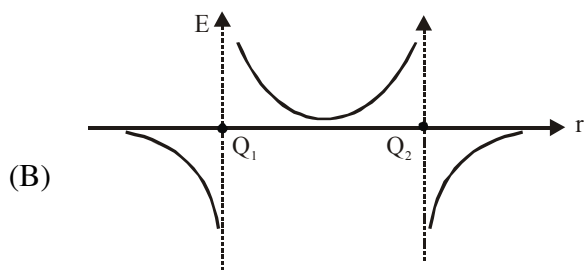
108. As shown in column I their are graphs of electric field (E) and potential (V) along the line joining charges Q_1 and Q_2 are drawn against distance (r) on x-axis for charges Q_1 and Q_2 . Take potential at infinity equal to zero. [Take direction of E in rightward direction as positive]

Column-I

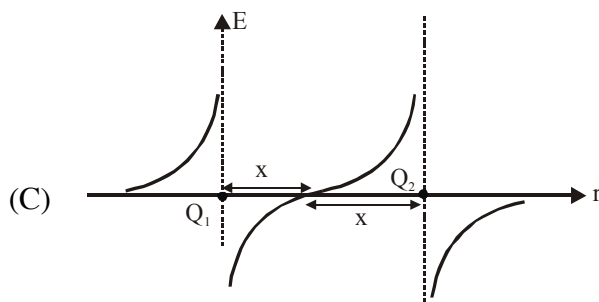
Column-II



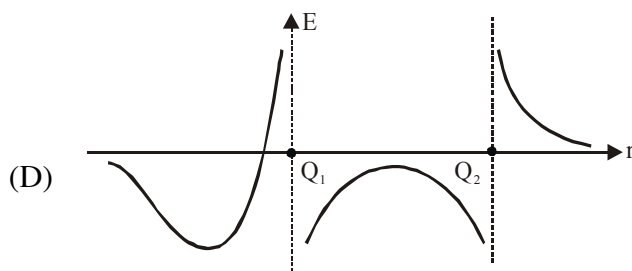
(P) $Q_1 < 0, Q_2 > 0$



(Q) $Q_1 > 0, Q_2 < 0$



(R) $|Q_1| > |Q_2|$



(S) $Q_1 < 0, Q_2 < 0$

(T) $|Q_1| = |Q_2|$

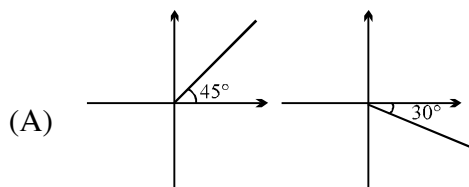
109. Column I shows graphs of electric potential V versus x and y in a certain region for four situations. Column II shows the range of angle which the electric field vector makes with positive x -direction.

Column I

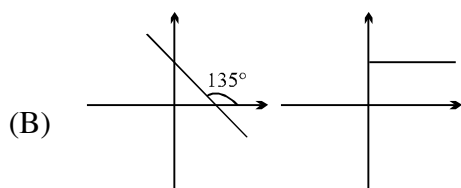
$V-x, V-y$

Column II

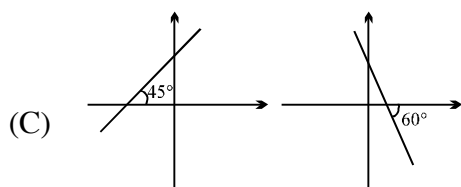
(Range of angle)



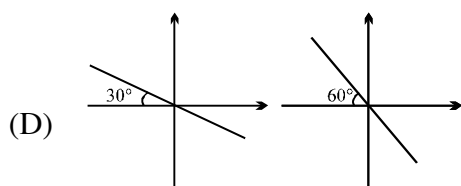
(P) $0 \leq \theta < 45^\circ$



(Q) $45^\circ \leq \theta < 90^\circ$



(R) $90^\circ \leq \theta < 135^\circ$



(S) $135^\circ \leq \theta \leq 180^\circ$

ES0157

110. A spherical metallic conductor has a spherical cavity. A positive charge is placed inside the cavity at its centre. Another positive charge is placed outside it. The conductor is initially electrically neutral.

Column I

(Cause)

Column II

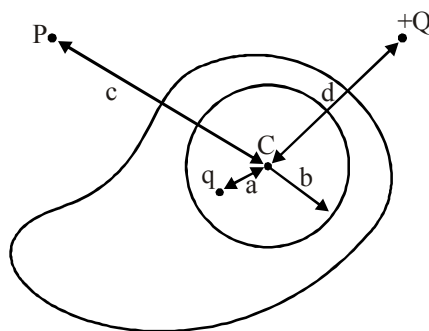
(Effect)

- (A) If outside charge is shifted to other position
(B) If inside charge is shifted to other position within cavity
(C) If magnitude of charge inside cavity is increased
(D) If conductor is earthed

- (P) distribution of charge on inner surface of cavity changes
(Q) distribution of charge on outer surface of conductor changes
(R) electric potential at centre of conductor due to charges present on outer surface of conductor changes
(S) force on the charge placed inside cavity changes

ES0158

111. In the shown figure the conductor is uncharged and a charge q is placed inside a spherical cavity at a distance a from its centre (C). Point P and charge $+Q$ are as shown. a, b, c, d are known.

**Column-I**

- (A) Electric field due to induced charges on the inner surface of cavity at point P
- (B) Electric potential due to charges on the inner surface of cavity and q at P
- (C) Electric field due to induced charges on the outer surface of conductor and Q at C
- (D) Electric potential due to induced charges on the inner surface of cavity at C

Column-II

- (P) zero
- (Q) non-zero
- (R) value can be stated with the given data.
- (S) value cannot be stated from the given data

ES0159

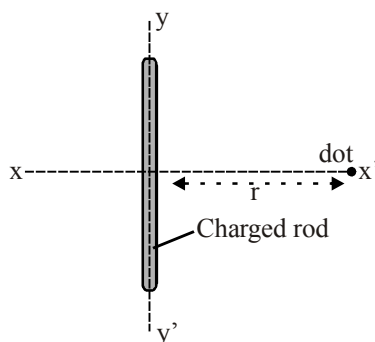
EXERCISE (O-2)

SINGLE CORRECT TYPE QUESTIONS

1. A square of side b centred at the origin with sides parallel to axes of x and y has surface charge density $\sigma(x, y) = \sigma_0 xy$ within its boundaries. Total charge on the square is
 (A) 0 (B) $\sigma_0 b^2$ (C) $2\sigma_0 b^2$ (D) $4\sigma_0 b^2$

ES0160

2. A uniformly charged rod is kept on y -axis with centre at origin, as shown. Which of the following actions will increase the electric field strength at the position of the dot?



- (A) make the rod longer without changing the charge
 (B) make the rod shorter without changing the charge
 (C) make the rod shorter without changing the linear charge density
 (D) rotate the rod about yy'

ES0161

3. A charged particle having some mass is resting in equilibrium at a height H above the centre of a uniformly charged non-conducting horizontal ring of radius R . The force of gravity acts downwards. The equilibrium of the particle will be stable

- (A) for all values of H (B) only if $H > \frac{R}{\sqrt{2}}$ (C) only if $H < \frac{R}{\sqrt{2}}$ (D) only if $H = \frac{R}{\sqrt{2}}$

ES0162

4. **Statement-1** : A positive point charge initially at rest in a uniform electric field starts moving along electric lines of forces. (Neglect all other forces except electric forces)

Statement-2 : Electric lines of force represents path of charged particle which is released from rest in it.

- (A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
 (B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
 (C) Statement-1 is true, statement-2 is false.
 (D) Statement-1 is false, statement-2 is true.

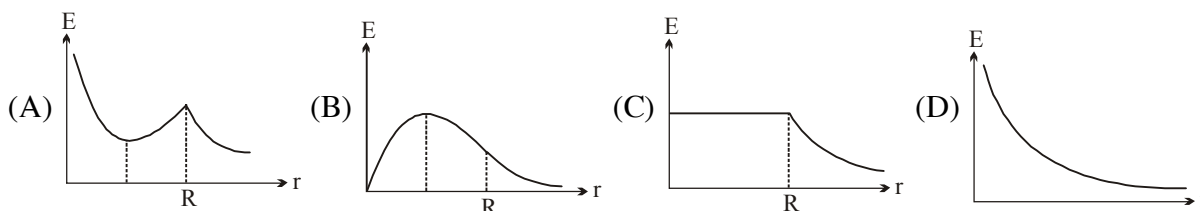
ES0163

5. A system consists of uniformly charged sphere of radius R and a surrounding medium filled by a charge with the volume density $\rho = \frac{\alpha}{r}$, where α is a positive constant and r is the distance from the centre of the sphere. The charge of the sphere for which electric field intensity E outside the sphere is independent of r is

(A) $\frac{\alpha}{2\epsilon_0}$ (B) $\frac{2}{\alpha\epsilon_0}$ (C) $2\pi\alpha R^2$ (D) αR^2

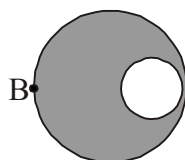
ES0164

6. A spherical insulator of radius R is charged uniformly with a charge Q throughout its volume and contains a point charge $\frac{Q}{16}$ located at its centre. Which of the following graphs best represent qualitatively, the variation of electric field intensity E with distance r from the centre.



ES0165

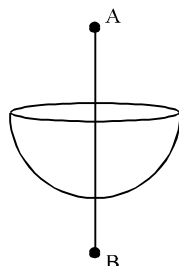
7. A positively charged sphere of radius r_0 carries a volume charge density ρ_E (Figure). A spherical cavity of radius $r_0/2$ is then scooped out and left empty, as shown. What is the direction and magnitude of the electric field at point B?



(A) $\frac{17\rho r_0}{54\epsilon_0}$ left (B) $\frac{\rho r_0}{6\epsilon_0}$ left (C) $\frac{17\rho r_0}{54\epsilon_0}$ right (D) $\frac{\rho r_0}{6\epsilon_0}$ right

ES0166

8. The diagram shows a uniformly charged hemisphere of radius R . It has volume charge density ρ . If the electric field at a point $2R$ distance above its center is E then what is the electric field at the point which is $2R$ below its center?



(A) $\rho R/6\epsilon_0 + E$ (B) $\rho R/12\epsilon_0 - E$ (C) $-\rho R/6\epsilon_0 + E$ (D) $\rho R/24\epsilon_0 + E$

ES0167